Pamoja tena ‘Together again’

African linguistics after COVID

Edited by

Vicki Carstens

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Contemporary African Linguistics

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Preface

It was a great pleasure to convene the 54th Annual Conference on African Linguistics at the University of Connecticut, Storrs, CT in June of 2023. This was the first in-person ACAL meeting since May of 2019, owing to the COVID-19 pandemic.

In addition to a full program of talks and posters we were delighted to welcome four excellent plenary speakers, two of whom flew in to join us from parts of Africa:

- Quentin Williams, University of the Western Cape:
  Toward an African Sociolinguistics of in difference: stance-taking on discrimination

- Eyasu Hailu Tamene, Addis Ababa University:
  Towards the official recognition of African sign languages

- Kathryn Franich, Harvard University:
  Co-speech gesture and prosodic structure: what African languages can tell us

- Jason Kandybowicz, CUNY Graduate Center:
  Escaping African 'Islands'

The twenty-two papers in this volume were among those presented at the conference. They provide descriptions and analyses of phenomena in a wide range of African languages. The foci include phonetics, phonology, morphology, syntax, semantics, and historical reconstruction. Their empirical basis is in many cases the outcome of painstaking fieldwork to document the myriad under-researched properties of Africa's linguistic diversity.

We gratefully acknowledge the units here at UConn which provided generous financial support for this conference. Without their funding, ACAL 54 could not have taken place.

- The Department of Linguistics

- The College of Liberal Arts and Sciences
- The Institute for Brain and Cognitive Sciences
- The Department of Anthropology

We are also very grateful to the efforts of our anonymous reviewers, without whom this proceedings volume would not have been possible.

The editors
June, 2025

Chapter 1

Labial-velar stop place identification in Igbo

Roslyn Burns[a] & Jason Shaw[a]
[a]Yale University

Previous research shows that simplex dorsals in the context of /i/-like vowels can be misperceived as coronal, a pattern which is more likely for voiceless segments than for voiced ones. We explore the effect of vowel context on the perception of labial-velar stops, pursing the hypothesis that the pattern observed for simplex velar stops will generalize to complex labial-velars. We opted for a cross-language perception study with Igbo participants listening to stimuli produced by a Nupe speaker. Our results show that while listeners mostly misperceived labial-velars as labials, voiced labial-velars are sometimes perceived as coronal before /i/-like vowels, similar to what is observed for simplex dorsals. Voiceless labial-velars, on the other hand, did not exhibit this pattern.

1 Introduction

Within West Africa, labial-velar plosives /g͡b/ and /k͡p/ often exhibit distributional restrictions based on vowel context. For example, in Dagbani (Niger-Congo: Gur), labial-velars are reported to become labial-palatals in the context before front vowels as shown in example (1) with [i] and [a].

(1)

Labial-velar		Labial-palatal		(Hudu 2014, Bennett 2014)
k͡páŋ́ (< k͡páŋ−gá)	‘guinea fowl’	c͡pín-î	‘guinea fowls.’	
g͡báŋ	‘squat’	ɟ͡bíŋm-g͡báŋ	‘elephant grass’	

Roslyn Burns & Jason Shaw. 2026. Labial-velar stop place identification in Igbo. In Vicki Carstens, Katherine Russell, Olawale Akingbade, Deborah Morton & Michael Diercks (eds.), *Pamoja tena ‘Together again’: African linguistics after COVID*, 1–19. Berlin: Language Science Press. DOI: ??

In (1), words sharing the same phonological form in the root undergo a vowel alternation to [i] in specific morphological contexts. When the vowel raises, it triggers palatalization on the preceding labial-velar resulting in a labial-palatal segment. The pattern of labial-velars becoming labial-palatal applies more broadly to all front vowels in Dagbani, not only [i] (Bennett 2014: 116).

In Fang (Niger-Congo: Bantoid), however, the high front vowel /i/ reportedly triggers labial-velars in noun class prefixes to become a simplex labial as shown in example (2).

	Labial-velar		Simplex Labial		(Mve et al. 2019, Burns 2023)
	k͡p–ú	'my cl.2'	p–í	'his/her cl.2'	
(2)	k͡p–ɛ́	'your sg. cl.2'			
	k͡p–ə́sə́	'our cl.2'			
	k͡p–ə́nə́	'your pl. cl.2'			

In (2), the class 2 prefix /k͡p–/ before vowel-initial possessive roots has two forms: [k͡p–] and [p–]. The [p–] allomorph only occurs before the high front vowel [i] which is known to be the most aggressive trigger of palatalization, yet instead of finding a labio-palatal in this environment, like in Dagbani, we find a simplex labial. Burns (2023) argues that an additional allomorph of /k͡p/ is found before consonant-initial roots, [bə–] < * k͡pi–. The reflex of this allmorph also has a simplex labial which developed from a labial-velar before a high front vowel.[1]

In addition to distributional restrictions and phonological alternations, like those shown above, upon conducting a preliminary survey of two major Niger-Congo languages spoken in Nigeria, we observed under-attestation of labial-velars in some vowel contexts. The static distributions of labial-velars across the lexicon suggest that labial-velars are less common in similar vowel contexts across Yoruba (Niger-Congo: Yoruboid, University Press 2001) and Igbo (Niger-Congo: Igboid, Echeruo 1998, Igwe 1999). Specifically, in Yoruba, labial-velars were underattested before both /i/ and /u/ (University Press 2001) whereas in Igbo they were under-attested before high front vowels (Echeruo 1998, Igwe 1999).

Production and perception biases often underlie the development of synchronic sound system patterns. Over time, these biases may lead to the diachronic development of specific distributions. Distributional patterns can be *phonologized*, resulting in a grammatical rule that generalizes to new forms. In this way, production and perception biases can serve as the historical source of phonological alternations of the sorts found in (1) and (2) or systematic gaps in the lexicon.

[1] In Fang, class prefix vowels neutralize to [ə] (Burns 2023: 22).

In this paper, we investigate whether a perceptual bias may play a role in the development of distributional restrictions on labial-velars across different vowel environments by means of an open-response cross-language listening task. The motivation for running an experiment testing the influence of vowel context on labial-velar stop-place identification can be found in past work, both experimental and typological, on *velar palatalization*. There are many definitions of velar palatalization. For our purposes, we define it as a contextual change whereby a velar comes to be pronounced further forward in the oral cavity, as a coronal.

Perceptually, velars are confusable with coronals in certain vowel contexts. Guion (1998) found that English listeners are more likely to confuse simplex velars with coronals in the context of /i/ than in the contexts of /ɑ/ and /u/. Misperception is even more likely if the consonant is voiceless (i.e. /k/) rather than voiced (i.e. /g/). Wilson (2006) reports an artifical grammar experiment, in which participants learned velar palatalization. Participants exposed to velar palatalization conditioned by lower front vowels generalized the process to higher front vowels. In contrast, those exposed to velar palatalization in the environment of the high front vowel did not generalize the process to lower front vowels. These results were interpreted as a substantive bias towards high front vowels as palatalization triggers.

The environments in which velars are likely to be misperceived as coronal correspond to the known typological patterns of palatalization with a primary place change (Bhat 1978, Bateman 2007). (3) summarizes the typological behavior of palatalization in simplex segments, as reported in Bateman (2007).

(3) a. Triggers
   i. Lower front vowels do not trigger palatalization unless higher front vowels do.
   ii. High back vowels do not trigger palatalization unless high front vowels do.

 b. Targets
   i. Voiced consonants do not undergo a primary place change unless voiceless consonants do.
   ii. Labials do not undergo a primary place change unless either coronals or dorsals do.

Notably missing from the typological behavior of targets in (3b) is what complex segments like labial-velars do. Labial-velars involve the coordination of at least two independent articulators (Ladefoged 1968): the lips and the tongue dorsum.[2] Burns (2023) hypothesizes that vowel-conditioned alternations for labial-velars have their basis in how labial-velars are misperceived in palatalizing contexts. Based on Connell's (1994) analysis of Ibibio (Niger-Congo: Cross-River) consonant place acoustics, Burns (2023) proposes that in palatalizing contexts, labial-velars may exhibit perceptual confusion with both labials and coronals. This contrasts with other vowel contexts, in which labial-velars are confused only with labials. This hypothesized pattern of confusion comes from the assumption that the tongue body will be fronted in palatalizing contexts, which has the effect of raising F2 in the formant transition, a typical cue to consonant place.

There are relatively few perceptual studies on West African languages. Some recent examples are Rose et al. (2023) and Ozburn et al. (to appear) on ATR harmony. We know of only one study on labial-velar perception, which reported data from just a handful of Yoruba speakers (Cahill 2006). Given the relative paucity of perceptual studies, our starting point in investigating labial-velars in different vowel contexts is centered around the typological literature on palatalization. To this end, our perceptual survey assess the questions in (4).

(4)
  a. Do labial-velars exhibit the most errors before /i/ (and progressively fewer as one moves away from the ideal palatalization trigger)?
  b. Do voiceless labial-velars exhibit more errors than voiced labial-velars?
  c. Do listeners believe that they are hearing a coronal in these contexts?
  d. Do the patterns of confusion for simplex components of labial-velars mirror the behavior of labial-velars?

If the answers to the research questions (4a-c) are affirmative and the same confusion patterns are observed with simplex velars, but not simplex labials, then it is likely that the synchronic patterns described above, e.g. (1) and (2), are due in part to the misperception of the dorsal constriction of the labial-velar.

The rest of this paper is organized as follows. Section 2.1 provides background information on the two languages involved in the cross-language study, Nupe and Igbo, focusing specifically on the differences in the consonant systems. Section 2.2 describes the Nupe materials used as stimuli. Section 2.3 describes the procedure that Igbo participants followed and our coding of the responses. Section 2.4 reports our methods of statistical analysis. Section 3 presents the findings of our study and how they compare to the questions presented in (4) above.

[2] Labial-velars fall into three major airstream mechanism types, two of which also involve coordinated laryngeal gestures.

Section 3.1 presents how vowel place influences the consonant place responses for labial-velars, thereby addressing (4a-c). Section 3.2 presents how vowel place influences consonant place responses for simplex labials and simplex velars and how these patterns compare to the labial-velars patterns, thereby addressing (4d). Section 3.3 summarizes how the Igbo listeners' responses for both simplex (labial and velar) and complex (labial-velar) segments align with the general palatalization typology outlined in (3) above. Section 4 discusses the implications of the findings for previous studies and future research. Finally, Section 5 closes by summarizing the overall findings of the study.

2 Methods

This section presents the methods for our cross-language study involving Nupe (Niger-Congo: Nupoid) stimuli and Igbo (Niger-Congo: Igboid) listeners. We opted for a cross-language design in order to increase the level of difficulty of the task. Although labial-velars are common across West African languages, the specific phonetic realization of these segments varies substantially across languages (Ladefoged 1968). We chose a somewhat less common language, Nupe, to be the language of the stimulus items to increase the level of difficulty of the task for Igbo listeners. Since we are interested in patterns of misperception, we needed to make the task difficult enough to induce errors. The difference between speaker language and listener language served this purpose.

2.1 Language background

The language of the stimuli, Nupe, has approximately 800,000 speakers (Lewis et al. 2019); the listener language, Igbo has approximately 29,000,000 speakers (Lewis et al. 2019). Both Nupe and Igbo have a full set of voicing contrasts across labial-velar, velar, and labial places: i.e. /k͡p, ɡ͡b, k, ɡ, p, b/. Nupe has a 5-vowel system and Igbo has an 8-vowel system with additional nasalized vowel contrasts (Ikekeonwu 1991, Uguru 2015, Moran & McCloy 2019).[3] In some varieties of Igbo, the labial-velars are reportedly plosives; in others, they are often realized as implosives /ɓ̥, ɓ/ (Moran & McCloy 2019, Ukaegbu 2023). The development of bilabial implosives from labial-velars is a common innovation in West Africa (Ladefoged 1968, Ponelis 1974) as the double articulation is sometimes coupled with a larynx lowering gesture and incomplete dorsal closure.

[3]In addition to these plosive contrasts, Igbo has aspiration contrasts, palatalization contrasts in labials, and labial-velarization contrasts in velars.

We make the assumption that Igbo listeners will hear the Nupe labial-velar as either a labial-velar with the same voicing or as a bilabial implosive with the same voicing.[4]

2.2 Materials

We developed a list of 96 disyllabic nonce words of the shape CVCV. These words were read by Nupe native-speaker linguist Ahmadu Kawu. The first C was either a labial, coronal, dorsal, or labial-velar plosive and varied in voicing (either voiced or voiceless). The first vowel was either /i, e, a, u/ and varied in tone (either low or high). The second syllable always began with a coronal and always ended in the vowel /a/ matching in tone with the first syllable. Kawu read these words in two Nupe carrier phrases outlined in (5).

(5) a. Wun ganan ___ be.
3SG speak ___ again.
'He said ___ again.'

b. Wun ka ___ be.
3SG write ___ again.
'He wrote ___ again.'

Kawu recorded the phrases on a Zoom H4n digital recorder at a sampling rate of 44.1 khz.

Recordings were annotated in Praat (Boersma & Weenink 2019) and aligned using the Montreal Forced Aligner (McAuliffe et al. 2017) with the English pronunciation model and Nupe dictionary created by the researchers. All forced alignment was manually checked and segment boundaries were moved to the nearest zero-crossing. Of the 192 target utterances, we selected 96 tokens for inclusion in the experiment based on (a) vowel peripherality (avoiding heavily centralized utterances), (b) duration (avoiding abnormally long or short tokens), (c) intensity (avoiding abnormally soft tokens), and (d) clarity.[5] After selecting the tokens for inclusion, we extracted the initial CV of the nonce word from the stop burst to the offset of the vowel. This method removed prevoicing and stop

[4]While voiceless implosives exhibit prevoicing, they are still differentiated from voiced implosives in having a 20–50 ms period of silence prior to the prevoicing (McLaughlin 2005: 207–209). As discussed in Section 2.2, based on how the stimuli for the experiment were processed, prevoicing cannot be a cue that listeners use to interpret the behavior of plosives in this study.

[5]Linear models evaluating the effect of carrier phrase on formant frequencies returned no significant effect of carrier phrase.

closures from the stimuli. Kawu produced a labial-velar glide instead of a plosive in the token /gu/, i.e. [wu] instead of [gu]. For this item, we excised the portion from the onset of glide voicing to the offset of the vowel. After extracting the initial CV, which was always less than 140 ms in duration, amplitude was normalized across all tokens in Praat to 70 dB.[6]

2.3 Procedures

51 Igbo listeners were recruited to participate in an experiment hosted on the Gorilla platform (https://gorilla.sc/). All consent forms and prompts were written in English by the authors and then translated from English to Igbo by Dr. Carol Anyagwa of the University of Lagos. After consenting, participants were instructed to listen to a steady tone (75 Hz, 70 dB) and adjust their speakers to a comfortable volume. After this, they were instructed to listen to the normalized CV stimuli (described in Section 2.2) and type a response indicating what they heard. Participants completed 7 practice trials with stimuli not used in the experiment, and then completed 96 randomized experimental trials. After completing the task, they were asked to fill out a demographic survey.

Responses were downloaded in a tab-delineated file and checked for interpretability (n = 4800). Answers with no response (n = 111), commentary about not being able to hear (n = 3), and uninterpretable orthographic representations (e.g. <q>, <c>, etc., n = 18) were excluded from analysis. One participant was excluded for failure to complete the task. In total, 4668 usable tokens were coded for voicing and response place based on the orthographic representation. The response places were coded as labial, coronal, dorsal, labial-velar, or neither. (6) shows the break-down of how each orthographic symbol was coded for place.

(6) Labial: f, v, m, b, p
Coronal: t, d, ch, l, r, y
Dorsal: k, g
Labial-velar: w, kp, gb, kw, gw, vb[7]
Neither: Vowel-initial, h-initial

Errors were coded based on whether the place represented in the response matched the Nupe stimuli place features. Responses to /gu/ were coded as cor-

[6]We performed acoustic analysis for stop place cues including investigating the maximum frequency at maximum amplitude in burst, F2 at 7% of the CV duration, and the change in F2 from 7% to 14% of the CV duration. For the results of this analysis see Burns & Shaw (2023).

[7]We treated <vb> as a typographical error for <gb> given the proximity of the <g> and <v> keys.

rect if Igbo listeners wrote either a velar or a labial-velar as Kawu produced [w] instead of [g] in these contexts (see Section 2.2).

2.4 Analysis

In order to evaluate the statistical reliability of the trends in the data, we fitted logistic regression models with mixed effects. We first investigated error rate. The dependent variable in this case was consonant place identification accuracy, coded as 0 for incorrect and 1 for correct. Our models contained a random intercept for participant. Our fixed effects coded for properties of the stimulus items: CONSONANT PLACE {labial, velar, labial-velar}, CONSONANT VOICING {voiceless, voiced}, and VOWEL {/a/, /e/, /i/, /u/}. We also tested for two-way interactions between CONSONANT PLACE * VOWEL and between CONSONANT VOICING * VOWEL and for the three-way interaction between CONSONANT PLACE * CONSONANT VOICING * VOWEL.

We evaluated statistical significance through nested model comparison, via ANOVA, and consideration of the Akaike Information Criterion (AIC). We considered a fixed factor to be statistically reliable if it both explained significantly more variance than a model that lacked it and lowered the AIC. In the case of significant interactions, we ran post-hoc models on each level of our fixed factors. For example, separate models with vowel as a fixed factor were fit to each combination of place and voicing, e.g. a model for /g͡b/, /k͡p/; /g/, /k/; /b/, /p/; and /d/, /t/. All of our fixed factors were treatment-coded. To assess the effect of each level of a fixed factor, such as VOWEL, we re-ordered the levels so that a different level served as the intercept. The level of a fixed factor was taken to be statistically reliable if it had a p-value equal to or below 0.05.[8]

In addition to error rate, we also fit logistic models assessing the probability of a coronal response, with coronal coded for 1 and other responses coded for 0. These models followed the same structure and procedure as those fit to error rate (described above).

3 Findings

This section presents the results of our experiment. The presentation follows the research questions in (4). We first examine research questions (4a-c) for labial-velars (Section 3.1) and then (4d) for the simplex components of the labial-velars (Section 3.2). We close the section with a summary of the findings (Section 3.3).

[8]P-values are based on an F-test using the Satterwaithe approximation method in the LMERTEST package.

3.1 Labial-velar error rates and consonant place responses

Among responses to labial-velar stimuli (/g͡b/ = 1165, /k͡p/ = 1171), /g͡b/ showed fewer errors than /k͡p/ (/g͡b/ errors = 372, /k͡p/ errors = 739). Figure 1 shows the error rates for voiced labial-velars (left panel) and voiceless labial-velars (right panel) in each vowel context. Error bars represent the standard error calculated across speakers. Statistical significance, determined according to the procedure described in Section 2.4, is indicated with an asterisk.

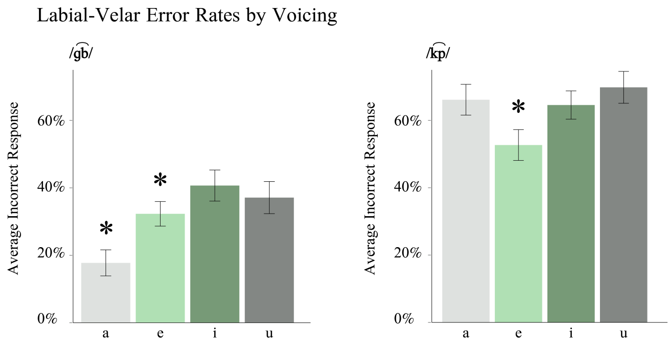

Figure 1: Average error rates for labial-velar stimuli

As shown in the left panel of Figure 1, for /g͡b/, /i/ has the highest error rate. Among the front vowels, there is a significant error rate decreases from /i/ to /e/ and then from /e/ to /a/ as indicated by the asterisk. Thus, among the front vowels, there is a gradual decrease in error rates with /i/ exhibiting the most errors and fewer errors as the front vowel becomes lower. Within the high vowels, /i/ has a numerically higher error rate than /u/ but this difference was not statistically reliable.

Among responses for /k͡p/, shown in the right panel of Figure 1, the difference in error rate between /i/, /a/, and /u/ was not statistically reliable. /e/, however, had a reliably lower error rate compared to all other vowels. In this respect, only /g͡b/ exhibits error patterns consistent with the previous literature on palatalization.

Turning next to the outcomes of erroneous responses, we find that both voiced and voiceless labial-velars are most frequently confused with simplex labials. Figure 2 presents the percentage of error types by voicing (separate panels) and vowel context (x-axis).

In Figure 2, we can see that simplex labial responses comprise over 60% of the errors in both /g͡b/ (n = 257) and /k͡p/ (n = 601) across all vowel contexts. For /g͡b/,

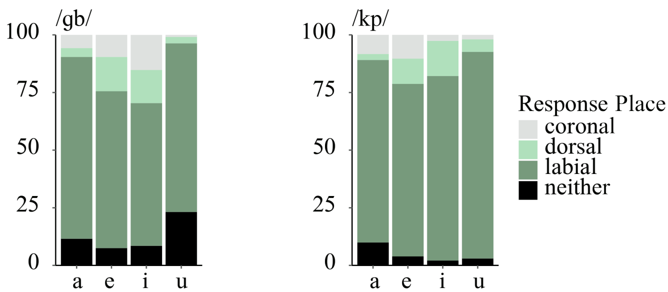

Figure 2: Proportions of incorrect labial-velar response types

the /i/context triggers a higher percentage of coronal errors (n = 31) than either /e/ or /a/, a difference which was statistically significant. There is also a difference in coronal responses between /e/ and /a/, although this difference was not statistically reliable in our sample. Similarly, coronal responses for /i/ make up a higher percentage of incorrect responses than coronal responses for /u/ (also statistically significant). For /k͡p/ coronal errors (n = 41), /i/ and /u/ pattern together (i.e. are not statistically different) with the lowest incidence of coronal errors. The other two vowel contexts, /a/ and /e/, also pattern together (not significantly different from each other), showing significantly more coronal errors than /i/ and /u/. This suggests that while coronal errors for /g͡b/ follow the typological patterns described for palatalization (i.e. lower front vowels do not trigger palatalization unless higher front vowels do, high back vowels do not trigger palatalization unless high front vowels do), /k͡p/ errors do not.

To follow up the differential effect of vowel context on voiced vs. voiceless labial-velars, we looked at the error types based on the listener's perception of voicing. This perspective on the data revealed a striking pattern. Figure 3 below shows the place errors of labial-velars grouped by the stimulus voicing property (columns) and the listener's accuracy in voicing perception (rows).[9]
As shown in the first row, if listeners hear /g͡b/ and correctly perceive that the consonant is voiced (n = 304), the pattern observed is the one wherein /i/ has the highest percentage of coronal responses and there are progressively fewer as the vowel moves away from /i/ (top right panel). If listeners hear /k͡p/ and correctly perceive that the consonant is voiceless (n = 603), /i/ exhibits no preference for coronal responses (top left panel). Looking at the second row, however, if listeners hear /k͡p/ and perceive it as voiced (i.e. /g͡b/, n = 136), the pattern

[9]These were not tested for statistical significance due to low occurrences across the different categories.

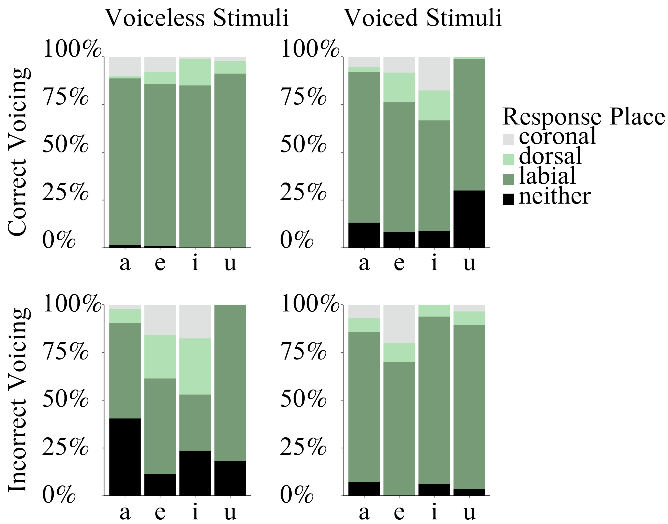

Figure 3: Proportions of incorrect labial-velar response types based on voicing perception

emerges wherein /i/ receives the highest percentage of coronal responses and other vowels exhibit progressively fewer as they move away from this ideal trigger of palatalization (bottom left panel). When listeners hear /g͡b/ and perceive it as voiceless (i.e. /k͡p/, n = 68) there is no preference for coronal responses before /i/ (bottom right panel). These findings suggest that the perception of voicing is important to how Igbo listeners process labial-velars in palatalizing contexts. Interestingly, the effect of perceived voicing patterns differently than expected from past work on the role of voicing in velar palatalization. We return to this result in the discussion (Section 3.3).

3.2 Simplex consonant errors

We close the presentation of the results with errors among simplex labial and simplex velar stimuli. Figure 4 shows the error rate for voiced (left panel) and voiceless (right panel) simplex stimuli by vowel context. The top row shows velar stimuli and the bottom row shows labial stimuli.

Statistically, for simplex /g/ (responses = 390, errors = 156), /i/ and /e/ pattern together with the fewest overall errors. /a/ had significantly more errors than other front vowels, /e/ and /i/. /u/ had significantly more errors than /a/, the most errors overall. For simplex /k/ (responses = 390, errors = 94), /i/ and /e/

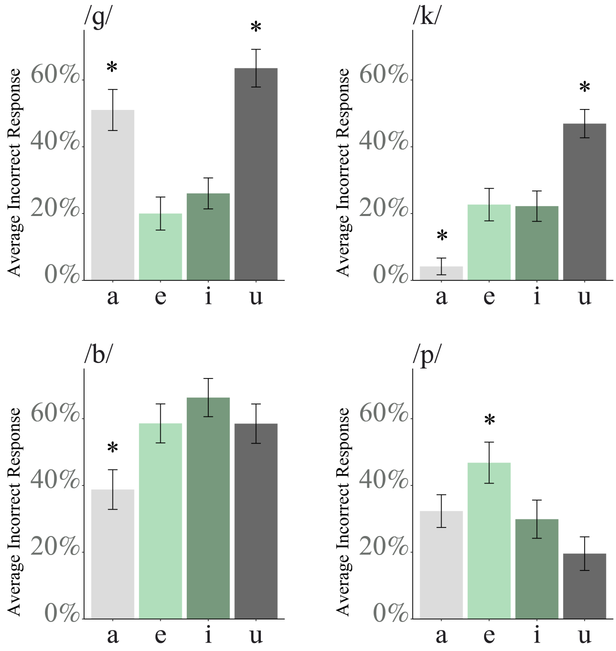

Figure 4: Simplex dorsal and simplex labial error counts

both have more errors than /a/, a difference which is statistically significant. /u/ has significantly more errors than the other contexts. For simplex /b/ (responses = 386, errors = 214), /a/ has the lowest error rate while /i/, /e/, and /u/ pattern together with a higher error rate. For simplex /p/ (responses = 387, errors = 124), /e/ has the highest error rate, a difference which was statistically significant. /i/ patterns together with /a/ and /u/ with lower error rates. In sum, of the four simplex stops examined, only voiced labials showed high error rates for /i/, which is typologically the most aggressive trigger of palatalization. The voiceless labial has high error rates for /e/, but no other vowel. Among velars, either /a/ or /u/ exhibit the highest error rates.

In order to interpret how the error rate patterns relate to the palatalization hierarchy described in (3), we need to know in particular how coronal responses are distributed across the different place, vowel, and voicing combinations. Figure 5 shows the rate of different place responses among the errors for simplex velars and simplex labials.

For simplex /g/ coronal errors (n = 49), all front vowels pattern together statistically having higher rates of coronal responses than /u/. For simplex /k/ coronal

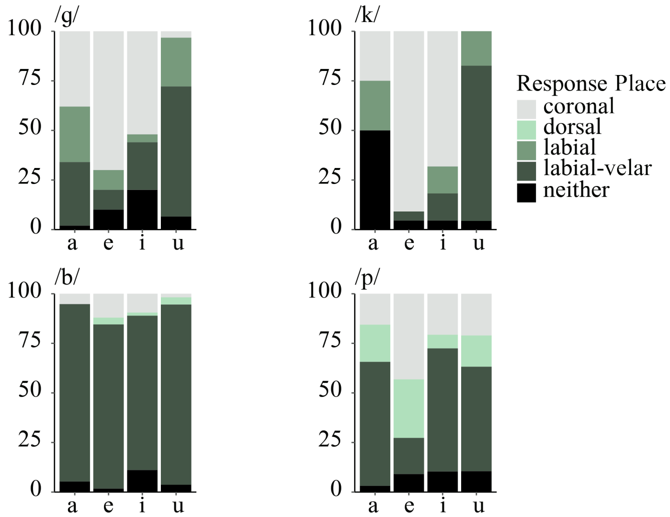

Figure 5: Proportions of incorrect simplex dorsal and simplex labial response types

errors (n = 36), /i/ and /e/ statistically exhibit the most coronal responses. There are significantly fewer coronal responses before /a/ and /u/ . In this respect, both /g/ and /k/ follow the palatalization hierarchy, but in different ways. For both, lower front vowels do not condition the percept of coronals to the exclusion of higher front vowels. This is consistent with the palatalization hierarchy. Additionally, the high back vowel does not condition the percept of coronals to the exclusion of the high front vowel, which is also consistent with the palatalization hierarchy. The difference between the two simplex velars is that coronal responses are more balanced across the front vowels for the voiced velar /g/ than for the voiceless velar /k/, which we did not expect.

When we look at the error pattern for simplex labials, the other closure component of the labial-velar, we find that with simplex /b/ coronal errors (n = 16), there is statistically no preferred vowel environment. For simplex /p/ coronal errors (n = 34), there were statistically more coronal responses in the context of /e/ than in the other vowel contexts. This indicates that simplex labials do not follow the patterns that have been described for the palatalization hierarchy outlined in (3), but simplex velars do.

3.3 Summary

The previous typological literature on palatalization identified target place, target voicing, and trigger vowel as important factors in observing palatalization (see example 3). Our investigation of how complex dorsals (i.e. labial-velars) factor into the known typology indicates that while vowel context is important to how listeners perceive labial-velars, there is a split in target voicing properties which runs counter to the known typological behavior. That is, the voiced labial-velar /g͡b/ is more consistent with the palatalization hierarchy than /k͡p/. This behavior appears to be connected specifically to how listeners perceive voicing, as opposed to the actual voicing of the segment.

When comparing complex dorsals (i.e. /g͡b/ and /k͡p/) to their simplex closure components (i.e. /g/, /k/, /b/, /p/), we find that /g͡b/, /g/, and /k/ statistically exhibit behaviors consistent with the palatalization hierarchy concerning coronal response rates. Neither /k͡p/, /b/, nor /p/ exhibit error patterns consistent with the literature on palatalization triggers. While /b/ exhibited no preferences by vowel for coronal responses, both /k͡p/ and /p/ showed a preference for coronal responses before /e/ but never /i/ (with /k͡p/, /e/ and /a/ patterned together).

4 Discussion

We sought to understand if misperception of labial-velars in certain vowel contexts mirrored patterns of underattestation in those same contexts. We tested this hypothesis through the lens of velar palatalization. Underattestation of labial-velars occurred in the same environments which have been reported to trigger changes from simplex dorsals to coronals, in both experimental and typological literature.

The results of our cross-language Nupe-Igbo study found that all labial-velars are associated with high confusability with simplex labials regardless of vowel context. When looking specifically at confusions of labial-velars as coronal consonants, we found patterns reminiscent of velar palatalization. These findings provide some support for Burns' (2023) hypothesis that labial-velars may have two outcomes in palatalizing contexts. In addition to becoming coronal in these contexts, they may also simplify to simplex labials. The latter outcome may develop to avoid confusing lexical items with labial-velars for lexical items with coronals, as labial-velar confusability with coronals increases specifically in palatalizing contexts. A change from a labial-velar to a coronal would increase contrast with a simplex labial in a palatalization context. As simplex labials are already confusable with labial-velars independent of vowel context (and vice versa), there is not much that is lost from an identification standpoint when a labial-velar is replaced by a simplex labial, but there is an added benefit in that simplex labials are not notably more confusable with coronals in palatalizing contexts. Thus, the alternation of labial-velars with labials in palatalizing contexts may support system-wide maintenance of contrast.

Although our study found that vowel contexts known to condition velar palatalization also conditioned the percept of a labial-velar consonant as coronal, this pattern surfaced for the voiced labial-velar but not the voiceless labial-velar, in an apparent reversal of the voicing hierarchy observed among simplex velars in previous literature. While Igbo perception of simplex velars did exhibit responses that are consistent with the palatalization hierarchy, simplex /g/ appears to exhibit more environments of palatalization (i.e. all front vowels) than simplex /k/. This follows the same apparent voicing reversal as observed for labial-velars. It should be noted that neither simplex labial conformed to the known behaviors of the typological literature on velar palatalization. This is in line with the target place hierarchy, wherein either coronals or dorsal may undergo the process of palatalization to the exclusion of labials, but the inverse does not occur. Interestingly, /k͡p/ appeared to pattern most closely with /p/. The split in the behavior of complex segments, where /ɡ͡b/ patterns with velars and /k͡p/ patterns with labials, leads to the question of whether or not there is some underlying motivation that would lead us to expect complex classes to exhibit disunity in which simplex component of the complex gesture that they pattern with.

Another key finding from our study, regarding the voicing hierarchy reversal, is that it seems to depend on listener perception of voicing. When Igbo listeners misperceived a Nupe voiced labial-velar stimuli as voiceless, the patterns of consonant place misidentification across vowel contexts matched the patterns found for voiceless labial-velar stimuli (3). The same was true for Nupe voiceless labial-velar stimuli voicing misperception. Voiceless labial-velar stimuli misperceived as voiced showed the patterns of consonant place misidentification across vowel contexts found for voiced labial-velar stimuli. This pattern could possibly relate to the distribution of voiced vs. voiceless labial-velars across vowel contexts in the lexicon. A cursory examination of these patterns, based on counts in two Igbo sources (Igwe 1999, Echeruo 1998), showed that there were differences by voicing. Voiced labial-velars occurred less often before /i/ than voiceless labial-velars. The same was true for the /e/ environment; voiced labial-velars occur less often before /e/ than voiceless labial-velars. Possibly, these distributional facts influenced listener expectations in our study. Listeners that perceived the vowel /i/

or /e/ and voicing may be biased against responding labial-velar, since voiced labial-velars are under-attested in this environment.[10]

One possible direction for future research is consideration of variation in laryngeal properties found across Nupe and Igbo. Many varieties of Igbo have a glottal gesture that is coupled with the oral gestures (see the discussion of phonemic system alignment in Section 2.1). In some varieties, the pairing of these articulatory gestures is such that /g͡b/ > /ɓ/ and /kp/ > /ɓ̥/. The primary cue differentiating voiceless implosives from voiced implosives is a short period of silence during the closure (Ladefoged 1976 as cited in McLaughlin 2005). This is different from the Nupe cues to voicing in labial-velars, which lack voicing during closure, at least in our materials, and contain robust cues to voicing in the burst. The perceptual stimuli we used in this study lack closure information that would allow Igbo listeners to identify the difference between voiced and voiceless implosives using the primary cue of voicing duration during the closure. Our stimuli did, however, retain burst information. For Igbo listeners with the voiced and voiceless implosive distinction, the burst information may differentiate implosives (both voiced and voiceless) from non-implosives. Thus, a cue to voicing in our materials may be reinterpreted as a cue to ingressive airstream for some Igbo users. Any reinterpretation of burst cues for voicing might interact with the effect of vowel quality on stop-place perception because of the distribution of voiced and voiceless labial-velars across vowels, as discussed above.

A possible implication of the differing cues to voicing is that voiceless /k͡p/ in Nupe could be perceived by certain Igbo listeners as /p/. On this note, there were some similarities in how the effect of vowel context patterned across /k͡p/ and /p/. For these two consonants, place identification accuracy was similar across /i/, /a/, and /u/ with /e/ being statistically different. /e/ showed higher place identification accuracy for /k͡p/ (Figure 1) and lower place identification accuracy for /p/ (Figure 4). Unfortunately, given that simplex dorsals also exhibit reversal of the voicing hierarchy, further studies, both perceptual and articulatory, are needed to determine the exact motivations underlying all voicing reversals. As a first step, it may be informative to conduct a version of our study which provides cues to voicing during closure to see if the voicing hierarchy reversals persist even with more robust cues to voicing. Past work has shown that voicing during closure is an important cue to laryngeal contrasts in some languages with labial-velars (Grawundera et al. 2011) and, moreover, excising this cue to voicing from the speech signal can lead to stop place misidentification (Cahill 2006).

[10] An anonymous reviewer noted that many West African languages lack labial-velars before rounded vowels likely due to their development from *KUV > K͡PV.

5 Conclusion

We presented the results of a cross-language perception study investigating how vowel context affects stop place perception. In an open response task, Igbo listeners heard eight Nupe stops, /k͡p, g͡b, k, g, p, b/, in four different vowel environments, /a, e, i, u/. Our primary theoretical motivation was a test of whether misperception of labial-velars as coronal would follow the three implicational relationships identified for velar palatalization. Results indicated that voiced labial-velars do indeed follow the velar palatalization hierarchy. There are more coronal percepts of voiced labial-velars in the environment of /i/ followed by /e/ and then /a/. Voiceless labial-velars, on the other hand, did not show this same pattern. This was surprising because it had been claimed that, for simplex velars, voiceless stops are more likely to palatalize than voiced stops. However, simplex velars in Igbo also flouted this generalization about voicing with /g/ exhibiting more coronal errors than /k/. Additionally, the coronal errors for /g/ were more balanced across all front vowel contexts than for /k/, which was also unexpected. A complete account of the voicing hierarchy reversals reported here requires future research, but our results indicate that voicing perception itself plays a key role in conditioning how vowel context impacts stop place perception.

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Chapter 2

Vowel alternations in Dagaare number marking

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Dagaare (Mabia) has two numbers, singular and plural, but nouns form the singular and plural in a variety of ways depending on their declension class. In this paper, I provide novel data from Northern Dagaare to investigate the vowel alternations exhibited by three of the classes: -r marked singulars, -bɛ marked plurals, and -r(ɪ) marked plurals. Following previous work on Central Dagaare (Anttila & Bodomo 2009, 2019), I conclude that the vowel alternations in the -r singulars are in fact true suffixes. However, Anttila & Bodomo (2009, 2019) propose that the vowel alternations in the -r(ɪ) plurals are due to epenthetic vowels inserted to meet a word minimality requirement. I argue that this explanation does not hold in Northern Dagaare. Rather I propose that the singular suffix within the -r(ɪ) plural class is an empty mora. Additionally, I unify the -bɛ and -mɛ plural classes in Northern Dagaare and provide a preliminary analysis to explain the compensatory lengthening that occurs in the former but not the latter.

1 Introduction

The Dagaare (Mabia)[1] language is made up of a dialect continuum which stretches from Northern Ghana to southern Burkina Faso. In this paper, I present novel data from the Nandome dialect of Northern Dagaare, which exhibits vowel alternations between several singular and plural nouns. Anttila & Bodomo (2009, 2019) found that similar patterns in Central Dagaare are driven by metrical foot

[1] Dagaare is often also called Dagaari, Dagara, Waale, and Birifor. The Mabia language family is sometimes referred to as Gur.

Cassandra Caragine. 2026. Vowel alternations in Dagaare number marking. In Vicki Carstens, Katherine Russell, Olawale Akingbade, Deborah Morton & Michael Diercks (eds.), *Pamoja tena 'Together again': African linguistics after COVID*, 21–41. Berlin: Language Science Press. DOI: ??

structure, minimal prosodic word requirements and other language general constraints. I investigate whether this analysis can be extended to data from Northern Dagaare. I argue that despite their success capturing the variation in Central Dagaare, this question in the larger Dagaare language is far from settled. While the Anttila and Bodomo analysis makes the correct predictions for the -r singulars, it does not accurately explain the variation seen in the -r(ɪ) plurals and it does not address vowel length alternations in the -bɛ plurals.

I maintain a simple affixation account of the -r singulars put forth by Anttila & Bodomo (2009), but I propose that within the -r(ɪ) plural declension class, the singulars receive a suffix consisting only of an empty mora. This mora is then filled by linking it to the features of the preceding vowel, resulting in vowel lengthening. Then language general constraints adopted from Anttila & Bodomo (2009) sometimes cause diphthongization. In the -bɛ plurals, I argue that there is compensatory lengthening of the stem vowel in the plural when the stem final consonant is deleted. When the final consonant is a nasal it coalesces with the [b] in the suffix to form an [m]. This [m] is ambisyllabic and thus no compensatory lengthening is triggered.

The paper is organized as follows. Section 2 introduces the Northern Dagaare dialect and its relevant linguistic features, Section 3 presents data of number marking, Section 4 works towards an analysis, and Section 5 concludes.

2 Northern Dagaare

2.1 Language background

Dagaare is an understudied and under-documented Mabia language (Niger-Congo) spoken in northwestern Ghana and southern Burkina Faso by between 1.5 and 2 million speakers (Ali et al. 2021, Lewis 2009). The Dagaare language is made up of a dialect continuum that, roughly speaking, can be divided into four main dialect groups – Northern, Southern, Western and Central – which show substantial variation (Bodomo 1997, Angsongna & Akinbo 2020, Ali et al. 2021). The map in Figure 1 shows the locations in which the dialect groups are spoken. This paper focuses on Nandome Dagaare, an under-documented sub-dialect of Northern Dagaare spoken in and around Nandom, Ghana.

A team of researchers at Georgetown University[2] and the University of Maryland[3] has spent over a year working with a native speaker to document and

[2]Michael Obiri-Yeboah, Sophie Migacz, and Jack Pruett

[3]Cassandra Caragine

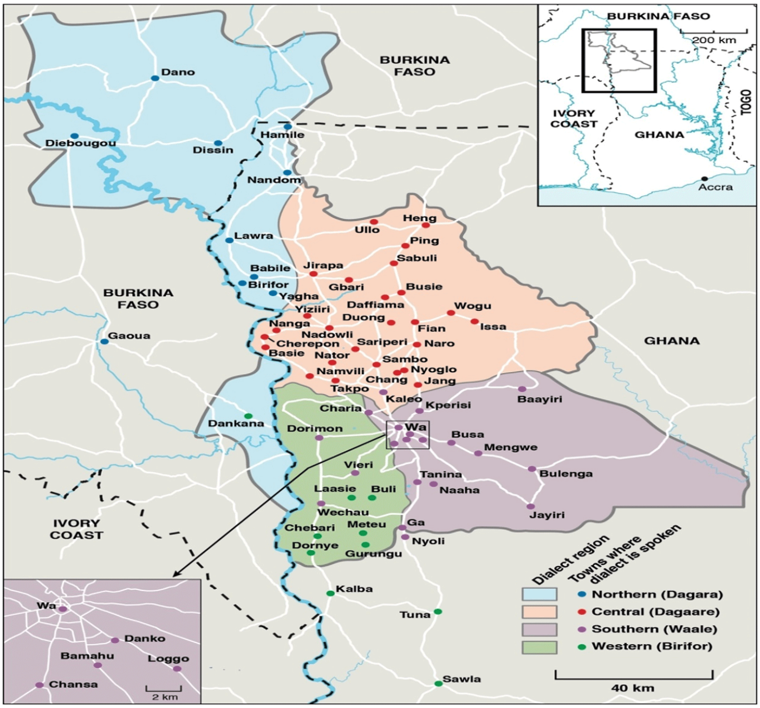

Figure 1: Map of the area where Dagaare is spoken (Angsongna & Akinbo 2020)

analyze this understudied dialect. The speaker is a male (27), who grew up in Kumasi, Ghana and speaks the Nandome dialect of Dagaare. Additionally, he speaks Twi, English, French, and Spanish. All interview sessions were recorded (about 40 hours) with a Zoom H4n Pro handheld recorder. Data was transcribed and uploaded to Twisted Tongues database management software. Researchers used Praat (Boersma & Weenink 2023) for confirmation and correction of initial transcriptions. This resulted in 68 singular-plural pairs in Northern Dagaare.

2.2 Phonological background

Northern Dagaare (ND) has a nine-vowel system consisting of four +/-ATR (Advanced Tongue Root) pairs and one low central vowel, Table 1. The low central vowel /a/ is likely under-specified for ATR; evidence for this is presented later in this section. Vowels also contrast in length ([nà] 'this' v. [nàà] 'chief'), tone ([báá] 'dog' v. [bàà] 'river'), and nasality ([sàà] 'rain' v. [sã̀ã̀] 'father').

Table 1: Dagaare Vowel Chart

	Front		Back	
	+ATR	-ATR	+ATR	-ATR
High	i	ɪ	u	ʊ
Mid	e	ɛ	o	ɔ
Low		a		

Additionally, ND exhibits ATR vowel harmony. In the nominal domain, this process is progressive. The low central vowel /a/ is invisible to this process; it does not undergo, trigger, or block vowel harmony. Thus, it is likely that /a/ is underspecified for ATR. For further discussion of vowel harmony in Dagaare see Bodomo (1997) and Ali et al. (2021).

Northern Dagaare has a (C)V(V)(C) syllable structure. It allows optional onsets and optional codas as well as short vowels, long vowels, and diphthongs. Complex onsets are currently unattested, and there is only one word in our data set which contains a complex coda: [tʊ̀βr] 'ears'. Syllable weight and moraic structure will be considered in Section 4.3.

3 Number marking

The Dagaare nominal system has two numbers: singular and plural. Dagaare is noted to have an 'inverse' or 'polarity' system, in which a single morpheme can act as either a singular or plural marker (Grimm 2012, 2021, Anttila & Bodomo 2009). As Table 2 shows, -r can mark either singulars or plurals depending on the declension class of the noun.

Table 2: Polarity Marking

Singular	Plural	Gloss
nóbíè	nóbí-r	'finger'
jí-r	jíé	'house'

However, -r is not the only number marker. There are several ways that number can be expressed on the noun depending on its declension class. The declension classes of Dagaare have been categorized in several ways. I use the six classes shown in Table 3, but the exact categorization is tangential to the current paper. Additionally, the labels used for each class have not be standardized and are provided for referential simplicity only. As noted in Section 2.2, vowel harmony is an active process in the language. The citation forms of each of the suffixes is given in the -ATR form as this is what occurs with stems whose only vowel is /a/, which is underspecified for ATR. The specific rules governing which number markers are used for a particular noun are outside the scope of this paper. For further discussion of this topic see Pruett (n.d.).

Table 3: ND Declension Classes

	Singular	Plural	Gloss
-r singulars	jí-r	jí-é	‘house’
-r(ɪ) plurals	bàà	bàà-r	‘river’
-bɛ plurals	dɛ́b̚	dɛ̀ɛ̀-βɛ̀	‘man’
l nominals	jéèpùlè	jéépùlì	‘sister’
-mɪnɛ plurals	sȁȁ	sȁȁ-mìnɛ́	‘father’
irregulars	wáàb	wɪ́ɪ́r	‘snake’

The declension classes and a representative example for each are included in 3. The first three of these classes (-r singulars, -r(ɪ) plurals, and -bɛ plurals) exhibit some sort of vowel alternation. Sections 3.1-3.3 will lay out the patterns seen in these noun classes in Northern Dagaare and compare them to their correspondents in Central Dagaare (CD).

3.1 -r singulars

The -r singular declension class is one of the classes exhibiting polarity marking with -r(ɪ). As can be seen in Table 4, this class overtly marks both the singular and plural with a suffix. An -r is suffixed in the singular and -ɛ is suffixed in the plural. The -ɛ plural suffix undergoes ATR vowel harmony to agree with the stem. When the -ɛ suffix results in vowel hiatus, there is no rearticulation. When the qualities are different for the two vowels, the phonetic yield is a diphthong, and when the two vowels have the same quality (eg. when -ɛ is added to an [ɛ] or [e] final stem), a long vowel is produced.

The left 2 columns of Table 4 provide novel singular/plural pairs in the -r singular declension class of Northern Dagaare. The right two columns provide the corresponding word in Central Dagaare.[4] Unless otherwise noted, here and throughout the paper, the Central Dagaare data come from Ali et al. (2021).

Table 4: -r singulars in Northern Dagaare and corresponding nouns in Central Dagaare. Central Dagaare data from Ali et al. (2021)

Northern Dagaare			Central Dagaare	
Singular	Plural	Gloss	Singular	Plural
bàpɛ́-r	bàpɛ́-ɛ́	‘male friend’	bábí-rì	*babi-e (‘best friend’)
g͡bɛ́-r	g͡bɛ́-ɛ́	‘leg’	*g͡bɛ́-rì	*g͡bɛ-ɛ
tʊ́β-r	tʊ̀β-ɛ́	‘ear’	tɔ̀ɔ̀-rí	*tobo
jí-r	jí-é	‘house’	jí-rì	*ji-e
dɛ̀-r	dɛ̀-ɛ̀	‘stair’	–	–
bȕbí-r	bȕbì-é	‘seed’	bʊ́mbí-rì	*bombi-e
mʊ̀ndɛ́-r	mʊ̀ndɛ́-ɛ́	‘spider’	bádɛ́-rí	*badɛr-rɪ
màrbí-r	màrbí-è	‘star’	ŋ͡máríbí-rì	*ŋ͡marebi-e

As can be seen in Table 4, the the singular -r suffix in Northern Dagaare always corresponds to a -rɪ suffix in Central Dagaare. ‘Ear’ and ‘spider’ appear to be irregular plurals in CD, but the remaining CD plurals are marked with a corresponding suffix to that used in ND.

3.2 -r(ɪ) plurals

The most common of the declension classes adds -r or -rɪ in the plural (18/68 noun pairs). Exactly what conditions the final [ɪ] is still unknown.[5] Other dialects

[4]In the original sources standard Dagaare orthography was used instead of IPA. For consistency and ease of comparison, the orthography has been converted to IPA. There is generally a one-to-one correspondence between the orthography and IPA. The only exceptions to this are <e> which can be either [e] or [ɪ] and <o> which can be either [o] or [ʊ]. I used whichever IPA symbol agreed in ATR with the rest of the word. When the context was ambiguous, I selected the symbol which more closely resembled the ND pronunciation. Tones are not recoverable from the orthography, so they have been omitted. These reconstructed examples have been marked with an asterisk (*). A dash (–) indicates that a translation could not be found for this word. When there is a semantic difference between the ND and CD words, the CD meaning is indicated in parenthesis.

[5]There may also be some synchronic variation. When used in a sentence context rather than in isolation, those forms which still “retain” the final [ɪ] may reduce/delete the final vowel [tí-rí] ~[tír] ‘tree’. This can alternate within a single speaker from one utterance to another. It is possible that since the -r is the only information needed to distinguish the singular from

seem to only have -rɪ (Anttila & Bodomo 2009, 2019, Bodomo & Marfo 2007) – see Tables 5-6 below for examples in CD. Interestingly, as was seen in Section 3.1, the [ɪ] has universally been lost when it marks the singular of the -r singular declension class even though their correlate in other dialecs always has a final [ɪ]. Table 5 shows some examples of plurals which are formed by the addition of -r(ɪ).

Table 5: -r(ɪ) plurals in Northern Dagaare and corresponding nouns in Central Dagaare

Northern Dagaare			Central Dagaare	
Singular	Plural	Gloss	Singular	Plural
báá	báá-r	'dog'	báá	*báá-rí (Bodomo 1997)
bàà	bàà-r	'river'	bàá	*bà-rí (Bodomo 1997)
dáá	dáá-r	'stick'	dàá	*dà-rí (Bodomo 1997)
nàfáʊ̀	nàfáʊ̀-ɾí	'shoe'	*nɔɔtɪ-rʊʊ	*nɔɔtɪ-ɛ
gàdò	gàdò-ɾí	'bed'	gádó	*gado-ri
mʊ̀ɔ́	mʊ̀ɔ́-r	'grass'	mɔ́-rí	*mɔ-ɛ

As seen in Table 5, in this declension class the singulars are unmarked and the plurals are marked by either -r or -ɾí.[6] When the plural marker has a final [ɪ], there is a general allophonic process in ND in which /r/ becomes [ɾ] in syllable onsets.

As can be seen in Table 6, some nouns in the -r(ɪ) plural declension class show variations in the vowels between the singular and plural forms. The singular nouns in ND have a vowel which does not appear in the plural. In some words, like [núú], the extra vowel in the singular is the same as the vowel in the plural stem [nú], but in others, for example [tíɛ̀], the vowel which appears in the singular is different from the vowel in the plural stem [tí]. There are also vowel alternations in the CD forms, but of a different sort. 'Hand' has an extra vowel in the plural rather than in the singular. Also there is a vowel quality difference between the singulars and plurals of 'tree', 'thief', and 'goat'. Another noteworthy fact is that the words 'eye', 'finger', and 'toe' are in different declension classes in the two dialects; while they are in the -rɪ plural class in ND, they are in the -r singular class in CD.

the plural, Northern Dagaare may be losing this final vowel in more common nouns. Further

Table 6: -r(í) plurals in Northern Dagaare and corresponding nouns in Central Dagaare

Northern Dagaare			Central Dagaare	
Singular	Plural	Gloss	Singular	Plural
nímíè	nímí-r	'eye'	nímí-rì	*nimi-e
nóbíè	nóbí-r	'finger'	núbì-rì	*nubi-e/*nubii-ri
g͡bébíè	g͡bébí-r	'toe'	gbɛ́bì-rí	*gbɛbi-e
tíɛ̀	tí-ɾí	'tree'	tìɛ́	tìì-rí (Bodomo 1997)
ɲȁɲúé	ɲȁɲú-r	'thief'	nànɲìgé	*nanɲigi-ri
bùó	bó-r	'goat'	bʊ́ʊ́ɔ́	bʊ́ʊ́-rí
núú	nú-r	'hand'	nû	*nuu-ri

3.3 -bɛ plurals

In the -bɛ plural declension class, the singular is unmarked and the plural is marked by -bɛ. The -bɛ plurals surface in two ways: -βɛ and -mɛ. We see -βɛ appear when the suffix is added to plosives final stems and -mɛ appears when it is added to nasal final stems. These appear to be the only stem types which take the -βɛ plural suffix.

As can be seen in Table 7, when -bɛ is added to a plosive-final stem, the plosive deletes. Additionally, there is a language general spirantization process which turns [b] into [β] between most sonorants.

Table 7: -bɛ plurals in Northern Dagaare and corresponding nouns in Central Dagaare. Central Dagaare data from Ali et al. (2021).

Northern Dagaare			Central Dagaare	
Singular	Plural	Gloss	Singular	Plural
dɛ́b̚	dɛ̀ɛ̀-βɛ̀	'man'	dɔ́ɔ́	*dɔ-bɔ
jɛ̀b̚	jɛ̀ɛ̀-βɛ́	'sibling'	jɔ́ɔ̀	jɔɔmine/*jɔɔre
pɔ́g̚	pɔ́ɔ́-βɛ̀	'woman'	pɔ́g-ɔ́	pɔ́g-bɔ́

research is required to determine exactly where and why the final [ɪ] is sometimes dropped.

[6] Note that in Tables 5-6 all stems which take the -r(ɪ) plural end in vowels. For discussion of how this may impact declension class, see Pruett (n.d.)

As shown in Table 7, there are vowel length alternations between the ND singulars and plurals in the -bɛ class.[7] The stem vowel is always short in the singular but long in the plural. However, this alternation is not universal in the class. There is a group of nouns in the -bɛ plural declension class which does not show any vowel alternations (Table 8).

Table 8: -mɛ plurals in Northern Dagaare and corresponding nouns in Central Dagaare. Central Dagaare data from Ali et al. 2021.

Northern Dagaare			Central Dagaare	
Singular	Plural	Gloss	Singular	Plural
nɛ́n	nɛ́mɛ́	‘meat’	nɛ́nì	nɛ́mɛ̀ (Bodomo 1997)
bʊ́n	bʊ̀mɛ́	‘thing’	bʊ́ní	*bʊma
pán	pàmɛ́	‘door’	pání	*pama (‘doorway’)
dàt͡ʃín	dàt͡ʃímé	‘wall’	dànʧíní	*dant͡ʃime
k͡pæ̀k͡pán	k͡pæ̀k͡pámɛ́	‘arm’	kpáŋkpání	*kpaŋkpama
zʊ́m	zʊ́mɛ̀	‘fish’	zʊ́mmʊ́	*zʊma
pìpán	pìpámé	‘rock’	píírí	*pie
gán	gámɛ́	‘book’	gánì	gámá
sáán	sáámɛ́	‘visitor’	sáánà	saama (Bodomo 1997)

As alluded to earlier, all of the nouns which end in a nasal consonant in the singular receive the -mɛ suffix in the plural. The -mɛ plural has historically been treated as a separate class (Bodomo & Marfo 2007). However, I argue it is an allomorph of the –bɛ plural when it is added to nasal final stems. When -bɛ́ is added to a nasal-final stem, the nasal and the voiced bilabial plosive [b] coalesce to form a bilabial nasal [m]. The nouns which have a nasal final consonant and take the -bɛ plural suffix do not undergo any sort of vowel alternation. The short vowels in the singular remain short when the plural suffix is added.

4 Towards an analysis

As shown in Section 3, several Northern Dagaare nouns exhibit alternations in vowel length and diphthongization between the singular and plural. Seeing data like this in a dialect of Central Dagaare spoken around the towns Jirapa and

[7] One may note that all of the words in Table 7 contain -ATR mid vowels and end in voiced consonants, but I assume this is a coincidence of the small dataset.

Ullo, Anttila & Bodomo (2009) formed an account for number marking based on minimal prosodic word requirements. In this section, I attempt to apply this account to our Northern Dagaare data, and analyze the instances in which it can capture the variation in our data and the places in which it fails. When their account cannot accurately capture the ND facts, I put forth an alternative analysis. Like Anttila & Bodomo (2009), throughout the course of this analysis, I take ATR harmony for granted and leave out tone for simplicity.

The section begins with a discussion of -r singulars in 4.1 which function identically to those in Central Dagaare. In 4.2 I show that the word minimallity requirement in CD may not be active in ND and propose an alternative analysis. Section 4.3 discusses the -bɛ plurals, not analyzed in Anttila & Bodomo (2009, 2019), and provides a potential direction for analysis of the class.

4.1 -r singulars

As shown earlier, in the -r singular declension class, nouns receive an -r suffix in the singular and an -ɛ suffix in the plural. The plural suffix often results in a dipthong or a long mid vowel. As motivation for a processes which will be discussed later, Anttila & Bodomo (2009) posit a constraint against derived long mid vowels. However, since -ɛ is a suffix and not an epenthesized vowel, the long mid vowel is allowed. They justify this in OT by having an IDENT(V) constraint ranked higher than *VV(mid). These constraints, and the widespread MAX constraint are defined below in (1).

(1) a. MAX: Assign one violation mark for each segment present in the input which does not have a correspondent in the output.
   b. *VV(mid): Assign one violation mark for each long mid vowel.
   c. IDENT(V): For a vowel in the input V_i which has a corresponding output vowel V_o, assign one violation for each feature value which is not shared between V_i and V_o.

Using these constraints, the derivation of the Northern Dagaare word 'legs' can be seen in Table 9 below. Thus far, this analysis seems to directly capture what we see in Northern Dagaare. The plural suffix -ɛ is added to the stem, resulting in a long mid vowel. Additionally, as the /ɛ/ is in the underlying form, the long mid vowel is allowed to surface.

Table 9: Tableau deriving [g͡bɛ́ɛ́] 'legs' in ND

/g͡bɛ-ɛ/	Ident(V)	Max	*VV(mid)
☞ a. g͡bɛɛ			*
b. g͡bɪɛ	*!		
c. g͡bɛɪ	*!		
d. g͡bɛ		*!	

4.2 -r(ɪ) plurals

In the -r(ɪ) plural declension class, the plural receives either the suffix -r or -ɾɪ while the singular is unmarked. As described in Section 3.2, some nouns have a long vowel or diphthong in the plural which is not present in the singular.

One could posit that one vowel in a VV sequence is deleted when the plural morpheme -r(ɪ) is added, and the lack of vowel shortening in Table 5 (repeated below in Table 10 for convenience) is due to a preference for low vowels, as we do not see any low vowels deleting. However, [nàfáʊ̀] ends in a high vowel which seems like it would be a good candidate to delete, yet it remains [nàfáʊ̀ɾí] in the plural. Thus, I follow A&B in assuming that this is an epenthesis process and the fact that only low vowels are long in the plural is merely an idiosyncrasy of this limited data set. This conclusion is further supported by considering the unpredictability of which vowel appears in the plural form. For example, when the singular form contains a diphthong, [tíɛ̀] and [bùó], sometimes the first vowel appears in the plural [tíɾí] and sometimes the second does [bór]. A deletion account could not explain this alternation.

Anttila & Bodomo (2009) state that in Central Dagaare, when a stem is monomoraic, a vowel is epenthesized in the singular to create a fully formed bimoraic foot. They state that this is a constraint seen across the language; all phonological words must minimally contain a fully formed prosodic foot. To satisfy this constraint, a vowel is added to the end of monomoraic stems. This vowel matches the quality of the previous vowel. However, as can be seen in Table 10, the vowels do not always match. Anttila and Bodomo argue that this is due to language general constraints against word final high vowels and long mid vowels (breifly mentioned above).

The following tableau are modeled after Anttila & Bodomo 2009, but the data are from Northern Dagaare. The relevant constraints they describe are as follows:

(2) a. Dep(V): Assign one violation mark for each vowel in the output which does not have a correspondent in the input.

Table 10: -r(ɪ) plurals

Singular	Plural	Gloss
báá	báá-r	‘dog’
bàà	bàà-r	‘river’
dàà	dáá-r	‘stick’
nàfáʊ̀	nàfáʊ̀-ɾí	‘shoe’
gàdò	gàdò-ɾí	‘bed’
nímíè	nímí-r	‘eye’
nóbíè	nóbí-r	‘finger’
g͡bébíè	g͡bébí-r	‘toe’
tíɛ̀	tí-ɾí	‘tree’
ɲȁɲúé	ɲȁɲú-r	‘thief’
bùó	bó-r	‘goat’
núú	nú-r	‘hand’

b. FtBin: Assign one violation mark for each foot which is not minimally bimoraic.

c. *I]: Assign one violation mark for each word-final [+high] vowel.

Table 11: Tableau deriving [tɪɛ] ‘tree’ in ND

/tɪ/	FtBin	Dep(V)	*I]
a. tɪ	*!		*
b. tɪɪ		*	*!
☞ c. tɪɛ		*	

Table 11 shows the derivation of [tíɛ̀] ‘tree’. The stem is monomoraic, so a vowel is epenthesized to satisfy FtBin. You would normally epenthesize a vowel with the same features as the stem vowel. However, to satisfy the constraint against word final high vowels, a mid vowel is added instead.

But, as can be seen in Table 10, there are several words which end in high vowels. Similar to the analysis in 4.1 of the *VV(mid) constraint, the constraint against word final high vowels only affects “derived vowels” – i.e., those which do not have a corespondent in the UR. This is again attained by a higher ranked

Table 12: Tableau deriving [gadoɾi] 'bed' in Northern Dagaare

/gado-ɾi/	IDENT(V)	DEP(V)	*I]
☞ a. gadoɾi			*
b. gadoɾe	*!		
c. gadoɾie		*!	

IDENT(V) which prevents vowel lowering if the violating high vowel has a correspondent in the input. A simple case of this can be seen in Table 12.
Despite violating *I], the [i] remains in the output because changing it would violate the higher ranked IDENT(V).

As noted in Section 3.2, the Central Dagaare plural suffix, -rɪ, often appears without the vowel in Northern Dagaare. This is important because while the plural marker straight forwardly satisfies the FTBIN constraint in Central Dagaare, in Northern Dagaare it is necessary to say that /r/ is moraic so as to satisfy the foot minimality requirement and not trigger vowel epenthesis in the plural. The moraic status of coda consonants will be discussed further in Section 4.3.

Table 13: Tableau deriving [buo] 'goat' in Northern Dagaare

/bo/	FTBIN	DEP(V)	*I]	*VV(MID)
a. bo	*!			
b. boo		*		*!
☞ c. buo		*		
d. bou		*	*!	

In Table 13, we see the derivation of [bùó] 'goat'. Again, the stem is monomoraic so a vowel must be epenthesized. The constraint against long mid vowels prevents the normal epenthesis of [o] and instead a high vowel is epenthesized. However, due to the constraint against word final high vowels, the epenthesized vowel cannot occur word finally as it has in the other examples. Rather, the high vowel [u] is epenthesized before the stem vowel.[8]

This accounts well for Bodomo and Anttila's data as well as the Northern Dagaare data discussed above, but when attempting to expand this theory to a

[8]One may note that there is nothing particular indicating where the epenthesized vowel appears or what quality it should take. This is a shortcoming of the original analysis. Since I will soon argue that this analysis does not work for ND and propose an alternative, I will not try to plug this hole. However, future work on CD would benefit from a more complete analysis of the nature and location of the epenthesized vowel.

larger set of ND nouns, it does not capture all the variation. The word minimality account does not explain the epenthetic vowel in the singulars of nouns with multi-moraic stems (Table 14).

Table 14: Multimoraic Stems of ND -rɪ plurals

Stem	Singular	Plural	Gloss
nóbì	nóbíé	nóbí-r	‘finger’
nímí	nímíè	nímí-r	‘eye’
g͡bébí	g͡bébíè	g͡bébí-r	‘toe’
tèɾét͡ʃì	tèɾét͡ʃié	tèɾét͡ʃì-ɾí	‘train’
ɲȁ̋ɲù	ɲȁ̋ɲùé	ɲȁ̋ɲù-ɾí	‘thief’

As the stems in Table 14 are already, at minimum, bimoraic, Anttila and Bodomo’s account does not provide an explanation for the epenthetic vowels in the singular. In fact, as seen in Table 15, their constraints would make the wrong prediction.

Table 15: Tableau of the failed derivation of [nóbíé] ‘finger’ in ND

/nobi/	FtBin	Dep(V)	*I]	*VV(mid)
💣 a. nobi			*	
☹ b. nobie		*!		
c. nobii		*!	*	

Finally, the prosodic account put forth by Anttila and Bodomo is dependent on the fact that Dagaare requires a fully formed foot for all prosodic words. They acknowledge that there are a few exceptions to this rule for Central Dagaare, specifically [bâ] ‘father’, [mǎ] ‘mother’, [nû] ‘hand’, [zû] ‘head’, all of which are very high frequency words. However, deficient feet seem to be much more common in Northern Dagaare. In addition to those mentioned for Central Dagaare, the Northern dialect also has [nú] ‘face,’ as well as many verbs including but not limited to [sʊ̀] ‘bathe’, [ɲʊ̀] ‘drink’, and [tʃɛ́] ‘slice.’

There may be a concern that the verbs do not in and of themselves form full prosodic units as they typically exist as part of a verbal paradigm including auxiliaries, tense markers, and object pronouns. However, we see that verbs can stand alone in imperatives (3).

(3) wá
'come'

Word minimality does not seem to be as active of a constraint in Northern Dagaare as it was in Central Dagaare. Thus, using it as a motivation for vowel epenthesis in nominals is less obvious. Additionally, epenthesis occurs in some words which already satisfy the proposed word minimality.

Rather, following work on quantity-manipulating morphology (Fujimura 1999, Trommer & Zimmermann 2014), I propose that there is an empty mora suffixed to all singular nouns in the -rɪ plural class. This mora is then filled by linking to the adjacent vowel. I will generally assume an autosegmental feature geometry such as those proposed in Clements (1985), but the specifics of the system are not central to the analysis. Having a mora be the affix which spells out singular number requires that all nouns in this class will have an "epenthetic" vowel due to Max-μ. This must be ranked over 1-to-1, a constraint against lengthening/diphthongization of the vowel.

(4) a. Max-μ: assign one violation mark for each mora in the input which is not filled in the output.
b. 1-to-1: Assign one violation mark for each segment that is linked to more than one prosodic node.

The same constraints proposed by Anttila & Bodomo 2009 can be used to account for the instances in which the vowel qualities do not match. If after linking your final vowel to the empty mora, the final vowel is high, the vowel is lowered to satisfy *I] (Table 16). If after linking, there is a long mid vowel, in violation of *VV(mid), the first element of the vowel sequence is raised (Table 17).

Table 16: Tableau illustrating high vowel lowering in [nóbíé] 'finger' in ND

/nobi$^{\mu-\mu}$/	Max-μ	*I]	1-to-1
a. nobi$^{\mu}$	*!	*	
☞ b. nobi$^{\mu}$e$^{\mu}$			
c. nobi$^{\mu\mu}$		*!	*

As shown in Table 18, not all singular nouns in this class undergo vowel lengthening. There is no lengthening in stems which already contain a long vowel. This can easily be explained, as there are no "superlong" vowels in Dagaare.

Table 17: Tableau illustrating mid vowel raising in [buo] 'goat' in ND

/bo$^{\mu - \mu}$/	Max-μ	*VV(mid)	1-to-1
a. bo$^{\mu}$	*!		
☞ b. bu$^{\mu}$o$^{\mu}$			
c. bo$^{\mu\mu}$		*!	*

Table 18: Tableau showing no lengthening of [daa] 'stick' in ND

/da$^{\mu\mu}$-μ/	*SuperLong	Max-μ
a. da$^{\mu\mu\mu}$	*!	
☞ b. da$^{\mu\mu}$		*

Under this analysis, in the -r(ɪ) plural class, nouns in the singular – so long as they do not already end in a long vowel – will always undergo vowel lengthening or diphthongization due to the added mora. This contrasts with Anttila and Bodomo's word minimallity analysis which would only predict vowel lengthening/diphthongization on stems that are less than a fully formed foot.

It should be noted that this analysis would not correctly predict the ND singulars [gàdò] 'bed' and [núú] 'hand' . Mora affixation accompanied by mid vowel raising would predict [gàdùò]. Additionally, the constraint *I] would predict high vowel lowering in 'hand' to produce [núó]. However, this would also be incorrectly predicted under Anttila and Bodomo's account.[9]

4.3 -bɛ́ plurals

Anttila and Bodomo did not address the -bɛ plurals in their work on Central Dagaare.[10] As shown in Table 7 the -bɛ́ plurals in ND have a short vowel in the singular and a long vowel in the plural. Logically, this could be due to shortening in the presence of a coda or lengthening with the deletion of an underlying coda. However, there are many words with long vowels and codas (all of the -r plurals with long vowels as well as several other words including [wáàb] 'snake', [bɛ́ɛ́n] 'one', and [lɛ̀ɛ̀m] 'taste') so it seems unlikely that the vowel is shortened in the presence of a coda.

[9] In CD [nû] 'hand' is also a difficult case which would not be predicted under Anttila and Bodomo's word minimality account.

[10] This in unsurprising since, as can be seen in Tables 7 and 8, it is likely that the -bɛ plurals do not actually form a unified class in CD.

Therefore, I conclude that the -bɛ plurals are undergoing a process of compensatory lengthening, in which one segment is lengthened to compensate for the loss of another. For example, /pɔ̀g-bɛ́/ 'women' becomes [pɔ́ɔ́βɛ̀]. Here the vowel lengthens to compensate for the loss of the [g]. Additionally, there is a language general process of /b/ spirantization between sonorants accounted for by a Spirant constraint. Compensatory lengthening cases such as these are frequently analyzed with a moraic account (Hayes 1989, Shaw 2008). A moraic coda, when deleted, leads to an empty mora which is then filled by lengthening of the preceding vowel. I consider this deletion to be triggered by a *CC constraint specified to ban adjacent [-sonorant] consonants. I propose this holds across the language, as we never see two adjacent obstruents. Additionally, the deletion of the first rather then the second obstruent is due to the ranking of Max$_{onset}$ over a general Max constraint.

(5) a. *CC: Assign one violation mark for each consonant cluster in which both segments are [-sonorant].

b. Max$_{onset}$: Assign one violation mark for each onset in the input which does not have a corespondent in the output.

c. Spirant: Assign one violation mark for each [-son] segment between two [+son] segments.

Table 19: Tableau deriving [pɔ̀ɔ̀βɛ́] 'women'

/pɔ$^{\mu}$g$^{\mu}$-bɛ/	*CC	Max$_{onset}$	Max-μ	Spirant	1-to-1	Max
a. pɔ$^{\mu}$g$^{\mu}$bɛ	*!					
b. pɔ$^{\mu\mu}$bɛ				*!	*	*
c. pɔ$^{\mu\mu}$ɣɛ		*!			*	*
☞ d. pɔ$^{\mu\mu}$βɛ					*	*
e. pɔ$^{\mu}$βɛ			*!			*

In Table 19 *CC in conjunction with Max$_{onset}$ causes the deletion of the coda [g]. In compensation for this, triggered by Max-μ, the vowel lengthens to fill the empty mora.

However, as seen in Section 3.3, when the -bɛ plural suffix attaches to nasal final roots, the /b/ and the final nasal coalesce into an [m] – [pán] 'door', [pàmɛ́] 'doors.' Thus, Uniformity, a constraint preventing coalescence, must be ranked below *NC, which triggers coalescence. Additionally, since the [nasal] and [labial] features both surface on the resulting segment when the [n] and [b] coalesce, I introduce Max$_{[nasal]}$ and Max$_{[lab]}$.

(6) a. Uniformity: assign one violation mark for each element in the output that has more than one correspondent in the input.

b. $\textsc{Max}_{[\text{nasal}]}$: assign one violation mark for each [nasal] feature in the input that has no correspondent in the output.

c. $\textsc{Max}_{[\text{lab}]}$: assign one violation mark for each [labial] feature in the input that has no correspondent in the output.

d. *NC: assign one violation mark for each nasal-consonant sequence.

Table 20: Tableau illustrating the failed derivation of [pàmɛ́] ‘door’

/pa$^{\mu}$n$_1^{\mu}$-b$_2$ɛ/	Max$_{[\text{LAB}]}$	Max$_{onset}$	*NC	Max$_{[\text{NASAL}]}$	Max-μ	1-to-1	Max	Uni
a. pa$^{\mu}$n$_1^{\mu}$.b$_2$ɛ			*!					
b. pa$^{\mu\mu}$.b$_2$ɛ				*!		*	*	
c. pa$^{\mu}$.b$_2$ɛ				*!	*		*	
d. pa$^{\mu\mu}$.n$_1$ɛ	*!	*				*	*	
☹ e. pa$^{\mu}$.m$_{12}$ɛ					*!			*
💣 f. pa$^{\mu\mu}$.m$_{12}$ɛ						*		*
g. pa$^{\mu\mu}$.n$_{12}$ɛ	*!					*		*

When a nasal and [b] coalesce, there is no change in vowel length. As can be seen in Table 20, the same ranking which correctly captured compensatory lengthening in obstruent final roots fails to correctly predict the behavior of nasal final roots. If we assume Onset Maximization (Ito 1989, Selkirk 1982) the [m] would be the onset of the second syllable, not the coda of the first. Therefore it would not be able to hold a mora, and [paamɛ] would incorrectly win to satisfy Max-μ. In some way, the retention of the nasal feature from the underlying coda must “count” as filling a second mora. One potential explanation for this would be that the resulting [m] is ambisyllabic; this way it could satisfy Onset Maximization but still be able to hold a mora. Table 21 shows what this could look like in an OT tableau.

Table 21: Tableau deriving [pàmɛ́] ‘door’

/pan$_1$-b$_2$ɛ/	*NC	Max-μ	1-to-1	Max	Uni
a. pa$^{\mu}$.m$_{12}$ɛ		*!			*
☞ b. pa$^{\mu}$(m$_{12}^{\mu}$)ɛ					*
c. pa$^{\mu\mu}$.m$_{12}$ɛ			*		*

In Table 21, Candidates (a) and (b) are segmentally identical, but in (b) one of the mora in the first syllable is filled by some of the feature information of the underlying /n/. If the [m] is ambisyllabic, filling both the coda of the first syllable and the onset of the second, we may expect to see some phonetic evidence. I leave investigation of this possibility to future research.

5 Conclusion

Anttila & Bodomo (2009) formulated an analysis of vowel alternations in Central Dagaare. Their analysis of the -r singulars seems to map directly to the -r singulars in Northern Dagaare. However, they argue that for Central Dagaare, vowels are epenthesized in the singular to fulfill a word minimality requirement for bimoraic feet. But, I have shown that in Northern Dagaare some words have epenthesized vowels even when they already contain bimoraic feet. Instead, I propose that the singular marker for the -r(ɪ) plural class is an empty mora. This mora suffix is then filled by linking to the adjacent vowel, and language general constraints formulated by Anttila and Bodomo modify the vowels to form the correct surface forms.

Additionally, I put forth a preliminary account for vowel length alternations in the -βɛ and -mɛ plurals, here analyzed as a single class, explaining why compensatory lengthening occurs in the former but not the latter. While more data is necessary to help clarify these patterns and formulate a more comprehensive analysis, I have provided a stepping off point for which to analyze the vowel alternations in Northern Dagaare number marking.

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Chapter 3

A negative alternation: Negation head movement allomorphy in Igala

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In Igala (Volta–Niger), pre-verbal negation changes forms depending on the syntactic environment. I argue that these distinct exponents occupy different syntactic positions, as in other Niger-Congo languages [*e.g.*, Kirundi (Bantu JD.62; Ndayiragije 1999) and Igbo (Volta–Niger; Amaechi 2019)], and propose that this is the result of Neg-to-C movement being blocked in certain syntactic environments. I show that this explains the distribution of negation in different constructions in Igala as well as embedded clauses in Kirundi (Great Lakes Bantu).

1 Introduction

In Igala (Volta–Niger), negation is bi-partite where it is expressed as two morphemes: a pre-verbal morpheme and a sentence-final particle, the first of which changes forms depending on the syntactic environment. Pre-verbal negation surfaces as extra-high tone on the subject in finite clauses similar to some dialects of Igbo (Emenanjo 1985, Ndimele 1995, Obiamalu 2013) and the Benue–Congo languages Efik (Mensah 2001) and Ibibio (Essien 1990), where negation can be marked by tone only, as in (1).

(1) a. ẹ̀ j(ẹ̄) ọ̀dà ọ̀nálẹ́.
2SG eat pear yesterday
'You ate a pear yesterday.'

b. ẹ̋ j(ẹ̄) ọ̀dà ọ̀nálẹ́ ń.
2SG.NEG eat pear yesterday SFP_{NEG}
'You did **not** eat a pear yesterday.'

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On the other hand, in clauses involving A'-movement (and inside nominalizations), it surfaces as the pre-verbal particle *mǟ*, shown in (2)[1].

(2) a. ẹ́n(ệ)$_i$ ẹ̀ mǟ lí t_i ń?
who 2SG NEG see SFP$_{NEG}$
'WHO did you **not** see?'

b. * ẹ́n(ệ)$_i$ ẹ̋ lí t_i ń?
who 2SG.NEG see SFP$_{NEG}$

Likewise, similar alternations are found in other Niger-Congo languages; for instance this is shown by Chaperon (2023) in Kirundi (Bantu JD.62) and Amaechi (2019) in Igbo (Volta–Niger). In Kirundi, negation usually surfaces prior to subject marking on the verb as the prefix *nti-*. In clauses where A'-movement occurs (or within subordinate clauses), negation surfaces as *ta-* following subject marking, as seen in (3).

(3) a. Yohani nti -a-a-funguye.
John NEG$_1$-1SM-REC.PST-eat.PFV
'Yohani did **not** eat.'

b. Ni-ndé$_i$ Yohani a-t(a) -a-bonye __$_i$?
COP-who John 1SM-NEG$_2$-REC.PST-see.PFV
'WHO did John **not** see?' (Kirundi; Chaperon 2023)

Similarly, in Igbo negation typically surfaces as the suffix *-ghẹí* on the verb. However, when A'-movement is involved, a pre-verbal particle *ná* must surface along with this suffix, as in (4).

(4) a. Úchè á-ꜝhẹú-ghẹí Òbí.
Uche PFX-see-NEG Obi
'Uche did **not** see Obi.'

b. Ònyé$_i$ kà Úchè ná á-ꜝhẹú-ghẹí __$_i$?
who SFP$_{FOC}$ Uche PRT PFX-see-NEG
'WHO did Uche **not** see?' (Igbo; Amaechi 2019)

Thus, this alternation between different forms of negation is not an independent phenomenon only found in Igala. Moreover, in both languages just presented, negation occurs in relatively distinct surface positions depending on the

[1]Foci are formatted using *small caps*

syntactic context. I use this as part of the evidence suggesting that the negation head moves to a different position in Igala. I argue that negation moves to C° where it surfaces as extra-high tone. In cases where it cannot, it surfaces as the particle *mā̋* instead. Subsequently, I use a contextual allomorphic approach to account for the different surface forms of the pre-verbal negation morphemes. More specifically, I argue that when negation moves to C, it is realized as an extra-high tone (usually on the subject), otherwise it surfaces as the particle *mā̋*.

In Section 2, I present a full and detailed distribution of negation in different syntactic environments. In Section 3, I propose that this distribution can be accounted for with head movement of negation to C°. Furthermore, I argue that the different exponents of negation are due to contextual allomorphy. In Section 4, I present additional evidence for this analysis first with conditionals and modals and then within embedded clauses, which I show can be extended to Kirundi (Great Lakes Bantu). Finally, in Section 5 I conclude with a summary along with some avenues for future research.

2 Distribution of negation

In this section, I show the distribution of pre-verbal negation. The extra-high tone which occurs on the subject will be referred to as *tonal negation*. When negation surfaces as the particle *mā̋*, it will be referred to as *particle negation*.

2.1 Tonal negation

Tonal negation occurs in declaratives, polar interrogatives, and imperatives, which Potsdam (2013) argues are the exact clauses where negation occurs in C in English. This form expones an extra-high tone on the last vowel of the subject, which always precedes the verb, as shown in the finite declarative clause in (5).

(5) a. Àchēnyọ̀ l(í) ájúwẹ̄ lẹ́.
Achenyo see chicken the
'Achenyo saw the chicken.'

b. Àchēnyọ̋ l(í) ájúwẹ̄ lẹ́ ń.
Achenyo.NEG see chicken the SFP$_{\text{NEG}}$
'Achenyo did **not** see the chicken.'

Additionally, *tonal negation* can appear in embedded finite clauses. In this case, the negative tone docks onto the embedded subject.

(6) a. ì kà [kàkíní ì mà].
3SG say COMPL 3SG know
'S/he said that s/he knows.'

b. ì kà [kàkíní ḯ mà ń].
3SG say COMPL 3SG.NEG know SFP$_{NEG}$
'S/he said that s/he does **not** know.'

Subject markers in Igala, as in (6), are pronouns and not agreement markers; they are in complementary distribution with R-expressions (see Ejeba 2016 for a full paradigm). Igala polar questions contain length clause-finally, which I mark as a question particle[2]. When negated, *tonal negation* is used; the interrogative length is added to the sentence-final particle *n*.

(7) a. ẹ̀ j(ẹ̄) ọ̀d(à) ọ̀nálẹ́ː .
2SG eat pear yesterday.Q
'Did you eat a pear yesterday?'

b. ẹ̋ j(ẹ̄) ọ̀d(à) ọ̀nálẹ́ ňː .
2SG.NEG eat pear yesterday SFP$_{NEG}$.Q
'Did**n't** you eat a pear yesterday?'

Additionally, *tonal negation* is used in imperatives. In Igala imperatives, the subject is usually not overt; otherwise it surfaces with the optative mid tone (see §4.1.2 for a broader discussion). However, the subject is required to surface in negative imperatives. I assume that it is because *tonal negation* needs to anchor somewhere, so the subject is realized overtly.

(8) a. (ē) jẹ̄!
2SG.OPT eat
'(You may) eat!'

b. *(ẹ̋) jẹ̄ ń!
2SG.NEG eat SFP$_{NEG}$
'Do**n't** eat!'

I have shown that tonal negation surfaces in declaratives, polar interrogatives, and imperatives. These clauses will be important in arguing for a shared position in the left periphery for negation and C.

[2] As in (7b), question length always occurs clause-finally, even in the presence of other sentence-final particles.

2.2 Particle negation

Next, the particle form of pre-verbal negation is used in clauses involving extraction and inside nominalizations. In these cases, it surfaces as the particle *mã̋* before the verb.

2.2.1 A'-movement

Particle negation is used when A'-extraction occurs: in *wh*-questions, with focus fronting, and inside relative clauses; these environments all pass the standard A'-movement tests (Martinović 2024). This is shown with subject focus in (9), non-subject focus in (10), and adjunct focus in (11).

(9) a. ẹ́nệ$_i$ __$_i$ lọ́ t(ẹ́) ójúkpọ́lọ́gwū í?
who go to park SFP$_{\text{FOC}}$
'WHO went to the park?'

b. ẹ́nệ$_i$ __$_i$ mã̋ lọ́ t(ẹ́) ójúkpọ́lọ́gwū ní ì?
who NEG go to park SFP$_{\text{NEG}}$ SFP$_{\text{FOC}}$
'WHO did **not** go to the park?'[3]

(10) a. ọ̀nwū$_i$ ẹ̀ fẹ̀dọ̀ __$_i$ í.
3SG.STR 2SG love SFP$_{\text{FOC}}$
'It's HIM you love.'

b. ọ̀nwū$_i$ ẹ̀ mã̋ fẹ̀dọ̀ __$_i$ ní ì.
3SG.STR 2SG NEG love SFP$_{\text{NEG}}$ SFP$_{\text{FOC}}$
'It's HIM you do **not** love.'

(11) ònálẹ́$_i$ ì mã̋ t(ẹ́) énè __$_i$ ń.
yesterday 3SG NEG ask question SFP$_{\text{NEG}}$
'It's YESTERDAY s/he did **not** ask a question.'

When long distance extraction occurs, *mã̋* only appears in the clauses which have been negated. In (12a), only the embedded clause is negated, in (12b) only the matrix clause is, and in (12c) both clauses are negated.

(12) a. ọ̀nwū$_i$ ípítà mà [kàkíní íjénì mã̋ lí __$_i$ ń] ì.
3SG.STR Peter know COMPL Jane NEG see SFP$_{\text{NEG}}$ SFP$_{\text{FOC}}$
'It's HIM$_i$ that Peter knows that Jane did **not** see.'

[3]The sentence-final particle 'i' is marked as *focus* as it appears when extracting to the left periphery; it is often optional, although its distribution is not yet well understood. Similarly, the sentence-final particle for negation can surface as /ni/ for emphasis.

b. ọ̀nwū$_i$ ípítà $\boxed{\text{mã́}}$ mà [kàkíní íjénì lí __$_i$] í ń.
3SG.STR Peter NEG know COMPL Jane see SFP$_{FOC}$ SFP$_{NEG}$
'It's HIM$_i$ that Peter does **not** know that Jane saw.'

c. ọ̀nwū$_i$ ípítà $\boxed{\text{mã́}}$ mà [kàkíní íjénì $\boxed{\text{mã́}}$ lí __$_i$] í ń.
3SG.STR Peter NEG know COMPL Peter NEG see SFP$_{FOC}$ SFP$_{NEG}$
'It's HIM$_i$ that Peter does **not** know that Jane did **not** see.'

In long distance extraction, not only does negation occur in the clause containing the extracted element's initial trace, but also in all clauses along the path of A'-movement. Thus, negation in Igala exhibits cyclic effects (Chomsky 1977, 1986, 1993).

2.2.2 Nominalizations

Finally, *particle negation* is used inside nominalizations, as in (13). Both the preverbal and sentence-final particle surface inside.

(13) a. [é ch(e) ìskúlù kpā] ch(e) ẹ̀nw(u) òmẹ̀mẹ̀lẹ̀.
NMLZ do school finish COP thing nice
'Finishing school is a good thing.'

b. [é $\boxed{\text{mã́}}$ ch(e) ìskúlù kpā $\boxed{\text{ń}}$] ch(e) ẹ̀nw(u) òmẹ̀mẹ̀lẹ̀.
NMLZ NEG do school finish SFP$_{NEG}$ COP thing nice
'**Not** finishing school is a good thing.'

I propose that nominalizations do not contain the C domain; I show that they can only take clauses up to *v*P or AspP (or NegP when negated). Two observations illustrate this point; first, they can contain inflectional elements like aspect, as in (14)[4].

(14) a. [é $\boxed{\text{f(u)}}$ ìskúlù chē kpā] ch(e) ẹ̄nwū ògbọ̄gágá.
NMLZ PFV school COP finish COP thing important
'[Having finished school] is an important thing.'

b. [é $\boxed{\text{nâ}}$ ch(e) ìskúlù kpā] ch(e) ẹ̄nwū ògbọ̄gágá.
NMLZ PROG COP school finish COP thing important
'[Finishing school] is an important thing.'

[4]Note that these are subject nominalizations (*i.e.*, located in the subject position).

Additionally, nominalizations cannot contain an overt subject, unless it is external. In the examples below, the two strategies used to circumvent this are shown; speakers can either use an external subject, as in (15a), or a possessor outside of the nominalized clause, as in (15b).

(15) a. [$\boxed{\text{ònw(u)}}$ [é l(a) ímōtò]] ch(e) ọ̀mẹ̀lẹ̀lẹ̄.
3SG.STR NMLZ buy car COP good
'[**For him** to buy a car] was good.'

b. [ímòtò [é lá] $\boxed{\text{nwū}}$] ch(e) ọ̀mẹ̀lẹ̀lẹ̄.
car NMLZ buy 3SG.POSS COP good
'[**His** buying of a car] was good.'

By hypothesis, nominalizations still contain a PRO subject in Spec,*v*P (Abney 1987, Kratzer 1996: among others). Overall, *mắ* surfaces both in negative clauses where extraction has occurred and inside nominalizations.

We examine this distribution more closely in the next section, where I propose an analysis to account for it. I argue that negation moves to C, but in clauses involving A'-movement its movement is blocked and within nominalization there is no C position for it to move to.

3 Neg-to-C movement

In this section, I offer an analysis of the two different instantiations of negation in Igala. I first assume that all finite clauses contain a CP (Chomsky 2007). I argue that negation heads its own functional projection in the inflectional domain and that it moves to C°. I claim that (i) when movement to C is blocked, negation surfaces as the pre-verbal particle *mắ*, (ii) when it does move to C, pre-verbal negation surfaces as an extra-high tone on the subject, and (iii) the different phonological forms of negation are the result of contextual allomorphy.

Particle negation surfaces *in-situ* in clauses involving A'-movement and inside nominalizations. I stipulate that these environments share a common property – the C-domain is not available for head movement. When A'-movement occurs, the [+*wh*] C° blocks movement to it, which blocks Neg-to-C movement[5]. I have also shown that nominalizations are not clausal (see Section 2.2.2), so negation has no C to move to. Both of these environments are unified in that the left periphery inaccessible.

[5]It has been argued that A'-movement and some types of negation are incompatible (Roberts 2018).

On the other hand, tonal negation occurs in declaratives, imperatives, and interrogatives. Potsdam (2013) argues that in English these exact clauses are examples in which negation occurs in C°. Similarly, Déchaine & Wiltschko (2002) argue that negation can be found in or close to C. In fact, they had previously argued that "the base position for NEG in Algonquian is pre-verbal (…) and that NEG may raise to higher positions outside of IP" (Déchaine & Wiltschko 2001). I take this as evidence that negation moves to C° (through I°) in these syntactic environments. This could also account for the different overt positions of negation shown earlier in Kirundi (Bantu JD.62; Chaperon 2023) and Igbo (Volta–Niger; Amaechi 2019).

Finally, I posit that the alternate forms of pre-verbal negation are due to contextual allomorphy (Embick 2010, Marantz 2013). Pre-verbal negation surfaces as an extra-high tone when it is located in C, otherwise it surfaces as *mã̋*. A more formal definition is shown in (16) below.

(16) Vocabulary entries for pre-verbal negation:

$$[\ \textsc{neg}\ ] \rightarrow \left[\begin{array}{c} \text{˝} \\ \text{mã̋} \end{array}\right] / \left\{\begin{array}{c} [_{\mathrm{C}}\ \_] \\ \ \end{array}\right\}$$

Henceforth, the remainder of this section illustrates that this analysis can account for all cases of negation shown throughout.

3.1 Blocked Neg-to-C and particle negation

In this section, I show two cases where pre-verbal negation does not move to C° and does not surface as extra-high tone on the subject, but instead as the particle *mã̋* – inside nominalizations and in clauses involving A'-movement.

3.1.1 Nominalizations

Here, I show an example derivation of a negated nominalized clause. A derivation for example (17) is shown in Figure (1)[6].

(17) [é mã̋ kpā ìskúlù ń] ch(e) ẹ̀nw(u) byẹ́nẹ̄.
NMLZ NEG finish school $\textsc{sfp}_{\textsc{neg}}$ COP thing bad
'**Not** finishing school is a bad thing.'[7]

[6] I follow Tremblay (2021) in assuming that Igala has an underlying SOV word order where the verb moves up to *v*, resulting in an SVO surface order (Koopman 1984).

[7] I omit sentence-final particles in trees for simplicity.

The nominalized clause is headed by the nominalizer *é* which can take clauses slightly bigger than *v*P (*e.g.*, negation). Given this, I assume that nominalizations involve a *PRO* (Abney 1987, Kratzer 1996: among others). The *PRO* subject is generated in spec,*v*P which I assume does not move given that no head within the nominalized clauses has an [EPP] feature. Pre-verbal negation does not move up to C° within nominalized clauses as they do not contain a CP; it instead surfaces as the particle *mấ*.

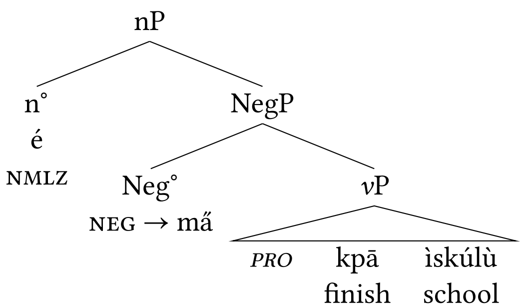

Figure 1: Negation inside a nominalized clause

3.1.2 A'-movement

Next, I show a derivation for a clause involving A'-movement. The derivation for example (18) below is shown in Figure (2).

(18) ẹ̀nwū$_i$ ẹ̀ mấ nyèjū __$_i$ ń ì?
what 2SG NEG like SFP$_{NEG}$ SFP$_{FOC}$
'WHAT do you **not** like?'

The subject generated in spec,*v*P moves to spec,IP to check the [EPP] feature on I°. The focused constituent (here the object) moves to the specifier of the [+*wh*] C head. In this case, since the [+*wh*] C head blocks movement of negation, pre-verbal negation can only move up to I°, where it surfaces as the *mấ* particle.

3.2 Neg-to-C and tonal negation

In this section, I show the derivation for contexts in which negation moves to C° and surfaces as *tonal negation*. More specifically, I only show a derivation for a

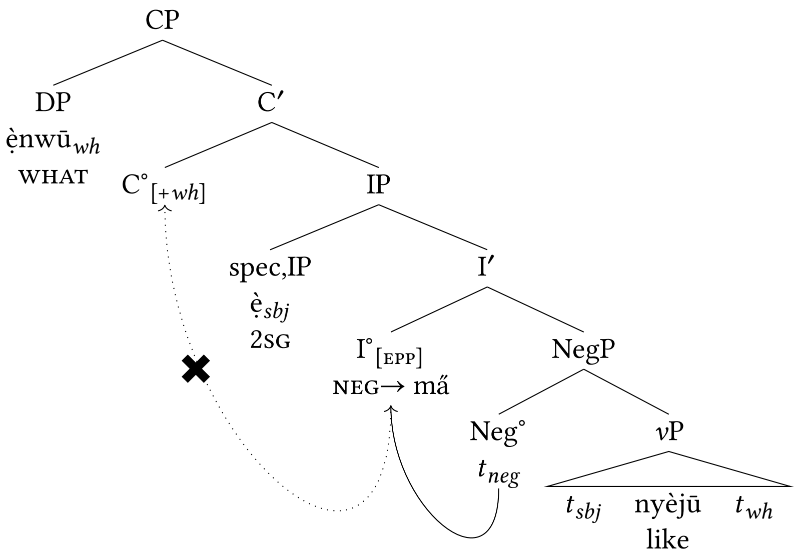

Figure 2: Negation in a clause involving A'-movement

declarative clause, as imperatives and polar questions would be derived similarly. In these cases, pre-verbal negation surfaces as an extra-high tone (on the subject).

I argue that both I and C have an [EPP] feature that must be checked by the subject (Chomsky 2000). Aboh (2006) argues that this is also the case in Gungbe (Volta-Niger). Hence, subjects first move to spec,IP and subsequently move to the specifier position of C. The derivation for (19) is shown in (3).

(19) ẹ̋ ny(i) ányí ń.
2SG.NEG laugh(V) laugh(N) SFP$_{\text{NEG}}$
'You did **not** laugh.'

Both the I and the C heads have an [EPP] feature which the subject generated in spec,*v*P must check. It first moves up to spec,IP and then to spec,CP. Finally, negation moves up to C° through I° and surfaces as extra-high tone (on the last vowel of the subject) since it is in the left periphery, as per its vocabulary entry rule in (16).

I assume that the same applies to polar questions and imperatives. In these clauses, negation also surfaces as tone on the subject; examples (7b) and (8b) are repeated in (20) and (21) respectively.

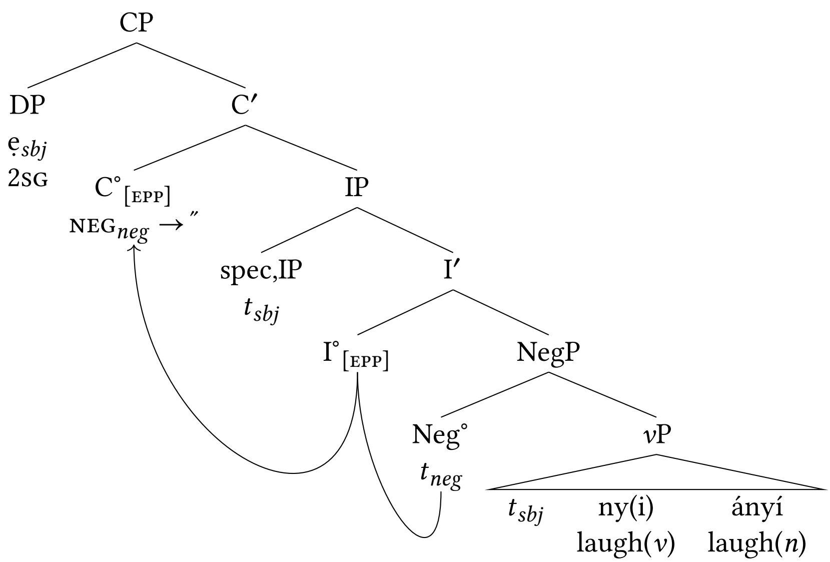

Figure 3: Negation in a finite matrix clause: declarative

(20) ẹ̋ j(ẹ̄) ọ̀d(à) ọ̀nálẹ́ ň: .
2sg.neg eat pear yesterday sfp$_{\text{NEG.Q}}$
'Did**n't** you eat a pear yesterday?'

(21) *(ẹ̋) jẹ̄ ń!
2sg.*neg* eat sfp$_{\text{NEG}}$
'Do**n't** eat!'

I assume that, as in declarative clauses, the C° head in these clauses also does not block negation from moving to it.

Given that subjects surface in the specifier to the left of C°, pre-verbal negation occurs immediately to its right. This accounts for why *tonal negation* anchors to the last syllable of subjects. More generally, *tonal negation* adjoins to the right-most vowel of its specifier. This is stated more formally in (22) below, although I leave the precise mechanism at hand to PF.

(22) *Linearization of tonal negation*
$[_{\text{CP}}[_{\text{spec,CP}}\ (\ldots\text{CV.C})\text{V}][_{\text{C'}}[_{\text{C}^\circ}\ ''][\ldots]]] \Longrightarrow (\ldots\text{CV.C})\text{V̋}$

In this section, I have shown that a head-movement analysis can account for the surface position and exponent of negation in different syntactic environments. I have argued that when movement to C is blocked – in clauses involving A'-movement and inside nominalizations – negation surfaces as the pre-verbal particle *mấ* and when it does move to C – in finite clauses – pre-verbal negation surfaces as an extra-high tone on the subject.

4 Additional evidence

In this section, I extend my analysis by showing that it can account for the interaction of negation with other elements in the left periphery. I first show that negation surfacing in C is supported by other tonal morphology – conditional and optative marking – also surfacing in this position. Additionally, I show that this analysis accounts for the exponent of negation inside embedded clauses.

4.1 Other left-peripheral tonal morphology

In this section, I demonstrate that two other instances of tonal morphology in Igala, which also appear on the subject, surface in C.

4.1.1 Conditionals

First, if extra-high tone surfaces on the subject without the presence of the negative sentence-final particle, the sentence is interpreted as a conditional, as in (23).

(23) ȉ nēkē l(a) īmótò, y=ǎ wá.
3SG.COND can buy car 3SG=IMPF come
'If s/he can buy a car, s/he will come.'

When such a conditional clause is negated, the negative particle *mấ* surfaces (along the sentence-final particle) even though no apparent element is in the left periphery or has been extracted, as in (24).

(24) ȉ mấ nēkē l(a) īmótò ń, y=ǎ wá.
3SG.COND NEG can buy car SFP_{NEG} 3SG=IMPF come
'If s/he can**not** buy a car, s/he will come.'

According to my analysis in §3, *mấ* surfaces as a result of an element in C blocking negation from head-moving to it; I posit that in the case at hand, it is blocked by the conditional head[8]. Boles & Socolof (2024) propose that conditional extra-high tone in Igala expones a head in the left periphery and involves operator movement. This is not a novel claim as it is commonly assumed that conditionals involve operator movement (*e.g.,* Haegeman 2010).

4.1.2 Modals

Next, I demonstrate that the behaviour of the optative (deontic) modal marker, along with its interaction with negation, supports my analysis. Subject clitics typically surface with a low tone; when they surface with a mid tone, an optative reading arises, as in (25).

(25) a. ì t(ẹ́) énè.
3SG ask question
'S/he asked a question.'

b. ī t(ẹ́) énè.
3SG.OPT ask question
'May s/he ask a question!'

Unlike with conditionals, there are two strategies to negate an optative clause: (i) using the optative particle *mā* (notice the mid tone) along with *tonal negation* or (ii) embedding negation within a subjunctive clause, as in (26a) and (26b) respectively.

(26) a. í̋ mā t(ẹ́) énè ń.
3SG.NEG OPT ask question SFP$_{\text{NEG}}$

b. ī [k(i)-í̋ t(ẹ́) énè ń].
3SG.OPT C.SBJV-3SG.NEG ask question SFP$_{\text{NEG}}$
'May s/he not ask a question!'

Similarly to conditionals, I claim that optative tone surfaces in C. A similar claim has been made by Aboh (2006), who follows Damonte (2002) and Aboh (2004) in arguing that Saramaccan (Niger-Congo derived English-Portuguese Creole) and Gungbe (Volta-Niger) deontic *fu* and *ní* respectively are modal complementizers that surface in Fin°.

I have proposed that both conditional and optative tonal morphology surface in C in Igala. I then showed that *tonal negation* cannot co-occur with both elements, other strategies must be used instead. This parallel behavior suggests that they all share the same position. This is in line with my proposal that *tonal negation* surfaces in C (via movement). More broadly, this shows that it is not only A'-movement which blocks negation from moving to C, but other left-peripheral material as well.

[8] I leave the question of the conditional tone no longer being apparent for future research.

4.2 Embedding complementizers

There are two types of languages: (i) those where the complementizer embeds the whole left periphery and projects higher than topic and focus (as *e.g.*, Wolof and Italian; Dunigan 1994, Rizzi 1997), and (ii) those where the complementizer only embeds IP and shares the A' slot (as *e.g.*, German and Old English; Gelderen 2004). This variation in left peripheral structure should affect the type of negation in embedded clauses; languages with high embedding complementizer should allow Neg-to-C movement and those with a low embedding complementizer should block this movement. In this section, I show that this is the exact contrast found between Igala and Kirundi (Bantu JD.62).

4.2.1 Igala

In this subsection, I show that Igala has a high embedding complementizer which allows Neg-to-C movement. In this language, constituents can be focus within embedded clauses, as in (27).

(27) a. ípítà má [kàkíní íjénì k(à) ọ̀là kp(àí) ánà].
Peter know COMPL Jane speak word with Anna
'Peter knows that Jane spoke to Anna.'

b. ípítà má [kàkíní ánà$_i$ íjénì k(à) ọ̀là kp(àí) ònwū$_i$ í].
Peter know COMPL Anna Jane speak word with 3SG.STR SFP$_{FOC}$
'Peter knows that it's ANNA$_i$ that Jane spoke to (her$_i$).'

This example shows that the complementizer in Igala can embed foci. Following a "split-CP" *á la* Rizzi (1997), I posit that *kakini* is a higher embedding complementizer in Force. It follows that Neg-to-C movement should be possible in embedded clauses, so negation should surface in the same form in embedded clauses as in matrix clauses – with *tonal negation*[9]. This prediction is borne out, as shown in (28) below.

(28) ì kà [kàkíní í̋ mà ń].
3SG say COMPL 3SG.NEG know SFP$_{NEG}$
'S/he said s/he did **not** know.'

I argue that the high embedding complementizer in Igala allows Neg-to-C movement, as it surfaces as extra-high tone on the embedded subject.

[9]Tonal negation appears on the subject and not the complementizer due to movement not being possible to a higher head.

4.2.2 Kirundi

Next, I show that Kirundi (Bantu JD.62) has a low embedding complementizer which blocks Neg-to-C movement. This language uses the complementizer *ko* to embedded clauses as in (29), and focus constructions arise from extraction of foci to the left periphery, where they follow the particle *ni* as in (30). Various claims have been made about *ni*; Gatchalian (2023) argues that it is a copula found in the left periphery.

(29) Keezá a-rá-zi [ko Juma a-somye igitabo].
Keezá 1SM-DJ-know COMPL Juma 1SM-read.PFV 7book
'Keeza knows that Juma read a book.'

(30) a. Yohani a-á-guze igitabo.
John 1SM-PST-buy.PFV 7book
'John bought a book.'

b. ni igitabo yohani a-á-guze.
COP 7book John 1SM-PST-buy.PFV
'It's A BOOK John bought.'

In contrast to Igala, foci, or the whole left periphery more generally, cannot be embedded in Kirundi, as shown in (31).

(31) * Keezá a-rá-zi [ko ni igitabo Juma a-somye].
Keezá 1SM-DJ-know COMPL COP 7book Juma 1SM-read.PFV
Intended: 'Keeza knows that it's A BOOK Juma read.'

I argue that the complementizer *ko* is located lower in C° and not in Force° as in Igala. As predicted, the lower non-matrix form (equiv. *particle negation*) must be used instead. In finite matrix clauses, negation surfaces as the prefix *nti-* on the verb, as in (32); in embedded clauses, it cannot be used, the prefix *ta-* following subject marking surfaces instead, as in (33).

(32) Yohani nti -a-a-funguye.
John NEG_1-1SM-REC.PST-eat.PFV
'Yohani did **not** eat.'

(33) a. * Keezá a-rá-zi [ko Juma nti -a-somye igitabo].
Keezá 1SM-DJ-know COMPL Juma NEG_1-1SM-read.PFV 7book

b. Keezá a-rá-zi [ko Juma a-ta -somye igitabo].
Keezá 1SM-DJ-know COMPL Juma 1SM-NEG$_2$-read.PFV 7book
'Keeza knows that Juma did **not** read a book.' (Chaperon 2023)

I argue that the low embedding complementizer in Kirundi blocks Neg-to-C movement. This is similar to V-to-C movement being blocked in Germanic subordinate clauses due to the complementizer filling the C° position (Kiparsky 1995).

I have shown that this analysis accounts for the height of complementizers. I have argued that in Igala, the high embedding complementizer in Force° allows negation to move to C°. On the other hand, in Kirundi the low embedding complementizer in C° blocks negation from moving to it. This accounts for both the surface positions and exponents of negation in embedded clauses in both languages.

5 Conclusion

To conclude, Igala has a bi-partite negation system: a pre-verbal form and a sentence-final particle. Pre-verbal negation can surface either as an extra-high tone on the subject or as the particle *mã̋*. In this paper, I have argued that these different exponents of pre-verbal negation are due to head movement (or its restriction) to C°. I claim that its different exponents are a result of contextual allomorphy. I propose that negation surfaces as extra-high tone when it moves to C° but surfaces as the particle *mã̋* when this movement is blocked. I show that this restriction occurs in two types of clauses. First, in A'-movement clauses due to the [+*wh*] head blocking movement to it. Additionally, in nominalizations, negation has nowhere to move to as C is not contained within them. Subsequently, I show that this analysis can account for other environments where negation occurs – with conditionals and modals. Furthermore, it accounts for what exponents (and position) negation is realized as in both Kirundi (Great Lakes Bantu) and Igala embedded clauses, which I argue have low and high complementizers respectively.

A shortcoming of this proposal is that it is not derived; a blocking property has to be stipulated or not for each separate C. For example, I have argued that in Igala the [+*wh*], conditional, and modal C°'s block the movement of negation to the left periphery. On the other hand, those in declaratives, imperatives, and polar questions do not block its movement. Hence, a blocking property has to be stipulated for all of these other C heads separately.

The next step in this research is to see how this could be captured in a more principled way. Given that this is a common phenomenon in Niger-Congo languages, investigating more languages would help in proposing a more attractive and consolidated analysis. Cross-linguistic variation is expected which would aid in distinguishing between all of the mechanisms occurring during the whole derivation. Another question that remains to be answered is the relationship between the pre-verbal and sentence-final forms of negation.

Abbreviations

1	noun class 1
7	noun class 7
COMPL	complementizer
COND	conditional
COP	copula
DJ	disjoint marker
FUT	future
IMPF	imperfective
NEG	negation
NEG_1	primary negation
NEG_2	secondary negation
NMLZ	nominalizer
OPT	optative
PST	past
PFV	perfective
PFX	prefix
PL	plural
PROG	progressive
PRT	particle
Q	question
REC	recent
REM	remote
SFP	sentence-final particle
SG	singular
SM	subject marker
STR	strong

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Chapter 4

Two types of verbal number in Kalenjin pluractionality

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This paper investigates pluractionality in Pokot and Kipsigis, two closely related varieties of Kalenjin, a macrolanguage belonging to the Southern Nilotic branch of Nilo-Saharan (Eberhard et al. 2023). Most typological literature (e.g. Corbett 2000), as well as generative analyses (Amato 2018, Thornton 2019), treat participant number and pluractionality (=event number) as one category, called verbal number. However, I show that these two categories cannot be conflated in Kalenjin by providing novel data, coming from original fieldwork conducted in Kenya. I further provide the first morphological and semantic description of pluractionality in both varieties.

1 Introduction

Kalenjin is a rather understudied language cluster. First grammar descriptions of the relevant varieties can be found in Toweett (1979), Rottland (1982), Crazzolara (1978) and Baroja (1989). Theoretical papers on Kipsigis include Kouneli (2021) on the tripartite number marking system in Kipsigis, Kouneli (2022) on the so called inflection classes in Kipsigis, and Bossi (2023) on epistemic modality. This paper provides the very first description of verbal number in two Kalenjin varieties, Kipsigis and Pokot. By doing so, this paper shows how Kalenjin verbal number is different from how verbal number works in many other languages. While many languages mark both event and participant number by the very same marking strategy, Kalenjin exhibits two distinct marking strategies for the two grammatical concepts.

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This paper is structured as follows: In section two, first, the topic of pluractionality is introduced briefly. The second section further introduces the languages and what pluractionality looks like in each of them. The variety of semantic functions of Kalenjin pluractionality is demonstrated via many examples from Pokot. The third section provides a morphological description of Kipsigis and Pokot pluractionality. The fourth section introduces the distinction of event and participant number and argues against the conflation of event and participant number in Kipsigis and Pokot and why this is problematic for existing generative accounts of verbal number. The last section concludes.

2 Description of pluractionality in Kipsigis and Pokot

While nominal number, an obviously related category, marks the number of individuals, verbal number marks the number of events (or situations, in the words of Mattiola 2019). Verbal number can be defined as in (1).

(1) **Verbal number**
“A phenomenon that marks plurality or multiplicity of the situations (i.e. states and events) encoded by the verb through any morphological mean that modifies the form of the verb itself.” (Mattiola 2019: 4)

Following Corbett (2000), verbal number can be observed in two forms: participant number (multiple participants create a multiplicity of events) and event number (a single participant creates/experiences a multiplicity of events). This distinction will be discussed in section Section 4 in detail. I follow Henderson 2012 and others in calling the second flavour (event number) *pluractionality* and also adapt his notation of calling the morphemes which derive this multiple event readings *pluractionals*.

Verbal number is most common in North America but can also be observed in several language families in South America, the Pacific region, Caucasus and all four major families of Africa (Veselinova 2013). In Africa, it can be observed most strongly in the Nilo-Saharan language family, which is the language family Kalenjin also belongs to. Kipsigis and Pokot are two varieties of Kalenjin, a dialect cluster belonging to the Southern Nilotic branch of Nilo-Saharan. Kipsigis is spoken by approximately 2 million speakers and Pokot by approximately 778,000 speakers in Kenya (Eberhard et al. 2023). Although the languages are closely related they are not mutually intelligible. If not otherwise indicated, all data from Kipsigis and Pokot in this paper come from fieldwork and were obtained in elicitations with five speakers in Kenya.

2.1 Kipsigis

Kipsigis verb stems can be reduplicated to derive a pluractional meaning. Pluractionality in Kipsigis is productive and can be used with Kipsigis words and also Swahili loan words. It doesn't seem to be limited to certain verb classes, but no Aktionsart-tests exist for Kipsigis yet, so a limitation to certain Aktionsart-classes can't be excluded at this point. In my field notes, only two verbs were unable to be reduplicated: *surie* 'to hate' and *wɛndat* 'to walk'. The meaning of Kipsigis reduplication differs slightly from root to root. Some of the meanings are 'to do something in stages', 'do something again and again', or 'do something on and off'. The core meaning all verbal reduplications share seems to be 'this event takes place more than once'. See (2) for a first Kipsigis example.

(2) Kipsigis[1]

a. í-ít-è
CL2-count-IPFV
'he counts'

b. í-ít~àː~ìt-ì
CL2-count~LK~count-IPFV
'He is counting several times.'
Comment: "You count and when you finish you start counting again. Maybe to confirm the sum you counted the first time."

In (2b) the verbal stem *ít* 'count' is reduplicated. In between the two reduplicative elements, a linking vowel *aː* appears, as is the case regularly with most short-vowel-stems in reduplication. The meaning of the reduplicated verb is given in the translation of (2b) and the speaker comment indicates that the whole event is repeated in this case and not only subparts of it. As shown in (3), Kipsigis' reduplications are incompatible with modifiers like *kɔnɪl ayɛːŋɛ* 'once'.

(3)

pir~ɑː~pir-e	laːkwɛːt	kibɛːt	(*kɔnɪl ayɛːŋɛ)
beat~LK~beat-IPFV	child	Kibeet	times one

'Kibeet is beating the child again and again (*once).'
Comment: "That does not work, for pir-aː-pire you don't know how often it happens, but not once."

This incompatibility, in addition to the example's translations and the speaker's comment, illustrates how the meaning of pluractionality is 'the event takes place more than once'.

[1]Tone is indicated whenever possible, with exception of a few examples which were obtained in online elicitations where transcription was difficult because of unstable audio quality.

2.2 Pokot

2.2.1 Basic facts of Pokot pluractionality

Unlike Kipsigis, Pokot has two pluractional morphemes: reduplication and -*sete*[2]. The morpheme -*sete* sometimes appears in slightly varying vowel quality due to phonological processes. Both morphemes can appear on the same stem, in both possible orders (verb-RED-*sete* or verb-*sete*-RED). The orders are abstractly illustrated in (4). For the same verb, pluractional expressions differ in meaning between the morphemes and sometimes between the possible combinations of them.

(4) co-occuring pluractional morphemes
   a. verb-RED-*sete*: verb~verb-*sete*
   b. verb-*sete*-RED: verb-*sete*~verb-*sete*

Both morphemes share a meaning like 'this event happens more than once', but the exact meaning of each of the morphemes and each of the combinations differs with the specific verb. See Section 3 for concrete examples of this variable ordering.

(5) Pokot
   a. ì-wù:t-éɲ-ì
      2SG-shoot-2SG-IPFV
      'You will shoot' or 'you are shooting.'
   b. ì:-wú:t~wú:t-é:ɲ-ì
      2SG-**shoot~shoot**-2SG-IPFV
      'You are just shooting and shooting without aiming.'
   c. í-wùt-sə́tə̀-nɛ́:ɲ-ì
      2SG-shoot-**sete**-2SG-IPFV
      'You are shooting so many times.' / 'It is your habit to shoot a lot.'

Example (5b) shows a reduplication of the verbal stem *wù:t*. The translation of (5b) indicates a repeated event of shooting. Modifying the verbal stem with the pluractional suffix -*sete* in (5c) also derives meanings where the event takes place several times. Both seem to differ in their interpretation only slightly and sometimes not at all, depending on the specific verb. Often they seem to differ in intensity, as in (37), in the specific amount of repetitions as in (27)/(28) or even in

[2] I am following Marantz 1982 here in assuming reduplication to be a morpheme.

telicity, as in (5). Both Pokot pluractionals, just as reduplication in Kipsigis, generally cannot co-occur with modifiers conveying the meaning 'once' (although, see an exceptional reading in example (9)). As (6) shows, neither reduplication nor *-sete* is compatible with *kəɲəl akoŋge* 'once.'

(6) Pokot

a. kì-pìs~à:~pís ɲíndə̀ (*kəɲəl akoŋge)
PST-split~LK~split PRON.3SG times one
'He split again and again (*once).'
Comment: "It can't be once because it is repetition."

b. kì-pìs-sàtə́ kwén (*kəɲəl akoŋge)
PST-split-sete firewood times one
'He split firewood again and again (*once).'

(6) also shows that the same linking vowel as in Kipsigis (*-a:-*) also appears with short-vowel-stems in Pokot.
Just as in Kipsigis, pluractionality in Pokot is productive and can be used with Pokot and also Swahili loan words. It doesn't seem to be limited to certain verb classes, but no Aktionsart-tests exist for Pokot yet, so a limitation to certain Aktionsart-classes can't be excluded at this point. In my field notes, there is no word that couldn't be modified by reduplication and *-sete*.

2.2.2 Semantic functions of Pokot pluractionality

The semantics of Kalenjin pluractionality varies from root to root. To illustrate the great variety of semantic functions I matched my data with the cross- linguistically observable semantic functions of pluractionality. Mattiola 2019 provides a typological investigation of pluractionality (246 languages, based on Veselinova 2013) and as a result provides a list of semantic functions that can be expressed by pluractionality. Although Kipsigis and Pokot exhibit a variety of semantic functions, for space reasons, only Pokot examples will be discussed in detail in the following to illustrate the semantic variety of Kalenjin pluractionality.
Pokot's two pluractional morphemes are able to express at least 5 out of the 17 possible functions Mattiola 2019 describes. Out of the several semantic functions observable cross-linguistically but not in Pokot (and Kipsigis), two functions will be discussed in some detail: Habituality is discussed because several translations seem to indicate this function for Pokot pluractionality, but Pokot actually uses imperfective aspect to express habituality. Participant number, on the other hand, is taken to be one of the two flavours of verbal number (e.g. Corbett 2000, Amato

2018, Thornton 2019) but can be shown to be two distinct categories in Pokot and Kipsigis. Arguments against conflating these categories in Kipsigis are shown in the next section.
In order to describe the semantic functions of pluractionality, Mattiola 2019 uses the following notations, based on Cusic (1981: 77), which will appear also in this paper:

(7) **event notation** (Mattiola 2019: 21)

  a. situation:
     an event or state
  b. phase:
     the subpart of a situation
  c. occasion:
     a specific occurrence of a situation: the situation with its specific time frame, place and involved participants

Iterativity

Iterativity, one of the core meanings of pluractionality, is described as the sequential repetition of an event on a particular occasion (Mattiola 2019: 22). Both, reduplication and *-sete* can express this meaning in Pokot.

(8) Pokot

  a. lùw-sə́tèː-nə́ mɛ́ːsà
     hit.IMP-sete-IPFV? table
     'hit the table repeatedly/continuously without a reason!'
  b. lùw~á~lúw-ɔ́ʰ mɛ́ːsà
     hit~LK~hit-IMP table
     'hit the table repeatedly for the sake of something!'
     Comment: e.g. to follow a rhythm or to a baby to get it to fall asleep.

Frequentative

Following Mattiola (2019: 24), *frequentative* is the repetition of a specific situation, performed over multiple different occasions. The downtime between the repetitions is long enough to understand each as separate occasions. This matches the so called *plurality of and in events*, as described by Cusic 1981. The translations of (9) show that Pokot reduplication and *-sete* are able to convey this meaning.

(9) Pokot

a. kì-tɛ́k~à~tɛ̀k-é-j kɔ̀ɲíl ákɔ̀ŋə́
PST-build~LK~build-3SG-IPFV times one
'There were several instances of him building once.'

b. kì-tɛ́k-sàtɔ́-j kɔ̀ɲíl ákɔ̀ŋə́
PST-build-sete-IPFV times one
'There were many instances of him building once.'

The building takes place only once at each occasion. Since this is repeated over a bigger time frame there is still a repetitive meaning, the *Frequentative* reading. This reading gives rise to the licit combination of pluractionality and *kɔ̀ɲíl ákɔ̀ŋə́* 'once' which is illicit for other verbs, like *pɪs* 'split' in (6). That a frequentative reading is not possible for all verbs, again shows how the interpretation of pluractionality is highly dependent on the lexical semantics of each root.

Spatial distribution

Spatial distribution describes multiple events which are distributed over several places (Mattiola 2019: 24). Mattiola 2019 doesn't explain if these interpretations are necessary with pluractionality in the languages he discusses or merely a possible situation in which the situation described by pluractionality can occur. For example, if a verb like 'to build' is reduplicated, spatial distribution is a necessary interpretation (if the thing being built is not built, torn down, and built again etc.). The Pokot example in (10) shows such an interpretation: The translation of the example with 'here and there' indicates a spatial distribution of the event.

(10) Pokot reduplication
kàríɓʊ̀ kì-jɛ́ɣ~à~jɛ́ɣ nà:djə́ sú:mì
almost PST.DIST-drink~LK~drink Nadja poison
'Nadja almost drank poison here and almost drank poison there.'

But spatial distribution is only one possible context in which the pluractionality can be interpreted. The context of (11) specifies a non-spatially-distributed reading and the very same pluractional form *kì:-jɛ́ɣ-à-jɛ́ɣ* '(s)he drank again and again' is felicitous in this context as well. This means that iterative or frequentative meanings are always conveyed by pluractionality, but spatial distribution is only a possible context of this repetition.

(11) Context: There is a person having several glasses of water in front of him. He is drinking them in a specific fashion: he drinks each glass really

really fast, it only takes him a few seconds. But then he waits a good while to drink the next glass. I ask you to describe this behaviour to me afterwards.

a. kìː-jɛ́ɣ~à~jɛ́ɣ lɔ̀kɔ̀dɛ́t-ì nàu píɣ làʷɛ́ʰl
PST.DIST-drink~LK~drink cup-PL of water fast
'He drank cups of water quickly.'

Interestingly, no *-sete* form appeared in my data which was translated with a similar construction indicating spatial distribution. Further investigations are needed to determine if the (optional) spatial distribution interpretation is maybe limited to reduplication.

Habituality

Mattiola (2019: 24), following Comrie (1976), defines *habituality* as a repeated action which is typical of a specific period of time. Several translations of pluractional sentences in Pokot suggest that Pokot pluractionality expresses this semantic function. See for example the a. sentences in (12) and (13), which have a clearly habitual translation.

(12) Context: We are talking about somebody who was a professional builder.

a. ki-tɛk-sata-j ɲində kɔ-rɪn
PST-build-sete-IPFV pron.3SG house-PL
'He used to build houses.'

b. #kì-tɛ́k-sàtà ɲində kɔ-rin
PST-build-sete pron.3SG house-PL
'He built several times.'

(13) Context: Somebody had a career as professional builder a while ago

a. kì-ték~à~tèk-é-j
PST-build~LK~build-3-IPFV
'He used to build a lot.'

b. #kì-tɛ́k~à~tɛ̀k
PST-build~LK~build
'He built many times.'

But neither reduplication, nor *-sete* can introduce habituality on its own. Overtly marked imperfectivity is needed additionally to express habituality, as the b. examples of (12) and (13) show respectively: Kipsigis and Pokot both mark imperfectivity, while unmarked verbs are perfective (Kouneli 2021, Toweett 1979). The aspectually unmarked, and therefore perfective, versions of the same pluractional constructions are infelicitous in a clearly habitual context. Further, habituality can even be introduced by imperfectivity on its own, not only in combination with reduplication or *-sete*:

(14) a. kì-rɑ́l-ə́-j
PST-cough-3-IPFV
'He used to cough.'
Comment: "He had this habit but got better"

b. ki-ral
PST-cough
'He coughed.' (once)

Habituality is, therefore, rather a property of imperfectivity, and not of pluractionality in Pokot.

Event-internal plurality

Event-internal plurality is defined as a single situation consisting of hard to distinguish sub-parts, e.g. "He whistles." (Mattiola 2019, see also Cusic 1981). The pluractional form of *támbìl* 'to swim' in (15) provides such an example. As the context shows, not the whole act of swimming from one point to another is repeated. Instead, and this is what the speaker's comment also implies, the indistinguishable subparts of swimming are repeated.

(15) Context: The person we are talking about is swimming from one side to the other without taking a rest.

a. tɑ́mbìl~ɑ̀~tɑ̀mbìl-éj pír
swim~LK~swim-IPFV water
'He swims repeatedly.'
Comment: "Swimming means to do the movement with your hands over and over again."

b. kì-támbìl-sètɔ́
PST-swim-sete
'He swims repeatedly.'

Intensity

The semantic function *intensity* is described as "a situation done with more effort or whose result is augmented with respect to the normal happening of the same situation" (Mattiola 2019: 36). The context of (16), as well as the translations of (16) and (17) indicate that both reduplication and *-sete* can express *intensity* via pluractionality.

(16) Context: The person we are talking about was really sick. They were coughing only once but that coughing lasted for 2 minutes.

a. kì-rál-sátà
PST-cough-sete
'he coughed really badly'

(17) kì-tɛ́k~à~tɛ̀k ki
PST-build~LK~build house
'he built a house faster than normal.'

Indefiniteness

Another function that can be observed cross-linguistically and also expressed by Pokot pluractionals is what Mattiola (2019) calls *indefiniteness*. Mattiola (2019: 40) describes this function as a lack of specific orientation or goal. Other analyses treat this function of pluractionality rather as a change in telicity, see e.g. Henderson (2012). (18) shows how the reduplicated verb *wú:t* 'shoot' expresses a non-telic reading ('without aiming'), while the form modified by *-sete* in (18b) doesn't express such a non-telic reading.

(18) Pokot

a. ì:-wú:t~wú:t-ɛ́:ɲ-ì
2SG-shoot~shoot-2SG-IPFV
'You are just shooting and shooting **without aiming**.'

b. í-wùt-sə́tə̀-nɛ́:ɲ-ì
2SG-shoot-sete-2SG-IPFV
'You are shooting regularly.'

On the other hand, (19) shows that *-sete* is able to express non-telicity/ indefiniteness as well. In this example, it is the form modified by *-sete* which expresses a non-telic version of the verb, while the reduplicated form of *lùw* 'hit' is translated with 'for the sake of something', clearly indicating a goal of the event.

(19) Pokot

a. lùw-sə́tèː-nə́ mɛ́ːsà
hit.IMP-sete-IPFV table
'hit the table repeatedly/continuously **without a reason**!'

b. lùw~á~lúw-ɔ́ʰ mɛ́ːsà
hit~LK~hit-IMP table
'Hit the table repeatedly for the sake of something.'
Comment: e.g. to follow a rhythm or to for a baby to make it fall asleep by listening to you.

Therefore, both reduplication and *-sete* are able to express this function, but apparently always only one of them, depending on the predicate.

2.2.3 Summary of the semantic functions

The previous examples show that Pokot is able to express at least 5 out of the 17 functions of pluractionality Mattiola (2019) identifies. Table 1 shows what morpheme is able to express which function and which functions are unattested. For all other functions, no examples came up in my field notes. This doesn't indicate that these functions don't exist in Pokot, but at least hints at the conclusion that these functions are not as readily available as the attested ones. *Spatial distribution* doesn't seem to be a function Pokot pluractionality expresses, but rather is one of several possible contexts in which repetition can be interpreted. Participant plurality, and how it is not expressed by reduplication or *-sete* is going to be discussed in detail in Section 4.

3 Morphological description of Kalenjin pluractionality

Cross-linguistically, affixation, reduplication and lexical alternation/suppletion are the most common marking strategies for pluractionality Mattiola (2019: 65). Pokot and Kipsigis therefore display two of the three most common strategies. Suppletion is also present in Kipsigis verbal number, but only to mark participant number, as will be shown in Section 4. Why Pokot employs two different marking strategies and how exactly their meaning differs, unfortunately, needs to remain a question for further research at this point. For both Pokot and Kipsigis, the reduplicative morpheme is structurally located below person, tense and aspect morphology, close to the verbal root but above little *v*, the verbalizing head in theories like distributed morphology (Halle & Marantz 1993). As (20) illustrates,

Table 1: Attested semantic functions of pluractional morphemes in Pokot

	-sete	RED
iterativity	✓	✓
frequentative	✓	✓
spatial distribution	(possible context)	no data
habitual	✗	✗
event-internal	✓	✓
intensity	✓	✓
indefiniteness (change in telicity)	✓	✓
successive events	no data	no data
completeness	no data	no data
singulactionality	no data	no data
emphasis	no data	no data
reciprocity	no data	no data
generic	no data	no data
continuative	no data	no data
antipassive	no data	no data
causative	no data	no data
participant plurality	no data	no data

reduplication only targets the verbal stem. Including person and aspect marking in the reduplication leads to ungrammaticality, as in (20c).[3]

(20) Pokot

a. ì-wù:t-é:ɲ-ì
2SG-shoot-**2SG-IPFV**
'You will shoot' or 'you are shooting.'

b. ì:-wú:t~wú:t-é:ɲ-ì
2SG-shoot~shoot-**2SG-IPFV**
'You are just shooting and shooting without aiming.'

c. *ì-wù:t-éɲ-ì~wù:t.éɲ.ì
2-shoot-**2-IPFV**~shoot.IPFV
Intended interpretation: 'He is shooting again and again.'

[3]If not indicated otherwise, all following examples are illustrative for the behaviour of Pokot *and* Kipsigis, even if only one language is shown explicitly.

(21) Kipsigis

a. ɑ́l-é kíbêːt tʃàbàtí
buy-IPFV Kibeet chapati
'Kibeet is buying chapati.'

b. *ɑ́l-é~ɑ́l.é kíbêːt tʃàbàtí
buy-IPFV~buy.IPFV Kibeet chapati
'Kibeet is buying chapati over and over again.'

c. ɑ́l~ɑ̀ːl-é kíbêːt tʃàbàtí
buy~buy-IPFV Kibeet chapati
'Kibeet is buying chapati over and over again.'

The same is true for Kipsigis, as (21) shows. The aspect marker *-e* cannot be reduplicated with the verbal stem (as in (21b)) but must be merged after the reduplication as in (21c). Pokot's *-sete* suffix, as well, appears closer to the stem than aspect and person marking as (5c) shows. Tense morphology, as well, cannot be reduplicated with the stem. Example (22) shows how doing so leads to ungrammaticality.

(22) Kipsigis

a. ki-bir~ɑː~bir-i
PST-beat~LK~beat-IPFV
'He was beating again and again.'

b. *ki-bir-(ɑː-)ki.bir-i
PST-beat~LK~PST.beat-IPFV
'He was beating again and again.'

Other verbal suffixes, as for example applicative morphology, also cannot be reduplicated with the stem as illustrated in (23) for Kipsigis. Analogously, (24) shows how in Pokot applicative morphology applies further away from the root than both reduplication and *-sete*.

(23) Kipsigis

a. kɔ̀ː-á-tɛ́m~áː~tɛ́m-ɛ̀ːn ìmbàr mògòːmbéːt
PST2-1SG-dig~LK~**dig-APPL** farm hoe
'I dug the farm with a hoe again and again.'

b. *kɔ̀ː-á-tɛ́m-ɛːn~(áː)~tɛ́m-ɛ̀ːn ìmbàr mògòːmbéːt
PST2-1SG-dig-**APPL**~LK~**dig** farm hoe
'I dug the farm with a hoe again and again.'

(24) Pokot
kὲ-líl~á~líl-sátá-tSíː
to-hate~LK~**hate-sete-APPL**
'to hate someone a lot.'

In contrast to the morphemes in the data above, the verbalizing head *-an* in both Kipsigis and Pokot can be reduplicated and appears closer to the root then *-sete*.

(25) Pokot
a. rɔ́p
rain
'the rain' (noun)
b. mak-an rɔ́p-**an**
want-1SG rain-VBLZ
'I want it to rain.'
c. míː rɔ̀p-**àn**-ə̀-j
COP rain-VBLZ-3-IPFV
'It is raining.'
d. rɔ́p-**àn**~ɑ̀~**rɔ̀p.àn**-ə́-j
rain-VBLZ~LK~rain.VBLZ-3-IPFV
'It was raining frequently/all the time.'
e. rɔ́p-**àn**-sàtɔ́-j
PST-rain-VBLZ-sete-IPFV
'It was raining a lot/frequently.'

(26) Kipsigis
a. rɔ̀ːp-tá
rain-SEC
'rain' (noun)
b. kò-ròːp-**ɑ́n**
PST-rain-VBLZ
'It rained.'
c. ko-roːp-**ɑn**~ɑ~roːp.**ɑn**-i
PST-rain-VBLZ~LK~rain.VBLZ
'It rained a lot/frequently.'

The Pokot data in (25), which are identical to the Kipsigis data in (26) (besides Kipsigis lacking the pluractional morpheme *-sete*), show how the noun *rɔ́p* 'rain' can be verbalized by the suffix *-an* (the b. examples respectively). To derive a pluractional meaning, the nominal stem plus its verbalizing suffix is reduplicated, as in (25d)/(26c) or modified by the suffix *-sete*, as in (25e).

In theories like DM, lexical categories like nouns and verbs are decomposed into a categorizing head and a root. Morphemes like *-an*, creating verbs from nouns can be analysed as such categorizing heads (Marantz 1997). In such an account, the data above show that the location of Kalenjin pluractionality is above little *v* (the verbalizing/categorizing head) but below all other morphology. The structure in Figure 1 illustrates this observation.[4] In Figure 1 and following structures, pluractionality (PLC) is taken to be adjoined to the *v*P and not project its own phrase. This is based on insights from Amato (2018), who treats pluractionality as an adverbial element (but see Thornton 2020 for an analysis where pluractionality is located on a head). At this point, I see no way to diagnose Kipsigis' and Pokots PLC elements as either heads or adjuncts and therefore will remain agnostic about this distinction.

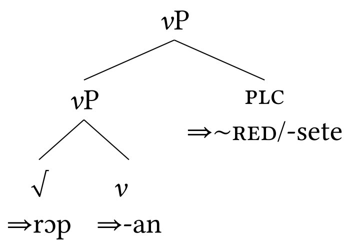

Figure 1: Location of pluractionality

The ordering of the two pluractional morphemes in Pokot appears to be variable, with both orderings showing up, resulting in varying surface orders. These varying orders are indicated in the structures in Figure 2 and Figure 3 below with the translations of the corresponding examples indicating their nuanced meaning differences ((27) and (28) respectively).

(27) verb ≺ RED ≺ *-sete*
kìː-rál~àː~ràl-sàtá
to-cough~LK~**cough-sete**
'to cough again and again' (maybe around 10 to 20 times)

[4] Alternatively, *-an* could verbalize an already nominalized structure *n*P at this point: [*v* [*n*P [$\sqrt{\text{ROP}}$ *n*]] Under both alternatives, pluractional morphology applies after the categorizing head(s).

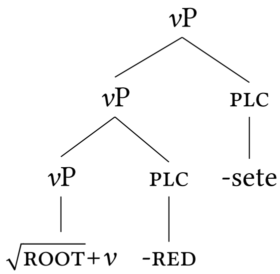

Figure 2: The structure of (27)

(28) verb ≺ *-sete* ≺ RED
kì:-rál-sàtà~ràl.sàtá
to-cough-sete~**cough-sete**
'to cough again and again' (at least 10 to 20 times)

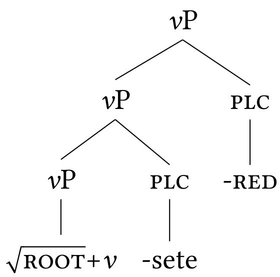

Figure 3: The structure of (28)

In (27), pluractionality by reduplication applies first. It therefore triggers the reduplication of the verbal stem (the root plus verbalizer) and the other pluractional morpheme *-sete* applies after that. In contrast to that, in (28), *-sete* modifies the verb before reduplication applies. The reduplication inducing morpheme therefore triggers the reduplication of the verb plus *-sete*. As the translations indicate, the meanings for this specific verb vary only slightly with their varying order: the meanings only seem to differ in the amount of times the event takes place. (29) provides another example where all possible orderings of reduplication and *-sete* occur.

(29) 'to fold' in all possible orderings (Pokot)

a. kə-ɲàkʊ̀m-àː-ɲàkʊ́m
PST-fold-a-RED
'to fold two or three times quickly'

b. ké-ɲákʊ́m-sàtà
INF-fold-sete
'to fold several times quickly'

c. kɛ̀ː-ɲàkùm-àː-ɲàkùm-sàtà
INF-fold-a-RED-sata
'to fold again and again (more than (29a) and (29b))'

d. kɛ̀ː-ɲàkʊ̀m-sàtà-ɲàkʊ̀m-sàtá
INF-fold-sete-a-RED
'to fold again and again (more than (29a) and (29b))'

In this case, the ordering of reduplication and *-sete* don't seem to make any difference, at least judging from the provided translations.[5]

Overall, both pluractional morphemes (*-sete* and reduplication) can be said to apply after little *v*P (and not before it as, for example, analysed by Thornton 2019) but below other, non-pluractional morphology.

4 Verbal number: participant and event plurality

Verbal number can be observed in two forms: participant and event number (Corbett 2000). Examples (31) and (30) illustrate these two cases respectively.

(30) event number
a. Peter pets a dog once.
b. Peter pets the dog again and again.

(31) participant number
a. Peter pets a dog.
b. Peter pets several dogs.

[5]Often, the subtle meaning differences can only be pinned down by asking the speakers to provide the forms in imperative forms (e.g. ɲàkʊ́m-á-ɲákʊ́m-sátɛ̀-n-ə 'fold again and again!'). The event is then acted out (folding a piece of paper two/three/fours times, folding it slowly/quickly and so on) and to the speaker is then asked if the order was fulfilled. Sometimes, meaning differences like in (27) vs. (28) can be found in this way, but in many other instances, no clear meaning differences were identified yet.

Event number differentiates if an event takes place once or several times. This distinction is illustrated by the contrast of (30a) and (30b). The dog is petted once in (30a), but several times in (30b). Participant number indicates the number of participants. Examples (31a) and (31b) are different from each other in the number of participants, in this case objects. Both distinctions (event and participant number) are described as verbal number, since in both cases an event takes place several times: in (30b) the event happens repeatedly with the same participants and in (31b) the event of petting happens repeatedly, distributed across different participants (dogs). In contrast to Corbett (2000), who classifies both of these phenomena as verbal number, other (especially semantic) literature considers event and participant number to be related but separate phenomena. In this literature event number is often called *pluractionality* (e.g. Henderson 2012, Wood 2007). Coming back to the definition given in (1), of course, I have to admit that the English examples in (30) and (31) are not really cases of verbal number, as they don't match the definition. The differences of the a. and b. examples are not encoded by morphologically modifying the verb, but instead by adverbial constructions. Let us therefore have a look at a more classical example of verbal number. Example (32) from Mupun illustrates verbal number marking encoded directly on the verb.

(32) Verbal number in Mupun (Afro-Asiatic, Chadic) (Frajzyngier 1993: 59-62)

a. wu $\boxed{\text{cit}}$ wur
   3M.SG hit.PST.**SG** 3M.**SG**
   'He hit him.'

b. *wu $\boxed{\text{cit}}$ mo
   3M.SG hit.PST.**SG** 3**PL**
   'He hit them.' (participant number)

c. wu $\boxed{\text{nas}}$ mo
   3M.SG hit.PST.**PL** 3**PL**
   'He hit them.' (participant number)

d. wu $\boxed{\text{nas}}$ wur
   3M.SG hit.PST.**PL** 3M.**SG**
   'He hit him **many times**.' (event number)

Mupun exhibits a suppletive verb pair for the word 'to hit': *cit* and *nas* for the singular and plural form respectively. The singular form *cit* is used with a singular object and a singular event reading, as in (32a). The form cannot be used with a plural object, as the ungrammaticality of (32b) illustrates. To express the hitting of a plural object, the plural form *nas* has to be used, as in (32c). The use of the plural form in this case is similar to the meaning difference described in (31): participant number. In contrast to that, in (32d) the plural verb in combination with a singular object expresses multiplicity of the same event happening to the same participant, similar to the meaning difference in (30): this is event number/pluractionality. The Mupun example therefore shows how both grammatical concepts related to verbal number are expressed by the very same marking strategy: both, participant number and pluractionality are expressed by the same suppletive form of a verb.

Although not all languages conflate event and participant number in this way (Kipsigis and Pokot will turn out to not do so), data like (32) are taken as evidence that both flavours of verbal number should be analysed by the same mechanism. Therefore, the only two existing generative analyses of verbal number morphology (Amato 2018 and Thornton 2019) both rely on the assumption that both flavours of verbal number trigger the same agreement on one head (v in Amato's and a #-head in Thornton's account). In these analyses the head is then spelled out by pluractional morphology, independent of whether it was triggered by event or participant number.

But Pokot and Kipsigis pluractionals behave rather different from languages like Mupun: plural participant number does not trigger or require pluractional marking, only plural events do. In (33b) the plural object *kwenik* 'wood$_{\text{PL}}$' is used with the unreduplicated form *ki:abete* 'I was splitting.'

(33) Kipsigis

a. ki:-ɑ-bet-e kwen-de:t
PST-1SG-split-IPFV wood-**SG**
'I was splitting a piece of wood.'

b. ki:-ɑ-bet-e kwen-ik
PST-1SG-split-IPFV wood-**PL**
'I was splitting pieces of wood.'

c. ki:-ɑ-bet~ɑ:~bet-i kwen-de:t
PST-1SG-split~LK~split-IPFV wood-**SG**
'I was splitting one piece of wood **again and again**.'

d. ki:-ɑ-bet~ɑ:~bet-i kwen-ik
PST-1SG-split~LK~split-IPFV wood-**PL**
'I was splitting many pieces of wood **again and again**.'

When the reduplicated (plural) form *ki:abeta:beti* is used, a meaning of 'the event takes place several times' is conveyed, with the number of the object being independent from that. This leads to the translation of several pieces of wood (plural object) being split several times (plural verb) in (33d) and to the meaning of a single piece of wood (singular object) being split several times (plural verb) in (33c). Plural objects are possible with reduplicated and unreduplicated verbs in Kipsigis irrespective of aspect. As the following, aspectually unmarked examples illustrate, reduplicated and unreduplicated forms are grammatical without perfective marking. Furthermore, the reduplicated form still exhibits a multiple event reading, just as with the imperfective form.

(34) Kipsigis

a. kì:-bɛ́t knwɛ̀:n-ík
PST-split firewood-PL
'He split wood.'

b. kì:-bɛ́t-à:-bɛ́t knwɛ̀:n-ík
PST-split-LK-RED wood-PL
'He split wood again and again.'

The same is true for both forms of Pokot pluractionality (only *-sete* shown here). As (35b) shows, the singular verb form *kìtɛ́k* 'he built' can be used freely with the plural object *kɔ̀:rín* 'houses'. The verb doesn't have to be modified by a pluractional marker like reduplication or *-sete*.

(35) Pokot

a. kì-$\boxed{\text{tɛ́k}}$ $\boxed{\text{kí}}$ kɔ̀ɲíl ákɔ̀ŋə́
PST-**build house.SG** times one
'There was one instance of him building a house.'

b. kì-$\boxed{\text{tɛ́k}}$ $\boxed{\text{kɔ̀:-rín}}$ kɔ̀ɲíl ákɔ̀ŋə́
PST-**build** house-**PL** times one
'There was one instance of him building houses.'

c. kì-$\boxed{\text{tɛ́k-sàtà}}$ ɲíndə̀ $\boxed{\text{kɔ̀:-rín}}$
PST-**build-sete** PRON.3SG **house-PL**
'He built houses several times.'

d. kì-$\boxed{\text{tɛ́k-sàtà}}$ ɲíndə̀ $\boxed{\text{kí}}$
PST-**build-sete** PRON.3SG **house.SG**
'He built a house very fast.'

If, as in (35c), a pluractional verb form is used, the sentence gets the meaning of 'building several houses several times'. This, again, is in contrast to the Mupun data in (32). In Mupun, the combination of plural verb and plural object doesn't receive a multiple event reading but instead only receives a multiple participant reading. The data above show that in both Kipsigis and Pokot reduplication and *-sete* only express one of the two categories associated with verbal number: event number, but not participant number.

Kipsigis and Pokot employ a different marking strategy for participant number. Some Kalenjin verbs are suppletive for participant number. Both of these suppletive forms can co-occur freely with reduplication and *-sete*.

(36) Kipsigis Kouneli 2020: 7

a. lɑbɑt-i lὰ:kwɛ̀:t.
run.SG-IPFV child.NOM
'The child is running.'

b. ruaj lὰ:gô:k.
run.PL.IPFV children.NOM
'The children are running.'

c. ruɑj~ruɑj lὰ:gô:k.
run.PL~**run.PL** children.NOM
'The children are running **over and over again**.'

The verb for 'run' in Kipsigis is either spelled out as *labat* or *ruaj*, depending on the number of the participant. In (36a), the form *labat* is chosen in the context of the singular participant/subject *lὰ:kwɛ̀:t* 'child'. In contrast to that, the form *ruaj* is chosen in the context of plural subjects like *lὰ:gô:k* 'children' in (36b). The reduplicative form *ruaj~ruὰj* in (36c) shows that participant number (marked by suppletion) and event number (marked by reduplication) can freely cooccur in Kipsigis and are not in complementary distribution, as would be expected if they were spelling out the same head. (37) shows the very same for the two suppletive Pokot verbs *rə̀p* 'run' (singular) and *rə́ʝɪɣ* 'run (plural)'. Both suppletive forms (*rə̀p* 'run (singular)' and *rə́ʝɪɣ* 'run (plural)') can freely occur with both exponents of pluractionality: reduplication and *-sete*.

(37) Pokot

a. kì-rə̀p-à~rə́p
PST-run.SG~LK~run
'S/he ran **repeatedly**.'

b. kì-rə̀p-sɛ̀tə́
PST-run.**SG**-sete
'S/he ran **repeatedly**.' (more often than (37a))

c. kì-rə́jíɣ~à~rəjɨ̌ɣ
PST-run.**PL**~LK~run
'They ran **repeatedly**.'

d. kì-rə́jíɣ-sɛ̀tɔ̀
PST-run.**PL**-sete
'They ran **repeatedly**.' (more often than (37c))

Kipsigis has an additional suffix to mark participant plurality in some cases. Just as with suppletion, this morpheme isn't in complementary distribution with event number marked by reduplication either, as (38) shows. The participant number suffix *-toːs* can freely modify an already reduplicated verb like *twal-twal* 'jump over and over again' in (38b).

(38) Kipsigis Kouneli 2020: 3

a. twɑl-toːs lɑ̀ːgôːk.
jump-**PL** children.NOM
'The children are jumping.'

b. twɑl~twɑl-toːs lɑ̀ːgôːk.
jump~**jump-PL children**.NOM
'The children are jumping over and over again.'

The above data indicate that in Kipsigis and Pokot, participant and event number are not two flavours of the same category as in Mupun. Instead, they are marked by different morphological strategies and are not in complementary distribution but can freely co-occur instead. Mupun and Kipsigis/Pokot therefore seem to form two extreme points of an continuum: on one end Mupun, where both categories are conflated fully and Kipsigis/Pokot on the other end, where both categories are fully distinct. It is of course expected that there are languages where the categories overlap in some aspects but not in others.

Although there is unfortunately not enough space to go into detail about the analyses of Amato (2018) and Thornton (2019), my data show that Kipsigis and Pokot pose a problem for any analysis, that relies on the fact that event and participant number are located and spelled out on the same head (like the aforementioned ones do).

5 Conclusion

This paper provided the very first structural description of pluractionality in two Kalenjin languages: Kipsigis and Pokot. The semantic functions of pluractionality were investigated in detail for Pokot: the pluractional morphemes in Pokot express several semantic functions. One of the functions expressed by pluractionals cross-linguistically, habituality, is not expressed by pluractionality in Pokot, but rather by imperfectivity.

This paper further showed that in both languages, the pluractionality morphemes are added to the structure after the verbalizing head little *v* but before any other, for example aspect related, morphology. In Pokot, where two pluractional morphemes can be observed, they co-occur in varying orders with respect to each other. Further, this paper investigated the relationship of event and participant number in both Kipsigis and Pokot, two categories that are often treated as flavours of the very same grammatical phenomenon: verbal number. It was possible to show that event and participant number are marked by different morphological strategies and are not in complementary distribution but can co-occur in both Kipsigis and Pokot. The results therefore suggest that participant and event number are two distinct categories at least in Kalenjin and cannot be accounted for by theories that rely on a conflation of both categories.

Further research needs to investigate if the distinction of both categories is a property of these two languages, the Kalenjin dialect cluster, or maybe their language family. Other questions for further investigations include the specific meaning differences of the two pluractional morphemes in Pokot and why Kipsigis has only one of them.

Abbreviations

Glosses follow the Leipzig glossing rules with the following additions:
the two Kalenjin pluractionality morphemes are glossed as RED and *-sete* respectively; LK = linker, PLC = pluractionality, SEC = secundary suffix, VBLZ = verbalizing head. Reduplicated morphemes are separated from their base with a tilde. Within reduplicants, morpheme boundaries that were present in the base reappear marked by a dot instead of the usual hyphen.

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Chapter 5

Interactions between lexical tone and intonation in Rɨ̀kpa'

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This paper investigates how lexical tone interacts with intonation in Rɨ̀kpá', a Cameroonian Bantu language. We present an acoustic analysis of intonation patterns of declarative sentences and polar questions, concluding that questions are produced with overall higher pitch, and that a right edge H% boundary tone is present, which leads to pitch raising at the right phrasal edge in questions. This boundary tone also interacts with lexical tone in key ways: while the boundary tone overrides and augments the pitch of a lexical high tone at the right edge of the sentence, there is little to no effect of raising for right-edge low tones.

1 Introduction

Though the tonal systems of African languages have long been of interest to phonologists, until recently, work on the intonation systems of these languages has been relatively sparse. Recent work has shown that many African tonal languages have robust intonational systems which interact with a range of grammatical properties, such as sentence type (e.g. questions vs. statements, Chai et al. 2022, Downing 2017, Gjersøe et al. 2016, Kula & Hamann 2017, Patin 2017, Rialland 2007, Rialland & Aborobongui 2017); focus (Inkelas 1998, Kügler 2017, Kisseberth 2017, Patin 2017) and syntactic boundaries (Chai et al. 2022). When it comes to marking questions, intonation can be seen to have an effect at either the local or global level (or both). Global intonation involves overall raising or lowering of pitch, a phenomenon which can also occur in conjunction with vowel raising/lowering and phonation changes, a host of properties sometimes

Fridah Gam & Kathryn Franich. 2026. Interactions between lexical tone and intonation in Rɨ̀kpa'. In Vicki Carstens, Katherine Russell, Olawale Akingbade, Deborah Morton & Michael Diercks (eds.), *Pamoja tena 'Together again': African linguistics after COVID*, 89–100. Berlin: Language Science Press. DOI: ??

characterized in terms of 'tense' vs. 'lax' prosody (Lindsey 1985, Rialland 2007, Downing & Rialland 2017). Local intonation involves raising or lowering of pitch on a specific word. When these effects are relegated to the left or the right edge of the phrase, they are typically seen to be caused by a boundary tone. There is variation across African tonal languages in terms of whether and which boundary tones are employed and which syllables they associate to. For example, a right-edge boundary tone might associate with the final syllable, penultimate syllable, or multiple syllables (Rialland 2007 and references therein). Boundary tones may also only surface with specific lexical tones. For example, a boundary tone may surface only when a final lexical high tone is present, but not in the absence of a high tone (Patin 2017). Here, we investigate both global and local properties of intonation in Rɨ̀kpá' to see how the language fits into the general typology of African intonational systems

2 Tone and intonation in Rɨ̀kpá'

Rɨ̀kpá' (A53) is a Bantu language spoken by approximately 60,000 speakers in the Center region of Cameroon. The language has SVO word order. Guarisma Popineau (1992) represents one of the most extensive descriptions of the Rɨ̀kpá' language, including the tonal system of Rɨ̀kpá', which includes both high and low, as well as contours which can be formed through the association of two distinct level tones to a single syllable (1).

(1) a. ká "look for"
  b. kà "serve"
  c. kà + PFV → kǎ "serve (perfective)"
  d. kpâ "speech"

Guarisma (2000) describes some aspects of the intonation of the language, including downstep and downdrift within declarative utterances. As can be seen in Figure 1 (pitch tracks based on the present study), high tones become progressively lower throughout the sentences, whether or not an intervening low tone occurs.

Beyond the effects of these downtrends, there appear to be additional intonational processes in Rɨ̀kpá' that are at play in marking sentence type. Specifically, there are clear intonational differences in the production of declarative statements versus polar questions even where the two constructions are otherwise morphologically identical. Figure 2 and 3 contain two declarative statements,

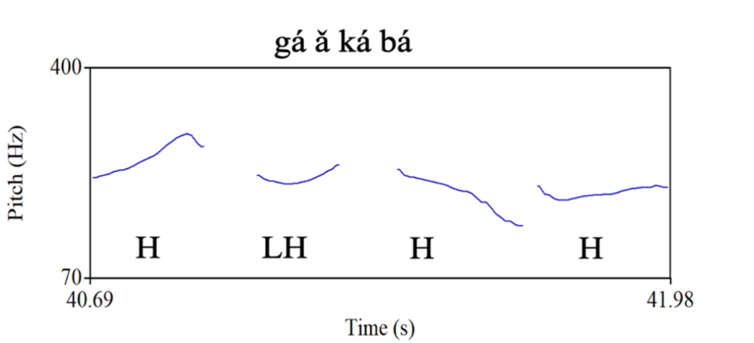

Figure 1: Downdrift in Rɨkpá’ *gá ǎ ká bá* ‘Mother looked for the meat’

with Figure 2 containing a high subject, verb, and object, as well as a rising tone subject marker in second position (*gá ǎ ká bá* ‘Mother looked for the meat’) and Figure 3 containing a low subject, verb, and object (*gà ǎ kà bà* ‘Ga served the sea salt’). Figure 4 and 5 contain corresponding polar questions (e.g. *gá ǎ ká bá?* ‘Did mother look for the meat?’) with the same lexical tone melodies. Note that the end of the high tone polar question in Figure 4 has a steep rise in final position, whereas the low tone polar question does not show as much of a rise.

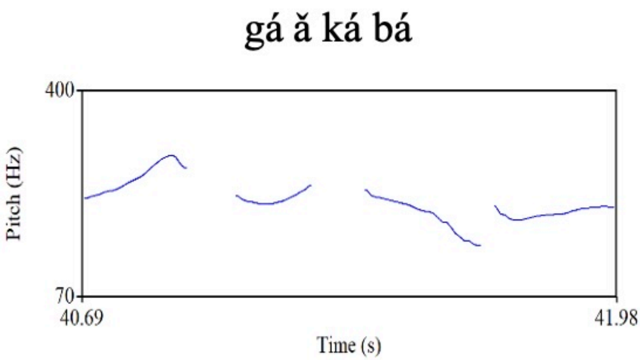

Figure 2: Declarative with H tones

The difference in patterning of the final word across Figure 2-5 suggests that a boundary tone may be present in at least some contexts, e.g. where the sentence ends with a word with a lexical high tone. We now aim to provide an acoustic analysis of f0 in Rɨkpá’ in order to establish 1) whether or not there is any evidence for the presence of a similar high boundary tone in questions ending with

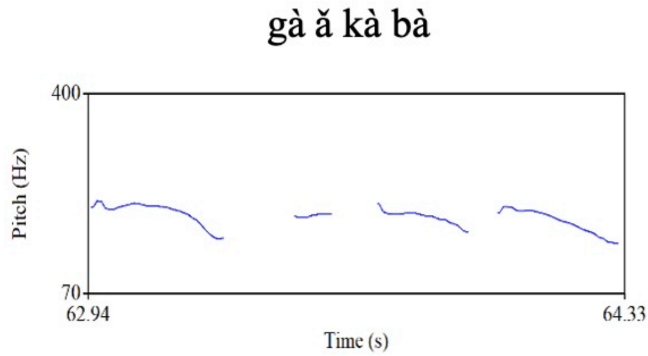

Figure 3: Declarative with L tones

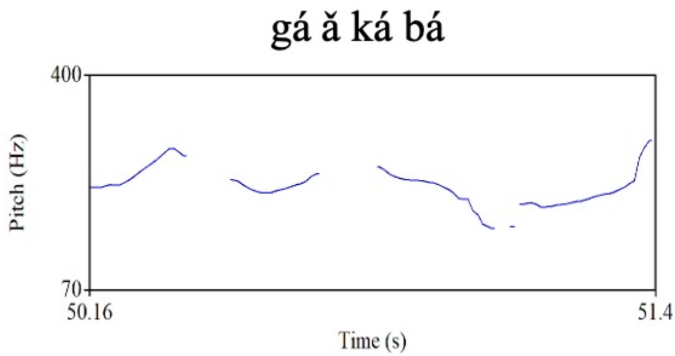

Figure 4: Polar question with H tones

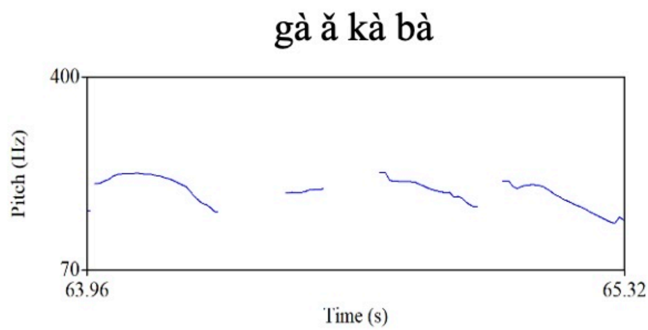

Figure 5: Polar question with L tones

a lexical low tone; 2) whether the boundary tone is limited to occur in constructions which end with a noun, like the examples above; and 3) whether Rɨ̀kpá' also shows evidence for more global intonation patterns, such as sentence-level pitch raising in polar questions.

3 Method

Data for this study come from a corpus of pairs of declaratives/yes-no questions in which the subject, verb, and object varied in terms of whether they carried a high or low lexical tone. The structure of the sentences was Subject – Subject Marker – Verb – Object. The subject marker in these sentences always had a LH rising tone. There were two different lexical items used for the subject and object for each tone, giving a total of 3 (sentence positions) × 2 (tones) × 2 (sentence types) × 2 (subject lexical items) × 2 (object lexical items) = 48 items. Sentences were repeated 3 times each and recorded in random order for each of six speakers, 3 identifying as male and 3 identifying as female. Sentences were recorded and annotated by the first author in Praat, and then pitch values were extracted from individual phones for analysis. F0 was z-scored by subject in order to control for individual variation in pitch. Linear mixed effects models including fixed factors of Word (subject, verb, or object), Lexical Tone (high or low), and Phrase Type (question or declarative), as well as a by-subject random intercept, were used to evaluate how these variables influenced fundamental frequency. Interactions between these variables were also examined. Main effects for each variable were assessed through a Type III Anova using Satterthwaite's method via the lmertest package in R.

4 Results

Table 1 shows the average pitch values (in Hz) for each word in the sentence by lexical tone and phrase type. Overall, the values for words in questions are higher than those in declarative statements.

We see differences in pitch across all words in the sentence by sentence type: polar questions are realized with higher pitch than declarative sentences. Of note, in particular, is the difference in pitch on the last word. We can see there is an especially large difference in pitch between the declarative and polar question for the final word when it carries a high tone. Here, there is an average difference of 44 Hz between conditions. For final low tones, this difference is only 10 Hz. The z-scored results for each word by tone and sentence type are shown in Figure 6.

Table 1: Average f0 (in Hz) for each word by tone and phrase type

Phrase Type	Subject	S.M.	Verb	Object
Declarative L	168.94	160.82	162.36	158.58
Question L	171.05	165.1	166.39	169.22
Declarative H	205.17	188.89	183.92	181.83
Question H	212.47	196.01	221.34	225.9

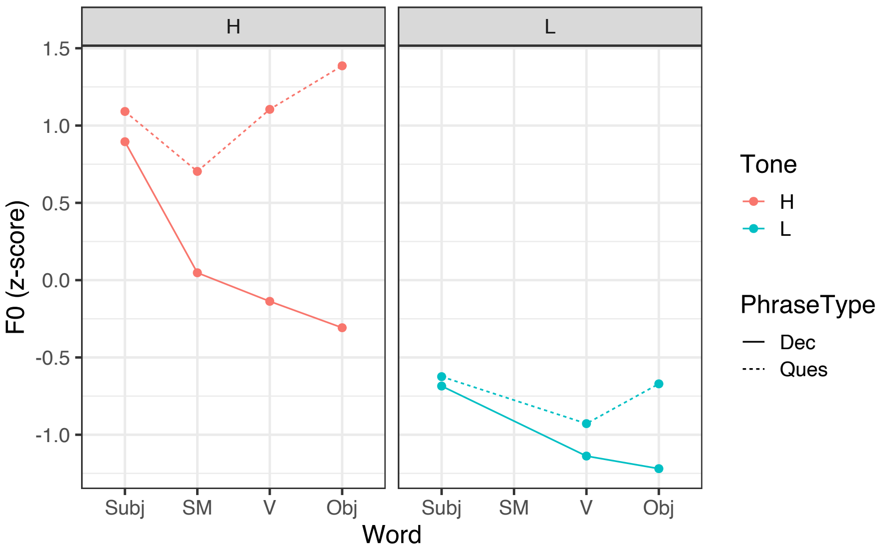

Figure 6: Pitch results by Word, Tone, and Phrase Type

Results of the linear mixed effects model are shown in Table 2. There was a main effect of Lexical Tone, as expected: high tones were, on average, higher than low tones. There was also a significant effect of Phrase Type, with questions generally produced at a higher pitch than statements. There was also a main effect of Word, with words earlier in the sentence realized with higher pitch values than those later in the sentence. A significant two-way interaction between Phrase Type and Lexical Tone reflected the fact that the effect of question raising was greater for words with lexical high tones than with low tones. An interaction between Phrase Type and Word reflected that declination effects seen in declarative sentences were not as pronounced – or even reversed – in questions, where the effects of the boundary tone can be seen to raise the pitch of the phrase-final word to a higher level than the initial word of the sentence. Finally, the three-way interaction between Phrase Type, Lexical Tone, and Word reflected the fact that differences in pitch across phrase type and sentence position were more pronounced where the underlying tone of the words was high than when they were low.

Table 2: Results of statistical analysis

PhraseType	$F(1, 31988) = 889.01$	$p < .001$
Lexical Tone	$F(4, 31988) = 116.74$	$p < .001$
Word	$F(4, 31988) = 61.20$	$p < .001$
PhraseType * Lexical Tone	$F(4, 31988) = 87.17$	$p < .001$
PhraseType * Word	$F(3, 31988) = 17.20$	$p < .001$
PhraseType * Lexical Tone * Word	$F(3, 31988) = 3.78$	$p < .001$

5 Follow-up analysis: Sentences ending in adverbs

So far, our analysis has only taken into account sentences of a specific structure, i.e. those involving a subject, subject marker, transitive verb, and an object. In order to explore whether the effects of the boundary tone can be seen more generally, we had participants record new stimuli which included the same material as the previous sentences, plus one of two additional adverbs, *gɛ̂nʌ́*, 'today' which has a HLH melody, and *dɪ̀ŋko*, 'yesterday' which has a LL melody. We followed the same procedure for the prior dataset.

Once again, we see an overall higher pitch level for questions as compared with declaratives. As before, we see a rise on the final word of the sentence for

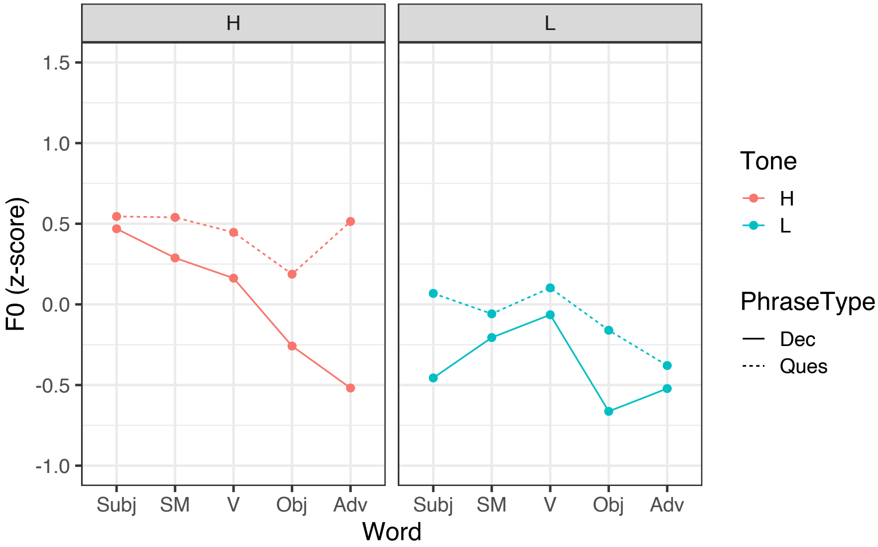

Figure 7: Pitch results by Word, Tone, and Phrase Type, with adverbs included

high tone words within questions, consistent with the presence of a boundary tone. We do not see the same rise on the last word for the low tone words in questions, though the difference in pitch between Phrase Type conditions does appear to increase somewhat at the object position.

We can now take a closer look at the two adverbs uttered in the two phrase type conditions to see exactly where the boundary tone is taking effect. In Figure 8, the x-axis represents the timepoint within the adverb as a percentage. We can see that the that the pitch rises toward the last third of the word and then plateaus, indicating the boundary tone likely docks to the last high tone syllable. For the adverb with the LL melody, we again do not see the same strong evidence for a boundary tone, though there is a very slight rise between the last two timepoints for questions.

Finally, we consider data for sentences ending with the LL adverb in order to see how the tone of the preceding object influences the presence or location of the high boundary tone. In other words, where we have a low tone in final position, do we find evidence that the high boundary tone in questions docks somewhere earlier in the sentence? The answer to this appears to be 'yes': whereas the highest pitch in questions is realized on the object (in penultimate position) when it is high, the highest pitch is realized on the verb (in antepenultimate position) when the object is low and the verb itself his high.

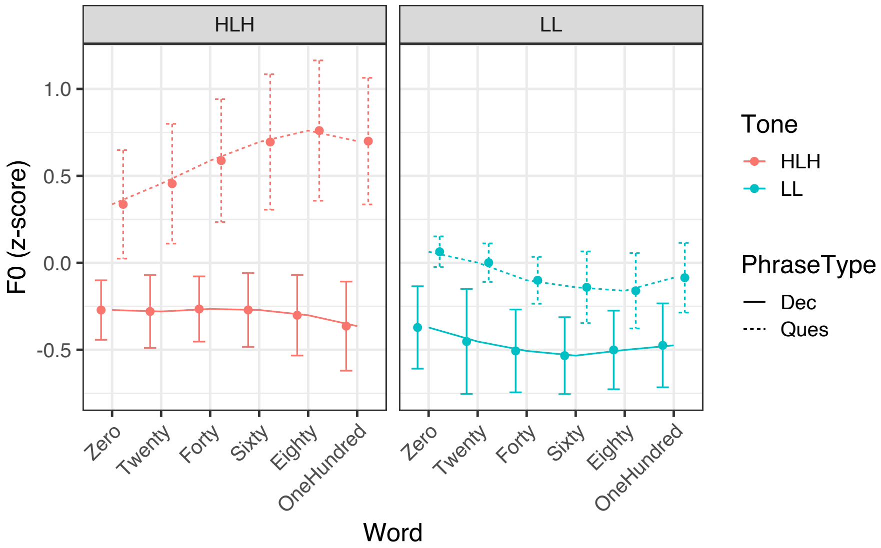

Figure 8: Pitch patterns on final adverb by Phrase Type and Timepoint

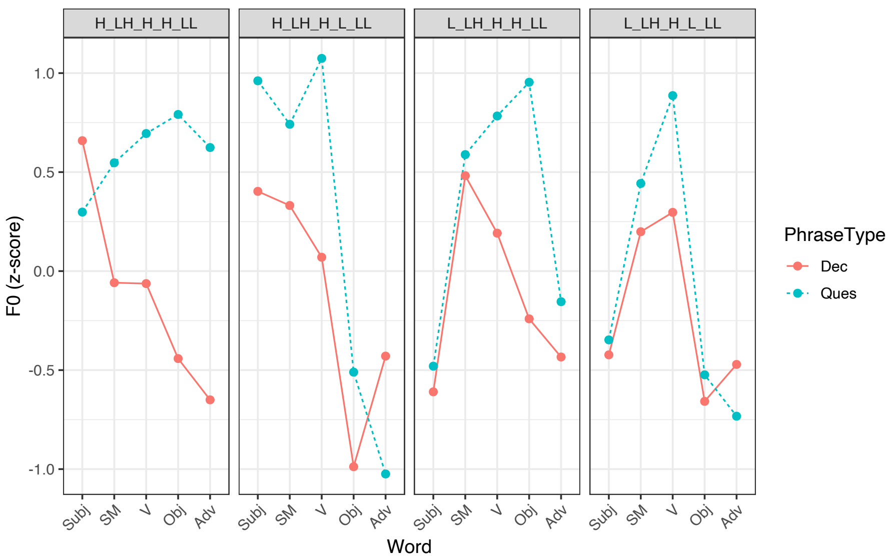

Figure 9: Change in boundary tone location based on location of final H lexical tone (object vs. verb)

6 Discussion

To sum up, Rɨkpa' shows phonetic effects which are characteristic of tense prosody in polar questions, such as overall higher pitch (Rialland 2007). In addition, we find local effects of pitch raising: a high, or even super-high, boundary tone is present at the right edge, attaching to the final high tone of the utterance and pushing the pitch of that syllable into a higher range than what is expected for lexical high tones. Final low tones show some raising, but not nearly as much as final high tones, and not as consistently. Therefore, although the boundary tone may be present in contexts with phrase-final lexical low tones, it appears to be mostly overridden by the lexical tone of the final word. The fact that there appear to be subtle phrase-final raising effects in at least some contexts (e.g. see Figure 6 for questions ending in an object) suggests that this effect of tonal override may be gradient, rather than categorical, in nature. Another possibility is that speakers might show variability in whether or not they show effects of a boundary tone in low environments. In this case, averaging over all speakers might obscure some more robust patterns of raising for certain speakers. We leave this question for future investigation.

The pattern of boundary tone association at the right edge in Rɨkpa' is similar to what is described for a range of other Bantu languages from different areas, including Ekoti (P30), Embosi (C25), Herero (R30) and Kinyarwanda (JD60) (Schadeberg & Mucanheia 2000, Möhlig et al. 2002, Kimenyi 1980). We note that ours is the first study to document this type of pattern in a Bantu language from Zone A. Similar to Shingazidja (G44a), as described by Patin, we find a mix of boundary tone overriding lexical tone vs. being blocked by lexical tone. However, the position and nature of boundary tones (e.g. whether they override or are overridden by lexical tones) is quite variable, and this variation is worthy of deeper investigation. It will also be necessary to investigate the behavior of the boundary tone in Rɨkpa' in order to understand its patterning in a broader array of sentence types and information structural contexts. A larger survey of the behavior of boundary tones in Bantu languages is also a promising next step in understanding how these tones are represented by speakers and how they interact with lexical tonology.

Finally, the present study has only considered a single quantitative measure – average pitch – in the patterning of different sentence types. This has been a good first step in investigating interactions of tone and intonation in Rɨkpa', as we see robust differences in average pitch across the conditions of interest. However, our pitch tracks also reveal considerable variation in the overall shape of the pitch contours for different words across tonal and sentential contexts.

Future work should focus on how other pitch-related factors such as tonal slope and pitch range may be implicated in the production and perception of lexical tones and boundary tones in Rɨkpa’.

Abbreviations

H: High lexical tone
L: Low lexical tone
H%: High boundary tone
PFV: Perfective
SM: Subject Marker

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Chapter 6

Towards a typology of Bantu complementizers

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University of Kansas

This paper provides a preliminary overview of the lexical sources and functions of complementizers used for finite, non-interrogative, syntactically selected embedded clauses in Narrow Bantu languages. It identifies five distinct sources for complementizers in Bantu languages: ‘say,’ ‘be,’ demonstratives, manner deictics, and pronouns. The paper further discusses languages which have more than one such complementizer, and investigates three factors that correlate with complementizer choice. Based on these initial findings, the paper concludes that there is a general dichotomy between complementizers, which is framed in terms of “perspectival” vs. “situational” anchoring.

1 Introduction

In this paper, I provide a preliminary typology of *complementizers* in the Narrow Bantu languages.[1] This typology covers three aspects. First, I address the lexical source of complementizers across Bantu languages, building on typological work in e.g., Cristofaro (2003), Kuteva et al. (2019). Second, Bantu complementizers are widely documented as having various “effects” on the utterance. I illustrate some of those effects, noting the range of syntactic, semantic, and morphological

[1]The classification “Narrow Bantu” applies to roughly 600 Bantoid languages, ranging from West to East Africa and into Southern Africa. Narrow Bantu specifically excludes the Grassfields Bantu languages of Western Africa. The classification of individual Bantu languages follows the Guthrie code system (updated in Hammarström 2019), which consists of one or two letters indicating sub-family and number indicating the specific language in that sub-family. In this paper, Guthrie codes are given following each language.

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variation. I finally conclude by looking briefly at some correlations between the lexical source of a complementizer and the effect it can have, observing that there is a general split among the Bantu complementizers.

Based on these preliminary findings, I suggest an informal classification in terms of *anchoring*, that is, how the embedded clause is integrated into the superordinate material. "Perspectival" complementizers anchor the embedded clause to a relevant *individual*; "situational" complementizers anchor the embedded clause to a relevant *situation* (or event). Ultimately, I suggest that all Bantu complementizers fall into one of these two camps – at least historically.

Before beginning, a note on terminology is in order. I use the term *complementizer* atheoretically to cover a number of potentially disparate categories, including clausal conjunction and (clausal) subordinator. I also assume that there is no formal distinction between a complementizer and a quotative marker (Güldemann 2002).

Finally, I am confining my investigation to finite noninterrogative clausal complementation. Thus, I put aside nonfinite complementation and interrogative complementation, as well as other clause-types, like relative clauses, causal clauses, temporal clauses, conditional clauses, etc. This is purely to constrain the scope of the study.[2]

2 Sources of complementizers

Crosslinguistically, a number of distinct sources for complementizers are recognized (Hopper & Traugott 1993, Güldemann 2008, Kuteva et al. 2019). Chappell (2008) summarizes the known sources in (1), with the bolding to indicate sources that Bantu languages instantiate.

(1) a. nouns such as 'thing', 'fact' or 'place'
    b. **demonstrative**, interrogative and **relative pronouns**
    c. dative, allative and locative case markers or prepositions
    d. **SAY verbs**

[2]The data presented here draws on over 100 languages (not all of them cited here for space) and is collected mainly from existing descriptive and theoretical sources on Bantu languages. Additional data comes from the authors' fieldnotes. I emphasize the preliminary nature of this study; more data are needed to paint an accurate picture. An attempt was made to include data from all of the Guthrie Zones (the classificatory system for Bantu languages; I will rely on Hammarström's (2019) updated Guthrie codes). However, Eastern Bantu is over-represented in the language sample due to the prevalence of material on those languages. When a citation is not present, the data was collected through fieldwork by myself.

e. similative verbs meaning 'resemble' or 'be like'

f. **similative manner adverbials and deictics**

adapted from Chappell (2008: 3)

As expected, in these data, we find considerable variation between languages (and even within languages) in the form of the complementizer and its diachronic or synchronic source. Bantu languages robustly instantiate at least three of these tendencies: demonstratives (including pronominal complementizers), *say*-verbs, and manner deictics. Bantu languages additionally employ a strategy not described in (1) above: *be*-complementizers, i.e., complementizers derived from or related to the copula. As these are distinct from similative verbs and particles (the latter also attested to a lesser extent in these data), I treat *be*-complementizers as a distinct lexical source. I note that my goal below is not to provide a diachronic trajectory for these transitions; I will not explain *why* certain lexical categories are recruited as complementizers. I believe that this motivation has been fleshed out for most categories in Kuteva et al. (2019) and related work – save *be*-complementizers, which I leave for future work.

2.1 *Say*-complementizers

The appearance of *say*-complementizers is widespread across the world's languages (cf. Hopper & Traugott 1993: 14; Kuteva et al. 2019: 375). Accordingly, a large number of Bantu languages employ a form of the verb meaning, or diachronically related to, 'say.' In (2a), Mwani's (G403) *kamba* is derived from proto-Bantu **-(g)amb* 'word, speak' (cf. Meeussen 1967: 94). In (2b), Bena's (G63) *uhutigíla* functions synchronically as a *say*-verb, as does Kisi's (G67) *kujobha*, in (2c). (Note that, despite the different lexicalizations, these are all Zone G languages.)[3]

(2) a. Mwani (G403) (LIDEMO 2010: 55)
Amadi akwijiwa kamba Ali kawa nao nzuruku.
Amadi knows COMP Ali be with money
'Amadi knows that Ali had money.'

b. Bena (G63) (Morrison 2011: 417)

[3] Here and throughout, examples are provided with the glossing as given in the source material. See Section 5 for a list of abbreviations.

a-va-anu va-i-pulih-a uhutigíla u-mu-yeesu
AUG.2-CL2-person CL2-PRES-hear-FV COMP AUG.1-CL1-our.friend
a-taagih-ile
CL1-die-FV
'People hear that our friend died.'

c. Kisi (G67) (Nicolle et al. 2018: 13)
Lɨngɨ.na.lɨngɨ. bha-ka-n'-jobh-el-a kujobha, "Bɨt-agh-e
after.a.while 3PL-CONS-3SG-say-APPL-FV QUOT go-IPFV-SBV
ka-kot-agh-e ku=bhulongolo
ITIVE-ask-IPFV-SBV 17.LOC=ahead
'After a while they said to him, "Go and ask up ahead..."'

The examples above illustrate that multiple languages have *independently* developed a *say*-complementizer from distinct sources. However, extending across nearly all subfamilies are derivatives of proto-Bantu **-tì*, which in many languages has evolved into an element meaning 'say' as well as a complementizer (Meeussen 1967, Güldemann 2002). Note that for many languages it is used primarily to introduce direct speech, and thus has a quotative function (Güldemann 2008). The examples in (3) demonstrate an infinitival, thus verbal, form of **-tì*.[4]

(3) a. Nyakyusa (M31) (Persohn 2017: 314)
ʊ-meenye ʊkʊtɪ Asia a-ka-kʊ-gan-a?
2SG-know.PFV COMP A. 1-NEG-2SG-love-FV
'Do you know that Asia doesn't love you?'

b. Zulu (S42) (Halpert 2016: 1)
ku-bonakala ukuthi uZinhle u-zo-xova ujeqe
17S-seem that AUG.1Zinhle 1S-FUT-make AUG.1steamed.bread
'It seems that Zinhle will make steamed bread.'

c. Fwe (K402) (Gunnink 2018: 432)
mbo-a̠-bo$_H$n-e̠ kutí ∅-ci-pepa bu-ryó
NEAR.FUT-SM$_1$-see-PFV.SBJV COMP COP-NP$_7$-paper NP$_{14}$-only
ci-bá-mu-dara
PP$_7$-NP$_2$-NP$_1$-old.man
'She will see that it is just a paper of her husband.'

[4] Gunnink (2018) explains that in Fwe, the form *kùtí* and the variant *kùtêyè* are "contractions of the verb *kùtá* 'to say', with the complementizer íyé."

In the data, derivatives of **-tì* are prevalent in Eastern and Southern families. Though often correlating with a verb of speech in modern languages, Güldemann (2002, 2008), following the reconstruction in Guthrie (1970b: 105), argues that the original function of **-tì* was that of a manner deictic, a demonstrative element that is anaphoric (or cataphoric) to an expression of manner or kind. Over time, in some languages, the manner deixis meaning transitioned to a speech verb (often defective). The manner deictic use of **-tì* is still found in a number of Bantu languages; I return to this in Section 2.3.2.

Still, it is undeniable that some languages have derivatives of **-tì* which are verbal.[5] This is clear from the fact that many *ti*-based complementizers may still bear vestiges of verbal inflectional marking, like tense, aspect, and voice morphology (Persohn 2017, Letsholo & Safir 2019, Dembetembe 1976).

However, it is sometimes unclear how much of this inflectional material is completely fossilized as part of the complementizer. For instance, should Xhosa's (S41) invariant complementizer *sengathi* be analyzed synchronically as consisting of *se-* 'aspect', *nga-* 'possibility modal', and *-thi* 'say' (cf. Halpert 2019: 33)? Moreover, it is also clear that some *say*-complementizers have nominal properties. This is the core finding of Pietraszko's (2019) work on the Ndebele (S44) complementizer *ukuthi* (again derived from **-tì*), which is argued to be housed inside of a nominal layer.

Finally, I note that Devos & Bostoen (2012) discuss a variant of the *say*-complementizer strategy: some Bantu languages employ a complementizer that is related to (again, either synchronically or diachronically) a word meaning 'do.'[6] This is shown for Shangaci (P312) in (4) with the verb *-ira*, translated variously as 'do, say, (be) like.' As reported in Devos & Bostoen (2012), the same polysemy is found in Lunda (L52), Nkore-Kiga (JE13/14), Rundi (JE62), Rwanda (JE61), and Mongo (C61).

(4) Shangaci (P312) (Devos & Bostoen 2012: 99)

a. ki-ao-ir-i <ni-law-e o-mu-ti>
SC$_{1SG}$-PST-QV-PFV <SC$_{1pl}$-go-SBJV NP$_{17}$-NP$_{3}$-town>
'I *said*: 'Let's go home.''

[5] A reviewer notes that there are multiple paths to being a verb for **-tì*. It could develop verbal properties and then become a complementizer, or alternatively, it could be co-opted as a quotative and then develop verbal properties subsequently.

[6] The data reported here draws heavily on the discussion in Devos & Bostoen (2012). Any errors in reporting and interpreting their data are my own.

b. ir-an-i tono mu-mu-thuul-e ontu
QV-IMP-PLA thus SC$_{2pl}$-OC$_{1}$-take-SBJV PP$_{1}$.DEM$_{i}$
'*Do* it as follows: Take this one…'

c. mu-khira o-awe o-sal-a o-ir-a khong'ongoo
PP$_{3}$-tail PP$_{3}$-POSS$_{1}$ PP$_{3}$-remain-PFV PP$_{3}$-QV-PSIT IDEO
'His tail remains *like* a stump.'

It is likely that there is a single proto-Bantu source for *do*-complementizers in Bantu: **-gìd-* (Bastin et al. 2002). As noted in Devos & Bostoen (2012), given the polysemy of **-gìd* across Bantu, it may ultimately be possible to list *do*-complementizers as a sub-type of *say*-complementizer.[7]

2.2 *Be*-complementizers

In this study, I found a number of Bantu languages which introduce embedded clauses using a form of the copula.[8] In the examples in (5), all the complementizers are *synchronically* multi-functional as infinitival forms of a verb meaning 'be.'

(5) a. Haavu (JD52)
Mugisha a-yubah-ire okuba o-lu-juki lu-a-mu-lume.
Mugisha 1SM-be.afraid.PFV COMP 11AUG-11NC-bee 11SM-FUT-1OM-sting
'Mugisha is afraid that the bee will sting her.'

b. Ikoma (JE45) (Roth 2018: 94)
nihó Núhu a-ka-mɛny-a kuβá amánche
so Noah 3SG-NARR-come.to.know-FV that waters
ɣa-aɣá-tiβok-a mose
6-NUCL-decrease-FV on.land
'So Noah knew that the waters had subsided from the earth.'

c. Xhosa (S41) (du Plessis 1989: 44)

[7] An interesting though perhaps exceptional case is Mwimbi-Muthambi (E531), which uses *taka*, plausibly derived from a verb meaning 'want' as the complementizer (Kaburo 2022). If this is truly the etymology of the complementizer, then it may also be grouped together with the other "action"-oriented complementizers like *say* and *do*.

[8] I refer the reader to the discussion in Gibson et al. (2018), Finholt (2024) concerning the varities of copular predication in Bantu. This is a complex topic. As it applies to complementizers, it is clear from the data collected that there are a limited number of *be*-complementizers, related to only a subset of possible copulas across Bantu.

ndafumana ukuba mandifunde isiXhosa
I.found that I.must.study Xhosa
'I found that I must study isiXhosa.'

d. Mbundu (H21a) (Johnson 1931: 77-78)
mwane wambe kuma "eme maza ngendele mu-nyanga"
he said that I aux hunt yesterday
'He said "I went hunting yesterday."'

The majority of *be*-complementizers in our sample appear to be derived from proto-Bantu **-bá* 'be, become' (Guthrie 1970a: 17; Meeussen 1967: 86), retaining a bilabial place feature (/b, β, m, w/) in all subsequent forms.[9] Still, there are some languages, particularly those in Zone E, which appear to have independently developed complementizers from other forms of the copula. Digo (E73) and Giryama (E72a) both use *kukala* 'be.' This verb is likely derived from proto-Bantu **-kàd* 'dwell, be' (Guthrie 1970a: 258) or **-yį̀kad* 'dwell, be' (Guthrie 1970b: 179), with the d/l alternation, already attested in proto-Bantu. Likewise, I believe that *kana*, found in a few disparate languages like Kamba (E55) has a similar origin, with *d>n.[10]

(6) a. Digo (E73) (Nicolle 2014: 55)
Ndipho atu a-chi-many-a kukala iye ndi=ye
then 2-people 3pl-cons-know-fv comp 3sg cop=1.ref
a-ri-ye-hend-a mambo higo.
3sg-pst-1.rel-do-fv 6.things 6.dem.np
'then people knew that it was her who did those things.'

b. Giryama (E72a) (Lax 1996: 273)
yuyu ŋ͡manamuče were walagiza kukala a-so-zik͡p-e
that woman aux instructed comp 1sm-neg-bury.pass-sbjv
'The woman had instructed that she should not be buried.'

c. Kamba (E55) (Myers 1975: 186)
aisye kana nũkũka ũmũnthi
he.say.past that he.come.pres tomorrow
'He said that he's coming tomorrow.'

[9] Johnson (1931: 78) observes that *kuma* is an "impersonal expression," generally used for "there/it is…" The source of the nasality here is not clearly *b however.

[10] The *d>n trajectory is not common, but it is not unheard of either. Guthrie (1970a) lists reflexes of **-yį̀kad/*-kàd* as **-įkan* and **-įan* in East Holoholo (D28b) and Bembe (D54) respectively.

Beyond the copular elements above, I found a few Bantu languages that use the morpheme *n(i)* as complementizer. In Bembe (D54) in (7a), *ni* appears with an invariant class 8 form of the relative pronoun *-bo* (and thus also falls under the pronominal category of complementizer, discussed in Section 2.3.3). In Luba-Kasai (L31a), I believe that *ne* is the cognate form in (7b).

(7) a. Bembe (D54)
Aachi a-le-ngany-a nibo Mswakeecha a-le mu-lwaala.
Aachi 1SM-PRES-think-FV COMP Mswakeecha 1SM-COP 1AGR-sick
'Aachi thinks that Mswakecha is sick.'

b. Luba-Kasai (L31a)
Kalombo mu-sw-e ne Mujinga a-y-e ku Tshinsansa.
Kalombo 1SM-want-FV COMP Mujinga 1SM-go-SBJV to Kinshasa
'Kalombo wants Mujinga to go to Kinshasa.'

Across Bantu, "[as] proclitic or prefix, [ni] has a number of functions easily relatable to the copula" (Nurse 2008: 53). I thus choose to list examples like (7) as *be*-complementizers. However, it is also observed that *ni* is strongly associated with *focus* (Nurse 2008, Güldemann 2013). Therefore, we may ultimately want to list *ni*-based complementizers as **focus complementizers**. Still, there is a tight connection between copulas and focus (Kuteva et al. 2019: 125), whereby focus markers are often grammaticalized from copulas. Therefore, it is not surprising that we see a link between focus and complementizers; however, it is unclear whether that link is always through the shared connection with copulas.

2.3 Deictic complementizers

Bantu languages contain a diverse range of complementizers related to deixis. I divide up the class of deictic complementizers into three distinct subclasses: demonstrative, manner, and pronominal deixis.

2.3.1 Demonstrative complementizers

Our data reveals that a number of Bantu languages employ a complementizer that elsewhere has (or had) a demonstrative function. All languages in Zone JD have as a complementizer a form of the class 15 form of the proximal demonstrative, realized as *uku* 'this$_{15}$' in Rwanda (J61).

(8) a. Shi (JD53) Aron Finholt (p.c.)

Mugisho a-lá:-waz-a ku Murhula a-li Bujumbura.
Mugisha 1SM-PRES-think-FV COMP Murhulla 1SM-COP Bujumbura
'Mugisho thinks that Murhula is in Bujumbura.'

b. Rundi (JD62) (Sabimana 1986: 200)
y-a-vuz-e ko Maria y-a-ri u-mu-nyeshuri
1-RPAST-say-ASP that Mary 1-FPAST-be A-1-student
'He said that Mary was a student.'

Lega's (D25) complementizer is homophonous with the proximal demonstrative *-nɔ* with class 14 marking, *bo-*.

(9) Lega (D25) (Botne 1995: 214)
ámbúndɛ bónɔ ɛ́kwɛndá ko Zaílɛ
3S.1S.tell.PST that 3S.PR-go to Zaire
'S/he told me that s/he is going to Zaire.'

In our data, deictic complementizers always show invariant noun class inflection, though the form is not predictable for any one language as far as I can tell. However, based on the available data, some generalizations can be made. Specifically, demonstrative complementizers broadly reflect noun classes that are otherwise used for abstract concepts and/or mass terms. For instance, Class 15 in JD60 is used elsewhere exclusively for infinitival forms of the verb. Class 14 in Lega is used for abstract nouns (and also gelatinous substances and body parts; Botne 2003). In Mbugwe (F34), the complementizer *kóko* is recorded as being identical to the class 17 locative form of the proximal demonstrative (Mous 2004).

2.3.2 Manner deictics

Included among the deictic complementizers are manner deictics. As observed originally in Guthrie (1970b: 105) (and argued for explicitly in Güldemann 2002, 2008), this is probably a relevant derivative function of proto-Bantu **-tì*. Though a speech verb in some Bantu languages (Section 2.1), in fact Guthrie traces the original meaning to 'that, namely', which later in some languages is "best expressed in English as 'saying'" (Guthrie 1970b: 105). In many Bantu languages, though, the manner deictic use persists. For instance, in Nyala East it is clear that *-chi*,[11] has the distribution of a manner deictic outside of embedding contexts.

(10) a. Nyala East (JE32f)

[11]This is, again, historically **-tì*. In much of JE30, stops (af)fricated over time (Guthrie 1970a)

o-mw-aana a-chi
1AUG-1NC-child 1SM-DEM
'a child like this'

b. Nyala East (JE32f) (Gluckman 2023)
Masika a-paar-a a-chi Wekesa ka-chi-a Nairobi
Masika 1SM-think-FV 1AGR-COMP Wekesa 1SM-go-FV Nairobi
'Masika thinks that Wekesa went to Nairobi.'

It is possible that manner deictics other than **-tì* have independently developed into complementizers. In Eton, *nâ* introduces selected embedded clauses as well as anaphorically describes a manner.

(11) a. Eton (A71) (Van de Velde 2008: 351)
à-Ltɛ́ L-kàd H b-òd nâ H-bə́-zù-L
I-PR INF-tell LT 2-person CMP SB-II-come-SB
'He tells the men to come.'

b. Eton (A71) (Van de Velde 2008: 170)
mə̀-Ltɛ́ kɔ̀m nâ
1SG-PR INF-do thus
'I do it this way.'

2.3.3 Pronominal complementizers

A number of Bantu languages have developed complementizers from ostensibly pronominal sources, a pattern that is well-attested outside of Bantu languages (Diessel & Breunesse 2020). This is explicitly argued by Botne (1995) to be true of the complementizers *ngo* and *mbo*, found throughout Bantu languages.

(12) a. Rwanda (JD61) (Rugege 1984: 38)
Mary ya-inubwa ngo abana ba-tah-e kare.
Mary 1SM-resent COMP 2NC.children 2NC-get.home-SBJV early
'Mary resents it that the children get home early.'

b. Nyambo (JE21) (Rugemalira 2005: 96)
akajira ngu muri abasúma
she.imagine.PAST COMP you.were thieves
'She imagined that you were thieves.'

c. Ruuli (JE103) (Sørenson & Witzlack-Makarevich 2020: 94)

naye abandi ba-kob-a mbu o-Bunyala bu-a-ikang-a
but AUG.2.other 2S-say-FV COMP AUG-Bunyala(14) 14S-PST-reach-FV
o-ku Nammanve
AUG-17.LOC Nammanve(9)
'But others says that the Bunyala reached Nammanve.'

d. Mongo (C61) (De Rop 1958: 126)
básanga mbɔ basúwa bǎosíl'ɔ̌kɛnda
2SM.claim that boats 2SM.left
'They claim that the boats have left.'[12]

Botne (1995) argues that these forms are historically related to (emphatic) pronouns. Note that, since they derive from pronouns, pronominal complementizers are often "agreeing," in that they co-vary with the subject of the embedding verb. Compare the "discourse markers," which can be used to introduced any kind of embedded clause (and are preceded by the "particle" *a*), with the pronominal forms in Table 1. An example of the discourse particle use is shown in (13).

Table 1: Ntomba (C35a) (Botne 1995: 209)

	Direct discourse markers		Personal pronouns	
	Singular	Plural	Singular	Plural
1	*a mi*	*a nso*	*mi*	*so*
2	*a we*	*a nyo*	*we*	*nyo*
3	*a nde*	*a mbo*	*nde*	*bo*

(13) Ntomba (C35a)] (Gilliard 1928: 69), cited in (Botne 1995: 208)
bayoye batepela, a mbo: ntaba inkuma isolota
3P-come-PST 3SP-say-FV PRT 3P 10-goats 10-all 10-PRF-run.off
'They came and said: all the goats have run off.'

Not all Bantu pronominal complementizers are related to *ngo* or *mbu*. In particular, Kawasha (2006, 2007) reports that complementizers in Chokwe (K11), Luchazi (K13), Lunda (L52) and Luvale (52) are directly related to the possessive pronouns in those languages.

[12] In original, *Ze beweren, dat de boten vertrokken zijn.*

2.4 Other categories

The above categories (*say*-complementizers, *be*-compementizers, deictics) were found fairly robustly throughout these data. I report below on two additional strategies that I feel should be mentioned, but did not appear as frequently in our language sample.

First, there are *similatives*, words meaning "like" or "as" Kuteva et al. (2019). In this study, we find derivatives of **-ŋgà* 'as, like' (Guthrie 1970b: 243) serving as complementizers, like Ruuli (JE103), shown in (14).

(14) Ruuli (JE103) (Sørenson & Witzlack-Makarevich 2020: 94)
a-baana a-iz-a ku-bon-a nga ba-ku-ikiriz-a
AUG-child(2) 2SG.S-AUX-FV INF-see-FV COMP 2S-2SG.O-believe-FV
'You will see the children believe you.'

Second, I feel obligated to mentioned that not all finite embedded clauses require an overt complementizer. In many cases, the complementizer can be omitted, with no discernable meaning difference, like in Kikuyu (E51), shown in (15).

(15) Kikuyu (E51) (Englebretson 2015: 150)
mũ-timia a-kũ-ĩtĩk-ĩt-i-e (atĩ) mũ-thuri
NC_1-woman SC_1-CR.PST-believe-PERF-TRNS-FV COMP NC_1-man
nĩ-a-∅-iy-ir-e N-gũkũ
FOC-SC_1-CR.PST-steal-COMPL-FV NC_9-chicken
'The woman believed (today) (that) the man stole the chicken.'

Very little documentary work addresses whether null complementizers are possible in a given language – though see Edelsten et al. (2022: section 3.10).[13] Nonetheless, in at least some languages, all finite, declarative embedded clauses must be headed by an overt complementizer. For instance, as reported in Masatu (2015: 8) for Simbiti (JE431), "Both direct and indirect speech is marked with the complementizer *igha* 'that.' This complementizer appears with almost every occurrence of direct or indirect speech, and it seems to be extremely ungrammatical to omit it."

[13] Edelsten et al. (2022) ultimately suggest that optionally expressed complementizers are more likely in East African Bantu languages. Thus, outside of East Africa, Bantu languages are more likely to have embedded clauses with obligatorily overt complementizers.

3 Functions of Bantu complementizers

We now turn to the functions of Bantu complementizers. This is a pertinent question because in our sample, many languages have multiple complementizers. Conservatively, 40 languages out of 103 have two or more complementizers used in finite non-interrogative embedding. What factors then affect the distribution of the complementizers within a language? And what factors influence when a speaker chooses one complementizer over another? I discuss here three factors that correlate with complementizer choice: commitment ("evidentiality"), information structure, and agreement. This is far from an exhaustive list. However, these three factors have been discussed in a number of sources, and so are amenable for making cross-linguistic comparison. Still, I emphasize that the documentary picture remains fairly weak when it comes to the functions of complementizers in Bantu languages, and so my findings here should be taken as preliminary. Nonetheless, based on the factors discussed below, I motivate a two-way distinction among Bantu complementizers. Across all three factors, *say*-complementizers, manner deictics, and pronominal complementizers pattern similarly, while *be*-complementizers and demonstratives pattern similarly.

3.1 Commitment ("evidentiality")

Many sources that document Bantu language complementizers mention "evidentiality" (see in particular Botne 1995, 2020), with the caveat that the term "evidentiality" covers a lot of ground. Rather than evidentiality *per se*, Bantu complementizers more often reflect an individual's – often the speaker's – commitment to the embedded proposition (cf. Gluckman 2021, 2023). For example, in Kihara (2017), Kikuyu's (E51) complementizer *atĩ* is argued to function as a dubitative marker (in some contexts). The speaker is "non-committal about the relayed information, rendering the information unreliable" (Kihara 2017: 114). In this way, *atĩ* is distinct from the "true" quotatives *atĩrĩ* and *atĩrĩrĩ*.

(16) Kikuyu (E51) (Kihara 2017: 114)

ndĩ-ra-igu-ir-e	atĩ	nĩ	ma-ra-cok-ir-e	ka-ao
1SG-RCPST-hear-PFV-FV	DUB	AM	2-RCPST-return-PFV-FV	16-theirs

'I heard that they returned to their home.'

In Givón & Kimenyi's (1974) study of Rwanda's (JD61) complementizer system, they note a contrast between *kó* and *ngo* (which are not in complementary distribution). Similar to Kikuyu above, the pronominal complementizer *ngo* gives rise to a reading of speaker-doubt.

(17) Rwanda (JD61) (Givón & Kimenyi 1974: 96)
ya-m-bgiye ngo u-a-koraga cyaana
he-PAST-me-tell ngo you-PAST-work-HAB hard
'He told me that you worked hard (but I personally doubt it).'

This contrasts with the demonstrative complementizer *kó*, which, in some contexts, correlates with stronger speaker belief. For example, with a verb like *kubeeka* 'to lie,' with *kó* (as opposed to *ngo*) in (18), the speaker's commitment to "the belief 'John didn't come' is the **weakest**" (Givón & Kimenyi 1974: 102, emphasis in original).

(18) Rwanda (JD61) (Givón & Kimenyi 1974: 102)
ya-mu-beešye kó Yohani yaaže
1SM-1OM-lie.PERF COMP John come.PERF
'She lied to him that John came.'

Thus, with *kó*, the speaker is more committed to the truth of the embedded clause – even contradicting the lexical semantics of the embedding predicate.

Another example of stronger speaker commitment is observed in Kamba (E55), for which Myers (1975) reports that the *be*-complementizer *kana* (as opposed to no complementizer) in (19) indicates "the speaker believes that the promise will be kept" (Myers 1975: 190).

(19) Kamba (E55) (Myers 1975: 190)
aisye kana nũkũka ũmũnthi
he.said that he.comes today
'He said that he is coming today.'

A parallel distinction is reported in Finholt & Gluckman (2023). In Swahili (G42), the *be*-complementizer *kuwa* contrasts with the *say*-complementizer *kwamba* under verbs of epistemic possibility. When the speaker is relatively sure of the outcome, *kuwa* is better (20); otherwise *kwamba* is preferred (21).

(20) *We're watching Tanzania play in a football [soccer] match. There are five minutes left to play, and Tanzania is up by three.*
i-na-onekan-a kuwa/#kwamba Tanzania i-ta-shind-a
9SM-PRES-seem-FV COMP/COMP 9Tanzania 9SM-FUT-win-FV
'It seems like Tanzania will win.'

(21) *We're watching Tanzania play in a football [soccer] match. It's halftime, and Tanzanian is up one to nil.*

i-na-onekan-a	kwamba/#kuwa	Tanzania	i-ta-shinda
15SM-PRES-seem-FV	COMP/COMP	9Tanzania	9SM-FUT-win

'It seems like Tanzania will win.'

We therefore find both weak and strong commitment readings associated with complementizer choice. Interestingly, there are correlations between the lexical source of the complementizer and the associated reading. When we find a weak reading, it is always in the presence of a *say*-complementizer, manner deictic, or pronominal complementizer. Strong readings only ever arise in the presence of *be*-complementizers and demonstratives.[14] Note that this does not entail that all complementizers have a "commitment" meaning component; the observation is simply that when such a meaning arises (in embedded clauses), the lexical source of complementizer is largely predictable.

3.2 Information structure

Complementizers often interact with the information structure of the embedded clause (cf. Djärv 2019). In Bantu languages, the clearest instance of such "embedded clause effects" is observed in the presence of *predicate focus*, i.e., a morphosyntactic configuration in which the predicate (or something inside of the predicate) is in focus. This distinction is often seen in Eastern and Southern Bantu languages. In some of those languages, certain complementizers prohibit the predicate focus operator *ni-* found in many Bantu languages (Güldemann 2003, 2013) (cf. Section 2.2). In Subi (JE26),[15] the predicate focus operator is banned in clauses introduced by *nkikwo*, but permitted in clauses introduced by *ngu*.[16]

[14] There is an interesting wrinkle here, however. *Be*-complementizers are often used to head embedded polar interrogatives ('if/whether'), which also entail a lack of commitment on the part of an individual. However, polar interrogatives differ in that they reflect *ignorance* rather than *doubt*. As mentioned earlier, we have to put aside embedded interrogatives.

[15] In Maho (2009), Hammarström (2019) Subi is listed as JD64. There appears to be some confusion about distinct languages called Subi in the region. The Subi discussed here is more likely in within JE20 (according to native speaker linguists), and so I have chosen to give it the code "JD26."

[16] Hyman & Watters (1984) assume that this restriction is "grammatical" (i.e., synchronically idiosyncratic). I respectfully find this to be an unsatisfying solution. Indeed, as Hyman & Watters note, such restrictions tend to occur in "backgrounded" (256) clauses, already suggesting a link to information structure. Moreover, the fact that this same restriction keeps appearing in unrelated languages begs a more general explanation.

(22) Subi (JE26) (Cyprian Vumilia, p.c.)

a. bha-ka-tu-gambira nkikwo tu-liku-bha-tel-era endulu
2sm-pst-1pl.om-tell comp 1pl.sm-pres-2om-make-appl noise
'They told us that we are making noise to them.'

b. bha-ka-tu-gambira ngu ni-tu-bha-tel-era endulu
2sm-pst-1pl.om-tell comp foc-1pl.sm-2om-make-appl noise
'They told us that we are making noise to them.'

An identical pattern is found in languages with a *conjoint/disjoint* system. While it remains debated about the function of the conjoint/disjoint alternation (it is likely to have multiple functions across languages), it is reasonably thought to be connected with predicate (internal) focus: disjoint forms are found when the predicate is in focus, while conjoint forms are found elsewhere (van der Wal 2016).

In languages with a conjoint/disjoint system, disjoint forms of the verb are often banned in certain embedding contexts. It is common to find restrictions inside of nonselected clauses (e.g., relatives and temporal clauses), but we also find that certain selected embedded clauses restrict the verbal morphology. For instance, in Rwanda (JD61) and Rundi (JD62), under the complementizer *kó*, only conjoint forms are found (Ngoboka & Zeller 2016, Nshemezimana & Bostoen 2016). The disjoint form (in this context, marked with *ra-*) is available under *ngo-* clauses.[17]

(23) a. Rwanda (JD61) (Ngoboka & Zeller 2016: 366)
a-ra-vúg-ye ngo a-ra-som-a
1sm-dj-say-pfv that 1sm-dj-read-fv
'He says that he reads.'

b. Rwanda (JD61) (Ngoboka & Zeller 2016: 365)
a-vúg-ye kó a-som-a
1sm-say-pfv that 1sm-read-fv
'He says that he reads.'

The prohibition on predicate focus in an embedded clause must stem from an interaction with constraints on information structure imposed by the clause itself – parallel to the fact that negation, a focus-sensitive operator, also blocks disjoint forms.

[17]Note that the main-clause verb is also affected, but the correlations between complementizer and main-clause morphology are less systematic, so I exclude them from the discussion here.

Interestingly, the same dichotomy described for commitment above holds here. Only selected embedded clauses headed by *be*-complementizers or demonstratives block focus operators in the embedded clause. Clauses headed by *say*-complementizers, manner deictics, and pronouns generally have no effect on the information structure – specifically predicate focus – in the embedded clause.[18] Again, I emphasize that not all *be*- or demonstrative complementizers prohibit predicate focus in selected embedded clause. But when it is blocked, the complementizer is always a *be*- or demonstrative complementizer.

3.3 Agreement

The final distinction I mention here is morpho-syntactic. It is well-known that a relatively large number of Bantu languages have so-called *agreeing complementizers*. These are complementizers which exhibit morphological concordance with an element in the matrix clause, almost always the embedding subject,[19] like Bukusu's (JE31c) well-discussed *-li*.

(24) Bukusu (JE31c) (Diercks 2013: 358)
ba-ba-ndu ba-bol-el-a Alfredi ba-li a-kha-khil-e
2-2-people 2S-said-AP-FV 1Alfred 2-that 1S-FUT-conquer
'The people told Alfred that he will win.'

There is no single lexical source for agreeing complementizers across Bantu languages; they are derived from disparate sources. While the majority of the cases in our sample are related to *-*tì*, there are numerous exceptions. Most prominently, there are agreeing complementizers derived from pronouns, discussed in Section 2.3.3.

While productive agreeing complementizers are robustly attested in Bantu, we also find that the agreement has fossilized on many Bantu complementizers. Some languages, like Nyengo (K16), have a complementizer inflected with what is presumably invariant 1SG morphology, (25). (*Nji* is also found throughout the K30 language group.)[20]

[18] der Wal (2014) makes a similar claim with respect to *situative* clauses; she is largely concerned with nonselected adverbial clauses.

[19] Gluckman (2023) reports that certain speakers of Nyala East (JE32f) permit an implicit subject to control the agreement of the complementizer *-chi*, though he notes that this is the exception, not the rule.

[20] Indeed, this appears to be an innovation in some dialects of Rwanda/Rundi (JD61/62). In the "standard" languages, *-ti* is fully agreeing. Based on conversations with speakers, however, some dialects only use *nti*, which can only plausibly be a fossilized 1st-person form.

(25) Nyengo (K16) (Lisimba 1982: 166)
tu-faa nji tu-ful-e
1pl.sm-want comp 2pl.sm-beat-sbjv
'We want to beat.'

In other languages, what is fossilized appears to be 3rd person agreement. In Lamba (M54) in (26a), *ati* likely bears 3rd singular (Class 1) agreement. *Baatɪ* in Nyakyusa (M31a) in (26b) shows 3rd person plural fossilized agreement.

(26) a. Lamba (M54) (Doke 1922: 144)
na li bwene ati ta ŵa pyaŋgile
I pst see comp neg they sweep
'I saw that they did not sweep.'

b. Nyakyusa (M31a) (Persohn 2017: 315)
… "keet-a ʊ-t-ile baatɪ n-heesya gw-ɪtu!? …"
look-fv 2sg-say-pfv hearsay 1-guest 1-poss.1pl
'…"Look, you said he is our guest!?"'

Pronominal complementizers, too, show similar variation. The invariant complementizers *mbo* and *ngo* discussed above are also complementizers with fossilized agreement. *Mbo* has 3rd plural (Class 2) invariant "agreement" while *ngo* is reconstructed in Botne (1995) as a complementizer/evidential particle with invariant 3rd singular (Class 1) morphology.[21]

In our sample, systemically excluded from the class of agreeing complementizers – including those with fossilized agreement – are *be*-complementizer and demonstrative complementizers.[22] For demonstratives, while these do reflect noun classes, they (i) never *change* noun class within a language, (ii) never reflect 1st, 2nd, or 3rd person (human) morphology, and (iii) never reflect agreement with something overt in the superordinate clause. More strikingly, the lack of person/number marking is particularly relevant for *be*-complementizers, which can and often do, exhibit agreement synchronically.

[21]Nduumo (B63) and Mbete (B61) have a more unique situation, as reported in Adam (1954). With 3rd person (singular/plural) subjects, invariant *ndi* is the complementizer used. With 1st and 2nd person (singular/plural) subjects, *mbidi* (for Nduumo) and *meti* (for Mbete) are used. Thus, these languages show partially fossilized agreement, collapsing the speech-act participants.

[22]Similatives also do not show evidence of co-variation with the main-clause subject in our language sample.

3.4 The (lack of) selection

Most typological studies of complementizers start by looking at the interaction between complementizer/complementation choice and the lexical semantics of the embedding predicate. Such studies are interested in whether there are selectional restrictions on complementation. Certainly, selection plays an important role in Bantu languages: we find a general divide between those predicates which embed nonfinite clauses (e.g., modals, aspectuals), and those which embed finite clauses (belief verbs, utterances predicates, etc.). Within the finite clauses, we also find that there are general divisions between interrogative and non-interrogative clauses. But in general, there is limited evidence that the lexical class of the embedding predicate plays a direct role in the choice of the complementizers described in this paper. Works that note correlations between complementizer choice and category of the embedding predicate are relatively sparse (cf. Givón & Kimenyi 1974; Van de Velde 2008; Finholt & Gluckman 2023).

I believe, in fact, that the importance of selection is often over-emphasized, or at the very least, not the main factor in complementizer selection. The issue is one of confirmation bias. The starting place for all investigations of Bantu complementizers is always lexical class of the embedding verb – a sensible place to start (see for instance Safir 2008). But this sometimes leads to the conclusion that any distinctions in complementizers that arise are solely attributable to the lexical class of the embedding predicate. But more often than not in the descriptive sources (even beyond Bantu languages) and in my own fieldwork, the same embedding predicate may appear with multiple kinds of embedded clause, yielding slightly distinct readings. Thus, Myers (1975: 189) reports that in Chewa (N31b) "the choice of complementizers is controlled in part by the syntactic elements of the sentence, and in part by the pragmatics of its use."

In other languages, there is essentially no effect of predicate class. In Kamba (E55) the two complementizers *kana* and *atĩ* are in free variation (Myers 1975: 187). Likewise, in Nyala East (JE32f), the complementizers *mbo*, *-chi*, and *vachi* are entirely interchangeable under all embedding verbs (and nouns).[23]

[23] It is worth noting that Mood of the embedded clause only has an indirect effect on the choice of complementizer as well. That is, while many complementizers correlate with subjunctive mood, the predicting factor for mood choice is almost always lexical class of the embedded verb. For instance, desire predicates like *want* strongly prefer subjunctive mood (particularly with disjoint subjects), regardless of the complementizer used to introduce the embedded clause. (See e.g., discussion concerning Gyeli (A801) in Grimm 2021.) Insomuch as the embedding verb influences the choice of complementizer, then it may appear that complementizer and mood are interacting.

Finally, there are additional factors that can influence complementizer choice, though these are restricted in their distribution. Letsholo & Safir (2019) discuss instances of Voice affecting the choice of complementizer in Nande (JD42) and Kalanga (S16). Similarly, Shona (S10) has both an active and passive complementizer, *kuti* and *kunzi*, respectively (Dembetembe 1976). In Iwoyo (H16d), as discussed in Mingas (1994), the *say*-complementizer *tí* appears in affirmative statements, but the complementizer *kàní* appears when the main verb is negated. In Fuliiru (JD63), the pronominal complementizer *mbu* is used in "frustrative" constructions (Van Otterloo 2011). And in Rufumbira (JD62a), as reported in Nassenstein (2016), the choice of *ngo* versus *mbu* is intermixed with language attitudes and identity. These additional distinctions deserve a more detailed crosslinguistic study than I can provide here.

4 Classifying the dichotomy

I summarize my initial findings in Table 2.

Table 2: Summary of the correlations between form and function

	say-comps	manner	pronouns	*be*-comps	demonstratives
Speaker/subject commitment	weak	weak	weak	strong	strong
Can restrict FOC. in emb. clause	✗	✗	✗	✓	✓
Can show agreement	✓	✓	✓	✗	✗

If these generalizations hold up, it suggests a striking correlation between form and function that deserves an explanation. I suggest the following tentative, intuitive explanation. We start with the assumption that embedded clauses can be related to main clause to various ways. I refer to this as *anchoring*. One way to anchor the embedded clause to the main clause is via an individual: the embedded clause reflects information conveyed by, or held to be true/false by, or rejected by (etc.), some individual, possibly the subject of the embedding verb, or possibly the speaker. For instance, quotation can be anchored to an individual: the speaker of the quoted utterance. This idea can extend to attitudinal embedding as well. The sentence *Maina believes that it's raining* is judged true if the proposition "It's raining" is among Maina's beliefs.

On the other hand, embedded clauses can be anchored to a situation (or eventuality). For example, a quotation can be anchored to "the event denoted by the quote" (Güldemann 2008: 362). Again this idea can extend beyond quotation to cover propositions which are judged true or false at a particular time and place. This is the approach taken across a number of theoretical works, including (Aboh 2004, Kratzer 2006, Moulton 2008, Hacquard 2009, Haegeman & Ürögdi 2010, Moulton 2013, Hegarty 2016). The basic idea across these works is that embedded clauses (can) describe propositional content that is evaluated with respect to an event(uality). Conceptually, according to the latter authors' view, in *Maina believes that it's raining*, the embedded proposition "It is raining" is evaluated relative to the beliefs held at the event described by *believe*. The attitude holder *Maina* only plays an indirect role in that it happens to be Maina's beliefs in this case.

Based on this dichotomy, I suggest that the division among Bantu complementizers reflects this individual/situation divide: some complementizers are "perspectival," anchoring the embedded clause to an individual. *Say*-complementizers, manner deictics, and pronouns all fall into this category: they are are all linked morphologically or semantically to an individual's perspective. This can be seen directly in the morphology, through agreement with a local subject. The effect of anchoring to an individual can have semantic effects as well. The weak commitment readings observed in Section 3.1 may be understood as the result a "subjective" attitudinal stance. That is, using a perspectival complementizer signals that the embedded clause is evaluated relative to a particular individual who is not the speaker, and is not common/shared knowledge.

The other complementizers, *be*-complementizers and demonstratives, are "situational," anchoring the embedded clause to a situation. The lack of agreement follows naturally. The interaction with predicate focus also follows on the assumption that if the complementizer introduces or foregrounds a salient situation, that situation can interacts with a forgrounded (focused) situation in the embedded clause. Indeed, clear evidence of the relationship between predicate focus and complementizers comes from shared morphology: the predicate focus operator in some languages *ni* is also a complementizer, as discussed in Section 2.2.

To account for stronger speaker commitment, we can make reference to *definite* situations, which commit the speaker to the belief in a situation that is relevant in evaluating the embedded clause. As discussed in Özyıldız (2017), Bochnak

& Hanink (2021) (for non-Bantu languages), presupposing a situation in which the embedded clause is true can lead to a factive inference, i.e., a "stronger" commitment towards the embedded proposition.

5 Conclusion

In this paper, I have provided a preliminary typology of Bantu complementizers. Even with limited data, this study reveals rich variation within the broader family. Additionally, I have explored three factors that correlate with complementizer choice in languages which have more than one copmlementizer. Based on the available data, I suggested a dichotomy between perspectival and situational complementizers. Though I caution again that these results need further evaluation, with closer inspection of Bantu complementizer systems, I nonetheless believe that the emerging patterns offer an important and novel window into the typology and function of complementizers crosslinguistically.

Abbreviations

- 1–20: noun classes/person
- AGR: agreement
- AM: assertive marker
- APPL/AP: applicative
- ASP: aspect
- AUG: augment
- AUX: auxiliary
- CL/NC/NP: noun class marker
- COMP/CMP: complementizer
- COMPL: completive
- CONS: consecutive
- COP: copula
- DEM: demonstrative
- DJ: disjoint
- DUB: dubitative
- FPAST: far past
- FUT: future
- FV: final vowel
- HAB: habitual
- IDEO: ideophone
- IMP: imperative
- INF: infinitive
- IPFV: imperfective
- ITIVE: itive
- LNK: linker

- LOC: locative
- LT: linking tone
- NEAR.FUT: near future
- NEG: negative
- NUCL: nucleative
- OC/OM/O: object marker/concord
- P: person marker
- PFV/PRF: perfective
- PL: plural
- POSS: possessive
- PP: pre-prefix or pronominal prefix
- PR: pronoun
- PRES: present
- PRT: particle
- PST: past
- QUOT/QV: quotative
- REF: referential
- RPAST/RCPAST: recent past
- S/SA/SC/SM: subject agreement/-concord/marker
- SB/SBV/SBJV: subjunctive
- SG: singular
- TAM: tense/aspect/mood

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Chapter 7

A reconstruction of Proto-Sawabantu

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The present paper aims at studying and comparing Bantu languages spoken along the Cameroonian, Equatorial Guinean and Gabonese coasts called the Sawabantu languages by reconstructing proto-Sawabantu (consonants and vowels) via the Comparative Method used in Historical Linguistics. The proto-Sawabantu reconstructions obtained are analyzed and compared with other proto-languages reconstructed for languages spoken in this region: proto-Bantu and proto-Manenguba in order to identify Sawabantu lexical innovations and phonological innovations. The resulting historical analyses shed new light on the fascinating history of the speakers of this particular region where the great Bantu Expansion started 5,000 years ago in southern Cameroon.

1 Introduction

Sawabantu Bantu languages are located along the Cameroonian, Equatorial Guinean and Gabonese coasts (Figure 1) and generally include Guthrie's (Guthrie 1967–1971, later updated by Maho n.d.) A10, A20 and A30 languages. Cameroon is well-known for its significant cultural and linguistic diversity with approximately 250 spoken languages (Eberhard et al. 2019) belonging to three different language families: Niger-Congo (190 languages including the Bantoid and Bantu languages), Nilo-Saharan (2 languages) and Afro-Asiatic (55 languages). Moreover, it is in southern Cameroon that the Bantu Expansion started 5,000 years ago with Bantu-speaking-populations migrating southward through Cameroon, Equatorial Guinea and Gabon expanding later across Central, Eastern and Southern Africa (Bostoen 2007, Grollemund et al. 2015, Koile et al. 2022).

Rebecca Grollemund, Gérard Philippson, Braden Brown & Chaeli Rule. 2026. A reconstruction of Proto-Sawabantu. In Vicki Carstens, Katherine Russell, Olawale Akingbade, Deborah Morton & Michael Diercks (eds.), *Pamoja tena 'Together again': African linguistics after COVID*, 129–168. Berlin: Language Science Press. DOI: ??

Several studies have been conducted on Sawabantu languages highlighting their relatedness, with A10-20-30 languages (minus Manenguba A15 and Bubi A31) forming a valid genetic group. Ebobissé (1989) proposed first a dialectometrical study of A20 and A30 Sawabantu languages for which he calculated the percentages of linguistic proximity based on the study of lexicon. His results divided the Sawabantu clade into two subgroups with A20 languages on one hand and A30 languages on the other hand. In 2015, Ebobissé (2015) furthered his study of Sawabantu languages by providing a detailed synchronic study of the same languages with a focus on phonology, morphology and lexicon confirming the two subgroups identified in 1989. The addition of A10 languages (minus Manenguba A15) sometimes called "Oroko" (Friesen 2002) to the Sawabantu clade was made by Nurse & Philippson (2003: 170) who established a classification of 80 Bantu languages based on the study of non-lexical data (30 phonological and morphological features). The study of lexicon but also morphological criteria such as the shape of the nominal stem that is generally CVCV in A10-20-30 as opposed to Manenguba A15 (Philippson 2022: 6-7) were decisive in their classification. Based on these findings, Philippson (2022) proposed to borrow the geographical term "Sawabantu" from Ebobissé (2015) to refer to these A10-20-30 languages that have a genetic value.

The purpose of this paper is to examine Sawabantu languages and reconstruct their putative ancestor Proto-Sawabantu (PSB) by the application of the Comparative Method (CM). PSB will be analyzed and compared with other proto-languages reconstructed in this region such as Proto-Manenguba (PM) reconstructed by Hedinger (1987) that includes A15 languages spoken near the Bantu cradle. The comparison between PSB and PM reconstructions reveals two different evolutionary trajectories with PSB being more similar to Proto-Bantu (PB) in certain aspects as opposed to PM confirming therefore the differences observed between Manenguba A15 and Sawabantu A10-A20-A30 languages. PSB reconstructions will also be compared to PB reconstructions taken from BLR3 (Bastin et al. 2002) in order to identify any diachronic phonological innovations that will shed new light on the peculiar evolution of the languages spoken in this region.

2 Data and methods

2.1 Languages

Based on the study of lexicon (Ebobissé 1989, 2015) and non-lexical data such as phonological or morphological features (Nurse & Philippson 2003: 70), the Sawabantu clade generally comprises between 20–27 Bantu languages belonging to Guthrie's zones A10, A20 and A30 (e.g., Table 1) with the exception of the A15 languages called Manenguba and A31 Bubi spoken on Bioko Island.

In order to start our comparative work, we selected the same A20 and A30 Sawabantu languages provided in Ebobissé (2015) study. Ebobissé (2015) divided the Sawabantu languages into three groups based on geographical locations: South (Banoho, Batanga, Yasa, Bapuku), Central (Duala, Bidiman, Oli, Pongo, Mongo, Mulimba) and Southwest (Mboke, Mokpe, Isu, Kole, Bubia) but his classification based on the study of lexicon published in 1989 showed that Central and Southwest were more closely related. We will use in this paper the same geographical distinctions established in Ebobissé (1989) and Ebobissé (2015) with the mention of South, Central, Southwest with the addition of the North region ("Oroko") represented in our study by the Kundu language (A10) that we have extracted from Friesen (2002) in order to get a more representative sample of the Sawabantu languages.

In his monumental study, Ebobissé (2015) collected, compared and analyzed 610 words for 15 languages. For the purpose of this study, we created an Excel database containing all 610 words for the same 15 languages with the addition of Kundu, an A10 language. Words from Ebobissé were translated from German to English. In addition to collecting the individual words, we searched for corresponding PB words drawn from BLR 3 (Bastin et al. 2002) for comparison purposes.

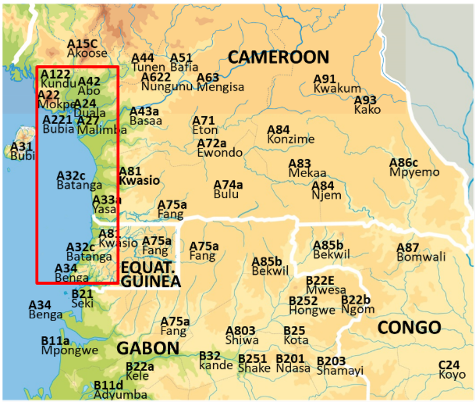

Figure 1: Approximate location of Sawabantu languages (in the red rectangle)

Table 1: Sawabantu languages in Ebobissé (2015) with selected languages for this study in italics with Maho's codes (n.d.) and ISO codes when available (some languages do not have ISO code because they are either understudied/unknown or considered as some variants of another language).

Region	Language	Maho (2009)	ISO 639-3
South	*Banoho (BAN)*	A32a	
	Batanga (BAT)	A32	bnm
	Yasa (YAS)	A33a	yko
	Bapuku (BAP)	A32b	bng
	Kombe	A33b	nui
	Benga	A34	bng
Central	*Duala (DUA)*	A24	dua
	Bidiman (BID)	A241	
	Oli (OLI)	A25	
	Pongo (PON)	A26	
	Mongo (MON)	A261	
	Mulimba (MUL)	A27	mzd
Southwest	*Mboko (MBO)*	A21	mdu
	Mokpe (KPE)	A22	bri
	Isu (ISU)	A23	srz
	Kole (KOL)	A231	kme
	Bubia (BUB)	A221	bbx
North	Londo	A11	
	Ngolo	A111	
	Bima	A112	
	Batanga	A113	
	Bakoko	A114	
	Londo	A115	
	Lue	A12	
	Mbonge	A121	
	Kundu	A122	
	Ekombe	A123	

2.2 Lexicon-based classifications of Sawabantu languages

Bantu languages spoken in southern Cameroon belong to the North-Western (NW) genetic group (Bastin et al. 1999, Grollemund 2012, Currie et al. 2013, Grollemund et al. 2015, Koile et al. 2022) that usually includes all the A languages (Guthrie's (1967–1971) referential zones) generally with B10-20-30 languages spoken in Gabon. Bastin et al. (1999), for example, include in their NW group A languages (without Mbam-Bubi composed of A31-some A40-50-60 languages) and B10-20-30 languages. According to Bastin et al. 1999's classification, Mbam-Bubi languages are the first group to diverge followed by the NW. Grollemund et al. (2015) shows that NW was composed of several clades that generally includes: Mbam-Bubi (+Jarawan), A10-20-30-40+A70-80-90 (labeled "NW Cameroon"), some B20+A80-90, remaining B20 and B10-30 (labeled "NW Gabon"). The latest phylogenetic classification established by Koile et al. (2022) proposes a different result wherein NW is composed of a sole node that includes only A languages and some of B20; whereas the remaining B20 and B10-30 languages are classified into a different clade. To sum up, Sawabantu languages belong to the first group of Bantu languages to diverge after the Mbam-Bubi group.

A closer look at the Bantu trees with a focus on the NW clades (Figure 2) established by Grollemund et al. (2015) and Koile et al. (2022) confirms that Mbam languages (A40-50-60) are the first Bantu languages to diverge followed by a group that includes A10-20-30-40-70 (in Grollemund et al. 2015) or only A10-20-30 (in Koile et al. 2022). In both trees, Sawabantu languages split after the Mbam languages and form a clade with Manenguba (A15) languages confirming their relatedness. In addition, both classifications highlight a noticeable division amongst Sawabantu languages with a first group composed of A24-A27-A33a-A34 (corresponding to Ebobissé's South and Central groups) and a second group composed of A22-A122 languages (Southwest) linked with two A10 languages Londo and Mbonge (North). This observation contradicts Ebobissé's (1989) results where Central and Southwest are more closely related but confirms Nurse and Philippson's (2003) classification with the inclusion of North Sawabantu languages within the Sawabantu group. Finally, we observe that this last subgroup (A22-A122-A11-A121) is linked to the Manenguba languages (A15) as a coordinated branch.

2.3 Method

In order to reconstruct PSB, we applied the Comparative Method (CM) and its series of steps. This upstream approach utilizes the systematic comparison of

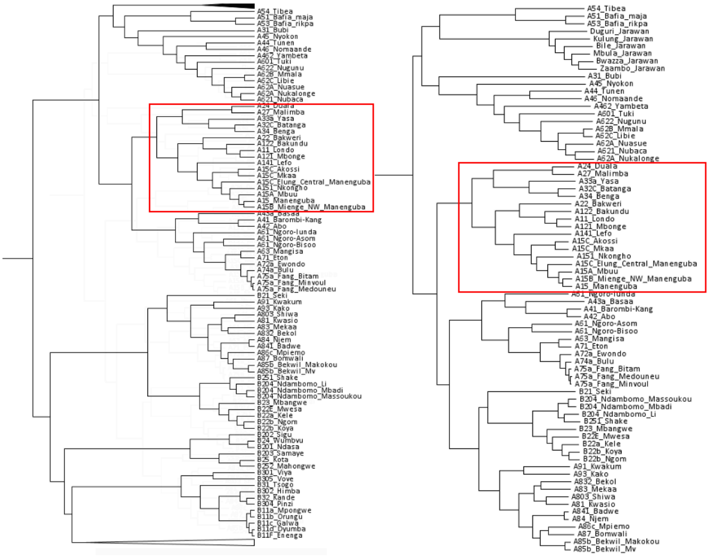

Figure 2: Comparison of the two NW groups in Grollemund et al. (2015) and Koile et al. (2022)

cognate words in related languages to reconstruct the linguistic properties of the putative ancestor with a great degree of certainty (Campbell 1998). In order to successfully apply the CM, several steps must be completed. First, the identification of cognates - words that are similar in form and meaning that are assumed to have been inherited from a common ancestral source - is paramount. We studied each meaning to identify cognate sets[1]. Most of the words studied yielded between one to three cognate sets (e.g., Table 2 'two', 'house' and 'surpass') whereas, in rare cases, meanings would yield as many as five different cognate sets (e.g., Table 2 'yesterday').

Once all the cognate sets were identified, we proceeded to the next step which consisted of the alignment of all the cognate Bantu roots in C1, V1, C2 and V2 positions in order to list all the sound correspondences present in the data. The sound correspondences (study of the phonological similarities between the languages studied) were then organized to facilitate the identification of regular sound correspondences. If a sound correspondence occurs several times, it suggests that the sound correspondence is regular which implies that the sounds could have

[1]Because of the space restriction, we are not able to present our data with our 600 comparative series. However, many examples are presented in this paper with approximately 100 different comparative series for the languages studied.

Table 2: Examples of cognacy judgement, numbers refer to cognate sets. The capital letters refer to the geographical denominations with “S” for South, “C” for Central, “SW” for Southwest and “N” for North. Empty cells mark missing data.

Meanings	*two*	*house*	*breast*	*surpass*	*yesterday*
BAN - S	ɓé-ɓà 1	ndáɓò 1	ɗì-ɓɛ́ɛ 1	i-ɓála 1	vákɛ́ 1
BAT - S	ɓé-ɓà 1	ndáɓò 1	ɗì-ɓɛ́ɛ 1	i-ɓála 1	váhɛ́ 1
YAS - S	ɓí-ɓà 1		i-ɓɛ́nɛ 1	e-ɓálɛ 1	
BAP - S	ɓé-ɓà 1	ndáɓò 1	i-ɓɛ́ɛ 1	i-ɓála 1	
DUA - C	ɓé-ɓàá 1	ndáɓò 1	di-ɓɛ́ɛ 1	búka 2	
BID - C	ɓí-ɓàá 1	ndáɓò 1	ɓɛ́ɛ 1	búa 2	ɓwéná 2
OLI - C	ɓí-ɓàá 1	ndáɓò 1	ɓɛ́ɛ 1	búa 2	ɓwéá 2
PON - C	ɓé-ɓàá 1	ndáɓò 1	i-ɓɛ́ɛ 1	i-búka 2	mbánā 2
MON - C	ɓé-ɓàá 1	ndáɓò 1	di-ɓɛ́ɛ 1	i-búka 2	kɛ́kɛ 3
MUL - C	ɓé-ɓàá 1	ndáɓò 1	di-ɓɛ́ɛ 1		cɛ́cɛ 3
MBO - SW	βé-βàá 1	ndáwò 1	li-βɛ́ɛ 1	laáka 3	
KPE - SW	βé-βàá 1	ndáwò 1	li-βɛ́ɛ 1	laáka 3	ŋmɛ́ɛlɛ́ɛ 4
ISU - SW	βé-ɓàá 1	ndáwò 1	li-βɛ́ɛ 1	yaáka 3	kɛ́ɛkɛ́ɛ 5
KOL - SW	bé-bàá 1	ndábò 1	di-bɛ́ɛ 1	yaáka 3	kɛ́kɛ́ɛ 5
BUB - SW	βé-βàá 1	ndáwò 1	li-βɛ́ɛ 1	láka 3	ŋméelɛ́ɛ 4
Kundu - N	βe-βɛ 1	ndáβo 1	di-bɛ̂ 1		jana 2

either been inherited from proto-sounds located at an ancestral level or innovated. For example, if in all the Sawabantu languages we found the sound /m/ in C1 and C2 positions, it is reasonable to assume that the PSB had *m in the same positions. The establishment of regular sound correspondences is then used to reconstruct ancestral sounds. Finally, once the regular sound changes have been understood via the study of regular sound correspondences, historical linguists can proceed with the reconstruction of words.

3 Results

3.1 Comparative tables

In this section, we present the consonant and vowel sound correspondences as well as the reconstructed sounds labeled “PSB”. The reconstructions made are primarily based on noun and verb stems which generally present a CVCV structure for nouns and CVC structure for verbs in Sawabantu languages.

3.1.1 Vowels in C1 and C2 positions

We reconstructed a system with seven vowels at the PSB level. Examining regular sound correspondences for vowels (presented in Table 3) shows that vowels are highly regular in Sawabantu languages. This regularity contrasts with Manenguba languages where vowels have undergone a number of changes since PM such as assimilation, raising or lowering which complicate the reconstruction according to Hedinger (1987: 44-45). Hedinger (1987: 45-46) also reconstructs long vowels for PM that are generally the result of two distinct processes: (i) loss of the PB *C2 in Manenguba languages has created CVV roots (PB *bèdú 'colanut' > PM *bèé) and (ii) compensatory lengthening in CVC roots after C2 was deleted (PM *fiɲ 'pus' > /-híí/ in Myenge).

The situation is less complicated in Sawabantu languages where we mostly have CVCV roots for noun stems. There are some languages where non-contrastive long vowels are attested. Generally, they are also the results of the loss of C2 (such as PSB *k) leading to the emergence of CVV roots. In such cases, long vowels or VV sequences can emerge such as in PSB *toko 'spoon' > /too/ or PSB *sako 'army ant' > /sao/ in some Central Sawabantu languages (where PSB *k2 > Ø). Therefore, because this phenomenon is caused by the deletion of a C2 that is still attested in the remaining Sawabantu languages (as opposed to PM where the PB *C2 has completely disappeared in all Manenguba daughter languages), we chose to not reconstruct long vowels.

The system reconstructed for PSB is a typical PB 7-vowel system with *ɪ and *ʊ for Bantu languages spoken in Central Africa (Guthrie 1967–1971, Meeussen 1967, Bastin et al. 2002). The sole disagreement that existed with the reconstruction of PB vowels was regarding the height of the high vowels *i, *ɪ, *u, *ʊ (Bastin et al. 2002) vs *į, *i, *ų, *u (Guthrie 1967–1971, Meeussen 1967). While Guthrie (1967–1971) and Meeussen (1967) propose that some PB vowels were super close (marked with a cedilla), Bastin et al. (2002) choose to have a contrast between tense and lax high PB vowels. Some modern-day Bantu languages present the tense/lax opposition ([ɪ] and [ʊ]) whereas other languages, such as Sawabantu languages, have mid vowels ([e] and [o]) instead.

3.1.2 Consonants

We present in this section the consonants reconstructed for PSB. When needed, a distinction between C1 (initial consonant) and C2 (intervocalic consonant) is made (e.g., *k_1 vs *k_2 in Table 5). First, we present the voiceless plosives in Table 5.

Table 3: Reconstruction of PSB Vowels

PSB	***i**	***e**	***ɛ**	***a**	***ɔ**	***o**	***u**
BAN - S	i	e	ɛ	a	ɔ	o	u
BAT - S	i	e	ɛ	a	ɔ	o	u
YAS - S	i	e	ɛ	a	ɔ	o	u
BAP - S	i	e	ɛ	a	ɔ	o	u
DUA - C	i	e	ɛ	a	ɔ	o	u
BID - C	i	e	ɛ	a	ɔ	o	u
OLI - C	i	e	ɛ	a	ɔ	o	u
PON - C	i	e	ɛ	a	ɔ	o	u
MON - C	i	e	ɛ	a	ɔ	o	u
MUL - C	i	e	ɛ	a	ɔ	o	u
MBO - SW	i	e	ɛ	a	ɔ	o	u
KPE - SW	i	e	ɛ	a	ɔ	o	u
ISU - SW	i	e	ɛ	a	ɔ	o	u
KOL - SW	i	e	ɛ	a	ɔ	o	u
BUB - SW	i	e	ɛ	a	ɔ	o	u
Kundu - N	i	e	ɛ	a	ɔ	o	u

For the voiceless plosives presented in Table 5 (examples are presented in Table 6), we reconstruct *p in C1 and C2 positions for the set p/ɸ with South and Central Sawabantu languages displaying /p/ whereas Southwest and North Sawabantu languages have the fricative /ɸ/. Based on the principles of majority wins used in CM and directionality (lenition is more common than fortition, (Bybee & Easterday 2019), *p is the logical choice.

We chose to reconstruct *t in C1 and C2 positions for the set t/r (Oli and Bidiman) because all the Sawabantu languages display /t/ in C1 and C2 positions except for Bidiman and Oli where $^*t_2$ undergoes lenition (rhotacism) only in C2 position (e.g., Table 7 for examples).

For the overlapping sets where we have k/c/s/Ø in C1 and C2 positions, we reconstructed one proto-sound *k based on several factors. First, some comparative sets can be explained by phonological contexts that triggered conditioned sound changes. For example, in C1 position, *k is maintained in Sawabantu languages when the stem presents another velar plosive (*k or *g) in C2 position (e.g., Table 8, examples 'to be sufficient/to grow' or 'molar') or when the words originally belonged to the PB class 9 (Hyman 2019) with the nasal prefix (e.g.,

Table 4: Examples for the reflexes of PSB vowels in V1 and V2 positions. Empty cells represent either missing data or non-cognate words.

PSB	***i**$_{12}$	***e**$_{12}$	***ɛ**$_{12}$	***a**$_{12}$	***ɔ**$_{12}$	***o**$_{12}$	***u**$_{12}$
meaning	***cloud***	***tree***	***stand***	***divide***	***knee***	***skin***	***turtle***
BAN - S	ɗì-vîndî	yelé	i-tɛ́mɛ	i-kaɓa	ɗì-ɓɔ́ŋgɔ́	ŋ̀-hoɓo	kúɗù
BAT - S	ɗì-vîndî	yelé	i-tɛ́mɛ	i-kaɓa	ɗì-ɓɔ́ŋgɔ́	ŋ̀-hoɓo	kúɗù
YAS - S	i-vîndó	e-yilé	e-tɛ́mɛ	e-kaɓa	ì-ɓɔ́ŋgɔ́	mo-koɓo	kúɗù
BAP - S	ɗì-vîndî	yelé	i-tɛ́mɛ	i-kaɓa	ì-ɓɔ́ŋgɔ́	ŋ̀-koɓo	kúɗù
DUA - C	di-wîndî	ɓw-elé	tɛ́mɛ	j-aɓa	dì-ɓɔ́ŋgɔ́	e-yoɓo	wúdù
BID - C	findîfindî	ɓw-elé	tɛ́mɛ	y-aɓa	ɗ-òŋgò	o-ɓo	wúlù
OLI - C	vîndîvîndî	ɓw-elé	tɛ́mɛ	y-aɓa	ɓɔ́ŋ	o-ɓo	wúlù
PON - C	i-wîndîwîndî	ɓw-elé	i-tɛ́mɛ	ij-aɓa	ì-ɓɔ́ŋgɔ́	e-yoɓo	wúdù
MON - C	di-wîndîwîndî	ɓw-elé	i-tɛ́mɛ	-kaɓanɛ	dì-ɓɔ́ŋgɔ́	e-koɓo	kúdù
MUL - C	di-îndî	ɓw-elé	i-tɛ́mɛ	i-kaɓa	dì-ɓɔ́ŋgɔ́	e-yoɓo	kúdù
MBO - SW	li-wîndó	eelé	li-tɛ́mɛ	li-βaá	lì-wɔ́ŋgɔ́ŋgɔ́	e-kowo	è-kúlɛ̀
KPE - SW	li-wîndî	gb-eé	li-tɛ́mɛ	li-βaá	lì-wɔ́ŋgɔ́ŋgɔ́	yowo	kúlù
ISU - SW	li-wîndîwîndî	gb-elé	i-tɛ́mɛ	i-kaβa	lì-wɔ́ŋgɔ́ŋgɔ́	e-kowo	è-kúlɛ̀kùlɛ̀
KOL - SW	di-wîndî	gb-eré	i-tɛ́mɛ		dì-bɔ́ŋgɔ́		è-kúrɛ̀
BUB - SW	li-îndó	eelé	li-tɛ́mɛ	li-kaβa	lì-wɔ́ŋgɔ́ŋgɔ́	m-owo	è-kúlɛ̀kùlɛ̀
Kundu - N		bo-le		-kaβa	di-bɔŋgɔ	e-kobo	ku

Table 5: Sound correspondences for unvoiced plosives. When two reflexes are presented, the first one is the most frequent. Bold characters mark sound changes from PSB to the daughter languages.

PSB	***p**$_{12}$	***t**$_{12}$	***k**$_{1}$	***k**$_{1}$ **/ _Vk**$_{2}$	***k**$_{1}$ **/_e**	***k**$_{1}$ **/_wV**	***k**$_{2}$
BAN - S	p	t	**h**~k	k	k~h	(w)	**h**~k
BAT - S	p	t	**h**~k	k	k~h	(w)	**h**~k
YAS - S	p	t	k	k	**c**	(w)	k
BAP - S	p	t	k	k	**s**	(w)	k
DUA - C	p	t	**Ø**~k	k	**Ø**	k(w)	**Ø**~k
BID - C	p	t / **r** (*t$_{2}$)	**Ø**~k	k	**Ø**	k(w)	**ɣ**~**Ø**
OLI - C	p	t / **r** (*t$_{2}$)	**Ø**~k	k	**Ø**	k(w)	**ɣ**~**Ø**
PON - C	p	t	**Ø**~k	k	**Ø**	k(w)	**Ø**~k
MON - C	p	t	k	k	k	k(w)	k
MUL - C	p	t	**Ø**~k	k	**c**	k(w)	**Ø**~k
MBO - SW	**ɸ**	t	k	k	k	**kp**	k
KPE - SW	**ɸ**	t	**Ø**~k	k	**Ø**~k	**kp**	**Ø**~k
ISU - SW	**ɸ**	t	k	k	k	**kp**	k
KOL - SW	**ɸ**	t	**Ø**~k	k	**Ø**~k	**kp**	**Ø**~k
BUB - SW	**ɸ**	t	**Ø**~k	k	**Ø**~k	**kp**	**Ø**~k
Kundu - N	**ɸ**	t	k	k	k	**kp**	k

Table 8 ‘pangolin’, reconstructed *kákà in PB, BLR 3 1684, classes 9/10). When followed by a front vowel, *k$_{12}$ is palatalized in Yasa, Bapuku, Mulimba while the remaining languages present the regular reflexes for *k$_{12}$ (e.g., Table 8 ‘molar’ and Table 9 ‘muteness’). Finally, we observe some sets for *k$_{1}$ with /h/ ~ /k/ in Banoho and Batanga, /k/ in Yasa, Bapuku, Mongo, Mboko, Isu and Kundu, /Ø/ ~ /k/ in Duala, Bidiman, Oli, Pongo, Mulimba, Kole, Mokpe and Bubia (e.g., Table 8, examples ‘divide’, ‘storm’, ‘spear’). The examination of the examples displaying these sound correspondences does not highlight a conditioning factor that would allow us to predict when we would have /k/, /h/ or Ø for the languages presenting two reflexes. Generally, in cases with similar partially overlapping correspondence sets with no phonological conditioning, the CM dictates that two proto-sounds should be reconstructed (e.g., *k$_{1}$ and *h$_{1}$). However, a closer examination of the data shows that for the languages with two reflexes, either /h/ ~ /k/ or /Ø/ ~ /k/, the sound sets with /k/ are attested in words that must have been acquired recently (e.g., ‘slave’, ‘cassava’) or words acquired by borrowings (e.g., ‘bag’, ‘sword’). In sum, words that present the sound set with /k/ attested in all the

Table 6: Examples for the reflexes of PSB *p in C1 and C2 positions. We can notice some voicing occurring in Bidiman with $^*p_2$ > /ɓ/ in the examples presented here (“lip”, “touch”), this phenomenon is irregular as we can find $^*p_2$ > /p/ in between vowels (“sweep”: [papa] in Bidiman).

PSB *meaning*	$^*p_1$ ***viper***	$^*p_1$ ***chalk***	$^*p_1$ ***fruit***	$^*p_1$ ***bury***	$^*p_{12}$ ***lip***	$^*p_2$ ***touch***	$^*p_2$ ***armpit***	$^*p_2$ ***wind***
BAN - S	**p**ée	**p**ɛ́mbɛ́	e-ɓumá	i-**p**uɗɛ	è-**p**ò**p**ò		jájá**p**ì	mpu**p**ɛ́
BAT - S	**p**ée	**p**ɛ́mbɛ́	e-**p**umá	i-**p**uɗɛ	è-**p**ò**p**ò	i-tá**p**a	jájá**p**ì	mpu**p**ɛ́
YAS - S	**p**éyi	**p**ɛ́mbɛ́	e-ɓumá				ɲɔ́ɲɔ́**p**ì	mpu**p**ɛ́
BAP - S	**p**ée	**p**ɛ́mbɛ́	e-ɓumá	i-**p**uɗɛ	è-**p**ò**p**ò		ɲá**ɸ**ɛ̀	mpu**p**ɛ́
DUA - C	**p**é	**p**ɛ́mbɛ́	e-**p**umá	**p**ulɛ	è-**p**ò**p**ò	tá**p**a	è-yá**p**ì	
BID - C	**p**é	**p**ɛ́mbɛ́	**p**umá		**p**ò**ɓ**ò	tá**ɓ**a		
OLI - C	**p**é	**p**ɛ́mbɛ́	**p**umá		**p**ò**p**ò	tá**p**a		
PON - C	**p**é	**p**ɛ́mbɛ́	e-**p**umá	i-**p**ulɛ	è-**p**ò**p**ò	i-tá**p**a	è-yá**p**ì	
MON - C	**p**é	**p**ɛ́mbɛ́	e-**p**umá	i-**p**ulɛ	è-**p**ò**p**ò	i-tá**p**a	è-yá**p**ì	
MUL - C	**p**é	**p**ɛ́mbɛ́	e-**p**umá	i-**p**uɗɛ	è-**p**ò**p**ò	i-tá**p**a	yàá**p**ì	mu-pu**p**é
MBO - SW	**ɸ**ée	**ɸ**ɛ́mbɛ́	e-**ɸ**umá				ɲá**ɸ**ɛ̀	
KPE - SW	**ɸ**ée	**ɸ**ɛ́mbɛ́	e-**ɸ**umá				ɲáɸà**ɸ**ɛ̀	
ISU - SW	**ɸ**é	**ɸ**ɛ́mbɛ́	e-**ɸ**umá	i-**ɸ**ulɛ	è-**ɸ**ò**ɸ**ò	i-tá**ɸ**à	ɲáɸà**ɸ**è	
KOL - SW	**ɸ**ée	**ɸ**ɛ́mbɛ́	e-**ɸ**umá	i-**ɸ**urɛ	è-**ɸ**ò**ɸ**ò	i-tá**ɸ**à	ɲáɸà**ɸ**ì	
BUB - SW	**ɸ**ée	**ɸ**ɛ́mbɛ́	e-**ɸ**umá				ɲáɸà**ɸ**è	
Kundu - N			e-**ɸ**uma	**ɸ**ura				di-ɸɛ**ɸ**ɛ

Table 7: Examples for the reflexes of PSB *t in C1 and C2 positions. Only one example displays $^*t_1$ > /r/ in Oli and Bidiman (e.g., "ear").

PSB *meaning*	$^*t_1$ *five*	$^*t_1$ *ear*	$^*t_1$ *drill*	$^*t_1$ *send*	$^*t_{12}$ *smoke*	$^*t_2$ *bag*	$^*t_2$ *hide*
BAN - S	ɓé-táno	ì-tóì	i-túɓa	i-tila	í-tùtù	ŋ̀-kùtà	i-huta
BAT - S	ɓé-táno	ì-tóì	i-túɓa	i-tila	í-tùtù	ŋ̀-kùtà	i-huta
YAS - S	ɓî-táno	dì-lɔ́ɔ̀	e-túɓa	e-tila	ví-tùtù	mò-kùtà	
BAP - S	ɓé-táno	dì-tɔ́	i-túɓa	i-tila	í-tùtù		i-kuta
DUA - C	ɓé-tánu	tóì	túɓa	tila	í-tùtù	mù-kùtà	wuta
BID - C	bî-táan	dì-rɔ́	túɓa	tila	mí-tùrù	ŋ̀-kùrà	wura
OLI - C	bî-táan	dì-ró	túɓa	tila	mí-tùrù	ŋ̀-kùrà	wura
PON - C	ɓé-tánu	ì-tóì	i-túɓa	i-tila	í-tùtù	mù-kùtà	i-wuta
MON - C	ɓé-tánu	ì-tóì	i-túɓa	i-tila	í-tùtù	mù-kùtà	i-kuta
MUL - C	ɓé-táan	ì-tóì	i-túɓa	i-tila	í-tùtù	mù-kùtà	i-kuta
MBO - SW	βé-táa	lì-tóò	li-túβa	li-tila	í-tùtù	mò-kùtà	
KPE - SW	βé-táa	lì-tóò	li-túβa	li-tilá	yú-tùtù	mò-kùtà	l-utá
ISU - SW	βé-tánu	ì-tóì	i-túβa	i-tila	yú-tùtù	mò-kùtà	i-kuta
KOL - SW	bé-tánu	ì-tóì	i-túba	i-tira	í-tùtù	mò-kùtà	i-wuta
BUB - SW	βé-táno	lì-tóò		li-tila	í-tùtù	mò-kùtà	l-uta
Kundu - N	be-ta	di-toi		tira	mo-tutu		

languages are words that cannot be traced back to PB as opposed to the set with /h/ or /Ø/ (e.g., 'hide', 'root', 'spear', 'rope', etc.), also attested in a larger number of words. The study of the reflexes /h/ and /Ø/ highlights that PSB *k is undergoing a process of lenition in some Sawabantu languages until its complete deletion in most of the Central languages and some Southwest: PSB $^*k_1$ > /h/ ~ Ø. This type of lenition chain has been described by Pacchiarotti & Bostoen (2022: 12) for West-Coastal Bantu languages spoken in southern Gabon, Congo and Democratic Republic of the Congo with Proto-West-Coastal Bantu $^*k_2$ > x/ɣ/ʁ/h > Ø. In addition, Grollemund & Philippson (to appear), in a paper devoted to the study of the reflexes of PB *k and *g in NW Bantu languages (including A and some B languages), have observed that PB *k tends to be deleted with the same stages of lenition with generally PB $^*k_{12}$ > /h/ > Ø. The lenition process is confirmed in C2 position with PSB $^*k_2$ being lenited in the same languages (Banoho, Batanga, Duala, Bidiman, Oli, Pongo, Mulimba, Mokpe, examples are presented in Table 9). For Bidiman and Oli, we can observe that $^*k_2$ > /ɣ/ (lenition) or Ø whereas in

C1, the velar fricative does not appear (because PSB $^*k_1$ has been deleted). In addition, we can notice that PSB $^*k_{12}$ is always maintained and stable in Yasa, Bapuku, Mongo, Mboko, Isu and Kundu. Despite the multiple reflexes observed for PSB $^*k_{12}$, our choice to reconstruct just one proto-sound was probably reasonable as we have observed that it was PSB *k that underwent lenition in some languages (while Yasa, Bapuku, Mongo, Mboko, Isu and Kundu only display /k/) and it is the same PSB *k that was retained in some possibly recently acquired words (that did not have the time to go through the lenition process yet). Finally, having two or more reflexes for a same proto-sound is a phenomenon attested in many Bantu languages with the double reflexes described by Guthrie (1967–1971), Stewart (1973, 1975), Jansssens (1991, 1992), Leynseele & Stewart (1980) (see Philippson 2022 for a detailed discussion on double reflexes found in NW Bantu languages) or multiple unconditioned reflexes (MUR) described by Pacchiarotti & Bostoen (2022) referring to a situation where a same proto-sound yields two or more reflexes (without apparent conditioning).

We present in Table 10 the reflexes and proto-sounds reconstructed for voiced plosives. We added the reflexes for PSB $^*l_{12}$ for analysis purposes.

We reconstruct the proto-sound $^*b_{12}$ for the partially overlapping comparative series ɓ/b/β and ɓ/b/w/β (e.g., Table 10) based on the principle of directionality (lenition is more common than fortition, see above). We observe that PSB $^*b_{12}$ > /ɓ/ in South languages, /b/ ~ /ɓ/ in Central languages, /b/ or /β/ in North languages[2] and PSB $^*b_{12}$ > the fricative /β/ in Southwest languages (lenition) except in Kole. The examples presented in Table 11 and Table 12 show that for Central languages, the distribution of /b/ and /ɓ/ is conditioned. The quality of the following vowel is going to trigger the sound to change: we will found the plosive /b/ before high vowels /_i/ and /_u/ (e.g., Table 11 and Table 12) whereas the implosive /ɓ/ will occur elsewhere (/_a, e, ɛ, ɔ, o/). This conditioning was already mentioned in Philippson (2022: 18) who demonstrated that PB *b does not present unconditioned double reflexes or MUR in Bantu A languages. For C1 and C2 in Southwest languages except for Kole, when followed by a back vowel (either _o or _u), PSB $^*b_{12}$ will be deleted with sometimes an epenthetic glide /w/ added (e.g., Table 11 and Table 12, examples ‘nine’, ‘snot’, ‘spider’, ‘house’, ‘hippo’).

For the similar partially overlapping comparative sets with ɗ/d/l/r attested in C1 and C2 positions (e.g., Table 13), we chose to reconstruct two proto-sounds $^*d_{12}$ and $^*l_{12}$. The choice was complicated because we can observe the same sets of reflexes for Southwest languages where PSB $^*d_{12}$ and $^*l_{12}$ yield to /l/ (and /r/

[2] Friesen 2002 often proposed the two variants with /b/ or /β/ for a same word.

Table 8: Examples for the reflexes of PSB *k in C1 position

PSB *meaning*	*k$_1$ *divide*	*k$_1$ *storm*	*k$_1$ *spear*	*k$_1$ /*N_ *tree trunk*	*k$_1$ / _*Vk$_2$/g$_2$ *sufficient/grow*	*k$_1$ / *N_ *pangolin*	*k$_1$ / _*ek$_2$/g$_2$ *molar*
BAN - S	i-kaɓa	ɓo-huɗi	di-hɔŋgɔ́	ŋ-kɔhɔ		káŋgà	ɗi-kɛhɔ
BAT - S	i-kaɓa	ɓo-huɗi	di-hɔŋgɔ́	ŋ-kɔhɔ	i-kowa	káŋgà	ɗi-kɛhɔ
YAS - S	e-kaɓa	ɓo-kuɗi	i-kɔŋgɔ́	mo-kɔkɔ	e-koka	káŋgà	i-cɔkɔ́
BAP - S	i-kaɓa	ɓo-kuɗi	i-kɔŋgɔ́	ŋ-kɔkɔ		káŋgà	i-sɛkɔ
DUA - C	j-aɓa	m-udi	j-ɔŋgɔ́	mu-kɔkɔ	koka	káàŋ	e-kikɔ
BID - C	y-aɓa	m-uli	j-ɔŋgɔ́	ŋ-kɔɣɔ	koɣa	káàŋ	kiɣi
OLI - C	y-aɓa	m-uli	j-ɔɣɔ́	ŋ-kɔhɔ	koɣa	káàŋ	kiɣi
PON - C	ij-aɓa	m-udi	j-ɔŋgɔ́	mu-kɔkɔ	i-koka	káàŋ	e-kikɔ
MON - C	-kaɓanɛ	mu-kudi	j-ɔŋgɔ́	mu-kɔkɔ	i-koka	káàŋ	e-kikɔ
MUL - C	i-kaɓa	m-udi	j-ɔŋgɔ́	mu-kɔɔ		káàŋ	e-cikɔ
MBO - SW			di-kɔŋgɔ́	mu-kɔɔkɔ	li-koka	káŋgɛ̀	e-kɛkɔ
KPE - SW		mu-uli	li-ɔŋgɔ́	mɔ-kɔkɔ	li-koka	káŋgɛ̀	ɛ-kɔkɔ
ISU - SW	i-kaβa	mu-kuli	di-kɔŋgɔ́	mo-kɔkɔ	i-koka	káŋgɛ̀	e-kikɔ
KOL - SW			j-ɔŋgɔ́	mo-kɔkɔ	i-koka	káàŋ	e-kikɔ
BUB - SW	li-kaβa		li-ɔŋgɔ́	mo-kɔkɔ	li-koka	káŋgɛ̀	ɛ-kɔkɔ
Kundu - N	-kaβa		di-kɔŋgɔ	ɛ-kɔkɔ (A11)			e-kekɔ

Table 9: Examples for the reflexes of PSB *k in C2 position

PSB *meaning*	***k**$_2$ ***elephant***	***k**$_2$ ***one***	***k**$_2$ ***scoop***	***k**$_2$ ***fog***	***k**$_2$ / _I ***muteness***	***k**$_2$ ***cow***	***k**$_2$ ***mortar***	***k**$_2$ ***spoon***
BAN - S	n-jɔhu	y-ɔhɔ́	i-tóka	mbàhà	ɗi-mbúhé	ɲaka	è-ɓòwà	tóhò
BAT - S	n-jɔhu	y-ɔhɔ́	i-tóka	mbàhà	ɗi-mbúhé	ɲaka	è-ɓòwà	tóhò
YAS - S	sɔku	e-vɔ́kɔ́		mbàkà	imbúcé	ɲaka	è-ɓòkà	tókò
BAP - S	n-jɔku	y-ɔkɔ́	i-tóka	mbàkà	imbúsé	ɲaka	è-ɓòkà	tókò
DUA - C	n-jɔu	e-wɔ́	tóa	mbà	mbúké	ɲaka	è-ɓòkí	tóò
BID - C	n-jɔ	fɔ́	tóa	mbà	mbúɣí	ɲaɣa	ɓòɣí	tóò
OLI - C	n-jɔ	wɔ́	tóa	mbà	mbúɣí	ɲaɣa	ɓòɣí	tóò
PON - C	n-jɔu	e-wɔ́	i-tóa	mbà	mbúké	ɲaka	è-ɓòkí	tóò
MON - C	n-jɔku	y-ɔkɔ́	i-tóka	mbàkà	mbúké	ɲaka	è-ɓòkí	tókò
MUL - C	n-jɔu	e-wɔ́	i-tóa	mbà	mbúcé	ɲaka	è-ɓòkí	tóò
MBO - SW	n-jɔku	y-ɔkɔ́	li-tóka	mbàkà	mbúké	ɲaka	è-wòkí	
KPE - SW	n-jɔku	y-ɔkɔ́	li-tóka	mbàkì	mbúké	ɲaka	è-wòkí	tókò
ISU - SW	n-jɔku	y-ɔkɔ́	i-tóka	mbàkà	mbúké	ɲaka	è-ɓòkí	
KOL - SW	n-jɔku	y-ɔ́kɔ́	i-tóka	mbàkà	mbúké	ɲaka	è-bòkí	tókò
BUB - SW	n-jɔku	ey-ɔ́kɔ́	li-tóka	mbàkà	mbúké	ɲaka		
Kundu - N	n-jɔku	ey-ɔkɔ				ɲaka		toko

Table 10: Sound correspondences for voiced plosives and the approximant lateral in C1 and C2 positions. When two or more reflexes are presented, the first one is the most frequent. Bold characters mark sound changes from PSB to the daughter languages.

PSB	***b$_{12}$ / _i, u**	***b$_{12}$ / _a,e,ɛ,ɔ,o**	***b$_{12}$ / _u, o**	***b$_1$ / _wV**	***d$_1$**	***d$_2$ / _i, u**	***d$_2$ / _a,e,ɛ,ɔ,o**	***l$_{12}$**
BAN - S	**ɓ**	**ɓ**	**ɓ**	**ɓ(w)**	**ɗ**	**ɗ**	**ɗ**	l
BAT - S	**ɓ**	**ɓ**	**ɓ**	**ɓ(w)**	**ɗ**	**ɗ**	**ɗ**	l
YAS - S	**ɓ**	**ɓ**	**ɓ**	**ɓ(w)**	**ɗ**	**ɗ**	**ɗ**	l
BAP - S	**ɓ**	**ɓ**	**ɓ**	**ɓ(w)**	**ɗ**	**ɗ**	**ɗ**	l
DUA - C	b	**ɓ**	b	**ɓ(w)**	d	d	**l**	l
BID - C	b	**ɓ**	b	**ɓ(w)**	d	**l**	**l**	l
OLI - C	b	**ɓ**	b	**ɓ(w)**	d	**l**	**l**	l
PON - C	b	**ɓ**	b	**ɓ(w)**	d	d	**l**	l
MON - C	b	**ɓ**	b	**ɓ(w)**	d	d	**l**	l
MUL - C	b	**ɓ**	b	**ɓ(w)**	d	d	**ɗ**	l
MBO - SW	**β**	**β**	**w ~Ø**	**gb**	**l**	**l**	**l**	l
KPE - SW	**β**	**β**	**w ~Ø**	**gb**	**l**	**l**	**l**	**Ø** ~l
ISU - SW	**β**	**β**	**w ~Ø**	**gb**	**l**	**l**	**l**	l
KOL - SW	b	b	b	**gb**	**r**	**r**	**r**	**r**
BUB - SW	**β**	**β**	**w ~Ø**	**gb**	**l**	**l**	**l**	l ~**Ø**
Kundu - N	b ~**β**	b ~**β**	b ~**β**	**gb**	**r** ~l	**r** ~l	**r** ~l	**r**

Table 11: Examples for the reflexes of PSB *b in C1 position

PSB *meaning*	**$*b_1$** ***marriage***	**$*b_1$** ***two***	**$*b_1$** ***add***	**$*b_1$** ***know***	**$*b_1$_ O, U** ***nine***	**$*b_1$ _O, U** ***snot***	**$*b_1$ _O, U** ***spider***
BAN - S	dì-**ɓ**áà	ɓé-**ɓ**à	ì-**ɓ**àɗɛ̀	i-**ɓ**îa	ɗì-**ɓ**ùá	ɗì-**ɓ**ɔ̀mbɔ́	di-**ɓ**o**ɓ**ɛ
BAT - S	dì-**ɓ**áà	ɓé-**ɓ**à	ì-**ɓ**àɗɛ̀	i-**ɓ**îa	ɗì-**ɓ**ùá	ɗì-**ɓ**ɔ̀mbɔ́	di-**ɓ**o**ɓ**ɛ
YAS - S	ì-**ɓ**áà	ɓí-**ɓ**à	è-**ɓ**àɗɛ̀		ɗì-**ɓ**ùá	ì-**ɓ**ɔ̀mbɔ́	di-**ɓ**o**ɓ**ɛ
BAP - S	ì-**ɓ**áà	ɓé-**ɓ**à	ì-**ɓ**àɗɛ̀		ɗì-**ɓ**ùá	ɗì-**ɓ**ɔ̀mbɔ́	di-**ɓ**o**ɓ**ɛ
DUA - C	dì-**ɓ**áà	ɓé-**ɓ**àá	**ɓ**átà	**b**îa	dì-**b**ùá	dì-**ɓ**ɔ̀mbɔ́	di-**ɓ**o**ɓ**ɛ
BID - C	**b**áà	ɓí-**ɓ**àá	**ɓ**árɛ̀	**b**îa	**b**ùa	**ɓ**ɔ̀mbɔ́	di-**ɓ**o**ɓ**ɛ
OLI - C	**b**áà	ɓí-**ɓ**àá	**ɓ**árɛ̀	**b**îa	**b**ùa	**ɓ**ɔ̀mbɔ́	di-**ɓ**o**ɓ**ɛ
PON - C	ì-**ɓ**áà	ɓé-**ɓ**àá	ì-**ɓ**átà	i-**b**îa	ì-**b**ùá	ì-**ɓ**ɔ̀mbɔ́	i-**ɓ**o**ɓ**ɛ
MON - C	dì-**ɓ**áà	ɓé-**ɓ**àá	ì-**ɓ**átà	i-**b**îa	dì-**b**ùká	dì-**ɓ**ɔ̀mbɔ́	di-**ɓ**o**ɓ**ɛ
MUL - C	dì-**ɓ**áà	ɓé-**ɓ**àá	ì-**ɓ**átà	i-**b**îa	dì-**b**ùá	dì-**ɓ**ɔ̀mbɔ́	di-**ɓ**o**ɓ**ɛ
MBO - SW	lì-**β**áà	βé-**β**àá	lì-**β**átà		lù-ùká	**w**ɔ̀mbɔ́	li-**w**o**β**ɛ
KPE - SW	lì-**β**áà	βé-**β**àá	lì-**β**átà	li-**β**îa	lù-ùwá	lì-ɔ̀mbɔ́	li-**w**o**β**ɛ
ISU - SW	lì-**β**áà	βé-**ɓ**àá	lì-**β**átà	i-**b**îa	lì-**b**ùká	lì-**w**ɔ̀mbɔ́	li-**w**o**β**ɛ
KOL - SW	dì-**b**áà	bé-**b**àá	ì-**b**átà	i-**b**îa	dì-**b**ùá	dì-**b**ɔ̀mbɔ́	di-**b**o**ɓ**ɛ
BUB - SW	lì-**β**áà	βé-**β**àá	lì-**β**átà	li-**β**îa	lì-gbá	lì-ɔ̀mbɔ́	li-**w**o**β**ɛ
Kundu - N		be-**b**e/ βe-βe					ri-**β**o**β**e

Table 12: Examples for the reflexes of PSB *b in C2 position

PSB	***b**$_2$	***b**$_2$	***b**$_2$	***b**$_2$	***b**$_2$ **/ _o, u**	***b**$_2$ **/ _o, u**
meaning	***six***	***divide***	***banana***	***wickedness***	***house***	***hippo***
BAN - S	n-tóɓá	i-kaɓa	di-húɓé	ɓoɓé	ndáɓo	ŋguɓú
BAT - S	n-tóɓá	i-kaɓa	di-húɓé	ɓoɓé	ndáɓo	ŋguɓú
YAS - S	mo-tóóɓá	e-kaɓa	vi-kúɓé			ŋguɓú
BAP - S	n-tóɓá	i-kaɓa	i-kúɓé	ɓoɓé	ndáɓo	ŋguɓú
DUA - C	mu-tóɓá	j-aɓa	di-kúbé	ɓoɓé	ndáɓo	ŋgubú
BID - C	n-tóɓá	y-aɓa		ɓobî	ndáɓo	ŋgubú
OLI - C	n-tóɓá	y-aɓa		ɓobî	ndáɓo	ŋgubú
PON - C	mu-tóɓá	ij-aɓa	i-kúɓé	ɓoɓé	ndáɓo	ŋgubú
MON - C	mu-tóɓá	-kaɓanɛ	e-kúɓékúɓé	ɓoɓée	ndáɓo	ŋgubú
MUL - C	mu-tóɓá	i-kaɓa	du-úɓé	ɓoɓé	ndáɓo	ŋgubú
MBO - SW	mo-tóβá			woβée	ndáwo	ŋguwú
KPE - SW	mo-tóβá			woβée	ndáwo	ŋguwú
ISU - SW	mo-tóβá	i-kaβa		woβée	ndáwo	ŋguwú
KOL - SW	mo-tóbá			bobé	ndábo	ŋgubú
BUB - SW	mo-tóβá	li-kaβa		woβée	ndáwo	ŋguwú
Kundu - N		-kaβa			ndabo/ndaβo	

in Kole). In addition, the study of the examples presented in Table 13 and Table 14 seems to display a phonological context that would require us to posit one proto-sound (PSB *d or *l) for these comparative sets with the impression that reflexes of $*d_{12}$ appear before high vowels (_i and _u) whereas the reflexes of $*l_{12}$ appear elsewhere. And finally, PB *d presenting two reflexes /d/~/l/ is a phenomenon that is attested in Bantu languages (Guthrie 1967–1971, Meeussen 1967, Stewart 1973, 1975, 1993, Jansssens 1992, Philippson 2022).

In order to untangle this situation, we first need to summarize the reflexes attested for PSB $*d_{12}$ and $*l_{12}$ (see Table 13 and Table 14). For South languages, we observe the reflexes /ɗ/ for $*d_{12}$ and /l/ for $*l_{12}$. For Central languages, PSB $*d_1$ > /d/ whereas in C2 position, we can find /d/ or /l/. In Duala, Pongo, Mongo, $*d_2$ > /d/ when followed by the high vowels / _i, u whereas /l/ is attested elsewhere. For Mulimba, $*d_2$ > /d/ when followed by high-vowels and /ɗ/ is attested elsewhere. For Bidiman and Oli, $*d_2$ > /l/ in all contexts. For PSB $*l_{12}$, we found the reflexes /l/ for these languages. For Southwest languages, the situation is simpler with $*d_{12}$ > /l/ and $*l_{12}$ > /l/ in all the languages except for Kole that presents /r/ for both proto-sounds. We can note that Mokpe and Bubia can present the reflexes Ø or /l/ for PSB $*l_{12}$. Finally, in Kundu (North), the data presented in Friesen 2002: 24 gave several reflexes with $*d_{12}$ > /l/ or /r/ (Friesen 2002: 25-27 explained that there is one phoneme /d/ in North languages that is realized [d] in front of the high vowels /i/ and /u/ whereas [l] is attested elsewhere with [r] occurring rarely). And $*l_{12}$ > /r/ in Kundu.

If we posit one proto-sound such as PSB $*d_{12}$ or $*l_{12}$ with the rule (conditioning) that the reflexes /d/ or /ɗ/ are attested in Sawabantu languages before high vowels whereas /l/ will be attested elsewhere, we would face several issues for the South and Central languages. First, for South languages, while we have observed that /ɗ/ is more often attested in front of high vowels whereas /l/ is more attested in front of non-high vowels, reconstructing one proto-sound would oblige us to posit double reflexes or MUR in C1 and C2 positions as /ɗ/ and /l/ can appear in similar phonological contexts (e.g., Table 13 examples ‘run’ and ‘read’; Table 15, examples ‘dig’, ‘be full’, ‘put in’, ‘lend’ or ‘oil’). Second, for Central languages, in C1 position, reconstructing one proto-sound is tempting as we found the reflexes /d/ only before high vowels and /l/ elsewhere. However, the situation in C2 with the conditioning described and more specifically the three reflexes observed in Mulimba with /d/, /ɗ/ and /l/ tipped the scales in favor of the reconstruction of two proto-sounds. In Mulimba, /d/ is found in front of high vowels (same as Duala, Pongo and Mongo) but /ɗ/ (as opposed to /l/ attested in Duala, Pongo and Mongo) is found in front of non-high vowels whereas /l/ is attested in many contexts, mostly in front of non-high vowels. The conditioning observed for /d/ and

Table 13: Examples for the reflexes of PSB *d and *l in C1 position

PSB *meaning*	***d**$_1$ ***extinguish***	***d**$_1$ ***old***	***d**$_1$ ***water***	***d**$_1$ ***run***	***l**$_1$ ***send***	***l**$_1$ ***read***	***l**$_2$ ***three***
BAN - S	i-**ɗ**ima	ì-**ɗ**ùnà	ma-**ɗ**îɓá	ì-**ɗ**àŋgwà	ì-**l**óma	ì-**l**áŋgà	ɓé-**l**álo
BAT - S	i-**ɗ**ima	ì-**ɗ**ùnà	ma-**ɗ**îɓá	ì-**ɗ**àŋgwà	ì-**l**óma	ì-**l**áŋgà	ɓé-**l**álo
YAS - S	e-**ɗ**ima	è-**ɗ**ùnà	me-héɓá		è-**l**óma	è-**l**áŋgà	ɓî-**l**álo
BAP - S	i-**ɗ**ima	ì-**ɗ**ùnà	mi-iɓa		ì-**l**óma	ì-**l**áŋgà	ɓé-**l**álo
DUA - C	**d**îma	**d**ùnà	ma-**d**îɓá	**ɗ**àŋgwà	è-**l**óma	**l**áŋgà	ɓé-**l**álo
BID - C	**d**îmɛ	**d**ùnà	man-**d**îɓa		è-**l**óma	**l**áŋgà	bî-**l**álo
OLI - C	**d**îmɛ	**d**ùnà	ma-**d**îɓá		è-**l**óma	**l**áɣà	bî-**l**álo
PON - C	i-**d**îma	ì-**d**ùnà	ma-**d**îɓá	ì-**ɗ**àŋgwà	ì-**l**óma	ì-**l**áŋgà	ɓé-**l**álo
MON - C	i-**d**îma	ì-**d**ùnà	ma-**d**îɓá		ì-**l**óma	ì-**l**áŋgà	ɓé-**l**álo
MUL - C	i-**d**îma	ì-**d**ùnà	ma-**d**îɓá	ì-**ɗ**àŋgwà	ì-**l**óma	ì-**l**áŋgà	ɓé-**l**álo
MBO - SW	li-**l**îma	lù-ùnà	ma-**d**îβá		lì-**l**ómà	lì-**l**áŋgà	βé-**l**álo
KPE - SW	li-**l**îma	lù-ùná	ma-**d**îβá		lò-óma	lì-**l**áŋgà	βé-yáo
ISU - SW	i-**l**îma	ì-**d**ùnà	ma-**d**îβá		lì-**l**ómà	ì-**l**áŋgà	βé-**l**álo
KOL - SW	i-**d**îma	ì-**d**ùnà	ma-**r**îβá		ì-**r**óma	ì-**r**áŋgà	bé-**l**áro
BUB - SW	li-**l**îmɛ	li-**l**ùnà	ma-**d**îβá		lì-yóma	lì-yáŋgà	βé-**l**álo
Kundu - N	**r**imɛ	unu	i-**l**iba/i-**r**iβa		**l**oma/**r**oma		be-**r**aro

Table 14: Examples for the reflexes of PSB *d and *l in C2 position

PSB	$^*\mathbf{d}_2$	$^*\mathbf{d}_2$	$^*\mathbf{d}_2$	$^*\mathbf{d}_2$	$^*\mathbf{d}_2$	$^*\mathbf{l}_2$	$^*\mathbf{l}_2$	$^*\mathbf{l}_2$
meaning	***turtle***	***birth***	***rope***	***goat***	***put inside***	***lend***	***write***	***beard***
BAN - S	kúɗù	i-yáɗi	ŋ-hɔɗî		ì-wɛ́ɗɛ̀	ì-ɓálɛ̀	i-tila	ɗi-hɛlú
BAT - S	kúɗù	i-yáɗi	ŋ-hɔɗî	mbóɗì	ì-vɛ́ɗɛ̀	ì-ɓálɛ̀	i-tila	ɗi-hɛlú
YAS - S	kúɗù	i-jáɗi	mo-kɔɗî		è-vɛ́lɛ̀	è-ɓálɛ̀	e-tila	sɛɗu
BAP - S	kúɗù	i-yáɗi	ŋ-kɔɗî		ì-wɛ́ɗɛ̀	ì-ɓálɛ̀	i-tila	i-hɛlú
DUA - C	wúdù	yádî	m-ɔdî	mbódì	wɛ́lɛ̀	ɓálɛ̀	tila	sɛdú
BID - C	wúlù	yálé	m-ɔlî	mbólì	fɛ́lɛ̀	ɓálɛ̀	tila	sɛlú
OLI - C	wúlù	yálî	m-ɔlî	mbólì	wɛ́lɛ̀	ɓálɛ̀	tila	sɛlú
PON - C	wúdù	jadi	m-ɔdî	mbódì	ì-wɛ́lɛ̀	ì-ɓálɛ̀	i-tila	i-sɛdú
MON - C	kúdù	jadi	mu-kɔɗî	mbódì	ì-wɛ́lɛ̀	ì-ɓálɛ̀	i-tila	i-sɛdú
MUL - C	kúdù	i-yádî	m-ɔdî	mbódì	ì-ɛ́ɗɛ̀	ì-ɓálɛ̀	i-tila	di-hɛlú
MBO - SW	è-kúlɛ̀	li-yálî	mo-kɔlî	mbólì	lì-wálɛ̀	lì-βálɛ̀	li-tila	njɛli
KPE - SW	kúlù	li-yálî	mɔ-ɔlî	mbólì		lì-βálɛ̀	li-tilá	njɛlu
ISU - SW	è-kúlɛ̀	jali	mo-kɔlî	mbólì		ì-ɓálɛ̀	i-tila	i-sɛlú
KOL - SW	è-kúrɛ̀	jádi	mo-kɔrî	mbórì	ì-wɛ́rɛ̀	ì-bárɛ̀	i-tira	i-sɛ̀rú
BUB - SW	è-kúlɛ̀	li-yálî	m-ɔlî	mbólì	lì-wálɛ̀	lì-ɓálɛ̀	li-tila	
Kundu - N	ku	ja	mo-kori	mboli/mbori	ule		tira	

/ɗ/ along with the similar contexts in which we can find /ɗ/ and /l/ in Mulimba shows that these sets of reflexes should be linked to two separate proto-sounds PSB *d and *l. An examination of Table 15 (along with Table 13 and Table 14) confirms the need of two distinct proto-sounds which resulted in two sound sets: /d/ /ɗ/ (< PSB *d) are attested in Mulimba when South languages also present /ɗ/ and reflexes /l/ (< PSB *l) are attested in Mulimba when South languages also present /l/.

Finally, there is some uncertainty as to the phonetic nature of the PSB (and PB) *b and *d as we cannot be absolutely sure that the original sounds were not implosives like in Southern and were later strengthened in some languages in front of first-degree vowels.

We do not reconstruct *g in PSB as the phoneme /g/ is absent in C1 and C2 positions in the daughter languages (except when preceded by a nasal homorganic prefix of class 9 such as in "moon" PSB *ŋ-gɔndɛ). This phenomenon is not unusual as it was noticed earlier by Möhlig (1981) and Nurse & Philippson (2003) who reported that Bantu languages spoken in the rainforest (generally A, B, C and parts of H) present this particular sound change where PB *g > /k/. In addition, Pacchiarotti & Bostoen (2020) showed that for West-Coastal languages (parts of B and H), there was a velar merger that led PB *k and *g to merge into /k/. Finally, Grollemund & Philippson (to appear) described that for NW Bantu languages (A and some B20), PB *k > Ø which triggered the PB *g > /k/. These sound changes modified the phonemic inventory of these Bantu languages yielding to the disappearance of /g/ in some languages.

We also chose not to reconstruct labial-velar stops consonants *kp and *gb attested in Southwest and North Sawabantu languages (e.g., Table 5 and Table 10) as they are irregular and generally the result of contact or the fortition of the PSB sequences *kwV and PSB *bwV. Labial-velar stops are rare in Bantu languages but they do exist in several Bantu languages belonging to zones A, B and C. It is assumed that Bantu languages have acquired these labial-velar stops through contact with either Ubangian, Central Sudanic or Southern Bantoid languages (Clements & Rialland 2008, Bostoen & Donzo 2013, Idiatov & de Velde 2021). Due the geographic location of Sawabantu languages in southern Cameroon, we could assume that Southwest and North Sawabantu languages have acquired these labial-velar stops by prolonged contact with Southern Bantoid languages located in the same region. While this hypothesis is highly plausible (and only a thorough study of these labial-velar stops will give us a clear answer), we have noticed some conditioning environments that might explain the presence of some labial-velar stops in Sawabantu languages. Bostoen & Donzo (2013: 450) examine the origin of labial-velar stops in Ngombe (C41), a Bantu language spoken in

Table 15: Comparison of PSB *d_2 and *l_2.

PSB	***bodi**	***kado**	***bolo**	***pud**	***ul**	***vɛd**	***bal**	***ula**
meaning	***goat***	***ladder***	***work***	***dig***	***be full***	***put in***	***lend***	***oil***
BAN - S		ɓò-hà**ɗ**ò		i-pu**ɗ**a	ȉ-hú**l**à	ì-wɛ́**ɗ**ɛ̀	ì-ɓá**l**ɛ̀	mà-ú**l**ɛ̀
BAT - S	mbó**ɗ**ì	ɓò-hà**ɗ**ò		i-pu**ɗ**a	ȉ-hú**l**à	ì-vɛ́**ɗ**ɛ̀	ì-ɓá**l**ɛ̀	mà-hú**l**ɛ̀
YAS - S			è-ɓó**l**ó	e-pu**ɗ**wa	è-hó**l**à	è-vɛ́**l**ɛ̀	è-ɓá**l**ɛ̀	màv-ú**l**ɛ̀
BAP - S		ɓò-kà**ɗ**ò		i-pu**ɗ**a	ȉ-hú**l**à	ì-wɛ́**ɗ**ɛ̀	ì-ɓá**l**ɛ̀	má-**l**ɛ̀
DUA - C	mbó**d**ì		è-ɓó**l**ó	pu**l**a	jú**l**à	wɛ́**l**ɛ̀	ɓá**l**ɛ̀	mù-ú**l**à
BID - C	mbó**l**ì		ɓò**l**ó		ó**l**à	fɛ́**l**ɛ̀	ɓá**l**ɛ̀	mù-ú**l**à
OLI - C	mbó**l**ì		ɓò**l**ó		ó**l**à	wɛ́**l**ɛ̀	ɓá**l**ɛ̀	mù-ú**l**à
PON - C	mbó**d**ì		è-ɓó**l**ó	i-pu**l**a	ì-jú**l**à	ì-wɛ́**l**ɛ̀	ì-ɓá**l**ɛ̀	mù-ú**l**à
MON - C	mbó**d**ì		è-ɓó**l**ó	i-pu**l**a	ì-jú**l**à	ì-wɛ́**l**ɛ̀	ì-ɓá**l**ɛ̀	mà-wú**l**à
MUL - C	mbó**d**ì	ɓò-kà**ɗ**ò	è-ɓó**l**ó	i-pu**ɗ**a	iú**l**à	ìɛ́**ɗ**ɛ̀	ì-ɓá**l**ɛ̀	mù-ú**l**ɛ̀
MBO - SW	mbó**l**ì		è-wò**l**ó		lù-ú**l**à	lì-wá**l**ɛ̀	lì-βá**l**ɛ̀	mà**l**-úwà
KPE - SW	mbó**l**ì		è-wò**l**ó		lù-ú**j**à		lì-βá**l**ɛ̀	mà-ú**j**à
ISU - SW	mbó**l**ì		è-wò**l**ó	i-ɸu**l**wa	yù-ú**l**à		ì-ɓá**l**ɛ̀	mà-ú**l**à
KOL - SW	mbó**r**ì		è-wò**r**ó	i-ɸu**l**a	i-wú**r**à	ì-wɛ́**r**ɛ̀	ì-bá**r**ɛ̀	mà-ú**r**à
BUB - SW	mbó**l**ì		è-wò**l**ó	li-ɸua		lì-wá**l**ɛ̀	lì-ɓá**l**ɛ̀	
Kundu - N	mboli					ule		

Democratic Republic of the Congo. While they do concur on the fact that labial-velar stops have been acquired by contact with neighboring Ubangian speakers, they also mention that conditioned sound changes can trigger labial-velar stops to develop. They demonstrate that labial-velars in Ngombe that do not have a Ubangian origin have developed because of the following rule: $C_{[+voiced]}$wV > gbV and $C_{[-voiced]}$wV > kpV, proposed first by Connell (1994), Connell (1998) and Cahill (1999). It seems that we have a similar phenomenon in Southwest and North Sawabantu languages where some labial-velar stops have possibly been acquired through contact while others seem to obey to this same conditioned sound change presented in Table 16. The status of these labial-velars stops /kp/ and /gb/ is still unclear and more data need to be analyzed.

Table 16: Examples of the labial-velar stops in Southwest and North groups.

PSB	South	Central	Southwest	North
*kwale "partridge"	/ŋgwae/	/kwale/	/**kp**ae/	
*kwedi "death"	/weɗi/	/kweli/	/**kp**eli/	
*bwa "dog"	/mbwa/	/mbwa/	/ŋ**gb**a/	/m**gb**á/

For the fricatives presented in Table 17, we have reconstructed *v and *s in C1 and C2 positions. For the correspondence set with v/w/f/Ø, we chose to reconstruct *v since weakening (*v > /w/ > Ø) is more likely than strengthening. We could also have proposed *f but we hesitated to do so on the basis of Bidiman alone, which is rather idiosyncratic on other counts. For the set with h/s/ʒ, we reconstructed *s in C1 and C2 positions based on the same principle (with PSB $*s_{12}$ undergoing lenition to /h/ in South languages and also Mulimba). The exact phonetic value of the Mokpe sound /ʒ/ is the subject of some debate in the literature (Atindogbe 2013: 9).

Nasal consonants *m, *n, *ɲ have been reconstructed in C1 and C2 position along with prenasalized consonants *mb, *nd, *ŋg (e.g., Table 19).

Prenasalized consonants can also be found in C1 position as they are the result of the combination of the nasal prefix of class 9 *N- (homorganic nasal) combined with a plosive consonant. For example, the word for 'crocodile' is reconstructed *gàndʊ́ in PB (BLR3 1326) with a class 9 which led to the PB root *N-gàndʊ́ > [ŋ-gàndó] in Banoho.

We also reconstructed the approximants *w and *y (presented in Table 22) in C1 and C2 positions.

Table 17: Sound correspondences for fricatives in C1 and C2 positions. Bold characters mark sound changes from PSB to the daughter languages.

PSB	$^*v_{12}$	$^*s_{12}$
BAN - S	v	**h**
BAT - S	v	**h**
YAS - S	v	**h**
BAP - S	v	**h**
DUA - C	**w**	s
BID - C	**f**	s
OLI - C	v	s
PON - C	**w**	s
MON - C	**w**	s
MUL - C	**Ø**	**h**
MBO - SW	**Ø**	s
KPE - SW	**Ø**	**ʒ**
ISU - SW	**Ø**	s
KOL - SW	**w**	s
BUB - SW	**w**	s
Kundu - N	**Ø**	s

The study of $^*w_{12}$ and $^*y_{12}$ shows that in many examples, /w/ and /y/ are epenthetic consonants in Sawabantu languages resulting from various processes occurring at morpheme boundaries. Generally, when PSB *k deletes, it leads to the addition of /w/ or /y/ (depending on the quality of the following vowel) in between the prefix and the root: PSB *kuba 'hen' > /wuɓa/ in Duala, Bidiman, Oli and Pongo. Also, when PSB *b > Ø / _o, u, it leads to the addition of /w/: PSB *bɔmbɔ 'snot' > /wɔmbɔ/ in Mbole. Finally, we also have examples where PSB *v > /w/ in some languages: PSB *vasa 'twin' > /wasa/ in Pongo. However, there are also cases where *w can be reconstructed in C1 and C2 position (e.g., Table 23) or in between C1 and V1 (e.g., PSB *bwa 'dog' > /imbwa/ in Batanga for example) or C2 and V2 (e.g., PSB *puŋgwa 'mix' > /i-puŋgwa/ in Banoko for example). Finally, we needed to reconstruct $^*y_1$ for comparative series where /y/ is attested in all Sawabantu languages (e.g., Table 23, examples PSB *ya 'give birth' or PSB *yoko 'penis'). We did not find non-epenthetic *y in C2 position.

We do find in Ebobissé's data (2015) the occurrence of /j/ (/ʤ/) that appears rarely and for this reason, we have decided to not reconstruct *j as it is often a

Table 18: Examples for the reflexes of PSB *v and *s in C1 and C2 positions.

PSB *meaning*	$^*v_1$ *wind*	$^*v_1$ *twin*	$^*v_2$ *itch*	$^*s_1$ *iron*	$^*s_1$ *army*	$^*s_2$ *corn*	$^*s_2$ *choose*
BAN - S	ɗi-vîndî	ɗì-váhà	yòvá	i-hîa	háo	mbáyí	
BAT - S	ɗi-vîndî	ɗì-váhà	yòvá	i-hîa		mbáhí	i-pɔhɔ
YAS - S	i-vîndó	ì-vágà	òvá	e-hîsa	i-yáko	mbàsì	
BAP - S	ɗi-vîndî	ì-váhà	yòvá	i-hîa	háo	mbáyí	
DUA - C	di-wîndî	dì-wásà		sîa	sáo	mbàsì	pɔsɔ
BID - C	findîfindî	fásà		sîa	sáo	mbàsì	pɔsɔ
OLI - C	vîndîvîndî	vásà		sîa	sáo	mbàsì	pɔsɔ
PON - C	i-wîndîwîndî	ì-wásà		i-sîa	sáo	mbàsì	i-pɔsɔ
MON - C	di-wîndîwîndî	dì-wásà		i-sîa	sáko	mbàsì	i-pɔsɔ
MUL - C	di-îndî	dì-áhà	yòwá	i-hîa	háo	mbàsì	i-pɔhɔ
MBO - SW	li-wîndó	lì-ásɛ̀		li-sîa	sáko		li-ɸɔsɔ́
KPE - SW	li-wîndî	mà-ɛ́ʒɛ̀		li-ʒîa	ʒiyaó	mbàʒì	li-ɸɔʒɔ́
ISU - SW	li-wîndîwîndî	mà-ɛ́sɛ̀		i-sîa	sáo	mbàsì	li-ɸɔsɔ
KOL - SW	di-wîndî	dì-wásà		i-sîa	síáko	mbàsì	li-ɸɔsɔ
BUB - SW	li-îndó	m-ásɛ̀		li-sîa	síyaó	mbàsì	li-ɸɔsɔ
Kundu - N		ri-asɛ					βɔsɔ

Table 19: Sound correspondences for nasals and prenasalized consonants. Bold characters mark sound changes from PSB to the daughter languages. We can note that denasalization is frequent in Oli.

PSB	***m_{12}**	***n_{12}**	***$ɲ_{12}$**	***mb**	***nd**	***ŋg**
BAN - S	m	n	ɲ	mb	nd	ŋg
BAT - S	m	n	ɲ	mb	nd	ŋg
YAS - S	m	n	ɲ	mb	nd	ŋg
BAP - S	m	n	ɲ	mb	nd	ŋg
DUA - C	m	n	ɲ	mb	nd	ŋg
BID - C	m	n	ɲ	mb	nd	ŋg
OLI - C	m	n	ɲ	mb	nd	**ɣ** ~ŋg
PON - C	m	n	ɲ	mb	nd	ŋg
MON - C	m	n	ɲ	mb	nd	ŋg
MUL - C	m	n	ɲ	mb	nd	ŋg
MBO - SW	m	n	ɲ	mb	nd	ŋg
KPE - SW	m	n	ɲ	mb	nd	ŋg
ISU - SW	m	n	ɲ	mb	nd	ŋg
KOL - SW	m	n	ɲ	mb	nd	ŋg
BUB - SW	m	n	ɲ	mb	nd	ŋg
Kundu - N	m	n	ɲ	mb	nd	ŋg

prefix attached to a root that starts with a vowel (e.g., Table 24 'ten' or 'mirror') or in free variation with /y/ for which we have reconstructed *y (e.g., Table 24 "wash").

We reconstruct in total 16 PSB consonants: *p, *t, *k, *b, *d, *l, *m, *n, *ɲ, *v, *s, *w, *y, *mb, *nd, *ŋg. The PB reconstructed by Guthrie (1967–1971) and Meeussen (1967) are very similar; although they include the addition of PB *g and *c (Guthrie also distinguished *j from *y, see Wills 2022 for details). In these PB system reconstructions, there are no fricative consonants reconstructed (see Bostoen 2019 for more details on PB).

Hedinger (1987) reconstructed for PM in C1 and C2 positions almost the same set of consonants with *p, *t, *k, *s, *b, *d, *j,*m, *n, *ny, *mb, *nd, *ŋg and *w. He also had *f (but not *v) and he reconstructed *g. A closer look at Hedinger's data showed that the reflexes of PM *g in C1 position are only attested when preceded by a nasal prefix (Hedinger 1987: 39). However, in C2, we observe that PM *g is lenited in some languages displaying either /ʔ/, /k˺/ or /g˺/. The choice made by Hedinger to reconstruct *g in PM is understandable as we find attestations of /g/

Table 20: Examples for the reflexes of PSB *m, *n and *ɲ in C1 and C2 positions.

PSB *meaning*	$^{*}\mathbf{m}_{1}$ ***finish***	$^{*}\mathbf{m}_{2}$ ***stand***	$^{*}\mathbf{n}_{12}$ ***bird***	$^{*}\mathbf{n}_{2}$ ***shame***	***ɲ**$_{1}$ ***surprised***	***ɲ**$_{1}$ ***happiness***	***ɲ**$_{2}$ ***shine***
BAN - S	ì-**m**à	i-tɛ́**m**ɛ	i-**n**ɔ**n**í	ì-yɔ́**n**ì		e-**ɲ**ɛŋgɛ	ì-pà**ɲ**à
BAT - S	ì-**m**à	i-tɛ́**m**ɛ	i-**n**ɔ**n**í	ì-hɔ́**n**ì		e-**ɲ**ɛŋgɛ	ì-pà**ɲ**à
YAS - S		e-tɛ́**m**ɛ	vi-ɲɔ**n**í	vì-yɔ́**n**ì		e-**ɲ**ɛŋgɛ	è-pà**ɲ**à
BAP - S	ì-**m**à	i-tɛ́**m**ɛ	i-**n**ɔ**n**í	ì-yɔ́**n**ì		e-**ɲ**ɛŋgɛ	ì-pà**ɲ**à
DUA - C		tɛ́**m**ɛ	i-**n**ɔɔ́**n**	ì-sɔ́ɔ̀**n**	**ɲ**áka	mu-**ɲ**ɛŋgɛ	pà**ɲ**a
BID - C	**m**à	tɛ́**m**ɛ	**n**ɔɔ́**n**	sɔ́ɔ̀**n**	**ɲ**áɣa	ɲ̀-**ɲ**ɛŋgɛ	pà
OLI - C	**m**à	tɛ́**m**ɛ	**n**ɔɔ́**n**	sɔ́ɔ̀**n**	**ɲ**áɣa	ɲ̀-**ɲ**ɛŋgɛ	pà
PON - C		i-tɛ́**m**ɛ	e-**n**ɔɔ́**n**	è-sɔ́ɔ̀**n**	i-**ɲ**áka	mu-**ɲ**ɛŋgɛ	ì-pà**ɲ**à
MON - C	ì-**m**à	i-tɛ́**m**ɛ	i-**n**ɔɔ́**n**	ì-sɔ́ɔ̀**n**	i-**ɲ**áka	mu-**ɲ**ɛŋgɛ	ì-pà**ɲ**à
MUL - C		i-tɛ́**m**ɛ	i-**n**ɔɔ́**n**	ì-hɔ́ɔ̀**n**	i-**ɲ**áka	mu-**ɲ**ɛnjɛ	ì-pà**ɲ**à
MBO - SW		li-tɛ́**m**ɛ	i-**n**ɔ**n**í	ì-sɔ́**n**ì	li-**ɲ**áka	mo-**ɲ**ɛŋgi	lì-kpà**ɲ**à
KPE - SW		li-tɛ́**m**ɛ	i-**n**ɔ**n**í	ì-ʒɔ́**n**ì	li-**ɲ**áka	mo-**ɲ**ɛŋgi	
ISU - SW	ì-**m**à	i-tɛ́**m**ɛ	i-**n**ɔ**n**í	ì-sɔ́ɔ̀**n**	i-**ɲ**áka	mo-**ɲ**ɛŋgi	ì-pà**ɲ**à
KOL - SW	ì-**m**à	i-tɛ́**m**ɛ	i-**n**ɔ**n**í	ì-sɔ́**n**ì	i-**ɲ**áka	mo-**ɲ**ɛŋgi	ì-pà**ɲ**à
BUB - SW		li-tɛ́**m**ɛ	i-**n**ɔ**n**í	ì-sɔ́**n**ì	li-**ɲ**áka	mo-**ɲ**ɛŋgi	
Kundu - N	**m**arɛ		i-**n**ɔ	i-sɔ	ma-**ɲ**aka		

Table 21: Examples for the reflexes of PSB *mb, *nd and *ŋg.

PSB *meaning*	***mb** ***namesake***	***nd** ***crocodile***	***ŋg** ***root***
BAN - S	**mb**ó**mb**ò	ŋgà**nd**ó	ŋ̀-hà**ŋg**á
BAT - S	**mb**ó**mb**ò	ŋgà**nd**ó	ŋ̀-hà**ŋg**á
YAS - S	**mb**ó**mb**ò	ŋgà**nd**ó	
BAP - S	**mb**ó**mb**ò	ŋgà**nd**ó	ŋ̀-kà**ŋg**á
DUA - C	**mb**ó**mb**ò	ŋgà**nd**ó	mw-à**ŋg**á
BID - C	**mb**ó**mb**ò	ŋgà**nd**ó	mw-à**ŋg**á
OLI - C	**mb**ó**mb**ò	ŋgà**nd**ó	mw-à**ɣ**á
PON - C	**mb**ó**mb**ò	ŋgà**nd**ó	mw-à**ŋg**á
MON - C	**mb**ó**mb**ò	ŋgà**nd**ó	mù-kà**ŋg**á
MUL - C	**mb**ó**mb**ò	ŋgà**nd**ó	mw-à**ŋg**á
MBO - SW	**mb**ó**mb**ò	ŋgà**nd**ó	ŋm-à**ŋg**á
KPE - SW	**mb**ó**mb**ò	ŋgà**nd**ó	ŋm-à**ŋg**á
ISU - SW	**mb**ó**mb**ò	ŋgà**nd**ó	mò-kà**ŋg**á
KOL - SW	**mb**ó**mb**ò	ŋgà**nd**ó	ŋw-à**ŋg**á
BUB - SW	**mb**ó**mb**ò	ŋgà**nd**ó	ŋm-à**ŋg**á
Kundu - N		ŋga**nd**o	

in C1 and C2 positions as opposed to Sawabantu languages where /g/ is absent in C2 (for example, we have in PM 'penis' *jòǵ > [n-jɔg ˺] in Akossi whereas we have *yoko in PSB).

The comparison of the two proto-systems reconstructed for Manenguba and Sawabantu (languages spoken in the same region, e.g., Figure 1 and classified within the same subgroup, e.g., Figure 2) illustrates some similarities between the two subgroups; however, we also observe some key differences. For example, the same sets of consonants have been reconstructed in both proto-languages (with the exception of *g and *v) but the study of the reflexes within the daughter languages shows different evolution patterns for the same proto-consonants in PM and PSB. The examination of Table 25 displaying the reflexes of PM's consonants (without the nasals) in C1 position shows that Manenguba languages seem more conservative (with respect to the protolanguage) than Sawabantu languages where multiple reflexes are attested for the same reconstructed consonants (e.g., PSB *d > /ɗ/, /d/, /l/ or /r/). However, the loss of *V2 in Manenguba languages led to the weakening of *C2 triggering the loss of the voicing distinction

Table 22: Sound correspondences for approximants *w and *y.

PSB	*w	*y
BAN - S	w	y
BAT - S	w	y
YAS - S	w	y
BAP - S	w	y
DUA - C	w	y
BID - C	w	y
OLI - C	w	y
PON - C	w	y
MON - C	w	y
MUL - C	w	y
MBO - SW	w	y
KPE - SW	w	y
ISU - SW	w	y
KOL - SW	w	y
BUB - SW	w	y
Kundu - N	w	y

for plosives whereas in Sawabantu languages, V2 is maintained (along with the voicing distinction). Finally, vowels in Manenguba languages underwent many changes (Hedinger 1987: 44-45) as opposed to Sawabantu languages.

3.2 Sound changes in Sawabantu languages and preliminary observations

The sound changes observed in Table 5, Table 10, Table 17, Table 19, and Table 22 have shown that Sawabantu languages underwent many changes with several PSB consonants undergoing lenition. The analysis of sound changes supports three genetic subgroups in accordance with the three geographical subgroups distinguished in Ebobissé (1989) and Ebobissé (2015) with the addition of Kundu (North) with (i) South: PSB $*b_{12}$ > /ɓ/, PSB $*d_{12}$ > /ɗ/ and PSB $*s_{12}$ > /h/, (ii) Central: we have a phonological conditioning found for PSB $*b_{12}$ and PSB $*d_{2}$ with also PSB $*k_{12}$ > Ø in Duala, Bidiman, Oli, Pongo and Mulimba, and (iii) Southwest and North: PSB $*d_{12}$ > /l/, the lenition of PSB $*p_{12}$ > /ɸ/ and PSB $*b_{12}$ > /β/. Therefore, the definition of our sub-groups differs from Ebobissé's (1989)

Table 23: Examples for the reflexes of PSB *w and *y in C1 and C2 positions.

PSB *meaning*	$^{*}\mathbf{w}_1$ ***die***	$^{*}\mathbf{w}_2$ ***coast***	$^{*}\mathbf{y}_1$ ***penis***	$^{*}\mathbf{y}_1$ ***give birth***
BAN - S	i-w**ɔ́ɔ**		yòó	i-yáa
BAT - S	i-wáa		yòó	i-yáa
YAS - S	e-wáa			e-jáa
BAP - S	i-wáa		yòó	i-yáa
DUA - C	wɔ́	sáwá	è-yòó	yáa
BID - C	wáa	sáwá	yòó	yáa
OLI - C	wáa	sáwá	yòó	yáa
PON - C	ij-ɔ́ɔ	sáwá	è-yòó	i-yáa
MON - C	i-wáa	sáwá	è-yòkó	i-yáa
MUL - C	i-wɔ́	sáwá	è-yòó	i-yáa
MBO - SW	li-wáa	sáwá	yòkó	li-jáa
KPE - SW	liŋm-áa	ʒáwá	yòkó	li-yáa
ISU - SW	i-wáa	sáwá	yòkó	i-yáa
KOL - SW	i-wáa	sáwá	è-yòkó	i-yáa
BUB - SW	li-wáa	sáwá	è-yòkó	li-yɛ́ɛ
Kundu - N	wa			

analysis in so far as he did not include Oroko (aka North) and so did not notice the proximity between the latter and Southwest.

In addition, the treatment of PSB *d_{12} (and *l_{12}) is paramount in Sawabantu languages for possibly establishing some putative chronology. With the three genetic subgroups distinguished (South, Central and Southwest+North), how PSB *d_{12} evolved in these languages is informative. For Southwest languages, PSB *d_{12} and *l_{12} have merged into /l/. This phonological merger defining Southwest and North languages must have occurred last because otherwise, all the PSB *d would be realized /l/ in the remaining subgroups. Instead, the distinction between PSB *d_{12} and *l_{12} is maintained in Central and South languages. The retention of PSB *d_1 in Central languages along with the conditioning observed in some Central languages for PSB *d_2 as opposed to the unconditioned sound change PSB *d_{12} > /ɗ/ in South languages confirms that we are dealing with two different subgroups.

Finally, the results observed are in accordance with the groupings distinguished in lexicon-based phylogenies with South and Central languages classi-

Table 24: Examples with /j/ in Sawabantu languages.

PSB	-	-	$^*y_1$
meaning	***ten***	***mirror***	***wash***
BAN - S	jóòm	jɛní	i-yoa
BAT - S	jóòm	jɛní	i-yoa
YAS - S	jóòm	jɛní	
BAP - S	jóòm	jɛní	i-yoa
DUA - C	ɗ-óòm	jɛnɛ́	jóa
BID - C	jóòm	jɛnɛ́	yoa
OLI - C	jóòm	jɛnɛ́	yoa
PON - C	ɗ-óòm	jɛnɛ́	i-joa
MON - C		jɛnɛ́	i-jóa
MUL - C	jóòm	jɛnɛ́	i-yoa
MBO - SW	dì-ómè		
KPE - SW	lì-yómè	li-ɛní	
ISU - SW		jɛní	i-yoa
KOL - SW	j̀ómì	jɛnɛ́	yoa
BUB - SW	dì-ómì	li-ɛɲí	
Kundu - N			

fied under a same node whereas Southwest and North are classified in a different subgroup linked to Manenguba languages (e.g., Figure 2).

3.3 Comparing Proto-Bantu and Proto-Sawabantu: new perspectives

In this section, we compared our reconstructions established for PSB with PB reconstructions in order to establish sound correspondences between the two proto-languages with a focus on plosive consonants in C1 and C2 positions. The aim is to bring a new perspective on the phenomena described in the previous section by studying Sawabantu languages in relation to Proto-Bantu.

The examination of Table 26 shows that PB *p yielded to two reflexes in PSB with *p and *v. Philippson 2022: 25-26 mentioned a possible partial conditioning *pVp that could trigger the strong reflexes of PB *p for NW Bantu languages which prevented him from positing double reflexes for PB *p. For PB *t, we have a case of double reflexes or MUR where PB *t > PSB *t or *l with no apparent conditioning. However, Philippson (2022: 30) did notice that the strong reflexes

Table 25: Reflexes of PM's plosives, fricatives and the approximant lateral in C1 position (Hedinger 1989: 38-39).

PM	$*p_1$	$*t_1$	$*k_1$	$*b_1$	$*d_1$	$*g_1$	$*s_1$	$*f_1$	$*l_1$
MBM	p	t	k	b	d	g	s	f / h	l / d
MBM	p	t	k	b	d	g	s / ʃ	f	l / d
MYE	p	t	k	b / bɦ	d	g	s	h	l
MBE	p	t	k	b	d	g	s / ʃ	h	l
ELU	p	t	k	b	d	g	s	h	l
NNE	p	t	k	b	d	g	s	h	l
AKO	p	t	k	b	d	g	s	h	l
MHE	p	t	k	b	d	g	s	h	l
MWK	p	t	k	b	d	g	s	h	l
MKA	p	t	k	b	d	g	s	h	l
BLN	p	t	k	b	d	g	s	h	l
BBO	p	t	k	b	d	g	s	h / f / v	y
LEF	p	t	k	b	d	g	s / ʃ	h	l / d
LEK	p	t	k	b / p	d	g	s	h	l / d

of PB *t tend to appear before high vowels in several NW Bantu languages. Our data shows that the reflexes of PSB *t and *l occur in the same phonological contexts confirming the status of double reflexes for PB *t with however one observation: the sequences /liC2V2/, /luC2V2/ or /C1V1li/, /C1V1lu/ are extremely rare in Sawabantu languages. As mentioned in the previous section, PB *k > Ø in Sawabantu languages but also in the remaining NW Bantu languages (A and some B20) which triggered PB *g > *k in the same languages (Grollemund & Philippson to appear). PB *b is retained in PSB. Finally, PB *d is deleted when followed by non-high vowels and retained (PSB *d) when followed by the vowels *i and *u, a phenomenon observed for many Bantu A languages (Philippson 2022: 30-31).

The comparison between PB and PSB shed a new light on the origin of the phoneme /l/ in Sawabantu languages. For Southwest languages, the reflexes /l/ originated from two different PB consonants: *t and *d (reconstructed *l and *d in PSB) which confirms why we had two distinct sound sets in Section 3.1.2. It also explains why the reflexes of PSB *d can only be found before high vowels as PB *d > Ø before non-high vowels. According to Philippson (2022: 34), the deletion of PB *d before non-high vowels occurred with an intermediate stage

Table 26: Comparison between PB and PSB for plosives in C1 and C2 positions.

PB	PSB	S	C	SW	N	Examples in PB (from BLR3)	PSB
						PB *pàpá "wing" (2407)	PSB *papa
	*p	p	p	ɸ	ɸ		
*p						PB *pácà "twin" (2348)	PSB *vasa
	*v	v	w ~v ~f	Ø ~w	Ø		
						PB *cʊ́pà "calabash" (746)	PSB *sova
						PB *bàdí "two" (9093)	PSB *baa
*b	*b	ɓ	b	β	b ~β		
						PB *tʊ́ʊ́bá "six" (3034)	PSB *toba
						PB *tʊ́ì "ear" (3030)	PSB *toi
	*t	t	t	t	t		
						PB *dùt "pull" (1267)	PSB *dut
*t							
						PB *téndé "palm tree" (2849)	PSB *lɛndɛ
	*l	l	l	l ~r	r		
						PB *tátʊ̀ "three" (2811)	PSB *lalo
						PB *dʊ́mè "husband" (1183)	PSB *ome
	Ø	Ø	Ø	Ø	Ø	PB *dànd "to buy" (849)	PSB *and
						PB *búdà "rain" (368)	PSB *bua
*d							
						PB *dìbà "water" (1025)	PSB *diba
	*d	ɗ	d ~l ~ɗ	l ~r	r	PB *dùt "pull" (1267)	PSB *dut
						PB *kúdù "tortoise" (2110)	PSB *kudu
						PB *kádí "wife" (1674)	PSB *adi
*k	Ø	Ø	Ø	Ø	Ø		
						PB *Nkákà "pangolin" (1684)	PSB *ka
						PB *gʊ̀bò "skin" (1479)	PSB *kobo
						PB *gòngá "spear" (1448)	PSB *kɔngɔ
*g	*k	h ~k	Ø ~k	Ø ~k	k		
						PB *jògù "elephant" (1607)	PSB *yɔku
						PB *jíg "learn" (3338)	PSB *yok

attested in some NW Bantu languages with PB *d > /l/ > Ø (whereas PB *d > /d/ or /ɗ/ before high vowels). We found this intermediate stage in some Central Sawabantu languages where we have observed for C2 PSB *d > /l/ before non-high vowels confirming this lenition chain which implied that PB $*d_2$ > /l/ or /ɗ/ or Ø before non-high vowels.

In addition, according to Philippson (2022: 37), the sound change PB *d > (*l) > Ø must have occurred before PB *t > PSB *l otherwise, most of the /l/ reflexes of PSB *l would have been deleted. Therefore, it seems that the loss of PB *d (> weak reflexes) triggered PB *t > PSB *l (as well as in some other NW Bantu languages, see Philippson 2022: 30-31) with the result of reintroducing the missing sound /l/. Interestingly, in Sawabantu languages, the reflexes of PSB *l are found in front of non-high vowels which could corroborate that the sound change from PB *t > PSB *l occurred to fill the void created by the rule PB *d > *l > Ø in the same context (but we need to recall that PB *t > PSB *t in all contexts so we do not have an explanation to why some words with PB *t > PSB *l before non-high vowels). Moreover, we can note that some languages such as Mokpe or Bubia present PSB *l > Ø or /l/ which could also indicate that the process of deleting PSB *l is continuing.

The same phenomenon is observed with PB *k. When PB *k is deleted, Sawabantu languages (as well as the remaining NW Bantu languages A and some B20) reintroduced the missing sound with the sound change PB *g > PSB *k. Interestingly, PSB *k is undergoing lenition again in Sawabantu languages.

In conclusion, the examination of Table 26 highlights interesting sound shifts from PB to Sawabantu languages with strategies of replacement for sounds that have been deleted.

4 Conclusion

Sawabantu languages are fascinating for several reasons highlighted in this study. First, the study and reconstruction of their consonants evidences some interesting phenomena occurring at the PSB level with some PSB consonants undergoing lenition (PSB *p > /ɸ/, *k > Ø, *b > /β/ or /w/, *d > /l/). The study of PSB sound changes divided the Sawabantu languages into three subgroups: South, Central and Southwest-North. At the PB level, some interesting sound changes are observed with PB *t > PSB *t and *l, PB *k > Ø, PB *g > PSB *k and PB *d > Ø when followed by non-high vowels confirming that languages spoken in this particular region underwent many sound changes that resulted in the modification of the shape of their lexicon in comparison with PB (e.g., PB *kútà

BLR3 2138 'oil' > PSB *ula but [ma-kuta] in F31 Nilamba (data from Yukawa 1989) - a Bantu language spoken in Tanzania, PB *dʊ̃k BLR3 1179 "vomit" > PSB *o but [kwi-lʊka] in F31 Nilamba). Finally, this study confirms the different status of Manenguba languages (A15) that underwent different sound changes in C1 and C2 positions which led Nurse & Philippson (2003) to exclude these languages from the Sawabantu clade.

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Chapter 8

Polar and *wh-* question formation in Ekhwa Adara

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This paper presents the first description of polar and *wh-* interrogatives in Ekhwa Adara. In Ekhwa Adara, polar questions are optionally marked by sentence-initial Q particles and involve final-vowel lengthening and L% boundary tones, while *wh-* question formation involves optional *wh-* movement where the *wh-* item obligatorily precedes a predetermined focus particle. Long-distance *wh-* movement is possible and requires pronominal resumption. Partial *wh-* movement is also possible but seemingly does not require pronominal resumption. Indirect questions are formed via relativization, except when embedded under 'ask'. Multiple *wh-* questions exhibit an absence of superiority effects and multiple *wh-* fronting might be available, albeit possibly constrained by an anti-superiority condition.

1 Introduction

Adara (ISO 639-3 [KAD]), also known as Eda, Edra, and Kadara, is an under-documented Benue-Congo language spoken by approximately 300,000 (Hon et al. 2018) to 500,000 people (Simons & Fennig 2018) in Kaduna and Niger states in Nigeria, as highlighted in Figure 1. Figure 2 provides a more detailed map of the Adara-speaking area of Nigeria taken from Hon et al. (2018), specifying the locations of the various Adara dialects.

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Figure 1: Map of Nigeria showing the Adara-speaking region

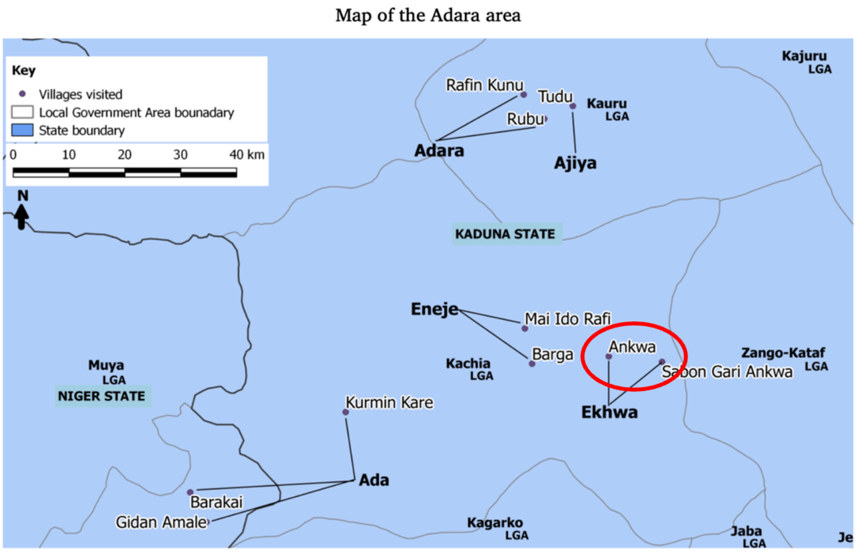

Figure 2: Map of the Adara-speaking area and its dialects (Hon et al. 2018)

Despite its large number of speakers, very little research has been done on Adara. This paper presents the first description of question formation in the language. Specifically, we investigate polar and *wh-* questions in Ekhwa Adara[1], the least researched of Adara's five dialects. Overall, we have found that polar questions in Ekhwa Adara involve final-vowel lengthening + L% boundary tones and that *wh-* movement in the language is optional. Where *wh-* movement does occur, the ex-situ *wh-* item obligatorily precedes a corresponding focus marker whose choice is determined by the animacy (human vs. non-human) and grammatical function (argument vs. adjunct) of the *wh-* item. Furthermore, both long-distance and partial *wh-* movement is possible in Ekhwa Adara. Lastly, and perhaps most remarkably, due to its (to the best of our knowledge) unattested status in the grammars of African languages, Ekhwa Adara may allow for multiple *wh-* fronting, but is possibly constrained by an anti-superiority condition.

This article is structured in the following way. In Section 2, we describe the formation of polar questions in Ekhwa Adara. Section 3 details our findings on *wh-* questions in Ekhwa Adara, including *wh-* in-situ phenomena, *wh-* movement (including partial and long-distance), embedded question formation, and multiple *wh-* questions. We conclude in Section 4 with a summary of our key findings and questions for future research.

2 Polar questions

Ekhwa Adara's primary polar question formation strategy is intonational in nature. Matrix polar questions involve final-vowel lengthening and L% tones, as is characteristic of the languages of the Sudanic belt region (Rialland 2007a, 2009, Cahill 2012, 2015). We see that the declarative sentences in (1a), (2a), and (3a) below all contain a short final vowel which is lengthened in their polar interrogative counterparts ((1b), (2b), and (3b), respectively).[2]

(1) a. ijâ sù kɹɔ́
Ija PROG cry
'Ija is crying.'

[1] The data and judgments presented in this paper come exclusively from fieldwork with the third author, a native speaker of Ekhwa Adara, in the context of a Field Methods class at the CUNY Graduate Center in Fall 2022. Data are presented in IPA. The following diacritics are used to mark surface tone: V́ = high, V̀ = low, V = mid; V̂ = falling, V̌ = rising.

[2] In response to one reviewer, we are currently unaware of any other prosodic differences between declaratives and interrogatives in Ekhwa Adara.

b. ijâ sù kɹɔ̂ː
Ija PROG cry.Q
'Is Ija crying?'

(2) a. omúsé ku ɹɔ́ utébur
Omuse PFV buy table
'Omuse bought a table.'

b. omúsé ku ɹɔ́ utébûːr
Omuse PFV buy table.Q
'Did Omuse buy a table?'

(3) a. omúsé ku kɹɔ́ oɹaì
Omuse PFV cry yesterday
'Omuse cried yesterday.'

b. omúsé ku kɹɔ́ oɹaìː
Omuse PFV cry yesterday.Q
'Did Omuse cry yesterday?'

Furthermore, we see that intonational phrase-final high and mid tones, such as in (1a) *kɹɔ́* 'cry' and (2a) *utébur* 'table', are both realized with falling pitch movements in their polar interrogative counterparts. In (1b), the phrase-final syllable *kɹɔ̂ː* exhibits a falling tone and similarly in (2b), the final syllable of *utébûːr* is also realized with a falling tone. On the other hand, intonational phrase-final L tones such as in (3a) *oɹaì* 'yesterday' are still realized as L in their polar interrogative counterparts, as seen in (3b). Crucially, they are not realized with falling pitch movements. We also see that the sentences in (1) make use of the progressive aspect, whereas the sentences in (2) make use of the perfective aspect. The data therefore indicate that final-vowel lengthening and L% tone insertion are not tense/aspect dependent. Additionally, given that the final vowel quality in each example differs, we conclude that polar questions are formed by true vowel lengthening and not simply the addition of a uniform vowel, which occurs in some languages of the region (Rialland 2007b, Cahill 2015).

Syntactically, polar questions may be marked by a clause-initial Q particle *kó*, which seems to be borrowed from Hausa (Bawa 2023). While this Q particle is optional in matrix polar questions (4a) and *wh*- in-situ constructions (4b), it is obligatory in embedded polar questions (4c).

(4) a. (kó) ijâ sù kɹɔ̂ː
Q Ija PROG cry.Q
'Is Ija crying?'

b. (kó) ijâ ku ɹɔ́ ɪncí
Q Ija PFV buy what
'What did Ija buy?'

c. omúsé ku ɹɔ́:ru *(kó) ijâ ku kɹɔ́
Omuse PFV ask Q Ija PFV cry
'Omuse asked if Ija cried.'

From (4c), we also see that in embedded polar questions, where the Q particle obligatorily appears in the embedded clause to mark the scope of the indirect question, there is no final-vowel lengthening or L% boundary tone. Thus, final-vowel lengthening and insertion of L% boundary tones represent main clause phenomena in the language, as in Ikpana (Kandybowicz et al. 2023).

3 *Wh-* questions

3.1 *Wh-* in-situ

All argument *wh-* items may appear in-situ in root clauses.[3] In (5), we observe the *wh-* item appearing in subject position (5a), direct object position (5b, 5d, 5e), and indirect object position (5c).[4]

(5) a. **iwé** sú ɹɔ egb͡bé
who FUT buy house
'Who will buy a house?'

b. ijâ ku ɹi **ɪncí/iwé**
Ija PFV see what/who
'What/who did Ija see?'

c. ijâ ku dʒe **iwé** onsɛ̌
Ija PFV give who name
'Who did Ija give a name to?'

d. ijâ ku dʒe awɛ́-ǹ **ɪncí**
Ija PFV give child-DEF what
'What did Ija give the child?'

[3]Though as discussed below, it is not clear whether focused subjects are in-situ or ex-situ at this stage of our research.

[4]From (5c, 5d), we observe that in Ekhwa Adara double object constructions, the indirect object immediately precedes the direct object.

e. ijâ ku ɹɔ́ **ocɪná** eg͡bé
Ija PFV buy which house
'Which house did Ija buy?'

As previously discussed in Section 2, the Q particle *kó* may optionally appear in root clause *wh-* in-situ questions. Currently, we only have evidence of its (optional) appearance in object root clause *wh-* in-situ questions such as (6a, 6b). We lack the data to determine whether Q may also appear in non-object root *wh-* in-situ constructions, but we suspect this to be the case.[5] Note that in (6b), we observe a focused *wh-* subject after *kó*. This, however, should not be considered an instance where *kó* is marking an ex-situ *wh-* subject question, because Q-marking does not occur in ex-situ *wh-* questions (note the absence of the Q particle in the data presented in Section 3.2). Rather, in this sentence, *kó* is marking the in-situ *wh-* object *ɪncí* 'what'. Given our consultant's emphasis on beginning an interrogative sentence with some indication of intent (either through the Q particle *kó* or an initial focused *wh-* phrase), the optional *kó* marker most likely signals that there is (at least) one in-situ interrogative expression following the subject.

(6) a. (kó) ijâ ku ɹɔ́ ɪncí
Q Ija PFV buy what
'What did Ija buy?'

b. (kó) iwé ŋu ku ɹɔ́ ɪncí
Q who FOC PFV buy what
'Who bought what?'

All adjunct *wh-* items may also appear in-situ in root clauses. This is demonstrated in (7) for a variety of adjuncts (locative (7a), temporal (7b), manner/quantity (7c) and reason (7d)).

(7) a. ijâ ku ɹɔ́ eg͡bé ɪmbí
Ija PFV buy house where
'Where did Ija buy a house?'

b. ijâ ku ɹɔ́ eg͡bé ocɪná atú/apa
Ija PFV buy house which day/time
'When (i.e., which day/time) did Ija buy a house?'

[5] Our consultant seems to prefer either ex-situ or focus marked in-situ *wh-* subject questions, at least in root contexts. This may explain why the Q particle is limited to *wh-* object questions in our dataset.

c. a sù nɛ otúmá-ŋ níní
3SG.NOM PROG do work-DEF how
'How is s/he doing the work?'
'For how much (money) is s/he doing the work?'

d. ijâ ku ɹɔ́ eg͡bé-ŋ domín ɪncí
Ija PFV buy house-DEF reason what
'Why did Ija buy the house?'

Long-distance *wh-* in-situ is also attested. All *wh-* items may appear in-situ in embedded clausal complements and take wide scope over the matrix clause. We observe this in (8), where the *wh-* items *iwé* 'who' (8a) and *ɪncí* 'what' (8b) appear as the subject and object of the embedded clause, respectively. Crucially, these sentences cannot receive indirect question interpretations.[6] Instead, the sentences are interpreted as long-distance in-situ interrogatives, with the *wh-* item in the complement clause taking matrix scope.[7]

(8) a. omúsé ga iwé ku ɹɔ́ eg͡bé
Omuse say who PFV buy house
'Who did Omuse say bought a house?'
NOT: 'Omuse said who bought a house.'

b. omúsé ga ijâ ku ɹɔ́ ɪncí
Omuse say Ija PFV buy what
'What did Omuse say Ija bought?'
NOT: 'Omuse said what Ija bought.'

Embedded in-situ adjunct *wh-* expressions may only be interpreted as originating in the embedded clause. We demonstrate this in (9), where the *wh-* items 'where,' 'when,' 'how much,' and 'why' can only be interpreted as being generated in the embedded clause, that is, modifying the verb *ɹɔ́* 'buy', as opposed to the matrix verb *ga* 'say'.

(9) a. omúsé ga ijâ ku ɹɔ́ eg͡bé ɪmbí
Omuse say Ija PFV buy house where
'Where did Omuse say Ija bought a house?'

[6] As discussed in (23) below, this would require relativization.

[7] The Q particle *kó* is unavailable in this construction to mark the scope of the *wh-* element. In fact, with the Q particle *kó* in sentence initial position, the *wh-* questions in (8) become quotative polar questions, expressing 'Did Omuse say: "Who bought a house?"?' and 'Did Omuse say: "What did Ija buy?"?' respectively.

✓'Where-buy'
* 'Where-say'

b. omúsé ga ijâ ku ɹɔ́ eg͡bé ocɪná apa
Omuse say Ija PFV buy house which time
'When did Omuse say Ija bought a house?'
✓'When-buy'
* 'When-say'

c. omúsé ga ijâ ku ɹɔ́ eg͡bé níní
Omuse say Ija PFV buy house how
'For how much did Omuse say Ija bought a house?'
✓'For how much-buy'
* 'For how much-say'

d. omúsé ga ijâ ku ɹɔ́ eg͡bé domín ɪncí
Omuse say Ija PFV buy house reason what
'Why did Omuse say Ija bought a house?'
✓'Why-buy'
* 'Why-say'

3.2 *Wh-* movement

3.2.1 Focus particles

Ekhwa Adara is an optional *wh-* movement language. As we saw in Section 3.1, all *wh-* arguments and adjuncts may remain in-situ with or without the initial Q particle *kó*. The interrogative phrase, however, may also be fronted to clause-initial position. This optional *wh-* movement requires a constituent focus particle, which immediately follows the *wh-* item and varies according to the animacy and type of the interrogative expression. When undergoing movement, we see that the human interrogative expression *iwé* 'who' must be followed by the particle *ŋu* (10a)[8], the non-human interrogative expression *ɪncí* 'what' must be followed by the particle *mo* (10b), and finally, the D-linked interrogative *ocɪná* (10c) 'which' and interrogative adjuncts such as *ɪmbí* 'where' (10d), *ocɪná apa* 'when' (10e), and *níní* 'how/for how much' (10f) must be followed by the particle *ku*. The only

[8]Unlike with long-distance *wh-* movement, there is no direct evidence in cases of monoclausal *wh-* questions that the subject is moved at all, since *wh-* items may be accompanied by a focus marker even when left in-situ, as shown in (16). There also seems to be a preference for the focus-marked construction when the interrogative phrase is the subject, while the same is not true for other arguments. Our point, in presenting (10a), is merely that, if moved, the *wh-* subject should behave like non-subject *wh-* items and require the focus particle.

exception is the *wh-* adjunct *domín ɪncí* 'why' (literally, 'what reason') which may either precede *ku*, like all other *wh-* adjuncts, or *mo*, like *ɪncí* 'what' (10g).

(10) a. iwé *(ŋu) __ ku ɹɔ́ eg͡bé
who FOC PFV buy house
'Who bought a house?'

b. ɪncí *(mo) ijâ ku ɹɔ́ __
what FOC Ija PFV buy
'What did Ija buy?'

c. ocɪná egbé *(ku) ijâ ku ɹɔ́ __
which house FOC Ija PFV buy
'Which house did Ija buy?'

d. ɪmbí *(ku) ijâ ku ɹɔ́ egbé __
what FOC Ija PFV buy house
'Where did Ija buy a house?'

e. ocɪná apa *(ku) ijâ ku ɹɔ́ egbé __
which time FOC Ija PFV buy house
'When did Ija buy a house?'

f. níní *(ku) a sù nɛ otúmá-ŋ __
how FOC 3SG.NOM PROG do work-DEF
'How/for how much is s/he doing the work?'

g. domín ɪncí *(ku/mo) ijâ ku ɹɔ́ eg͡bé __
reason what FOC Ija PFV buy house
'Why did Ija buy a house?'

The ungrammaticality of the examples in (11), in which the focus particles that follow the *wh-* phrase are switched, demonstrate that these markers are compatible only with specific *wh-* items – human *iwé* 'who' must be paired with *ŋu*, rather than *mo* or *ku*; non-human *ɪncí* 'what' must be paired with *mo*, rather than *ŋu* or *ku*; lastly, the D-linked interrogative *ocɪná* 'which' and *wh-* adjuncts, with the exception of *domín ɪncí* 'why', must be paired with *ku*, rather than *ŋu* or *mo*.

(11) a. * iwé mo/ku __ ku ɹɔ́ eg͡bé
who FOC PFV buy house
Intended: 'Who bought a house?'

b. * ɪncí ŋu/ku ijâ ku ɹɔ́ __
what FOC Ija PFV buy
Intended: 'What did Ija buy?'

c. * ocɪná egbé ŋu/mo ijâ ku ɹɔ́ __
which house FOC Ija PFV buy
Intended: 'Which house did Ija buy?'

d. * ɪmbí ŋu/mo ijâ ku ɹɔ́ egbé __
what FOC Ija PFV buy house
Intended: 'Where did Ija buy a house?'

e. * ocɪná apa ŋu/mo ijâ ku ɹɔ́ egbé __
which time FOC Ija PFV buy house
Intended: 'When did Ija buy a house?'

f. * níní ŋu/mo a sù nɛ otúmá-ŋ __
how FOC 3SG.NOM PROG do work-DEF
Intended: 'How/for how much is s/he doing the work?'

g. * domín ɪncí ŋu ijâ ku ɹɔ́ eg͡bé __
reason what FOC Ija PFV buy house
Intended: 'Why did Ija buy a house?'

Furthermore, the examples in (12) show that the choice of *ŋu* or *mo* is truly dependent on the interrogative item itself and not on its syntactic relationship to the verb (e.g., case). As we can see, ex-situ *iwé* still requires *ŋu* when thematically linked to non-subject positions like direct objects (12a) and possessors of objects (12b). Likewise, ex-situ *ɪncí* selects the marker *mo* even when linked to the subject position (12c).

(12) a. iwé ŋu ijâ ku dá __
who FOC Ija PFV touch
'Who has Ija touched?'

b. iwé ŋu ijâ ku ɹɔ́ eg͡bé-ŋ __
who FOC Ija PFV buy house-DEF
'Who did Ija buy a house from?'

c. ɪncí mo __ ku dá ijâ
what FOC PFV touch Ija
'What has touched Ija?'

In addition to appearing in ex-situ *wh-* questions, these constituent focus particles may occur in answer fragments. Here, we see that (13b) is a grammatical response to (13a).

(13) a. iwé ŋu mín
who FOC here
'Who is this?'

b. omúsé ŋu
Omuse FOC
'(This is) Omuse.'

Both *ŋu* and *ku*[9] also appear as focus particles in declarative sentences with focus fronting. Like in ex-situ interrogative constructions, focus markers follow left peripheral foci and the realization of the surfacing focus particle depends on the constituent it focus-marks. Consider the facts in (14) and (15), which illustrate.[10]

(14) a. ijâ ŋu/*ku ɹɔ́ eg͡bé
Ija FOC buy house
'IJA bought a house.'

b. ijâ ŋu/*ku omúsé sú mani __ ɔnʃó
Ija FOC Omuse FUT teach song
'Omuse will teach a song (to) IJA.'

[9]The constituent focus particles *ŋu* and *ku* also occur in allomorphic variation with *ŋo* and *ko*, respectively. Further research is necessary to see whether this variation is phonologically conditioned. Note that the constituent focus particle variant *ko*, which bears a mid tone, is tonally (not segmentally) contrastive with the high tone-bearing initial interrogative particle *kó*.

[10]We initially glossed (14-15) with English *it*-cleft translations (e.g., (14a) was glossed as 'It is IJA who bought a house'). Thanks to one of our reviewers, however, we realized that these translations might not be semantically appropriate given that *it*-cleft sentences in English have exhaustive properties which the sentences in (14-15) do not possess. For example, our consultant notes that in the case where Omuse will teach both Ija and Awen a song, (14b) can only be used if the hearer is unaware of Awen's existence. We assume this does not constitute an exhaustive interpretation since Awen's existence is not part of the common ground shared by both speakers. Thus, Awen is not in the domain of individuals whose "learning of the song from Omuse" we need to negate. An exhaustive interpretation of (14b) is only available when both Ija and Awen are relevant to the context (i.e., in the common ground) but only Ija will be taught the song (possibly among other students who, similar to Awen in the previous case, are not part of the common ground and thus, the alternative propositions involving them need not be negated).

c. ená ŋu/??ku omúsé sú mani ___ ɔnʃó
cow FOC Omuse FUT teach song
'Omuse will teach a song to a COW.'

(15) a. ɔnʃó ku/*ŋu omúsé sú mani-ŋ ijâ ___
song FOC Omuse FUT teach-3SG.ACC Ija
'Omuse will teach Ija a SONG.'

b. eg͡bé ku/*ŋu/*mo ijâ ku ɹɔ́ ___
house FOC Ija PFV buy
'Ija bought a HOUSE.'

The data in (14) and (15) suggest that the choice of focus marker in declarative sentences is related to animacy – focused animate DPs appear with *ŋu* (14), while focused inanimates are accompanied by *ku* (15). As before with ex-situ *wh-* questions, the choice of either *ŋu* or *ku* in declarative sentences does not seem to depend on the relationship of the moved phrase to other elements of the sentence (e.g., case, thematic role, etc.).

The focus markers introduced in this section are not limited to contexts in which an interrogative undergoes movement. *Wh-* items that remain in-situ may optionally precede these same particles (16).

(16) a. ijâ ku dʒe iwé (ŋu) onsɛ̆
Ija PFV give who FOC name
'Who did Ija give a name to?'

b. ijâ ku ɹi ɪncí (mo)
Ija PFV see what FOC
'What did Ija see?'

c. ijâ ku ɹɔ́ eg͡bé ɪmbí (ku)
Ija PFV buy house where FOC
'Where did Ija buy a house?'

d. ijâ ku ɹɔ́ eg͡bé ocɪná apa (ku)
Ija PFV buy house which time FOC
'When did Ija buy a house?'

e. a sù nɛ otúmá-ŋ níní (ku)
3SG.NOM PROG do work-DEF how FOC
'How/for how much is s/he doing the work?'

f. ijâ ku ɹɔ́ eg͡bé-ŋ domín ɪncí (ku/mo)
Ija PFV buy house-DEF reason what FOC
'Why did Ija buy the house?'

A list of Ekhwa Adara *wh-* items and the focus particles they are compatible with is presented in Table 1 below.

Table 1: Ekhwa Adara *wh-* items and their accompanying focus particles

Wh- item	Accompanying focus particle
iwé 'who'	*ŋu*
ɪncí 'what'	*mo*
ɪmbí 'where'	*ku*
ocɪná 'which'	*ku*
ocɪná atú/apa 'when'	*ku*
níní 'how'	*ku*
domín ɪncí 'why'	*ku/mo*

3.2.2 Long-distance movement

Long-distance *wh-* movement in Ekhwa Adara exhibits the same dependency between the moved phrase and its accompanying focus marker. Additionally, however, we see an asymmetry between subjects and non-subjects. The subject of an embedded clause, when fronted to sentence-initial position, must be resumed by an agreeing pronominal (17a), while resumption of displaced embedded objects is ungrammatical (17b).[11] This is typical of languages in the region, such as Nupe (Kandybowicz 2004, 2008) and Igbo (Amaechi & Georgi 2019).

(17) a. iwé ŋu omúsé ga *(a) ku ɹɔ́ eg͡bé
who FOC Omuse say 3SG.NOM PFV buy house
'Who did Omuse say bought a house?'

b. ɪncí mo omúsé ga ijâ ku ɹɔ́ (*ŋ)
what FOC Omuse say Ija PFV buy 3SG.ACC
'What did Omuse say Ija bought?'

[11]We did not observe resumption in the indirect object or adjunct positions in long-distance *wh-*movement. Furthermore, to the best of our knowledge, the language does not have any null resumptive forms. That is, it only has overt pronominals.

Examples (18a)-(18c) demonstrate that movement of a *wh-* adjunct in biclausal questions does not require pronominal resumption or any other indication of the position in which the *wh-* adjunct was generated. Accordingly, the sentences may receive either matrix or embedded interpretations, questioning the saying event or the buying event, respectively. This is in contrast to examples containing in-situ *wh-* adjuncts, as well as in-situ *wh-* arguments. As we saw earlier in (9), long-distance (i.e., embedded) *wh-* in-situ with adjuncts forces embedded interpretations, meaning that the *wh-* phrase can only be interpreted as modifying the lower clause verb, whereas in (8), long-distance *wh-* in-situ with arguments forces matrix interpretations, meaning that such sentences must be interpreted as long-distance in-situ *wh-* interrogatives as opposed to indirect questions.

(18) a. ɪmbí ku omúsé ga ijâ ku ɹɔ́ eg͡bé
where FOC Omuse say Ija PFV buy house
'Where did Omuse say Ija bought a house?'
✓'Where-buy'
✓'Where-say'

b. ocɪná apa ku omúsé ga ijâ ku ɹɔ́ eg͡bé
which time FOC Omuse say Ija PFV buy house
'When did Omuse say Ija bought a house?'
✓'When-buy'
✓'When-say'

c. domín ɪncí ku/mo omúsé ga ijâ ku ɹɔ́ eg͡bé
reason what FOC Omuse say Ija PFV buy house
'Why did Omuse say Ija bought a house?'
✓'Why-buy'
✓'Why-say'

3.2.3 Partial movement

Ekhwa Adara also allows partial *wh-* movement, that is, movement of an embedded interrogative to the initial position of a lower clause. When partially moved, *wh-* is accompanied by its corresponding focus particle (as discussed in Section 3.2.1). Unlike in long-distance *wh-* questions, partially moved subjects in Ekhwa Adara do not seem to require pronominal resumption (19a), erasing the subject–non-subject asymmetry that was exhibited in (17a, 17b) above.

(19) a. omúsé ga iwé ŋu __ ku ɹɔ́ eg͡bé
Omuse say who FOC PFV buy house
'Who did Omuse say bought a house?'

b. omúsé ga ɪncí mo ijâ ku ɹɔ́ __
Omuse say what FOC Ija PFV buy
'What did Omuse say Ija bought?'

c. omúsé ga ɪmbí ku ijâ ku ɹɔ́ eg͡bé __
Omuse say where FOC Ija PFV buy house
'Where did Omuse say Ija bought a house?'
✓'Where-buy'
* 'Where-say'

d. omúsé ga ocɪná apa ku ijâ ku ɹɔ́ eg͡bé __
Omuse say which time FOC Ija PFV buy house
'When did Omuse say Ija bought a house?'
✓'When-buy'
* 'When-say'

e. omúsé ga níní ku ijâ ku ɹɔ́ eg͡bé __
Omuse say how FOC Ija PFV buy house
'For how much did Omuse say Ija bought a house?'
✓'How much-buy'
* 'How much-say'

f. omúsé ga domín ɪncí mo ijâ ku ɹɔ́ eg͡bé __
Omuse say reason what FOC Ija PFV buy house
'Why did Omuse say Ija bought a house?'
✓'Why-buy'
* 'Why-say'

As with embedded in-situ questions, partial movement of the *wh-* adjuncts in (19c)–(19f) can only be interpreted as originating in the immediately-containing clause (at least in biclausal contexts). Questions formed in this way do not require long-distance movement to sentence-initial position to mark scope, nor do they require (or allow) the clause-initial Q marker *kó*. We can thus classify Ekhwa Adara as a Simple/Naked partial *wh-* movement language, referencing Fanselow 2006's typology. In this way, partial *wh-* movement in Ekhwa Adara resembles partial *wh-* fronting in other West African languages such as Krachi (Torrence & Kandybowicz 2013, 2015, Kandybowicz & Torrence 2015); Bono and Wasa (Kandybowicz 2017, 2020, Kandybowicz & Torrence 2015); Avatime (Devlin et al. 2021); and Ikpana (Kandybowicz et al. 2021, 2023).

Interestingly, there is seemingly a limit on how far a *wh-* item can move in partial movement structures. In (20), where there are two levels of embedding, we see that *wh-* items can only either partially move within the originating clause (20a) without the resumptive pronoun or move long-distance to the matrix operator position (20b) with the resumptive pronoun. They cannot move to any intermediate position. That is, a *wh-* item originating in the lowest embedded clause cannot move to the left periphery of an intermediate clause, with or without a resumptive pronoun (20c).

(20) a. awɛ̂n ga (aɪ) ijâ ga-ŋ (aɪ) iwé ŋu __ ku ɹɔ́ eg͡be
Awen say COMP Ija say-3SG.ACC COMP who FOC PFV buy house
'Who did Awen say that Ija said bought a house?'
Lit: 'Who did Awen say that Ija told her (Awen) bought a house?'

b. iwé ŋu awɛ̂n ga (aɪ) ijâ ga-ŋ (aɪ) a ku ɹɔ́ eg͡be
who FOC Awen say COMP Ija say-3SG.ACC COMP 3SG.NOM PFV buy house
'Who did Awen say that Ija said bought a house?'
Lit: 'Who did Awen say that Ija told her (Awen) bought a house?'

c. *awɛ̂n ga (aɪ) iwé ŋu ijâ ga-ŋ (aɪ) (a) ku ɹɔ́ eg͡be
Awen say COMP who FOC Ija say-3SG.ACC COMP 3SG.NOM PFV buy house
Intended: 'Who did Awen say that Ija said bought a house?'

3.3 Embedded questions

Wh- movement to the left periphery of the lower clause also occurs in the formation of embedded questions under the matrix verb 'ask' (21).[12] This construction is identical to the partial *wh-* movement construction described above – both subjects and objects are moved without use of a resumptive pronoun and the clause-initial Q particle is absent. The scope of the *wh-* item in this case is limited to the lower clause due (presumably) to the semantic properties of 'ask.'

[12]The lower clauses in (21) are indeed embedded questions under the matrix verb 'ask', rather than, as per a reviewer's concern, quotations. In (i), the embedded first-person object refers to the speaker, as expected of an embedded question interpretation, rather than the matrix subject Omuse, which is what we would expect of a quotative interpretation.

(i) omúsé$_i$ ku ɹɔ́:ru iwé ŋu __ ri-i$_{SP/*i}$
Omuse PFV ask who FOC see-1SG.ACC
'Omuse$_i$ asked who saw me$_{SP/*i}$'

(21) a. omúsé ku ɹɔ́:ru iwé ŋu __ ku ɹɔ́ eg͡bé
Omuse PFV ask who FOC PFV buy house
'Omuse asked who bought a house.'

b. omúsé ku ɹɔ́:ru ɪncí mo ijâ ku ɹɔ́ __
Omuse PFV ask what FOC Ija PFV buy
'Omuse asked what Ija bought.'

Indirect questions with 'ask' do not, however, require *wh-* movement in the embedded clause. (22) shows an example of an embedded *wh-* in-situ question under 'ask'.

(22) omúsé ku ɹɔ́:ru-i ijâ ku rɔ́ ɪncí mo
Omuse PFV ask-1SG Ija PFV buy what FOC
'Omuse asked me what Ija bought.'[13]

Indirect question complements of other embedding verbs are formed via relativization, generating what can be described as "concealed questions". The sentence in (23a) is nearly identical to that in (21a), apart from the matrix verb and aspect marking[14]; however, only (21a) is grammatical. Local fronting of the *wh-*item within the lower clause does not achieve the intended reading in (23a) – i.e., it is not interpreted as an indirect question.[15] In order to attain the intended reading, it is necessary to use a relative clause construction. In the case of an embedded 'who' question, the relative clause is headed by *anú-ǹ* 'the person' (23b), replacing the *wh-* phrase *iwé ŋu* 'who'. In the case of an indirect 'what' question, the relative clause is headed by *ɔŋgă-ŋ̀* 'the thing' (23c) replacing *ɪncí mo* 'what'.[16]

[13]Our consultant notes that in (22), without the matrix first-person object *i*, the speaker would expect an answer (except in certain contexts), which defeats the purpose of the reported speech we are aiming to document.

[14]In many cases, we found a preference for certain sentences to be produced without overt aspect marking, even if the corresponding sentence with a marker was judged to be grammatical. Absence of the aspectual element in (23a), therefore, would not be expected to negatively affect its acceptability. If anything, we might expect it to improve it.

[15]Currently, we are unsure whether (23a) can be parsed as an instance of partial *wh-* movement, with the moved phrase taking scope over the matrix clause. Given that 'know' can embed a clause (see (i) in footnote 15 below), if the partial *wh-* parse is not available, this may suggest that 'know' predicates in Ekhwa Adara can form factive islands.

[16]Although the verb 'know' is followed by a noun phrase in every example in (23), we do not analyze *k͡péɹi* 'know' as an acquaintance verb similar to French *connaitre* because it can be followed by a clause, as seen in (i).

(i) omúsé k͡péɹi aɪ ijâ ɹɔ́ egbé
Omuse know COMP Ija buy house
'Omuse knows that Ija bought a house.'

As indicated below, these structures translate more literally as ‘the person/thing that…’. Subject and object relative clauses of this type are both generally grammatical in Ekhwa Adara, using this exact construction.

(23) a. * omúsé k͡pé.ɹi iwé ŋu ku ɹɔ́ eg͡bé
Omuse know who FOC PFV buy house
Intended: ‘Omuse knows who bought a house.’

b. omúsé k͡pé.ɹi anú-ǹ da ku ɹɔ́ eg͡bé
Omuse know person-DEF REL PFV buy house
‘Omuse knows who bought a house.’
Literally: ‘Omuse knows the person who bought a house.’

c. omúsé k͡pé.ɹi ɔŋgǎ-ŋ̀ da ijâ ku ɹɔ́
Omuse know thing-DEF REL Ija PFV buy
‘Omuse knows what Ija bought.’
Literally: ‘Omuse knows the thing that Ija bought.’

This limitation of true embedded questions to ‘ask’ complement clauses connects Ekhwa Adara to Krachi (Torrence & Kandybowicz 2015), where identical facts obtain.

3.4 Multiple *wh-* questions

At present, we have observed three potential strategies for the formation of multiple *wh-* questions in Ekhwa Adara: (i) all *wh-* items remain in-situ; (ii) one *wh-* item is fronted to clause-initial position, while the other(s)[17] remain(s) in-situ; and (iii) all *wh-* items are fronted (i.e., multiple *wh-* fronting). As expected, focus particles appearing in each of these three construction types must be the ones selected by the accompanying *wh-* item. We are certain that strategies (i) and (ii) exist in the grammar. We are less certain about the reality of strategy (iii) given unstable judgments observed over a one-year period, which we discuss below.

Beginning with strategy (i), example (24) demonstrates that multiple *wh-* questions may be formed simply by leaving all interrogative expressions in-situ.

(24) (*kó) omúsé sú mani iwé (ŋu) ɪncí (mo)
Q Omuse FUT teach who FOC what FOC
‘What will Omuse teach to who?’

[17] We do not have any data pertaining to questions involving more than two *wh-* elements, though we suspect that such questions would also use this strategy.

Note that multiple *wh-* questions formed in this way may not be marked by the clause-initial Q particle *kó*[18], despite the fact that the Q marker may optionally appear when a single *wh-* item is left in-situ (6). Although focus markers may optionally accompany each in-situ interrogative expression, there is a preference for realizing the focus particle after the first *wh-* item, but not the second.

Turning next to strategy (ii), example (25) demonstrates that multiple *wh-* questions formed by moving a single interrogative item may optionally be marked by the Q particle *kó*, unlike certain other types of *wh-* movement constructions in Ekhwa Adara discussed in this paper (e.g., long-distance *wh-* movement, partial *wh-* movement, and possibly multiple *wh-* fronting, as described below). In example (25), we also observe that the hierarchically inferior *wh-* item (the object 'what') is fronted, crossing over the higher in-situ interrogative (the subject 'who'), an apparent violation of the superiority condition (Kuno & Robinson 1972, Chomsky 1973, 1977).

(25) (kó) ɪncí mo iwé ku ɹɔ́ __
Q what FOC who PFV buy
'Who bought what?'

Similar behavior in the distribution of moved interrogative phrases in multiple *wh-* fronting constructions (see below) also suggests an absence of superiority effects in the language. In this way, Ekhwa Adara is similar to a number of other West African languages that have been shown to lack superiority effects outright (Saah 1994, Adesola 2005, 2006, Torrence & Kandybowicz 2015, Kandybowicz et al. 2023, Schurr et al. 2024).

The sentences in (26) below demonstrate strategy (iii) – fronting of multiple *wh-* phrases.[19] When more than one *wh-* item is moved to initial position[20], the items are pronounced in an order inverse to the hierarchical position of their extraction sites.

[18] A sequence of two clause-initial occurrences of *kó* (i.e., one for each in-situ *wh-* item) is also prohibited in structures like (24).

[19] Since focused in-situ subjects are indistinguishable from their moved counterparts, we avoid presenting multiple *wh-* movement structures in which one interrogative is the subject. We attempted to elicit multiple *wh-* fronting structures in which one of the moved elements was an adjunct, but the types of marginal judgements that we received for these require further investigation.

[20] We have only collected monoclausal multiple *wh-* questions so far. 'Initial' here, therefore, means both clause- and sentence-initial, but we have not yet determined that this must be the case.

(26) a. omúsé sú mani ijâ ɔnʃó
Omuse FUT teach Ija song
'Omuse will teach Ija a song.'

b. ɪncí$_i$ mo iwé$_j$ ŋu omúsé sú mani-*(ŋ)$_j$ t$_i$
what FOC who FOC Omuse FUT teach-3SG.ACC
'What will Omuse teach to who?'

c. * iwé$_j$ ŋu ɪncí$_i$ mo omúsé sú mani-*(ŋ)$_j$ t$_i$
who FOC what FOC Omuse FUT teach-3SG.ACC
Intended: 'What will Omuse teach to who?'

The canonical ordering of the two lower arguments in a double-object construction places the indirect object in a position immediately preceding the direct object (26a). When both of these arguments are fronted, as in (26b), however, the direct object interrogative ('what') must precede the indirect object ('who'). The reverse order, which would both respect the superiority condition and remain closer in form to the canonical Ekhwa Adara word order, was consistently judged ungrammatical (26c). Assuming the accuracy of these judgment contrasts, we therefore observe a kind of anti-superiority effect in Ekhwa Adara multiple *wh-* movement constructions – the initial fronted *wh-* item must be hierarchically inferior to the other(s). The acceptability of sentences like (26b) might also suggest a recursive FocP domain in the Ekhwa Adara left periphery (contra Rizzi (1997)), unlike what we see in other African languages (e.g., Aboh 2004). However, the availability of focus-marking on in-situ *wh-* elements in the language (see (16)) likely supports an alternative analysis in which the peripheral focus particles that surface in multiple *wh-* fronting structures are more accurately "term focus" markers.

Further research is needed to determine whether multiple *wh-* fronting truly exists in the grammar as a productive multiple question formation strategy. The judgments reported in (26) were observed stably from late 2022 to summer 2023, leading to the initial conclusion that multiple *wh-* fronting exists in the language. However, in fall 2023, structures like those in (26b) were consistently judged ungrammatical, leading to our present reluctance to fully commit to the existence of multiple *wh-* fronting in the grammar. The third author of this paper (our language consultant) speculates that the availability of multiple *wh-* fronting may have an inter-generational dimension, reporting that older Ekhwa Adara speakers like his mother or grandparents seemingly allow/produce multiple *wh-* fronting structures. If all three multiple *wh-* question formation strategies are indeed available in the grammar, strategies (i) and (ii) are clearly preferred over strategy (iii). Of relevance to the question of whether multiple *wh-* fronting truly exists, we note that multiple focus movement does not seem to be available in declarative sentences, as shown in (27).

(27) a. * ɔnʃó$_i$ ku ijâ$_j$ ŋu omúsé sú mani —$_j$ —$_i$
song FOC Ija FOC Omuse FUT teach
Intended: 'It is a SONG that Omuse will teach IJA.'

b. * ijâ$_i$ ŋu/ku ɔnʃó$_j$ ku omúsé sú mani —$_i$ —$_j$
Ija FOC song FOC Omuse FUT teach
Intended: 'It is a SONG that Omuse will teach IJA.'

In order to emphasize more than one non-interrogative argument in the clause, the preferred strategy is to leave one item in-situ with a change in intonation (represented by all caps), as shown in (28).[21]

(28) ɔnʃó ku omúsé sú mani IJÂ
song FOC Omuse FUT teach Ija
'It is a SONG that Omuse will teach IJA.'

The absence of multiple focus-fronting in this context potentially casts doubt on the existence of multiple *wh-* fronting in the language (or at least in the grammar of the third author/others of his generation). If it does exist, something we leave for future research, then Ekhwa Adara is exceptional in its behavior regarding multiple *wh-* questions, as it employs a multiple fronting strategy that is not found, as far as we know, in any other African language.

4 Conclusion

We summarize our key findings on question formation in Ekhwa Adara as follows. Polar questions in Ekhwa Adara are primarily formed using final-vowel lengthening and L% boundary tone insertion. Furthermore, they are optionally marked by the sentence-initial Q particle *kó*. The only exception to this rule is found in embedded polar questions, which require the presence of an embedded left peripheral Q particle and the absence of final-vowel lengthening + L%

[21] A reviewer noted that the change in intonation in (28) may serve a different focus function besides the emphasis of a non-interrogative argument as documented here (for example, marking an exhaustive argument). We acknowledge this point and leave the exploration of this possibility for future research.

boundary tones. Regarding movement, Ekhwa Adara is an optional *wh-* movement language. When there is movement, the *wh-* item obligatorily precedes a focus marker, which can also optionally be used with in-situ *wh-* phrases. There is a selectional dependency between the *wh-* item and the accompanying focus marker in all contexts. Ekhwa Adara allows for long-distance *wh-* movement as well as simple (naked) partial *wh-* movement. Indirect questions in the language are formed via relativization, except when embedded under the verb 'ask', which involves *wh-* movement to the left periphery of the embedded clause. Ekhwa Adara potentially has three distinct strategies for forming multiple *wh-* questions: (i) all *wh-* items remain in-situ; (ii) one *wh-* item is fronted to clause-initial position, while the other(s) remain(s) in-situ; and (iii) multiple *wh-* fronting, a phenomenon that to the best of our knowledge is (currently) unattested in any other African language. Multiple *wh-* movement, to the extent that it truly exists in the grammar, is constrained – the initial fronted *wh-* item must be hierarchically inferior to the other(s), an effect we have characterized as an anti-superiority effect. Despite the existence of the anti-superiority effect in multiple *wh-* fronting constructions, *wh-* movement in Ekhwa Adara is not constrained by the superiority condition that limits *wh-* movement in languages like English.

Our investigation into Ekhwa Adara question syntax is ongoing. Future work will address the following issues (among others) related to the content of this paper: a fuller description of the distribution of optional Q particle *kó*; a better understanding of the various focus particles and the selectional dependency between them and *wh-* items; and an assessment of whether multiple *wh-* fronting truly exists in the grammar and if so, a structural analysis of multiple *wh-* fronting that derives the fact that the initial fronted *wh-* item must be hierarchically inferior to the other(s), given the absence of superiority effects in the language. We hope our description of the Ekhwa Adara interrogative system inspires further research on the grammar of this considerably under-documented language.

Abbreviations

1 = first person
3 = third person
ACC = accusative
COMP = complementizer
DEF = definite
FOC = focus

FUT = future
NOM = nominative
PFV = perfective
PROG = progressive
Q = question morpheme
REL = relative marker
SG = singular
SP = speaker

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Chapter 9

A critique of areal-only approaches to Bantu spirantization

Brandon Kieffer

Some scholars, such as Janson (2007), have claimed that Bantu Spirantization was spread primarily by contact, and to such an extent that it is not a useful tool for establishing the taxonomic structure within Bantu. Such areal approaches fail to account for some distribution patterns and spirantization reflexes, especially among languages around Lake Victoria. A preferable explanation is that Bantu Spirantization was spread by a mix of inheritance and contact.

1 Introduction

What is traditionally called Bantu Spirantization (BS) is a complex series of sound changes, whose history and distributions are strongly debated. As part of this debate, some have dismissed BS as a tool for establishing taxonomic relations within Bantu, claiming instead that the history and distribution of BS are overwhelmingly contact-based. The quote below, from Janson (2007: 99), exemplifies this view:

> Wave-like changes usually can tell nothing or little about prehistoric splits or migrations. [...] spirantisation is such a change. Consequently, as it cannot be used an indicator of major groupings within Bantu, as has been stated before, most recently by Nurse & Philippson (2003: 174)[1]

[1]A reviewer points out that Janson's (2007) reference to Nurse & Philippson (2003) is a misleading representation of their stated position. The reviewer, like myself, interpreted Nurse & Philippson (2003) as arguing for a mixture of tree and wave, much as I do in this paper.

Brandon Kieffer. 2026. A critique of areal-only approaches to Bantu spirantization. In Vicki Carstens, Katherine Russell, Olawale Akingbade, Deborah Morton & Michael Diercks (eds.), *Pamoja tena 'Together again': African linguistics after COVID*, 195–214. Berlin: Language Science Press. DOI: ??

While it is true that BS as a singular entity proved unreliable for establishing the primary branches of Bantu and immediate daughters of Proto-Bantu (PB), this does not necessitate that the changes it encompasses are useless for establishing the links within and between other levels of Bantu taxonomy. Indeed, components of BS have been successfully utilized as a taxonomic indicator for families within Bantu such as the Kikongo Language Cluster (Bostoen & Goes 2019) and Northeast Coast Bantu (Hinnebusch 1981) (see Section 6).

Janson, in a 2007 paper where he proposes a model for the areal distribution of BS, dismisses the utility of BS for taxonomy. In this paper I argue instead that we should acknowledge and utilize the shifts collectively identified as BS as a useful tool for taxonomy in Bantu subfamilies, and that we should move past the theory of BS's distribution being exclusively contact-based. We should look instead to different hybrid models that involve both inheritance and contact as vectors of distribution, which can better account for the data.

While the term BS is used a number of times in this paper, it should be understood that BS is not a single phenomenon, but a conventional term used by Bantuists to describe a set of distinct yet interacting sound shifts in a large number of languages. The BS changes are made up of an initial group of changes targeting a common environment in PB, and the other changes that were fed by the initial shifts. Due to the intertwined histories of these shifts and the opacities this caused, it is hard to isolate the shifts from each other, and so researchers look at them collectively. This does not mean that BS is unitary in any way however. Additionally, one should not take the "spirantization" part of "Bantu Spirantization" to have the same definition it would have in other language traditions, but rather as a component of the conventional name.

2 Background

PB is reconstructed as having 11 consonantal phonemes (Bastin et al. 2002), but only the six stops are of interest to us, as only they are involved in BS. While it is possible that the PB sounds *c and *j were also stops, and sometimes became affricates and fricatives (as in Luganda and Kinyarwanda, for example Katamba 1974, Kieffer 2021), we will not be considering them, as most of the BS literature ignores them (Janson 2007, Bostoen 2008, Labroussi 2000). PB is also reconstructed as having 7 vowels, of four heights. The highest two vowels, *i and *u are of the most interest in discussions of BS.

The traditional story is that BS shifted the PB stops into affricates and fricatives when they precede *i and *u. This can be seen in the Kinyarwanda data in Table 1 below.

Table 1: BS in Kinyarwanda

(Bastin et al. 2002, Kieffer 2021, Seymour 2017)

Proto-Bantu	Kinyarwanda	Gloss
*gid	/-ziɾ-/	be.taboo
*gɪd	/-giɾ-/	have
*kund	/-pfund-/	become.fuller
*kʊnd	/-kund-/	like

There is a lot of variation in the BS reflexes between modern languages. The consonant-vowel sequences *pi, *bi, *tu, and *du show the most variation in their modern reflexes, with little regularity across languages in their place of articulation, and in some languages these sequences even retained their status as stops, like in Kinyarwanda (Kieffer 2021) and Shi (Polak-Bynon 1975). Many languages had their BS reflexes undergo further changes, such as deaffrication, devoicing, debuccalization, and elision (Janson 2007). The diversity of reflexes can be seen in Table 2 below.

Table 2: BS Reflexes in some East Bantu Languages

(Bastin et al. 2002, Katamba 1974, Kieffer 2021, Poletto 1998, Hinnebusch 1981, Janson 2007)

PB	Rwanda		Nkore		Ganda		Swahili		Saamia		Ngoni		Ruihi	
	*_i	*_u	*_i	*_u	*_i	*_u	*_i	*_u	*_i	*_u	*_i	*_u	*_i	*_u
*p	ç	f	h	h	s	f	f	f	f	f	h	h	∅	∅
*t	s	ʈʂ	s	tʃ	s	f	s	f	s	f	h~s	h	∅	∅
*k	ts	pf	s	f	s	f	ʃ	f	s	f	h	h	∅	∅
*b	bɟ,	v	z	ʒ	z	v	v	v	f	f	h	h	∅	∅
*d	z	v	z	ʒ	z	v	z	v	s	f	h	h	∅	∅
*g	z	v	z	ʒ	z	v	z	v	s	f	h	h	∅	∅

Most of the BS languages also underwent the 7>5 vowel merger, which merged the high and near-high vowels of PB, ie: *i,*ɪ>i and *u,*ʊ>u. By merging the BS-triggering highest vowels with non-triggering near-high vowels, the merger obscured that which triggers the initial shifts of BS. Thus it is well agreed upon that the initial BS shifts must have predated the merger, since the merger obscured the triggering environment (Schadeberg 1994).

It should be made explicit that the high vowels in five-vowel languages, along with the near-high vowels in seven-vowel languages, have never been found to trigger BS-like rules (Schadeberg 1994). It is because of this commonality that earlier work on BS represented the near-high vowels (*ɪ and *ʊ in this work) as <i> and <u>, and used more esoteric symbols for the highest two. Due to the debate over the phonetic identity of the highest PB vowels, there is still a lot of variation in how they are represented orthographically (Janson 2007). I have adapted Schadeberg's (1994) system, due to its similarity to attested Bantu vowels inventories. His system does raise a question of why *[i] and *[u] triggered BS shifts in seven-vowel languages, yet [i] and [u] cannot in five-vowel languages. While this is a topic ripe for debate, we will not be delving into it in this paper.

BS shifts and the vowel merger are now found in around half of the Bantu languages. Figure 1 below shows the distribution and isoglosses for both.[2] One should note the presence of languages without either shift in South Africa and the areas east of Lake Victoria in addition to the large patch in the northwest. Seven vowel languages that underwent BS shifts can be found on either side of the Great Lakes and Lake Corridor areas. Two BS-free, five-vowel languages can be seen as well, in western Zambia and the north of the DRC. It should be noted that many of the languages said to have BS do not have the vowel merger, and languages with the vowel merger are a subset of languages with BS. This indicates (as comparative and theoretical evidence prove) that the initial BS shifts began earlier, and the vowel merger later spread within the BS-area. Based on the vowel merger's cross-clade distribution, and the relative simplicity of this single shift, it seems likely that the merger was spread predominantly through contact.[3]

Figure 2 below, from Bostoen (2008), shows both the approximate extent of the Congo rainforest and the locations of the (according to Bostoen 2008) six primary daughter-clades of PB. By comparing Figure 1 and Figure 2, we see that the BS languages are predominantly found within the East Bantu, Southwest Bantu, and West Coastal Bantu clades, and that very few of those languages are found within the Congo rainforest.

Figure 3 below shows the distribution of the subfamilies of East Bantu. By comparing Figure 1 and Figure 3 we can see that the distribution of BS languages and

[2]The isogloss for BS is an isogloss of the languages that Janson (2007) describes as having "Full Spirantisation", which will be explained in Section 3

[3]I devised this map to show the isoglosses of BS and the vowel merger. My list of which languages fall into which category is based on data from Schadeberg (1994), Janson (2007), and Bostoen (2008). I used the map in Nurse & Tucker (2001) to approximate the locations of the dots to represent the different languages. The base map comes from Wikimedia, (Thuvack, <https://creativecommons.org/licenses/by-sa/3.0>, via Wikimedia Commons).

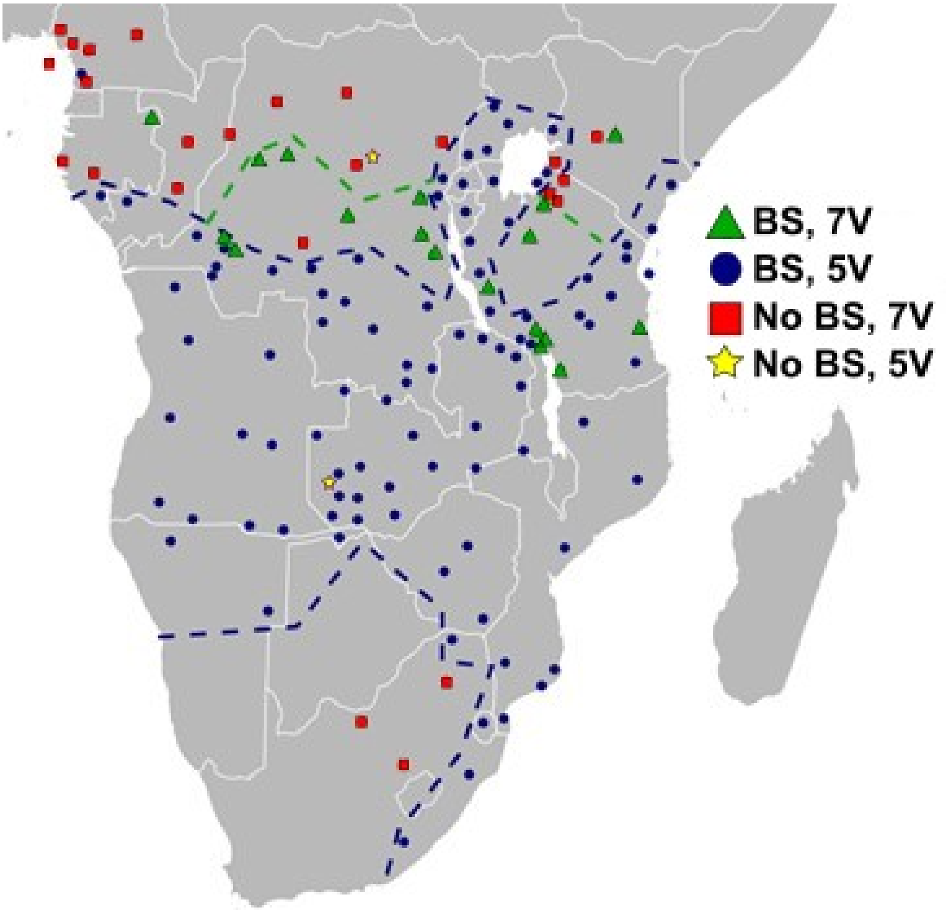

Figure 1: Isoglosses of BS and Vowel Merger

the isogloss of the vowel merger do sometimes line up with the clades. The Central Kenya clade is outside both isoglosses, while the North Mozambique clade is inside both. The Rufiji-Ruvuma clade is said to have BS in all of its languages, but, as comparing the maps shows, only some of its members underwent the vowel merger. By comparing the two maps again, we can see that while most of Southern Bantu is said to have BS, there are some members that do not, namely Sotho-Tswana.[4]

[4]This map, of my own making, derives from one in Nurse & Tucker (2001). The identity and make up of the East Bantu subfamilies comes from Nurse (1999). Since there is debate on the extent of individual languages, and on the makeup of the subfamilies, this map should be viewed as a visual aid to the approximate locations of the clades rather than a precise survey.

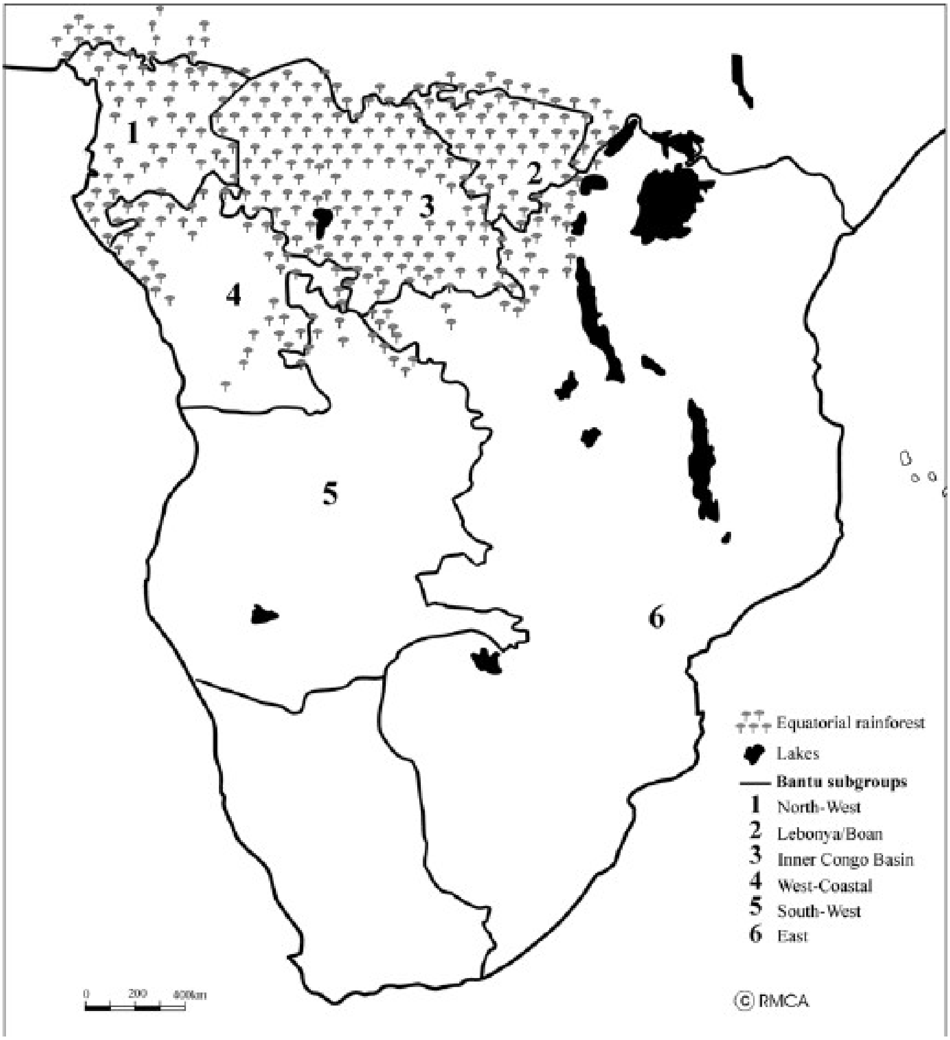

Figure 2: Immediate Daughter-Clades of Proto-Bantu, and Extent of Congo Rainforest

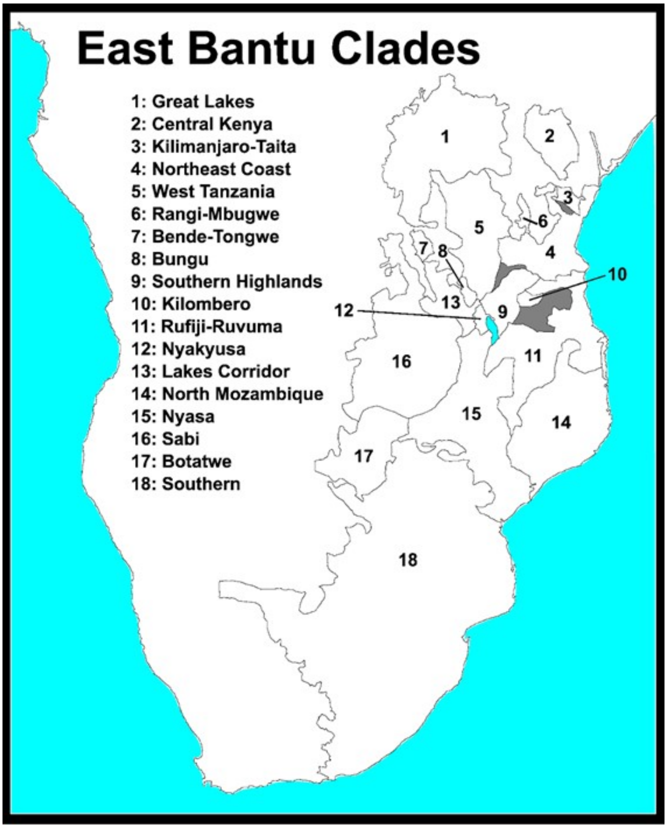

Figure 3: Clades of East Bantu

2.1 Issues with the traditional tree model of BS

One possible explanation for the distribution of BS and BS-free languages is that BS was an innovation of an early daughter of PB, from which all BS languages are descended, in contrast with the BS-free languages. Such an idea is disproven by some of the established cladistic results, where BS and BS-free languages have been shown to share a recent common ancestor. These clades cut across the imagined division between BS and BS-free languages.

One example is Central Kenya Bantu (E50). Central Kenya Bantu has been shown to be closely related to several neighboring clades such as Great Lakes Bantu (Zone J), Chaga-Taita (E60+E74), and Northeast Coast Bantu (G10-40+E70). The sisterhood of these clades with Central Kenya Bantu is supported by both lexical evidence (Ehret 1973, 1999, Bastin et al. 1999) and morpho-phonological works (Nurse 1999, Nurse & Philippson 2003). Central Kenya Bantu is entirely

BS-free (Nurse 1999), while its sister clades are mostly or entirely made up of BS languages (Nurse 1999, Janson 2007).

If it were correct that BS reflexes were inherited from a single common ancestor which innovated them, then the presence of BS reflexes in Chaga-Taita would entail that an ancestor of Central Kenya Bantu inherited BS reflexes as well. In order to derive the attested phonologies in Central Kenya Bantu, we would need to posit a Duke of York effect, where BS was undone in Central Kenya Bantu. If a Duke of York effect had actually occurred in Central Kenya Bantu, we would expect it to have effected non-BS-derived affricates and fricatives as well, such as in loans or reflexes of PB *c and *j. I am not aware of any data supporting or works claiming such an across the board fortition rule in Central Kenya Bantu.

Additional factors, like the diversity of BS reflexes (Janson 2007) both within and across clades, pose issues for the single common ancestor model. If BS had occurred in an early daughter of PB, we would expect to see evidence of its BS reflexes across most or all BS languages. However, as Janson documented, there is a lot of variation in BS reflexes. Without positing any Duke of York effects, one cannot reconstruct a common ancestral BS reflex for all of the BS languages. Even for the PB sequence *ti, which underwent BS quite early according to Janson, one cannot reconstruct a single ancestral BS reflex, as some BS languages, such as Siha (E621C), retain a coronal stop reflex for *ti, but have fricative reflexes for other BS-triggering sequences (Nurse 1981).

There is one taxonomic model which is sometimes claimed to roughly line up with the distribution of BS: the Savanna-Forest theory. This theory proposes a Savannah Bantu family which includes East Bantu (Zones E, F, G, J, K20+40, M, N, P, S), Lega (D20), Luban (L20-40), and South-West Bantu (H20, K10+30, L10+50+60, R) (Ehret 1999). Nurse & Philippson (2003: 174) describe the connection between BS and Savanna Bantu as follows: "The great majority of Forest languages show no signs of BS […] This seems at first to support a split of Forest vs. Savanna". As they later point out however, the proposed Savanna Bantu includes several BS-free languages, such as Central Kenya Bantu, Tswana-Sotho, and some of the languages in Zones F and J. Savanna Bantu, as defined by Ehret, also fails to incorporate the Kikongo Language Cluster which has BS. While evaluating Savanna Bantu, Nurse & Philippson (2003: 174) proceed to say they "found nothing non-lexical supporting the notion of a Savanna branch of Bantu". Due to their opinions on the weakness of the lexical data supporting the Savanna-Forest theory, they argue against its validity.

In response to the failure of the initial Tree-model approach, an areal, Wave-model of distribution was championed by Janson (2007). We will look at his argument in more depth in the next section.

3 Janson's argument

Janson (2007) makes two main points. The first is a critical look at which types of changes get called BS and the theoretical mechanics of such changes. He also discusses the frequency and distribution of individual changes, and reconstructs an order in which the changes must have happened. This discussion of the minutiae of the components of BS was an important innovation in how BS is discussed, and, in my opinion, contains several breakthroughs. The other point of the paper is to explain the complex data through Wave Theory. I do not aim to argue against the first point on the nature, mechanics, and history of the BS changes, only against Janson's conclusion that that data requires or indicates that BS must have been spread areally, and that the BS changes are therefore not useful in determining groupings and clades within Bantu.

Janson (2007: 84) describes BS as being a phenomenon made up of many shifts, as shown by the quote below:

> Since many changes are involved, and many languages show evidence of only some of the changes, it is in some cases debatable whether a language has undergone spirantisation or not.

This view differs slightly from my own, in that Janson still seems to consider BS to be an entity in the phonology, instead of an academic concept to describe and collectivize a number of interacting and related shifts. In Janson (2007), then, BS consists of affricating and spirantizing rules, and their reflexes often underwent further shifts, such as devoicing, debuccalization, and elision. If we follow Janson's (2007) model for areal distribution then, there must be many waves as well, one or more for each individual shift within BS. Janson seems to suggest that all of the BS changes originated in the Lake Corridor, but if BS was spread by many waves, then we have no reason to think these waves had identical isoglosses and origin points. And indeed, when looking at the distribution for different changes, we can see that their isoglosses do differ. Janson suggested the Lake Corridor as the origin point, not just because of its relatively central location, but because the BS reflexes there are "full" and "homogenous". A collection of shifts like BS cannot truly be called "complete" or "full", it is just that some languages have undergone different sets of innovations than others. Additionally, the homogeneity of outputs in a region is not strongly indicative of how long ago a change started. It simply means that the languages in the area failed to undergo any shifts that differentiated their BS reflexes, or they underwent some shifts that aligned their reflexes. Homogeneity of output does not indicate much useful information on the dating of changes.

Janson explains the variation and distribution of affricate and fricative reflexes by hypothesizing that the initial stages of BS involved affrication, and that fricative reflexes derive from a spirantization of these proposed affricates. For instance, *ki and *ku would have first had the shift of *ki>*tsi and *ku>*pfu, respectively, before undergoing the shifts *tsi>si and *pfu>fu. In the "*k" row of Table 2, we can see that Rwanda maintained these intermediate forms, while the others underwent deaffrication. Janson also noted that the languages with affricates are concentrated around the borders of the BS area and claims this is because the deaffrication shifts failed to reach these edges.

I would argue that the largest contribution of Janson (2007) was the discussion of the different reflexes, histories, and orderings of the various PB consonant–vowel pairs. Janson describes how PB consonant–vowel pairs that agree on the features [+Labial] or [+Coronal], such as *pu or *ti, are the least likely to have an affricate reflex. These agreeing pairs also maintain their places of articulation, such as *pu>fu or *ti>si. In contrast, the PB consonant–vowel pairs that disagree in their [±Labial] and [±Coronal] features, like *pi and *tu, are the most likely to be affricates, and the most likely to have not shifted at all. Additionally, there is a lot of cross-Bantu variation in whether the disagreeing pairs have a labial or coronal reflex, as Table 2 shows. The dorsal–vowel pairs, like *ki and *ku, maintain the place of the PB vowel, and sometimes have affricate reflexes. The "Rwanda" column in Table 2 exemplifies some of these generalizations.

Since the agreeing pairs are the most likely to have fricative reflexes, Janson hypothesizes that these pairs shifted before the others, as they have rarely failed to undergo a deaffrication. Since the disagreeing pairs somewhat frequently have plosive or affricate reflexes, he hypothesizes that these shifted most recently, since in many languages they have not undergone BS shifts at all, or have not undergone as many subsequent shifts. The dorsal pairs would therefore have shifted after the agreeing pairs, but before the disagreeing pairs. While there are individual languages whose BS reflexes seem to challenge this hypothesis, it does line up well with the majority of the data, and I find it convincing.

The implication of this ordering of the shifting of different pairs is that each deaffrication for each pair was its own independent phonological operation,[5] and that the initial affrication for each pair was its own independent phonological operation too. If we accept Janson's areal model, this would mean there must have been dozens of BS waves.

Janson's model also implies that each instance of a shift happened once on a BS-wide scale, such that the deaffrication of *ts (deriving from *ki) in Kikongo

[5] I avoid the more common terms "sound change" or "phonological rule" here in an attempt to remain agnostic as to whether this is a synchronic or diachronic change.

and Kiswahili were both part of the same wave. However, it is also possible that the same shifts were innovated multiple times, in multiple locales, and spread from multiple centers.

Janson provides several maps of BS languages by the type of BS reflexes they have, focusing on affricate versus fricative reflexes; debuccalization, devoicing, and elision; and whether languages have "Full Spirantisation" or not. Before we can get into the details of these maps, however, we need to understand what exactly Janson (2007: 84) means by "Full Spirantisation". He defines "Full Spirantisation" as below:

> I have chosen to use one simple criterion for whether a language has undergone Full Spirantisation, namely that both [*ku] and [*ki] have reflexes that are not stops and are not velar

This, of course, presents issues, since many languages have undergone BS shifts, but fail to meet the above criteria. An example is Gogo, which has the reflexes shown in Table 3. Gogo has classically BS reflexes for all of its consonant-vowel pairs, except *ki, which never shifted.

Table 3: BS Reflexes of Gogo (G11)

(Janson 2007)

	*p	*t	*k	*b	*d	*g
*u	f	f	f	bv	bv	bv
*i	f	s	**k**	bv	z	z

Janson's concept of "Full Spirantisation" is quite prominent is his maps, since it is the metric he uses for determining whether to label a language as BS-free or not. He does acknowledge, though, that his maps "really describe broad categories of changes (and not individual changes)" Janson (2007: 90), meaning that none of his maps represents a single change and its isogloss, but rather amalgamations of many shifts.

I compiled the data from Janson's maps to produce the one below, Figure 4, in lieu of including each of Janson's maps. The solid and checkered dark blue sections represent areas that Janson describes as not having "Full Spirantisation". The checkered sections are those with at least one affricate BS reflex. The solid yellow represents the most common pattern, "Full Spirantisation" without any affricates and missing any of the devoicing/debuccalizing shifts. The light blue polka dots are languages which have devoiced their BS reflexes, the green horizontal lines are languages which have reduced their BS reflexes to /h/, and the vertical pink lines are the languages which have elided their BS reflexes.

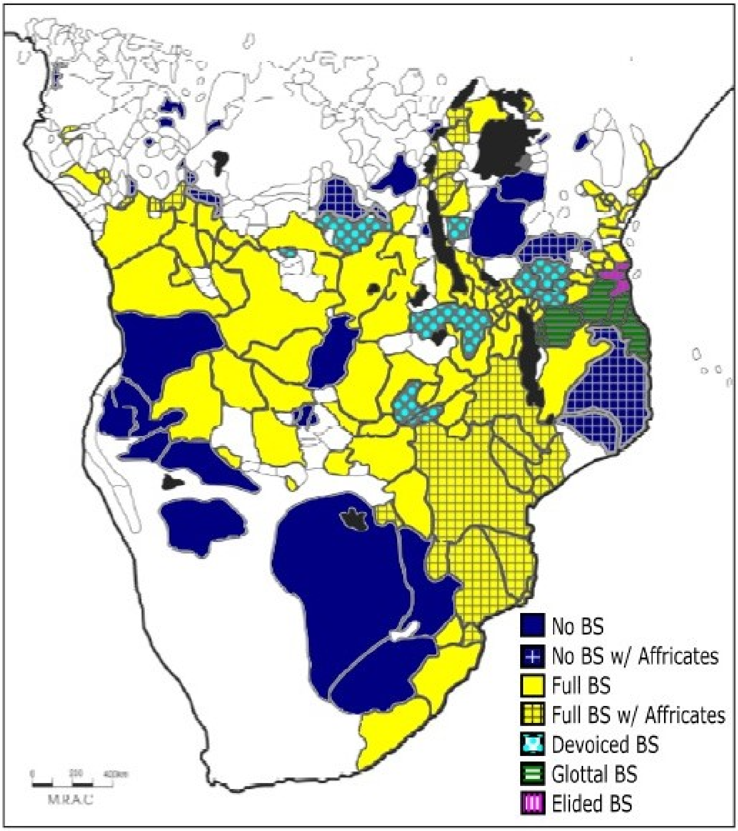

Figure 4: Composite Map of Janson (2007)'s Spirantisation Maps

As mentioned earlier, I do take issue with Janson's model for the distribution of BS changes and his assertion that BS shifts "cannot be used as an indicator of major groupings within Bantu" (Janson 2007: 99). The following section looks more closely at his areal model, its implications, and my counterarguments.

4 Issues with Janson's model

If the BS shifts were spread as a spirantization/affrication rule from one language to its neighbors, we would have certain expectations about the languages' BS reflexes: they should have similar BS reflexes to their neighbors'; we should be able to trace paths of contact; and we should find evidence of a dialect continuum of sorts of BS reflexes. And while one can find this in some areas, other groups of BS languages disprove it.

The BS languages around Lake Victoria provide a good example with which to problematize areal BS. To the east of the lake are BS-free, 7-vowel languages like Kuria (JE43), Gusii (JE42), and the other East Nyanza languages (Mould 1981). Northeast of the lake is the Luhya dialect continuum (JE30), the majority of which has BS shifts and the vowel merger (Mould 1981). North of the lake is Luganda (JE15), a North Nyanza language; northwest is Nkore-Chiga (JE13/14), a Rutara language; and west is Kinyarwanda (JD61), a Kivu language; all of which have BS shifts and the merger. The Victoria region is bound by non-Bantu languages to the north and east, and BS-free Bantu languages to the west, east, and southeast.[6] If the BS changes were indeed spread areally, it must have entered the Victoria area from the southwest, coming through the Kivu, Rutara, and North Nyanza subfamilies before finally entering Luhya. Kinyarwanda, Nkore-Chiga, Luganda, and Saamia are, respectively, representative examples of Kivu, Rutara, North Nyanza, and Luhya subfamilies.

With such a clear path of transmission, one would predict that if the languages adopted BS changes areally from each other that they would have similar BS reflexes or connected evolutions of their BS reflexes. Table 4 below lays out the BS reflexes of the voiceless stops for the four languages mentioned, taking its data from Kieffer (2021), Poletto (1998), and Mould (1981).

One notices that the reflexes for *ti are all identical, and those for *ku and *ki are almost identical, except for Kinyarwanda's affricates. In Nkore-Chiga we see that both *pi and *pu have /h/ as their reflex; this is due to P-debuccalization, an East-Bantu phenomenon that affected *p in non-nasal environments (Nurse 1999). Unlike other Great Lakes Bantu languages, Nkore-Chiga seems to have progressed further in its P-debuccalization at the advent of the BS shifts of *pu, since it debuccalized *p before *i and *u in addition to the other five PB vowels. The debuccalization in Nkore-Chiga is unconnected to the debuccalization seen in Rufiji-Ruvuma, since it is not a debuccalization of a post-BS fricative, but of a pre-BS stop.

One should also note the split in reflexes for *tu. Kinyarwanda and Nkore-Chiga have posterior /tʃ/ and /ʈʂ/, while Luganda and Saamia have labiodental

[6] Southeast of Lake Victoria is Zone F. Zone F is made up of BS-free languages and, in the west, languages with unusual BS patterns. It is unclear from Janson's notation in his appendix ("ku, fu"), if BS shifts are optional allophony, lexically specific, or some other type of "partial" spirantization.

Table 4: Voiceless BS Reflexes of Languages around Lake Victoria (Kieffer 2021, Mould 1981, Poletto 1998)

PB	*pi	*ti	*ki	*pu	*tu	*ku
Kinyarwanda (Kivu)	/ç/	/s/	/ts/	/f/	/ʈʂ/	/pf/
Nkore-Chiga (Rutara)	/h/	/s/	/s/	/h/	/tʃ/	/f/
Luganda (North Nyanza)	/s/	/s/	/s/	/f/	/f/	/f/
Saamia (Luhya)	/f/	/s/	/s/	/f/	/f/	/f/

/f/. Additionally, each language has a different reflex for *pi, with Luganda having a coronal reflex, Saamia having a labial one, and Kinyarwanda having a palatal one.

If the BS shifts were indeed spread from Kinyarwanda, to Nkore-Chiga, to Luganda, to the Luhya continuum, we should see the innovations in the earlier shifts carried along to the later ones. Since Nkore-Chiga did not have a BS reflex for *pi, how could Luganda have gotten its *pi>/s/ rule from Nkore-Chiga? Additionally, how could Luhya have gotten the rule *pi>/f/ from Luganda's *pi>/s/ rule? There is no labiality left on the Luganda /si/ sequences, nor remembrance of its historical labiality, to motivate labialization. The /tʃ/–/f/ differences in the reflexes of *tu present similar issues, but there is at least a labial vowel in the /tʃu/ sequences that retains the labiality of the PB sequence *tu. The BS languages around Lake Victoria show that not all BS shifts could have spread as a simple affrication or spirantization rule, due to the contrast in their reflexes.

The differences in reflexes among neighbors exemplifies how what we call BS was not a simple shift, nor made up of exclusively areal shifts. In many cases, such as those for *tu and *pi, cross-Bantu BS does not have a clear target, but rather assibilation in some direction. It is hard to imagine a situation where contact with a language with the rule /p/⟶ [s] /_i would prompt its neighbor to adopt the rule /p/⟶[f] /_i.

While Janson may be correct about some of the BS shifts being areal, that does not mean that all are. That some changes, like debuccalization, line up nicely with strong clades like Rufiji-Ruvuma, indicates that inheritance plays a role in some BS shifts as well. Each shift needs to be evaluated distinctly before one can claim it was spread by wave or tree.

Additionally if there were many waves of BS shifts that spread across most of Zones E-S, that would constitute a large and sustained contact zone. We would expect there to be other linguistic innovations spread in such a zone. I am not aware of any study which suggests the existence of such a contact zone.

To summarize, while there are regions where the spread of BS shifts areally can be modeled clearly, there are several regions which pose major challenges to Janson's model. In areas with a greater diversity of BS reflexes, such as around Lake Victoria, it is not clear how all of the BS shifts could be spread from neighbor to neighbor using Janson's model. The existence of some areal BS shifts does not necessitate that all BS shifts must have been areal too. And lastly, in order to strengthen Janson's approach, one would need to demonstrate that there is other evidence for the large contact zone it implies.

5 Alternative models

There has been support for a model of BS that incorporates both contact and inheritance since Hinnebusch (1981) at the least. While it has not been established what exact mix is needed, it seems clear that the BS shifts were distributed through both. There are many different ways for them to combine, and I shall outline a few below.

5.1 Parallel evolution

Bostoen (2008: 309) proposes that BS was innovated and propagated multiple times, in multiple areas, saying:

> The inherited properties of the PB highest vowels obviously allowed BS to be triggered repeatedly and independently and at different times in different Bantu subgroups, producing not only varied results across language, but also frequently identical outcomes.

The current landscape of BS reflexes could then be attributed to inheritance (many ancestors innovated and passed along the shifts), to contact (BS was innovated in many locals, and spread from these various foci), or a mix. While Parallel Evolution does imply that the innovations were separate, Bostoen does float the idea that something in an earlier stage primed the Bantu languages to undergo the initial BS changes.

5.2 A phonetic rule

If the Bantu languages were primed for BS at an earlier stage as Bostoen (2008) suggested, I believe this priming would have been a phonetic rule. By phonetic rule, I mean a language-specific difference in articulation that has yet to become part of the phonological system. I propose, then, that before any of the BS shifts started, there was a phonetic rule common to much of Bantu, where stops were articulated differently in front of the highest vowels. This could be the "noisiness" Hyman (2003) suggested as the phonetic origin of BS changes, the onglides Janson (2007) suggested, aspiration, or any other valid phonetic origin that may be devised. This phonetic rule would then have evolved differently, and entered the phonology differently in the different BS languages, and when combined with the other sound shifts that occurred in the post-PB period, produced the variety of BS reflexes found today.

As to the origin and distribution of the phonetic rule, one could argue for the Tree model, the Wave Model, or a mixture of the two. This phonetic rule may have started around the PB era, been passed down to PB's daughters, and only stabilized and phonologized in the daughters described as having BS. As for the BS-free daughters, the rule may have been lost, or may still be present today. While I am not aware of any phonetic studies attesting such a pattern in Bantu, I predict that it can be found in some modern BS-free languages, and hope to verify this prediction in future research.

One could alternatively argue that the initial propagation and distribution of the proposed phonetic rule could have happened via wave. Said phonetic rule could have been innovated in a BS-language, and then spread via contact to other BS-languages. The phonologization and stabilization of this phonetic rule could have been propagated on a wave as well.

6 Instances of BS being used for cladistics

As was mentioned earlier, there are a few cases where BS has been used to support taxonomic claims, namely Bostoen & Goes (2019) and Hinnebusch (1981) for the Kikongo Language Cluster and Northeast Coast Bantu, respectively. In this section I will summarize these case studies, as further evidence that BS can be useful to taxonomy.

6.1 BS in Kikongo Language Cluster

BS is used as a taxonomic diagnostic in two scenarios in Bostoen & Goes (2019). In the first and more explicit case, they show how a specific BS innovation lends support to De Schryver et al.'s 2015 phonological and phylogenetic evidence for a branch of West Kikongo that they refer to as the "northwestern cluster of West Kikongo" (B40+H12+H16c). In these languages, BS reflexes that are in the second syllable of a root are always devoiced. The isogloss for this change in West Kikongo also aligns with the distribution of an affrication rule where stem final /si/ regularly becomes /tsi/. The distribution of a component of BS, namely the voicing or devoicing of fricatives, is thus used as evidence for a taxonomic claim.

The other instance of BS as a taxonomic diagnostic in Bostoen & Goes (2019) does not specifically claim BS as a diagnostic, but does conform to the concept. Bostoen & Goes are able to reconstruct the BS system of the Kikongo Language Cluster (KLC) back to Proto-Kikongo. This reconstruction has fricative realizations for all 12 possible CV triggers of BS. Since BS is found throughout all of the KLC languages, and BS can be reconstructed into their last common ancestor, this would strongly imply that BS is both a feature of the KLC and a diagnostic for the clade. The reconstructed BS for Proto-Kikongo contrasts sharply with the BS systems in related clades.

KLC is considered to be part of the larger West Coastal Bantu family (Bostoen & Goes 2019, Bostoen & de Schryver 2018). Bostoen & de Schryver (2018) include the languages Nzebi (B52), Mbaama (B62), Tege-Kali (B71a), and Yans (B85) in their West Coastal node. Table 5 below shows some of the BS reflexes for these four languages. One can see how BS has not affected the velar stops at all in these languages. While the highest vowels have affected *d, they have not triggered affrication or spirantization at all, but rather blocked lenition. These BS reflexes contrast heavily with those in Proto-Kikongo, which has fricative reflexes for all of the Proto-Bantu CV pairs which trigger BS. That the BS reflexes of the KLC are so homogeneous, and contrast so heavily with some other West Coastal Bantu languages, indicates that BS is a useful taxonomic diagnostic for the KLC, since it sets this branch apart.

6.2 BS in Northeast Coast Bantu

Hinnebusch (1981) uses BS to distinguish Northeast Coast Bantu (NEC) from some of its East Bantu neighbors. He notes the lack of BS in Central Kenya Bantu (E50) as distinguishing the clade from NEC. The uniqueness of the BS reflexes in Sukuma-Nyamwezi (F21-22), Chaga-Taita (E60+E74), and Pare-Tavaita (G21-22) leads him to conclude that "the phenomenon is a unique event within these languages" Hinnebusch (1981: 40), a line which, to me, seems to claim that BS underwent a different history in these languages than the path taken by NEC and its

Table 5: Selected BS Reflexes in some West Coastal Bantu languages (Guthrie 1967–1971)

	*k	*ki	*ku	*g	*gi	*gu	*d	*di	*du
Nzebi (B52)	/k/	/k/	/k/	/k/	/k/	/k/	/l/	/ɗ/	/ɗ/
Mbaama (B62)	/k/	/k/	/k/	/k/	/k/	/k/	/l/	/d/	/d/
Tege-Kali (B71a)	/k/	/k/	/k/	/k/	/k/	/k/	/l/	/d/	/d/
Yans (B85)	/k/	/kj/	/k/	/k/	/kj/	/k/	/l/	/d/	/d/

southern neighbors. These differences would imply that Proto-NEC underwent its BS processes after separating from Central Kenya Bantu, Sukuma-Nyamwezi, Chaga-Taita, and Pare-Tavaita.

Hinnebusch contrasts NEC from other neighboring BS languages on the basis of the devoicing and debuccalization rules found in languages to the south. Nyakusa (M30), Kilombero (G50), and Southern Highlands (G60) have devoiced all of their BS reflexes, while the Rufiji-Ruvuma (N10+P10-20) languages went even further and debuccalized many of their BS reflexes (Hinnebusch 1981). This implies that these languages underwent innovations in BS after their divergence from NEC.

Hinnebusch is thus able to use BS to cladistically separate NEC from all of the neighboring clades except Lake Corridor Bantu (M10-20). It should be noted, however, that since he was not able to distinguish NEC from all of the neighboring clades exclusively using BS, he considers it a failed method. Since BS is common to all NEC members, and NEC's BS is significantly different from that found in most neighboring clades, I would consider it a successful method.

7 Conclusion

The history and evolution of the shifts we collectively call Bantu Spirantization are too complex to be conceptualized as a single phenomenon, or be explained with simple tree splits or waves. Neither the traditional tree model nor Janson's areal model is able to perfectly account for the patterns of BS reflexes and existing cladistics, such as those found in the languages around Lake Victoria. This leads us towards a model which utilizes both trees and waves. Such a model allows for the use of BS in cladistic endeavors, as seen in Hinnebusch (1981) and Bostoen & Goes (2019).

Abbreviations

PB : Proto-Bantu
BS : Bantu Spirantization
PK : Proto-Kikongo
KLC : Kikongo Language Cluster
NEC: Northeast Coast Bantu

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Chapter 10

The interaction between tone and intonation in Hausa *wh-* questions

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Newman (2000) suggests that in Hausa, question-word questions (WHQs) are marked by a sentence-final Q-morpheme which lengthens a final short vowel and add a L(ow) tone to the final H(igh)-toned vowel. The current work seeks phonetic evidence of the L-tone and the vowel-lengthening effects potentially induced by the Q-morpheme in WHQs. A production test was conducted to compare final syllables ending in CV, CVV, and CVO syllables bearing H, L, and HL tones in both statements and WHQs. The results indicate that while WHQs generally display a register raise from statements, only final H-toned syllables show an F0 drop compared to the penultimate syllable. Moreover, all final syllables, except for CVV-HL, experience lengthening. The lengthening of CV syllables, which neutralizes the contrast between short and long vowels, is due to the length component of the Q-morpheme, while the lengthening of CVO and CVV syllables is likely the result of phonetic regulation. The investigation shows that the Hausa WHQ pattern is influenced by both phonological effect (the Q-morpheme) and phonetic effects (final lengthening and overall pitch raise).

1 Introduction

The interaction between tone and intonation has been an important topic in the interface study of phonology and phonetics: how does a tone language regulate the pitch patterns associated with intonational melodies while implementing the pitch target for word-level tones? Studies have shown that African tone languages also use intonation at the sentence level, which sometimes interacts with tones, for example, in Embosi (Rialland & Aborobongui 2016), Shingazidja

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(Patin 2017) and Akan (Kügler 2017). Downing & Rialland (2016: p.2) argue that intonation in tone languages is commonly realized by register change, downdrift, expansion, and boundary tone to mark focus, interrogatives, or syntactic boundaries at the sentence level (see also Cruttenden 1997, Yip 2002, and Ladd 2008).

The current study examines the interaction of tone and intonation in Hausa, a tone language spoken in West Africa by more than 50 million speakers. Specifically, I will focus on the intonation of question-word questions (commonly shortened as WHQs). Newman (2000) describes the intonation of Hausa WHQs as having a declarative contour (an overall downward slope) and an optional pitch raising (see also Kraft & Kirk-Greene 1973). Compared with statements, a sentence-final H tone in WHQs has a lower pitch, and a final short vowel is lengthened into a long vowel.

Newman (2000) argues that the final falling end and vowel lengthening in Hausa WHQs are due to a separate morpheme attaching to the end, known as the *Q-morpheme* (also referred to as the *Question Morpheme q* in Newman & Newman (1981)). The affixation of the Q-morpheme will lengthen a final short vowel (1a and 1b), and add a L tone if the final vowel carries a H tone (1a and 1c). If neither condition is met, the Q-morpheme will apply vacuously (1d).

	Underlying Form	WHQ-final Form	Translation
	a. fìtá	wâ: zâi **fìtâ:**	"Who will go out?"
(1)	b. háihù	yàushè tá **háihù:**	"When will she give birth?"
	c. sáyár̃	mè: súkà **sáyâr̃**	"What did they sell?"
	d. yá:ròn	ìná: ká gá **yá:ròn**	"Where did you see the boy?"

The Q-morpheme analysis is different from other works where the falling end of WHQs is analyzed as a downdrift intonation with modification on the final tone (Miller & Tench 1980, Lindau 1986). One analysis from a prosodic perspective that also reflects Newman's Q-morpheme is "lax question prosody" (Clements & Rialland 2008, Rialland 2009). This term refers to a set of yes/no question markers typically characterized by a falling contour, open vowels, vowel lengthening, and a breathy voice, which is considered an areal feature in the Macro-Sudan Belt (Rialland 2009, Güldemann 2008, 2018). Among these features, the final falling contour and the vowel lengthening effects align with Newman's Q-morpheme, even though these prosodic characteristics are associated with YNQs while the Q-morpheme is specifically obligatory for WHQs.

Both analyses predict vowel lengthening and pitch-lowering effects but differ in their scope of application. The Q-morpheme proposal suggests that the L-tone effect applies exclusively to H-toned final vowels, while the lengthening effect is limited to vowels in CV syllables. However, if it is a prosodic or intonational effect, these two effects would not be restricted to H-toned vowels or CV syllables. Despite empirical studies on Hausa, the factors influencing the WHQ pattern remain unclear. On one hand, the results on pitch contours are inconsistent, with some studies reporting rising intonation in WHQs (Dresel 1977, Meyers 1976). On the other hand, there is limited research on duration changes in final vowels in WHQs.

The current work investigates the fit between the patterns of vowel length and pitch predicted by Newman's Q-morpheme analysis, and seeks phonetic evidence from eight native speakers of Hausa. Following Newman's proposal, I hypothesize that when a statement is converted into a WHQ, pitch lowering occurs exclusively among H-toned final syllables, and a duration increase occurs only among final CV syllables. An acoustic analysis was carried out to compare the pitch and duration of the final vowels in WHQs with those in corresponding statements.

The linguistic data presented in this paper are transcribed in orthography according to Newman (2000) and the Hausa database (Awagana et al. 2009) in the World Loanword Database (WOLD) (Haspelmath & Tadmor 2009). Long monophthongs are indicated by a colon, e.g. *shá:* "tea". L tone is marked with a grave accent *à*, H with an acute accent *á*, and HL with a circumflex caret *â*. Tones of CVV and CVC syllables are marked on the nucleus vowel, e.g. *mâi* "oil", *yàushè* "when", *bâs* "bus".

2 Tone and intonation in Hausa

2.1 Tone and syllable structure

Hausa has five basic vowels with short and long counterparts /i, i:, e, e:, a, a:, u, u:, o, o:/ and two diphthongs /ai, au/. There are three different types of syllable structures: CV, CVV, CVC. Among them, a CVV syllable contains either a long monophthong or a diphthong, and a CVC syllable consists of a short vowel with a consonant coda which could either be a sonorant coda such as /n, r̃/ or a voiceless obstruent /k, t, p, s/ in limited situations.

In terms of the tone system, Hausa has two contrastive tones: H and L, both of which can appear on all syllable types. Tones function to differentiate lexical meanings as in (2) and grammatical meanings as in (3).

(2) kápá “gap”
kàpá “to hang”

(3) tá: “she (completive aspect)”
tâ: “she (future tense)”

When H and L appear on a single heavy syllable, they will be phonetically realized as a falling (HL or F) tone. Therefore, HL tones only occur in heavy syllables (CVV or CVC) shown in (4-5). CVC syllables bearing HL closed by an voiceless obstruent coda /k, t, p, s/ are usually loanwords from English or French as in (5). Monosyllabic rising or three-toned contours are not permitted.

(4) mâi “oil”
yâ:rá: “children”
nân “here”
sádáúkâr̃wáá “the sacrifice”

(5) kwât “coat”
bâs “bus”
jîp “skirt” (jupe)
bîs “screw” (vis)

These facts support that the tone-bearing unit (TBU) in Hausa is the mora, and coda consonants are moraic. Since Hausa syllables can only bear up to two moras, contour tones can only occur in bimoraic syllables. Table 1, adapted from Dresel (1977), shows the relationship between the syllable weight and the vowel duration. The bimoraic syllable CVC, although considered as heavy, contains a short vowel referred to as a “half vowel” by Newman (2000: p. 711).

Table 1: Syllable weight and duration

<table>
<tr><th>Phonological weight</th><th>Syllable</th><th>Phonetic duration</th><th>Tone</th></tr>
<tr><td>light (monomoraic)</td><td>CV</td><td rowspan="2">short</td><td>H, L</td></tr>
<tr><td rowspan="2">heavy (bimoraic)</td><td>CVC</td><td rowspan="2">H, L, HL</td></tr>
<tr><td>CVV</td><td>long</td></tr>
</table>

2.2 WHQ pattern in Hausa

Newman (2000: p. 612) summarizes that Hausa applies four intonational patterns at the sentence level: declarative, interrogative, sympathetic address, and vocative, of which the first two are relevant to this study. The declarative pattern, common in both statements and WHQs in African tone languages, is characterized by a gradual downward slope in pitch (known as *downdrift*) (Downing & Rialland 2016). Specifically, several studies have reported that the final H tone often changes into a falling tone in WHQs (Kraft & Kirk-Greene 1973, Miller & Tench 1980, Newman 2000). However, as shown in Table 2, some observations report nearly opposite findings, with the final syllable described as either having a rising pattern (Dresel 1977) or being raised overall (Meyers 1976).

Table 2: Descriptions of intonation patterns

Source	WHQ patterns
Kraft & Kirk-Greene (1973)	overall higher-pitched declarative pattern with final H becoming F
Meyers (1976)	raised pitch in the final syllable
Dresel (1977: p.101)	final rising
Lindau (1986)	less slopes and optional final rising
Miller & Tench (1980)	final H becomes F
Newman (2000: p. 613)	mandatory Q-morpheme and optional key raising

Different than other analyses, Newman proposes a separate morpheme, the *Q-morpheme*, to account for the final syllable change in WHQs.[1] This Q-morpheme has two effects (6) on the final syllable to which it is attached.

(6) a. L-tone: The Q-morpheme will add a L tone if the final syllable bears a H tone, thereby producing a HL. If the final syllable ends in L or HL, the L-tone component will apply vacuously.

b. Lengthening: The Q-morpheme lengthens a short vowel in a final CV syllable; otherwise, the length component has no surface effect.

Newman (2000: p. 497) argues that historically the Q-morpheme had a segmental shape, possibly **à*, of which the vowel length and L tone are the only vestiges of the modern-day Hausa. Similar markers can be found in other languages in West and Central Africa such as /yà/ in Galambu (I.A.2) and Ron (I.A.4). Table 3 lists how syllable structures interact with the Q-morpheme in WHQs where the CV-H syllable has both effects, whereas CVC-L has neither.

Both the lengthening and the L-tone effect largely reflect what is termed as *lax question prosody* by Rialland (2009), which makes use of falling pitch contour, breathy termination, final open vowel, and vowel lengthening to prosodically signal YNQs. This prosody is commonly found along the Macro-Sudan Belt, and specifically among Niger-Congo, Nilo-Saharan, and Afro-Asiatic families (Güldemann 2008). Hausa has not been reported to use lax prosody in YNQs. The effects

[1]In Newman & Newman (1981), the Q-morpheme is proposed to appear in both YNQs and WHQs with the same effects. In the later work, Newman (2000: p. 497) clarifies that the L tone component is optional for YNQs in Standard Hausa but mandatory for WHQs.

Table 3: Q-morpheme effects on final vowels in WHQs

Final syllable	UR → SR after Q affixation	Q effect(s)
CV-H	*wà: zâì* ***fìtá*** *→* ***fìtâ:*** "Who will go out?"	length + L tone
CV-L	*yàùshè tá* ***háìhù*** *→* ***háìhù:*** "When did she give birth?"	length
CVV-H	*kúɗín wánnàn dáwà* ***né:*** *→* ***nê:*** "How much does it cost?"	L tone
CVC-L	*ìná: ká gá* ***yá:ròn*** *→* ***yá:ròn*** "Where did you see the boy?"	vacuous addition

induced by the Q-morpheme, although similar to lax prosodic features, are only obligatory in WHQs.

There is no definitive conclusion from previous phonetic studies on whether the falling glide is an intonational effect or due to the potential Q-morpheme. In fact, Dresel (1977) notes that there is a significant rise in the final syllable followed by a breathy voice at the end of WHQs, which might have given the impression of a fall. In addition, there is no comparison concerning the duration of the final vowel between WHQs and statements.

2.3 The current study

Building upon Newman's Q-morpheme, this study aims to test two effects potentially induced by the Q-morpheme. The first hypothesis (H1) posits that if the Q-morpheme consists of a L-tone component, the pitch-lowering effect will only apply to the H-toned vowel at the end of WHQs. Therefore, the pitch values of the final H-toned vowel in WHQs will be lower than those in the corresponding statements. To verify H1, the pitch values of final vowels in WHQs will be compared to those in statements.

The second hypothesis (H2) suggests that the Q-morpheme triggers a vowel-lengthening effect that only surfaces on final short vowels in CV syllables, thereby neutralizing the contrast between long and short vowels. To verify H2, the duration of final vowels will be measured and compared between statements and WHQs, as well as across different syllable types.

3 Phonetic experiment

3.1 Participants

Eight Hausa speakers participated in the study. Two male speakers (age 25–30) were recorded in a soundproof booth in the Phonetics Lab in the Linguistics Department at Stony Brook University. Participants were given enough time for preparation to get familiar with the reading sample. The rest of the speakers (six females, age between 25–40) were asked to record themselves remotely by using their own phone in a quiet room and save their audio as .wav files.

3.2 Material

The reading material selected for recording consisted of 12 WHQs and 12 corresponding statements. These two types of sentences differ only in the first lexical item: statements featured the content word *kàntí:* "(in the) store", whereas WHQs included the question word *ìná:* "where". To avoid placing the target word in the utterance-final position, a carrier sentence *tá cê:* "she said" was used following the target word. Two examples are provided in Table 4.

Table 4: Examples from the reading material

WHQ	Statement
ìná: tá gá ***àgó:gó****, tá cê:* where she see-PT clock-SG? she say-PT "where did she see a clock?" she said.	*kàntí: tá gá* ***àgó:gó****, tá cê:* Store she see-PT clock-SG, she say-PT "In the store, she saw a clock." she said.

The target word list consists of 12 words varying in their final syllable structures (4 with CV, 6 with CVV, and 2 with CVO) and final tones (4 for each H, L, HL group), as shown in Table 5. The penultimate syllables in the target words were kept consistent with a H-toned heavy syllable. While efforts were made to create a balanced list, three combinations are missing from the list, namely CV-HL, CVO-H and CVO-L. The absence of CV-HL is due to the fact that the contour HL is only associated with heavy syllables. CVO-H and CVO-L are absent because word-final consonant codas are rare in Hausa and seldom carry H or L tones. There are only three CVO-H/L lexical items out of 1668 core words in WOLD, none of which have a H-toned penultimate heavy syllable or can be properly embedded into the carrier sentences. The total number of tokens is 192 (= 8 speakers × 12 words × 2 sentence types).

Table 5: List of target words selected for the reading material

Syllable	H	L	HL
CV	*àgó:gó* "clock" *àbíncí* "meal"	*sàrbé:tì* "towel" *bá:bà* "aunt"	–
CVV	*ká:tá:kó:* "board" *bú:tú:tú:* "chimney"	*zó:bè:* "ring" *bú:kè:* "helmet"	*àbín shâ:* "drinks" *bákín sâ:* "black ox"
CVO	–	–	*bákín bût* "black boot" *bákín kwât* "black coat"

Although vowel quality should ideally be controlled, as it might interact with vowel duration, it is difficult to balance vowel quality, syllable structure, and tone while also maintaining sufficient tokens. Additionally, intrinsic duration differences due to vowel quality are less significant in a language such as Hausa where vowel length is contrastive.

3.3 Data collection

The data collection involved three major steps: auto-segmentation, manual inspection, and data extraction. Firstly, the Montreal Forced Aligner (MFA) tool (McAuliffe et al. 2017) was used for automatic phonetic alignment. Once the segmental annotation was completed by MFA, TextGrid files were automatically generated based on this segmentation.

Secondly, a manual inspection was conducted using Praat (Boersma & Weenink 2022) to verify the accuracy of the segmentation boundaries. The inspection process followed a basic principle: the onset and offset of the vowel were defined as the start and end points of the region where at least two formants could be clearly traced, and a stable formant band with relatively high amplitude can be observed in between. Aspiration and transitional phases of preceding consonants were excluded from the vowel phase. For CVO syllables, the vowel was identified as ending before the silent phase of the obstruent.

Finally, data extraction was completed by a Praat script (Arnhold 2000). The target token for analysis was the last vowel in each sentence. The script extracted duration, intensity, and F0 at 10 equidistant points per sonorant phase. All extracted data were saved in a .csv format file and further analyzed in R (R Core Team 2022). Outlier data points were filtered out, such as F0 values below 100Hz or above 350Hz, and duration longer than 200 ms, leaving 83.5% of the original data. Linear mixed-effects models were used for statistical analysis, with acoustic parameters (F0 or duration) as the response variable and participants treated as the random effect. Fixed effects included sentence type (Q or S), tone (HL, H, or L), and syllable type (CV, CVO, CVV).

4 Results

4.1 Overall contours

Figure 1 displays the contours of the final syllables from eight speakers. The horizontal panels represent H, L, and HL tones, and the vertical panels indicate the individual speakers. The first observation is that, although there are individual differences among the speakers, none of them shows a noticeable rising contour, except Speaker 4, who shows a slight rising "upflip" towards the end of HL-ending statements. All speakers have an overall falling contour for both statements and WHQs, though the steepness of the slope differs across tones and speakers. Three speakers (S4, S6, S7) demonstrate a steeper falling slope for all three tones. Five speakers (S1, S2, S3, S5, S8) show a relatively less steep slope for the level tones, especially for the H tone.

The second observation is an overall higher pitch in WHQs compared to statements, especially among H-tone vowels. Speakers S3, S4, S6, S7, and S8 display a notably higher pitch for WHQs, particularly in H-tone endings. In the L-tone columns, S3, S6, and S7 also have higher pitch patterns in WHQs. There is greater variation among the HL-toned vowels: one speaker (S1) has a higher pitch in statements, while others have the opposite pattern (S2, S3, S7), and some do not show remarkable differences. Generally, the pitch contours for WHQs and statements are largely similar, but WHQs tend to have a slightly higher pitch overall.

Figure 2 shows two distinct patterns on the sentence level from two speakers (S2 male and S7 female). The speaker S2 shows little difference between two sentence types while S7 displays a pitch raise in WHQs. Two sentence pairs, ending with a CV-H syllable *àgó:gó* and a CVV-H syllable *ká:tá:kó:* respectively, were selected to see whether a final falling can be observed in the H-toned WHQs as Newman stated. The first lexical items in the sentence pair, *ìná:* in WHQs and *kàntí:* in statements, were excluded in Figure 2. It can be seen that the starting point of the WHQ (blue) is consistently positioned above the statement (red) for both speakers in both sentence pairs. However, for the male speaker S2, the final syllables in both sentence pairs do not differ markedly from each other. For the female speaker S7, a greater pitch difference can be seen throughout the sentences, and a sharp fall is captured at the end of both sentences, especially

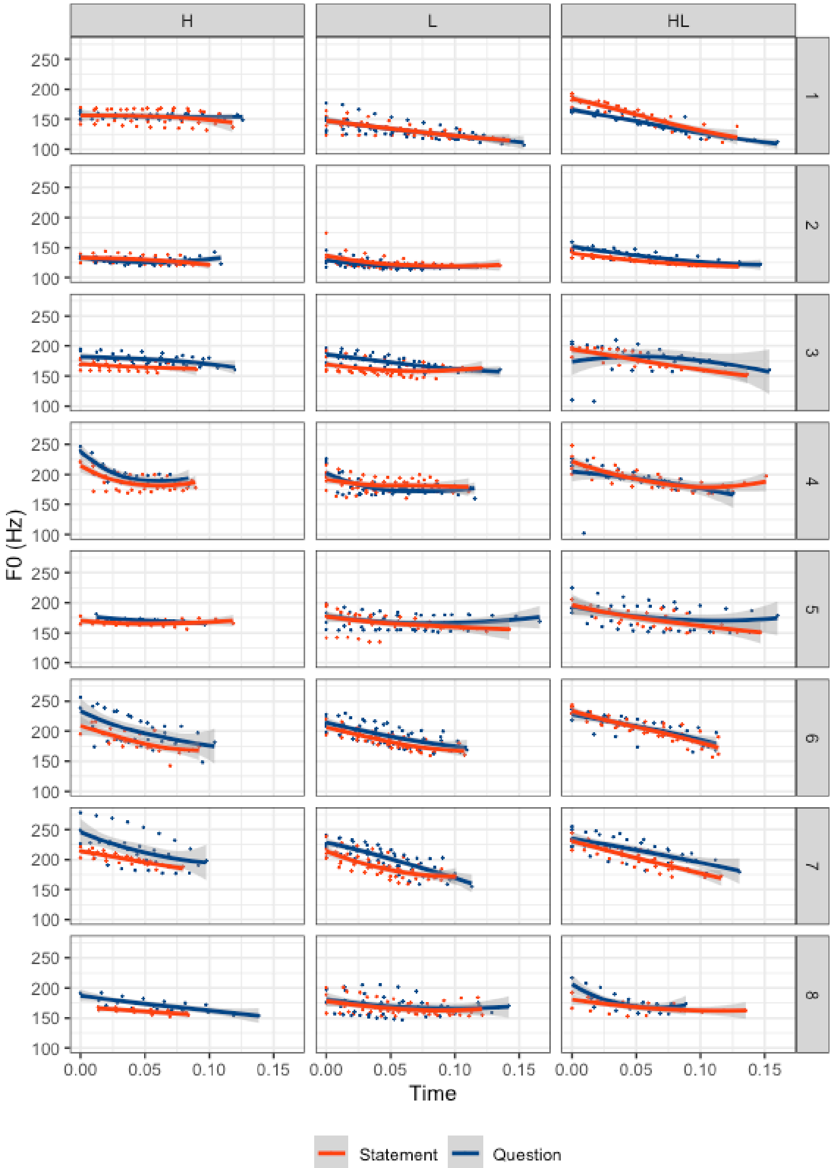

Figure 1: Final vowel contours from eight speakers

in the sentence ending with *ká:tá:kó:*. In addition, the higher pitch of WHQs is most apparent at the beginning of the pitch contour corresponding to the lexical items *tá gá* ("she saw"), particularly for the male speaker S2.[2]

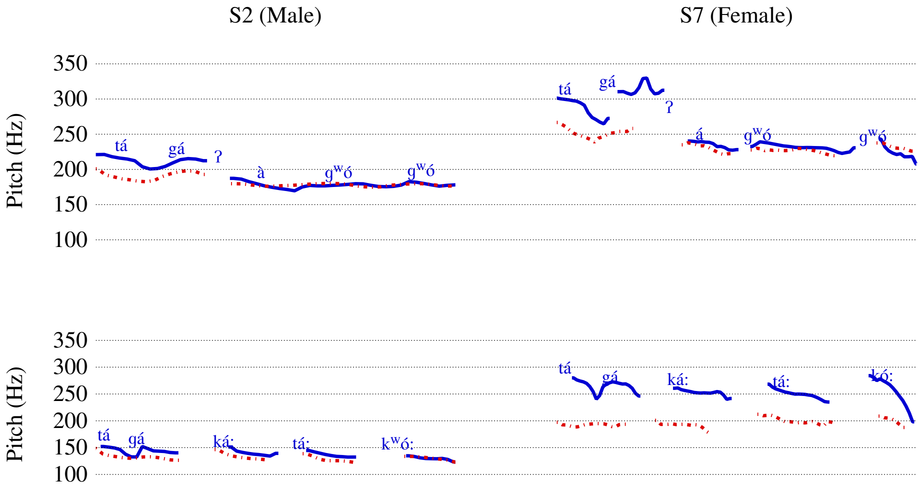

Figure 2: Questions (blue) and statements (red) ended with a CV-H syllable *àgó:gó* (upper) and a CVV-H syllable *ká:tá:kó:* (bottom) from a male speaker S2 (left) and a female speaker S7 (right)

4.2 F0 Results

Two different types of F0 values of the final syllable were measured: absolute F0 values $F0_{abs}$, which are obtained directly from data, and referenced F0 values $F0_{ref}$, which are derived from the ratio of the F0 of the final syllable to the F0 of the penultimate syllable.

Both $F0_{abs}$ and $F0_{ref}$ were fitted into a linear mixed effect model, where the fixed effects included *Tone* (H, L, HL), *Sentence Type* (*Q* for WHQs, *S* for statements), and *Duration* (ms) of the vowels. The random effect was *Participants*. The best-fitted model was selected by comparing different models using ANOVA with the lowest AIC values and the highest likelihood.

The best fitted linear mixed-effects model with the lowest AIC values for F0 analysis revealed that *Sentence Type* and *Tone* were significant predictors (p=

[2]The author, who is not a native speaker of Hausa, noted that it was generally possible to identify a recording as statement or WHQ based on the pitch of *tá gá*.

0.004** and <2.2e-16***, respectively). Table 6 presents the results of the post-hoc analysis using Tukey's test for comparing the difference between Q and S in three different tone groups. In all three types of final tones, the difference between Q and S (indicated by *Diff Q-S* in Table 6) has always been positive values (estimate >0), meaning the pitch value for Q is higher than S. Specifically, H-toned syllables were found to have a greatest difference between Q-S (5.79Hz), while HL-toned syllables have the least (2.55Hz). L-toned syllables, being in the middle, has a Q-S difference of 3.07 Hz.

Table 6: Absolute F0 difference between Q and S with Tukey's test

Diff Q-S	estimate	SE	df	t.ratio	p.value
H	5.79	1.283	1868	4.513	<.0001***
L	3.07	0.777	1868	3.951	0.0001***
HL	2.55	1.029	1868	2.475	0.0134*

The comparison of the pitch in final syllables appears to contradict Newman's predictions that a L-tone morpheme is added in WHQs. Rather, all final syllables have a higher pitch in WHQs than statements. This is more similar to what has been called a register raise (Inkelas & Leben 1990) or key raising (Newman 2000) documented in Hausa intonation. The raising is quite common for YNQs in many tone languages such as Mandarin (Shih 1986) and Embosi (Rialland & Aborobongui 2016).

Since the overall raising might cancel out the pitch lowering potentially induced by the Q-morpheme, a referenced F0 ratio was also examined to observe how the pitch of final syllables varies relative to the preceding syllable. This is feasible because in all tokens, the penultimate syllables were controlled as a bimoraic H-toned syllable.

$$F0_{ref} = \frac{final\ mean\ f0\ (Hz)}{penultimate\ mean\ f0\ (Hz)}$$

Likewise, $F0_{ref}$ was fitted into the linear mixed-effects model where *Tone* was a significant predictor (p =9.680e-06***). The post-hoc analysis in Table 7 reveals a significant $F0_{ref}$ difference between Q and S exclusively in H-tone group. This can be seen from the estimate of -0.037 in Table 7, which indicates that H-toned vowels in WHQs have a lower pitch compared to those in statements. Figure 3 shows two types of F0 values across three tone groups where the significance levels indicate the differences between WHQs and statements.

Table 7: Referenced F0 difference between Q and S with Tukey's test

Diff Q-S	estimate	SE	df	t.ratio	p.value
H	-0.037	0.013	1873	-2.839	0.005**
L	-0.002	0.008	1870	-0.318	0.751
HL	0.007	0.010	1870	0.703	0.482

The result from the $F0_{ref}$ aligns with Newman's proposal that only H-toned final syllables are subject to the L-tone effect when a Q-morpheme is attached. An overall pitch raise was observed across all three groups, with the H-toned group showing the greatest increase and the L-toned group the least. It is likely that the pitch lowering of H-toned syllables could be obscured by the overall raising if not compared with the penultimate syllable.

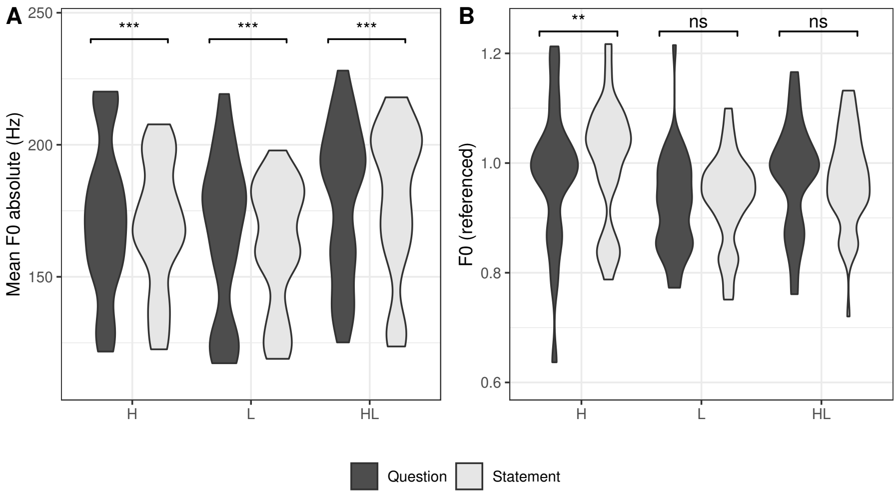

Figure 3: Violin plots showing Absolute and Referenced F0 values across three tone groups. Significance levels in both plots correspond to the p-values from Tables 6 and 7, indicating the F0 differences between WHQs and statements.

4.3 Duration

To verify whether the Q-morpheme causes lengthening of a final short vowel, a referenced duration was used to balance individual speech rates and account for the influence of tone shape (Gordon 2001, Zhang 2001) and syllable position (Dresel 1977) on vowel duration. Referenced duration Dur_{ref} is defined as the proportion of the duration of the final sonorous phase relative to the total duration of the last two sonorous phases. For instance, the referenced duration of the final vowel in *ká:tá:kó:* is calculated by dividing the duration of the final /ó:/ by the sum of the duration of both the final /ó:/ and the penultimate vowel /á:/.

$$Dur_{ref} = \frac{final\ sonourance\ Duration\ (ms)}{penultimate\ sonourance\ Duration\ (ms) + final\ sonourance\ Duration\ (ms)}$$

The best fitted linear mixed-effects model indicated significant main effects of *Sentence*, *Tone*, and *Syllable* (p <2.2e-16*** for all three factors). Additionally, interaction effects were observed between *Tone~Sentence* (p =1.744e-05***), and *Sentence~Syllable* (p =9.536e-07***). Table 8 and Figure 4 demonstrate that all groups, except CVV-HL, show a significantly greater duration in WHQs than statements. In terms of the Q-S differences among syllable types, the largest duration increase was observed in CV-H (0.054) followed by CVO (0.043). CV-L, CVV-H, and CVV-L have very close Q-S differences (from 0.023 to 0.027). CVV-HL is the only group that does not show a significant Q-S difference, with Q being slightly lower than S, hence the negative estimate (-0.008).

Table 8: Duration changes in tone and syllable combinations

Diff Q-S	estimate	SE	df	t.ratio	p.value
CV-H	0.054	0.008	1868	7.053	<.0001***
CV-L	0.023	0.007	1868	3.325	0.0009***
CVO-HL	0.043	0.009	1868	4.935	<.0001***
CVV-H	0.027	0.007	1868	3.602	0.0003***
CVV-L	0.020	0.005	1868	3.954	0.0001***
CVV-HL	-0.008	0.006	1868	-1.324	0.1856

Moreover, to examine if there is any length neutralization, Table 9 compares short vowels in CV and CVO syllables (in both questions and statements, labeled as CV(Q), CVO(Q), CV(S), and CVO(S), respectively) and long vowels in CVV syllables (only in statements, labeled as CVV(S)). For both H- and L-toned vowels, there is a significant duration difference between CV(S) and CVV(S), indicating that in neutral conditions (before the possible affixation of the Q-morpheme), CV syllables have a shorter vowel duration than CVV syllables. However, when

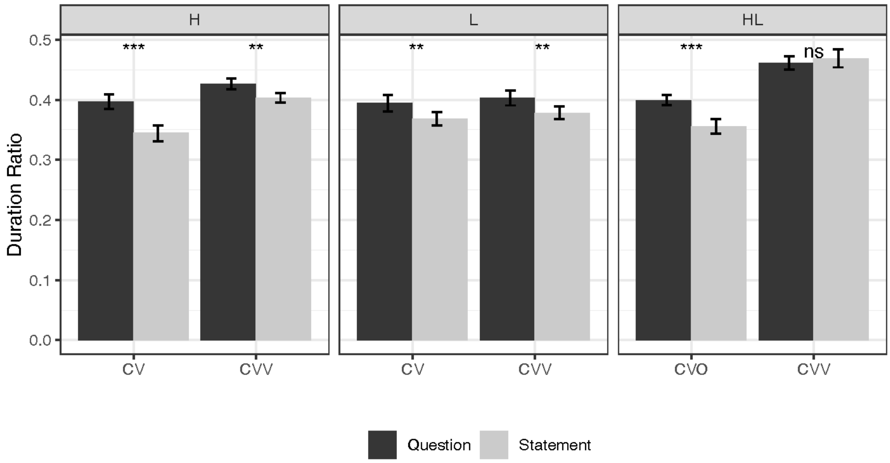

Figure 4: Bar plots showing Referenced Duration across all tone and syllable combinations. Significance levels correspond to the p-values from Table 8, indicating the duration differences between WHQs and statements.

the short vowels appear in WHQs sentence-finally, the difference between CV(Q) and CVV(S) is neutralized (p-value = 1 for H-toned, and 0.651 for L-toned CV(Q)-CVV(S)), possibly due to the vowel-lengthening effect of the Q-morpheme.

For the HL-toned group, the neutralization pattern was not observed. The duration difference between CVO(S) and CVV(S) confirms that the vowel in CVO is significantly shorter than that in a CVV syllable in a neutral context. However, unlike the H- and L-toned groups, this difference was not neutralized after the CVO was placed in WHQs, as indicated by the p-value of HL-toned CVO(Q) - CVV(S) still remaining <0.001.

This confirms that the Q-morpheme lengthens the short vowels in CV syllables in WHQs regardless of tone quality, thereby neutralizing the duration difference between short and long vowels. This aligns with Newman's predictions, and further suggests that the Q-morpheme adds one mora of length to the CV syllable. Although the vowels in CVO syllables are also phonetically short and get lengthened in WHQs, they do not undergo a length neutralization.

5 Discussion

Two hypotheses based on the Q-morpheme were tested. Hypothesis H1 states that the pitch of the final vowel in WHQs will be lower than the statement coun-

Table 9: Comparisons between monomoraic syllable in WHQs and bimoraic syllables in Statement with the same tone

Tone	Comparison	Estimate	SE	df	t.ratio	p.value
H	CV(S) - CVV(S)	-0.056	0.008	1868	-7.350	<.001***
	CV(Q) - CVV(S)	-0.002	0.008	1868	-0.195	1.000
L	CV(S) - CVV(S)	-0.036	0.006	1868	-5.927	<.001***
	CV(Q) - CVV(S)	-0.013	0.006	1868	-2.060	0.651
HL	CVO(S) - CVV(S)	-0.094	0.008	1868	-12.003	<.001***
	CVO(Q) - CVV(S)	-0.051	0.007	1868	-6.979	<.001***

terparts due to the L-tone effect of the Q-morpheme. To verify this, two types of F0 values, $F0_{abs}$ and $F0_{ref}$, were examined. The results show that $F0_{abs}$ in WHQs is consistently higher than statements across three different tone groups (Diff Q-S = 5.79Hz for H; 3.07Hz for L; and 2.55Hz for HL). This confirms that an overall pitch raise applies to WHQs, with different tones reacting differently, but it does not directly verify the L-tone effect of the Q-morpheme. A referenced F0, which compares the pitch of the final vowel to that of the preceding syllable, was also measured. The result shows that $F0_{ref}$ is lower in WHQs than in statements exclusively among H-toned vowels. Therefore, it shows that WHQs exhibit both a key raising as an intonation effect and a local L-tone effect due to the Q-morpheme. However, the pitch-lowering effect might be obscured by the overall pitch raise. The F0 results reflect two possible tone-intonation interactions: intonation interacts with underlying lexical tones and with tones introduced by phonological affixation.

Hypothesis H2 predicts that the final vowel duration will be longer in WHQs than in statement counterparts, and this prolonging can only be observable for short vowels. The result, however, shows a significant duration increase occurs in all groups except for the CVV-HL group. Notably, different types of syllables have varying degrees of duration increase: CV has undergone the greatest duration increase (mostly CV-H), followed by CVO syllables, and lastly CVV. I will argue that there are two causes of vowel lengthening. One is the phonological increase induced by the affixation of the Q-morpheme, which is categorical and can result in length neutralization. The other is phonetic lengthening, likely from intonational effects like final lengthening, which does not change short vowels into long ones but only increases the duration that the current syllable can accommodate. The Q-morpheme only applies to CV syllables, while the increase seen in CVO and CVV is caused by phonetic regulation.

First of all, the duration increase seen in the CV group in WHQs is likely caused by the mora affixation from the Q-morpheme. An important duration neutralization is found between CV(Q) and CVV(S), which supports Newman & Newman (1981)'s observation that the length component of the Q-morpheme produces neutralization of the contrast between long and short final vowels. Therefore, the lengthening of CV can be ascribed to the length component of the Q-morpheme. Moreover, the neutralization indicates that the Q-morpheme adds one mora length to the final CV syllable that it attaches to.

In the CVV group, a lengthening effect is observed in simple-toned CVV syllables (H- and L-toned) but not in contour-toned CVV syllables (HL-toned). Regarding the absence of lengthening in CVV-HL syllables, Figure 4 shows that CVV-HL syllables in statements have the longest duration among all groups. These syllables may have reached their maximum duration limit both phonologically and phonetically, which prevents any further lengthening from the Q-morpheme or additional phonetic adjustments.

In contrast, an increase was found among the H-toned and L-toned CVV groups, which does not align with Newman's prediction. For simple-toned CVV syllables, the length component (an extra mora) of the Q-morpheme remains unrealized, as seen in the CVV-HL group, because the syllable is fully occupied and leaves no space for affixation. However, there may be a phonetic lengthening that increases vowel duration without altering the current syllable structure. This could also explain why CVV-HL does not undergo lengthening, as the vowels bearing the HL contours are already at their maximum duration.

Similar to simple-toned CVV syllables, the duration increase in CVO syllables is likely due to a phonetic effect rather than a phonological lengthening induced by the Q-morpheme. This is supported by the fact that vowels in CVO syllables in WHQs remain significantly shorter than long vowels in CVV syllables in statements. The phonetic lengthening is possible because CVO syllables have an obstruent coda, which occupies one mora but may not be optimal for bearing a tone. This allows for greater flexibility in lengthening the short vowel without altering the current syllable structure, for example, by shortening or removing the coda. Future research could explore the duration changes of consonants to better understand how CVO syllables respond to WHQ intonation.

Overall, the production test confirms a local L-tone effect induced by the Q-morpheme on the H-toned vowels at the end of WHQs, which also undergo a pitch raise at the same time. Regarding duration, varying degrees of lengthening are observed in all groups except for the CVV-HL. The lengthening of the CV group is likely due to the Q-morpheme's length component, which results in length neutralization. The lengthening observed in the CVO and simple-toned CVV groups is likely due to phonetic adjustment that does not alter syllable weight but only lengthens the vowel to the extent that the current syllable can accommodate.

6 Conclusion

The present study focused on Newman's Q-morpheme in Hausa, which stands for a good example of how phonology intersects with phonetics in shaping intonation in a tonal language. The production test shows that the Q-morpheme lowers the pitch of the final H-toned vowel relative to the preceding syllable. Additionally, there is an overall rise in pitch in WHQs, possibly due to key raising at the sentence level. This demonstrates that a local L-tone effect interacts with the pitch-raising effect, which likely explains why the L-tone effect is not easily captured at the sentence level.

For the duration change, I argued that there are two types of lengthening in Hausa WHQs. One is phonological and categorical lengthening, which applies to CV syllables and neutralizes the contrast between short and long vowels, as predicted by Newman & Newman (1981). The other type of lengthening, seen in CVO and simple-toned CVV syllables, should not be considered as the same lengthening effect from the Q-morpheme, as it does not alter the syllable weight. Rather, it may be an intonational effect that phonetically increases the final vowel duration while still fitting within the current syllable structure.

These results confirm that the WHQ pattern in Hausa cannot be attributed solely to the Q-morpheme since the pitch-lowering affects only H-toned vowels but lengthening is not limited to CV syllables. Besides, an overall pitch raise was observed in syllables bearing all three tones, and a non-categorical lengthening was significant in both CVO and simple-toned CVV syllables. These observations suggest that, in addition to the Q-morpheme, other phonetic adjustments or intonational effects are influencing the WHQ patterns in Hausa. Although the current work does not determine which method is more prominent, it reveals that Hausa WHQs are influenced by both phonetic and phonological factors, and indicates an interaction between tone and intonation.

There are possibly other factors contributing to the observed patterns of Hausa WHQs. One of them could be the focus realization or the word order in the statement carrier sentences: the statement carrier sentences used in this study are similar to "In the store, she saw…" rather than "She saw a… in the store." One speaker confirmed that this word order is correct but perhaps sounds literary. Besides, the current data also show that H-toned syllables are generally shorter than L-toned syllables in Hausa, suggesting that the correlation between tone and length should be considered when examining syllable duration. Whether this is individual variation or a systematic pattern is worth further exploration.

In future research, it would be valuable to investigate breathy voice, as reported by Dresel (1977), which is considered a key feature of lax prosody. Additionally, a perception test on WHQs could help determine whether phonological affixation or phonetic adjustment has a more dominant effect on perception.

Abbreviations

F	Falling tone
F0	Fundamental frequency
H	High tone
L	Low tone
PT	Past tense
Q	Question
SG	Singular
WHQ	Question-word sentence
YNQ	Yes/no question

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Chapter 11

An overview of Nduga phonology

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I present an overview of the phonology of Nduga, a Sara-Bongo-Bagirmi language in Central African Republic. Nduga's consonant system includes implosives, prenasalized plosives, prenasalized fricatives, and labial-velar plosives. It has two rhotics, /r ɽ/. Nduga has eight oral monophthongs, one oral diphthong /ᵘa/, and five nasal vowels. Vowel length is contrastive. Oral vowels exhibit vowel harmony between /e o/ and /ɛ ɔ/. Final vowels often reduce to schwa or elide completely. Nduga has three tone levels. All nine combinations of tones occur on bisyllabic nominals. Verb infinitives have lexical tone, with the patterns HH, HM, MM, ML, and LL occurring on bisyllabic verbs.

1 Introduction

In this chapter, I present an overview of the phonology of Nduga (pronounced [ndū.gā]), a speech variety found in the northcentral part of Central African Republic. It is found mostly in the Nana-Grébizi prefecture (Kaga Bandoro subprefecture), but it is also found on the eastern border of Ouham prefecture (Kabo subprefecture). More precisely, it is spoken in and around the villages of Ouandago, Konvi, and Bissekebou (aka Bissikibou) (Moéhama 2021: 7).

Ethnologue considers Nduga to be a dialect of the Luto language (ISO 639-3 code: [ndy]), which as a whole is spoken by about 19,000 people in Central African Republic and Chad (Eberhard et al. 2024). *Ethnologue* lists five dialects of the Luto language: Luto, Nduga, Ndokoa, Wada, and Konga. The first four of these likely correspond to what Boyeldieu et al. (2006) refer to as Luto (lto), Nduga (ndg), Ndoka (ndk), and Wad (wad), respectively. That source provides

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extensive wordlist data for these four varieties.[1] The Nduga data therein were collected by Pierre Nougayrol and include about 460 lexical items.

Nduga is currently classified as part of the Sara-Bongo-Bagirmi (SBB) group within the Central Sudanic family. Dimmendaal et al. (2019: 343–344) consider SBB to be essentially the same as "West Central Sudanic." More general information can be found in *Ethnologue*[2], as well as *Glottolog*[3] (Hammarström et al. 2024).

Some previous research exists on the various varieties of the Luto language. Linguistic descriptions include Moundo (1975) (Luto variety), Ndoko (1991) (Luto variety), and Mololi (2011) (Ndokoa variety). In addition, there is a language survey (Moéhama 2021), demographic information in Nougayrol (1990: 75–77), lexical data in Boyeldieu et al. (2006), a comparative paper looking at the evolution of labial-velar consonants (Olson 2024) (Luto and Nduga varieties), and brief mention in several other articles by Boyeldieu and Nougayrol, as well as Tucker & Bryan (1956: 17) (Ndokoa dialect) and Bender (1997: 34). To my knowledge, the current work is the first to focus on Nduga phonology.

My research was conducted in Bangui, Central African Republic, during three visits from 2012 to 2017, for a total of about four weeks. I worked with a team of three speakers, who are fluent in both the Luto and Nduga dialects. Unless otherwise noted, the data in the paper are from our consulting sessions. Part of the research was done in collaboration with Paul & Jo Murrell and Brad & Maria Festen.

We collected mostly lexical data, but also some paradigms and texts. The wordlist in the appendix is based on the 204-item wordlist tool in Moñino (1988).

I discuss consonants in Section 2, vowels in Section 3, tone in Section 4, syllable patterns in Section 5, and word patterns in Section 6. Additional discussion is provided in Section 7, including addressing the question of how Nduga relates to the rest of SBB. Concluding remarks are given in Section 8.

Phonemic forms are enclosed by forward slashes, e.g. /ᵑg͡b/, and phonetic forms are enclosed by square brackets, e.g. [ŋgb]. Superscripts and tie bars are used in phonemic forms in order to help distinguish them visually from phonetic forms, as well as to help show the unitary nature of the phonemic digraphs and trigraphs. The phonetic data are transcribed using the extant International Phonetic Alphabet (IPA 2006).

[1] The data from Boyeldieu et al. (2006) can be downloaded from the following URL: https://en.wiktionary.org/wiki/Appendix:Proto-Sara-Bongo-Bagirmi_reconstructions.

[2] https://www.ethnologue.com/language/ndy

[3] https://glottolog.org/resource/languoid/id/luto1241

2 Consonants

Nduga has 28 consonant phonemes, as shown in Table 1.

Table 1: Consonant phonemes in Nduga

	Labial	Alveolar	Retr./Pal.	Velar	Labial-velar	Glottal
Implosive	ɓ	ɗ				
Plosive	(p)	t		k	(k͡p)	
	(b)	d		g	g͡b	
	(ᵐb)	ⁿd		ᵑg	(ᵑg͡b)	
Fricative	(f)	s				(h)
	v	z				
	ᵐv	ⁿz				
Nasal	m	n				
Trill/flap		r	ɽ			
Lateral		l				
Approximant			j		w	

Labial-velar and prenasalized consonants are written with two or three symbols, but they pattern as unitary phonemes in Nduga. Parentheses indicate phonemes that are marginal to the phonological system.

Sample contrasts between the consonants within the same broad category of place of articulation are given in Table 2. Insofar as possible, I provide contrast between nouns in word-initial position with a following /a/.

The period (full stop) symbol < . > indicates a syllable break. The hyphen < - > indicates a morpheme break. Glosses were obtained in French and translated into English for this paper. Corresponding entries in the wordlist are indicated by the number sign < # > followed by the entry number. Corresponding entries in the Boyeldieu et al. (2006) wordlist are given as N/[number] for nouns and V/[number] for verbs. Transcriptions from that source are unmodified.

Several phonemes are rare: /p, ᵐb, h/ do not occur in the wordlist, while /b, f, k͡p, ᵑg͡b/ occur only once or twice. These phonemes are also the rarest ones in Nougayrol's data, with 10 or fewer occurrences of each phoneme. The implosives /ɓ, ɗ/ are more common than their plain counterparts /b, d/ in the wordlist.

Boyeldieu et al. (2006) include a retroflex lateral [ɭ] in their transcriptions of the Luto varieties. I initially employed the same transcription. However, examination of the acoustic properties (cf. Figure 1) showed that the sound exhibits a short

Table 2: Consonant contrasts in Nduga

	Phoneme	Phone	Example	Gloss	No.
Labial	/ɓ/	[ɓ]	[ɓā.rà]	rainy season	# 163
	/p/	[p]	[pì.píī]	hot pepper	
	/b/	[b]	[bā.ŋgàm]	sweet potato	
	/ᵐb/	[mb]	[mbà.mbò]	one hundred	
	/f/	[f]	[fã́ã̄]	type of tuber	
	/v/	[v]	[và.ɗù]	pig	
	/ᵐv/	[ɱv]	[ɱvá.lè]	flute (from horn)	
	/m/	[m]	[mà.nɛ̀]	water	# 54
	/w/	[w]	[wā]	granary	
Alveolar	/ɗ/	[ɗ]	[ɗá.rə́ tà.rú]	to stand	# 47
	/t/	[t]	[tā.rà]	mouth	# 14
	/d/	[d]	[dā.lā]	drum	
	/ⁿd/	[nd]	[ndà.rā]	skin	# 131
	/s/	[s]	[sà.ná]	type of basket	
	/z/	[z]	[zā.rù]	civet	
	/ⁿz/	[nz]	[nzā.mè]	ground squirrel	
	/n/	[n]	[nā.nā] ~ [nā.nū]	maternal uncle	# 126
	/l/	[l]	[là.mú]	clothing	
	/r/	[r]	[rā]	sky	
Retr./Pal.	/ɽ/	[ɽ]	[ɽā.ɗà]	type of net	
	/j/	[j]	[jà.kú]	quick, fast	
Velar	/k/	[k]	[kā.gā]	tree	# 9
	/g/	[g]	[gā.zù]	horn	# 36
	/ᵑg/	[ŋg]	[ŋgá.ɽā]	ground	# 178
Lab.-velar	/k͡p/	[kp]	[kpā.rù]	poison	
	/g͡b/	[gb]	[gbā.gū]	wing	# 3
	/ᵑg͡b/	[ŋgb]	[ŋgbā.ɽā]	assegai	# 162
Glottal	/h/	[h]	[hà]	pour	

closure period of about 30 ms, which is indicative of a tap or flap (Catford 1977: 130) rather than a lateral. In addition, there is a significant reduction in intensity during closure. These properties lead me to consider the sound to be a retroflex flap [ɽ] and transcribe it accordingly.

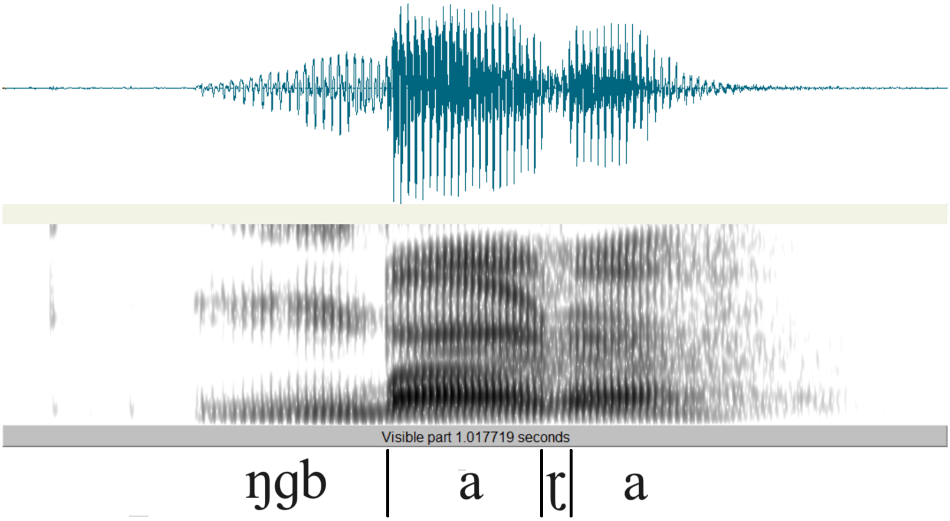

Figure 1: Waveform and spectrogram of the word [ŋgbā.ɽā] 'assegai' (# 162)

In word-initial position, the retroflex flap [ɽ] is preceded by a short vowel-like segment, as shown in Figure 2. This prothetic segment is discussed by Ladefoged & Maddieson (1996: 240–241) for a post-alveolar flap in Walpiri. They state that the "segment assists in setting up the acoustic cues for the flap, which thus become comparable in both initial and intervocalic environments" (p. 240).

A similar prothetic segment has been observed for the word-initial labiodental flap [ⱱ] in Mono (Olson 2005: 126). It has also been attested for alveolar trills in several languages (Bolter 2021) (cf. Ladefoged & Maddieson 1996: 219 for Italian). However, the alveolar trill /r/ in Nduga is not accompanied by this prothetic segment.

The phoneme /j/ is nasalized before a nasal vowel, as shown in (1):

(1) Nasalization of /j/
/j/ → [j̃] / _Ṽ

The nasalized allophone [j̃] could potentially be interpreted instead as a phoneme /ɲ/ which causes a following vowel to become nasalized. Boyeldieu

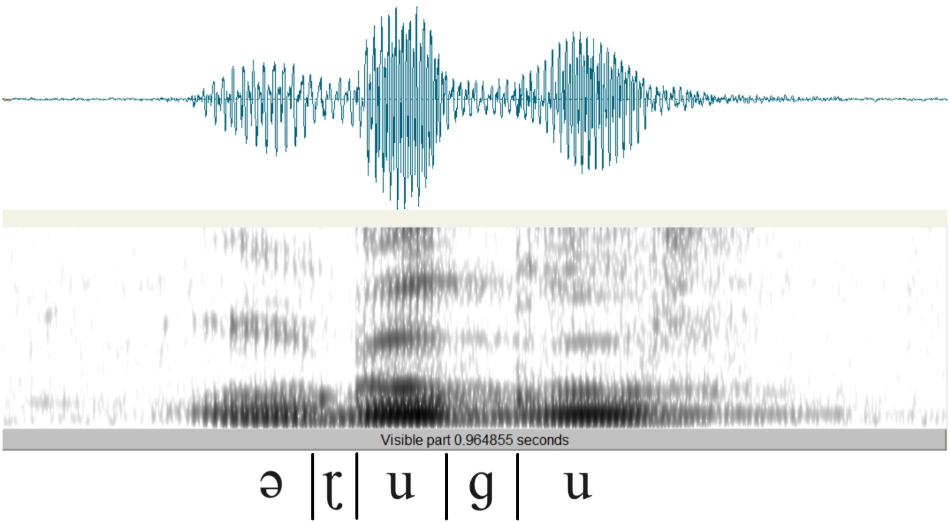

Figure 2: Waveform and spectrogram of the word [ɽū.gū] 'to burn (transitive)' (# 18), with prothetic vowel-like segment

et al. (2006) implicitly follow this analysis as they employ the symbol [ɲ] for the sound in their transcriptions of the Luto varieties.

However, I opt to interpret [j̃] as an allophone of /j/ for a several reasons. First, nasal vowel phonemes are independently motivated in Nduga (cf. Section 3). Second, I did not find other cases of nasalization on vowels occurring adjacent to a nasal consonant. Third, the speakers' intuitions were that [j] and [j̃] comprise a single phoneme. Fourth, the Nduga speakers noted that the tongue does not touch the roof of the mouth during the articulation of the sound. Phonetically, the sound is [j̃], not [ɲ].

Sample words that show the nasalization of /j/ are given in Table 3:

Table 3: Sample words with nasalized [j̃]

Phonemic	Phonetic	Gloss	Number
/vī.jā̃/	[vī.j̃ā̃]	goat	# 27
/zɔ̄.jɔ̃̀/	[zɔ̄.j̃ɔ̃̀]	to bite	# 110
/jɛ̃́.lɛ̃̀/	[j̃ɛ̃́.lɛ̃̀]	wind	# 195
/jà̃.lú/	[j̃à̃.lú]	rheum	
/jà̃.ɓū/	[j̃à̃.ɓū]	shea fruit	

There is possibly a comparable rule for /w/, but our data are limited to one example: /kũ̄.wũ̏/ [kũ̄.w̃ũ̏] ‘to drink’ (# 12).

3 Vowels

Nduga has nine oral vowels and five nasal vowels. The oral vowels include eight monophthongs and one diphthong, as shown in Table 4.

Table 4: Vowel phonemes in Nduga

	Front	Central	Back
Oral	i		u
	e	ə	o
	ɛ		ɔ
		a	(ᵘa)
Nasal	(ĩ)		(ũ)
	(ɛ̃)		(ɔ̃)
		(ã)	

Contrast between the oral vowels is given in Table 5. I provide contrast between vowels immediately following a word-initial /k/:

Table 5: Oral vowel contrasts in Nduga

Phoneme	Phone	Transcr.	Gloss	Number
/i/	[i]	[k-ī.lì]	to be black	# 117
/e/	[e]	[kē.mè]	stomach	# 196
/ɛ/	[ɛ]	[k-ɛ́.ɽɛ̄]	to collect water, to bail	
/ə/	[ə]	[kə̀]	(grammatical particle)	
/a/	[a]	[kā.là]	alone, separately	
/u/	[u]	[kú.lù]	charcoal	# 24
/o/	[o]	[kō.lō]	type of tree	
/ɔ/	[ɔ]	[kɔ̄.lɔ̀]	giraffe	
/ᵘa/	[u̯a]	[k-u̯à]	to die	# 113

Nduga exhibits vowel harmony: The vowels /e o/ do not cooccur with /ɛ ɔ/ in the same root. There is very little affixation in Nduga, so the root is usually the same as the word.

The vowels /ə/ and /ᵘa/ are somewhat rare in the wordlist, only occurring in a handful of words, as shown in Table 6.

Table 6: Cases of /ə/ and /ᵘa/ in the wordlist

Phoneme	Transcription	Gloss	Number
/ə/	[ɗá.rə́ tà.rú]	to stand (stay + upright)	# 47
	[mə̄]	1SG.SBJ	# 96
	[ŋgé.ŋgə̄ sá.kū]	panther (lion + little)	# 130
/ᵘa/	[k-ū.gu̯à]	to cut (with an axe)	# 39
	[nzū.ŋgu̯à]	twin	# 97
	[k-u̯à]	to die	# 113
	[ní.ŋgá ku̯â]	snake (thing + die)	# 172

The rarity of /ə/ in the wordlist is due to its absence in monomorphemic words in major grammatical categories. Most of its occurrences involve the reduction of another vowel to schwa in word-final position before other words in a noun or verb phrase, as shown in (2).

(2) Vowel Reduction (before another word within a phrase)
$V \rightarrow ə$ / _# word

Examples of Vowel Reduction are shown in Table 7.

Table 7: Examples of Vowel Reduction

UR	SR	No.
[kā.gā] ‘tree’ + [gē] ‘PL’	⟶ [kā.gə̄ gē] ‘trees’	
[bē.gī] ‘shoulder’ + [má] ‘1SG.POSS’	⟶ [bē.gə̄ má] ‘my shoulder’	
[ŋgé.ŋgē] ‘lion’ + [sá.kū] ‘little’	⟶ [ŋgé.ŋgə̄ sá.kū] ‘panther’	# 130

In some cases, the reduced vowel is instead completely elided, as shown in Table 8. Mololi (2011: 106) describes a similar process involving the plural morpheme in the Ndokoa variety.

Table 8: Example of Vowel Elision

UR		SR	No.
/ɗá.rá/ ‘remain’ (V/094) + /tà.rú/ ‘standing’ (N/063)	⟶	[ɗá.rə́ tà.rú] ‘to stand’ ~ [ɗár tà.rú]	# 47

Table 9: Schwa in grammatical function words

Phonetic	Gloss	Number
[mə̄]	1SG.SBJ	# 96
[nzə̀]	PFV	
[má.nə̄]	FUT	
[sə̀]	this	
[kə̀]	(grammatical particle)	

The only cases where schwa is not a reduced form of another vowel involve grammatical function words. Examples are shown in Table 9.

This situation, in which a phoneme has a robust presence only in grammatical particles, is attested elsewhere. Parker (2009) mentions three languages in which a consonant phoneme only occurs in an affix, and never in a lexical root: Arammba [stk] (PNG), Awara [awx] (PNG), and Guambiano [gum] (Colombia), cf. Quigley (2003), Branks & Branks (1973).

I consider /ᵘa/ [u̯a] to be a diphthong for several reasons. First, there are no other syllable medial approximant-vowel phonetic sequences in Nduga. Second, the approximant portion [u̯] of the phoneme is phonetically short. Third, on the two words in the wordlist that have falling tones on the phoneme ([k-u̯à] ‘to die’ # 113, [ní.ŋgá ku̯â] ‘snake’ # 172), the tone occurs on the [a] portion of the phoneme.

Figure 3 shows an F1 vs. F2 plot of the Nduga oral monophthongs. Ten tokens of each vowel are plotted. A vowel symbol is given for each individual token. The axes are marked in Hertz but scaled on the Bark scale. The ellipses are centered on the mean for each vowel and have radii of two standard deviations (Maddieson & Anderson 1994). There is reasonable separation between most of the vowels, indicating that the values of F1 and F2 are sufficient acoustic properties for distinguishing the vowels.

In addition to the oral vowels, Nduga also has a series of nasal vowels [Ṽ]

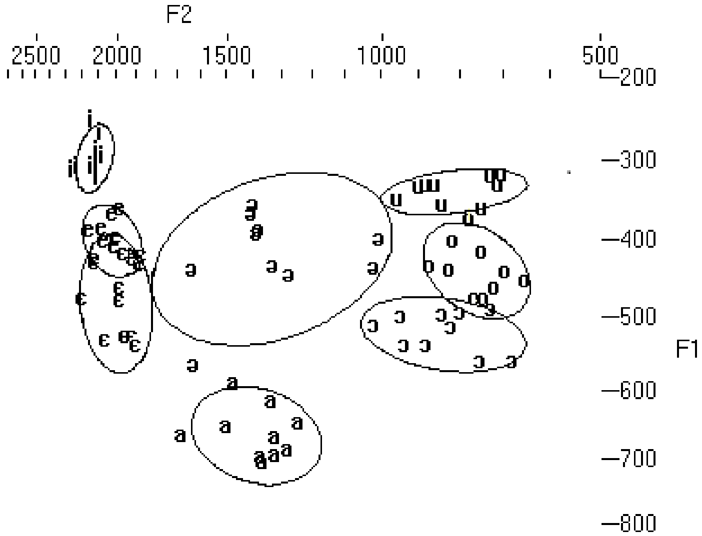

Figure 3: Plot of F1 vs. F2 for Nduga oral vowels

and long vowels [Vː]. Not enough data were collected to provide contrast in the same environment for all possible nasal and long vowels. Tables 10 and 11 provide samples of the sounds.

Table 10: Nasal vowels in Nduga

Phoneme	Transcription	Gloss	Number
/ĩ/	[ȷ̃ĩ̄ĩ̀]	scorpion	
/ɛ̃/	[ȷ̃ɛ̃́.lɛ̀]	wind	# 195
/ã/	[zã̄]	animal	# 6
	[hã̀]	galago	
	[fã́ã̄]	type of tuber	
/ũ/	[k-ũ̄.w̃ũ̀]	to drink	# 12
/ɔ̃/	[zɔ̄.ȷ̃ɔ̃̀]	to bite	# 110
	[nɔ̃̄ɔ̃̄]	tears	# 141
	[nɔ̃́]	other	

Two lexical items in our data contain vowels which are both nasalized and lengthened, shown in Table 12.

Table 11: Long vowels in Nduga

Phoneme	Transcription	Gloss	Number
/iː/	[mìí]	five	# 30
/eː/	—		
/ɛː/	—		
/aː/	[nzāà]	white	# 11
	[sāà]	smoke	# 82
	[māā]	thigh	# 43
	[bāà]	papa	# 132
	[kàà]	mother-in-law	
	[ɱvāà]	tsetse fly	
/uː/	[k-ūù]	to swell	# 59
/oː/	[kóò]	mother	# 108
	[tōō]	to blow	# 174
	[kóó tì.gì]	bee (mother + honey)	# 1
/ɔː/	[ɗɔ̄ɔ̄]	war	# 89
	[sɔ̀ɔ́]	four	# 151
	[sɔ̄ɔ̄]	type of trap	

Table 12: Long nasal vowels in Nduga

Phoneme	Transcription	Gloss	Number
/ãː/	[fã́ã̄]	type of tuber	
/ɔ̃ː/	[nɔ̃̄ɔ̃̄]	tears	# 141

The long vowels could be interpreted as either phonemic long vowels (Vː) or sequences of identical vowels (V_1V_1). I have chosen the first option, because there are no cases of dissimilar vowel sequences (i.e. V_1V_2) in Nduga, given that [u̯a] is considered to be a diphthong.

The long vowels often bear two distinct level tones (cf. Table 11). The tone patterns attested on long vowels are: HH HM HL MM ML LH LL. Since long vowels can bear two tones, I generally write them with two identical vowel symbols in order to allow marking the two tones in a clear manner.

Short vowels sometimes have contour tones, which I consider to be sequences of level tones. This is discussed further in Section 4.

4 Tone

Nduga has three robust level tones: high (H), mid (M), and low (L). Sample contrasts are shown in Table 13.

Table 13: Tone contrasts in Nduga

	Transcription	Gloss	Number
H vs. M	[kpɔ́.rɔ̀]	flavor of a salt substitute	
	[kpɔ̄.rɔ̀]	path	
H vs. L	[kī.lá]	tail	# 152
	[kū.là]	work	# 187
M vs. L	[kī.rī]	bundle of sticks	# 67
	[kē.rì]	type of bird	

In addition, several contour tones occur on short vowels. If these are analyzed as sequences of level tones, the patterns HM HL and ML are attested, as shown in Table 14.

Table 14: Contour tones in Nduga

	Transcription	Gloss	Number
HM	[ja᷇]	new	# 115
HL	[wâ]	sand	# 161
	[kí.ɽí.kî]	knee	# 84
	[ní.ŋgá ku̯â]	snake	# 172
ML	[k-u̯a᷆]	to die	# 113
	[na᷆]	who	# 153
	[k-ū.lì ɱva᷆]	to nurse	# 180
	[ɓi᷆]	to fly	# 200

In the subsections that follow, I present several observations about tone in Nduga. There are likely other tonal phenomena in the language, which I leave for future research.

4.1 Lexical tone on nouns

Nouns have lexical tone. Eight of the nine possible combinations of tones occur on bisyllabic nouns, as shown in Table 15. The additional pattern (HH) is not attested on nouns, but it does occur on the numeral [zí.jó] 'two' (# 49).

Table 15: Tone patterns on bisyllabic nominals

	Transcription	Gloss	Number
HH	[zí.jó]	two	# 49
HM	[kú.nū]	nose	# 116
HL	[kú.lù]	charcoal	# 24
MH	[kū.nzá]	chicken	# 146
MM	[ŋgā.ŋgā]	tooth	# 48
ML	[kā.ɓè]	egg	# 124
LH	[mà.sú]	blood	# 166
LM	[nì.jē]	female	# 70
LL	[mà.nɛ̀]	water	# 54

4.2 Lexical tone on verbs

Verbs exhibit lexical tone on infinitives. Five of the nine possible combinations of tones occur on bisyllabic verb infinitives, as shown in Table 16.

Table 16: Tone patterns on bisyllabic verb infinitives

	Transcription	Gloss	Number
HH	[tá.zá]	to count	# 33
HM	[nzí.lē]	to say	# 50
HL	–		
MH	–		
MM	[ndī.ɗī]	to cook	# 42
ML	[tɔ̄.ɗɔ̀]	to sleep	# 52
LH	–		
LM	–		
LL	[gè.ɽì]	to know	# 34

The most common patterns on infinitives are ML and LL. We did not find any infinitives that exhibit a pattern in which the second tone is higher than the first.

4.3 Tone on pronouns

Subject and object pronouns bear M tones on all syllables. For the corresponding possessive pronouns, the first syllable is H instead of M, as shown in Table 17.

Table 17: Tone patterns on pronouns

Person	Subject	Object	Inalienable Possessive	Alienable Possessive
1SG	mə̄	mā	NOUN + má	NOUN + nā má
2SG	jī	jī	NOUN + i	NOUN + nā jí
3SG	rō	rō	NOUN + ró	NOUN + nā ró
1PL.INCL	zī.gī	zī.gī	NOUN + zí.gī	NOUN + nā zí.gī
1PL.EXCL	zē	zē	NOUN + zé	NOUN + nā zé
2PL	sē.gē	sē.gē	NOUN + sé.gē	NOUN + nā sé.gē
3PL	rō.gē	rō.gē	NOUN + ró.gē	NOUN + nā ró.gē

For the 2SG inalienable possessive pronoun, the vowel /i/ replaces the final vowel of the noun and takes its tone, e.g. [kú.nū] 'nose' + [i] '2SG.POSS' ⟶ [kú.nī] 'your (sg.) nose'.

4.4 H-spread

For nouns with a /HL/ tone pattern, the L changes to H when the plural clitic /gē/ is attached, as shown in Table 18.

Table 18: Examples of H-spread in Nduga

Singular	Plural	Gloss	Number
[mú.mù]	[mú.mú gē]	fog	# 16
[kú.lù]	[kú.lú gē]	charcoal	# 24
[ɗɔ́.vɔ̀]	[ɗɔ́.vɔ́ gē]	road	# 26
[bí.sì]	[bí.sí gē]	dog	# 28
[ká.ɱvà]	[ká.ɱvá gē]	leaf	# 73

In an autosegmental analysis, this would likely be construed as the spread of the H from the first syllable to the second syllable, accompanied by the delinking of the L.

More research is necessary to determine if this process is more widespread in Nduga.

5 Syllable patterns

Nduga has the following syllable patterns: V, CV, CṼ, CVː, CṼː, Cu̯a, and CVC, as shown in Table 19. The most common syllable pattern is CV.

Table 19: Nduga syllable patterns

	Transcription	Gloss	Number
V	[á.nɛ̄]	this	# 19
CV	[zī]	arm	# 15
CṼ	[zã̄]	animal	# 6
CVː	[mìí]	five	# 30
CṼː	[fã́ã̄]	type of tuber	
Cu̯a	[k-u̯à]	to die	# 113
	[nzū.ŋgu̯à]	twin	# 97
CVC	[kèm]	in	
	[kàm]	in front of	
	[bà.ŋgàm]	sweet potato	N/757
	[gbèr.ŋgē]	support (put on head to carry wood)	

Since Cu̯a includes a diphthong (as discussed in Section 3), it could be considered to fit in the CV syllable pattern. The syllable patterns Cu̯a and CVC do not occur with nasal or long vowels.

Most cases of the CVC syllable pattern result from the loss of a word-final vowel in a CV.CV word pattern, as shown in Table 20.

The coda position in the CVC syllable pattern is usually filled by a sonorant. In one case in the wordlist, a fricative occurs in the coda position due to the elision of a word-final vowel: [k-ō.sò] ‘pierce’ (# 186) + [ká] ‘action of crying’ ⟶ [k-ōs.-ká] ‘to cry’ (# 141).

Consonant clusters are very rare in Nduga, and they always span a syllable boundary. The only example I found that may be monomorphemic was [gbèr.ŋgē] ‘support’.

Table 20: Sample origins of words with CVC syllable pattern

Transcription	Gloss	Source
[kèm]	in	< [kè.mè] ‘stomach’ (# 196)
[kàm]	in front of, spread out on	< [kā.mù] ‘eye’ (# 123)
[ɓól̄] ~ [ɓó.lō]	to be bitter (# 5)	< SBB *ɓɔlɔ (N/370)
[ɓāl] ~ [ɓā.là]	year (# 7)	< SBB *ɓala (N/355)

An alternative analysis that might fit the data would involve treating the CV: and CṼ: patterns as bisyllabic. In this case, CV: would be divided into CV and V, and CṼ: would be divided into CṼ and Ṽ. This would reduce the number of syllable patterns in the language by one while maintaining the same number of word patterns. One advantage of my analysis is that it matches native speaker intuitions.

A couple of additional phonetic syllable patterns occur very rarely: [Ci̯a] and [Cu̯i]. These occur as the result of the shortening of some two-syllable words to one syllable in casual speech, as shown in Table 21.

Table 21: Examples of [Ci̯a] and [Cu̯i] syllables

		Gloss
[ŋgī.jà mā]	⟶ [ŋgi̯à mā]	myself
[ɽī.jā]	⟶ [ɽi̯ā]	type of gourd
[gū.jī]	⟶ [gu̯ī]	beyond

6 Word patterns

Nduga has the word patterns shown in Table 22. The most common word pattern is CV.CV.

There is a robust set of monosyllabic lexical words in Nduga. Examples are shown in Table 23. This indicates that there is no minimal word restriction in the language.

The V syllable pattern only occurs word-initially. The CV: and Cu̯a syllable patterns only occur word-finally. Many of the words that we originally thought had more than two syllables turned out to be multimorphemic. For example,

Table 22: Word patterns in Nduga

	Transcription	Gloss	No.
CV	[zī]	arm/hand	# 15
CṼ	[zã̄]	animal	# 6
CVː	[mìí]	five	# 30
CṼː	[nɔ̃̄ɔ̃̄]	tears	# 141
Cu̯a	[k-u̯à]	to die	# 113
CVC	[kèm]	in	
V.CV	[ā.nɛ̀]	this	# 19
V.Cu̯a	[ù.ku̯à]	take (cf. V/088)	# 80
CV.CV	[kā.gā]	tree	# 9
CṼ.CV	[ɟɛ̃́.lɛ̀]	wind	# 195
CV.CṼ	[vī.ɟã̄]	goat	# 27
CV.CVː	[pì.píī]	hot pepper	
CV.Cu̯a	[k-ū.gu̯à]	to chop (axe)	# 39
CVC.CV	[gbèr.ŋgē]	support (cloth put on head to carry wood)	
CV.CVC	[bā.ŋgàm]	sweet potato	
CV.CV.CV	[kí.ndà.gú]	star	# 63

Table 23: Monosyllabic lexical words in Nduga

Transcription	Gloss	No.
[zã̄]	animal	# 6
[zī]	arm	# 15
[tá]	to finish	# 75
[wà]	heavy	# 102
[ɽī]	name	# 118
[dè]	person	# 133
[wâ]	sand	# 161
[zò]	head	# 179
[ɓē]	village	# 198

[nzà.kú.zì] ‘bottom’ (# 44) likely derives from [nzà] ‘foot’ (# 136) + [kú.zì] ‘house’ (# 104). Four 3-syllable words in the wordlist appear to be monomorphemic: [kí.ndà.gú] ‘star’ (# 63), [kí.ɽí.kî] ‘knee’ (# 84), [dà.vú.là] ‘man’ (# 91), and [ɽí.kí.zī] ‘nail/claw’ (# 127).

As mentioned in Section 5, most words with a CVC word pattern can be traced to words with a CV.CV word pattern. For example, the function words [kèm] ‘in’ and [kàm] ‘in front of’ appear to come from [kē.mè] ‘stomach’ (# 196) and [kā.mù] ‘eye’ (# 123), respectively. Lexical words with an optional CVC pattern are more commonly pronounced CV.CV, e.g. [ɓól̄] ~ [ɓó.lō] ‘be bitter (# 5) and [ɓāl] ~ [ɓā.là] ‘year’ (# 7). The Proto-SBB pattern for these words was CV.CV: *ɓɔlɔ (N/370) and *ɓala (N/355), respectively.

7 Discussion

Nduga is well-situated within the SBB family (Boyeldieu 2006; Dimmendaal et al. 2019). It has prenasalized stops, labial-velar stops, and implosives, all of which are common in SBB. It has closed syllables that result from the erosion of a final vowel *CVCV > CVC. One unique aspect of its consonant inventory is the absence of /tʃ/ and /dʒ/.

Nduga shares some characteristics with Sara and others with Bongo-Bagirmi. Like Sara, it has contrastive nasal vowels and a three-tone system. Like Bongo-Bagirmi, on the other hand, it has vowel harmony and a more robust set of fricatives than is typical of Sara. This harmonizes with Boyeldieu (2006: 4)’s view that Nduga is closely related to Sara, but it forms a separate branch.

Most verbs in our data begin with [k]. The same verbs in the wordlist found in Boyeldieu et al. (2006) are given as vowel initial, without the [k]. Some forms from the two sources are compared in Table 24.

This suggests that when the root has an initial vowel, the infinitive is marked by a prefix /k-/, and when the root has an initial consonant it is marked as /∅-/.

The very last example in Table 24 shows a case where the initial /k/ appears to be an integral part of the verb root and not a prefix.

8 Conclusion

This paper makes a significant contribution to our understanding of a speech variety that has received very little attention in the literature. I’ve provided information about Nduga consonants, vowels, tone, syllable patterns, and word patterns.

Table 24: Nduga words that begin with /k-/ in our wordlist, with corresponding forms in Boyeldieu et al. (2006)

Our data	Boyeldieu et al.	Gloss	Number	Boyeldieu ref. no.
k-à.ɓà	àɓ	to go	# 4	V/051
k-ù.zà	ùzà (parfois ūzà)	to attach	# 10	V/011
k-ū.jà	ūjà	to cut	# 38	V/098
k-ū.ɗì	ūɗì	to dig	# 41	V/241
k-à.dà	àd	to give	# 51	V/024
k-ō.jò	ɔ̄y`	to give birth	# 58	V/147
k-ɔ̄	ò (ò: ?)	to hear	# 60	V/018
k-ū.là	ūlà	to send	# 61	V/085
k-ì.nà	ìnà	to throw	# 98	V/203
ké.sī	ké.sū	to cough	# 184	V/110

At the same time, it is a modest first step in the large task of describing a language. There are many possible topics for further research. The tonal system of Nduga warrants a more in-depth investigation, especially how tone is realized in the tense-aspect-mood system. A deeper examination of H-spreading could determine if the process extends to other environments. Examining in more detail the relationship between the extant Nduga forms and those of Proto-SBB would inform our understanding of the historical development of the language. The occurrence of vowel reduction/elision at the phrasal level suggests the possibility of metrical structure in the language. The H-spreading process suggests that tonal feet may be present. Finally, it would be fruitful to examine morpho-syntactically complex forms in more detail to uncover possible additional phonological alternations in the language.

Abbreviations

Abbreviations in this chapter follow the Leipzig Glossing Rules, with the following additions.

H - High tone
L - Low tone
M - Mid tone
C - Consonant
Lab. - Labial
n. - Noun
No. - Number
Pal. - Palatal

PNG - Papua New Guinea
ref. - Reference
Retr. - Retroflex
SBB - Sara-Bongo-Bagirmi
SR - Surface representation
UR - Underlying representation
V - Vowel
vb. - Verb

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Appendix A Nduga Wordlist

In this appendix, I present a 204-item wordlist of Nduga (ISO 639-3 code [ndy]). It is based on the 204-item wordlist tool from Moñino (1988). The French glosses are from the source and were used as prompts in collecting the data. The English translations are mine.

Table 25: 204-item wordlist (001–034)

	French	English	Phonetic form	Notes
001	abeille	bee	kóó tì.gì	‘mother’ + ‘honey’
002	acide (vb.)	to be tart	ɗú.sū	
003	aile	wing	gbā.gū	
004	aller	to go	k-à.ɓà	
005	amer (vb.)	to be bitter	ɓó.lō	
006	animal	animal	zā̃	
007	année	year	ɓā.là	
008	appeler	to call	ɓɔ̄.rɔ̄	
009	arbre	tree	kā.gā	
010	attacher; lier	to attach	k-ù.zà	
011	blanc	to be white	nzāà	
012	boire	to drink	k-ū̃.w̃ù̃	
013	bon (être)	to be good	gà.ɽà	
014	bouche	mouth	tā.rà	
015	bras	arm	zī	
016	brouillard	mist; fog	mú.mù	
017	brûler (INTR)	to burn (INTR)	ɽū.gū	
018	brûler (TR)	to burn (TR)	ɽū.gū	
019	ceci	this	á.nɛ̄	
020	cela	that	á.nɛ̄ ɗá	
021	ce (en question)	this	á.nɛ̄	
022	cendres	ash	vī.tī	
023	champ	garden	mē.gè	
024	charbon	charcoal	kú.lù	
025	chaud	to be hot	tù.mà ~ k-ù.mà	
026	chemin	path	ɗɔ́.vɔ̀	
027	chèvre / bouc	goat	vī.ɟā̃	
028	chien	dog	bí.sì	
029	chose	thing	ní.ŋgà	
030	cinq	five	mìí	
031	cœur	heart	ŋgā.lā	
032	colline	hill	kó.ŋgō	
033	compter	to count	tá.zá	
034	connaître	to know	gè.ɽì	

Table 26: 204-item wordlist (035–065)

	French	English	Phonetic form	Notes
035	corde	cord	kū.là	
036	corne	horn	gā.zù	
037	cou	neck	mí.ndì	
038	couper (couteau)	to cut (knife)	k-ū.jà	
039	couper (hache)	to chop (axe)	k-ū.gṳà	
040	cracher	to spit	tī.ɓī	
041	creuser	to dig	k-ū.ɗì	
042	cuire	to cook	ndī.ɗī	
043	cuisse	thigh	māā	
044	cul; fondement	bottom	nzà kú.zì	‘foot’ + ‘house’
045	cultiver	to cultivate	nzà.kà	
046	danser	to dance	k-ū.zì	
047	debout (être)	to stand	ɗá.rə́ tà.rú ~ ɗár tà.rú	‘stay’ + ‘upright’
048	dent	tooth	ŋgā.ŋgā gō.gò	incisors premolars, molars
049	deux	two	zí.jó	
050	dire	to say	nzí.lē	
051	donner	to give	k-à.dà	
052	dormir	to sleep	tɔ̄.ɗɔ̀	
053	droite (à)	right	kā vú.lā	
054	eau	water	mà.nɛ̀	
055	écorce	bark	mà.mù	
056	éléphant	elephant	kì.zì	
057	enfant	child	ŋgō.nō	
058	enfanter	to give birth	k-ō.jò	
059	enfler	to swell	k-ūù	
060	entendre	to hear	k-ɔ̀	
061	envoyer	to send	k-ū.là	
062	épine	thorn	kó.nō, mɔ́.ɗɔ̀	
063	étoile	star	kí.ndà.gú	
064	étranger	stranger	ŋgbā	
065	excréments	excrement	kí.zī	

Table 27: 204-item wordlist (066–093)

	French	English	Phonetic form	Notes
066	façonner (pot)	to make (clay pot)	k-ù.ɓà	
067	fagot	firewood	kī.rī	
068	faim	hunger	ɓō	
069	farine	flour	ndù.zú	
070	femme	female	nī.jé	
071	fer	iron	tē.ɱvē	
072	feu	fire	fú.ɗū	
073	feuille	leaf	ká.ɱvà	
074	filet	net	bá.ndà	
075	finir	to finish	tá	
076	flèche	arrow	kù.rà	
077	foie	liver	wū.rù	
078	forger	to forge	k-ō.kù	
079	frapper; battre	to hit	k-ū.ndà	
080	froid (être)	to be cold	kū.lī ù.ku̯à dè	
081	fructifier	to bear fruit	k-à.nù	bear fruit (trees, grain...)
			vā.kà	become numerous
082	fumée	smoke	sāà	
			sì.lí	
083	gauche (à)	left	kā gí.lī	
084	genou	knee	kí.ɽí.kî	
085	graisse	fat	kē.ɱvì	
	huile	oil	kù.ɓù	
086	grand	big	ɓō.ɽì	
087	gratter (se)	to scrape	k-ū.gù	
088	griller	to grill	ɽū.gū	
	faire frire	to fry	nzī.kò	
089	guerre	war	ɗɔ̄ɔ̄	
090	herbe	grass	ŋgì.lì	
091	homme	male	dà vú.là	
092	hyène	hyena	ɓɔ̀.ŋgɔ̀	
093	il	he/she	rō	

Table 28: 204-item wordlist (094–125)

	French	English	Phonetic form	Notes
094	ils	they	rō.gē	
095	intestins	intestines	tì.kpì	
096	je	I (SBJ)	mə̄	
	me	me (OBJ)	mā	
097	jumeaux	twins	nzū.ŋgu̯à	
098	lancer; jeter	to throw	k-ì.nà	
099	langue (organe)	tongue	nzò	
	langue (langage)	language	tà.rà ɓē	
100	laver	to wash	nzì.gbò	
101	long	tall	kā.wù	
102	lourd	heavy	wà	
103	lune	moon	nzó.jì	
104	maison	house	kú.zì	
105	manger	to eat	k-ū.sà	
106	marcher	to walk	ndí.gā	
107	mauvais (être)	to be bad	kà.sù	
108	mère	mother	kóò	
109	montagne	mountain	kó.ŋgō	
110	mordre	to bite	zɔ̄.ɟɔ̃̀	
111	mortier	mortar	vì.ɗì	
112	mouche	fly	jú.lū	
113	mourir	to die	k-u̯à	
114	ne...pas	not	gù.nú	
115	neuf; nouveau	new	jã́	
116	nez	nose	kó.nū	
117	noir	to be black	k-ī.lì	
118	nom	name	ɽī	
119	nombreux (être)	to be many	vā.kà	
120	nombril	navel	kū.mū	
121	nous (INCL)	we	zī.gī	
122	nuit	night	ndū.rí	
123	œil/visage	eye/face	kā.mù	
124	œuf	egg	kā.ɓè	
125	oiseau	bird	jì.lì	

Table 29: 204-item wordlist (126–156)

	French	English	Phonetic form	Notes
126	oncle (maternel)	uncle	nā.nā ~ nā.nū	
127	ongle/griffe	nail	ɽí.kí.zī	
128	oreille	ear	ɱvī	
129	os	bone	jī.ŋgō	
130	panthère	panther	ŋgé.ŋgə̄ sá.kū	‘lion’ + ‘little’
131	peau	skin/fur	ndà.rā	
132	père	father	vū.ɽà	
133	personne	person	dè	
134	petit	small	sá.kū	
135	peur	fear	ɓè.lè	
136	pied	foot	nzà	
137	pierre	rock	kɔ̄.sɔ̄	
138	plaie	wound	dī.jò	
139	planter (arbre)	to plant (tree)	k-ū.dù	
	semer	to sow (seed)	k-ù.sì	
140	plein	full	ɗɔ̄.sɔ̀	
141	pleurer (pleurs)	to cry	k-ōs-ká	‘pierce’ + ‘action of crying’
	larme/pleurs	tears	nɔ̃̄ɔ̃̄	
142	pluie	rain	nzī	
143	poil/plume	hair/feathers	vè.lé	
144	poisson	fish	kā.nzē	
145	pou	louse	ŋgí.sā	
146	poule	chicken	kū.nzá	
147	pourrir	to rot	nzū.mù	
148	prendre	to take	k-ū.nù	
149	profond (être)	to be deep	k-ū.lì	
150	puiser	to draw (water)	k-ō.dò	
151	quatre	four	sɔ̄ɔ́	
152	queue	tail	kī.lá	
153	qui ?	who	nà̄	
154	quoi ?	what	ɗī	
155	racine	root	ŋgī.rà	
156	respirer	to breathe	k-à.vò	

Table 30: 204-item wordlist (157–186)

	French	English	Phonetic form	Notes
157	rester	to stay	k-ī.nzì	
	s'asseoir	to sit down	k-ī.nzì ŋgá.ɽā	'stay' + 'ground'
158	rire	to laugh	tí.gbō	
159	rosée	dew	tā.lù	
160	rouge	to be red	k-ā.kɛ̀	
161	sable	sand	wâ	
162	sagaie	spear	ŋgbā.ɽā	
163	saison des pluies	rainy season	ɓā.rà	
164	saison seche	dry season	ɓā.lā	
165	salive	saliva	wɔ̄.rɔ̄	
166	sang	blood	má.sù	
167	sec (être)	to be dry	nzū.tù	
168	sein	breast	ɱvà	
169	sel	salt	kà.tà	
170	semence/graine	seed/grain	kī.jō	
171	sentir (INTR)	to smell	k-ā.tù	
172	serpent	snake	ní.ŋgá ku̯â	'thing' + 'die'
173	soleil	sun	kā.zà	
174	souffler	to blow	tōō j̃ɛ̃́.lɛ̀	'blow' (V/186) + 'wind'
175	sucer	to suck	tá.nū	
176	sucré (être)	to be sweet	ndī.tɛ̄	
177	termite	termite	ŋgɔ̄.ɽɔ̀	
178	terre; sol	earth; soil	ŋgá.ɽā	
179	tête	head	zò	
180	téter	nurse	k-ū.lì ɱvà	'suckle' (V/095) + 'breast'
181	tomber	fall	k-ì.sò	
182	tortue	turtle	jí.kò	
183	tous; tout	all	dū.tū	
184	tousser	to cough	ké.sī	
185	tranchant (être)	to be sharp	k-ā.tù	
186	transpercer	to pierce; to stab	k-ō.sò	

Table 31: 204-item wordlist (187–204)

	French	English	Phonetic form	Notes
187	travail	work	kū.là	
188	trois	three	mù.tá	
189	trou	hole	gɔ̄.tɔ̀	
190	tu	you (2SG)	jī	
191	tuer	to kill	tā.lā	
192	un	one	ɗɔ́.jí	
193	urine	urine	kí.zī kà.rɛ̄	‘urine’ + ?
194	venir	to come	k-ī.ɗè	
195	vent	wind	j̃ɛ̃́.lɛ̀	
196	ventre	stomach	kē.mè	
197	viande/chair	meat/flesh	zã̄, tá.ɗù	
198	village	village	ɓē	
199	voir	to see	k-ā.kù	
200	voler (oiseau)	to fly	ɓĩ̀	
201	voler; dérober	to steal	ɓō.gò	
202	vomir	to vomit	tú.ɗē	
203	vouloir	to want	ndī.gì	
204	vous	you (2PL)	sè.gè	

Chapter 12

Middle verbs in Maa: Alternating, *media tantum*, and (spurious) look-alikes

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Based on semantics, morphophonology, morphosyntax, and some diachronic evidence, this study distinguishes four types of Maa (Maasai) verbs that might initially appear to be middle forms: those which alternate with simple transitive stems, non-alternating middles (sometimes called *media tantum* in the literature), verbs partially assimilated to the *media tantum* category, and spurious "look-alikes." Compared to alternating middles, a greater proportion of *media tantum* verbs express human related or human experience concepts. The study sets the stage to understand how irregularities in a system may arise. The investigation is based on text and lexicography materials, collected collaboratively with native speakers.

1 The problem: Maa verbs ending in *a* and *o*

Maa (Eastern Nilotic of Kenya and Tanzania; also known in the literature as *Masai* and *Maasai*; ISO mas) has a set of verbs or verb stems which Tucker & Mpaayei (1955: 134-139) refer to as "neuter" forms. These verbs can or sometimes must end in *a* or *o*, and have reflexive/reciprocal, non-active, or in some cases intransitive translational movement meaning. Tucker & Mpaayei (1955: 134) analyzed *-a*/*-o* as a "present neuter" suffix, stating that it alternated with *-ayu*/*-oyu* in the "future", and *-ɛ*/*-e* in the "past." From a typological perspective, this paper argues that verbs which, at least at first glance, fit the relevant meaning types and end with *a*/*o* do not form a homogeneous group. Most do fit the profile of what are called "middle" verbs in other linguistic literature. However, this study shows that there are four subgroups of apparent middle verbs, based on semantics, morphophonology, morphosyntax, and some diachronic evidence:

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- Alternating or "oppositional" (Grestenberger 2018, Inglese 2022a,b) middles: These productively alternate with simple transitive stems.
- Maa-style *media tantum*: These are non-alternating or "non-oppositional" stems, which in citation form can only occur as a middle.
- Apparent middle stems which may be partially assimilated to the *media tantum* category.
- Spurious, though phonological and semantic look-alikes to *media tantum* verbs.

The study thus elaborates on Tucker & Mpaayei's description, and sets the stage to help us better understand the historical development of middle systems and how irregularities may arise. The investigation is, however, not primarily an historical one.

Data were culled from a list of about 540 verbs which initially appeared to be middles, drawn from lexicography and text material collected collaboratively with native speakers. The paper is organized as follows: Section 2 gives a brief treatment of typological approaches to defining "middle." Section 3 introduces basic information on Maa and relevant morphology. Section 4–Section 7 describe the four subgroups of verbs listed above. Section 8 undertakes a descriptive statistical study of the semantic profiles of alternating versus *media tantum* verbs, in an effort to understand motivations for why *media tantum* verb stems might arise. Section 9 draws brief conclusions about why categories like "middle verbs," or others, might "leak."

2 Definition of "middle"

Three characterizations of "middle" will figure in our discussion of Maa.[1] Lyons (1968: 373) defines the middle largely semantically as a construction where "the 'action' or 'state' affects the subject of the verb or his interests." In contrast, in her typological study of about 32 languages, Kemmer (1993: 3) states that "At present there is no generally accepted definition or characterization of the middle voice." That work goes on to argue for a general definition of "middle voice" as involving situation types with a relatively low degree of event elaboration; that is, there is low conceptual difference between the Initiator and the Endpoint of

[1]To maintain a coherent focus, I do not address formalist approaches to characterizing middle verbs; see Embick (2004) and Grestenberger (2018) for some introduction to that literature.

the situation. Kemmer's discussion can be read as positing the reflexive situation as the prototype of the middle category.

In more recent investigation of the nature and polyfunctionalality of middle voice systems, Inglese (2022a: 490) writes, "The greatest obstacle in the typological study of the middle is that there is no agreement on what cross-linguistically counts as a middle marker." Inglese finds Kemmer's and others' definitions still lacking for typological work and thus he proposes a multi-part definition of a "middle system" as involving a construction which:[2]

(i) Occurs with multivalent verbs to encode one or more of the following valency reducing operations: PASSIVE, REFLEXIVE, RECIPROCAL, ANTIPASSIVE, ANTICAUSATIVE. (This excludes patient-oriented argument marking in split-intransitive systems from being considered middle morphology.)

(ii) Is obligatory with some monovalent verbs; that is, they cannot occur without a "middle marker." (This ensures that some stems are lexicalized to carry the morphology; i.e., *media tantum* forms occur.)[3]

As these features hold for the Ancient Greek middle inflection, Inglese states that Ancient Greek has a "middle system."[4] Citing Delbrück (1897: 417-425) and Romagno (2010: 431-432), he notes that *-omai* 1SG.MID(DLE) occurs with verbs that alternate with non-middle *-ō* 1SG.ACT(IVE) to mark valency change, as in (1); and it occurs with verbs that exclusively occur as middles, as in (2). Valence-reducing and situational functions for the middle forms are in small caps in the right-most columns of these examples. Those in (1) are found in Inglese (2022b: 490), and I have added those in (2).

(1) alternating

[2]The parenthetical comments are my attempt to understand why Inglese (2022b) includes the criterion.

[3]Inglese (2022b) actually includes a third criterion: The semantics [or function] of at least some verbs in (i) doesn't match that of those in (ii), or vice versa. (In part, this excludes monofunctional morphemes.) The reader is referred to Inglese (2022b) for further details; this criterion is not repeated in Inglese (2023).

[4]Fortuitously, we might say, for the history of linguistic terminology.

a.	*daí-ō*	*daí-**omai***	ANTICAUSATIVE	
	burn-ACT	burn-MID		
	'burn (tr.)'	'burn (intr.)'		
b.	*kópt-ō*	*kópt-**omai***	REFLEXIVE	
	hit-ACT	hit-MID		
	'hit (tr.)'	'hit oneself'		
c.	*dikáz-ō*	*dikáz-**omai***	PASSIVE	
	judge-ACT	judge-MID		
	'judge (tr.)'	'be judged'		

(2) non-alternating (*media tantum*)

*hê**mai***	'sit'	BODY POSTURE
*kínu**mai***	'move (intr.)'	TRANSLATIONAL MOTION
*skúz**omai***	'be angry at'	EMOTION
*mákh**omai***	'fight'	RECIPROCAL

Kemmer discussed the REFLEXIVE function as perhaps the central function of a middle construction. Inglese finds ANTICAUSATIVE is the most central function in a multi-dimensional mapping study, followed (in order) by REFLEXIVE, PASSIVE, RECIPROCAL, SELF-BENEFACTIVE, ANTIPASSIVE, FACILITATIVE and IMPERSONAL.

Somewhat separately from thinking about middles only in terms of valency reducing functions, both Kemmer and Inglese discuss a range of what I call semantic "situation types" that middles commonly express. Some of these are included in the list of functions (in small caps) for the examples just above and for the Italian *si* construction illustrated in (3).

(3) Italian *si* construction, with situation types; adapted from Inglese (2022b: 492)

radersi	'shave (self)'	GROOMING/REFLEXIVE
alzarsi	'stand up'	CHANGE IN BODY POSTURE
girarsi	'turn (intr.)'	NON-TRANSLATIONAL MOTION
spostarsi	'move (intr.)'	TRANSLATIONAL MOTION
arrabbiarsi	'get angry'	EMOTION
immaginarsi	'envisage'	COGNITION
sciogliersi	'melt (intr.)'	SPONTANEOUS/ANTICAUSATIVE
combattersi	'fight'	RECIPROCAL
colpirsi	'hit oneself'	REFLEXIVE
si va	'one goes'	IMPERSONAL
si vende	'is sold'	PASSIVE
si taglia (facilmente)	'is (easily) cut'	FACILITATIVE

With this brief background on approaches to defining middle constructions and systems, I now turn to the Maa data.

3 Background on Maa

Maa is generally described as a verb initial language, though order of major constituents can vary. It is a mixed head- and dependent-marking language: verbs mark person and number of subjects and speech-act-participant objects, while post-verb nominal phrases and free pronouns display a marked nominative case system (König 2006), expressed by tone melodies which vary by lexical item (Tucker & Mpaayei 1955). Many morphemes have multiple allomorphs due to advanced or retracted tongue-root (ATR) vowel harmony, grammatical and lexical tone, and other morphophonemic processes.[5]

At a minimum, Maa verbs must index person (or number of subject for infinitives) via prefixes, and (roughly) finiteness distinctions via tone. But they can also be morphologically quite complex (Tucker & Mpaayei 1955: 51-95). Due to three intersecting parameters, a single maximal verb template does not exist. Briefly, verb roots and stems divide into two morphological classes, known as Class 1 and Class 2 (Tucker & Mpaayei 1955: 52), which differ in how causatives and perfect(ive) aspect are marked. Transitivity and stative versus dynamic lexical aspect (*Aktionsart*) also affect what morphology a root or stem can take (Payne 2015b). For example: Already intransitive stems cannot further take middle or antipassive morphology; already stative stems do not normally take directionals (without shifting to a non-directional meaning), but can take the inchoative; dynamic stems can take directionals but not the inchoative.

Maa is rich in valence morphology. Valence-increasing morphology includes the dative applicative (expressing 'goal-reached, benefactive, recipient'), the instrumental applicative ('instrument, location, manner'), and causatives which differ for Class 1 versus Class 2.[6] Valence-decreasing constructions include the impersonal (Greenberg 1959, Payne 2011), antipassive (Payne 2021b), and middles. These are illustrated where relevant in the paper.

Mood and grammatical aspect are salient categories. Morphological aspects

[5]In examples, an accute accent marks High tone, a macron marks Downstep High, a circumflex marks Falling, and Low tone is unmarked.

[6]The instrumental applicative doubles as the Class 2 causative, where an animate "instrument" is a causee. Additionally, an external possession construction increases syntactic valence but does not involve any verb morphology (Payne 1997).

include non-perfect(ive) (generally unglossed in examples), perfect(ive),[7] and progressive (König 1993, Payne 2015a). The itive ('motion away') and ventive ('motion towards') directionals also have some aspectual, as well as other grammatical functions (Payne 2021a).

We will see that Maa middles interact in particular ways with certain other morphemes. For convenience, I refer to middle plus these other relevant items as "core" affixes and in this paper simply stipulate them as the items in Table 1. They include causative, antipassive, applicative, directional, and aspect morphology. All are suffixes except the Class 1 Causative.[8]

Table 1: Core affixes (major allomorphs)

All middle forms	(see Table 2)
Progressive	*-ɪta/-ito*
Perfect(ive)	*(tV-)...-a(k)/-o(k)*
Antipassive	*-ɪshɔ(r)/-isho(r)*
Itive directional	(see Table 3)
Ventive directional	*-ʊ(n)/-u(n)*
Dative applicative	*-akɪ(n)/-oki(n)*
Instrument applicative	*-ie(k)*
Class 1 Causative	*ɪta-/ito-*

Table 2 presents middle and middle-complex endings. Note that some alternate for perfect(ive) versus non-perfect(ive) aspect themselves. None of the middle affixes or affix complexes co-occur with progressive or antipassive affixes. This is likely for semantic reasons, as the progressive and most uses of the antipassive assume dynamic readings, contrasting with the general tendency for middle functions and senses to be non-dynamic (but notably, dynamicity is typical for reflexive and reciprocal functions of the middles). The middle endings can occur on directional, applicative, and causative stems, as partially seen in Table 2; or as we will see in the case of *media tantum* verbs, they may completely replace them. Some middle forms vary for subject number and for aspect but it appears that

[7] König (1993) argues that the relevant forms are 'perfective' in function; Payne (2015a) shows they function in discourse like a 'perfect'. Here I use the term "perfect(ive)" to cover the range of functions discussed in those two works.

[8] "Core" morphology is probably old as affixal material. It mostly corresponds to what some might call "derivation" but it includes productive aspect morphology which some might consider inflectional. Participant number is expressed both in some core and non-core affixes. See Payne (2015b) for fuller discussion.

applicative-plus-middle and directional-plus-middle combinations do not display number and aspect contrasts (aside perhaps from tone).

Table 2: Middle morphology (major allomorphs)

	Singular	Plural
Non-Perfect(ive) Middle (MID.NPF)	*-a(r), -o(r), -ɔ(r)*	*-ata, -oto, -ɔtɔ*
Perfect(ive) Middle (MID.PF)	*-ɛ, -e*	*-atɛ, -ote*
Ventive+Middle	*-ʊnyɛ, -unie*	
Itive+Middle	*-ari, -ori*	
Dative+Middle	*-akino, -okino*	
Middle+Instrument	*-arɛ, -ore*	
Middle+Inchoative	*-ayu, -oyu*	

In Maa middle constructions, the only expressable argument (which would be the patient or "internal" argument of any corresponding transitive stem) is indexed as subject on the verb and occurs in the nominative case if expressed in post-verbal position by a lexical phrase or free pronoun. Compare the tonal case marking on 'heart' in (4), where 'heart' is in the absolute case (unglossed) as the transitive object, with its form in (5)–(7), where it has nominative case (glossed NOM). In (7), note that 'heart' is the only argument of a middle verb.

(4) k-ɛ́-ɛ́ta ɔl=**táʊ́** o-gól ɛlɛ́ tʊ́ŋání
CN2-3-have MSG=heart REL.MSG-be.hard that.MSG.NOM elder.NOM
'That man has a hard heart.' (e.g., doesn't show concern when s.o. dies) [*ɔltáʊ́* = transitive object in absolute case]

(5) n-áa-lilîû ɔl=**tâʊ̂** nalɛ́ŋ
CN1-3>1SG-stink MSG=heart.NOM very
'I feel like vomiting.' (lit. 'My heart stinks a lot.') [*ɔltâʊ̂* = transitive subject in nominative case][9]

[9]Example (5) is an external possession construction in which the 1SG object is interpreted as the possessor of the subject (Payne 1997). Even more literally it would read 'The heart stinks me very much.'

(6) ɔl=tʊŋáni ɔ-wáŋ ɔl=**tâʊ̂**
MSG=person REL.MSG-be.bright MSG=heart.NOM
'a person whose heart shines' (i.e., a person who has no bad intentions)
[*ɔltâʊ̂* = intransitive subject of relativized verb, in nominative case]

(7) n-é-dūŋ-ō **tâʊ̂**, n-é-yē
CN1-3-cut-MID.NPF heart.NOM CN1-3-die
'(Her) heart stops (lit. is cut/cuts), and she dies.' (nochild.034) [*tâʊ̂* = argument of middle construction in nominative case]

Finally, some middle allomorphs are quite similar phonologically to those of the perfect(ive) and itive. Thus, for clarity, Table 3 presents major allomorphs of the perfect(ive) and itive categories.

Table 3: Major perfect(ive) and itive allomorphs

Perfect(ive)	*-a(k), -o(k)*
Itive	*-aa, -oo, -ar(a), -or(o), -aya, -oyo*

4 Alternating Maa middles

I now illustrate alternating middle behavior. Middles of this type have non-middle transitive roots (or stems). Dynamic transitive roots quite productively take middle endings to express reflexive, reciprocal (in plural contexts), anticausative or spontaneous (Haspelmath 1987), and result-state functions.[10] Depending on semantics of the root, some alternating middles may also express movement, emotion, cognition, etc., as well as some quite idiosyncratic (hence, clearly lexicalized) meanings. Consider the range of senses that arise with middle forms of the dynamic transitive verb root *duŋ* 'cut' in (8); the listed senses for the intransitive middle forms are not necessarily exhaustive.

[10]By RESULT-STATE, I mean the state of an undergoer as the result of an action-process. The middle form itself expresses a state, while the transitive non-middle root would express an action-process potentially resulting in that state. Note that RESULT-STATE is not intended to (necessarily) imply a passive reading where any sense of an AGENT is even semantically retained. Also, some Maa middle forms express a STATE where it is hard to see that state as the result of a transitive action-process.

(8) a. a-dúŋ
INF.SG-cut
'to cut' DYNAMIC, TRANSITIVE

b. a-duŋ-ó
INF.SG-cut-MID.NPF
'to be in a cut state, demarcated, separated' RESULT-STATE
'to cut self, divide self (into groups)' REFLEXIVE
'to cease/disrupted flow' ANTICAUSATIVE, RESULT-STATE
(of electricity, water, cows from heaven, heart beat, etc.)
'to be past menopause' SPONTANEOUS, STATE
'to be dried up'[11] SPONTANEOUS, STATE

c. áa-duŋ-o
INF.PL-cut-MID.NPF
'to cut each other' RECIPROCAL
'to be divided (into portions)' RESULT-STATE

Additional examples of alternation between dynamic transitive and middle stems follow. In (9), the middle has quite predictable senses. In (10b), we see both predictable and metaphorical senses. In (11), the middle has a rather unpredictable sense of '(spontaneously) coagulate', even though the operative valence decreasing function is a highly productive ANTICAUSATIVE one.

(9) a. a-ɪnyɪnyɪrʊ́
INF.SG-exude.liquid.VEN
'to let out tiny amounts of liquid' (e.g., sweat) DYNAMIC/SPONTANEOUS, TRANSITIVE

b. a-ɪnyɪnyɪrʊnyɛ́
INF.SG-exude.liquid.VEN.MID
'to be let out in tiny amounts' ANTICAUSATIVE, SPONTANEOUS

(10) a. a-jíŋ
INF.SG-enter
'to enter (a place)' DYNAMIC, TRANSITIVE

b. a-jɪŋ-á
INF.SG-enter-MID.NPF
'to be entered (of a place)' RESULT-STATE

[11] Note that the transitive root *duŋ* does not have a sense 'dry sth.'; a completely unrelated transitive stem *itoi* means 'dry sth'.

'to be possessed by evil' (RESULT-)STATE
'to be(come) mentally ill' COGNITION

(11) a. a-ɪbʊ́ŋ
INF.SG-catch
'to catch (sth. moving), arrest, touch'
'to rape, stick to following a path' DYNAMIC, TRANSITIVE

b. a-ɪbʊŋ-á
INF.SG-catch-MID.NPF
'to coagulate, be coagulated' SPONTANEOUS, STATE

The preceding examples show that the various valence-reducing functions are not necessarily distinct when it comes to particular verbs: a single middle verb like *a-duŋó* can have different functions like REFLEXIVE or ANTICAUSATIVE. Also, given complex lexical semantics, a single middle stem can simultaneously express multiple situation types such as MOTION+POSTURE, EMOTION+COGNITION, EMOTION plus some other aspect of behavior, etc.

Finally, in light of Inglese (2022b)'s findings, one might ask whether Maa alternating middles do not express PASSIVE, FACILITATIVE, and ANTIPASSIVE valence-decreasing functions. In some contexts, the middle+inchoative form *-ayu/-oyu* can express an 'able to/will become' sense, which is close to, if not the same as, facilitative. This combination is seen in (12)–(13).

(12) en=tóki na-j-î m-e-shúk-**ōy-ū**
FSG=thing REL.FSG-say-IMPS NEG-3-return.sth-MID.NPF-INCH
'a thing that is/was called unreturnable' (renoi.038)

(13) n-é-pūō-ī o-m-e-tú-duŋ-**oy-ú** ɔl=mʊ́rráni
CN1-3-go.PL-PL until-POT-3-SBJV-cut-MID.NPF-INCH MSG=warrior
ɛn=ámʊkɛ
FSG=sandal.NOM
'They go until the warrior's shoe becomes cut (frayed, torn, separated).' (enamuke1.0002)

To my understanding, Maa does not really use the middle for the passive function because the impersonal typically expresses the idea that there is some unexpressed (and non-expressable) agentive "people" that do the action, which is the passive sense (Payne 2011); this is evident in *najî* 'that is called/which "people" call' in (12) just above. The Maa middle – in contrast to the impersonal – completely erases the sense of an agent or causer, leaving just the undergoer. Finally, Maa has a dedicated antipassive suffix *-ɪshɔ(r)/-isho(r)* (Payne 2021b), which can be seen in (14) below. The dedicated impersonal and antipassive forms carve out valence-reducing functions that the middle forms do not extend into (see also Inglese (2022a) for some discussion).

5 Maa-style *media tantum*

In citation or their simplest person-inflected form, some Maa verbs always contain one or another of the middle endings in Table 2. Such verbs are reminiscent of Greek *media tantum* verbs, but potentially differ in that they can alternate between the middle non-perfect(ive) (mid.npf) and middle perfect(ive) (mid.pf) aspect endings; or they drop the middle ending when they carry certain "core" affixes. However, they cannot occur without a middle or some other core affix.

For example, consider *a-mɛrá* 'to be drunk/intoxicated, get (more) drunk'. There is no synchronically recognizable verb **a-mɛ́r* without a final *a*. It thus contrasts with alternating middle verbs described in the preceding section.

Now, even though modern Maa has no simple verb **a-mɛ́r*, the final *a* disappears in (14). The antipassive suffix *-íshɔ̄(r)* presumably occurs on the bracketed stem *ítɛ-mɛ́r* 'cause to be drunk', but the relevant point is that the antipassive immediately follows what looks like a non-middle root form *mɛ́r*.

(14) n-é-ōk il=keék ɔɔ-[ítɛ-mɛ́r]-íshɔ̄
cn1-3-drink mpl=medicines rel.mpl-caus-be.drunk-apas
'They take drugs that intoxicate.' (malk01.1083)

The antipassive suffix begins with a vowel; so one might ask whether *a* disappears from *mɛra* due to phonological simplification of a vowel cluster *aɪ*. The answer is "no." Maa has no prohibition against vocalic clusters; at least 42 cluster types, including rising and falling diphthongs, are attested (Griscom & Payne 2017). Also, epenthetic [y] and [w] can break up certain vowel sequences. So deletion to simplify a vowel cluster is an unlikely explanation.

Most tellingly, a Class 1 causative prefix *ɪta-*, alone, can result in dropping the final vowel from a *media tantum* verb, demonstrated by (15). The contrasts in (16) show the same patterns for *a-nʊtá* 'to be pregnant'. Similar alternation patterns occur for other Maa-style *media tantum* verbs.[12]

[12]Syntactic repercussions vary in accord with what core affix "displaces" the historical middle (i.e., with the top-most category of a derived stem): an antipassive stem correlates with intransitive syntax, a causative stem with transitive syntax, etc. For reasons of space, I cannot deal much with derivational layers here.

(15) k=ɛ́-ítɛ́-mɛr
CN2=3-CAUS-be.drunk
'S/he will make him/her drunk.'

(16) a. a-nʊtá
INF.SG-be.pregnant
'to be pregnant'

b. *a-nʊ́t

c. a-ɪtʊ-nʊ́t
INF.SG-CAUS-be.pregnant
'to impregnate (a human)'

For *a-mɛrá*, historical evidence suggests there may have been a root **mɛr* at some point in Nilotic history. Vossen (1982: 357) reconstructs **-mɛr-a* (parsing off the final **a*) for Proto-Teso-Lotuko-Maa, as his data shows all cognates within this sub-branch of Eastern Nilotic as having a middle-like ending or suffix. Notably, he presents the Ongamo form *a-tɛ́-mɛ́r-ɛ̀*, which would correspond exactly to a Maa MID.PF form of a Class 1 (but synchronically non-existing) root **mɛr*. The alternation between *a* and *ɛ* on Eastern Nilotic cognates of 'be drunk, intoxicated' argues that the final *a* in Maa *a-mɛrá* corresponds (historically) to the MID.NPF affix *-a(r)*. That is, *mɛra* is not a historically basic stative root which just coincidentally happens to end in the phoneme *a*.

In sum, there is evidence that Maa-style *media tantum* verbs contain a now-lexicalized historical *-a/-o* middle ending. This ending is part and parcel of such a verb in its most basic form, such that native speakers recognize no meaning without it. But the fact that it disappears when certain other core affixes occur is a tell-tale sign that the *a~o* is the historical middle ending. (In Section 7 we will see that ending in an *a~o* vowel, even if having the "right" kind of semantics, does not necessarily mean a verb belongs to the *media tantum* set.)

To give a better sense of the range of semantics found with Maa-style *media tantum* verbs, Table 4 lists a few additional such verbs with their situation types; Section 8 will further address semantic issues.

6 Partially assimilated middles

I now turn to verbs which appear to be partially assimilated into the *media tantum* group. *A-ɪgará* 'to obscure, hide sth. behind sth.' behaves like a *media tantum* verb in that the form **a-ɪgár* does not occur, but the final *a* is dropped when other

Table 4: Selected *media tantum* verbs and situation types

a-rrị̂ạ́	'to fall down'	TRANSLATIONAL MOVEMENT
a-isiijó	'to ferment'	SPONTANEOUS
a-ɪbalá	'to be conspicuous; unambiguous; likely to happen; true'	STATE, FACULTATIVE
a-ineníá	'to be piled up, crammed'	(RESULT) STATE
a-lioó	'to be visible, apparent'	STATE
a-lubó	'to be underfed (cattle)'	STATE
a-narikinó	'to match; be advisable'	STATE, FACULTATIVE
a-ɪkɪrɪkɪrá	'to quake (earth)'	NON-HUMAN PROCESS
a-sɛshá	'to be sickly'	BODY-STATE
a-ɪgʊtʊmá	'to squat, sit'	BODY-POSTURE
a-gʊrrʊmá	'to vomit'	BODY-PROCESS
a-dalá	'to be inattentive; play joyfully'	EMOTION/COGNITION, HUMAN BEHAVIOR
a-ɪmasó	'to be proud'	EMOTION/COGNITION
a-ŋɪdá	'to be happy; haughty; naughty'	EMOTION, HUMAN PROPENSITY, HUMAN BEHAVIOR

core affixes are added. Unlike the vast majority of alternating middle and *media tantum* stems, *a-ɪgará* can function in a transitive sentence frame, even with the final *a*, as in (17).[13]

(17) ɛ-ɪgára ɛnk=áji en=kíné
3-obstruct FSG=house.NOM FSG=goat
'The house will obstruct/obstructs the goat (from view).'

Like alternating middles, *a-ɪgará* can have a reflexive sense, as in (18). This suggests the final *a* might correspond to the historical middle suffix (note the tone difference in the verbs in (17) versus (18)). And like *media tantum* verbs, final *a* is dropped if a core affix is added. We see this in (19) when the Class 2 causative/instrumental is added. Further, the progressive is normally disallowed

[13]Such mismatch between intransitive morphological form and transitive syntactic behavior is known as "deponency" (Grestenberger 2023). Aside from *a-ɪgará*, I do not know of deponent behavior among other Maa *media tantum* verbs.

with a stative base, but it occurs in (20) directly on the form *ɪgar*.[14]

(18) ɛ-ɪgárā ɔl=páyian (ɔl=chaní) ŋolé
3-hide.PF MSG=elder.NOM MSG=tree yesterday
'The man hid himself (behind a tree) yesterday.'

(19) a-ɪgar-íé em=búku ol=órika
INF.SG-hide-CAUS FSG=book MSG=chair
'to hide a book behind a chair'

(20) ɛ-ígar-íta ɛnk=áji en=kíné
3-hide-PROG FSG=house.NOM FSG=goat
'The house is obstructing the goat (from view).'

These facts suggest that *ɪgar* was (historically) a dynamic root and the final *a* might be the historical middle, allowing either a reflexive/reciprocal sense or the rather stative (though transitive) notion of 'obstruct from view, block'. (Subjects of this verb are easily inanimate, also potentially correlating with a less dynamic profile.) More specifically, these facts might suggest that *a-ɪgará* is a Maa-style *media tantum* verb. Where it notably differs from that group of verbs, however, is that it does not take the MID.PF suffix *-ɛ*. Rather, its PF counterpart varies tonally from the non-PF form. Thus, the PF form looks like a Class 2 non-middle verb. Compare (17) above with (21):

(21) ɛ-ɪgárā ɛnk=áji en=kíné
3-hide.PF FSG=house.NOM FSG=goat
'The house obstructed the goat from view.'

The PF form suggests the hypothesis that there was a regular (historical) transitive root **ɪgar*, which still uses the productive Class 2 perfect(ive) *-a(k)*, surfacing here with Downstep High tone (marked by the macron). In other respects, *a-ɪgará* appears to be part of the *media tantum* group.

As another type of partially assimilated *media tantum* verb, consider *a-sɪnyá* 'to be holy, perfect, spotless', seen in a relativized form in (22).

(22) ɔl=kɪtɛ́ŋ ɔ-sínya oshî ɛ-pɪk-í in=taléŋo
MSG=bovine REL.MSG-be.holy always 3-put-IMPS FPL=ceremonies
'An ox which has no physical fault is put (slaughtered) in ceremonies.'
[pure black or white, not spotted, nor an "unsaintly" colour]

[14]This example is from a speaker of IlKeekonyokie Maa. Another speaker of Purko Maa states that the verb form makes sense though he "hasn't used it before".

Mol (1996: 372) states this verb occurs in the "present tense" (i.e., non-PF aspect) only. That is, the final *a* does not alternate with the MID.PF ending *-ɛ*. In other ways, the base behaves as a Maa-type *media tantum* verb: The sequence *sɪny* is immediately followed by the ventive+middle form *-ʊnyɛ* in (23); and there is no final *a* in the causative form in (24).

(23) A-sɪny-ʊnyɛ́ naá doí e=síáâî ná-tií
3-be.holy-VEN.MID FOCUS indeed FSG=work.NOM REL.FSG.NOM-be.at
ɛnk=arruoísho atúā?
FSG=conflict inside
'Can work grow blessedly which has conflict (chaos) in it?'

(24) a-ɪtɪ-síny
INF.SG-CAUS-be.holy
'to make holy, cleanse'

A-sɪnyá is clearly related (though in some as-yet unclear diachronic way) to the adjectives *sinyáti*/*sinyát* 'holy, blameless (SG/PL)'. This, plus the fact that there is no perfect(ive) form – neither *-ɛ* nor a PF tonal form like in (18) and (21) – suggests that the diachronic route for how *a-sɪnyá* may have entered or be entering the *media tantum* group may differ from that of *a-ɪgará*. Conceivably, however, its partial media tantum behavior is motivated by semantic similarity to other *media tantum* verbs (an issue discussed in Section 8).

7 Spurious "look alikes"

Sections 5 and 6 have argued that various stative citation-form verbs ending in *a* or *o* are *media tantum* verbs. However, not all stative citation-form verbs ending phonologically in this way belong to this group. A case in point is *a-ɪrowúá* 'to be hot'. Though semantically stative, it is not a *media tantum* verb. First, there is no alternation with *-ɛ* for PF aspect; instead, it takes the regular Class 2 perfect(ive) suffix *-a(k)*, as in (25). In this respect, it might seem like *a-igará* 'to obscure', discussed above. But secondly, the relevant *a* remains when various core affixes are added, as shown in (26). Indeed, various suffixes show that the root has a final (historical) *dʒ* (orthographic *j*) – rather than a final *r* which would be expected if it carried the (historical) non-PF middle.

(25) ɛ-ɪrɔ́wūāj-ā táatá
3-be.hot-PF today
'It became hot today.' / 'It was hot today.' / 'It has been hot today.'

(26) a-irowuaj-íé
INF.SG-be.hot-INST/CAUS
'to heat with' / 'to make sth. hot'

Also, the inchoative suffix can be added, as is true generally for stative verbs, but not the middle+inchoative form *-ayu/-oyu*:

(27) a-irowuaj-ú
INF.SG-be.hot-INCH
'to become hot'

Some vowel-final stative verbs can be shown to end in a historical *r* but nevertheless do not pattern as *media tantum* forms. Consider *a-adɔ́* 'to be linearly extended' in (28). The (historical) *r* is shown by the inchoative derivation – making one think the otherwise final *ɔ* might be a middle allomorph. However, it is neither a middle nor *media tantum* verb: Its final vowel does not disappear in the causative, or with any other core affix, nor does it alternate with *ɛ* in PF aspect.

(28) a. a-adɔ́
INF.SG-be.linearly.extended
'to be long/tall'

b. ɛ-adór-ū ɔl=chánī
3-be.linearly.extended-INCH MSG=tree
'The tree will become tall.'

c. í-ntɔ́-ɔ́dɔ
2-CAUS-be.linearly.extended
'You make him/her/it be tall/long.'

Another spurious case is *a-naná* 'to be tender, soft, gentle' which semantically would appear to be a prime middle candidate. However, it lacks a perfect(ive) counterpart with *-ɛ*, distinguishing it from regular middle/*media tantum* verbs (Mol 1996), and it keeps final *a* in conditions that trigger loss in *media tantum* verbs; note the causative in (30).

(29) in=kírí naá-nana
FPL=meats REL.FPL-be.soft
'meats that are soft' (inkiri.057)

(30) k=é-lō a-ɪta-naná
CN2=3-go INF.SG-CAUS-be.soft
'S/he will make it (e.g., a hide) soft.'

Unlike with ‘to be hot’, no final consonant surfaces in the inchoative derivation of ‘be soft’; compare (31) with (27) above.

(31) a-naná-ú
INF.SG-be.soft-INCH
‘to become soft, etc.’

8 What motivates development of *media tantum* forms?

To explore what might motivate the development of *media tantum* verbs, I tabulated senses and situation types of about 450 alternating middle and Maa *media tantum* verbs.[15] The specific situation types for different senses are listed in Table 5.

Table 5: Situation types evaluated for Maa middle verbs

Body-state	Body-posture
Body-process	Non-human process
Emotion/Cognition	Facilitative (most with inchoative)
Human behavior	Human propensity
(e.g., once)	(typical characteristic)
Reflexive (grooming, action, etc.)	Reciprocal
Motion-translational	Motion-nontranslational
(most with directional)	
Result-State	State (other; not from process)
Spontaneous	

To appreciate how senses were categorized for situation type, Table 6 presents a few middle stems, their senses, and corresponding situation types. This was done for all 450 middle stems in the data. (For reasons of space, the middle stems in Table 6 are not full word forms, as they lack an infinitive or person prefix.)

Frequencies of situation types relative to alternating and *media tantum* stems were then examined. Eventually, I grouped those situation types judged to be more typically human-related (HR), versus those typically less human-related (“other”; OTH). The HR types were BODY-STATE, BODY-POSTURE, BODY-PROCESS,

[15] These were middle forms that happened to be in a lexicography database, with well over 2750 senses involving verb forms. Since middle formation from active transitive verbs is highly productive, many potential alternating middles happen to not be included in the database.

Table 6: Middle verb situation types and senses (selected examples and categories)

middle stem	BODY-PROCESS	EMOTION/COGNITION	HUMAN PROPENSITY	HUMAN BEHAVIOR
raya		emotionally cling to, have exclusive relationship with	love to be s.where or have sth. against contrary forces	elope (of a girl)
ŋɪdarɛ		be proud of		boast of
peepeeno			be trouble-maker	loiter
wuapa			be insane, abnormal due to drugs	overreact angrily, “snap”
oko				sing about exploits
gɪra	do slowly, listlessly		be naturally quiet, soft-spoken	be silent, quiet
liyio		be lonely		

cognition/emotion, human propensity, and human behavior. The oth types were non-translational motion, spontaneous, state, result-state, facilitative, and non-human process. Reflexive/reciprocal and translational motion were now excluded from further statistical evaluation, as the former is so highly productive with nearly all dynamic transitive stems and the latter is so highly productive with derived directional stems. The reflexive/reciprocal functions are dominantly human-related,[16] and the middles of translational motion are dominantly non-human related, and I wanted to evaluate any potential skewing among the other situation types. This left 402 middle stems.

The distribution of situation-type tokens in the sample of 402 middle stems is given in Table 7. Since a given verb stem can have more than one sense and instantiate more than one situation type, each token of a situation type is reflected in the table: 313 total for alternating middle stems, and 113 total for *media tantum* stems (N=426). That is, the N's in the table do not represent numbers of middle verb stems, but rather instances of situation types. The difference in how often *media tantum* forms correlate with typically human-related situations versus "other" situations is statistically significant ($\chi 2 = .00365$, $p < .01$).

It may be hard to see the tendencies from Table 7. Thus, Figure 1 depicts the "other" (OTH) versus typically human-related (HR) situation types relative to alternating (Alt) versus *media tantum* (MT) types. The figure shows that a higher percentage of Maa *media tantum* verbs correlate with human-related situations, compared to the percentage of all alternating middles that correlate with human-related situations. This pattern accords with Lyons (1968)'s general characterization of what middle verb forms are about, namely, that they express concepts that concern or affect the interests of the subject of the verb. (But perhaps contra Lyons, the pattern in Figure 1 does not support the hypothesis that **all** middles have this semantic profile, as the percentage of alternating verbs with "other" functions outweighs the percentage of human-related functions.)

Quite clearly, the *media tantum* verbs do not have a preponderance of reflexive uses; so at least the Maa *media tantum* group seems to generally counter Kemmer (1993)'s characterization of the middle prototype as involving the reflexive function. However, reflexive, reciprocal, and anticausative (potentially leading to result-state) functions are productive for alternating middles.[17]

[16]Of the 28 reflexive functions that happened to be included in the senses of 450 middle stems, only four or five had grooming-type senses and all pertained to alternating middles: *a-itukúó* 'to bathe self', *a-ɪsʊjá* 'to bathe, wash self/hands', *a-ɪsɪkarrá* 'to adorn self', *a-ishopó* 'to dress self', and if one considers it to be a grooming event also *a-girró* 'to scratch self (e.g., due to itching)'.

[17]For Acholi (Western Nilotic), Kemmer (1993: 194–196) traces a middle marker *-ɛ̂* to a reflexive

Table 7: Typically human-related (HR) and other (OTH) situation types relative to alternating versus media tantum stems

SITUATION	ALTERNATING N=313	MEDIA TANTUM N=113
Less typically human-related (OTH)		
Motion-nontranslational	18	1
Spontaneous	18	4
State/Result-State	111	26
Facilitative	8	2
Non-human process	13	2
(Subtotals)	(168)	(35)
Typically human-related (HR)		
Body-state	25	12
Body-posture	13	15
Body-process	26	16
Emotion/Cognition	30	15
Human propensity	30	15
Human behavior	21	9
(Subtotals)	(145)	(78)

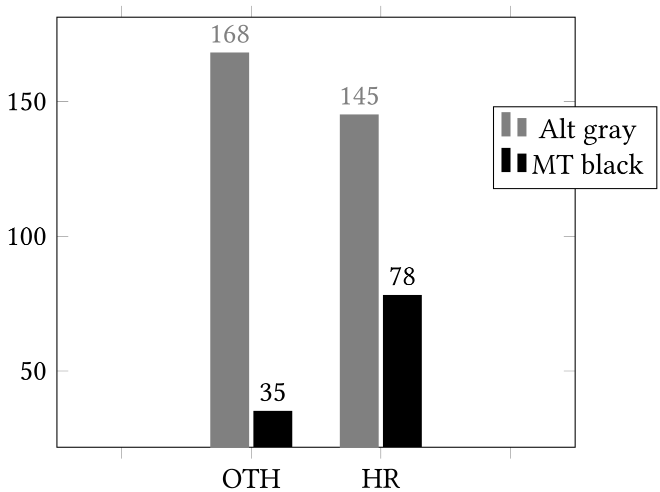

Figure 1: “Other” (OTH) versus typically “Human Related” (HR) Situation Types in Alternating (Alt) and *media tantum* (MT) Verbs

The pattern in Figure 1 leads to the hypothesis that higher usage frequency of middle forms expressing typical human situations may promote their lexicalization in such situations, followed by ultimate loss of the dynamic transitive counterparts. For some comparison, Zombolou & Alexiadou (2014) find that in a corpus study of non-alternating middle-form verbs in Modern Greek, a majority (89%) have an "experiencier" meaning (Zombolou & Alexiadou 2014: 337); and an "experiencer" is typically human. Of course, the semantics of "human situations" does not explain all of the *media tantum* instances in Maa; but it is likely the explanation for lexicalization of forms like *a-mɛrá* 'to be drunk, intoxicated'.

9 Conclusions

This study concurs that Maa has a middle system in the sense of Inglese (2022b) with alternating middle and *media tantum* verbs, both with a range of functions. It goes beyond previous work by demonstrating the specific nature of "Maa-type *media tantum*" verbs and showing that some forms share only certain *media tantum* features. The analysis thus is also a case study of a linguistic category with permeable boundaries. A category consists of a web of signs (constructions), each member with its own semantic profile and frequency of use. New items can get (partially) pulled into a category based on perceived similarity in form and meaning. Alternatively, items can become marginalized in a category, lose full productivity, and may eventually disappear from a category. This may be happening with verbs like *a-ɪgará* 'to obscure', which appear to be partially assimilated to the *media tantum* category, as their simple root forms have lost full productivity as members of the alternating middle set.

ending which she represents as **-ri* (but without actual reconstruction). To my knowledge, no one has reconstructed a middle suffix for Eastern Nilotic, but grammars of two Eastern Nilotic languages closely related to Maa, Lopit and Ateso, note middle (Moodie 2019) and reflexive (Barasa 2017) suffixes, respectively, with the form *-a/-o*. See Inglese (2023) for a broader discussion of the diachronic typology of middle markers.

Abbreviations

ACT	active	MSG	masculine singular
APAS	antipassive	NEG	negative
CAUS	causative	NOM	nominative case
CN1	connective 1 (high continuity)	NPF	non-perfect(ive)
CN2	connective 2	PF	perfect(ive)
FPL	feminine plural	PL	plural
FSG	feminine singular	POT	potential
IMPS	impersonal	PROG	progressive
INCH	inchoative	REL	relative (clause)
INF	infinitive	SBJV	subjunctive
INST	instrument	SG	singular
intr.	intransitive	tr.	transitive
MID	middle	VEN	ventive
MPL	masculine plural		

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Chapter 13

Non-prototypical usages of the Lugwere causative extension

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Lugwere is an under-studied Great Lakes Bantu language spoken in Eastern Uganda, and this paper presents a thorough investigation of non-prototypical usages of the causative suffix in Lugwere, including usages that would be classified as applicative. Causative and applicative isomorphism has been reported for a range of language cross-linguistically (Peterson 2006), and within Bantu, reflexes of the inherited Proto-Bantu causative **-ici* has instrumental usages in a handful of languages (Jerro 2017, Schadeberg & Bostoen 2019). This is also found in Lugwere, alongside examples of a comitative usage that has not been reported elsewhere. There are also examples where the causative suffix does not affect either the semantic or the syntactic argument structure.

1 Introduction

Lugwere (JE.17) is a Bantu language spoken by around 621 000 people in Eastern Uganda (Eberhard et al. 2021). It belongs to the Great Lakes branch of the Bantu family, and is closely related to Luganda and Lusoga. Lugwere retains reflexes of many of the Proto-Bantu extensions; a group of derivational suffixes that often, but not always, affect the valency of the verb.

This article focuses of the Lugwere reflex of the Proto-Bantu causative suffix **-ici*. The affix still functions as a causative, but has additional usages as well, some of which could be termed applicative. The goal of this article is to provide a descriptive account of some of these non-canonical usages of the causative morphology, and I will not propose a systematic analysis of these phenomena here.

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To this end, Section 2 lays out definitions for prototypical causatives and applicatives, while Section 3 gives an overview of the Lugwere reflexes of relevant Proto-Bantu morphology, and describes the different strategies for causativization in Lugwere. Sections 4–6 then look at different non-causative usages of the causative suffix, and Section 7 discusses the possible diachronic paths to the current Lugwere usages of the causative suffix.

2 Prototypical causatives and applicatives

Bantu languages frequently have a causativising and applicativising affixes, and this is a morphological feature that has been reconstructed back to Proto Bantu (Schadeberg & Bostoen 2019). Mostly commonly, modern languages have applicative morphemes that are reflexes of Proto-Bantu **-ɪd* and causative morphemes that are reflexes of Proto-Bantu **-i* and **-ici*.

Frequently, Bantu languages use one applicative marker to promote a range of different adjuncts to the object position, such as beneficiary, instrument, location, cause, goal and instrument. Pacchiarotti (2020) shows that there is wide variation in the functions and semantics of the reflexes of Proto-Bantu **-ɪd*. Reflexes of **-ici* tend to function as a causative, but also have instrumental usages in quite a few languages, which will be illustrated for Lugwere below (and see Section 7 for more on this). The affixes also differ in productivity between languages. The reflex of **-ici* has for instance become fully lexicalized in Mbuun (Bostoen & Mundeke 2011).

The literature has also reported cases of deponency and “pseudo” forms involving these affixes (Good 2007, Pacchiarotti 2020). That is, lexicalized verbs that appear to have the relevant affixes, but where no base form exists as an independent verb, and where the meaning can no longer easily be interpreted as applicativising or causativising. While such forms are fascinating, and some also found in Lugwere (especially with the applicative *-ir*), this article will focus on forms where the causativized form exists alongside a base verb without a causative suffix.

While there is a lot of variation in how the term ‘causative’ is used, this paper will rely on the definition given in Zúñiga & Kittilä (2019), which lays out a list of characteristics of the prototypical causative, reproduced below.

(1) **Prototypical causatives:**

- The syntactic valency of a causative clause is one higher than that of the base, non-causativized, clause.

- A new A (the causer) is added into the semantic argument structure.
- The new A (the causer) is introduced as the subject of the causative clause; the base subject of the non-causative clause (the causee) may be a core argument or an adjunct in the causative clause.
- Causativization is formally coded on the predicate complex.

This definition makes it clear that the causative includes changes both to the syntactic and the semantic argument structure of the verb, and in order to qualify there must be formal coding on the predicate, which would exclude for instance the English causative alternation from inclusion ("The door closed"/"She closed the door").

Zúñiga & Kittilä's (2019) definition of the applicative, reproduced below, is quite similar, but differs in a couple of ways. In both cases a new argument is added both to the semantic and the syntactic argument structure. However, in the case of the causative the new argument is an agent, and is installed into the subject position. In the case of the applicative, the new argument can have a lot of different thematic roles (beneficiary, location, instrument), and is installed as the primary object.

(2) **Prototypical applicatives:**
- Its syntactic valency is one more than that of the non-applicative diathesis.
- Its subject corresponds to the subject of the non-applicative diathesis
- It's primary/direct object corresponds to an adjunct or non-core argument in the non-applicative voice, or to a participant that is introduced to the clause as primary/direct object
- Applicativization is formally coded on the predicate complex

The next section will give examples of causatives from Lugwere, which fit the definition given above. Sections 4 – 6 will then give some examples of cases where the inherited causative marker does not behave like a causative; in some cases functioning more like a prototypical applicative (Section 5), in other cases having no effect on the argument structure (Section 6.2).

3 Causative strategies in Lugwere

3.1 Methodology

This study is based on a combination of elicitation and the use of a corpus. The corpus consists of a combination of written texts (a translation of the New Testament, some storybooks and teaching materials for children) and various recordings from 2021-2023 which have been transcribed and translated by a native speaker. The recordings are of a range of genres, from natural conversations, narratives and monologues, to radio programs and songs, as well as some more structured narratives (based on the stories provided by *Totem Field Storyboards* 2024) and some conversations where participants were asked to act out different scenarios (going to the doctor, shopping at the market, gossiping about a neighbour). Altogether the corpus has just below 435 000 words. FieldWorks 9.1 (SIL 2003) was used to search through the corpus for relevant examples. The searches looked for the causative suffix *-isy-* or *-isisy-* by searching for these either in combination with the final vowel *-a* or the past tense suffix *-ire* (the combination of the causative and the past tense suffix becomes *-isirye*). A similar search for the applicative suffix was also conducted.

Additionally, there is a dictionary of Lugwere (Nzogi & Diprose 2012) which has proven helpful in finding words to investigate, and for investigating the lexicalized meanings of verbs with the causative suffix.

The elicitation was carried out on field trips to Uganda in 2022 and 2023, as well as online. The elicited examples given in this paper come from two speakers, both from the Nakooma village in Kibuku district, in their twenties and with higher education. Examples found in the corpus were tested in elicitation to make sure that the linguist had an accurate understanding of the sentence in the context.

The examples given in this paper are taken both from the corpus and from elicitation sessions. Whether an example is from the corpus or from elicitation will be indicated by a letter (C) or a letter (E) in parentheses following the first line of the example. Examples from elicitation are either translations of sentences given in English, or Lugwere sentences that have been considered by native speakers to be acceptable when asked for grammaticality judgement.

3.2 Proto-Bantu extensions and Lugwere reflexes

Proto-Bantu has been reconstructed to have had 11 extensions (Schadeberg & Bostoen 2019). These are suffixes that follow the verb root and typically have an effect on the valency of the verb. Table 1 shows the extensions reconstructed by Schadeberg & Bostoen (2019), alongside the Lugwere reflexes.

Most relevant for the current study are the causative suffixes. Two affixes have been reconstructed for Proto-Bantu, which are considered to have been in complementary distribution, where **-i* followed a consonant and **-ici* followed a vowel (Schadeberg & Bostoen 2019). Considering the shape of Proto-Bantu verb

Table 1: Proto-Bantu extensions and Lugwere reflexes

	Proto-Bantu	Lugwere
Causative	*-i/*-ici	-y,-isy
Applicative	*-ɪd	-ir
Impostive	*-ɪk	-ik
Neuter	*-ɪk	-ik
Positional	*-am	-am
Reciprocal	*-an	-an, -angan
Tentive	*-at	-at
Separative	*-ʊd, *-ʊk	-ul, -uk
Repetitive	*-ag ~ -ang	-iag, -ang
Passive	*-ʊ/-ibʊ	-w/-ibw

roots, this meant that for the most part the longer form was found on CV roots, while the shorter extension was used after stems of a -CVC(-VC)- shape, although later developments have typically extended the context in which the longer suffix is used. (Schadeberg & Bostoen 2019).

A similar variation exists in Lugwere between *-y* and *-isy*. Additionally, there is a reduplicated version of the longer suffix; *-isisy*. Their distribution is only partially decided by the shape of the stem that the suffixes attach to. For CV stems, the *-y* extension is never found, while both the *-isy* and the *-isisy* affixes are possible. For the most part, other factors determine the distribution of the different extensional suffixes, and this will be discussed in more detail below.

Lugwere also has a reflex of the Proto-Bantu applicative suffix, and a reduplicated version of this also exists: *-irir*. In both Lugwere and Proto-Bantu one (or more) extensions would come after the verb root, and be followed by another suffix, often called the 'final vowel'. This suffix relates to tense, aspect and/or mood.

3.3 Lugwere causative strategies

Lugwere has five different strategies for causative formation. Firstly, it is possible to create a analytic causative by use of the verb *kuleta* 'to bring', as illustrated in (3).

(3) a. *Nga ontama, baleta ontama* (C)

Nga o-n-tama ba-let-a o-n-tama.
NAR AUG-NF9-sheep 2S-bring-FV AUG-NF9-sheep
'When the sheep, they brought the sheep.'

b. *(...) ekyakuletera okukobera omuntu ogondi* (C)
e-ky-a-ku-let-er-a o-ku-kob-er-a
REL-7S-FUT-2sgO-bring-APPL-FV AUG-INF-say-APPL-FV
o-mu-ntu o-g-ondi.
AUG-NF1-person AUG-AGR1-OTHER
'... which will make you tell someone else.'

c. *Andeteire okulya* (E)
A-n-let-er-ere o-ku-ly-a.
1S-1sgO-bring-APPL-PST AUG-INF-eat-FV
'He made me eat.'

Then there are the three suffixes discussed above, *-y*, *-isy* and *-isisy*. The vowels of the latter two harmonize with the last vowel of the verb root. If there is a mid vowel, the affixes are pronounced *-esy* and *-esesy*, while following any non-mid vowel the suffixes have the high vowel /i/.

-isy is perhaps the most widespread of the causative suffixes. In the corpus, it is most commonly found with verbs of learning and with the verb *kulya* 'to eat'. Examples of the first are *kwega* 'to learn' and *kusoma* 'to read, to study', which become *kwegesya* and *kusomesya*, both meaning 'to teach'. *Kulisya* can mean 'to make someone eat', but it also commonly used to mean 'to graze'. Elicitation has shown that *-isy* is very productive, and can be used with most verbs. (4) gives some examples of causative sentences with *-isy*.

(4) a. *Mbu oodi owamugangulisya amaizi nti pa* (C)
Mmbu oo-di ow-a-mu-gangul-isy-a a-ma-izi nti
HRS AGR1-DEM.DIST SBR-1S-1O-overturn-CAUS-FV NF6-water like
pa ...
ONOM
'When the other makes her throw down the water like 'pah'...'

b. *Musomesya asomesya primary seven* (C)
mu-somesya a-som-esy-a primary seven
NF1-teacher REL.1S-learn-CAUS-FV primary seven
'A teacher who teaches primary seven'

c. *Emumala bababisya ameizi* (C)

E-mu-mal-a ba-ba-ab-isy-a a-ma-izi.
SBR-2PLS-finish-FV 2S-2PLO-go-CAUS-FV AUG-NF6-water
'When you finish, they make you fetch water.' (lit. go water)

-isisy also seems to be quite productive, and can be used with most verbs, while *-y* is somewhat more restrictive in its distribution. Examples of each are given in (5).

(5) a. *Kaisi nomwirukya aina? Omwidwaliro.* (C)
Kaisi n-o-mw-iruk-y-a aina? O-mw-i-dwaliro.
then NAR-2SGS-1O-run-CAUS-FV where AUG-NF18-NF5-hospital
'Then where do you rush him? To the hospital.'

b. *Amugingisisya amaizi* (E)
A-mu-ging-isisy-a a-ma-izi.
1S-1O-carry-CAUS-FV AUG-NF6-water
'S/he makes her/him carry water.'

In many cases, speakers report that these can all be used, without change to the meaning. For instance, the causative of *kubimba* 'to swell' can be either *kubimbya, kubimbisya*, or *kubimbisisya* all meaning 'to cause something to swell'. In other cases, there might be differences. The causative of *kunaaba* 'to bathe' is *kunaabisisya* or *kunaabisya*, while *kunaabya* means 'to do the dishes'. Particularly the *-y* has sometimes become lexicalized, so while *kukoba* means 'to say', *kukobya* means 'to court/flirt'.

While there is a lot of overlap in the usage of each of these three suffixes, there are also some differences. One of the usages of *-y* is to turn verbs for entering a state (*kukala* 'to dry' *kudoda* 'to get wet') into causatives, as in (6).

(6) a. *Olugoye lukalire* (E)
O-lu-goye lu-kal-ire.
AUG-NF11-dress 11S-dry-PST
'The dress dried.' (earlier today)

b. *Eisana likaliriye olugoye mangu* (E)
E-i-sana li-kal-ir-y-e o-lu-goye mangu.
AUG-NF5-sun 5S-dry-PST-CAUS-PST AUG-NF11-dress quickly
'The sun dried the dress quickly.' (earlier today)

-isy and *-isisy* can often be used in the same contexts, but there are some differences between them. *-isisy* is sometimes used when more than one additional

argument is introduced, as in (7) below. However, note that while the speaker originally gave a sentence with *-isisy* when asked to translate (7b), he also accepted a sentence with *-isy* when asked for grammaticality judgments. Note also that the preposition *ne* can optionally be included even when the verb has the causative suffix. In these examples we see the causative suffix used to promote an instrumental argument, something that will be discussed further in Section 5.1. (8) shows a sentence where *-isisy* is used, probably due to there being another level of causation. If teaching is 'causing someone to learn', then paying school fees could be considered 'causing someone to cause someone to learn'.

(7) a. *Asalisya ebijo (n=a)kambe* (E)
A-sal-isy-a e-bi-jo (n=a-)ka-mbe.
1S-cut-CAUS-FV AUG-NF8-sweet-potato (with-AUG-)NF12-knife
'S/he cuts sweet potatoes with a knife.'

b. *Imusalisisya ebijo (n=a)kambe* (E)
I-mu-sal-isisy-a e-bi-jo (n=a-)ka-mbe.
1SGS-1O-cut-CAUS-FV AUG-NF8-sweet.potato (with=AUG-)NF12-knife
'I make her/him cut potatoes with a knife.'

(8) a. *Kukola niki? Kusomesesya baana* (C)
Ku-kol-a niki? Ku-som-esesy-a ba-ana.
INF-do-FV what INF-learn-CAUS-FV NF2-child
'To do what? To pay fees for children.'

Additionally, in elicitation speakers report that forms with *-isisy* can in some contexts be interpreted as less direct than forms with *-isy*, so that in (9), sentence (9a) was interpreted as someone physically putting a child down, while in (9b) it could be either a matter of physically making the child in question sit, or by giving it instructions to sit. Similarly, *kulisya* is interpreted as physically feeding someone, while *kulisisya* can mean making someone eat in a more indirect way.

(9) a. *Otyamisye omwaana ooyo* (E)
O-tyam-isy-e o-mw-ana oo-yo.
2SGS-sit-CAUS-SBJ AUG-NF1-child AUG-AGR1-DEM.MID
'Make that child sit!' (by physically putting them down)

b. *Otyamisisye omwaana ooyo* (E)
O-tyam-isisy-e o-mw-ana oo-yo.
2SGS-sit-CAUS-SBJ AUG-NF1-child AUG-AGR1-DEM.MID
'Make that child sit!'

For simplicity, in the remainder of the paper *-isy* will be referred to as ‘the causative suffix’, even though of course the purpose of this paper is to show that that particular label is not sufficient to describe the full range of usage of the suffix. The other two suffixes will be referred to by their form, i.e. *-y* and *-isisy*.

A final causative strategy relates to the separative extensions. Lugwere has two separative extensions, *-uk* and *-ul*, where the latter often functions as a causativised version of the former. A couple of examples are given in (10).

(10) a. *kusinduuka* ‘to become loose’
b. *kusinduula* ‘to loosen’
c. *kusetuka* ‘arise, get up’
d. *kusetula* ‘lift, raise’

3.4 Subject- and objecthood in Lugwere

Before moving on to look at the non-causative usages of these affixes, however, it is worth first briefly looking at what the properties of an object or a subject are in Lugwere.

As is common for Bantu languages, Lugwere has no case marking on the nouns themselves. The clearest evidence of a nouns subject- or objecthood are therefore found in the word order, the indexation on the verb, and the presence or absence of prepositions. Additionally, a direct object can be the subject of a passivised verb.

Lugwere has SVO word order, and so we therefore expect to see the subject preceding the verb and the object(s) following it. Verbs obligatorily agree with their subjects, and determining what the subject is, is therefore fairly straightforward. Objects can also be indexed on the verb, and this most commonly happens with pronominal objects. For nominal objects, there is typically no indexation. Lugwere allows for up to three object markers, and the ordering of these is very flexible, as example (11) shows.

(11) a. *Jenny asumbisya oShanita matooke (ne) egaasi* (E)
Jenny a-sumb-isy-a o-Shanita ma-tooke (ne)
PN.F 1S-cook-CAUS-FV AUG-PN.F NF6-matooke (with)
e-gaasi.
AUG-9.gas.cooker
‘Jenny makes/helps Shanita cook matooke with the gas cooker.’

b. *Jenny agigamusumbisya* (E)
Jenny a-gi-ga-mu-sumb-isy-a.
pn.f 1S-9O-6O-1O-cook-caus-fv

c. *Jenny agagimusumbisya* (E)
Jenny a-ga-gi-mu-sumb-isy-a.
pn.f 1S-6O-9O-1O-cook-caus-fv

d. *Jenny agamugisumbisya*(E)
Jenny a-ga-mu-gi-sumb-isy-a.
pn.f 1S-6O-1O-9O-cook-caus-fv

e. *Jenny agimugasumbisya* (E)
Jenny a-gi-mu-ga-sumb-isy-a.
pn.f 1S-9O-1O-6O-cook-caus-fv
(b.) – (e.): 'Jenny helps/makes her cook it with it.'

As is also common for Bantu languages, Lugwere does not have a lot of prepositions, and instead relies to a large extent on locative nominal classes or locative verbal suffixes. The most common preposition is *ne*, which has a range of functions, some similar to English 'with, by'. The preposition is needed to introduce certain arguments when there is no applicative morphology on the verb, and the preposition can still be present even with such applicative morphology, as was illustrated in example (7) above.

Causee objects of causativised verbs can also become the subject of a passive verb, as shown by (12c). (12a)-(12b) show that the two objects can both occur right after the verb.

(12) a. *Omusawo amenyeserye omwana ekikopu.* (E)
O-mu-sawo a-meny-es-erye o-mw-ana e-ki-kopu.
aug-nf1-doctor 1S-break-caus-pst.caus aug-nf1-child aug-nf7-cup
'The doctor made the child break the cup.' (earlier today)

b. *Omusawo amenyeserye ekikopu omwana* (E)
O-mu-sawo a-meny-es-erye e-ki-kopu o-mw-ana.
aug-nf1-doctor 1S-break-caus-pst.caus aug-nf7-cup aug-nf1-child
'The doctor made the child break the cup.' (earlier today)

c. *Omwana amenyesewirwe ekikopu n'omusawo* (E)
O-mw-ana a-meny-es-ew-irwe e-ki-kopu
aug-nf1-child 1S-break-caus-pass-pst.pass aug-nf7-cup
n=o-mu-sawo.
by=aug-nf1-doctor
'The child was made to break the cup by the doctor.' (earlier today)

4 Sociative usage of *-isy*

The Lugwere causative can have a sociative interpretation. That is, the causative suffix can be used to indicate that an action was done alongside someone or as an effort to help someone. An example is given in (13). This usage seems to be fairly productive, and given the right context most causative verbs can be interpreted this way.

(13) a. *Ooba ofunireku abaala basatu bakusesesyaku eidebbe* (C)
Ooba o-fun-ire-ku a-ba-ala ba-satu
or 2SGS-get-PST-LOC AUG-NF2-girl AGR2-three
ba-ku-s-esesy-a-ku e-i-debbe.
2S-2SGO-grind-CAUS-FV-LOC AUG-NF5-container
'Or you got three girls and they help you grind in the container.'

Example (14) below can be said in a context where your sibling has lost a friend, and you come along with her to the funeral. It is therefore clear that the 'primary mourner' is the sibling, and in (14b) the speaker is accompanying her or helping her in her grief. The sibling would therefore be the subject of the base verb, as in (14a), where the direct object is the theme - the person that is being mourned. The sociative in (14b) installs a new subject, and the direct object is now an agent, much like what happens in prototypical causatives. The main difference is, that in sociative constructions the subject does not necessarily directly or indirectly cause the object to perform the action, but the focus is rather on the fact that the new subject performs the action alongside or in aid of the base subject, which is now the direct object.

(14) a. *mwonyoko wange akunga mukagwa we.* (E)
Mw-onyoko w-a-nge a-kung-a mu-kagwa w-e.
NF1-cross.sibling AGR1-POSS-1SG 1S-cry-FV NF1-friend AGR1-AGR1.POSS
'My cross-sibling is mourning her/his friend.'

b. *Nkungisya mwonyoko wange.* (E)
N-kung-isy-a mw-onyoko w-a-nge.
1SGS-cry-CAUS-FV NF1-cross.sibling AGR1-POSS-1SG
'I am mourning someone on behalf of/for my cross-sibling.'

Sociative constructions do often utilize the same morphology as the causative, and it is considered a semantic subtype of the causative, occupying an intermediary space between direct and indirect causation (Shibatani & Pardeshi

2002, Zúñiga & Kittilä 2019). While the sociative resembles more prototypical causatives in many ways, they differ from the definition given in Section 2 in that the newly installed subject is not a prototypical 'causer'.

In Lugwere these sociative readings are context dependent, as the form is ambiguous with a more direct causative meaning. The semantic range of causatives in Lugwere is quite broad, and the *-isy* marker can be used for both direct and indirect causation, sociative constructions like in (13), and can be used in contexts where the newly installed agent is simply facilitating an action, as in (15) below, where the relevant person is bringing drinks, causing drinking in so far as they make drinking possible. In contrast, example (16) can be interpreted as someone giving the child milk from a bottle, thus very directly making it drink.

(15) a. *Nti onani niiye ayanwisya* (C)
Nti o-nani ni-iye a-y-a-nw-isy-a.
that AUG-who COP.ID-1.PRO REL-1S-FUT-drink-CAUS-FV
'This one is the one who will provide drink.'

(16) a. *Anwisirye omwana amata* (E)
A-nw-is-irye o-mw-ana a-ma-ta.
1S-drink-CAUS-PST.CAUS AUG-NF1-child AUG-NF6-milk
'S/he made the child drink milk.'

5 Applicative usages of *-isy*

Dixon (2012) provides a classification of the thematic roles of applied objects cross-linguistically, and recognizes the four broad categories of Goal, Instrument, Comitative and Locative. Two of these, the Instrument and Comitative, can be promoted to direct object position by *-isy*. These two will be discussed separately below, and as will be seen, the examples fit perfectly within the definition of a prototypical applicative as given above. The other two types, Goal and Locative, can only be promoted by use of the *-ir* applicative.

Pacchiarotti (2020) gives many examples of Bantu languages using **-ɪd* with Goal, Instrument and Locative arguments. She also gives examples of functions that fall outside of Dixon's categories, such as indicating repetition, completeness, or thoroughness of the action. I have not seen references to **-ɪd* being used for comitatives elsewhere, so this migh be a new development in Lugwere. It should be mentioned that in Lugwere this is not a very productive function, and only happens with certain predicates.

5.1 Instrumental

A fairly common function for the *-isy* suffix is to introduce an instrumental argument. The suffix can be used productively to introduce an instrument, and this reading seems to be quite highly available to speakers, who will often translate a verb with the *-isy* suffix out of context as some variant of 'to V with' or 'use something to V"

Dixon (2012) categorizes instrumental applicative arguments as falling into five main semantic groups. The causative suffix can be used to introduce instrumental arguments from all of these categories.

The first of Dixon's (2012) categories is what calls 'Implement'. This refers to a tool, weapon or other implement that physically changes the state of the original object. Examples of predicates that can take such an instrumental argument would be 'hit' or 'break'.

(17a) shows how the causative suffix can be used to promote such an instrumental argument to the direct object position, while (17b) shows an example of the applicative marker fulfilling the same function. Going back to the definitions in Section 2, (17a) clearly does not fit the description of a prototypical causative. The new argument introduced, the instrument *nkumbi* 'hoe', appears as a direct object: it is following the verb and without the preposition *ne*. Semantically it is not a causer, but rather an instrument. In fact, this instance of valency change fulfills all the criteria for a prototypical applicative.

(17) a. *Tulimisyanga enkumbi egyo* (C)
Tu-lim-isy-a-nga e-n-kumbi e-gy-o.
1PLS-dig-CAUS-FV-HAB AUG-NF10-hoe AUG-AGR10-DEM.MID
'We used to dig with those hoes.'

b. *Bamukubbira omwigo buli lunaku* (E)
Ba-mu-kubb-ir-a o-mw-igo buli lu-naku.
2S-1O-hit-APPL-FV with=AUG-NF3-stick every NF11-day
'They hit her/him with a stick every day.'

The next group are those that have what Dixon (2012) terms a 'surface effect', and includes instruments used with predicates like 'wipe', 'sweep' and 'touch', which only affects the surface of the original object. (18) gives examples of this kind of instrument being promoted to direct object by the causative suffix (a) and the applicative suffix (b). The examples also show that in these cases *ne* can optionally be used.

(18) a. *Asimulisya emenza (n'e-)kitimbo* (E)
A-simul-isy-a e-menza (n=e-)ki-timbo.
1S-wipe-caus-fv aug-table (with=aug-) nf7-cloth
'S/he wipes the table with a cloth.'

b. *Asimulira emenza (n'e-)kitimbo* (E)
A-simul-ir-a e-menza (n=e-)ki-timbo.
1S-wipe-appl-fv aug-table (with=aug-)nf7-cloth
'S/he wipes the table with a cloth.'

The next category is what Dixon (2012) terms 'something which assists', which includes cases such as 'ascend with a rope' or 'cook with pan.' Examples of both the causative and the applicative suffix promoting such arguments to direct object position are found in (19) below, where the relevant predicates are 'to calculate with the brain' and 'to lie using money'.

(19) a. *Nasomere nga iswe imeite nti okubala otekwa kubalisya buwongo* (C)
N-a-som-ere nga iswe i-meite nti
1sgS-pst-study-pst rel.tns 1pl.pro 1sgS-know.pres sbr
o-ku-bal-a o-teek-w-a ku-bal-isy-a bu-wongo.
aug-inf-countfv 2sgS-put-pass-fv inf-count-caus-fv nf14-brain
'I studied knowing that for us math, you must calculate with the brain.'

b. *Ate imwe bababbeyera niki? Sente* (C)
Ate imwe ba-ba-bbey-er-a niki sente.
and 2pl.pro 2S-2plO-lie-appl-fv what money
'And for you, what do they use to lie to you? It is money.'

The next group is 'materials used', which would include the instrument in phrases such as 'build with bricks' or 'cover with a shawl'. Examples of this kind of instrumental arguments are found in (20) below.

(20) a. *Nga muzimbisya bisaale …* (C)
Nga mu-zimb-isy-a bi-saale …
rel.tns 2plS-build-caus-fv nf8-tree
'When you would build with wood…'

b. *Abantu bazimbira amanyumba matafali* (E)
A-ba-ntu ba-zimb-ir-a a-ma-nyumba ma-tafali.
aug-nf2-person 2S-build-appl-fv aug-nf6-house nf6-brick
'People build houses out of bricks.'

Finally, the last category specified by Dixon (2012) is 'reason/cause' which specifies the reason, cause or motive for the action. While this seems to be rather different from the other categories in that it is not what one would think of as a typical instrument, Dixon 2012 categorises it as a type of instrumental applicative, as languages tend to use similar strategies and morphology with reasons and instruments. In Lugwere as well this is the case, and the causative suffix can be used in these contexts. (21) illustrates the causative and applicative suffixes promoting a reason or cause to direct object position.

(21) a. *Era mudi iiye tete atandika okifuula nga talowoza nti kinu ebyokola binu obikolesya mukago* (C)

Era	mu-di	iiye	tete	a-tandik-a	o-ki-fuul-a
and	AGR18-DEM.DIST	1.PRO	again	1S-begin-FV	AUG-7O-overturn-FV

nga	t-a-lowoz-a	nti	ki-nu	e-by-o-kol-a
REL.TNS	NEG-1S-think	SBR	AGR7-DEM.PROX	REL-8O.REL-2SGS-do-FV

bi-nu	o-bi-kol-esy-a	mu-kago.
AGR8-DEM.PROX	2SGS-8O-do-CAUS-FV	NF3-friendship

'For her there again she starts to change as she doesn't understand that this you are doing, you do it for love.'

b. *Asabira enzala* (E)

A-sab-ir-a	e-n-zala.
1S-beg-APPL-FV	AUG-1

'S/he begs because of hunger.'

Another verb where the causative suffix can promote a reason to direct object is *kwesiima* 'to be proud'. In the base form, it can take a CP an argument, either headed by the subordinator *nti* or not. To take a NP argument, however, the *-isy* suffix is required, as illustrated in (22). As (22c) shows, *kwesiima* cannot take a direct object without an extension, and (22d) shows that with the causative suffix it can no longer take a CP argument.

(22) a. *Nesiima (nti) nasomere ino* (E)

N-e-siim-a	(nti)	n-a-som-ere	ino.
1SGS-REFL-proud-FV	(SBR)	1SGS-PST-study-PST	very

'I am proud that I studied a lot.'

b. *Nesiimisya empapula gyange* (E)

N-e-siim-isy-a	e-m-papula	gy-a-nge.
1SGS-REFL-proud-CAUS-FV	AUG-NF10-paper	AGR10-POSS-1SG

'I am proud of my papers.'

c. **Nesiima empapula gyange*
N-e-siim-a e-m-papula gy-a-nge.
1sgS-refl-proud-fv aug-nf10-paper agr10-poss-1sg
Intended: 'I am proud of my papers.'

d. **Nesiimisya (nti) nasomere ino*
N-e-siim-isy-a nti n-a-som-ere ino.
1sgS-refl-proud-caus-fv sbr 1sgS-pst-study-pst very
Intended: 'I am proud that I studied a lot.'

The two other causative suffixes can also be used to introduce instrumental arguments, as illustrated in (23).

(23) a. *Asibya ombuli muguwa* (E)
A-sib-y-a o-m-buli mu-guwa.
1S-tie-caus-fv aug-nf9-goat nf3-rope
'S/he ties the goat with a rope.'

b. *Alumisisya maino* (E)
A-lum-isisy-a ma-ino.
aug-bite-caus-fv nf6-tooth
'S/he bites with the teeth.'

The syntactic causative construction, on the other hand, can not be used in this way. This is illustrated in (24), where an attempt to use this construction to express something like 'he plays with the doll' can only give the somewhat odd interpretation 'he makes the doll play'.

(24) a. *a-zeny-esy-a odole* (E)
A-zeny-esy-a o-dole.
1S-play-caus-fv aug-doll
'S/he is playing with the doll.'

b. *Aletera omwana okuzenya* (E)
A-let-er-a o-mw-ana o-ku-zeny-a.
1S-bring-appl-fv aug-nf1-child aug-inf-play-fv
'S/he makes the child play.'

c. *?Aletera odole okuzenya*
A-let-er-a o-dole o-ku-zeny-a.
1S-bring-appl-fv aug-doll aug-inf-play-fv
'?He makes the doll play'
Intended: 'S/he plays with the doll.'

Like the causee object shown in (12), these instrumental objects also display all the properties of a direct object in Lugwere. (25) shows that the new instrumental object can follow the verb without any preposition (although *ne* can optionally be added), and that the new instrumental object can be indexed on the verb. (26) shows that both the instrumental and the original objects can be the subject of a passive sentence.

(25) a. *Alisya ngalo gye.*
A-l-isy-a n-galo gy-e.
1S-eat-CAUS-FV NF10-hand AGR10-POSS
'He eats with his hand.'

b. *Agilisya.*
A-gi-l-isy-a.
1S-10O-eat-CAUS-FV
'He eats with them.'

(26) a. *Omugai gumenyesiwirwe kikopu n'omwana* (E)
O-mu-gai gu-meny-es-iw-irwe ki-kopu
AUG-NF3-mingling.stick 3S-break-CAUS-PASS-PST.PASS NF7-cup
n=o-mw-ana.
by=AUG-NF1-child
'The mingling stick was used by the child to break the cup.'

b. *Ekikopu kimenyesiwirwe mugai n'omwana* (E)
E-ki-kopu ki-meny-es-iw-irwe mu-gai
AUG-NF7-cup 7S-break-CAUS-PASS-PST.PASS AUG-NF3-mingling.stick
n=o-mw-ana.
by=AUG-NF1-child
'The cup was broken with a mingling stick by the child.'

5.2 Comitative

There is a group of verbs where the causative suffix can be used to promote a comitative argument to direct object position. Many of the relevant verbs describe actions that by their nature require more than one agent, such as *kulwana* 'to fight', *kukaayana* 'to argue' and *kusyana* 'to reconcile.'[1] In the base form, the

[1]Note that all the relevant verbs end in *-an*. This is a marker of reciprocity (see the reconstructed form for Proto-Bantu in Table 1), lexicalised on quite a few verbs that have reciprocity as an integarl part of their semantics. However, it is no longer prouductive, and the productive suffix for reciprocity is *-angan*.

person that the action is done with (i.e. the one that is argued *with*) can be introduced as an adjunct headed by *ne* 'and, with', as in (27a). With the causative suffix, however, that argument can be a direct object, as in (27b). (27c) shows that without the causative suffix, there cannot be a direct object.

(27) a. *Lwaki muli kukaayana nomukagwa wanyu?* (E)
Lwaki mu-li ku-kaayan-a ne o-mu-kagwa w-a-nyu.
why 2plS-COP INF-argue-FV with AUG-NF1-friend AGR1-POSS-2PL
'Why are you arguing with your friend?'

b. *Lwaki mukaayanisya omukagwa wanyu?* (E)
Lwaki mu-kaayan-isy-a o-mu-kagwa w-a-nyu.
why 2PLS-argue-CAUS-FV AUG-NF1-friend AGR1-POSS-2PL
'Why are you arguing with your friend?'

c. * *Lwaki muli kukaayana omukagwa wanyu?*
Lwaki mu-li ku-kaayan-a o-mu-kagwa w-a-nyu.
why 2plS-COP INF-argue-FV AUG-NF1-friend AGR1-POSS-2PL
Intended: 'Why are you arguing with your friend?'

An example of a verb that does not necessarily include more than one participant, but where a similar process can be applied, is *kubina* 'to dance', as in (28).

(28) a. *Abina nomwala, amubinisya.* (C)
A-bin-a n=o-mw-ala a-mu-bin-isy-a.
1S-dance-FV with=AUG-NF1-girl 1S-1O-dance-CAUS-FV
'He dances with a girl, he dances with her.'

The applicative marker can not be used in this way with these verbs, and the object instead gets a beneficiary interpretation, as in (29).

(29) a. *Alwanira muganda we* (E)
A-lwan-ir-a mu-ganda w-e.
1S-fight-APPL-FV NF1-parallel.sibling AGR1-AGR1.POSS
'S/he fights for her/his parallel sibling.'

It is worth noting that these cases differ from the sociative examples discussed in (13) above. In the comitative cases, a new direct object is installed, while the subject remains the same as in the base form. There is also no change to the semantic argument structure. The sociative usage, however, installs a new subject, and is therefore closer to a prototypical causative. It also changes the semantic argument structure by introducing a new agent.

6 Other non-prototypical usages

6.1 Other changes to valency

There are also other cases where the addition of *-isy* does have an effect on either the semantic or syntactic argument structure of the verb, but which do not fit into the categories discussed so far. Here we will look at a couple of examples.

One particularly interesting case of valency change is what happens with the verb *kufa* 'to die'. In its base form, this is a one argument predicate that takes the experiencer as the subject. When the causative suffix is added, we get the somewhat unexpected meaning 'to lose someone', where the subject is the experiencer, and the direct object is the theme, this is illustrated in (30). Syntactically, this is what would be expected from a causative: the new argument that is added is installed as subject, and the base subject has been demoted to object. Semantically, however, this is odd, as the thematic role of the new subject is as experiencer, rather than agent, and the subject is in no sense the 'causer'. There is a separate lexical item for 'to cause someone to die', i.e. *kwita* 'to kill'.

(30) a. *ne nfisya abanyoko bange* (C)
Ne n-f-isy-a a-ba-nyoko b-a-nge.
NAR 1SGS-die-CAUS-FV AUG-NF2-cross.sibling AGR2-POSS-1SG
'I lost my cross-siblings.'

Another example is found with the verb *kweta* 'to call'. With the causative suffix, as in (31) we get the meaning 'call for something', as in a restaurant when you request food. (31) is referring to a husband that goes into town to eat. As the argument that is introduced is an object, this is closer to the applicative definition in Section 2 than the causative definition. However, the argument that is introduced is not really an adjunct, as its semantic role is the theme, it is therefore a more central argument than what is generally found with the applicative.

(31) a. *N'atuuka eeyo nayetesya ochapati, nayetesya aamo bwita, muceere, nayetesya aawo obumere mere nyama* (C)
N=a-tuuk-a eeyo n=a-yet-esy-a o-chapati
NAR=1S-reach-FV there NAR=1S-call-CAUS-FV AUG-chapati
n=a-yet-esy-a aamo bw-ita mu-ceere,
NAR=1S-callCAUS-FV together NF14-bwita NF3-rice
n=a-yet-esy-a aawo o-bu-mere-mere ny-ama.
NAR=1S-call-CAUS-FV there AUG-NF14food-NAR=1S-call-RDP NF9-meat
'He reaches the place, and he calls for chapati, rice, bwita and meat.'

6.2 No change in argument structure

There are also examples where adding *-isy* does not change either the semantic or the syntactic argument structure. In (32), for instance, both the subject and the object of the sentences remain the same, regardless of whether the causative suffix is present or not. The two sentences are ditransitive, in that they have two direct objects. The subject and the objects retain their thematic roles as agent, goal and theme. Semantically the sentences are very similar, although one speaker who was consulted suggested that in (32b) the friend being introduced needs to be in front of them, whereas in (32a) it could be a matter of pointing at someone from a distance. Another speaker said there was no difference.

(32) a. *Njaba kumulaga omukagwa wange* (E)
Nj-ab-a ku-mu-lag-a o-mu-kagwa w-a-nge.
1sgS-go-fv inf-1O-show-fv aug-nf1-friend agr1-poss-1sg
'I am going to show him my friend.'

b. *Njaba kumulagisya mukagwa wange* (E)
Nj-ab-a ku-mu-lag-isy-a o-mu-kagwa w-a-nge.
1sgS-go-fv inf-1O-show-caus-fv aug-nf1-friend agr1-poss-1sg
'I am going to show him my friend.'

The explanation for why the causative suffix does not change the argument structure in this particular case is probably linked to the fact that 'to show' has an inherent causative meaning, i.e. 'to make someone see'. *Kwita* 'to kill', another verb with an inherent causative meaning, is likewise often found with the causative suffix in the corpus, as in (33). *Kulema* and *kulemesya*, both meaning 'to hinder' also seems to allow similar variation.

(33) a. *Omufu aitisya omwomi* (C)
O-mu-fu a-it-isy-a o-mw-omi.
aug-nf1-dead.person 1S-kill-caus-fv aug-nf1-living.person
'The dead person kills the living person.'

It is not only verbs that are inherently causative that can behave this way, however. (34a) was found in the corpus, and a consultant accepted (34b) as possible and meaning the same thing. The only difference seems to be an intensifying effect of the causative suffix, as (34a) was seen as emphasising the fact that one was just starting for real more than in (34b).

(34) a. *Ne nga okutandikisya okusoma okunyerenyere…* (C)

Ne nga o-ku-tandik-isy-a o-ku-som-a
but when AUG-INF-start-CAUS-FV AUG-INF-study-FV
o-ku=nyere-nyere ...
AUG-LOC=real-RDP
'But starting to study for real...'

b. *Ne nga okutandika okusoma okunyerenyere...* (E)
Ne nga o-ku-tandik-a o-ku-som-a o-ku=nyere-nyere ...
but when AUG-INF-start--FV AUG-INF-study-FV AUG-LOC=real-RDP
'But starting to study for real...'

Another example is seen in (35) where (35a) comes from the corpus, and (35b) was accepted by a speaker as more or less equivalent. Again the syntactic and semantic argument structure remains the same, however, the speaker said that in the (35a) example it sounds more like the person that was found had left because they were guilty, and subsequently been found, where as in (35b) this connotation is missing. Again, we might say that there is something of an 'intensifying' effect of the causative suffix, in that the finding event of someone you have been looking for because they are guilty of something might in a sense be more intense than finding someone without those surrounding circumstances.

(35) a. *o'Langha nanjagirisya owenairire* (C)
o-Langha n=a-nj-agir-isy-a owe-n-a-ir-ire.
AUG-L. NAR-1S-1SGO-find-CAUS-FV SBR-1SGS-PST-return-PST
'Langha found me, when I returned.'

b. *o'Langha nanjagirya owenairire* (E)
o-Langha n=a-nj-agir-a owe-n-a-ir-ire.
AUG-L. NAR-1S-1SGO-find-FV SBR-1SGS-PST-return-PST
'Langha found me, when I returned.'

There are other cases where the semantic and the syntactic argument structure stays the same, but the causative verb has a different meaning to the base form. This is the case for *kupanga* 'to arrange' and *kupangisya* 'to rent', illustrated in (36).

(36) a. *Twaba kupanga amatafali aago* (E)
Tw-ab-a ku-pang-a a-ma-tafali aa-g-o.
1PLS-go-FV INF-arrange-FV AUG-NF6-brick AUG-AGR6-DEM.MID
'We are going to arrange those bricks.'

b. *Napangisirye enyumba* (E)
N-a-pang-is-irye e-nyumba.
1SGS-PST-arrange-CAUS-CAUS.PST AUG-house
'I rented the house.'

A few other examples from the dictionary (Nzogi & Diprose 2012) are listed in (37). In these cases the causative verb has a meaning that is a more specific or more intense version of the base meaning. Another example, although less straightforward, is *kuwulira* 'to hear', and *kuwulisisya* 'to listen'. There is no base form **kuwula*, and the first form has the applicative suffix. However, this also appears to be a case where causative morphology is associated with a more intense meaning, to listen rather than just hear.

(37) a. *kubala* 'count, add up.'
b. *kubalisya* 'calculate, evaluate, budget, weigh, count, take into account'
c. *kugezya* 'to try'
d. *kugezesya* 'to experiment'
e. *kubebenya* 'destroy, damage, contaminate, destruct, corrupt, defile, '
f. *kubebenyesya* 'waste, squander.'

7 Discussion

The preceding sections have shown that the Lugwere reflexes of the Proto-Bantu causative suffix have a range of functions that do not adhere to the characteristics of a prototypical causative, as outlined in Section 2. The next questions is then to consider what diachronic forces might have resulted in the synchronic situation described in previous sections, and whether or not there are any unifying features of the various usages of *-isy*.

7.1 The typology of causative-applicative isomorphism

A causative-applicative isomorphism is not unique to Lugwere, but rather is found in multiple languages worldwide, as observed by large cross-lingustic studies of applicatives and causatives such as Peterson (2006) and Dixon (2012). Both Peterson (2006) and Dixon (2012) note that it is not uncommon for languages to have one valency increasing strategy that gives causative readings with monovalent verbs and applicative readings with bivalent verbs.

This, however, is not descriptive of the situation in Lugwere, as there are examples of both monovalent verbs (*kulima* 'to dig', see example (17)) and bivalent verbs (*kusimula* 'to wipe' see example (18)) having applicative (instrumental) readings.

The situation is different if we look at the comitative. Most of the examples of comitative usage of the causative suffix that have been observed have been with verbs that semantically require more than one participant (i.e. *kulwana* 'to fight', *kukaayana* 'to argue'), however syntactically they are intransitive, as in (38). Example (28) also showed that the intransitive verb *kubina* 'to dance' can get a comitative reading. I have not found any clear examples of an underlyingly transitive verb with a comitative meaning.

(38) a. *Bali kulwana* (E)
Ba-li ku-lwan-a.
2S-COP INF-fight-FV
'They are fighting.'

According to Peterson (2006) the most common kind of causative-applicative isomorphism is with beneficiary/maleficiary arguments. A less common form, he notes, is isomorphism between causatives and comitative/instrumental applicatives, which is what we have in Lugwere.

Peterson (2006) notes that causatives with an inanimate causee are by their very nature semantically close to instrumentals. 'I made the key open the door' and 'I opened the door with the key' describe the same set of events. Through such bridging contexts it is easy to understand how the instrumental connotation might then later become part of the semantics of the causative strategy. In Lugwere we find examples such as (39), which although similar to something like 'I made the key open the door,' seem to be firmly within the camp of the instrumental, as it would not make sense to say 'He makes the stick walk' or 'He makes his hands eat,' as one would not consider the stick to be 'walking' or the hand to be 'eating'.

(39) a. *Atambulisya mwigo* (E)
A-tambul-isy-a mw-igo.
1S-walk-CAUS-FV NF4-stick
'S/he walks with a walking stick.'

b. *Alisya ngalo gye* (E)
A-l-isy-a n-galo gy-e.
1S-eat-CAUS-FV NF10-hand AGR10-1.POSS
'S/he eats with her/his hands.'

For Kinyarwanda, a language where there is a similar synchretism between a causative and instrumental *-ish*, Jerro (2017) proposes an analysis of underspecification, where the marker "adds a new causal subevent and argument to the argument structure of a verb, but the exact position of the subevent licensed by *-ish* is variable, although constrained by general and verb-specific constraints on possible verb meanings as well as general constraints on argument realization" (Jerro2017:785). This analysis also seems applicable to Lugwere.

Peterson (2006) also notes that causatives can become markers of a comitative construction through the development of sociative causative semantics. As pointed out in Section 5.2 the syntax of the comitative and the sociative are different, in that the comitative installs a new object, while the sociative installs a new subject. However, by the very nature of these usages, the thematic role of both subject and object are very agent-like. In the comitative usage, with predicates like *kulwana* 'to fight' *kubina* 'to dance' and *kukayaana* 'to argue', both the subject and the object need to be active participants in the action. Similarly, with the sociative, both the subject and the object are engaging in the relevant activity. It may therefore be that the syntax-semantics mapping could have shifted historically, giving us the two different processes we observe today.

The change from causative to comitative can perhaps also be explained in terms of the subject initiating the action, thus causing the other participant to engage in it. If I am fighting with you, I am, perhaps, in a sense also making you fight. The syntactic-semantic mapping of the arguments in the comitative would fit with such a bridging context.

7.2 Non-prototypical usages of the causative in Bantu

Section 7.1 examined the typology of causative-applicative isomorphism, and investigated some plausible semantic paths from causative to the applicative functions observed for *-isy* in Lugwere. This section will examine what is known about this phenomena from other Bantu languages, to see what that can tell us about the diachrony of the phenomenon in Lugwere.

There are reports in the literature of cognates of PB **-ici* having instrumental functions in Rwanda (JD.61), Ganda (JE15), Soga (JE16), and Haya (JE22) (Pacchiarotti 2020), Ruuli (JE.103, based on corpus data shared with me by Alena Witzlack-Makarevich), as well as Shona (S11-15) (Peterson 2006), Cuwabo (P34) (Pacchiarotti 2020), Nen (A44) Ila (M63) Tonga (M64) Kwanyama (R21), Zulu (S42) and Kagulu (G12) (Schadeberg & Bostoen 2019). It would seem then, that this kind of isomorphism is very common in zone J languages, to which Lugwere belongs, but it is also reported in zones P, A, M, R, G and S. Zone M is close to zones J, S, and G, which in turn is adjacent to zone P. Zone S and zone R are also adjacent to each other, but A is quite far from the other ones. Note also that in many langauges where reflexes of **-ici* are used with instrumental arguments, it is no longer possible to use the **-ɪd* applicative in these contexts (Pacchiarotti 2020).

The suffix has always been considered to have been causative in PB, and cognates with causative functions are found even outside of Narrow Bantu, in Bantoid languages such as Cicipu (Kainji), Sama Mum (Dakoid) and Noone (East Beboid) (Blench 2022), so it would seem that the causative function is the oldest, and that the instrumental functions have developed later. However, the fact that it is found is so many parts of the Bantu area, alongside the fact that Peterson (2006) reports causative/instrumental isomorphism to be a less common form of causative/applicative isomorphism (in fact, the only example he gives is from Bantu), would suggest that this might be inherited. It is possible that the instrumental usage has existed alongside the causative for quite some time. However, the question would then be why this particular function of the causative extension has been lost in zones where there are no reports of such an isomorphism. One might also wonder why the instrumental usage has been lost in so many cases, while I am unaware of any cases where the causative function has been lost and only the instrumental remains. As many Bantu languages remain understudied, future research might show that this kind of isomorphism is more common than what we currently have data for.

Alternatively, we could be looking at cases of parallel innovation. Mbuun is another language that is reported to have a causative/applicative synchronism (Bostoen & Mundeke 2011). In this case, however, there are lexical causatives that use the same morphology as the productive applicative. This therefore seems like a different case from the instrumental usages of **-ici* seen in Lugwere and many other languages, and can probably safely be assumed to have come about through parallell innovation. It is possible that the causative/instrumental **-ici* syncretism could also have been invented in a few separate branches, and spread areally. In zone J, however, it is so widespread that it is hard to imagine that the usage developed independently in the different languages. At least in zone J, then, we can assume that we are looking at either a shared inheritance or an areal feature that has spread through contact, or there might have been elements of both. The fact that the instrumental/causative isomorphism has likely been around for a while, would suggest that the situation can be fairly stable.

The comitative usage of the causative suffix has not been reported in any of the accounts that I have seen, and Peterson (2006) writes that he has not come across any case of the Bantu **-ici* used with comitative arguments. This might therefore represent a new development in Lugwere.

According to Schadeberg & Bostoen (2019) sociative functions, particularly the adjutive (helping someone do something) can be found in other Bantu languages as well, particularly in the South, such as in Zulu (S42). As Zulu and other southern languages are quite distantly related to Lugwere, it might be a case of parallel developments, as it is not uncommon for causatives to develop sociative meanings.

Finally, the examples given in Section 6.2 where there is no change to the argument structure, but where there often is an intensifying sense to the version with the causative suffix, also has parallels in other languages. Schadeberg & Bostoen (2019) write that the intensive is 'another recurrent meaning of the causative extension'. In some languages there is a formal distinction between causatives and intensives. In Shona (S10) and Zulu (S42) , for instance, a reduplicated causative suffix is used for the intensive meaning. In other languages the reduplicated applicative has this function. In Lugwere, as we have seen, both the reduplicated (*-isisy*) and the simple causative (*-isy*) can have intensifying functions, and they can both function as prototypical causatives.

8 Conclusion

This article has illustrated different causative strategies in Lugwere, and shown how the inherited causative suffixes *-isy* and *-y*, as well as innovative *-isisy*, can be used in several ways that do not fit the definition of a prototypical causative. The suffix can be used to introduce instrumental object arguments, as well as comitative objects, two functions that fall under the umbrella of the applicative. The causative suffix can also be used in context where it has a sociative meaning. Additionally, there are cases where the presence of the causative suffix does not actually change the semantic or syntactic argument structure of the verb.

Most of these functions of the causative extension have been observed in other Bantu languages. The comitative usage, however, has not been mentioned elsewhere, and might be a new development in Lugwere.

Abbreviations

AUG	augment
APPL	applicative
CAUS	causative
COP	copula

COP.ID	identificational copula
DEM.DIST	distal demonstrative
DEM.MED	medial demonstrative
DEM.PROX	proximal demonstrative
FV	final vowel (default verbal ending used when no other suffix appears)
INF	infinitive
NAR	narrative tense
NEG	negation
NF1, NF2, NF3 (...)	Nominal form class - indicates what agreement class the nominal prefix typically corresponds to, but there are mismatches between nominal morphology and agreements.
ONOM	onomatopoeia
PASS	passive
PL	plural (only used for 1st and 2nd person. For 3rd person number is indicated by the prefix/agreement class)
PN	personal name
POSS	possessive pronoun
PST	past
PST.CAUS	past suffix used with causative stems
PST.PASS	past suffix used with passive stems
PRO	personal pronouns
REL	relativizer
REFL	reflexive
S	subject agreement
REL.OBJ	object relative marker
SG	singular (only used for 1st and 2nd person. For 3rd person number is indicated by the prefix/agreement class)
SBR	subordinator
O	object agreement
1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15=17, 16, 18, 20, 22	Agreement classes

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Chapter 14

Beyond the transitive prototype: Bodily-condition predications in Emai

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We assess transitivity prominence from the perspective of its periphery for an Edoid language of Nigeria. Emai exhibits a set of simple verb predications that assume a two-position transitive shape and express a bodily condition or an abstract condition. They permit a human participant in one position and a body-part noun or abstract noun in the other. They deviate from the transitive prototype, whose two participants exhibit meanings that are maximally distinct: a volitional instigator vs. a non-volitional entity showing material effect. Deviation is evident in the failure of participant distinctness and failure of material effect. Only the general condition on two-participant form remains of the transitive prototype. We also consider whether bodily-condition predications might belong to a grammatical domain other than transitivity. Copulas in Emai are considered since English translation of most examples employ a form of BE. There are five copula constructions. Their frames are not a match. None of their restrictions hold for bodily/abstract condition predications. We conclude by considering whether nominals other than those for body parts or abstractions can contribute to loss of the asymmetric force-dynamic relationship between participants that is central to the transitive prototype. Key words: transitivity prominence, peripheral predications, Edoid, Emai

1 Introduction

The linguistic concept of transitivity remains underexplored despite a wealth of studies. At its prototypical core, transitivity has received rather extensive attention (Hopper & Thompson 1980, Tsunoda 1985, Rice 1987, Kittilä 2002, 2008, Næss 2007, Haspelmath 2015). The periphery of the transitivity concept, however, has

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been examined much less. This is especially so for languages in West Africa, where "hyper-transitivity" has been asserted (Ameka & Essegbey 2006, Ameka 2013, Muysken & Smith 2015) but not defined or located relative to verb forms and their functions nor to nominal forms and their contribution to transitivity.

For this paper we examine features of transitivity prominence in Emai, not at its core but at its periphery. Emai, an Edoid language of Nigeria Elugbe (1989), is SVO with lexical and grammatical tone, the latter relying on floating tones that are either high or low. It displays verbs that combine in series or that permit either postverbal aspectual particles or postverbal prepositions. Although these complex predicate types occur, we direct attention to simplex predicates, i.e. simple verb constructions, that assume a transitive shape, employ two arguments, and express a condition pertaining to humans and their bodies.

2 Background

Haspelmath (2015) suggests that the perception of transitivity prominence may be more than impressionistic. He conducted a quantitative assessment of transitivity use, following a path initially explored with small verb sets and language sets by Müller-Gotama (1994) and Bossong (1998).

To assess transitivity prominence, Haspelmath sought to satisfy three criteria:

- A systematic database that would be reflective of languages from around the world;
- Verbs that would be diverse in meanings expressed and arguments used yet reasonably frequent in language use and therefore expected in most societies;
- A rigorous definition of transitivity.

These criteria were met in the Valency Patterns Leipzig electronic-database of Hartmann et al. (2013). The ValPaL database had a relatively large sample of verbs from thirty-five languages found around the globe. Verbs were defined as transitive if, like English 'break,' they had Agent and Patient arguments that correlated with the micro-roles "breaker" and "broken thing." Eighty verb meanings were documented in ValPaL; their meanings were diverse in kind, conveyed event experiences common to human societies, and appeared to be reasonably frequent in everyday usage.

Haspelmath employed a single measure of transitivity prominence across the ValPaL database. He computed the percentage of verbs in ValPaL that showed a transitive encoding of their meanings. Contrary to conventional wisdom about transitivity prominence in Central European languages, Haspelmath found that there were languages in Asia and Africa that had higher or much higher transitivity prominence. Some representative percentages from Haspelmath include the following:

- Europe: Bezhta= .46; Russian= .56; English and German= .60; Italian= .64;
- Asia: Mandarin= .50; Jaminjung= .62; Japanese= .65; Balinese= .68; Chintang= .77;
- Africa: Yoruba= .54; Standard Arabic= .62; N‖uu= .64; Mandinka= .64; Emai= .73.

Among European languages, English and German had percentage scores somewhat above 50%, whereas in Asia and Africa scores were found that exceeded 70%, which no European language exhibited. Included among the percentage scores from Africa were those of Emai (73%). Other languages from Africa did not have such a high score for the same set of meanings, e.g. Yoruba at 54%. At least for the verbs and languages compared by Haspelmath, it appeared that a significant number of meanings in Emai employed a transitive coding and that such a number was among the highest recorded. In large part we undertake the following study to explore further the nature and scope of transitivity prominence in Emai.

As for the set of meanings in English that stimulated the ValPaL database from thirty-five languages, they are presented in Table 1, taken from Haspelmath. Even in English, many of these meanings are coded by transitive verbs, but a number are coded with an intransitive form or a copula.

3 Periphery of transitivity prominence in Emai

There is more to transitivity and its prominence than the rather nebulous characterization “hyper-transitivity” that is found in the Africanist literature. Transitivity can be examined from the perspective of not only its core or prototype but also its periphery. But how does one explore the character of transitivity at its periphery? Where does one begin? Is there a domain that might be particularly revealing about the peripheral character of transitivity?

Over the last decade or so a series of publications have investigated the human body and its coding by linguistic means (e.g. Ruthrof 2015, Zariquley & Valenzuela 2022). Among them is Brenzinger & Kraska-Szlenk (2014) with its focus on

Table 1: The 80 verb meanings of ValPaL

eat	tell	put	die
hug	say	pour	play
look at	name	cover	be sad
see	build	fill	be hungry
smell	break	load	roll (intr.)
fear	kill	blink	sink (intr.)
frighten	beat	cough	burn (intr.)
like	hit	climb	be dry
know	touch	run	rain
think	cut	sit	be a hunter
search for	take	sit down	grind
wash	tear	jump	wipe
dress (tr.)	peel	sing	dig
shave	hide	go	push
help	show	leave	bring
follow	give	live	steal
meet	send	laugh	teach
talk	carry	scream	hear
ask for	throw	feel pain	cook
shout at	tie	feel cold	boil (intr.)

Africa. These studies address linguistic coding of the human body primarily by nominal forms; verbal forms that employ a body-part participant and encode a condition pertaining to a human participant have received far less attention.

In the following we direct attention to two-participant predications that express experiential conditions applicable to humans and their bodily makeup. Subsequently, we consider two-participant predications that express activities of motion and bodily emissions. Both derive from fieldwork of various kinds with Emai, an Edoid language of central Nigeria. Through translation of Emai into English in various projects, we have compiled an extensive comparative corpus, including spontaneous texts from oral tradition and databases from directed elicitation fostering dictionary and grammar activities.

Relative to illustrations from English, Emai bodily-condition predications deviate from the transitive prototype, as articulated by Næss (2007). Her principle of Maximally Distinguished Arguments (MDA) holds that a canonical transitive clause exhibits two participants whose meanings are maximally distinct. Essentially, NP-1 is a volitional instigator, whereas NP-2 is a non-volitional, affected entity that registers a material effect.

Deviation from the transitive prototype in Emai is evident in three primary predication types. At one level of analysis, these types differ with respect to the nature of their participants, which can be a human noun, an abstract noun, or a body-part noun. At a second level, they differ with respect to the positioning of the participant types. NP-1 can be a human noun or an abstract noun. NP-2 can be either a human noun, an abstract noun, or a body-part noun. Regardless of nominal nature or order, verb constructions expressing bodily-conditions constitute a two-participant predication (TPP), where a human participant occupies one position and a body-part noun or abstract noun the other position. Deviation from the prototype is evident in the failure of participant distinctness and the failure of material effect rather than simple overuse of a two-place predicate.

TPPs of one subtype diverge from the MDA principle based on the participant type in NP-2 position. The examples from Emai in (1) show that NP-1 is a human noun and NP-2 is an abstract noun that conveys a physical or non-physical bodily condition. Linking the nominals under this subtype is the high-telicity verb *de* 'reach.'

(1) a. Òjè dé ókìn.
Oje:PRX PST:reach:PFV unconsciousness
'Oje is unconscious.'

b. Òjè dé ɛ́khìì.
Oje:PRX PST:reach:PFV crippledness
'Oje is crippled.'

c. Ójé dé ɛ̀hò.
Oje:PRX PST:reach:PFV laziness
'Oje is lazy.'

A second subtype differs from the MDA based on non-distinctness of the entities involved in the predication. NP-1 and NP-2 do not refer to distinct entities. In the examples of (2) and (3), NP-1 is a human noun. NP-2 is a body part of NP-1. As a consequence, NP-1 and NP-2 do not stand as entities that are referentially distinct. Nonetheless, the examples in (2) and (3) differ in their meaning. In (2), for example, the verbs that link participants express a change of position: 'pull,' 'reposition,' or 'extend.' The NP-2 direct object for each is a body-part noun, respectively, *éhɔ̀n* 'ear,' *ɛ̀ò* 'eye,' and *òɛ̀* 'leg.'

(2) a. Òjè yí éhɔ̀n.
Oje:PRX PST:pull:PFV ear
'Oje is deaf.'

b. Òjè gbé ɛ̀ò.
Oje:PRX PST:reposition:PFV eye
'Oje blinked.'

c. Ójé ɔ́ ɔ̀ fì òɛ̀.
Oje:DST SI:DST PRS extend:IPFV leg
'Oje limps.'

The meanings assigned to the participants in the examples of (3) also do not refer to distinct entities. NP-2 is a body-part noun of the NP-1 participant. Although the meanings referred to involve a repositioning of a body part or the psychological state of a body, they all apply to a condition where the NP-1 referent is assumed to be interacting with the participants of a larger psycho-social/cultural matrix. Relative to this larger context of participants, the NP-1 referent is nosy, listening, or being patient.

(3) a. Ójé ɔ́ ɔ̀ hòò ɛ̀ò.
Oje:DST SI:DST PRS search:IPFV eyes
'Oje is nosy.'

b. Òjè gbé éhɔ́n kpɛ́kpɛ́kpɛ́.
Oje:PRX PST:reposition:PFV ear attentively
'Oje listened attentively.'

c. Òjè zíɛ́n égbè.
Oje:PRX PST:endure-PFV body
'Oje was patient.'

TPPS of a final subtype diverge from the MDA principle in that NP-1 is an abstract instigator and NP-2 an affected human entity. NP-2 experiences the abstract condition instigated by NP-1. Nominals are linked by verbs of contact with fictive effect.

(4) a. ɛ̀khɔ̀ì ó òjè.
shame:PRX PST:enter:PFV Oje
'Oje is ashamed.'

b. Ófɛ̀n ɔ̀ ɔ́ nwù òjè.
fear:PRX SI:PRX PRS take.hold:IPFV Oje
'Oje is becoming afraid.'

c. Úììn ɔ̀ ɔ́ gbè òjè.
fever:PRX SI:PRX PRS catch:IPFV Oje
'Oje is feverish.'

d. Òhànmì ɔ̀ ɔ́ gbè òjè.
hunger:PRX SI:PRX PRS catch:IPFV Oje
'Oje is hungry.'

With these examples we hope to have illustrated one feature of transitivity prominence in the Edoid language Emai. Across four sets of predications, members deviate from the transitive prototype but are still coded as transitive. We refer to these predications as peripheral members of the category transitive. Can we be satisfied with this general conclusion? There may be some room for doubt.

How uniform are the predications in (1) - (4) in their failure to attest the transitive prototype and so to conflict with some definitional feature of transitivity? Suppose we say that the predications in sets (1) - (4) conflict in some fashion with our two definitions of transitivity: Næss (2007) and Haspelmath (2015). First there is the principle of Maximally Distinguished Arguments (MDA) advanced by Næss (2007). It held that a canonical transitive clause exhibited two participants with maximally distinct meanings. NP-1 should be a volitional instigator; NP-2 should be a non-volitional entity showing material effect.

Across the four predication sets from Emai, there is no material effect, i.e., material change of state. There is no volitional instigator as NP-1 in any of the examples found in sets (1) - (4). Regarding the strong association of volition with human entities, there is a human noun as psychologically affected entity, NP-2 in set (1). It is not a non-human entity and it is not volitional for the predication in question. It is repeated as (5).

(5) Òhànmì ɔ̀ ɔ́ gbè òjè.
hunger:PRX SI:PRX PRS catch:IPFV Oje
'Oje is hungry.'

None of the predications in (1) - (4) expresses change of material state of the 'break' type, which Haspelmath viewed as central to the transitive prototype. Perhaps physical effect among these predications is more like 'hit,' where the NP-2 argument is affected but does not undergo a material change of state. Or perhaps we can allow that each set expresses a very general sense of change vis-à-vis some sense of cultural normalcy, as in the instance of reaching a crippled condition: (6) repeated from the first subtype in (1). But each of these possible interpretations takes us well beyond a straightforward change in material effect.

(6) Òjè dé ɛ́khìì.
Oje:PRX PST:reach:PFV crippledness
'Oje is crippled.'

4 Transitivity scaling of verb meanings

At this point it seems useful to review Haspelmath's approach to the scaling of verb meanings by Tsunoda (1985) and Malchukov (2005). Tsunoda provided a unidimensional scale of verb meanings and their likelihood of being coded as transitive. His scale can be represented as in Table 2 with small letters representing verb meaning types and small caps indicating specific verbs that instantiate these meaning types. According to this scale, transitive coding is more likely for verb meanings on the left than for verb meanings on the right.

Table 2: Unidimensional scale by Tsunoda for transitivity coding of verb types

direct effect	>	perception	>	pursuit	>	cognition	>	emotion
BREAK:HIT		SEE:LOOK		SEARCH		KNOW		LIKE

Subsequently, Tsunoda's unidimensional scale was restructured as predominantly two-dimensional by Malchukov, as in Table 3. He devised a split-scale for verb meanings that followed the meaning represented by BREAK, which a host of prior studies associated with the transitive prototype (Rice 1987, Croft 1993, Kittilä 2008). Malchukov's split-scale had an activity dimension and an experiential dimension, each of which consisted of different verb meaning types and the likelihood of their meaning being encoded as transitive. On the activity dimension, HIT meanings were presented as more likely to be coded as transitive than GO meanings. Between these two verb-meaning types, Malchukov positioned SEARCH meanings. Similarly, on the experiential dimension, the meanings for the SEE and LOOK type were more likely to be coded as transitive than those for the ACHE type, with FEAR type meanings assigned a midpoint on the experiential scale.

Guided by Malchukov's two-dimensional scale, Haspelmath computed the number of languages in the ValPaL database that coded a given verb meaning as transitive. The resulting numbers were then converted into percentage figures for each meaning type. The combination of meaning type and percentage as

Table 3: Two-dimensional scale by Malchukov for transitivity coding of verb types

	HIT	SEARCH	GO	(activity)
BREAK				
	SEE:KNOW	FEAR	ACHE	(experiential)

Table 4: Multi-layered scale by Haspelmath for transitivity coding of verb types

	HIT .94	SEARCH .89	GO .06
BREAK 1.0			
	SEE .92 : KNOW .88	FEAR .55	ACHE .12

likelihood for transitivity coding were then arranged in the multi-layered scale shown in Table 4.

The verb-meaning 'break' in the ValPaL database was coded as transitive by 100% of the thirty-five languages surveyed by Haspelmath. Activity-verb meanings aligned with HIT and SEARCH were coded as transitive at the relatively high rates of 94% and 89%, respectively. On the other hand, the activity-verb meaning GO in ValPaL was coded as transitive by only 6% of the languages. On the experiential layer, we find that verb meanings related to SEE and KNOW, respectively, were coded as transitive by 92% and 88% of the ValPaL languages. Verb meanings of the FEAR type manifested transitive coding in 55% of the languages. Only 12% coded verb meanings related to 'ache' as transitive.

With respect to Haspelmath's multi-layered scale, where do the Emai meanings expressed in example sets (1) - (4) fall? The predications in sets (1) - (4) fall exclusively onto the experiential dimension. Most reflect the extreme rightward point on the experiential dimension designated by ACHE, which we assume refers to experiential conditions that are internal to a human individual and pertain to that individual's mental makeup. These conditions may have physical as well as psychological manifestations. This seems true for the predications in (1) where we find the meanings 'be unconscious,' 'be crippled,' and 'be lazy.' The predications in (2), 'be deaf,' 'blink,' and 'limp,' also appear to be experiential conditions associated with ACHE, although 'blink' seems not to have the negative or non-normative qualities associated with the other conditions in (2). As for the predications in (3), they align with the most leftward point on the experiential dimension. The verb meanings 'be nosy,' 'listen,' and 'be patient,' appear interactional and assume the involvement of other members or features of a culture. They express psycho-social conditions that operate with respect to other members of a society. The experiential scalar point most nearly aligned with these conditions is designated by SEE and KNOW. As for the predications in (4), 'be ashamed,' 'be afraid,' 'be hungry,' and 'be feverish,' they align with the midpoint of the experiential dimension designated by FEAR. They represent conditions that are internal and most often non-normative relative to basic human experiences, but they are also particular to an individual's mental makeup and explicitly call attention to their effect on an individual human.

Much more could probably be said about each of these example sets and our initial attempt to distinguish them as subtypes. Experiential meanings of this sort are not often discussed in the semantically oriented literature of linguistics nor are the features or particulars that distinguish one experiential meaning from another given much attention, although meanings related to perception, most recently Norcliff & Majid (2024), and emotion (Lindquist 2021) point in promising directions. Over the years much more linguistic analysis has been directed at verbs that capture events in the physical world rather than in the non-physical world. For the moment, we ask that the reader indulge us on this matter of analytic shortcomings.

Having tentatively assigned our Emai predication sets to various meaning points provided by Haspelmath's experiential scale, we ask another question. Is the semantic space favorable to the transitivity coding of meanings at the periphery limited to the experiential dimension?

What of the most rightward portion of the activity dimension, the scalar point designated by the verb form GO? It, too, manifests a very low likelihood of transitivity coding since it was coded as transitive by only 6% of the languages in ValPaL.

Thus far we have not provided any evidence that transitivity coding in Emai is associated with the activity dimension, in particular activities related to meanings designated by GO. We take this point on Haspelmath's multi-layered scale to bear on activities of movement and motion where one entity moves through space, regardless of whether explicit mention is made of a second entity relative to which movement took place.

To consider further the scalar point identified by GO we turn to the likelihood of transitive coding for human body activities. A relevant domain concerns bodily emissions, which manifest two essential subtypes. Emissions from the human body can be characterized as either fluid/gaseous or sonorous (sound). In Emai, activities of bodily emission tend to be coded as transitive.

The first activity type we consider denotes fluid or gaseous emissions. Emai meanings represented by such verbs as *fɛna* 'excrete,' *roo* 'release,' *vbia* 'discharge,' *nɛ* 'pass,' and *fi* 'exhale' articulate as transitive structures (7). They combine with nouns in direct object position, respectively *ìsɔ̀n* 'feces,' *áàhìɛ̀n* 'urine,' *évìɛ̀* 'tears,' *èsɛ̀ìn* 'spittle,' *ìhɔ̀n* 'fart,' and *étìn* 'breadth.' Each verb-noun combination in construction with a subject noun refers to the emission of a fluid or gaseous substance and encodes as transitive.

(7) a. Òjè fɛ́ná ìsɔ̀n.
Oje:PRX PST:excrete:PFV feces
'Oje has excreted his feces/has defecated.'

b. Òjè fɛ́ná áàhìɛ̀n.
Oje:PRX PST:excrete:PFV urine
'Oje has excreted his urine/has urinated.'

c. Àlèkè róó évìɛ̀.
Aleke:PRX PST:release:PFV tears
'Aleke has shed tears/has teared up.'

d. Òjè vbíá èsɛ̀ìn.
Oje:PRX PST:discharge:PFV spittle
'Oje has discharged spittle/has spat.'

e. Òjè nɛ́ ìhɔ̀n.
Oje:PRX PST:pass:PFV fart
'Oje passed a fart/has farted.'

f. Òjè ɔ̀ ɔ́ fì étìn.
Oje:PRX SI:PRX PRS exhale:IPFV breath
'Oje is breathing.'

Second, we identify expressions for activities that code a sound emission. In Emai some of the relevant verbs include *ta* 'speak,' *so* 'sing,' and *zɛ* 'express.' Each occurs with a following noun in direct object position, sometimes with a cognate noun (*étà* 'words') but not usually (*íòò* 'song,' or 'thought,' *ùnyɔ̀* 'grumble'). Together with a subject noun, these verb-noun combinations are coded in a transitive pattern.

(8) a. Òjè tá étà.
Oje:PRX PST:speak:PFV words
'Oje has spoken.'

b. Òjè ɔ̀ ɔ́ sò íòò.
Oje:PRX SI:PRX PRS sing:IPFV song
'Oje is singing a song/ singing.'

c. Òjè ɔ̀ ɔ́ zɛ̀ ùnyɔ̀.
Oje:PRX SI:PRX PRS express:IPFV grumble
'Oje is grumbling.'

d. Yàn á zɛ̀ ìòò.
3PL:PRX PRS express:IPFV thought
'They are conversing.'

To be sure there are bodily emission meanings that are coded as intransitive. A few of the relevant verbs are seen in (9), where we find the meanings for *tiho* 'sneeze' and *oo* 'ooze' coded as intransitive. They do not participate in transitive coding patterns.

(9) a. Òjè tíhó-ì.
Oje:PRX PST:sneeze-PFV
'Oje has sneezed.'

b. Élì ɛ̀màì óó-ì.
ART boil:PRX PST:ooze-PFV
'The boils have oozed.'

Transitive coding relevant to Haspelmath's activity dimension is not limited to expressions of bodily emission. There are motion activities, specifically of the motion+path type, that Emai codes as transitive. Representative examples are shown in (10). They include the motion+path verbs *ye* 'move toward,' *lagaa* 'move around, encircle,' *shan* 'move through,' *hɛɛn* 'ascend,' and *kpɔɔn* 'descend.' Each combines with a noun of location in direct object position as well as a subject noun.

(10) a. Àlèkè yé èkó.
Aleke:PRX PST:move.toward:PFV Lagos
'The woman has moved toward Lagos.'

b. Àlèkè ɔ̀ ɔ́ làgáá ùhàì.
Aleke:PRX SI:PRX PRS move.around:IPFV well
'Aleke is moving around / circling the well.'

c. Àlèkè shán égbóà.
Aleke:PRX PST:move.through:PFV backyard
'Aleke has moved through the backyard.'

d. Àlèkè hɛ́ɛ́n ɔ́lí ɔ́kɔ̀ɔ́.
Aleke:PRX PST:ascend:PFV ART hill
'Aleke has ascended the hill.'

e. Àlèkè kpɔ́ɔ́n ɔ́lí ɔ́kɔ̀ɔ́.
Aleke:PRX PST:descend:PFV ART hill
'Aleke has descended the hill.'

The transitive coding of the motion+path verbs in (10) contrasts with the intransitive coding shown by the Emai verb corresponding to the designator point GO on Haspelmath's activity dimension. Corresponding to GO is the Emai verb *lodɛ* 'go' that is coded as intransitive. Its following noun phrase is coded as oblique by locative marker *vbi*.

(11) Àlèkè lòdɛ́ vbì ìwè.
Aleke:PRX PRS:go:IPFV LOC house
'Aleke is going to the house.'

Emai shows a similar coding pattern that employs oblique noun phrases for a few motion+path verbs. They are presented in (12). Both *o* 'enter' and *sɛ* 'move as far as' are coded as intransitive, as was the Emai counterpart of GO.

(12) a. Àlèkè ó vbí ɛ́kɔ́à.
Aleke:PRX PST:enter:PFV LOC room
'Aleke has entered the room.'

b. Àlèkè sɛ́ vbí ɛ́dà.
Aleke:PRX PST:move.up.to:PFV LOC river
'Aleke has moved as far as the river.'

5 Discussion

Given the preceding, one can ask what definitional elements remain from our attempt to identify predications at the periphery of transitivity in Emai? Recall that these elements were framed by the principle of Maximally Distinguished Arguments (MDA), which held that a canonical transitive clause exhibited two participants with maximally distinct meanings. NP-1 was a volitional instigator; NP-2 was a non-volitional, affected entity displaying material effect. Neither of these conditions on the status of transitive clause participants held for the simplex predications of Emai bodily-conditions reviewed in (1) - (4). It appears that it is only the general condition on form that remains of the MDA: A transitive verb must code a predication with two participants.

Let's consider another possibility. Could the examples or perhaps some of the examples in sets (1) - (4) belong to a grammatical domain other than transitivity? The copula domain comes to mind. Although copula status is perhaps far-fetched, the English translations of most examples in (1) - (4) employ a form of BE.

Emai has five constructions in its copula domain. Respectively, they assign a nominal to a state of existence (13a), clarify the discourse relation between copula nominals (13b), assign a nominal to a property (13c), assign a nominal to a location (13d), and assign a nominal to a class (13e).

(13) a. Óì kí ɔ́ɔ̀.
thief NCF be
'It isn't a thief.'

b. ɔ́áìn lí í khì àgbògbòràn.
that.one PCF ID be woodpecker
'That one is a woodpecker.'

c. ɔ́lì ɛ̀kpɛ̀n ú nwɛ̀nɛ́nwɛ̀nɛ́.
ART leopard:PRX PST:be:PFV spotted
'The leopard is spotted.'

d. ɔ́lí ɔ́vbèkhàn rîì vbí ìwè.
ART youth:PRX PST.be LOC house
'The youth is in the house.'

e. ɔ́lí ɔ́mɔ̀hè í ì vbì óì.
ART man SI NEG be thief
'The man is not a thief.'

Not all BE constructions provide an acceptable syntactic frame for the verbs in sets (1) - (4) or the sets that came later in the paper. Four of the five BE forms exhibit a two-participant structure. The specifier has only one participant (13a). One of the five with a two-participant structure requires a pronoun of identity as participant one and relates this pronoun to a preceding nominal in focus position. This complex structure is obligatory for the equational identity copula (13b). Two of the five limit the second participant to a particular type of semantic category: property term for the adjective in (13c) and locative for the spatial referent in (13d). One copula, class membership in (13e), shows two-participant nominal structures. However, it exhibits a strong preference for negative polarity and displays affirmative structures only in exceptional discourse contexts.

No copula seems to fit. None of the restrictions on copulas hold for the predication sets (1) - (4) from Emai or, for that matter, the predication sets in (5) through (12).

Copula assignment might be considered for set (1), which involved verb *de* 'reach.' But *de* is not a copula. Normatively, it allows two-participant structures, where the second participant is a location, as in (14).

(14) Òjè dé ókhúnmí ɔ́kɔ̀ɔ́.
Oje:PRX PST:reach:PFV top hill
'Oje reached the top of the hill.'

Perhaps one could propose that *de* is a copula in the making. But copular status is not where the meaning of *de* stands today. It seems time to conclude that there is no other grammatical domain other than the periphery of transitivity to which sets (1) - (4) might belong.

Moving on, the analysis undertaken in this paper, although leaving some matters unresolved, directs attention to potential areas of future investigation concerning transitivity coding. What we know from the present analysis of Emai is that the patterns of transitivity coding are not limited to one side of the scalar dimensions presented by Malchukov and Haspelmath. That is the side closest to BREAK.

Furthermore, we have seen that Emai transitivity coding is found across both scalar dimensions: activity and experiential. Relatively often in the past it has been assumed that transitivity coding is less likely for items in the experiential domain where meanings for mental events can be identified (Croft 1993). Nonetheless, the way transitivity coding for mental experiences diverges from the BREAK prototype has not been explicitly addressed in the linguistic literature. Nor has the literature explored the manner of divergence for non-mental experiences, i.e. physical activities such as motion+path verbs. Both strategies of divergence point to potential areas of future investigation.

For the moment let us return to the BREAK prototype. It consists of the elements proposed by the MDA Principle of Næss: two participants, participants that are maximally distinct (volitional instigator as NP-1 and non-volitional affected entity as NP-2), and realization of a material effect or material change of state.

Which of these elements is more susceptible to failure for verb meanings that do not reflect the transitivity prototype. In the case of Emai these meanings are on the experiential dimension or the activity dimension. Do all transitivity elements fail uniformly and across the board? Might there be a certain asymmetry among these elements of transitivity in their failed application to the activity dimension as opposed to the experiential dimension? Answers to such questions may provide us with additional insight into the nature of linguistic transitivity and for that matter intransitivity.

Generalizations that can be drawn from the present study of Emai are twofold. They pertain to the features of maximal distinctness and realization of material effect. Experiential predications failed regarding the MDA features maximal distinctness and realization of material effect. Both were absent. Activity predications failed concerning the MDA feature realization of material effect. Regarding the feature maximal distinctness, activity predications showed a split. Meanings for motion+path verbs manifested maximally distinct participants, whereas meanings for bodily emission verbs did not. Motion+path verbs (e.g. *hɛɛn* 'ascend,' *lagaa* 'encircle') that were coded as transitive exhibited a moving object as NP-1 and a location as NP-2. The referents for these noun phrases are arguably maximally distinct, especially when compared to bodily emission verbs. The latter, for example *fɛna* 'excrete,' and *vbia* 'discharge,' employ noun phrases that are not maximally distinct. NP-1 refers to a human noun and NP-2 refers to a substance that is retained within that human referent. In this respect, bodily emission verbs are comparable to bodily condition verbs, e.g. *fi òɛ̀* 'limp' from 'extend leg.'

Additionally, the realization of material effect feature relates in an interesting way to activity verb meanings. Neither activity type, motion+path or bodily emission, denotes a material change of state, a material effect. However, each refers to a change of location for an entity: NP-1 for verbs of motion+path and NP-2 for verbs of bodily emission. Furthermore, NP-1 for motion+path and bodily emission verbs could be viewed as a volitional instigator, or at least instigator, of their respective changes of location. Common to both the transitive prototype and the verbs of motion+path activity is change. Change of state in the former and change of location in the latter. Viewed from this perspective, the propensity for transitivity coding on the activity dimension seems less distant from the MDA prototype than does transitivity coding on the experiential dimension.

The preceding discussion has focused exclusively on transitive coding of meanings. Intransitive coding has remained in the background. A strong focus on intransitivity, however, has the potential to increase our understanding of transitivity. For example, little sustained linguistic investigation has been directed toward whether there are meanings confined exclusively to intransitive coding within and across languages. In other words, are there bedrock meanings that code as intransitive, just as there are bedrock meanings that code as transitive, e.g. BREAK? It is our belief that meanings that cross-linguistically tend to show exclusively intransitive coding might provide a greater understanding of how feature complexes underlying transitivity, such as those designed for the BREAK type, may be extended beyond the transitive prototype and so to the way transitivity is characterized.

As a final comment, the Emai data sets suggest an additional line of inquiry. Further investigation of transitivity prominence in West Africa, especially for meanings aligned with the experiential dimension, might usefully question whether nominals other than those for body parts or abstractions contribute to the failure of the asymmetric force-dynamic relationship between participants that is the linchpin of the transitive prototype (Rice 1987).

Abbreviations

ART	article	DST	distal temporal distance
ID	identity pronoun	IMPF	imperfective viewpoint aspect
LOC	locative	NEG	negation
NCF	negative contrastive focus	PCF	positive contrastive focus
PFV	perfective viewpoint aspect	PRX	proximal temporal distance
PRS	present tense	PST	past tense
SI	subject index		

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Chapter 15

A collaborative methodology for grammatical tone acquisition in Seenku

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Most research on L1 acquisition of tone focuses on East Asian languages, which are typologically distinct from tone systems in other parts of the world, such as Africa. One important difference is the greater prevalence of grammatical tone in African tone systems. This paper reports on a collaborative longitudinal study of the acquisition of Seenku (Mande, Burkina Faso), with a focus on the methodology for building a naturalistic, longitudinal and cross-sectional corpus, at a distance through community collaboration. While analysis of the tonal patterns is ongoing, preliminary findings corroborate the conclusion that tone is acquired early, with the novel finding that this may extend to grammatical tone.

1 Introduction

First language acquisition research has long privileged Indo-European languages (Kidd & Garcia 2022), leaving many linguistic structures understudied due to their absence or rarity in the language family. Among these is tone. Though recent decades have seen an increase in studies on the acquisition of tone, the vast majority of this work has focused on high-resource East Asian languages such as Mandarin or Cantonese. This poses an issue not just for diversity in language acquisition research but also skews our understanding of tonal acquisition, since East Asian tone systems are typologically distinct from tone in many other parts of the world, including Africa. For instance, East Asian lexical tone systems are often characterized by a larger number of contour tones, which may act as indivisible units, while African tone systems tend to privilege level tones, with contours

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more transparently made up of combinations thereof. Another important difference is that African tone systems often make extensive use of grammatical tone, which is comparatively rare in East Asia. This may have important ramifications for tonal acquisition, since tone is not simply a part of lexical representations but also an active part of the language's morphosyntax.

A large part of the bias towards European and other high-resource languages comes down to where the researchers are based (primarily Europe, North America and other wealthy nations) and what their primary languages are (Kidd & Garcia 2022). Studying children's speech requires a good grasp of the adult language, making research on understudied languages a challenge given the underrepresentation of first-language speakers of these languages in academia and the years of prior research required for non-speakers to reach a sufficient level of proficiency. Further, many child language studies are carried out in lab settings that may be inaccessible or unfamiliar to minority language communities; instead, studies of the acquisition of understudied languages tend to rely on naturalistic data collection, which on the one hand provides a better understanding of the child's linguistic input and surroundings but on the other hand makes it more difficult to target particular linguistic structures.

In recent years, there has been a push to diversify child language acquisition research, including by harnessing the methodology of language documentation. For instance, the Sketch Acquisition Project lays out a methodology for building a five-hour language acquisition corpus based on the speech of 2–4 year-olds as well as guidelines for an accompanying sketch of language acquisition (Hellwig et al. 2021, Defina et al. 2023a,b). The goal is to make language acquisition more accessible to a broader range of researchers, including field linguists and community members. This paper likewise seeks to broaden participation in language acquisition research by laying out a collaborative methodology that can be used by linguists and community members to build a longitudinal corpus of child speech. I report on an ongoing project in Burkina Faso to document the acquisition of Seenku, a northwestern Mande language spoken by roughly 15,000 people. The ultimate analytical goal is to understand how a complex tone system, replete with grammatical tone, is acquired, but the naturalistic data corpus can be used to investigate myriad topics in child language acquisition, from the emergence of morphosyntax to child-directed speech. For similar conclusions on the multipurpose nature of large, naturalistic acquisition corpora, see Snyder (2021) and Demuth (2022). It should be noted that this acquisition project is ongoing, and as such, data analysis is in its early stages. The focus of this paper will be methodological, though I highlight some of the research questions the corpus aims to answer and sketch out some preliminary results. In-depth tonal analysis will await future work.

The paper is laid out as follows: In Section 2, I provide an overview of previous work on the acquisition of tone and African languages to highlight the need for further studies. In Section 3, I introduce the target language, Seenku, and its tonal characteristics. The heart of the paper is in Section 4, which carefully describes the methodology used such that it could be replicated in other linguistic communities. Section 5 sketches out some preliminary findings with regard to grammatical tone acquisition, and Section 6 concludes.

2 Previous work on African languages

As noted in the introduction, languages of the “Global South” are underrepresented in the literature on child language acquisition (Kelly et al. 2015, Kidd & Garcia 2022). African languages are no exception. Kunene (1979) represents an early influential study in Bantu language acquisition focusing on the acquisition of the Siswati noun class system. This work was followed up by similar studies of nominal morphology in a range of other Bantu languages, including IsiZulu (Suzman 1991), Setswana (Tsonope 1987), Sesotho (Connelly 1984, Demuth 1988, 2000), and Sangu (Idiata 1998). Other morphosyntactic phenomena studied in Bantu languages include agreement (IsiZulu, Suzman 1982; Sesotho, Demuth 1988, 2000, Demuth et al. 2009; Swahili, Deen 2005), passives (IsiZulu, Suzman 1985, 1987; Sesotho, Demuth 1989, 1990, Demuth et al. 2010, Crawford 2012), applicatives (Sesotho, Demuth et al. 2005), and negation (Chichewa, Chimombo 1981). For an overview of the Bantu acquisition literature, see Demuth (2003). Studies on the acquisition of morphosyntax outside of Bantu are rare, but see Sagna et al. (2022) for the acquisition of demonstratives in Jóola-Eegimaa.

Various studies have focused on broader questions of phonological acquisition in African languages, both Bantu and, more recently, non-Bantu. Among Bantu studies, we find Mowrer & Burger (1991) on the acquisition of IsiXhosa consonants, with Lewis (1994) and Lewis & Roux (1996) looking more specifically at click consonants. Demuth (1992) likewise includes a study of consonant acquisition in Sesotho. Outside of Bantu languages, Cissé (2014) looks at segment acquisition in Bambara- and Fulfulde-speaking children, both monolingual and bilingual in the two languages. Amoako et al. (2020) study the phonological acquisition of Akan, and Kpogo et al. (2021) focus on the acquisition of labiovelars in Ga.

There are comparatively fewer studies of the acquisition of African tone. Chimombo & Mtenje (1989), who look at the acquisition of negation in Chichewa, make a mention of grammatical tone in particular, showing that tone patterns associated with negative constructions are acquired earlier than segments, but that the specific tone rules for different constructions are not in place until age 2;6. In a study of the acquisition of Sesotho verbal tone, Demuth (1989, 1993) found that the tone of subject agreement prefixes was acquired earlier than lexical tone on verbs, which may be due to the fact that verb tone is often obscured by postlexical tone processes like H tone doubling or H tone delinking. Likewise, Suzman (1991) reports that in IsiZulu, noun tone is acquired before verbal tone, likely for the same reasons. Outside of Bantu, the only other tonal studies I am aware of on L1 tone acquisition are Bakare (1995) and Harrison (2000), who investigated infants' acquisition of Yorùbá tone.

From this brief survey, it is clear that much more work is needed, as the vast majority of African language families remain completely unstudied, and even the extant studies focus on just a tiny slice of the language's structure. My goal in sharing our methodology is to encourage other scholars working with African languages to help fill these lacunae.

3 Seenku and its tone system

The target language of our acquisition study is Seenku, a northwestern Mande language spoken by approximately 15,000 people in southwestern Burkina Faso (McPherson 2020). It is part of the Samogo group of languages, which also includes Jowulu (Djilla et al. 2004), Dzùùngoo (Solomiac 2014), Duungoma (Hochstetler 1994), and Bankagooma. All of the children in our study are being raised in the village of Bouendé (Gbéné-gṵ̏), and many of the primary caretakers are bilingual in Jula, the local lingua franca, though the children are exposed primarily to Seenku.

From a morphosyntactic perspective, Seenku is a typical Mande language. It displays S-Aux-O-V-X word order and has largely isolating morphology (outside of grammatical tone):

(1) mənı̋ nǎ bı̋ sɔ̋ɔ səgı̰̂ı̰ nɛ̏
woman PROSP goat.PL sell.IRR market LOC
'The woman will sell goats at the market.'

It has no noun classes and no verbal or nominal agreement. Further, most words in Seenku are either monosyllabic or sesquisyllabic (Matisoff 1990), consisting of a syllable and a half, as illustrated by *mənı̋* 'woman' or *səgı̰̂ı̰* 'market' above.

This morphosyntactic simplicity is counterbalanced by a highly complex tone system. Seenku displays four contrastive levels of tone, as evidenced by the near-minimal set in Table 1.

Table 1: Seenku’s four level tones

Tone	Example	Gloss
Super-high (S)	sí̋	‘tree sp.’
High (H)	(í) sí	RECIPROCAL
Low (L)	sì	‘first son (proper name)
Extra-low (X)	sȉ	‘water jar’

These four tones can combine to create a wide array of two- and three-tone contours, including HX (e.g., *bî* ‘goat’), LS (e.g., *nǎ* PROSPECTIVE), SX (e.g., sɔ̋ɔ̏ ‘sold PERF’), HS (e.g., *ŋáá̋* ‘yawn’), XHX (e.g., *gɔ̏ɔ̂n* ‘sorrel’), LSX (e.g., *nàä* ‘come PERF’), etc. Two-tone contours can be found on both short and long vowels, though three-tone contours require a long vowel to be realized.

Seenku’s tone system presents an interesting case study in acquisition, as it has characteristics of both African and Asian tone systems. Like East Asian tone systems, Seenku employs a large number of contour tones on largely monosyllabic vocabulary. Like other African tone systems, however, there is clear evidence that these contour tones are composed of combinations of level tones and, importantly to our study, it makes extensive use of grammatical tone.

We can divide Seenku’s grammatical tone patterns into two broad categories: morphological tone, i.e., cases in which tone is a primary marker of a morphosyntactic feature, and argument-head tone sandhi, in which tone is sensitive to morphosyntactic structure but does not directly expone any feature. As a shorthand, we refer to these categories as “meaningful” grammatical tone and “meaningless” grammatical tone. Examples of the former include plural tone raising (e.g. *bɛ̏ɛ* ‘pig’ vs. *bɛ̀ɛ* ‘pigs’, McPherson 2017a), perfective tone lowering (e.g. *jɔ̋* ‘brew’ vs. *jɔ́* ‘brewed’, McPherson 2017b), and perfect formation through the suffixation of -SX (e.g. *jɔ̋* ‘brew’ vs. *jɔ̈* ‘has brewed’, McPherson 2020).

Argument-head tone sandhi involves tone changes on a syntactic head triggered by its internal argument. It is found between an inalienable noun and its preceding possessor, a postposition and its preceding NP argument, and between a verb and its direct object. Sandhi is sensitive to more than simply syntactic structure, however, as verbal sandhi only occurs when the verb is in its irrealis form (prospective, imperative, etc.). The tone changes themselves are not always phonologically predictable, with different outputs depending on whether the argument is a noun or a pronoun and even small differences in outputs depending on the syntactic category of the head. For details, see McPherson (2019).

Given this tone system, the questions guiding our acquisition study include:

1. How do Seenku-speaking children acquire the tone system?
2. When is grammatical tone acquired?
3. Are there differences between the acquisition of morphological tone vs. morphosyntactically-sensitive sandhi?
4. Can children's acquisition of grammatical tone patterns shed any light on where sandhi belongs in the adult grammar (phonological, morphology, lexicon, etc.)?

Further analysis of the data corpus is required to answer these questions. We offer them here to provide insight into how we designed our study and how we hope to use the data we have gathered. Once analyzed, the Seenku data will enrich the literature on the acquisition of tone sandhi, as the tone changes are paradigmatic like Chinese sandhi (see, e.g., Tang et al. 2019) but sensitive to morphosyntactic information like Bantu tone (see, e.g., Demuth 1993).

4 A collaborative methodology

In this section, I turn to the collaborative methodology our team has used to build a yearlong corpus that will allow us to answer these and other questions in Seenku L1 acquisition. The collaboration involves an international team, including Burkinabe and American university students. The methodology we use was designed in consultation with Dr. Ibrahima Abdoul Hayou Cissé (formerly University of Bamako, now UNESCO), who compiled a similar corpus through his dissertation work on Bambara and Fulfulde acquisition (Cissé 2014) and helped train the Burkinabe acquisition team, as laid out below.

4.1 Training the team

The project began with identifying a team of six Seenku-speaking university students who would commit to gathering data twice a month for a year. I had already been working with the Seenku-speaking community for 8 years, and so I left this identification to one of my primary consultants, Emma Traoré, who is also part of the acquisition team. The other members of the team are Sibiri Traoré, Louise Traoré, Siaka Traoré, Gérard Traoré, and Abou Traoré. It is important to note that none of these students study linguistics (their studies range from nursing to economics), so a lack of linguistics students should not be seen as a hindrance to doing this type of work.

In October 2021, I held a training workshop for the team at our project base in Bobo Dioulasso, Burkina Faso. On the first day, we covered some basic linguistic background on Seenku, focusing on transcription; Seenku does not have a widely adopted orthographic tradition, and the combination of complex tone, a contrast between nasal vowels and a nasal coda, and a rich vowel system makes orthography development a challenge. I trained the team on the transcription system I use in my own work, using handouts and exercises. As part of this training, I introduced them to the Keyman IPA keyboard on the Windows laptops purchased for the project.

The second day focused on technology: how to use the video recorders (Canon Vixia HF R62), audio recorders (Zoom H4N), lavalier microphones (Shure SM93, plugged into the Zoom H4N), and tripods, and how to set this equipment up for a recording session with a child. The Zoom H4N is placed into a toddler backpack, with the lavalier mic clipped to the front of one of the straps (a technique learned from Barbara Kelly at the 2019 workshop "Talking about child language documentation: Experiences, barriers, methods & outcomes" at the International Conference on Language Documentation and Confirmation). After practicing recording each other, we covered the use of external hard drives, file naming conventions, and adding files to Google Drive. In the second half of the day, the team took the equipment out to record practice sessions with small children in their neighborhood to work with the following day.

The final day of the workshop added ELAN to the workflow, teaching the team how to create new files from a template and transcribe child and caregiver speech on different tiers, including a French translation.

The original plan was to run this workshop together with Dr. Ibrahima Cissé, but due to a number of logistical and scheduling challenges, his portion of the training was held in Bamako, Mali in early March 2022. He complemented the students' training with a theoretical overview of child language acquisition, the ethics of working with children and families, practical strategies for creating a natural environment for children to speak, and pre-study surveys of children and their caretakers to better understand their linguistic ecosystem (who is the primary caregiver, what languages are spoken in the child's environment, etc.).

4.2 Data collection and corpus design

With this training in place, the team was ready to begin data collection in late March 2022. The goal was to build a corpus that is both cross-sectional, including children of different age groups, as well as longitudinal, looking at the same children over time. To capture both the emergence of lexical tone as well as more complex patterns of grammatical tone, we chose to include children between the ages of 1;0 and 3;0 at the start of data collection. We identified 10 subjects in Bouendé, two children in each age group (1;0, 1;6, 2;0, 2;6, and 3;0) at the start of data collection. One subject in the 3;0 group never got comfortable in front of the camera and was replaced after two months with another child of a similar age. Four of the children are female and six are male (seven including the first 3;0 child, who was replaced with another male child). For most children, the primary caregiver is the mother, though for one child it is the grandmother. All are predominantly Seenku-speaking, though a few of the caregivers report the use of Jula, which does appear in some of the recordings.

Each child was recorded twice a month over the course of a year (March 2022 -through March 2023), with each recording consisting of one hour of interaction with their primary caregiver. The students traveled to Bouendé in pairs to make the recordings and would leave the camera running on a tripod (with the Zoom H4N in the child's backpack), letting the child and caregiver interact without a direct observer. Families received financial compensation of 2000 CFA (roughly $4 USD) per recording session, with a final bonus of 10,000 CFA at the end of the year.

With the number of recordings ranging from 22 to 25 per child, the final corpus consists of around 236 hours of audio and video child language data.

4.3 Data management

One of the challenges of an international project such as this is data management and sharing. With such a large corpus, it was also imperative to stay organized as we went. The first step was for the Burkinabe team to transfer and rename all recordings to external hard drives for safe keeping. File names consisted of the following format: Child's initials_YYYY_MM_DD (though this was eventually changed when archiving for consistency with the rest of the collection). Twice a year, hard drives with copies of all original files were brought back to the US by a colleague or shipped from Burkina Faso. However, in order to facilitate access to data throughout the year, Sibiri Traore of the Burkinabe team would reduce the file sizes of the audio and video and upload them to a private Google Drive with permissions set only to team members. As I have unlimited Drive space through my institution, I would make copies of all of these files so the originals could be deleted from Sibiri's account.

At this stage, with available data in the US, a team of undergraduate research assistants at Dartmouth College took over data management. This included managing an organizational spreadsheet of the data in Google Sheets, as well as producing IMDI metadata using Lameta (Hatton et al. 2021). The spreadsheet allows us to double check for missing files and keep track of transcription progress, both on the Burkinabe side for the original transcriptions and on our side for tone transcriptions; we turn to transcription in the next subsection. US team members to date have included Yuanhao Chen, Daniel Chen, Simon Lamontagne, Sofia Lee, and Zhuoxin Ma.

4.4 Transcription

For the recordings to be usable for analysis, they must be transcribed, and as any fieldworker knows, this is the bottleneck. Transcription is time consuming under the best of circumstances, and child language data is even more challenging, especially for the younger children in the corpus. Nevertheless, the Burkinabe team was able to transcribe a large number of recordings before funding ran out, with 93 out of 236 recordings transcribed (nearly 40% of the corpus).

Before transcribing the sessions, the Burkinabe team combined the two video files for the session (the camera automatically splits the file once it reaches a certain size) and clipped the audio such that it aligns with the video file. This ensures that both files can be added to ELAN, and the (typically) cleaner audio produced by the Zoom H4N can be listened to in conjunction with the video.

Transcriptions were done in ELAN, using a simple template with two participants: child and caregiver. Each participant has a transcription tier, where the Seenku is transcribed using the practical orthography I discussed in Section 4.1 above. The French translation is then put on a daughter of this tier. Where the child's speech is unintelligible, the Burkinabe team puts *pas compris* 'not understood'. By and large, the Burkinabe team's transcriptions represent the adult targets rather than phonetically reflecting the child's pronunciations, though they occasionally pointed out divergences between the child's speech and adult speech.

Once these transcriptions are uploaded, the US team adds tiers for children's tone (phonetic), children's tone (phonological categories), adult tone (phonological categories), and grammatical notes. The phonetic transcriptions are done using Chao tone numbers (where 5 is the highest pitch and 1 is the lowest). This was chosen for convenience, as the three undergraduate tone transcribers are speakers of Mandarin/Shanghainese and are familiar with the transcription system, but it also allows a more neutral rendition of pitch contours rather than trying to fit the f0 contours directly to Seenku tone targets. This is especially useful for young children's speech, which often includes pitch contours that are not always easily identifiable as an adult target. None of the undergraduate transcribers had any familiarity with the Seenku tone system, eliminating the possibility of bias (i.e., that they would hear the tone they expected to hear). For the first couple recordings, I would independently produce phonetic transcriptions, and comparison of our transcriptions revealed a very high degree of reliability, with most discrepancies related to pitch scaling rather than overall contours. As a final step, I go back through all transcriptions and produce a tier for the adult tone target of the child's speech (correcting any inconsistencies from the Burkinabe team) and a tier mapping the child's phonetic contours to their closest phonological category match, a non-trivial task that can involve difficult judgment calls in such a complex tone system (e.g. was that falling tone the child aiming for HX but falling short, or did the child miscategorize the word as HL?). Such methodological questions for tonal acquisition will be addressed in future work.

At the time of writing, the team has produced 11 hours of phonetic tone transcription, with three hours mapped to phonological categories. In other words, each step of the transcription process presents a new bottleneck, but with time and, crucially, collaboration, a large corpus of tonal acquisition data for Seenku is possible.

5 Preliminary findings

The main focus of this paper is the collaborative methodology, described with enough detail to serve as a "how-to" guide for others interested in expanding their work to encompass child speech. We are still in the early stages of data analysis as we continue to transcribe enough tone data to be able to systematically address the emergence of phonemic categories, the honing of f0 contours, and the relative timing of when grammatical tone appears as compared to lexical tone. Nevertheless, we offer some preliminary results here based on the available data thus far.

The results draw from 11 hours that have been transcribed phonetically for tone. This includes at least one hour for each age quartile between 1;0 and 2;0, plus one hour for each of 2;9 and 3;0; we currently have no tone transcribed for children around the ages 2;3 or 2;6.

Examination of our youngest participants reveals evidence for lexical tone acquisition. Of course, to be able to identify whether or not a child has accurately acquired lexical tone, we need identifiable words. I provide here some examples from FRST, aged 1;0 at the start of data collection. At 1;2, he produced 63 utterances, of which all but 6 were interjections without a clear lexical tone target. He independently produced a short phrase, *kɛ́ tɛ́* 'who is it?', and both repetitions of the phrase showed level tone consistent with the adult H H sequence. His pronunciation of *tàtá* 'auntie' likewise shows a rising sequence of pitches, though with a smaller interval than his mother's repetition of the word; nevertheless, he demonstrates knowledge of its overall lexical tone contour. He even demonstrates some evidence for grammatical tone in a short phrase, responding to an older child he is playing with. Consider the following exchange:

(2) a. Ń bá̰!
1SG hit.IRREAL
'Hit me!'

b. Nű bá̰ . (adult target: ń !nű á bá̰)
1SG.PROSP hit.IRREAL
'I'm going to hit you.'

The imperative and prospective are constructions that show argument-head tone sandhi, with the tone of the verb changing depending upon the tone of its argument. Here, the H-toned pronoun causes the lexically S-toned verb *bá̰̋* to lower to H. FRST's short phrase elides the initial moraic nasal of the 1sg pronoun, and the fall from S on the prospective *nű* to H on the 2sg pronoun *á* is rendered as simply S. However, the verb is pronounced with the correct H tone for this environment. It is likely at this age that he is simply repeating back the tone produced by the older child rather than productively applying the rules of sandhi (which I turn to below), but more data are required to test this hypothesis. The only lexical tone error FRST produces at this age is the proper name *Sà*; a L-toned name, FRST instead pronounces it with a high falling pitch (phonetically 43).

By 1;5, FRST displays a lot of mimicking of his mother's speech, with quite good pitch mimicry. He still does not independently produce a lot of words; out of 170 utterances, only one consists of independent identifiable words. However, this short phrase shows surprisingly good tone. Consider the adult target in (3a) versus what the child produces in (3b):

(3) a. Ń !nű nân nɔ̏.
1SG PROSP sauce eat.IRREAL
'I will eat sauce.'

b. Náắ nän nɔ̏.
1sg.prosp sauce eat.irreal
'I will eat sauce.'

The child produces a rise from H to S seen in adult speech between *ń* and *nắ*, though for the child, this rise is produced on a lengthened vowel instead. The child likewise produces a fall on 'sauce', though while this is a HX contour in adult speech, the child begins the fall on S instead. Finally, the irrealis verb takes on the X from the end of the object (argument-head tone sandhi), and this is reflected in this child's speech; I will return to more cases of grammatical tone below. In short, this example shows that at 1;5, FRST may begin to command both lexical and phrasal tone rules, though the exact phonetic implementation is not yet adult-like.

If we look at a child in the next age bracket (DET, 1;6 at the start of the study), we see more lexical items as well as more phrases, giving us the opportunity to examine grammatical tone. At 1;8, DET produced 88 utterances, ranging from interjections to mimicry to independent novel phrases. By and large, his lexical tone is accurate, though he did display a few errors and inconsistencies. Specifically, he shows some confusion between the lexical melodies L-S, X-S, and X-HX; all three involve a lower first syllable and a higher second syllable, though for X-HX, the second syllable falls. For instance, the L-S word *sàkɪ̋* 'bag' (borrowed from French *sac*) is pronounced twice as something close to L-S (3-4 and 4-5 phonetically) and once as X-HX (3-42). The word *tɔ̏ntɔ̋n* 'uncle' shows similar variation; out of 14 tokens, 5 show a larger pitch interval consistent with the melody X-S (2-4, 3-5, 1-3, etc.), 4 show a smaller interval consistent with the melody L-S (3-4, 4-5), one shows a pronunciation 2-31, consistent with the melody X-HX, and the remainder show rising contour tones on either the first or second syllable, such that the overall pitch contour is rising (like the adult target) but with the alignment of the pitch rise incorrectly mapped to syllable boundaries. Interestingly, the two X-HX pronunciations of *sàkɪ̋* and *tɔ̏ntɔ̋n* occur just seconds apart in the recording, suggesting a confusion of lexical tone categories rather than individual lexical items.

At this age, we see more short phrases where we can investigate the (non-)application of grammatical tone processes, both "meaningful" and "meaningless". In terms of "meaningful", i.e., morphological, grammatical tone, most of what we see at this age is verbal tone, especially perfect verbs (suffixed with a -SX contour); there are surprisingly few plurals in the data we have transcribed thus far. Even in mimicking, grammatical tone application is not always consistent for DET. He repeats a short phrase back and forth with his mother, and while the adult pronunciation is consistent, as shown in (4a), his pronunciation of the verb's tone varies, as shown in (4b):

(4) a. Jɛ̀ dó-dő̃n kàä.
first.girl RED-child go.PERF
'Little Jɛ (birth order name) has left.'

b. Jɛ̀-dő̃n kàä ~ *ká ~ *káà.
first.girl-child go.PERF
'Little Jɛ has left.'

The adult pronunciation of the verb is a LSX tri-tone contour (the result of suffixing -SX to an underlyingly X-toned verb). In one repetition, the child correctly produces this morphological tone. In a second, he shows no contour whatsoever with a H-toned pronunciation that reflects neither lexical tone nor grammatical tone. In a final pronunciation, he shows a HL fall, perhaps approximating the grammatical tone melody, though without the concatenation with lexical tone. Later in the recording, DET produces a perfect verb unprompted with correct falling tone in the phrase *tɔ̀ntɔ̋n sïɔ* 'uncle has arrived' (though DET produces incorrect segments for the verb, rendering it as *küɔ*). Since the verb in question is H-toned rather than X-toned, the tonal realization is simply the grammatical melody SX rather than the tritone LSX, so we cannot explicitly see in this case whether DET is aware of the verb's lexical tone or not.

Finally, we take a brief look at argument-head tone sandhi, which from an outside perspective would appear to be a more challenging pattern for children to acquire, as it carries no discernible meaning and is dependent on a host of factors, including lexical tone, syntactic category, and aspect/mood (in the case of verbs). At 1;8, 8 of DET's utterances contain contexts for tone sandhi. His application is inconsistent, with 3 correct applications and 5 incorrect or unanalyzable due to segmental and other errors. For instance, we can consider the phrase in (5), which contains two environments for sandhi, the first between the 2sg pronoun *á* and the postposition *wɛ̏* (which would become *wɛ̋*), and the second between the 3sg pronoun *ȁ* and the verb *nɔ̋* 'eat' (which would become *nɔ́*). Lexical tone is shown in slashes in the top line:

(5) /ń nǎ cɛ́ á wɛ̏ ȁ nɔ̋/
(ń) !nǎ cɛ́ á ***wɛ̏** ȁ **nɔ́**. (correct: …á wɛ̋ ȁ nɔ́)
1SG PROSP give 2SG with 3SG eat.IRREAL
'I will give it to you to eat.'

DET fails to apply tone sandhi to the postposition *wɛ̏* but appears to apply sandhi on the verb 'eat', with a smaller interval between the object and verb than we would expect for a lexical X to S sequence; however, the phonetic implementation is not adult-like, with a small rise the verb rather than a level H tone.

In another context for object-verb tone sandhi, DET commands his mother 'give (me) candy', which in adult speech would be *bɔ̏nbɔ̋n bɛ̋*, with the X-toned verb *bɛ̏* 'give/put' taking on the final S of *bɔ̏nbɔ̋n* 'candy'. However, in his speech, the verb is instead pronounced as *bɛ́*, that is, with neither lexical tone nor correct sandhi tone. One hypothesis is that this reflects interference from the verb's lower lexical tone, i.e., the raising to S is incomplete, but we will need more examples to see if this is a broader pattern or a single error.

By 2;0, however, DET's Object+Verb tone is categorically good; that is, he appears to have internalized the rules of changing verb tone depending on the tone of the object, even if his phonetic implementation is not yet adult-like. Out of 37 tokens of sandhi contexts, 29 were clearly correct, 3 were borderline, and 1 was incorrect; 4 were impossible to assess due to either missing segmental material or uncertainty about the adult tone target.

Another child at 2;1, SRDT, shows evidence of tone alternations resulting from sandhi. For instance, consider the following two phrases produced in close proximity; once again, the first line represents lexical tone while the second line represents the child's utterance with sandhi rules applying:

(6) a. /tərɛ́ kʊ́ bʊ́ ń cɛ̏/
Tərɛ́ kʊ́ **bʊ̏** ń cɛ̋.
REL.PRO D.DEF take.out.IRREAL 1SG give
'Take that one and give it to me.'

b. /tsɛ̋ bʊ́ ń cɛ̏/
Tsɛ̋ **bʊ̋** ń cɛ̋.
INDEF take.out.IRREAL 1SG give
'Take one and give it to me.'

In (6a), the object of the verb *bʊ́* 'take out' is the discourse definite pronoun *kʊ́*; this triggers X tone on the verb, which SRDT produces. In (6b), however, the object is the indefinite pronoun *tsɛ̋* which spreads its S tone onto the verb. SRDT correctly produces both of these forms.

We should note, however, that all children acquire language at their own pace. While DET and SRDT are producing complex grammatical tone, another participant at 2;1 is still speaking in single words. This demonstrates the importance of having a sufficiently large sample size of participants lest we draw conclusions from a single child, or a couple of children, who may be either ahead or behind the language acquisition curve.

Turning to an older child, DEMT, we see that by around 2;8, the grammatical tone system is largely in place, including both morphological tone and tone sandhi. Preliminary analysis of over 400 utterances shows few tonal errors, either lexical or grammatical. The following example demonstrates DEMT's grasp of complex interacting grammatical tone processes:

(7) /ń ȁ dzḭ̋ ń gȕ nɛ́/
Ń ȁ dzḭ̈ ń gű nɛ̋.
1SG 3SG put.PERF 1SG neck LOC
'I have put it on my neck.'

In the top line, we see the lexical tone of each word. First, the verb *dzḭ̋* 'put' is realized as *dzḭ̈* (SX falling tone) to mark perfect aspect. Because the verb is not irrealis, there is no interaction between it and the preceding object pronoun *ȁ*. Second, we see argument-head tone sandhi between the 1SG *ń* and the noun *gȕ*, rendering it S-toned *gű*. Finally, this S tone then spreads onto the locative postposition *nɛ́*, rendering it *nɛ̋*. The child accurately produces all of these complex grammatical tone changes.

While these are only preliminary examples, the acquisition trajectory that they suggest echoes the literature on East Asian tone languages, namely that tone is acquired earlier than segments but that adult-like pronunciations take more time (Wong 2012). In other words, tone is acquired both early and late. With Seenku's complex grammatical tone system, we can add the finding that both lexical and grammatical tone appear to be in place by around 2;6–2;8, with morphological tone acquired earlier than argument-head tone sandhi. Future work will test this hypothesis, with quantitative results drawn from a larger set of transcribed data.

6 Conclusions

In this paper, I have laid out the motivations and the methodology for a collaborative approach to children's acquisition of Seenku tone. An international team consisting of trained native speaker students of Seenku as well as linguistics students and faculty in the US allows data collection, transcription, and analysis to persist across time and national boundaries and has resulted in a yearlong longitudinal and cross-sectional corpus. To my knowledge, at 236 hours of recorded speech, this represents the largest extant corpus of African language acquisition.

Acquisition of African languages remains radically understudied and yet it stands to shed light on a range of important topics for how children acquire the language faculty, from multilingualism to noun class systems to complex grammatical tone. This lack of attention paid to languages of the Global South likely comes from a vicious cycle, whereby most language acquisition research is focused on high resource languages, and so those interested in languages of Africa or other parts of the world gravitate to other areas of linguistics (phonology, syntax, language documentation); hence these gaps in the literature are never filled, and few mentors are available to inspire the next generation's work. This is thankfully beginning to change, with initiatives such as the Sketch Acquisition Project laying out a path for those interested in language documentation to make a meaningful contribution to the study of child language, but more work must be done. Here, I hope to have shown how those already working in language communities but without formal training in language acquisition (as was the case for myself) can still set up a project to collect child language data. Doing so not only helps fill the gaps in child language acquisition research but also enriches a documentary corpus by including children's speech and child-directed speech patterns.

The Seenku acquisition project may have completed its data collection phase, but much work remains to be done. As the number of transcribed sessions grows, we will be able to get more quantitative measurements to better understand the trajectory of tonal acquisition. For instance, we could ask what percentage of lexical and grammatical tones are accurately produced, and how do these numbers increase over time? Does accuracy differ by tonal category, such as higher accuracy for H tones (as shown by Suzman 1991 for IsiZulu or Li & Thompson 1977 for Mandarin)? With access to both longitudinal and cross-sectional data, we can also study when errors begin to decrease for a particular child, in what order, and whether these patterns are similar across children. Finally, as the data corpus contains hours of naturalistic speech, it can serve as a resource to study a range of other topics in language acquisition, from segmental phonology to morphosyntax, to how caregivers interact with their children.

Abbreviations

D.DEF	discourse definite
INDEF	indefinite
H	high tone
IRREAL	irrealis
L	low tone
LOC	locative
PERF	perfect
PL	plural
PRO	pronoun
PROSP	prospective
RED	reduplicant
REL	relative
S	super-high tone
SG	singular
X	extra-low tone

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Chapter 16

Phonosemantic conditioning in Kirundi loanword noun class allocation

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This paper examines the phonological and semantic factors in the noun class allocation of loanwords in Kirundi. I provide a formal account of the criteria used in determining the noun class of loanwords during the lexicalization process. The data indicates an interaction between phonological and semantic factors. I formalize the relevant factors as Optimality Theory constraints and describe a ranking capable of predicting the noun class allocation of novel loanwords in Kirundi. The analysis taken here calls for greater unity between the disciplines of semantics and phonology, traditionally taken to be distinct save for certain phenomena, such as prosody.

1 Introduction

This paper examines the noun class allocation of loanwords in Kirundi, a Bantu language of Burundi, providing a semantic and phonological account of the noun class loanwords are assigned. Kirundi employs a noun class system expressed through noun class prefixes. Noun class pairings are pairs of noun classes that nouns typically alternate between to express number. Generally, Kirundi loanwords appear in class 9/10 (Zorc & Nibagwire 2007, Niyondagara 1993, Niyonkuru 1987).[1] However, there are several other semantic and phonologically motivated

[1]The data used for this paper was taken from a variety of sources. Cox's (1969) dictionary, Zorc & Nibagwire's (2007) grammar, Niyondagara's (1993) dissertation, and Rwamo & Constantin (2019) supplied a large portion of the data on loanwords. In addition, I consulted native speakers for their judgement of loanwords discussed in those sources. Extra data points were found

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factors in determining noun class. I present examples where semantic factors override phonological criteria. Restrictions on semantic content within a noun class may block nouns whose phonological form would otherwise condition them to appear with a different prefix. I demonstrate instances where phonological factors place nouns in noun classes inconsistent with their semantic properties. Neither phonological, perceptive, nor semantic factors solely account for the allocation of noun class in Kirundi loanwords. The data indicate a clear interaction between phonological and semantic constraints on faithfulness and markedness. A formal ranking is provided, capable of predicting the noun class allocation of novel loanwords in Kirundi.

Section 2 provides a brief background of Kirundi's noun class system. Section 3 presents data on loanwords in Kirundi and identifies core morphological and phonological factors in determining noun class of Kirundi loanwords. In Section 4, I formalize these morphophonological factors in an Optimality Theory (OT) framework by supplying constraints and justifying their ranking. Section 5 presents apparent irregularities in allocating Kirundi loanwords to the noun class system, where semantic aspects of noun classes affect phonological principles. Section 6 accounts for semantic effects by providing formal semantic constraints ranked critically with morphophonological constraints discussed in Section 4. I show a clear interaction between semantic, morphological, and phonological constraints and advocate for a single merged interface accounting for noun class allocation, the implications of which are discussed in Section 7. In Section 8, I summarize the findings and provide a conclusion.

2 The noun class system

A prominent feature of many Bantu languages is their noun class system. Van de Velde (2019) defines the Bantu noun class system as a grouping of nouns which trigger the same agreement pattern. Bantu noun classes are often seen as analogous to grammatical gender. Maho (2009) notes that, on average, most Bantu languages have about fifteen noun classes. This number excludes locative noun classes, generally glossed as any noun class greater than 15. I do not consider locative noun classes in my analysis.

for Kirundi loanwords by referring to Kayigema (2018) for Kinyarwanda loanwords and confirming with a native speaker of Kirundi that those loans exist in Kirundi as well. I would like to extend a sincere thank you to native speakers of Kirundi who have helped me on this project. This would not have been possible without their generosity.

In Kirundi, noun classes are realized overtly by a noun class prefix. These appear on most nouns, although some proper and human-denoting nouns may appear without the prefix. The Kirundi noun class prefixes are shown in Table 1, where single noun classes appear on the left and plural/ non-individuated prefixes appear on the right[2]. I display the augment in brackets in Table 1 to capture that it does not appear in all environments, discussed below in Section 2.1.[3] I omit the augment from later discussion for sake of clarity.[4] Nouns are often inherently associated with a particular noun class but may shift for derivational purposes (Givón 1972; as cited in Aunio 2006). I later discuss this in Section 5.3.

Table 1: Kirundi Noun Class Prefixes

Class	Prefix	Class	Prefix
1	(u)mu-	2	(a)βa-
3	(u)mu-	4	(i)mi-
5	(i)Ø / (i)ri-	6	(a)ma-
7	(i)ki- / (i)gi-	8	(i)βi-
9	(i)N- / (i)Ø-		
10	(i)N- / (i)Ø-		
11	(u)ru-		
12	(a)ka- / (a)ga-	13	(u)tu- / (u)du-
14	(u)βu-		

Kirundi expresses number opposition through noun class shift. Odd-numbered noun class prefixes generally refer to singular entities, while corresponding even-numbered noun class prefixes express plurality (Ndayiragije et al. 2012, Van de

[2]Note that I use N to refer to a homorganic nasal unspecified for place of articulation. This is either realized as prenasalization or a nasal stop whose place is determined by the following vowel.

[3]As noted by an anonymous reviewer, the appearance or lack of the augment is conditioned by the syntactic and/or phonological environment it appears in. While the augment may not be truly 'optional' in that nouns in certain environments may appear augmentless, the crucial consideration for my analysis is that the augment is not a part of the noun in the same way as noun class prefixes. All nouns in Kirundi surface bearing a noun class prefix (though it may be covert) regardless of their syntactic environment. The same cannot be said for the augment, as there are certain environments where nouns must be augmentless. I refer the reader to Ndayiragije et al. (2012) and Shanks (2023) for in-depth discussions on factors conditioning the appearance of the augment in Kirundi.

[4]While some may object to the omission of the augment from later discussion, the fact that the augment is conditioned by its syntactic environment lead me to assume it is not a fundamental aspect of nouns in the same way as noun class prefixes.

Velde 2019, Zorc & Nibagwire 2007: , a.o.)[5]. In (1), we see an alternation between singular and plural interpretations of the noun with class 1 and class 2 prefixes. In the proceeding sections, I refer to individual noun classes simply as ‘class 1’, for example, and the corresponding noun class pairing as ‘class 1/2’.

(1) a. u-mu-gaβo
A-CL1-man
‘man’

b. a-βa-gaβo
A-CL2-man
‘men’

2.1 The augment

The last relevant piece of background for Bantu noun class systems is the augment. The exact nature of the augment in Bantu languages is contentious, and Van de Velde (2019) claims the role and function of the augment must be evaluated on the basis of individual Bantu languages. All that must be said of the augment now is that its form is of an initial vowel appearing before the noun class prefix on nouns. The augment’s realization in Kirundi is determined by the following noun class prefix, copying the vowel quality of the vowel in the noun class prefix. The sole exception is in the class 9/10 prefixes, which appear either as prenasalization or as a null element. In this case, the form of the augment is [i].

With this background being established, the next section describes the set of phonological generalizations regarding noun class allocation for loanwords. I first present the clear phonological patterns in Section 3 and propose an initial analysis for them in Section 4 before moving on to discuss the interaction of semantic factors in Section 5.

3 Phonological conditioning in noun class allocation

Most Kirundi loanwords are lexicalized in class 9/10, as seen in the examples in Table 2 below. For clarity, I present all forms without the augment. I assume that the augment is added subsequent to noun class prefixes in the derivation.

An analysis of Kirundi loanword allocation would be quite simple if loanword nouns always appear in class 9/10. There are exceptions to this pattern, however. The first situation is loanwords with initial segments resembling existing

[5] For an alternative analysis, see Mufwene (1980).

Table 2: Kirundi loanwords in class 9/10

	Kirundi	Gloss	Source
a.	Ø_9-maʃiːne	‘machine’	French, /maʃin/
b.	Ø_9-sukari	‘sugar’	French, /sykr/
c.	Ø_9-farine	‘flour’	French, /farin/
d.	Ø_9-fero	‘iron’	French, /fɛr/
e.	Ø_9-kurudʒeti	‘zucchini’	French, /kurʒɛt/
f.	Ø_9-ko$^{\text{n}}$dʒe	vacation	French, /kõʒe/

noun class prefixes in Kirundi. In such cases, the initial segment sequence in the loanword is “reanalyzed” as a noun class prefix, allocating the loanword to the corresponding noun class (Niyondagara 1993, Niyonkuru 1987). Table 3 shows examples of this, where the initial segments are repurposed.

Table 3: Reanalysis of initial segment as noun class prefix

	Kirundi	Gloss	Source
a.	mu_3-sigiti	‘mosque’	Arabic, /masdʒid/
b.	ma_6-joneze	‘mayonnaise’	French, /mɛjonɛs/
c.	ki_7-ro	‘kilogram’	French, /kilo/
d.	βu_{14}-re$^{\text{n}}$geti	‘blanket’	English, /bleɪnkɪt/
e.	ki_7-nini	‘pill’	French, /kinin/

Kirundi also disallows consonant clusters and has several strategies for resolving them in loanwords, such as vowel epenthesis. Different criteria may determine the form of the epenthesized vowel. In some cases, the epenthesized vowel is a copy of an adjacent vowel or is derived from place features of the preceding consonant. However, we see instances of the epenthesized vowel differing from what’s expected if another vowel could form a recognizable noun class prefix. I provide a deeper analysis of epenthetic vowel forms in Section 4.3.

The reader may immediately notice the apparent disparity between the forms (a) and (b) in Table (3). Both source words begin with the same CV sequence, yet only (b) in Table 3 allows that sequence to be repurposed as a noun class prefix. I will argue that it is the different semantic categories of the two nouns that allow repurposing of (b) but not (a). Specifically, I argue in Section 5.2 that *ma_6-joneze* is realized as class 6 due to its having cumulative reference, while *Ø-maʃiːne* is blocked from class 6 for its lack of cumulative reference.

In the proceeding section, I argue that a simple ranking of alignment constraints and constraints on the realization of morphemes can account for this described behaviour.

4 A formal account of phonological factors

4.1 Spellout and boundaries

An observed commonality between the nouns in Table 2 and those in Table 3 has to do with word boundaries. The word-initial segments in all examples from the source language's input maintain their position as left-aligned to word boundaries in the loanword, only preceded by the augment. Compare the forms in Table 2 and Table 3 with the ill-formed candidates in Table 4.

Table 4: Ill-formed candidates

	Kirundi	Gloss	Source
a.	*mu_3-maʃiːne	'machine'	French, /maʃin/
b.	*mu_3-sukari	'sugar'	French, /sykr/
c.	*ki_7-farine	'flour'	French, /farin/
d.	*ki_7-fero	'iron'	French, /fɛr/
e.	*ki_7-kurudʒeti	'zucchini'	French, /kurʒɛt/
f.	*mu_3-ko^ndʒe	vacation	French, /kõdʒe/

The ill-formed candidates in Table 4 have extra phonological material between the source word's input and the word edge in the form of an overt noun class prefix. I posit the ill-formedness of these candidates is explained as the violation of an alignment constraint maintaining word boundaries. This constraint is formalized as:

(2) Faith-Align-SB
Maintain word boundaries from the source language in the base language (i.e. the receiving language)[6]

[6]An anonymous reviewer asks if repurposing forms such as those in Table 3 violate Faith-Align-SB. For the alignment constraint in (2) to work as intended, there is a necessary assumption that Kirundi nominals only form a complete prosodic word when prefixed with noun class prefixes. Thus, the forms in Table 3 would constitute a violation of Faith-Align-SB only if the root were a prosodic unit. However, I assume that Faith-Align-SB is evaluated of a prosodic word with the noun class prefix already attached.

This simple constraint predicts which class most loanwords fall. Class 9/10 is the most common noun class for loanwords, as its prefix is phonologically null in most cases. Class 9/10's null prefixes avoid violation of FAITH-ALIGN-SB while ensuring the noun still receives proper nominal morphology. FAITH-ALIGN-SB explains the relative prominence of loanword allocation to class 9/10 when no existing segments are available for repurposing.

This constraint alone cannot account for the forms seen in Table 3. It would certainly be notably easier to assign all loanwords to a single noun class. Thus, it must be explained why existing segments are reinterpreted as class prefixes when applicable rather than simply assigning every loanword to class 9/10. The analysis should account for the ill-formedness of Table 5 compared to Table 3 for not effectively utilizing the existing phonological material given by the source language as a prefix.

Table 5: Ill-formed candidates

	Kirundi	Gloss	Source
a.	*$Ø_9$-musigiti	'mosque'	Arabic, /masdʒid/
b.	*$Ø_9$-majoneze	'mayonnaise'	French, /mɛjonɛs/
c.	*$Ø_9$-kiro	'kilogram'	French, /kilo/
d.	*$Ø_9$-βure^{n}geti	'blanket'	English, /bleɪnkɪt/
e.	*$Ø_9$-kinini	'pill'	French, /kinin/

To resolve this issue, I follow Aissen's (1999) proposal for a constraint *Ø, which penalizes null morphology. This constraint states that morphological categories must be overtly realized.

(3) *Ø
No null morphology

I account for Kirundi's preference to utilize existing phonological material for class prefixes with the constraint *Ø. FAITH-ALIGN-SB must outrank *Ø to explain why violations of *Ø are licensed to avoid violations of FAITH-ALIGN-SB, as demonstrated in Table 6 and Table 7. If *Ø were higher, we would expect every loanword to have an overt class prefix.

4.2 Consonant and vowel alternation

Compare the attested forms in Table 8 (I) and (II) with the ill-formed candidates in Table 8 (III) and (IV).

Table 6: Evaluation of /kilo/

/kilo/	Faith-Align-SB	*Ø
☞ a. ki_7-ro		
b. $Ø_9$-kiro		*!
c. mu_3-kiro	*!	
d. ki_7-kiro	*!	

Table 7: Evaluation of /bætəɹi/

/bætəɹi/	Faith-Align-SB	*Ø
☞ a. $Ø_9$-βatiri		*
b. ki_7-βatiri	*!	
c. mu_3-βatiri	*!	

Table 8: Attested and unattested candidates

		Kirundi	Gloss	Source
I.	a.	$Ø_9$-farine	'flour'	French, /farin/
	b.	$Ø_9$-pine	'tire'	French, /pn∅/
	c.	$Ø_9$-faːta	'Fanta'	English, /fænta/
II.	a.	*ma_6-rine	'flour'	French, /farin/
	b.	*$βi_8$-ne	'tire'	French, /pn∅/
	c.	*ma_6-ːta	'Fanta'	English, /fænta/
III.	a.	mu_3-dari	'medal'	French, /mədajə/
	b.	mu_3-sigiti	'mosque'	Arabic, /masdʒid/
	c.	mu_3-doka	'car'	English, /moɾəɹkɑɹ/
	e.	$Ø_9$-kurudʒeti	'zucchini'	French, /kurʒɛt/
	d.	$Ø_9$-ko^{n}dʒe	vacation	French, /kõʒe/
IV.	a.	*ki_7-rudʒeti	'zucchini'	French, /kurʒɛt/
	b.	*ki_7-^{n}dʒe	vacation	French, /kõʒe/

To avoid the potential problematic candidates in Table 8 (III), it is crucial to determine the ranking of faithfulness constraints relative to the aforementioned FAITH-ALIGN-SB and *Ø. Kirundi loanwords could conceivably alter some segments in the input for it to resemble a class prefix. Indeed, forms such as those in Table 8 (II, a-c) indicate this happens with certain vowels. This does not happen to all vowels and does not happen for any consonants, however. There is no evidence that rounding is a contrastive feature in Kirundi vowels, so it is nonessential for our analysis. I propose the constraints IDENTV(BK)-SB, IDENTV(HI)-SB, and IDENTC-SB, for back and high vowels and consonants, respectively.

(4) a. **IDENTV(BK)-SB**
Do not change the backness of any vowels from the source language in the base language

b. **IDENTV(HI)-SB**
Do not change the height of any vowels from the source language in the base language

c. **IDENTC-SB**
Do not change any consonants from the source language in the base language

As the consonant never changes to match a class prefix, IDENTC-SB must outrank both FAITH-ALIGN-SB and *Ø. Given that vowel quality can change to match some existing class prefix, some vowel faithfulness constraint should be ranked lower than FAITH-ALIGN-SB and *Ø. In Table 8 (II), we see only vowel height change and not backness, suggesting *Ø outranks IDENTV(BK)-SB. This is demonstrated in the tableau below.

Table 9: Evaluation of /kilo/

/kilo/	IDENTC	FAITH-ALIGN	*Ø	IDENTV(BK)
☞ a. $Ø_9$-kiro			*	
b. ma_6-kiro	*!			
c. $βa_2$-kiro	*!			

However, we do find some forms with variation. For example, (a) and (b) in Table 10 show it is possible to change the vowel to fit some existing class prefix, but also acceptable to keep a faithful vowel while adding covert class prefixes. We can stipulate that there is some level of variable ranking between *Ø and IDENTV(HI) amongst speakers. This would mean that for different speakers (and possibly different varieties), either candidate in Table 10 may be optimal. The tableau in Table 11 shows that both possible forms are attested.

Table 10: Variation in forms

	Kirundi	Gloss	Source
a.	Ø$_9$-modoka	'car'	English, /motɚkaɹ/
b.	mu$_3$-doka	'car'	English, /motɚkaɹ/

Table 11: Evaluation of /motɚkaɹ/

/motɚkaɹ/	IdentC-SB	Faith-Align-SV	*Ø	IdentV(Hi)-SB
☞ a. mu$_3$-doka				*
☞ b. Ø$_9$-modoka			*	
c. ki$_7$-modoka		*!		
d. βu$_{14}$-modoka	*!			

I account for the observed asymmetry between back and front vowels by distinguishing IdentV(Bk)-SB and IdentV(Hi)-SB. In Table 8 (II, a-c), back vowels can change in height to resemble a class prefix. Examples in Table 8 (II, d-e) and Table ?? (IV, a-b) show that back vowels cannot be fronted for the same result. Thus, IdentV(Bk)-SB outranks at least *Ø, as demonstrated in the tableaux below.

Table 12: Evaluation of /kurʒɛt/

/kurʒɛt/	IdentC-SB	Faith-Align-SV	*Ø	IdentV(Hi)-SB
☞ a. Ø$_9$-kurudʒeti			*	
b. ki$_7$-rudʒeti		*!		

4.3 Epenthetic vowels

Kirundi loanwords often use vowel epenthesis to break up complex syllable sequences (Niyondagara 1993). Niyondagara (1993) claims that the Kirundi syllable structure is best characterized as (C)V(V), the second V being an additional mora in long vowels. Kirundi also allows glides to follow consonants in onsets. Table 13 shows examples of vowel epenthesis breaking up syllable clusters.

Table 13: Vowel epenthesis

	Kirundi	Gloss	Source
a.	$Ø_9$-petero:ri	'gasoline'	French, /pɛtrɔl/
b.	$Ø_9$-po:sita	'post office'	French, /pɔstə/

Epenthetic vowels in Kirundi can appear in several potential forms. The two most common strategies involve copying an adjacent vowel or using the epenthetic vowel [i]. Example (a) in Table 13 shows the epenthesized vowel [e] as a copy of the previous syllable's nucleus. Kirundi loanwords show that the copied vowel can be on the left or right side of the epenthetic vowel. The other strategy, shown in (b) in Table 13, uses an epenthetic vowel [i]. Niyondagara (1993) notes that the most common epenthetic vowel in Kirundi is [i] and is the form seen when the consonant that would form the potential onset of a newly created syllable is /s/. We also see some instances in Kirundi where the epenthetic vowel is variable, giving two possible forms derived from the same source word. This is seen in (a-b) in Table 14, both acceptable and attested forms. Thus, we must account for (a) and (b) in Table 14 being variable. I deal with this variability in Section 5.

Table 14: Vowel epenthesis

	Kirundi	Gloss	Source
a.	βu_{14}-re^{n}geti	'blanket'	English, /bleɪnkɪt/
b.	βi_8-re^{n}geti	'blanket'	English, /bleɪnkɪt/

I refer to Rose & Demuth (2006)'s constraints and ranking to account for epenthesis in Kirundi loanwords. Rose & Demuth focus on Sesotho consonant cluster resolution, but I briefly show that their analysis can be successfully applied to Kirundi with minor modifications. The relevant constraints and ranking they suggest for Sesotho are:

(5) **OK(σ) » CrispEdge(CV, σ) » Dep(VPl) » AgrL(VPl) » AgrL(CPl) » AgrR(VPl)**

Dep(VPl) states that no new vowel place features may be added. In other words, if a vowel is added to the source word's input, it should avoid new features by using the same features of some adjacent segment. AgrL(VPl) and AgrR(VPl) are constraints indicating which side's vowel the epenthetic position is a copy of. AgrL(CPl) states that segments should agree with consonants to the left. CrispEdge(CV, σ) (shown in tableaux as CE((CV, σ)) prevents consonant-vowel agreement across syllable boundaries. Thus vowels only agree with the features of consonants in their syllable's onset. These constraints are given in (6a-6f) and the success of this ranking is shown in the tableaux in Table 15.

(6) a. **Dep(VPl)** (Rose & Demuth 2006)
Every output VPlace specification has an input correspondent

b. **AgrL(VPl)**
The place feature of the epenthetic vowel must agree with the place feature of the vowel immediately to its left.

c. **AgrL(VPl)**
The place feature of the epenthetic vowel must agree with the place feature of the vowel immediately to its left.

d. **AgrL(CPl)**
The place feature of the epenthetic vowel must agree with the place feature of the consonant immediately to its left.

e. **CrispEdge(CV, σ)**
Consonant–vowel feature sharing cannot occur across syllables.

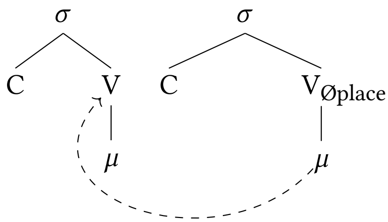
f. **OK(σ)**
Portmanteau constraint ensuring syllable well-formedness in output (adapted) forms

Table 15: Evaluation of /pɛtrɔl/

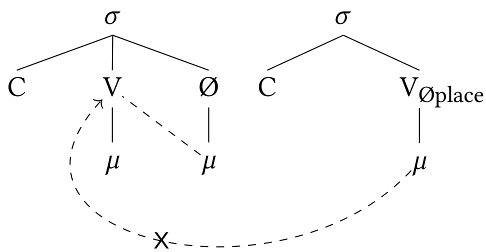
/pɛtrɔl/	OKσ	CE(CV,σ)	Dep(VPl)	AgrL(VPl)	AgrL(CPl)	AgrR(VPl)
☞ a. Ø$_9$-petero:ri						*
b. Ø$_9$-petro:ri	*!					
c. Ø$_9$-petoro:ri				*!	*	
d. Ø$_9$-petiro:ri				*!		*
e. Ø$_9$-petaro:ri			*!	*	*	*
f. Ø$_9$-peturo:ri			*!	*	*	*

Rose & Demuth (2006) do not discuss the role of vowel length in epenthetic forms. From the data, Kirundi epenthetic vowels do not seem to agree with long vowels to their left. To understand this, I adopt the theory of syllable structure and agreement relations shown in (7). In this structure, long vowels consist of two morae, the latter floating without being inherently associated to a vowel but becoming linked to the preceding segment (Zec 1995).

(7) a.



b.



The structure in (7a) shows an acceptable agreement relationship between two vowels. The structure in (7b) shows an ill-formed agree relationship between two vowels. By understanding long vowels as a vowel associated with two morae, we add the boundary of an additional mora that is not inherently associated with a vowel between the epenthetic site and the preceding vowel. This unspecified mora blocks the epenthetic vowel from probing backwards for a vowel to copy its place features from. Agreement between the epenthetic vowel and the left adjacent vowel is no longer possible, and a new agreement target is sought. This evaluation is shown in Table 16. No candidates trigger a violation of AgrL(VPl) as the input does not see any vowel immediately left of the epenthetic site[7].

Table 16: Evaluation of /pɔstə/

/pɔstə/	OKσ	CE(CV,σ)	Dep(VPl)	AgrL(VPl)	AgrL(CPl)	AgrR(VPl)
☞ a. $Ø_9$-po:sita						*
b. $Ø_9$-po:sta	*!					
c. $Ø_9$-po:sata				*!	*	
d. $Ø_9$-po:sota				*!		*

[7]I leave the augment in tableaux to make the noun class more explicit, but it can be omitted and should not be seen as a potential goal for agreement.

Table 17

Kirundi	Gloss	Source
mu$_1$-perezida	'president'	French, /prezidə/

I propose a final constraint to account for forms such as those in Table 17. The class prefix *mu-* is not a potential goal for vowel agreement, avoiding the form [mupurezida]. I propose the constraint *AgrMorphBd(Pl). This rules out the epenthetic vowel agreeing with the noun class prefix vowel, shown in the tableaux in Table 18. This constraint outranks AgrL(VPl), but its relative ranking to lower constraints is not crucial.

(8) ***AgrMorphBd(Pl)**
Do not agree with the place of segments across morpheme boundaries.

Table 18: Evaluation of /prezidə/

/prezidə/	OKσ	CE(CV,σ)	*AgrMoBd(Pl)	Dep(VPl)	AgrL(VPl)	AgrL(CPl)	AgrR(VPl)
☞ a. mu$_9$-perezida					*	*	
b. mu$_9$-prezida	*!						
c. mu$_9$-purezida							

5 Semantic factors in noun class allocation

Kirundi noun class allocation of loanwords is additionally conditioned by various semantic factors. Although Bantu noun classes are rarely entirely semantically coherent, we see strong restrictions and general tendencies affecting a noun's class allocation. Thus, we cannot account for the distribution of noun class prefixes to loanwords with phonological criteria alone.

In this section, I outline notable patterns in the semantic effects of noun class allocation in Kirundi loanwords. In Section 6, I propose an analysis reconciling the conflict between semantics and phonology, demonstrating a capability for one unified interface.

5.1 Animacy

I first view the effect of animacy on Kirundi noun class allocation for a simple reason: [±human] is the only semantic feature that correlates perfectly with a particular noun class. Zorc & Nibagwire (2007; among others) note that nouns of class 1/2 always denote human entities. Not all human nouns belong in class 1/2, but every noun in this class is a human.

There is a general tendency for Kirundi loanwords to be allocated to class 1/2 when denoting a human. Often, this involves adding the class 1 prefix to an existing input to mark that noun as belonging to that noun class, as seen in Table 19. In (a) in Table 19, the overt class 1 prefix, /umu-/ is added to the source word.

In other cases, loanwords with initial segment sequences resembling a class 1/2 prefix may have their initial CV sequence repurposed when denoting human nouns. In (d) in Table 19, the loanword /mʊs.ləm/ is incorporated to the language as /mu$_1$-silamu/, using phonological material from the source word for the class 1 prefix.

Table 19

	Kirundi	Gloss	Source
a.	mu$_1$-perezida	'president'	French, /prezidə/
b.	mu$_1$-furatiri	'brother'	Latin, /frater/
c.	mu$_1$-kirisito	'christian'	Christian
d.	mu$_1$-silamu	'muslim'	Arabic, /mʊs.ləm/
e.	mu$_1$-seːneri	'mister'	French, /məsjØ/

Because class 1/2 is reserved solely for human nouns, loanwords with initial segment sequences resembling a class 1/2 prefix are put into other classes. This is shown in (a) in Table 20, where the homophonous class 3 prefix is used instead of the class 1 prefix. In (20b), the class 2 prefix is avoided.

Table 20

	Kirundi	Gloss	Source
a.	mu$_3$-sigiti	'mosque'	Arabic, /mas.dʒid/
b.	Ø$_{10}$-βata	'duck'	Swahili, /bata/

We can establish that those nouns in Table 19 are given class 1 prefixes rather than the class 3 prefixes on the basis of noun class agreement with other elements in the sentence and their non-singular noun class alternation.

(9) a. U-mu-se:neri a-fis-e a-βa-ána be-nshi.
A-CL1-mister 1SM-have-PFV A-CL2-child CL2-many
'The man has many children.'

b. A-βa-se:neri ba-fis-e a-βa-ána be-nshi.
A-CL2-mister 2SM-have-PFV A-CL2-child CL2-many
'The men have many children.'

(10) a. U-mu-sigiti mu-fis-e a-βa-ntu be-nshi.
A-CL3-mosque 3SM-have-PFV A-CL2-person CL2-many
'The mosque has many people.'

b. I-mi-sigiti i-fis-e a-βa-ntu be-nshi.
A-CL4-mosque 4SM-have-PFV A-CL2-person CL2-many
'The mosques have many people.'

In (9a), *umuse:neri* triggers class 1 verbal subject agreement. In (9b), the same noun is given a class 2 prefix, triggering class 2 subject agreement on the verb. In (10a), *umusigiti* triggers class 3 subject agreement on the verb and alternates with the class 4 prefix when plural in (10b).

5.2 Cumulativity

Another mostly semantically coherent aspect of noun class is cumulativity. Oliver & Kinchsular (2022) note that even-numbered noun classes in Kirundi all denote entities with cumulative reference[8]. Only pluralities or non-atomic entities are found in these classes. Class 6 is often where substance-denoting nouns are found and is understood as a 'polyplural' class in Bantu languages (Maho 2009, Zorc & Nibagwire 2007, Oliver & Kinchsular 2022). Cumulative reference is shared by individuated plural nouns and non-individuated/mass nouns.

Instances where nouns with cumulative reference have initial segment sequences reanalyzed as noun class prefixes are seen in Table 21 (I). Other noun classes beyond class 6 may also show cumulative reference. There are also instances where loanwords with sounds resembling a non-individuated noun class prefix may not be interpreted as those class prefixes when lacking cumulative reference. Instead, these nouns either take an overt prefix or take the covert class 9 prefix. This is seen in Table 21 (II). Unlike human-denoting nouns allocated to class 1, we do not see instances of nouns with cumulative reference being automatically allocated to class 6.

[8] Noun classes above 10 are exceptions to this rule. However, this is a consequence of them being derivational classes and thus should not be understood as regular in regard to the

Table 21

		Kirundi	Gloss	Source
I.	a.	ma_6-joneze	‘mayonnaise’	French, /mɛjonɛs/
	b.	$Ø_{10}$-βata	‘ducks’	Swahili, /bata/
II.	a.	$Ø_9$-mandata	‘mandate’	French, /mɑ̃da/
	b.	$Ø_9$-maʃi:ne	‘machine’	French, /maʃin/

5.3 Abstract and diminutive nouns

Less semantically coherent but still salient semantic categories are those of abstract nouns and diminutives. Zorc & Nibagwire (2007) observe that class 14 is largely comprised of abstract nouns, while classes 12 and 13 are primarily nouns with diminutive meanings.

(11) a. i-gi-tabo
A-CL7-book
‘book’

b. a-ga-tabo
A-CL12-book
‘pamphlet’

6 Phonosemantic interface

In this section, I use an optimality theoretic approach, with crucially ranked phonological and semantic constraints to account for the interaction of phonological and semantic factors. Kirundi’s noun classes are not usually fully semantically cohesive. However, an optimality theoretic ranking system is well-equipped to handle the lack of complete semantic cohesion, allowing for violations of noun class constraints to prevent a violation of some higher-ranked constraint.

In sections 6.1-6.3, I propose constraints and crucial rankings that account for the semantic factors described above. In 7, I consider and argue against an alternative analysis where a division between the semantic and phonological interfaces is upheld.

individuated/non-individuated contrast.

6.1 Human constraints

I first propose a constraint to derive the human semantics of class 1/2. To capture that human nouns are strongly preferred in class 1/2 and that only human nouns can comprise class 1/2, I use the following constraint:

(12) **Human↔1/2**
All and only human nouns in class 1/2.

As noted in 5.1, there are instances where human-denoting nouns beginning with segments resembling class 1/2 prefixes may have those segments reinterpreted as class 1/2 prefixes. Some examples of this are repeated in Table 22.

Table 22

	Kirundi	Gloss	Source
a.	mu_1-silamu	'Muslim'	Arabic, /mʊs.ləm/
b.	mu_1-seːneri	'mister'	French, /məsjØ/

I have also noted cases where human-denoting nouns without any segments that resemble class 1/2 prefixes can have an overt prefix assigned. This happens when human denoting loanwords have no initial sequence resembling a class 1/2 prefix. Examples are given in Table 23.

Table 23

	Kirundi	Gloss	Source
a.	mu_1-perezida	'president'	French, /prezidə/
b.	mu_1-furatiri	'brother'	Latin, /frater/
c.	mu_1-kirisito	'christian'	Christian

It becomes clear from this evidence that our semantic constraint must interact with the previous constraints proposed. The preference to have human nouns in class 1/2 apparently takes precedent over the preference for loanwords to appear in class 9/10 without disrupting word boundaries. To capture this, we need only rank Human↔1/2 higher than *ø and Faith-Align-SB. In [umuperezida], for example, a violation of Faith-Align-SB is incurred to avoid triggering a potential violation of Human↔1/2 that would arise without assigning the noun to class 1/2. If Faith-Align-SB were ranked higher, we would expect the noun to be allocated to class 9/10 with a covert noun class marker.

Table 24: Evaluation of /prezidə/

/prezidə/	Hum↔1/2	IdentC-SB	*Ø	FaithAlign-SB	IdentV-SB
☞ a. mu_9-perezida				*	
b. $Ø_9$-perezida	*!		*		

6.2 Cumulative constraints

To contain the discussion of correlation between even numbered and cumulative noun classes, I only consider class 6 loanwords. Class 6 houses many substance-denoting nouns and serves as the irregular plural for many. I introduce the following constraint to account for this property of class 6;

(13) **Cumulative↔6**
All and only nouns with cumulative reference are allocated to class 6

Not all substance-denoting loanwords are in class 6, as seen in Table 25, where the substance-denoting noun [isukari] ('sugar') is realized with a class 10 prefix. A possible alternative would have this noun be realized as [*amasukari]. That [isukari] is the preferred form suggests that the constraint penalizing candidates with unfaithful word boundaries outranks our semantic constraint for class 6.

Table 25: Evaluation of /sykr/

/sykr/	IdentC-SB	FaithAlign-SB	Cumul↔6	*Ø	IdentV-SB
☞ a. $Ø_9$-sukari			*	*	
b. mu_9-sukari		*!			

Cumulative-reference nouns with initial segments resembling the class 6 prefix can have initial segments repurposed as a class prefix. Nouns without cumulative reference resembling the class 6 prefix cannot, as seen in Table 22. This suggests that Cumulative↔6 outranks *Ø, as violations of *Ø can occur to avoid violations of Cumulative↔6. The difference is shown in Table 26 and Table 27.

Table 26: Evaluation of /mɛjonɛsə/

/mɛjonɛsə/	IdentC-SB	FaithAlign-SB	Cumul↔6	*Ø	IdentV-SB
☞ a. ma_6-joneze					*
b. $Ø_9$-mejoneze			*!	*	

Table 27: Evaluation of /maʃinə/

/maʃinə/	IdentC-SB	FaithAlign-SB	Cumul↔6	*Ø	IdentV-SB
☞ a. $Ø_9$-maʃine				*	
b. ma_6-ʃine			*!		

6.3 Abstract and diminutive constraints

I now consider some of the derivational noun classes in Kirundi. Classes 12/13 and 14 behave differently from other Kirundi noun classes. Class 12/13 is the diminutive noun class. As demonstrated previously in (11), nouns in class 12/13 are given diminutive interpretations. Class 14 is mostly reserved for abstract nouns. Abstract qualities can be expressed by replacing the inherent class prefix of other roots with the class 14 prefix.

The number of nouns inherently belonging to either class is highly limited and are often lexicalized nouns that can be traced back to some derived form (Zorc & Nibagwire 2007). I discuss these noun classes as their derivation use makes them clearly semantically cohesive. Following similar treatment for previous noun classes considered, I introduce the following two constraints to account for these classes' semantic cohesion:

(14) **Diminutive↔12/13**
All and only diminutive nouns in class 12/13

(15) **Abstract↔14**
All and only abstract nouns in class 14

We have examples of loanwords that begin with sequences of segments resembling class 12/13 or class 14 prefixes that are not semantically cohesiveness with other nouns in those classes. With [imaʃiːne], the semantic constraint overrides the phonological constraint to ensure the noun is not inappropriately allocated to class 6. We see the opposite with class 12/13 and 14 loanwords, where semantic constraints are clearly violated. For example, in Table 28, [akaβati] ('cabinet') is allocated to class 12 without having any diminutive sense, and [$uβu_{14}re^{n}geti$] ('blanket') is allocated to class 14, despite being a concrete noun. The semantic constraints for class 12/13 and class 14 therefore must be outranked by phonological constraints. This is shown in Table 29 and Table 30.

Semantic and phonological constraints relevant to noun class allocation can vary in ranking, with the hierarchy in (16). This further supports the proposal that the semantic and phonology interface work as one in noun class allocation.

Table 28

	Kirundi	Gloss	Source
a.	ka_{12}-βati	'cabinet'	English, /kæbnɪt/
b.	ka_{12}-βare	'cabaret'	English, /kæbəreɪ/
c.	$βu_{14}$-re^{n}geti	'blanket'	English, /bleɪnkɪt/

Table 29: Evaluation of /bleɪnkɪt/

/bleɪnkɪt/	OK(σ)	FaithAlign-SB	*Ø	AgrL(CPl)	Abst↔14	Dim↔12/13
☞ a. $βu_{14}$-βure^{n}geti				*		
b. $Ø_{9}$-βure^{n}geti			*!			
c. ki_{7}-re^{n}geti		*!				

(16) **Human↔1/2, IdentC-SB » FaithAlign-SB » Cumulative↔6 » *Ø » Abstract↔14, Diminutive↔12/13**

Considering the abstract class 14, we also gain insight into the relative ranking of semantic and phonological constraints determining vowel epenthesis. As noted in Section 3, [$uβu_{14}$re^{n}geti] and [$iβi_{8}$re^{n}geti] are both accepted forms in Kirundi. As a vowel must be epenthesized to resolve the consonant cluster in 'blanket', the relevant phonological constraint for determining the form of that vowel is AgrL(CPl). In [$uβu_{14}$re^{n}geti], the epenthetic vowel is rounded to agree with the consonant, but [$iβi_{8}$re^{n}geti] triggers a violation of AgrL(CPl). Considering semantic constraints, [$uβu_{14}$re^{n}geti] incurs a violation of Abstract↔14, as the noun is not abstract in nature, while [$iβi_{8}$re^{n}geti] avoids violating the constraint by allocating the noun to a different class.

Accounting for variation between these candidates, we can assert there is no critical ranking between AgrL(CPl) and Abstract↔14. This is demonstrated in Table 31.

Table 30: Evaluation of /kæbnɪt/

/kæbnɪt/	OK(σ)	FaithAlign-SB	*Ø	Abst↔14	Dim↔12/13
☞ a. ka_{14}-βati					*
b. $Ø_{9}$-kaβati			*!		

Table 31: Evaluation of /bleɪnkɪt/

/bleɪnkɪt/	OK(σ)	Hum↔1/2	FaithAlign-SB	*Ø	AgrL(VPl)	Abst↔14	AgrL(CPl)
☞ a. βu$_{14}$-βure^{n}geti						*	
b. β$_{8}$-re^{n}geti							*
c. Ø$_{9}$-βure^{n}geti				*!			
d. Ø$_{9}$-βre^{n}geti	*!						
e. βu$_{2}$-re^{n}geti		*!					
f. βi$_{8}$-βure^{n}geti			*!				

7 Interface implications

In the preceding sections, I have argued that semantics and phonology can be construed as using one and the same interface. Using an Optimality Theory framework, I have demonstrated that semantic and phonological factors and constraints can be ranked in relation to each other in the same way that simple phonological constraints can. I consider an alternative analysis for the data presented. An alternate analysis may posit that phonology and semantics do not interact directly, but one 'feeds' the other, giving it input for evaluation. I evaluate the shortcomings of this approach and show that a unified phonosemantic interface is favourable. I return to [imaʃiːne] as an example to illustrate an important behaviour lending support to a unified system.

If we consider the analysis where phonology feeds the semantic interface, we could explain much of the data observed as simple cases of derivation. In such a hypothetical analysis, we might assume that loanwords are first allocated to a particular noun class based on phonological principles. Once allocated, some semantic evaluation system determines whether that loanword class allocation is acceptable based on its lexical content. If deemed acceptable, the loanword maintains the given class prefix. If unacceptable, the noun is reallocated to a more suitable noun class. Crucially, this would involve phonology incorporating the noun into the morphological system before the semantic interface could evaluate the noun class allocation. However, I argue this cannot be the case, and phonology and semantic considerations must work in tandem to determine noun class allocation. Consider how such an account would apply to the loanword [imaʃiːne].

In [imaʃiːne], the sequence resembling the class 6 prefix is retained, and the source word is simply augmented with a class 9 prefix. The approach described above cannot account for this behaviour. We must realize that both semantic and phonological constraints can be violated in favour of the other.

If there were indeed two distinct interfaces, we would expect [*iʃiːne]. The phonological interface could first analyze the existing sequence of segments in

/maʃinə/. After the appropriate sound changes occur, we would expect the morphophonological form to enter the Kirundi lexicon as /ma$_6$-ʃiːne/. The tableaux in Table 32 and Table 33 demonstrate EVAL within this approach, where semantic constraints do not interact with phonology. This would separate the loanword into root and affix before any semantic factors are considered. The semantic interface could then evaluate the lexical item [ma$_6$-ʃiːne], determine that class 6 is not a semantically appropriate class for the item in question, and replace the prefix with any prefix that does not indicate cumulative reference. This could give us the form [Ø$_9$-ʃiːne], to which the augment would be added afterward.

Table 32: Phonology Eval

/maʃinə/	IdentC-SB	FaithAlign-SB	*Ø	IdentV-SB
☞ a. Ø$_9$-maʃine			*!	
b. ma$_6$-ʃine				

Table 33: Semantic Eval

/maʃinə/	Human↔1/2	Cumulative↔6
☞ a. Ø$_5$-ʃine		
b. ma$_6$-ʃine		*!
c. mu$_1$-ʃine	*!	

The forms we see suggest that specific phonological and semantic constraints can be overridden by the other. The only way to properly account for these forms is by concluding one of two things. I propose that semantics and phonology either have the same interface, where semantic and phonological constraints are crucially ranked with respect to each other, or the distinct interfaces are intertwined to the degree that it becomes undesirable and ineffective to view them separately.

8 Conclusion

In the preceding sections, I hope to have shown an analysis of Kirundi loanwords that demonstrates a unity between semantics and phonology is possible within an optimality theory framework. I have focused on how semantic and phonological factors contribute to the allocation of Kirundi loanwords into the rich noun class system.

In Section 3 and Section 4, I first focused on phonological patterns in Kirundi noun class allocation of loanwords. The tendency for loanwords in Kirundi to be allocated primarily to noun classes with null prefix morphemes or to use existing phonological material in the source language can be understood as primarily the result of constraints seeking to maintain faithful word boundaries from source to the base language and constraints requiring that morphology must be overt. These constraints, in addition to other faithfulness constraints penalizing the alteration of segments, can fairly reliably predict the designation of Kirundi loanwords to the noun class system.

In addition, epenthesis and syllable structure constraints interact with these faithfulness constraints. The restrictions set on well-formed syllables in Kirundi often lead loanwords to require some sort of syllable restructuring. In the process of incorporating these loanwords into the noun class system, Kirundi can exploit some syllabification processes to ensure that the output fits well within the noun class system.

In Section 5, I showed that semantic conditioning and noun class restrictions factor into determining how a loanword is assigned a noun class. Semantic factors of the loanword can cause irregularities in the phonological patterns of noun class allocation described in the previous section. In Section 6, I addressed this by providing an account of Kirundi loanword allocation that takes both semantic and phonological factors into consideration. The analysis I proposed is one where phonology and semantic conditioning of noun classes essentially play different roles in the same system.

An asymmetric ranking of semantic noun class constraints within an optimality theoretic approach accounts for the observed asymmetry in the semantic cohesion of Kirundi noun classes. Therefore, certain tendencies of Kirundi noun classes to be stronger than others are explained by constraints that may be violated to avoid violating higher-ranked constraints, whether semantic or phonological. There are numerous instances both of higher-ranked semantic constraints leading to the violation of some phonological constraints and the violation of lower semantic constraints incurred by avoiding the violation of phonological constraints.

This paper is by no means a conclusive study of the phonology of Kirundi noun class prefixes or the semantic factors of Kirundi noun classes. I hope to have made some small contribution towards an understanding that semantics and phonology can be interconnected to a greater degree than is regularly granted. The analysis I have proposed lends credence to an argument that semantic and phonological patterns can be accounted for within the same system, and there is a clear benefit to doing such, as any alternative can not as effectively capture the interaction between the two.

Abbreviations

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Chapter 17

A scattered deletion account of Chichewa partial dislocation

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A Chichewa object DP with [N modifier] structure can be made discontinuous by displacing the noun to the left periphery and leaving the modifier postverbal (Mchombo 2006). Contra Branan & Davis (2022), I argue based on novel data that the whole DP moves, but only the leftmost noun is pronounced in the highest copy. This explains why a quantifier that appears to be stranded postverbally has obligatory wide scope over negation: It raises with the noun but is pronounced in situ. I propose that this follows from a requirement that the verb form a phonological phrase with its local complement. I also show that for the same reason, Chichewa *wh*-objects surface in situ but show island effects, indicating that they undergo movement not reflected in surface word order.

1 Introduction

This paper examines the status of partial dislocation (henceforth PD) of Chichewa objects in (1c). I will argue that (1c) involves *scattered deletion* (Bošković 2001; Fanselow & Ćavar 2002; Bošković & Nunes 2007, inter alia): The adjective *zákúda* moves with the noun from the postverbal position in the syntax, but is pronounced in situ. Note that the basic order of a Chichewa sentence is svo (1a) and Chichewa DPs are N-initial; also notice the total dislocation (henceforth TD) option is disallowed in this context (1b):[1]

[1] I follow Downing & Mtenje's 2017 system in transcribing Chichewa data. The acutes marking high tones reflect the tone system of the Ntcheu variety. The penult vowel of a prosodic phrase-final element is automatically lengthened (indicated as vowel doubling in the examples).

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(1) a. A-tsíkána á=mfúumu a=a=gul-á **mbúzi** **zákúuda**
2-girls 2.ASSOC=9.chief 2SM=PERF=buy-FV 10.goats 10.black
'The chief's girls have bought black goats.'

b. * **Mbúzi** **zákúuda** a-tsíkána á=mfúumu a=a=guúl-á
10.goats 10.black 2-girls 2.ASSOC=9.chief 2SM=PERF=buy-FV
intended: 'Black goats, the chief's girls have bought.'

c. **Mbúuzi** a-tsíkána á=mfúumu a=a=gul-á **zákúuda**
10.goats 2-girls 2.ASSOC=9.chief 2SM=PERF=buy-FV 10.black
lit. 'Goats, the chief's girls have bought black [ones].'

d. * **Zákúuda** a-tsíkána á=mfúumu a=a=gul-á **mbúuzi**
10.black 2-girls 2.ASSOC=9.chief 2SM=PERF=buy-FV 10.goats

(Mchombo 2006; also personal fieldnotes)[2]

Since the copy theory of movement was incorporated into the modern syntactic theory from Chomsky (1993, 1995), it has been generally accepted that what is left behind by a moved element is not a *trace*, but a *copy* of the element itself, which by definition must bear all the features contained in that element at the point when the copy is created. It is also often assumed that the phonological and semantic features of all but one copy of the moved element are deleted at the interfaces. On the PF side, where the phonological features are dealt with, the standard assumption is that the deletion is highly restrictive: Only those of the *highest* copy survive. Multiple authors have argued that this is so due to reasons of economy (Franks 1998, Nunes 2004, among many others; for a thorough literature review on this topic see Bošković & Nunes 2007): Pronouncing the highest copy is claimed to be the most economical option and thus is preferred. Importantly, this line of research gives the prediction that pronouncing a lower copy should in principle be possible in contexts where the pronouncing-the-highest-copy option is ruled out for independent PF reasons. Such cases are indeed attested. One clear example from Romanian is discussed in Bošković (2002):

(2) a. Cine ce precede?
who what precedes

b. * Cine precede ce?
who precedes what
'Who precedes what?'

(3) a. * Ce ce precede?
what what precedes

b. Ce precede ce?
what precedes what
'What precedes what?'

[2] Data from the literature are also confirmed by my own fieldnotes; examples in the rest of the paper are solely from elicitation notes if no reference is given.

As in (2), Romanian is a multiple *wh*-movement language, where all *wh*-phrases obligatorily move to the left of a clause. However, if the movement would create a consecutive homophonous sequence **ce ce*, the multiple fronting is ruled out (3a); instead, the lower *wh*-phrase may occur in situ as shown in (3b). Bošković (2002) proposes that the second *ce* nevertheless moves in (3b) in narrow syntax; the apparently in-situ *ce* is in fact the pronunciation of a lower copy. It is a low-level PF constraint excluding (3a) (i.e., to avoid a *ce ce* sequence) that makes the otherwise less economical (3b) possible:

(4) ce ~~ce~~$_i$ precede ce$_i$?

It should be mentioned that there is direct syntactic evidence that the lower *ce* in (3b) indeed moves: It licenses parasitic gaps. This is illustrated in (5) (notice the literal English translation is ungrammatical):

(5)

Ce	precede	ce	fără	să	influenţeze?
what	precedes	what	without	SUBJ.particle	influence.3P.SG

lit. '*What precedes what without influencing?' (Bošković 2002)

Within this line of research, scattered deletion refers to cases where different pieces of an element are pronounced in different copies, as illustrated in (6). (6a) and (6b) differ in the direction of the deletion. Suppose in both that the copy on the left is the structurally higher one. I will call the two cases as leftward and rightward deletion, respectively:

(6) a. <~~X~~ Y>$_i$ … <X ~~Y~~>$_i$ b. <X ~~Y~~>$_i$ … <~~X~~ Y>$_i$

Things may get more complicated if more pieces and more copies of an element are involved. In any case, it should be borne in mind that scattered deletion is a costly option: It only becomes possible if other more economical options are ruled out, for independent PF reasons. I will show that this is exactly the case of Chichewa PD. In addition, if the scattered deletion account is indeed on the right track, as I will argue in detail, we need to explain why (1d) (a case of leftward deletion) cannot be derived while rightward deletion cases like (1c) are possible. I will show that this direction constraint is expected under the analysis to be spelled out in this paper.

The remainder of the paper is organized as follows. I first argue in Section 2 that DP splits in Chichewa do not involve base-generated topics in the left periphery but call for a movement account. However, it will be shown that a direct what-you-see-is-what-you-get analysis does not work: It is not just the noun or the smallest projection of it that is moved. In Section 3, the scattered deletion approach to Chichewa PD is discussed in detail, which I show not only captures the distribution of PD, but also manages to explain why PD should keep the original N>modifier order. A language-particular PF constraint is proposed in this section to restrict scattered deletion. More evidence for the account is provided in Section 4: In a PD case, a low-pronounced quantifier cannot be interpreted low, a fact that I suggest can only be explained by the analysis illustrated in Section 3. Section 5 gives some brief comments on (i) the absence in Chichewa of the so-called conjoint-disjoint alternation found in many other Bantu languages, and (ii) *wh*-in-situ in Chichewa, suggesting the latter is shaped by the same PF constraint proposed in Section 3. Section 6 concludes.

2 The discontinuity puzzles

As mentioned, it is in principle possible for a Chichewa object DP to split. One other example of this phenomenon is given in (7c):[3]

(7) a. * **Ci-thúnzi cá=óphunziila** ndi=ná=pééz-á
7-picture 7.ASSOC=1.student 1P.SG=PST=find-FV
intended: 'The/a picture of the student, I found.'

b. * **Cá=óphunziila** ndi=ná=péz-á **ci-thúunzi**
7.ASSOC=1.student 1P.SG=PST=find-FV 7-picture

c. **Ci-thúunzi** ndi=ná=péz-á **cá=óphunziila**
7-picture 1P.SG=PST=find-FV 7.ASSOC=1.student

Again, TD is ruled out in the same context (7a), and dislocating the modifier, rather than the noun, also causes ungrammaticality (7b). Mchombo (2006) observes that in Chichewa 'discontinuity ... seems to require preservation of the 'base' order.' Since Chichewa DPs are N-initial, in cases where there is one noun *plus* one modifier, it is always the noun that is dislocated, so the N>modifier order is kept. Because (7b) gives the modifier>N order, it violates this order preservation constraint (it should be mentioned that the type of the modifier (e.g., adjectives, possessives, numerals, etc.) does not really affect the PD pattern).[4]

[3] In this paper I will only be concerned with PD cases without object markers, which essentially will change the dislocation pattern. It has been shown by Bresnan & Mchombo 1987 that the object marker in Chichewa is a (non-doubling) incorporated pronominal clitic, i.e., the real object; a lexical DP associated with it is always adjoined to the clause.

[4] There seem to be some exceptions. For example, clitic demonstratives are phonologically weak

One may wonder if the PD cases like (1c) and (7c) in fact involve a hanging/aboutness topic base-generated at the left periphery, as inferred by the English translation of (1c): '(As for) goats, the chief's girls have bought black ones.' I suggest that PD is a result of syntactic movement. The evidence comes from island effects. (8) shows that PD cannot happen out of an adjunct; (9) indicates that a noun cannot be dislocated from a complex DP. By contrast, (10) demonstrates that the dislocation can be long-distance as long as islands are not involved, as expected under a movement analysis.

(8) * **Mweezi** ndi=ná=gúl-a búukhu **wáa=tha**
3.month 1P.SG=PST=buy-FV 5.book 3.ASSOC=last
intended: 'I bought a book last month.'

(9) * **Vúuto** cikoondi a=ná=péz-á yankho li-méné
5.problem 1.Chikondi 1SM=PST=find-FV 5.answer 5-COMP
lí=ma=kónz-á **lá=kále** **lii-ja**
5SM=HAB=solve-FV 5.ASSOC=old 5-that
intended: 'Chikondi found an answer which solves that old problem.'

(10) **Mbúuzi** mavúuto a=ku=gáníz-a kuti cikoondi
10.goats 1.Mavuto 1SM=PROG=think-FV that 1.Chikondi
a=a=gul-á **zákúuda**
1SM=PERF=buy-FV 10.black
lit. 'Goats, Mavuto thinks that Chikondi bought black [ones].'

The above facts argue that PD does involve movement, but they do not tell us what is moving. Now, the direct what-you-see-is-what-you-get solution is to say that PD involves extraction of the DP-initial N(P) (as explicitly assumed by Branan & Davis 2022). However, it is generally assumed that N-initiality in Bantu is a result of N-to-D head movement (I will not go into this rather independent issue; see Carstens 1991, 2008, 2010, Zeller 2013 for arguments; Carstens 1997 proposes N-to-D specifically for Chichewa): If this is so, the noun in D° does not even form a phrasal constituent by itself. It becomes not immediately clear how a head can be extracted from the DP to the clause-initial position.

Besides, to the best of my knowledge, it is cross-linguistically very rare for a language to have PD of the object while ruling TD out. For example, Serbo-Croatian allows both (11a) and (11b):

(they are monosyllabic whereas independent prosodic words in Chichewa are minimally disyllabic; see Downing & Mtenje 2017) and cannot be left alone postverbally. I will not deal with these cases in the current study.

(11) a. **Koju** Jovan mrzi **knjigu**?
which Jovan hates book
lit. 'Which does Jovan hate book?'

b. **Koju** **knjigu** Jovan mrzi?
which book Jovan hates
'Which book does Jovan hate?'

Recall that Chichewa, by contrast, in general does not allow TD (1b) (but see immediately below), a fact that is left unexplained if one simply posits that the noun can move by itself from the DP. Why cannot the DP move?

Perhaps more interestingly, there are in fact cases where TD becomes possible, and it is exactly in those cases where PD is ruled out. As pointed out by Downing & Mtenje (2017), (12a) as a case of TD is totally acceptable:

(12) a. **Galásí lá=lí-kúlú iili** kankhaa=ni
5.glass 5.ASSOC=5-big 5.this push=PL
'Push this big glass.'

b. * **Galáasí** kankhaa=ni **lá=lí-kúlú iili**
5.glass push=PL 5.ASSOC=5-big 5.this

Notice that (12) involves an imperative where the enclitic *=ni* expresses politeness towards the addressee (it is glossed as PL since it is also used to mark 2nd person plural objects in declaratives; note plural-as-politeness is a typologically well-attested strategy). What is important to us here is that this is exactly when PD becomes impossible (12b).

A similar case where TD becomes possible involves *wh*-adjuncts. As shown by Downing & Mtenje (2017), a non-subject *wh*-word can optionally occur in the so-called Immediately After the Verb (IAV) position and form a phonological phrase with the verb.[5] In (13a), for instance, the *wh*-element *liti* 'when' occurs immediately after the verb stem:

(13) a. Mavúuto a=ná=kónz-a liiti **gálimoto lá=tsópaánó**?
1.Mavuto 1SM=PST=fix-FV when 5.car 5.ASSOC=new
'When did Mavuto fix the car?'

b. **Gálimoto lá=tsópaánó** Mavúuto a=ná=kónz-a liiti ?
5.car 5.ASSOC=new 1.Mavuto 1SM=PST=fix-FV when

[5]Moving *wh*-non-subjects to the IAV position is an obligatory operation in many Bantu languages but is optional in Chichewa.

c. * **Gálimooto** Mavúuto a=ná=kónz-a liiti **lá=tsópaánó**?
   5.car 1.Mavuto 1SM=PST=fix-FV when 5.ASSOC=new

As in (13b), TD is good when the IAV position is filled in, and PD of the object DP will lead to ungrammaticality (13c). It can be concluded that PD and TD in Chichewa show complementary distribution, a fact that is unexpected under a what-you-see-is-what-you-get account (i.e., PD involves sub-extraction of the initial N(P) from the object DP). I now show that all these puzzles discussed in this section can be solved or simply disappear under the scattered deletion account.

3 A scattered deletion analysis

I first propose a language-particular PF constraint (14) for Chichewa:

(14) A verb and its internal argument must be adjacent sub-constituents of a phonological phrase if they share phi-features.

(14) is inspired by Clemens's (2014, 2016) *argument condition on phonological phrasing* (ARG-φ). It states that if a verb is transitive in Chichewa, it must form a prosodic phrase with its complement that immediately follows it. (14) will not apply if the verb of concern is intransitive, since there is independent evidence that only transitive verbs phi-agree with the internal argument in Chichewa, as illustrated by the contrast between (15) and (16) (the latter involves an unaccusative verb and locative inversion): Only in (15) can the verb stem be attached to by an object marker, which I take to be a consequence of Agree ((16) is grammatical without the object marker):[6]

(15) Mi-káango i=ku=zí=saak-a zi-gawéenga
   4-lions 4SM=PROG=8OM=hunt-FV 8-terrorists
   'The lions are hunting them, the terrorists.' (Mchombo 2004; adapted)

(16) * Pa=mu-dzí pá=dá=í=gw-a njaala
   16.in=3-village 16SM=PST=9OM=fall-FV 9.hunger
   intended: 'Famine ravaged the land.' (*lit.* 'In the village fell hunger.')

[6]Recall from footnote 3 that the object marker is not a pure agreement marker in Chichewa, but a clitic pronoun. At any rate, the latter must also be a result of Agree; see Baker 2018 for directly relevant discussion. Again, in this paper we set aside the dislocation patterns with the occurrence of object markers.

(Mchombo 2004; adapted)

Unsurprisingly, the internal argument of an intransitive verb can occur preverbally freely:

(17) Njovu i=náa=gw-a
9.elephant 9SM=PST=fall-FV
'An elephant fell.' (Mchombo 2004; adapted)

(18) Ma-úngú a=ku=phík-iidw-a
6-pumpkins 6SM=PROG=cook-PASS-FV
'The pumpkins are being cooked.' (Mchombo 2004; adapted)

Turning back to transitives, (14) immediately rules out TD cases such as (1b) and (7a). The following is another such example:

(19) * **A-leenje** njúuci zi=ná=luum-a
2-hunters 10.bees 10SM=PST=bite-FV
intended: 'The hunters, the bees bit.' (Bresnan & Mchombo 1987)

Assuming a multiple spell-out framework (Uriagereka 1999, Chomsky 2000) and following a number of recent studies arguing that what is sent to spell-out is phases, not phasal complements (Bošković 2016, 2017, Cecchetto & Donati 2022, among others), I further argue that (14) becomes effective when the vP phase is spelled out. Suppose an element to be extracted from a phase has to move to the edge of that phase first, in the current case being SpecvP. At the point of the spell-out of vP, The structure of (19) will be like (20a):

(20) a. [$_{\text{vP}}$ alenje$_{\text{i}}$ luma alenje$_{\text{i}}$]

Now, the PF interface will decide which copy of *alenje* 'hunters' is to be deleted. There are three options, which we discuss in turn:

(20) b. [$_{\text{vP}}$ alenje$_{\text{i}}$ luma ~~alenje~~$_{\text{i}}$]
c. [$_{\text{vP}}$ ~~alenje~~$_{\text{i}}$ luma alenje$_{\text{i}}$]
d. [$_{\text{vP}}$ alenje$_{\text{i}}$ luma alenje$_{\text{i}}$]

Firstly, though the higher copy of the DP and the verb stem are still adjacent in (20b), I would like to suggest that it is already problematic regarding (14) at this point. This is simply because a preverbal lexical DP is unable to form a prosodic phrase with the verb following it. Due to the agglutinative nature and the prefixal/proclitical preference of Chichewa/Bantu, what can occur preverbally and form a prosodic phrase with the verb stem should always be a prefix/proclitic. Note that proclitic objects, namely object markers, do occur in this position (the object marker and the stem form what Bantuists call the macrostem):[7]

(21) Mjúuci zi=ná= [$_{\text{macrostem}}$ wá= luum-a]
10.bees 10SM=PST= 2OM=bite-FV
'The bees bit them.'

Secondly, I suggest that the second option (20c) is in fact not problematic with regard to (14). However, the surface PF form it derives is just the same as that where the movement has never occurred. Thus (20c) may cause a severe learnability problem: Children have no evidence to learn it. Finally, consider (20d), which involves pronouncing multiple copies of the same element. I assume that this option is excluded by Fox & Pesetsky's (2005) proposal that the ordering of elements should be established at the point of spell-out. In (20d) *alenje* precedes and follows the verb stem simultaneously, causing a self-contradictory situation, i.e., it is not pronounceable. Note that this approach will rule out all multiple pronunciation cases in general.

Now we turn to cases with DP splits. Take (7) as an example. Suppose that the whole object DP undergoes topicalization. At the point of vP spell-out, the relevant structure is illustrated as follows:

(22) a. [$_{\text{vP}}$ < cithúnzi cá=óphunzila>$_i$ pezá < cithúnzi cá=óphunzila>$_i$]
picture of=student find picture of=student

Bošković (2002) argues that when determining whether a higher/lower copy is pronounced when a PF violation requires a lower copy pronunciation, the

[7] Downing 2018 reports that there is a human/non-human asymmetry in object marking in some modern colloquial Chichewa varieties: With a human object the object marker is always present, whether the object is dislocated or not, whereas the object marker is not necessary for a non-human object even when the object undergoes TD. This pattern is in contradiction with the dislocation pattern described in Bresnan & Mchombo 1987 and in the current paper, the latter two being mutually compatible (where the object marker is a clitic pronoun). Object markers in the system described by Downing 2018 are purely agreement markers, which she argues are historically derived from a pronominal clitic system. I assume that there may be dialectal variation regarding the status of object markers (and how they affect the PD/TD pattern) in Chichewa, though the data from my consultants seem to only reflect the arguably more conservative pronominal clitic system. I thank an anonymous reviewer for raising this issue.

structure is scanned left to right, with the decision of which copy to pronounce made locally, without look-ahead. That is, the PF interface will first decide where to pronounce the left piece, namely *cithúnzi*, in (22a).[8] Assuming that the default option is to realize the phonological features in the highest copy, we get (22b):

(22) b. [$_{vP}$ <cithúnzi cá=óphunzila>$_i$ pezá <~~cithúnzi~~ cá=óphunzila>$_i$]
c. * [$_{vP}$ <cithúnzi cá=óphunzila>$_i$ pezá <~~cithúnzi cá=óphunzila~~>$_i$]
d. [$_{vP}$ <cithúnzi ~~cá=óphunzila~~>$_i$ pezá <~~cithúnzi~~ cá=óphunzila>$_i$]

Then the next piece *cá=óphunzila* is scanned. At this point, the pronouncing-the-highest-copy option for it is no longer available since that will essentially give (22c) and violate (14). In this case (22d) becomes the only available option: The phonological features of the lower piece of *cá=óphunzila* is kept, and PD is derived, as expected.

The current approach explains directly why PD in Chichewa shows the order preservation constraint: Deleting the left piece in the highest copy is simply never a possibility at PF, since the preferred option, i.e., pronouncing that piece, causes no PF violations and is obligatorily selected (thus a structure like (22e) is not considered at any point at PF). After that, things only get wrong if the right piece of the highest copy is also pronounced. To put it in another way, we have explained why scattered deletion can only occur rightward in Chichewa.[9]

(22) e. * [$_{vP}$ <~~cithúnzi~~ cá=óphunzila>$_i$ pezá <cithúnzi ~~cá=óphunzila~~>$_i$]

Recall that (12) and (13) show a reversed pattern, where TD is possible and PD is ruled out. I propose that the PF constraint (14) is simply not applicable in those cases because of the presence of an intervening element: (i) The enclitic *=ni* in (12), and (ii) the *wh*-adjunct in the IAV position in (13). They intervene between the verb stem and the complement when the vP is transferred to spell-out, as illustrated below:

(23) a. [$_{vP}$ < galású lá=líkúlú ili>$_i$ kankha$\boxed{\text{=ni}}$ < galású lá=líkúlú ili>$_i$]
glass of=big this open=PL glass of=big this

[8]It needs to be noted that 'pronunciation' can only be understood abstractly under the current multiple spell-out framework. Since (22a) as the spell-out of vP is not the ultimate pronunciation of a sentence, 'pronunciation' simply refers to the realization/reservation of phonological features, which are still abstract.

[9]This of course does not mean that scattered deletion is universally rightward. The direction issue is an empirical one, subject to specific PF constraints. See Bošković 2001 for a leftward deletion case in Bulgarian.

b. [$_{vP}$ <galásí lá=líkúlú iili>$_i$ kankha=ni <~~galásí lá=líkúlú ili~~>$_i$]

c. * [$_{vP}$ <galásí ~~lá=líkúlú ili~~>$_i$ kankha=ni <~~galásí~~ lá=líkúlú iili>$_i$]

(24) a. [$_{vP}$ < gálimoto lá=tsópanó>$_i$ kónza liti < gálimoto lá=tsópanó>$_i$]
car of=new fix when car of=new

b. [$_{vP}$ <gálimoto lá=tsópaánó>$_i$ kónza liti <~~gálimoto lá=tsópanó~~>$_i$]

c. * [$_{vP}$ <gálimooto ~~lá=tsópanó~~>$_i$ kónza liti <~~gálimoto~~ lá=tsópaánó>$_i$]

In (23a), although the verb stem and its complement might have been adjacent when they merge in the first place, they are not adjacent when (14) comes into play. Consequently, the default option (23b) is selected, and (23c) is automatically ruled out. The same logic applies to (24a), where I argue that the *wh*-adjunct already moves into the IAV position when vP is spelled out.[10] Since at this point the verb stem simply does not have a linearly adjacent complement, (24b) causes no problem and is thus grammatical, and (24c) is not a possibility.

An anonymous reviewer points out that the current account predicts that, in ditransitive constructions, where only one of the objects can be adjacent to the verb, TD of the other non-linearly adjacent object should be possible. The following data from my consultants shows that this prediction is borne out:

(25) a. Ndi=na=páts-á cikondi búukhu
1P.SG=PST=give-FV 1.Chikondi 5.book
'I gave Chikondi a book.'

b. **Búukhu** ndi=na=páts-á cikoondi
5.book 1P.SG=PST=give-FV 1.Chikondi

c. *? **Cikoondi** ndi=na=páts-á búukhu
1.Chikondi 1P.SG=PST=give-FV 5.book

While (25c) is ruled out by (14) in the same way as (19), (25b) is grammatical. Since the direct object is not adjacent to the verb as in the V>IO>DO order, this pattern is unsurprising. However, it is interesting to mention that the relative order of DO and IO in Chichewa is in fact flexible; (25d) is grammatical alongside (25a):

(25) d. Ndi=na=páts-á búkhu cikoondi
1P.SG=PST=give-FV 5.book 1.Chikondi
'I gave a book to Chikondi.'

[10] I leave the question open on how to analyze the exact syntactic status of the IAV position, which is an independently important issue. What is crucial to us is that this position may intervene between the verb and the in-situ complement at the vP level.

Note, however, that Chichewa is a typical 'asymmetrical' language in that only the IO (which always c-commands the DO regardless of the linear order) shows typical object behaviors (Mchombo 2004, Van der Wal 2022). What is important to us is the fact that only the IO agrees with the verb:

(26)

a.

A-leenje	a=ku=phík-íl-á	a-nyaní	zí-túmbúuwa
2-hunters	2sm=prog=cook-appl-fv	2-baboons	8-pancakes

'The hunters are cooking pancakes for the baboons.'

b.

A-leenje	a=ku=$\boxed{\text{wá=}}$phík-íl-á	zí-túmbúuwa
2-hunters	2sm=prog=2om=cook-appl-fv	8-pancakes

c.

* A-leenje	a=ku=$\boxed{\text{zí=}}$phík-íl-á	a-nyaání
2-hunters	2sm=prog=8om=cook-appl-fv	2-baboons

(Mchombo 2004; adapted)

If what is implied by the ungrammaticality of (26c) is that DO is unable to agree with the verb in ditransitives in any case, we can conclude that (14) (which is stated in terms of feature sharing) should not apply to DO in general, so the TD of DO should always be possible, as in (25b). The DO>IO order in (25d) is at any rate not base-generated (and it does not feed an Agree relationship between DO and the verb).[11]

In summary, it is clear now how the puzzles discussed in Section 2 can be resolved under the scattered deletion account with the language-particular PF constraint (14) being posited. First, the potential non-constituent movement puzzle does not arise, since what undergoes syntactic movement is the whole DP, not the head noun.[12] Second, the fact that Chichewa (in most cases) allows object PD but disallows TD is no longer a typological mystery: It is in fact expected because PD as a case of scattered deletion only becomes possible if the pronouncing-the-highest-copy option (which gives TD) is excluded. We have seen that TD is

[11]It is not immediately obvious why (25c) is still degraded, since it may start from (i) (which will give (25d) if topicalization does not happen):

(i) $[_{vP}$ cikondi$^{IO}{}_{i}$ patsa^{V} búkhu^{DO} cikondi$^{IO}{}_{i}$ $]$

Here the IO is not adjacent to the verb and thus should be possible to undergo TD. One possibility is that (i) is in fact not a licit structure per se (i.e., postverbally only IO>DO is possible at the point vP is spelled out), the DO>IO order in (25d) being derived via further order readjustment after the entire structure is built. I will not go further into this issue.

[12]Assuming N-initiality involves NP-raising in Chichewa is also compatible with the scattered deletion account, so the current discussion by itself is orthogonal to the head vs. phrasal movement debate.

possible in some cases in Chichewa, and that TD and PD are crucially in complementary distribution. Finally, the so-called order preservation constraint is accounted for: Only rightward deletion is possible regarding Chichewa PD because the pieces of a copy are scanned left to right at PF.

4 Scattered deletion and scopal relations

I have argued that the scattered deletion account of Chichewa PD derives cases like (1c) and (7c) as in the following:

(27) < mbúuzi ~~zákúda~~>$_i$ atsíkána á=mfúumu a=a=gulá < ~~mbúzi~~ zákúuda>$_i$
goats black girls of=chief they.have.bought goats black

(28) < cithúunzi ~~cá=óphunzila~~>$_i$ ndi=ná=pézá < ~~cithúnzi~~ cá=óphunziila>$_i$
picture of=student I.found picture of=student

Notice that all cases of scattered deletion, or lower copy pronunciation in general, may involve some syntax-phonology or semantic-phonology mismatch, since pronouncing (part of) the lower copy is purely a PF phenomenon, and should not affect the formal and semantic features of an element. For example, in (27) the phonological features of *zákúda* are not realized in the highest copy, but its semantic features that are not deleted earlier in the derivation should be visible at LF at this position. That is, the current approach predicts that the dislocated topic here is *mbúzi zákúda* 'black goats', rather than *mbúzi* 'goats'. Besides, recall from (5) that an in-situ *wh* element in Romanian licenses parasitic gaps, indicating that it is just left behind at PF, but in fact structurally high in narrow syntax.[13]

In this section I provide scopal facts that strongly support the scattered deletion approach to Chichewa PD. Consider first (29), which shows a typical PD pattern:

(29) a. Sí=ndí=na=ón-é **a-phunzitsi óonse** mu=kaláasi
NEG=1P.SG=PST=see-FV 2-teachers 2-all 18=5.classroom
lit. 'I didn't see all the teachers in the classroom.' (¬>∀; (*)∀>¬)

[13]There is unfortunately no evidence of this sort in Chichewa, where parasitic gaps are not independently attested.

b. **A-phunziitsi** sí=ndí=na=ón-é **óonse** mu=kaláasi
2-teachers NEG=1P.SG=PST=see-FV 2-all 18=5.classroom
(*¬>∀; ∀>¬)

Interestingly, the interpretation of (29a) and (29b) is very different. (29a) can used either in contexts where I saw no teachers in the classroom, or in contexts where I saw some, but not all of the teachers. By contrast, (29b) can only mean that all the teachers are such that I did not see them, i.e., I did not see any teachers in the classroom. This indicates that the universal *ónse* 'all' must scope over negation in (29b). On the other hand, however, it is not readily evident whether (29a) shows real scopal ambiguity, i.e., *ónse* 'all' can either scope over negation or not. This is because in terms of truth conditions, ¬>∀ holds as long as there are some teachers that I did not see, and it is just one possibility that I by chance did not see any of them. That is, while the ¬>∀ reading must be available in (29a), the status of ∀>¬ is not clear. I thus put an asterisk in parentheses for the ∀>¬ reading in (29a). What is important here is the observation that although in both cases *ónse* 'all' occurs postverbally on the surface, its interpretation changes radically.

Now consider a slightly more complicated case to illustrate the point further. In (30), where the noun has two modifiers, the noun can be dislocated alone (30b) or with one modifier immediately following it (30c):[14]

(30) a. Sí=ndí=na=ón-é **a-phunzitsi ónse á=nzéélú**
NEG=1P.SG=PST=see-FV 2-teachers 2-all 2.ASSOC=10.intelligence
mu=kaláasi
18=5.classroom
lit. 'I didn't see all intelligent teachers in the classroom.' (¬>∀; (*)∀>¬)

b. **A-phunziitsi** sí=ndí=na=ón-é **ónse á=nzéélú**
2-teachers NEG=1P.SG=PST=see-FV 2-all 2.ASSOC=10.intelligence
mu=kaláasi
18=5.classroom
(*¬>∀; ∀>¬)

[14]The PD pattern keeps more or less similar with some complication when more than one modifier is involved. If both (30b) and (30c) are a result of topicalization of the whole object DP *plus* scattered deletion, we apparently are observing some optionality here. Both cases can be seen as rightward deletion (i.e., they are both cases of (6b)), but differ in how the two pieces are grouped. I tentatively suggest that the two are only different in phonological phrasing, but leave this important issue for future study.

c. **A-phunzitsi óonse** sí=ndí=na=ón-é **á=nzéélú**
2-teachers 2-all NEG=1P.SG=PST=see-FV 2.ASSOC=10.intelligence
mu=kaláasi
18=5.classroom

(*¬>∀; ∀>¬)

Interestingly, while the in-situ case (30a) allows narrow scope for *ónse*, only the wide scope option is available in the PD cases (30b–30c).

This pattern strongly suggests that in both cases (30b–30c) *ónse* moves in narrow syntax. The position in which *ónse* is pronounced is different in (30b) and (30c), i.e., in (30b) *ónse* is pronounced after the verb while in (30c) it occurs preverbally along with the noun, but the interpretation keeps the same. I consider this a decisive argument for the scattered deletion approach to PD. The structures of (30b) and (30c) are illustrated in (31a) and (31b), respectively. Note I am assuming here that only the highest copy of *ónse* is responsible for its scope property (as for (29a) and (30a), where *ónse* is in situ in narrow syntax, the inverse ∀>¬ reading, if available at all, may result from QR in LF).

(31) a. <aphunziitsi ~~**ónse** á=nzélú~~>$_i$ sí=ndí=na=óné <~~aphunzitsi~~ ónse
↑interpreted at LF
á=nzéélú>$_i$ mu=kaláasi

b. <aphunzitsi **óonse** ~~á=nzélú~~>$_i$ sí=ndí=na=óné <~~aphunzitsi ónse~~
↑interpreted at LF
á=nzéélú>$_i$ mu=kaláasi

5 The loss of conjoint-disjoint alternation and *wh*-in-Situ in Chichewa

Recall the PF-constraint (14) proposed in Section 3, which is crucial for the current scattered deletion account. It essentially rules out TD in cases where PD is attested, explaining the complementary distribution of the two. In this section, I provide two further independent arguments for the presence of the constraint. The first argument, being perhaps indirect and speculative, comes from the conjoint-disjoint (CJ/DJ) alternation widely attested in Bantu. As is well-known, verbs in many Bantu languages may have different forms depending on whether there is a complement following the verb. In Kirundi, for example, the verb takes the CJ form if it has a complement directly following it (32a); otherwise it surfaces in the DJ form (32b). Importantly, the CJ form is zero-marked while the DJ form is marked by an extra affix:

(32) a. Imuúngu [$_{vP}$ zi=ry-a i-gǐti]
10.woodworms 10SM=eat-FV 7-wood
'(The) woodworms eat wood (and nothing but wood).'

b. Imuúngu [$_{vP}$ zi=**ra**-ry-á] uruugi
10.woodworms 10SM=PRES.DJ=eat-FV 11-door
'(The) woodworms eat through the door.'

(Kirundi, Meeussen 1959: 216; adapted)

On the one hand, Chichewa lacks the CJ/DJ alternation; on the other hand, the alternation is found in Bantu languages that are not necessarily geographically adjacent, and multiple scholars have argued that the alternation existed in Proto-Bantu (Meeussen 1967, Nurse 2008). This means that Chichewa used to be a CJ/DJ language at an earlier historical stage, but has lost the alternation entirely. In other words, Chichewa has lost the DJ morpheme.[15]

The conjecture, then, is that the PF constraint (14) or something similar to it is directly correlated to the CJ/DJ alternation. In a comparative sense, a Chichewa verb trivially takes the CJ form, so (14) can be rewritten as follows:

(33) A verb *in the conjoint/unmarked form* and its internal argument must be adjacent sub-constituents of a phonological phrase if they share phi-features.

(33) can be viewed as a statement of the distribution of the CJ form. Suppose it is not only active in Chichewa, but is in fact present in some other Bantu languages as well, perhaps with some micro-variation. One can say that the DJ morpheme is added exactly when the verb does not form a prosodic phrase with its complement, in which case the DJ morpheme is used to satisfy/avoid (33). I suggest the reason why a lot of Bantu languages have the CJ/DJ alternation is precisely because of the presence of this PF constraint (or something alike: The constraints of course do not have to be stated in the same way in different languages). In Chichewa, where the alternation is lost, TD is ruled out in

[15] Or one could say that the loss of CJ/DJ is a templatic one, because the DJ markers in different Bantu languages do not necessarily share the same lexical origin. Chichewa has lost the morphological slot that could host the DJ marking.

general by (33), and PD becomes possible via scattered deletion. Consequently, some syntax-phonology or semantic-phonology mismatches are attested in Section 4. If the current thought is on the right track, on the one hand, there is good reason for the striking Bantu CJ/DJ alternation to exist—to avoid potential syntax-phonology or semantic-phonology mismatches produced by (33); on the other hand, we have good historical reasons to explain why Chichewa DP splits show such a complex pattern: It is a result of the morphological loss of DJ. In any case, the picture is far from crystal clear, and a thorough comparative study among Bantu is needed. I leave this issue for future research.

The second argument is probably more direct and telling. Chichewa shows an interesting subject/object asymmetry that subject *wh*-words cannot occur in situ (they must be clefted) while object *wh*-words normally do.[16] One may take it for granted that Chichewa *wh*-objects in this respect are just like *wh*-objects in a 'canonical' *wh*-in-situ language such as Chinese or Japanese. However, I propose that this is not the case. In-situ *wh* objects in Chichewa essentially show island effects:[17]

(34) * Mu=ná=kúman-a ndí=mú-nthu a-méné á=ma=phuzíts-á
2P.PL=PST=meet-FV with=1-person 1-COMP 1SM=HAB=teach-FV
ci-yáani?
7-what
'*What did you meet a person who teaches?'

(35) * Ku=ímb-il-a **ndaání** ndi=kósávúuta?
INF=call-APPL-FV who COP=not.hard
'*Whom is to call easy?'

(36) * Mavúuto a=ná=gúl-a gálimooto ci-fukwá w=a=mang-a
1.Mavuto 1SM=PST=buy-FV 5-car 7-for.reason 1SM=PERF=build-FV
ciyáani?
7-what
'*What did Mavuto buy a car because he built?'

[16] Recall that non-subject *wh*-words can optionally occur in the so-called IAV position. I will not go into this issue in this paper.

[17] Note that this is not the case in many other *wh*-in-situ Bantu languages, though Bergvall 1983 claims that *wh*-in-situ in Kikuyu also shows island sensitivity. It is also interesting to mention that beyond Bantu, Vietnamese is another language where *wh*-in-situ is shown to be sensitive to islands; see Bruening & Tran 2006.

As exemplified above, an object cannot be questioned if it occurs in a complex DP (34), a subject (35), or an adjunct (36).[18] By contrast, the counterparts of these cases are grammatical in 'canonical' *wh*-in-situ languages like Chinese or Japanese. I take this contrast to mean that *wh*-objects in Chichewa in fact move in narrow syntax, while in Chinese or Japanese they do not. Therefore, *wh*-in-situ in Chichewa is no more than a superficial phenomenon: The seemingly in-situ *wh*-phrase reflects the pronunciation of a lower copy. It is (14/33) that excludes overt fronting of the *wh*-phrase. Though *wh*-in-situ in Chichewa does not directly involve PD, it strongly suggests that the proposed PF constraint exists, and eventually supports the analysis offered in this paper.

6 Conclusion

Discontinuous DPs in Chichewa without object markers have been examined carefully in this paper. I conclude that the patterns concerned can only be captured by a scattered deletion account: The whole object DP undergoes topicalization, but pronouncing it in the highest copy as a whole will violate a language-particular PF-constraint (14/33), so as a last resort part of it is pronounced low. The approach explains why in most cases a left-dislocated object DP with more than one phonological piece has to split (total dislocation in Chichewa is rare and is in complementary distribution with partial dislocation). The observation that the proposed deletion can only be rightward is also expected.

The scattered deletion approach is strongly supported by the scope properties of the universal quantifier *ónse*, which may take narrow scope in the object position but only allows the wide scope reading if the apparently in-situ quantifier is part of a DP split. I also showed that the proposed PF constraint (14/33) is empirically well-supported by *wh*-in-situ in Chichewa, which shows island effects and should involve *wh*-fronting in syntax and lower copy pronunciation at PF. By briefly discussing the CJ/DJ alternation in Bantu, I tentatively suggested that the PF constraint may also be active in some other Bantu languages.

Abbreviations

1/2/3P = 1st/2nd/3rd person; APPL=applicative; ASSOC = associative; COMP = complementizer; COP = copula; DJ = disjoint; FV = final vowel; HAB = habitual; INF= infinitive; NEG = negation; OM = object marker; PASS = passive; PERF = perfective; PRES = present; PST = past; PROG = progressive; SG = singular; SM = subject

[18](34–36) become grammatical if the *wh*-object receives an echo reading, which I ignore here.

marker; SUBJ = subject; **numbers** indicate noun classes and agreement/concord associated with noun classes.

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Chapter 18

Interpretive properties of perception verbs in Tiriki: Hyperraising and copy-raising

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Tiriki allows hyperraising constructions, similar to many other Bantu languages (Carstens & Diercks 2013, Halpert 2019). In this paper, we document a range of nuanced interpretive properties that arise in parallel constructions in Tiriki hyperraising and copy-raising contexts. We show that there are connectivity effects in apparent copy-raising constructions, while those same constructions nonetheless behave as if the matrix predicate assigns a thematic role to the matrix subject. Assignment of a matrix thematic role to a subject is traditionally thought to be in complementary distribution with movement from the embedded clause. Instead, we suggest that all perception verbs assign thematic roles to their subjects, even when those subjects move from an embedded position where they were already assigned a first thematic role.

1 Introduction

1.1 A copy-raising puzzle

Perception verbs have long played a central role in syntactic theory. Most attention has been on so-called subject-to-subject raising constructions, but there is a growing focus on a variety of related constructions such as hyperraising and copy-raising. English allows raising and copy-raising constructions. In raising constructions, the subject appears to be shared between the main and embedded clauses (e.g., (1a)). There also tends to be coreference between subjects in

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copy-raising constructions with finite embedded clauses such as (1b), but in these constructions there is also a pronoun in the embedded clause.

(1) a. Tania seems ~~Tania~~ to be sick. Raising
b. Tania seems like she is sick. Copy-raising

On canonical analyses of raising constructions, the subject moves from the embedded clause to a position in the main clause (Davies & Dubinsky 2008, Polinsky 2013). Raising is allowed from non-finite clauses, as (2) illustrates: the subject is generated inside the embedded clause (2a) and then raises to the matrix clause (2b).

(2) Raising allowed from non-finite clause
a. __ seems [$_{\text{TP}}$ ~~Tania~~ to be happy] .
b. Tania seems [$_{\text{TP}}$ ~~Tania~~ to be happy] .

A central part of the canonical analysis of (2b) is that raising predicates do not assign a thematic role to an external argument; they are unaccusative predicates with an internal argument (the embedded proposition) but no external argument. This analysis ensures that these derivations abide by the Theta Criterion, wherein arguments must be assigned exactly one thematic role and no more (Chomsky 1993).

In contrast, (3) illustrates that raising from a finite clause (i.e., hyperraising) is not allowed in English.[1] In (3a), the subject remains in the finite embedded clause and is acceptable. In these constructions, the canonical analysis is that a non-thematic subject (expletive *it*) appears in the matrix clause. But when the subject is moved out of the finite embedded clause, as in (3b), the result is unacceptable.

(3) Raising not allowed from finite clause
a. It seems [$_{\text{CP}}$ that Tania is happy] .
b. *Tania seems [$_{\text{CP}}$ (that) ~~Tania~~ is happy] .

In canonical copy-raising constructions—the other kind of raising-type construction that English allows—an overt noun phrase is present in the matrix clause, and an anaphorically linked pronoun appears in the embedded clause.

[1]See Greeson (2023) for claims that certain hyperraising constructions are in fact acceptable in English. We briefly discuss this proposal in §6.2.

A commonly assumed analysis of copy-raising is that the matrix subject is base-generated in the matrix clause as a thematic argument of *seems* (Potsdam & Runner 2001).[2]

(4) "Copy-raising:" Complement clause is finite
Tania$_k$ seems [$_{CP}$ like she$_k$ is happy] .

This traditional story predicts a complementary distribution between connectivity effects and assigment of thematic roles to perception predicates' subjects. On this traditional account, arguments can only be assigned one thematic role. Thus, if a matrix predicate assigns a thematic role, the argument that receives it must have originated in the matrix clause (and the construction will therefore not exhibit connectivity effects). If the matrix predicate does not assign a thematic role, the subject of this predicate must have raised from the embedded clause, where it was assigned a theta role (leading to connectivity effects). The primary contribution of this paper is an observation that this traditionally-proposed complementary distribution is not empirically supported.

To illustrate the relevant patterns in English, consider the constructions in which the idiomatic meaning of (5) is (un)available (see Potsdam & Runner 2001 for some early comments on this topic).

(5) The cat is out of the bag. = *The secret has been revealed.*

When the subject remains in the embedded clause, as in (6a) and (6b), the idiomatic interpretation is natural.

(6) a. It **seems that** the cat is out of the bag. Raising Predicate
b. It **looks like** the cat is out of the bag. Copy-raising Predicate

The idiomatic interpretation remains natural in a canonical raising construction such as (7a), despite the fact that the subject of the idiom is not adjacent to the rest of the idiom. In raised constructions with copy-raising predicates such as (7b) and (7c), the idiomatic reading is also still accessible for many speakers, although in (7c) *looks like* places some empirical restrictions on the speaker's perception (i.e., their reason for thinking the secret is out must come from visual evidence). The judgments reported here reflect the relative naturalness of the idiomatic reading (not grammatical acceptability).

[2]This comment undersells the complexity of the analytical situation, but it suffices for this introduction. See Landau (2009, 2011), Asudeh & Toivonen (2012), and den Dikken (2017) for additional commentary.

(7) a. The cat **seems to** be out of the bag. Raising
b. %The cat **seems like** it is out of the bag. Copy-raising
c. %The cat **looks like** it is out of the bag. Copy-raising

The pattern becomes more complex with different predicates. When the perception predicate encodes more limitations on the source of information, the idiomatic interpretation tends to be less accessible. The idiomatic reading in (8a) is somewhat accessible as long as the evidence is via sound (either directly, or via someone narrating an account of a secret coming out). But (8b) makes it much harder to access the idiomatic reading and instead lends itself to an interpretation of a smelly feline that has escaped.

(8) a. %The cat **sounds like** it is out of the bag. Copy-raising
b. ??The cat **smells like** it is out of the bag. Copy-raising

So idiomatic interpretations (as we will see below) seem to be possible even in copy-raising constructions where it is not obvious that the subject has moved from the embedded clause (i.e., there is no gap in the lower clause). But despite the presence of connectivity effects, the matrix predicate nonetheless seems to place thematic restrictions on the subject, suggesting that there is also a matrix thematic role that is assigned to the subject.

These kinds of effects are not well-researched, to our knowledge.[3] That is, there is empirical complexity amongst the range of perception verbs that is still under-explored: there seems to be variation in idiomatic interpretations' accessibility based on the particular lexical predicate that is used. In this paper, we explore a complex array of connections between morphosyntactic variation and interpretive distinctions in Tiriki perception verb constructions. These patterns will point to a way that the English facts above can be considered. This paper mainly makes empirical and analytical contributions to lay the foundation to better understanding perception verb constructions in Bantu languages and cross-linguistically, but in Section 6.2 we also offer an initial analysis of hyperraising constructions that takes these findings into account.

1.2 Raising constructions in Tiriki

Tiriki (Luyia, Bantu, Kenya) also allows various raising constructions. (9a) shows that unraised constructions in Tiriki, as in English, are acceptable. But hyperraising, wherein the embedded clause that the subject moves from is finite, is

[3]The availability of idiomatic readings in copy-raising is discussed by Potsdam & Runner (2001), Asudeh & Toivonen (2012), and den Dikken (2017), but to our knowledge the interaction with different perception predicates is not.

also acceptable; this is shown in (9b). (9b) is an instance of *agreeing* hyperraising: the matrix verb exhibits agreement with the hyperraised subject.[4]

(9) a. Unraised

Ka-/i-lolekh-a	khuli	va-ana	va-tukh-i.
6SM-/9SM-seem-FV	that	2-child	2SM-arrive-PST

‘It seems that the children arrived.’
(Diercks et al. 2022: 97)

b. Agreeing raising

Va-ana	$\boxed{\text{va-}}$ lolekh-a	khuli	~~vaana~~	va-tukh-i.
2-child	2SM-seem-FV	that		2SM-arrive-PST

‘The children seem to have arrived.’
(Lit: ‘The children seem that arrived.’)
(Adapted from Diercks et al. 2022: 98)

Notably, Tiriki also exhibits *non-agreeing* hyperraising constructions, in which the matrix clause exhibits atypical subject agreement that does not match the DP subject’s φ-features.[5] In (10), observe that the raised subject is class 2, but the matrix verb can be marked with a class 6 or class 9 subject marker.[6]

(10) Non-agreeing raising

Va-ana	$\boxed{\text{ka-/i-}}$ lolekh-a	khuli	~~vaana~~	va-tukh-i.
2-child	6SM-/9SM-seem-FV	that		2SM-arrive-PST

‘The children seem to have arrived.’
(Adapted from Diercks et al. 2022: 98)

Diercks et al. (2022) address many central questions that arise from hyperraising constructions. They show that in Tiriki, both agreeing and non-agreeing raising are instances of A-movement to typical subject position in the matrix clause. This conclusion is based on a variety of typical raising diagnostics (retention of idiomatic interpretations, perceptual source readings, creation of new positions for binding, subject/non-subject diagnostics, etc.). We summarize these findings in Section 2. Crucially, Diercks et al. (2022) show that these hyperraising

[4]Judgments and commentary on Tiriki data come from our research assistant and language consultant, Kelvin Alulu.
[5]Similar agreeing and non-agreeing raising patterns have been documented in Zulu (Halpert 2019), Logoori (Gluckman & Bowler 2017), and Wanga (Diercks field notes).
[6]Both class 6 and class 9 are available in non-agreeing constructions. Throughout, we typically give the class 6 non-agreeing form for simplicity.

constructions are not covertly copy-raising constructions with a null subject in the embedded clause (i.e., not 'The children$_k$ seem like *pro*$_k$ arrived.'). Likewise, they demonstrate that hyperraising is also not left-dislocation of the embedded subject: the subject in (9b) and in (10) behaves like a typical matrix subject.

This paper focuses on additional questions prompted by these hyperraising constructions. Section 3 introduces apparent copy-raising constructions and explores questions regarding those agreeing and non-agreeing constructions, specifically around issues similar to those introduced in Section 1.1 where connectivity effects seem to vary in different instances of the (apparent) same kind of grammatical construction. We ultimately conclude that what we call "copy-raising" constructions in Tiriki are in fact structurally identical to hyperraising constructions (i.e., they involve movement rather than a coreferent *pro* in the embedded clause). To be clear about how these relate to previous work on the issue, we still refer to these constructions as copy-raising constructions throughout.

Section 4 addresses the question of what the class 6 and class 9 subject markers agree with in non-agreeing raising constructions, concluding that they agree with null arguments that are expletive-like in some ways but are in fact referential subjects (we will refer to them as quasi-expletives). Section 4's conclusion is that the non-agreeing raising constructions are in fact multiple subject constructions (one subject is the raised lexical DP, the other is the null quasi-expletive). Section 5 shows that raised subjects have a topicality reading that unraised subjects lack.

We use these conclusions in Section 6 to sketch a direction of analysis in a Minimalist analytical framework (Chomsky 2000, 2001). While the paper primarily makes an empirical contribution rather than a theoretical one, Section 6.2 considers how we can explain the empirical distinction between agreeing and non-agreeing raising while also capturing what drives movement of the embedded subject to the matrix clause. This is an initial proposal, however, and more work is necessary to solidify an analysis of Tiriki hyperraising.[7] Our main contribution is to show the various interpretive effects that are present in fine-grained variations of Tiriki perception verb constructions; the main takeaway is that, contrary

[7]There are a number of critical issues that are not discussed in this paper. The main one is the question of how the hyperraised subject escapes the finite embedded clause. It is somewhat ironic that we don't address this, as it is a fundamental puzzle about hyperraising and is the precise issue that most of the literature is dedicated to. It may well be that the issues we tackle here will lead to insights about escaping finite clauses, but we leave the question for future research. For relevant discussion, see: Martins & Nunes (2005), Nunes (2008), Boeckx et al. (2010), Martins & Nunes (2010), Obata & Epstein (2011), Carstens & Diercks (2013), Fong (2018, 2019), Deal (2017), Zyman (2023).

to traditional assumptions, perception verbs seem to assign matrix thematic roles to their subjects even in contexts that show evidence of A-movement out of another theta-role-assigning position in the embedded clause. We suggest that this conclusion points to an analytical solution for hyperraising, although more work is necessary to exhaustively articulate and defend such an analysis.

2 Diagnostics for Tiriki hyperraising (Diercks et al. 2022)

There are two salient alternative analyses for apparent hyperraising constructions. First, apparent agreeing hyperraising could in principle be covert copy-raising. Tiriki is a null subject language, so *pro* could be a "copy" pronoun in the embedded clause, which would derive the same word order seen in hyperraising constructions.

(11) a. Hyperraising analysis
[SUBJ$_k$ seems [$_{CP}$ that ~~SUBJ~~$_k$ [$_{TP}$ ...]]]
b. Copy-raising analysis
[SUBJ$_k$ seems [$_{CP}$ that *pro*$_k$ [$_{TP}$ ...]]]

Second, apparent non-agreeing raising could in principle involve a left-dislocated argument with an expletive subject, as opposed to hyperraising to canonical subject position with noncanonical subject agreement.

(12) a. Non-Agreeing raising analysis
[$_{TP}$ SUBJ$_k$ **ka**$_i$-seems [$_{CP}$ that t_k [$_{TP}$ ...]]]
b. Expletive + Dislocation analysis
[$_{CP}$ SUBJ$_k$ [$_{TP}$ **expl**$_i$ **ka**$_i$-seems [$_{CP}$ that t_k [$_{TP}$ ...]]]]

However, Diercks et al. (2022) argue at length that both agreeing and non-agreeing raising constructions in Tiriki are true hyperraising constructions: they are derived via movement of the embedded subject from the embedded clause to matrix subject position.

(13) Properties of Tiriki hyperraising (Diercks et al. 2022)

- **Not left-dislocation**: Hyperraising is possible in contexts where left-dislocation is not, suggesting that apparent hyperraising constructions are not in fact left-dislocation constructions.
- **A-movement**: Hyperraising behaves like A-movement to canonical subject position, rather than like Ā-movement.

- **Connectivity effects**: Idiomatic readings are maintained in hyperraising constructions, but they are not retained in left-dislocation or control constructions. (Other connectivity effects hold as well.)

Space constraints preclude a full summary of these effects, but we illustrate idiom connectivity effects here to set the stage for this paper's contributions. By assumption, idioms must enter syntactic constructions as a constituent to maintain their idiomatic interpretations. Raising constructions retain idiomatic interpretations because the raised subject is originally part of an idiom inside the embedded clause. The same does not happen with control constructions: idiomatic readings are lost because the main clause subject is base-generated in the main clause (on standard assumptions), and as such the idiom does not enter the sentence as a constituent.

(14) a. Raising construction (Traditional analysis)
The cat seems to ~~the cat~~ be out of the bag.
Lit: An actual cat seems to be out of an actual bag.
Fig: The secret seems to have come out.

b. Control Construction (Traditional Analysis)
The cat wants PRO to be out of the bag.
Lit: An actual cat desires to be out of an actual bag.
?? on the figurative reading

The Tiriki constructions that we claim are hyperraising constructions retain idiomatic interpretations: the idiom in (15) retains its meaning in both agreeing and non-agreeing hyperraising constructions, as in (16).[8]

(15) **Tiriki Idiom**
I-mbisi i-hurir-e mu-riro.
9-hyena 9SM-feel-FV 3-fire
Lit: 'The hyena has felt the fire.'
Fig: 'Someone has eaten too much.'
(Diercks et al. 2022: 99)

(16) shows that with the hyperraising predicate *-lolekha*, the idiomatic reading is still acceptable. Hyperraising consistently shows these connectivity effects

[8] Halpert (2019) shows the same for Zulu, and Diercks et al. (2022) show the same for Logoori.

(wherein idiomatic meanings are maintained in hyperraising constructions), indicating that the matrix subject raises from the idiomatic constituent in the embedded clause rather than originating in the matrix clause.

(16) a. Hyperraising (Agreeing)

I-mbisi	i-lolekh-a	khuli	i-hurir-e	mu-riro.
9-hyena	9SM-seem-FV	that	9SM-feel-FV	3-fire

Lit: 'The hyena seems to have felt the fire.'
Fig: 'Someone seems to have overeaten.'

b. Hyperraising (Non-agreeing)

I-mbisi	ka-lolekh-a	khuli	i-hurir-e	mu-riro.
9-hyena	6SM-seem-FV	that	9SM-feel-FV	3-fire

Lit: 'The hyena seems to have felt the fire.'
Fig: 'Someone seems to have overeaten.'

The idiom does not retain its figurative reading in (17) with the apparent copy-raising predicate *-manyia* 'show/look like' (Diercks et al. 2022: 28-29). (Although it is worth noting that we will introduce more complication on this point in Section 3.)

(17) Copy-raising Construction

I-mbisi	i-manyi-a	khuli	i-hurir-e	mu-riro.
9-hyena	9SM-show-FV	that	9SM-feel-FV	3-fire

Lit: 'The hyena looks like it felt the fire.'
*Fig: 'Someone looks like they have overeaten.'
(Diercks et al. 2022: 102)

Likewise, control constructions also lose the idiomatic interpretation.

(18) Control construction

I-mbisi	i-cherits-a	khu-hurir-e	mu-riro.
9-hyena	9SM-try-FV	15-feel-FV	3-fire

Lit: 'The hyena tried to feel the fire.'
*Fig: 'Someone tried to overeat.'
(Diercks et al. 2022: 100)

Idioms also generally resist participating in left-dislocation constructions:

(19) a. Canonical word order

Isaka a-vor-i khuli i-mbisi i-hurir-e mu-riro.
Isaka 1SM-say-FV that 9-hyena 9SM-feel-FV 3-fire
Lit: 'Isaka said that the hyena has felt the fire.'
Fig: 'Isaka said that someone has overeaten.'

b. Left-dislocation construction

I-mbisi, Isaka a-vor-i khuli i-hurir-e mu-riro.
9-hyena Isaka 1SM-say-FV that 9SM-feel-FV 3-fire
Lit: 'The hyena, Isaka said that it has felt the fire.'
*Fig: 'Isaka said that someone has overeaten.'

This suggests that both agreeing and non-agreeing raising constructions are movement constructions. A summary of Tiriki's hyperraising vs. left-dislocation behavior is given in (20):

(20) Summary: Raising diagnostics by construction

	-lolekha			**LD Phrases**
Diagnostic	**AGR-**	**ka-**	**i-**	
Idiomatic reading retained	✓	✓	✓	*
Fronted DP can be new information	✓	✓	✓	*
Fronted quantified DP	✓	✓	✓	*
Cleft constructions: AGR-*a*?	*	opt	opt	n/a
Possible inside relative clause?	✓	✓	✓	*
Principle C: coreference?	*	*	*	n/a
Cyclic raising	✓	✓	✓	*

(Adapted from Diercks et al. 2022: 130)

These diagnostics are built on Halpert's (2019) argumentation about Zulu hyperraising, which shows the same properties. In short, Zulu likewise allows non-agreeing raising, wherein the raised subject behaves like a main clause subject despite not triggering canonical subject agreement. The conclusion from (20) is that both agreeing and non-agreeing hyperraising exist in Tiriki (as in Zulu, Logoori, and Wanga) and that plausible alternative analyses (base generation of subjects in matrix clauses, left-dislocation of subjects) do not match the evidence.

Specifically relevant for our purposes is that hyperraising shows connectivity effects with the lower clause; the evidence we illustrated here is that idioms introduced in the lower clause maintain their figurative readings in raising constructions. This specifically contrasts with copy-raising predicates and control predicates, where idiomatic readings are generally lost or degraded in parallel constructions.

3 Apparent copy-raising predicates show variable connectivity effects

Copy-raising constructions generally do not show the same connectivity effects with the embedded clause that hyperraising constructions do. However, an interesting distinction appears between agreeing and non-agreeing constructions with copy-raising predicates: some copy-raising constructions do in fact show connectivity effects.

3.1 Idiomatic meanings in apparent copy-raising constructions

Consider the examples below with the apparent copy-raising predicate *-sasa* 'seem/appear like' and the idiom in (21).

(21) I-nyungu ya-atikh-a.
9-pot 9SM-break-FV
Lit: 'The pot broke.'
Fig: 'The secret came out.'

(22a) demonstrates that the idiomatic reading is available in an unraised construction, as expected. (22b) shows the lack of connectivity effect that is typical of copy-raising constructions.

(22) a. Ka-sas-a i-nyungu ya-atikh-a. **Unraised**
6SM-appear-FV 9-pot 9SM-break-FV
Lit: 'It appears like the pot broke.'
Fig: 'It appears like the secret came out.'

b. I-nyungu i-sas-a ya-atikh-a. **Raised (Agreeing)**
9-pot 9SM-appear-FV 9SM-break-FV
Lit: 'The pot appears like it broke.'
??Fig: 'The secret appears to have come out.'

In a non-agreeing raising context, however, the idiomatic reading surprisingly reemerges.

(23) I-nyungu ka- sas-a ya-atikh-a. **Raised (Non-agreeing)**
9-pot 6SM-appear-FV 9SM-break-FV
Lit: 'The pot appears like it broke.'
Fig: 'The secret appears to have come out.'

The same pattern appears with the copy-raising predicate *-hulikha* 'sound like.'

(24) a. I-nyungu i- hulikh-a ya-atikh-e. **Raised (Agreeing)**
9-pot 9SM-sound.like-FV 9SM-break-PST
Lit: 'The pot sounds like it broke.'
*Fig: 'The secret sounds like it came out.'

b. I-nyungu ka- hulikh-a ya-atikh-e. **Raised (Non-agreeing)**
9-pot 6SM-sound.like-FV 9SM-break-PST
Lit: 'The pot sounds like it broke.'
Fig: 'The secret sounds like it came out.'

The agreeing "copy-raising" construction behaves as expected: the matrix subject acts like a thematic argument of the main clause (i.e., apparent non-movement, lacking connectivity effects that are present in hyperraising constructions in Tiriki). But non-agreeing constructions with the same predicates surprisingly exhibit connectivity effects (which are typical of movement constructions). It would be surprising for the same raising predicate to sometimes assign a theta role to its subject (in the constructions without connectivity effects and thus with apparent base-generation of the argument in matrix Spec,*v*P), but sometimes not assign a theta role (in the constructions with connectivity effects and thus with apparent movement) when they otherwise appear to be structurally identical.[9]

To address this puzzle, we advance the hypothesis that predicates such as *-sasa* "appear like" and *-hulikha* "sound like" (as well as other raising predicates in Tiriki) do in fact always assign theta roles to their subjects. These thematic roles can be assigned to either a raised lexical DP (in the agreeing constructions) or to

[9]In English, the difference between *seems* and *seems like* presents a similar conundrum. A common analysis of English *seems* is that there is simply homophony between two separate predicates (one that assigns an external thematic role and one that does not). In this paper, however, we argue that such an analysis misses a broader generalization that appears in English with *seems* and in Tiriki with apparent copy-raising predicates: often the same predicate can behave like a raising predicate in one context and a copy-raising predicate in another. Here, we propose that this difference in behavior is not caused by the existence of two different homophonous predicates, but by the different constructions in which a single predicate can appear.

the null argument that the class 6 and class 9 subject markers agree with (in the non-agreeing constructions). Of course, this means that we are proposing that a DP can be assigned two theta roles in the agreeing raising constructions (one from the matrix predicate and one from the embedded predicate); see Boeckx et al. (2010) for similar claims within the Movement Theory of Control.

3.2 Perceptual sources: Interpretive effects with apparent copy-raising predicates

This section offers another argument that apparent copy-raising predicates place thematic restrictions on their overt lexical DP subjects in agreeing constructions but not in non-agreeing constructions. In general, copy-raising constructions induce a perceptual-source reading of their subjects, requiring that the speaker have direct perception of the referent of the matrix subject; raising constructions do not (Carstens & Diercks 2013, Potsdam & Runner 2001). Therefore a non-specific subject like *somebody* is not possible with a copy-raising construction (e.g., (25c)), which requires that a speaker have direct perception of the referent of the lexical subject and therefore presupposes an identifiable referent of the subject.

(25) *Context: You look in the refrigerator, only to find that it is empty.*

a. It seems like somebody has eaten all the food! **Unraised**

b. Somebody seems to have eaten all the food! **Raising**

c. #Somebody seems like they have eaten all the food! **Copy-raising**
(*requires direct perception of the food-eater*)
(Adapted from Carstens & Diercks 2013)

We can thus use the availability of perceptual source readings in apparent copy-raising constructions to diagnose a thematic relationship between the raising predicate and its grammatical subject.

Returning to Tiriki, (26) shows a similar pattern to the idiom pattern seen in Section 3.1, this time with perceptual sources. In unraised (26a) and non-agreeing raising (26c) constructions, the subject is less clearly thematically-linked to the raising predicate.[10] In the examples below, a narrator reports variable constructions meaning something akin to *it sounds like a monster is walking in the forest.* vs. *a monster sounds like it is walking in the forest.* In English, the copy-raising

[10] We demonstrate these patterns with the class 6 non-agreeing subject marker *ka-*, although the class 9 non-agreeing subject marker *i-* behaves the same way.

construction induces an odd effect in a non-fictional narrative, as it presupposes that the monster is real and perceivable. The same effect is evident in the agreeing construction in Tiriki as well.[11]

(26) a. Ka-hulikh-a khuli li-nani li-chend-a mu-mu-rirhu.
6SM-sound.like-FV that 5-monster 5SM-walk-FV 18-3-forest
'It sounds like a monster is walking in the forest.'
Monsters are not necessarily real

b. Li-nani li- hulikh-a khuli li-chend-a mu-mu-rirhu.
5-monster 5SM-sound.like-FV that 5SM-walk-FV 18-3-forest
'A monster sounds like it is walking in the forest.'
Monsters must be real

c. Li-nani ka- hulikh-a khuli li-chend-a mu-mu-rirhu.
5-monster 6SM-sound.like-FV that 5SM-walk-FV 18-3-forest
'A monster sounds like it is walking in the forest.'
Monsters are not necessarily real

In (26a) and (26c), the "monster" does not literally have to be the perceptual source: that is to say, the monster does not have to be real in order for the sentence to be felicitous (as in English *It sounds like a monster is walking in the forest*, which makes no presupposition or claim that monsters exist). The construction in (26b), however, is only licit if the speaker is saying that monsters are real: a real monster must be doing the "sounding like." This is an instance of apparent copy-raising placing a perceptual-source requirement on its subject as is typically expected, meaning that there is a thematic relationship between *-hulikha* 'sound like' and its surface subject. In the non-agreeing raising construction in (26c) with the same predicate, it is surprising that the perceptual source requirement disappears and that again (like in the unraised construction), it is possible for this to be a way of describing what the noises in the forest sound like (without claiming or assuming that monsters exist).

We see that apparent copy-raising predicates place a perceptual source requirement on their subjects when their subject marker agrees with the subject, distinguishing them from hyperraising predicates like *-lolekha* 'seem' and *-fwaana* 'seem/appear.' But they can also appear with the non-agreeing subject markers (class 6 and 9) in apparent non-agreeing raising constructions, and in these instances the perceptual source reading disappears.

[11] *-sasa* 'seem/appear like' behaves in the exact same way.

Our interpretation of these facts is akin to what we concluded for the idiom effects: in the so-called non-agreeing constructions that appear with class 6 and class 9 subject markers, those subject markers agree with null arguments that are expletive-like. But these quasi-expletives are not nonreferential subjects: we propose that they are referential (referring to direct/indirect evidence – see Section 4.2), and when they appear, they are the thematic subject of the perception verb. This suggestion that the quasi-expletives are referential presupposes that verbs that appear in these constructions (both hyperraising predicates and apparent copy-raising predicates) do in fact have thematic subjects, assigning a "lightweight" thematic role that is something like "provides evidence."

What we suggest below is that non-agreeing raising constructions are in fact multiple subject constructions: the quasi-expletive is the thematic and grammatical subject, and the raised lexical DP is another subject that takes on the topic role that subjects normally take. We suggest that this holds for both non-agreeing hyperraising constructions like (10) and non-agreeing apparent copy-raising constructions like (26c) (i.e., there is no syntactic difference between these constructions).

4 Thematic properties of raising predicates: Evidence from quasi-expletives

In this section, we look more closely at the class 6 and class 9 subject markers that we have suggested agree with null quasi-expletives (arguments that are expletive-like in some ways but not in others).

4.1 Interpretation of agreeing vs. non-agreeing raising

There are subtle interpretive differences between the agreeing and non-agreeing variants of raising constructions themselves. Non-agreeing raising draws attention to the situation/evidence being addressed, while agreeing raising tends to be interpreted relatively neutrally. We will argue that this difference in interpretation is a result of the thematic properties of perception predicates and of the quasi-expletives that appear in non-agreeing constructions.

In (27), the context is designed to require an inference based on the available evidence from context, and non-agreeing raising is more natural than agreeing raising. Our language consultant, Kelvin Alulu, comments that (27b) "addresses the context directly," while (27a) does not.

(27) *Context: You walk into the house and find that it is in good order. You are used to finding it messy when children are around. From what you see, you can tell that the children have left, because if they were still here they would have created a mess. So the situation presents evidence that the children have left.*

a. ?Va-ana va- lolekh-a va-tsiir-e. **Agreeing Raising**
2-child 2SM-seem-FV 2SM-leave-PST
'The children seem to have left.'

b. Va-ana ka- lolekh-a va-tsiir-e. **Non-agreeing Raising**
2-child 6SM-seem-FV 2SM-leave-PST
'The children seem to have left.'

In (28), the context is designed to draw attention in a different direction, and the preferred raising construction changes as well. There is still no direct observation of the children, but the evidence is specifically of things that the children do.

(28) *Context: You walk into the house and find that it is disorganized. You know well that when children play in the house, they disorganize a lot of things. From the evidence, you can tell that the children are the ones who were here but have left.*

a. Va-ana va- lolekh-a va-tsiir-e.
2-child 2SM-seem-FV 2SM-leave-PST
'The children seem to have left.'

b. ?Va-ana ka- lolekh-a va-tsiir-e.
2-child 6SM-seem-FV 2SM-leave-PST
'The children seem to have left.'

In (28), Alulu comments that "you are relying more on what you know about the children. This sounds more natural to put more emphasis on the children. You're not so bothered with the state of the room, it's about the children's behavior. So [(28a)] is [more natural] than [(28b)]."

We therefore see that there are demonstrable empirical distinctions between agreeing and non-agreeing raising constructions that are otherwise identical (*contra* the predictions of Halpert's 2019 analysis of Zulu if it were applied directly to Tiriki). In particular, non-agreeing raising emphasizes the evidence/situation, while agreeing raising is more neutral. The reason for this difference will become clearer based on what we see in the following section.

4.2 Quasi-expletives and their interpretive properties

As we have already mentioned, Tiriki has two different non-agreeing subject markers: the class 6 subject marker (*ka-*) and the class 9 subject marker (*i-*). As reported by Diercks et al. 2022, these different subject markers prompt interpretive differences. *ka-* (class 6) is associated with indirect evidence, while *i-* (class 9) is associated with more direct evidence. These effects are also not exclusive to Tiriki: Gluckman & Bowler (2017) and Gluckman (2021) document similar effects in Logoori, Gluckman (2023) does so for Nyala East, and the second author has done so for Wanga as well (Diercks, field notes).

We have been referring to these subject markers as agreeing with null "quasi-expletives." These quasi-expletives are expletive-like in that they occur where expletives occur in many other languages (e.g., as subjects of perception verbs). They are not expletive-like in that they are clearly associated with interpretations. For example, when the context specifies that the speaker has indirect evidence of their claim, *ka-* is more natural than *i-*.

(29) *Context: You hear the children making noise as they are leaving school (but you don't see them directly).*

a. Va-ana $\boxed{\text{ka-}}$ lolekh-a khuli va-mal-i kasi y-a mu-sukulu.
2-child 6SM-seem-FV that 2SM-finish-PST 9work 9-ASSC 18-school
'The children seem to have finished their schoolwork.'

b. #Va-ana $\boxed{\text{i-}}$ lolekh-a khuli va-mal-i kasi y-a mu-sukulu.
2-child 9SM-seem-FV that 2SM-finish-PST 9work 9-ASSC 18-school
'The children seem to have finished their schoolwork.'
(Diercks et al. 2022: 151)

In contrast, the speaker's relatively direct evidence in (30) makes *i-* more natural than *ka-*.

(30) *Context: You come across the students leaving the gate of the school.*

a. #Va-ana $\boxed{\text{ka-}}$ lolekh-a khuli va-mal-i kasi y-a mu-sukulu.
2-child 6SM-seem-FV that 2SM-finish-PST 9work 9-ASSC 18-school
'The children seem to have finished their schoolwork.'

b. Va-ana $\boxed{\text{i-}}$ lolekh-a khuli va-mal-i kasi y-a mu-sukulu.
2-child 9SM-seem-FV that 2SM-finish-PST 9work 9-ASSC 18-school
'The children seem to have finished their schoolwork.'
(Diercks et al. 2022: 150)

It is worth noting that these interpretive differences are not exclusive to hyperraising: they also appear in unraised constructions and other kinds of modal constructions, with the same evidence-based interpretive effects.

We assume (as is widespread for Bantuist syntacticians) that in null subject contexts, subject markers agree with null pronominal elements (*pro*). Here, we assume that the class 6 and class 9 subject markers agree with null quasi-expletives (*pro*$_{EXPL}$) which are in fact referential DPs that refer to different kinds of evidence. On this approach, the quasi-expletives are the thematic subjects of the perception verbs they can appear with. In effect, in constructions such as (29) and (30), the evidence is doing the "seeming."

(31) a. *ka-* = agreement with *pro*$_{EXPL.INDIRECT.EVIDENCE}$
b. *i-* = agreement with *pro*$_{EXPL.DIRECT.EVIDENCE}$

4.3 Thematic quasi-expletives as perceptual sources

For some time, it appeared that copy-raising constructions were in part typified by a requirement of the so-called "copy" (a pronoun coferential with the matrix subject) in the embedded clause. Constructions lacking such a copy are often unacceptable:

(32) a. *Bill seems as if Mary is intelligent.
(Lappin 1984: (16b))
b. *Tina seems like Chris has been baking sticky buns.
(Asudeh & Toivonen 2012: (14a))

One of the observations of Landau (2009, 2011) is that the term "copy-raising" is a misnomer: so-called copy-raising constructions may optionally involve *neither* raising *nor* a copy in the embedded clause. For example, consider a situation where it is known that the speaker's mother is not a very good cook. If they see smoke in the house, they can say in both English and Tiriki:

(33) I-nzu i-manyiny-a khuli Mama a-tekh-aang-a.
9-house 9SM-look.like-FV that 1Mama 1SM-cook-HAB-FV.
'The house looks like Mama is cooking.'

In such a construction, the subject of the embedded clause is crucially not a copy: it is not coreferent with the matrix subject. Landau (2009, 2011) notes similar patterns in Hebrew and English, exemplified in (34):

(34) a. The car sounds like a trip to the mechanic is necessary.

b. The house looks like the kids had a party last night.
(Based on Landau 2011)

As Landau points out, however, there are also instances where pronominal copies in the embedded clause are obligatory. Landau (2011) proposes that in copy-raising constructions, a coreferent pronoun in the embedded clause is optional in general but is required in one specific context—in the event that the subject of the copy-raising construction is not the speaker's perceptual source for the perceptual report.

(35) *The P-source-Copy Generalization* (Landau 2011: (26))
Given a sentence "DP_i V_{perc} (to DP_j) *like* CP",
where $V_{perc} \in$ {*seem, appear, look, sound, feel, smell, taste*}:
A copy (= pronoun coindexed with DP_i) is necessary in CP iff DP_i is not a P-source.

As Landau points out, two sorts of evidence support this claim. "First, [copy-raising] examples whose subject can be construed as a P-source *allow* a copy-less complement. Second, [copy-raising] examples whose subject cannot be construed as a P-source *require* a copy in their complement." Therefore, for Hebrew and English (the main languages Landau considers), each copy-raising predicate appears in three variations of the copy-raising construction that are distinguished by the properties of the matrix subject—an expletive, a non P-source DP, or a P-source DP—which are illustrated in the examples below.

(36) a. Expletive Subject (*sound*$_1$)
Context: I read about the nutritional merits of Tibetan food, and remark:
It *sounds*$_1$ like Tibetans are healthier than us.

b. Non P-source DP Subject (*sound*$_2$)
Context: I read about the nutritional merits of tsampa, the Tibetan flour (made of roasted barley and butter tea) and remark:
Tsampa *sounds*$_2$ like Tibetans are healthier *(eating $\boxed{\text{it}}$).

c. P-source DP Subject (*sound*$_3$)
Context: My friend tells me about the nutritional merits of Tibetan food. I respond:
You *sound*$_3$ like Tibetans are healthier than us.
(Landau 2011: 41)

Landau (2011) constructs a set of paradigms to test his P-source generalization wherein the presence of a required copy in the embedded clause in copy-raising constructions is dependent on the thematic status of the matrix subject. Specifically, when the matrix subject is *not* the perceptual source of the perception that the lower clause reports, then a copy is required; we replicate his diagnostics from Hebrew and English in Tiriki here. In (37), we can see that in a context where the matrix subject is the source of the perceptual report, it is possible to have a complement clause with no overt or covert pronominal that is coreferent with the matrix clause in both Tiriki and English:

(37) *Context: Your friend describes to you how well-kept and flourishing her garden is. You remark:*

Li-vola lyolyo li -hulikh-a khuli mu-seemberi y-its-a
5-description your.5 5SM-sound.like-FV that 1-gardener 1SM-come-FV
vuri li-tukhu.
every 5-day
'Your description sounds like a gardener comes every day.'

We see a distinction, however, when the matrix subject is not the perceptual source of the report. In the context below, the speaker has received a report about the garden in question but has not directly observed the garden themselves. Therefore, even though *murimi kwokwo* 'your garden' is the subject of the copy-raising construction, it is not the perceptual source of the reported information because the speaker in this context has not directly observed the garden. In such an instance, it is impossible to have a complement clause that lacks an overt or covert pronominal element that is coreferent with the matrix clause, as (38) shows. If such a pronominal is added by including additional information in the sentence, then the sentence becomes perfectly natural (precisely as Landau's 2011 P-source generalization in (35) predicts).

(38) *Context: Your friend describes to you how well-kept and flourishing her garden is. You remark:*

Mu-rimi kwokwo ku-hulikh-a khuli mu-seemberi y-its-a
3-garden your.3 3SM-sound.like-FV that 1-gardener 1SM-come-FV
vuri li-tukhu *(khu-ku- seember-a).
every 5-day 15SM-3OM-maintain-FV
'Your garden sounds like a gardener comes every day *(to maintain it).'

It is instructive to apply this paradigm to the non-agreeing constructions. The context shown in (??) is one in which the speaker has not directly perceived the garden that is being described, but the matrix subject (*livola lyolyo* 'your description') is nonetheless a P-source. Per Landau, a construction with a P-source subject in the matrix clause does not require a pronoun copy in the lower clause. Here, we see no distinction between the agreeing (37) and non-agreeing (39) constructions. Given the "indirect evidence" interpretation of the class 6 expletive with perception verbs, this overlapping distribution is unsurprising.

(39) *Context: Your friend describes to you how well-kept and flourishing her garden is. You remark:*
Li-vola lyolyo ka- hulikh-a khuli mu-seemberi y-its-a
5-description your.5 6SM-sound.like-FV that 1-gardener 1SM-come-FV
vuri li-tukhu.
every 5-day
'Your description sounds like a gardener comes every day.'

An important contrast emerges, however, when the matrix subject is a non-P-source, as in the context in (40). We see the same fact shown above in (40b), where a pronominal copy is required in the lower clause when the matrix subject is not the perceptual source of the embedded report (precisely as predicted by Landau 2011). The curious fact that emerges is in (40c): this is the same sentence as in (40b) with the exception that it utilizes the non-agreeing class 6 subject marker. In this instance, the pronominal copy in the lower clause becomes optional again.

(40) *Context: Your friend describes to you how well-kept and flourishing her garden is. You remark:*

a. Ka-hulikh-a khuli mu-seemberi y-its-a vuri li-tukhu
6SM-sound.like-FV that 1-gardener 1SM-come-FV every 5-day
(khu-seember-a mu-rimi kwokwo).
15SM-maintain-FV 3-garden your.3
'It sounds like a gardener comes every day to maintain your garden.'

b. Mu-rimi kwokwo ku-hulikh-a khuli mu-seemberi
3-garden your.3 3SM-sound.like-FV that 1-gardener
y-its-a vuri li-tukhu *(khu- ku- seember-a).
1SM-come-FV every 5-day 15SM-3OM-maintain-FV
'Your garden sounds like a gardener comes every day *(to maintain it).'

c. Mu-rimi kwokwo ka-hulikh-a khuli mu-seemberi
3-garden your.3 6SM-sound.like-FV that 1-gardener
y-its-a vuri li-tukhu (khu- ku- seember-a).
1SM-come-FV every 5-day 15SM-3OM-maintain-FV
'Your garden sounds like a gardener comes every day (to maintain it).'

If Landau's (2011) P-source generalization is on the right track, then why is it possible for the coreferent pronominal in (40c) to be omitted, when the lexical DP matrix subject is not a P-source? We suggest that there is in fact a P-source argument of the matrix copy-raising predicate in (40c): the "non-agreeing" subject markers agree with null arguments that are expletive-like but are in fact thematic arguments of the verb and carry their own interpretations. This analysis follows the conclusions of Gluckman & Bowler (2017) for Logoori and is also consistent with Ruys' (2010) analysis of CP-linked expletives in English and Dutch. It also shares some similarities with how Greeson (2023) analyzes expletive *it* in English constructions that he argues are hyperraising constructions.

5 Interpretation of raised vs. unraised subjects

Before we present a direction of analysis, we turn to the question of what motivates hyperraising in Tiriki. This section demonstrates that there is an interpretive difference between raised and unraised subjects: raised subjects are topical. We propose that this topicality is what prompts them to raise.

5.1 Background: What canonically motivates raising?

A key proposal in analyses of raising constructions is that the argument moves to the matrix clause when the lower clause is defective or truncated in some way. For example, in English, the embedded subject arguably cannot be Case-licensed within a non-finite embedded clause and therefore must move to the subject position of the finite matrix clause to be Case-licensed, as seen in (41).

(41) Tania seems [~~Tania~~ to be happy] .

When the embedded clause is not defective (in English, when it is finite and can license nominative Case), raising out of the embedded clause is illicit.

(42) *Tania seems [(that) ~~Tania~~ is happy] .

There is arguably no motivation (for Case-licensing or otherwise) for the embedded subject to raise out of the embedded clause in situations like (42) (in addition to other barriers to such movement, such as an intervening phase).

5.2 Interpretation of non-agreeing raising in Tiriki

Given that it is possible for the embedded subject to remain in the lower clause in Tiriki, as in (43a), why does the subject move to the main clause in non-agreeing hyperraising constructions like (43b), when everything else about the construction appears identical?[12]

(43) a. Unraised
Ka-lolekh-a khuli va-ana va-tukh-i.
6SM-seem-FV that 2-child 2SM-arrive-FV
'It seems that the children arrived.'

b. Non-Agreeing Raising
Va-ana ka-lolekh-a khuli va-tukh-i.
2-child 6SM-seem-FV that 2SM-arrive-FV
'The children seem to have arrived.'
(Lit: 'The children seem that arrived.')

If there is an interpretive difference between potential hyperraising constructions based on whether the subject is in the main clause or the embedded clause, that could point towards an explanation of what motivates raising. The concern is especially salient in Tiriki (and Logoori and Zulu) hyperraising due to the presence of non-agreeing hyperraising constructions, where there is no evident morphological difference between the unraised and raised constructions (apart from the position of the subject).

In Tiriki, there is such a property: raised subjects tend to be interpreted as topical. For example, in (44), the speaker declares what they are going to talk about (*vaana* 'the children'), establishing that as the aboutness topic. In this context, raising the topical subject is more natural than leaving it in the embedded clause, even though (44a) is typically a perfectly acceptable sentence.

(44) Nd-eny-a khu-vol-er-e khu-va-ana.
1SG.SM-want-FV 15SM-say-APPL-FV 17-2-children
'I want to tell you about the children.'

a. Unraised
#? ... Ka-lolekh-a va-ana va-tsiir-e.
6SM-seem-FV 2-child 2SM-leave-PST
'It seems that the children have left.'

[12]The same question exists for agreeing hyperraising, but it is even more salient in the non-agreeing constructions where there is no obvious morphological change in the raising contexts.

b. Raised
... Va-ana ka-/va-lolekh-a va-tsiir-e.
2-child 6SM-/2SM-seem-FV 2SM-leave-PST
'The children seem to have left.'

The inverse is also true: in a context where the embedded subject is not topical, an unraised construction is preferred.

(45) *Context: You have been out in the fields all morning. You come back and the living room is empty of people, but you see a tea tray with dirty mugs and some bread and fruit along with empty plates and crumbs. You have not been expecting guests, but it appears that guests have been here and been entertained.*

a. Unraised
Ka-lolekh-a va-cheni va-tukh-i.
6SM-seem-FV 2-guest 2SM-arrive-PST
'It seems that guests have arrived.'

b. Raised
?Va-cheni ka-/va-lolekh-a va-tukh-i.
2-guest 6SM-/2SM-seem-FV 2SM-arrive-PST
'Guests seem to have arrived.'

In the words of Alulu, "[In (45a)], based on the situation of the room, you are able to tell that the guests have arrived. ... You would use [(45b)] more naturally if you were expecting the guests to arrive." Agreeing and non-agreeing raising behave identically in this respect.

In summary, in these particular configurations of hyperraising constructions, topical subjects more naturally raise to the main clause, while non-topical subjects more naturally remain in the embedded clause. We consider one way to analytically derive this pattern in Section 6.2.

6 Empirical summary and directions of analysis

There is still much to be done to document the properties of Tiriki raising constructions, so a comprehensive analysis is yet to be completed. Here, we summarize the state of our empirical conclusions about Tiriki raising and suggest the direction of analysis that we find most promising.

6.1 Properties of Tiriki hyperraising

Diercks et al. (2022) conclude that apparent hyperraising in Tiriki is true hyperraising, not copy-raising or left-dislocation of the subject. Agreeing and non-agreeing hyperraising constructions involve movement, per connectivity effects of idioms and non-perceptual source readings (Diercks et al. 2022). Both agreeing and non-agreeing hyperraising constructions are A-movement to canonical subject position (Diercks et al. 2022).

(46) **Empirical contributions from Diercks et al. (2022)**

- **Not left-dislocation**: Hyperraising is possible in contexts where left-dislocation is not, suggesting that apparent hyperraising constructions are not in fact left-dislocation constructions.
- **A-movement**: Hyperraising behaves like A-movement to canonical subject position, rather than like Ā-movement.
- **Connectivity effects**: Idiomatic readings are maintained in hyperraising constructions, but they are not retained in left-dislocation or control constructions. (Other connectivity effects hold as well.)

(47) **Empirical contributions from this paper**

- There is no uniform distinction between hyperraising predicates and copy-raising predicates in Tikiri. Some predicates lack connectivity effects in their agreeing versions but maintain connectivity effects constructions with non-agreeing subject markers (class 6/9) (Section 4).
- Non-agreeing raising draws attention to circumstances and evidence in a way that agreeing raising does not (Section 4.1).
- Class 6 and 9 subject markers (when not agreeing with the overt subject) agree with null quasi-expletives, which have evidential interpretations (Section 4.2) and are thematic arguments of perception predicates.
- Raised subjects are topical, while unraised subjects are not (Section 5).

6.2 Direction of analysis

In this section, we sketch a direction of analysis of Tiriki hyperraising that incorporates these conclusions. We begin by focusing on hyperraising predicates, and in the end of this section we extend the same analysis to the apparent copy-raising predicates that we have discussed throughout the paper, unifying all of these predicates as hyperraising predicates. This analysis is not exhaustive in either details or empirical defense, but it is a way of conceptualizing hyperraising constructions that is consistent with the evidence presented here. The intuition that we pursue here is that subjecthood in Tiriki is a composite of three syntactic functions that are all represented grammatically: topicality, grammatical subjecthood (marked by agreement on the verb), and thematic subjecthood. Perceptual verb constructions make available the possibility that these three functions can be shared between multiple subjects.

First, we propose that all raising verbs have thematic subjects—either an overt lexical subject or a null quasi-expletive. The thematic roles assigned to these subjects are all some variant of EVIDENCE (i.e., the thematic role from the predicate specifies that the referent of the argument provides evidence relevant to the perception). Quasi-expletives differ in the kinds of evidence that they denote; class 6 quasi-expletives denote indirect evidence, while class 9 quasi-expletives denote direct evidence. And likewise, predicates can differ with respect to how robust of a perceptual requirement they place on their subjects, which we comment on below. Second, we suggest that a topic feature on T°, [TOP], drives movement to the matrix clause (see Grishin 2018 and Greeson 2023 for similar proposals).[13,14]

Agreeing raising works as follows: the embedded subject raises to matrix Spec,*v*P and is assigned a theta role by the matrix predicate (EVIDENCE).[15] The subject then raises further, to matrix Spec,TP, to satisfy the topic feature on matrix T°. We assume an interaction/satisfaction model of Agree for reasons that will become apparent in the following analysis of non-agreeing raising (Deal

[13]We also assume there is an EPP quality associated with T°: some phrase must raise to Spec,TP.

[14]It is worth noting that information structure features in the syntax are not uncontroversial: there is a rich literature arguing against information structure in syntax (Fanselow 2006, Neeleman & van de Koot 2008, Horvath 2010). This is in no small part due to Minimalist assumptions that narrow syntax acts on lexical items to produce sentences; information structure features are plausibly *not* assigned to lexical items in the lexicon, leaving open the question of how they would end up in the syntax. But this state of affairs is paired with an equally rich literature arguing *for* the presence of information structure features in the narrow syntax (Rizzi 1997, Kallulli 2008, Aboh 2010, Kratzer & Selkirk 2020). These references barely scratch the surface, and the empirically-focused nature of this paper precludes a proper theoretical exploration of the point. We do assume that information structure features are assigned to lexical items sometime *after* the lexicon; for present purposes, it is sufficient to assume that information structure features are assigned to lexical items in the Numeration.

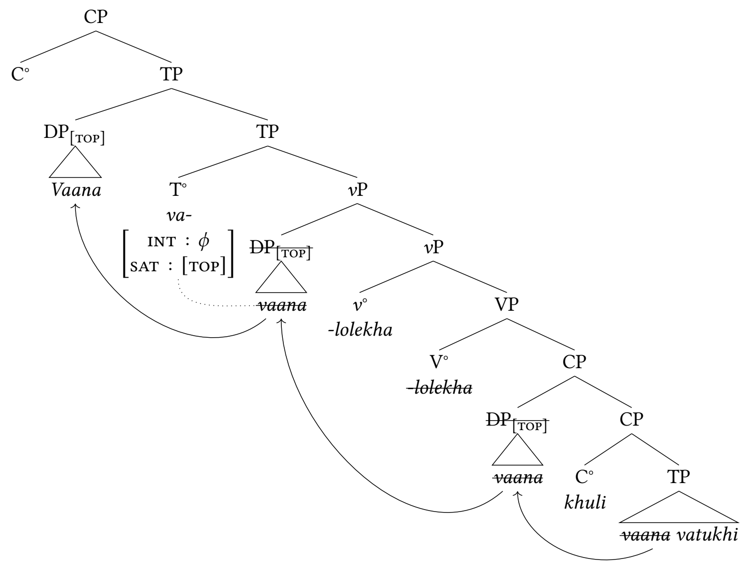
[15]This analysis allows a single argument to be assigned multiple theta roles, as in the Movement Theory of Control (Boeckx et al. 2010).

2015).[16]

(48) Agreeing hyperraising

a. Va-ana va-lolekh-a khuli va-tukh-i.
   2-child 2SM-seem-FV that 2SM-arrive-FV
   'The children seem to have arrived.'
   (Adapted from Diercks et al. 2022: (98))

b.



Non-agreeing raising introduces an additional complication.[17] A null quasi-expletive is merged in matrix Spec,vP and receives a theta role from the matrix predicate. When T° probes, the probe first Agrees with the expletive in Spec,vP and raises it to Spec,TP. When the quasi-expletive is not topical, the probe's satisfaction condition, [TOP]. has not been met.[18] So the probe continues searching

[16]For the purposes of this paper, we set aside the question of how DPs can A-move out of phases. We follow Fong (2018, 2019) in assuming successive-cyclic A-movement through CP is possible. There is much more to be said on this topic, but we must leave the question for future research.

[17]While space precludes appropriate discussion, Halpert's (2019) analysis of similar constructions in Zulu as agreement with the embedded CP cannot be extended to Tiriki; a variety of the Tiriki facts are incompatible with her approach.

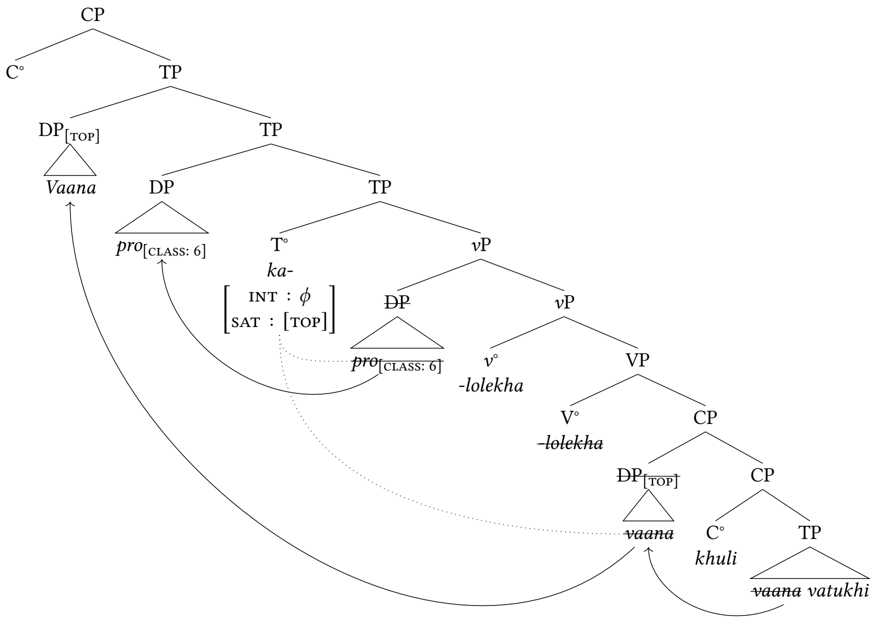
[18]Non-raising constructions may be derived when the quasi-expletive is itself topical, or it may be the case that quasi-expletives cannot be topical (similar to how non-specific DPs are generally unable to function as sentence topics – see Erteschik-Shir 2007) and that non-raising constructions thus result from either a different probe on T° that does not probe for [TOP] or from matrix T°'s probe remaining unsatisfied for [TOP].

until it finds the topical DP in embedded Spec,CP and raises that DP to matrix Spec,TP.[19]

(49) Non-agreeing Hyperraising

a. Va-ana ka-lolekh-a khuli va-tukh-i.
   2-child 6SM-seem-FV that 2SM-arrive-FV
   'The children seem to have arrived.'
   (Adapted from Diercks et al. 2022: (98))

b.



The result, then, is that non-agreeing raising constructions are multiple subject constructions. The thematic and grammatical subject of the matrix clause is the quasi-expletive, and the topic-subject is the DP that has raised from the embedded subject position of the complement clause.

On this approach, there is no structural distinction between hyperraising and copy-raising in Tiriki. Both are constructions that raise embedded subjects to a

[19]This analysis assumes multiple specifiers of TP.

matrix subject position, similar to control constructions (on the Movement Theory of Control: Boeckx et al. 2010).[20] Our suggestion is that the distinct empirical behaviors between them are based on the relative narrowness of the entailments imposed by the perception predicate on the referent of its subject. Hyperraising predicates such as *-lolekha* 'seem' convey something like "evidence exists," which is fairly readily compatible with any raised argument. In contrast, an apparent copy-raising predicate such as *-hulikha* 'sound like' places more specific entailments on the kind of evidence offered by the referent of its thematic subject (specifically, the source of evidence must be sound-related). In general, therefore, connectivity effects more readily arise in contexts where there are not strict requirements placed by the matrix thematic predicate (*seems*/*appears*-type verbs), because there are fewer potential incompatibilities between the raised argument and the thematic role of the matrix predicate. (This is the same approach to explaining distinctions between raising and control on the Movement Theory of Control: Boeckx et al. 2010.)

It is worth noting that the approach sketched here has a lot in common with the proposal set forward by Greeson (2023) for a newly documented English construction that parallels what we have proposed for Tiriki. Greeson argues that constructions such as (50) are in fact hyperraising constructions akin to what we have suggested for Tiriki: multiple subject constructions where the expletive is a thematic subject of the matrix predicate and the DP subject has raised from the embedded clause to another subject position in the matrix clause.

(50) a. Mei$_k$ it seems t_k is happy.
b. The kids$_k$ it seems all t_k know what they're doing.
(Greeson 2023)

Greeson (2023) suggests various possible analytical approaches to these constructions, including a topic-driven analysis of the moved DP like we have suggested for Tiriki. Further work on English and Tiriki (and related Bantu languages with similar patterns, for example Zulu, Lubukusu, Wanga, and Logoori) is necessary in order to evaluate the empirical similarities and differences, but these parallels are intriguing.

[20] By "matrix subject position," we mean a grammatical, thematic, and topical subject position in the agreeing raising constructions, or only a topical subject position in non-agreeing raising constructions.

6.3 Analytical/theoretical conclusions

We have shown that there are fine-grained interpretive differences between agreeing and non-agreeing raising (and unraised constructions) in Tiriki, especially when predicates with different thematic properties are considered. Raised subjects are topical, whereas their parallels in non-raising constructions are not. We also showed that there is a difference in interpretation between agreeing raising constructions and non-agreeing raising constructions: non-agreeing raising constructions induce a reading where some salient evidence (rather than the referent of the raised subject itself) leads to the conclusion, whereas agreeing raising constructions are more neutral. This pattern accords with the interpretations of quasi-expletives more generally, which have to do with (in)directness of evidence. Finally, we showed that there appears to be a thematic relationship between raising predicates and their subjects by showing that even apparent copy-raising constructions have non-agreeing variants that show the same connectivity effects that typical hyperraising predicates do. In short, we have suggested that less distinguishes hyperraising and copy-raising than is typically thought. All of these predicates assign thematic roles to their subjects, and these thematic roles have to do with the evidence being perceived. However, there is variation in the degree of restriction imposed by the perception predicate: underspecified (hyperraising) or more highly specified (copy-raising). This analysis contributes to the unification of raising and control predicates under the Movement Theory of Control (Boeckx et al. 2010).

Concerning perception predicates in particular, this helps to explain how some predicates can behave like canonical copy-raising predicates when they agree with the lexical DP subject (lacking connectivity effects, requiring perceptual source readings of their subject) but can behave the opposite way when appearing with class 6 and class 9 non-agreeing subject markers. We suggest that canonical raising predicates (like *seems*) assign thematic roles that are bleached of most semantic entailments, so these predicates give the appearance of lacking thematic roles for their subjects. The traditional differences between movement and non-movement constructions (raising vs. control/copy-raising) could be explained based on the properties of these thematic roles. That is, the lack of connectivity effects in control and so-called copy-raising constructions can be explained by the relative incompatibility of the interpretation of the matrix thematic role with the interpretation of the raised subject (whether it is non-referential as part of an idiom or is a referential subject that is inconsistent with the selectional restrictions of the matrix verb). But if that is the case, we expect variability between predicates based on nuances of their selectional restrictions.

This prediction is borne out, as seen in the differences between the predicates that behave like canonical hyperraising predicates (e.g., *-lolekha*) and those that behave like canonical copy-raising predicates (e.g., *-hulikha*). Of course, more work will be needed to further explore and formalize this analysis.

Abbreviations

APPL = applicative, FV = final vowel, HAB = habitual, OM = object marker, PST = past, SG = singular, SM = subject marker

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Chapter 19

A speech corpus of Kom verbal arts and its applications

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The move towards large-scale analysis of corpora in phonetics and phonology requires new approaches to data collection, both for African languages specifically and for less-described languages more generally. We argue that short-form verbal arts, such as proverbs and riddles, offer a convenient and culturally sensitive way to build out suitable language resources in the African context, particularly for less-documented languages. As a proof of concept that use of the verbal arts to construct corpora is an effective strategy, we describe the construction of a corpus of Kom speech (Grassfields Bantu, Cameroon) from more than 150 proverbs (*ngàynsɨ*) and riddles (*mitɨtɨ̂*). We illustrate the potential uses of the corpus with two case studies: vowel production under positional prominence and a variably applied phonological process of monophthongization which is sensitive to upcoming segmental content. Throughout, we consider the affordances of proverbs and other short verbal arts for corpus development, how these relate to the typical materials for read-speech corpora, and the potential role of these corpora in the development of language technology.

1 Introduction

1.1 Corpus approaches and Africanist linguistics

African languages have long been influential in the fields of phonetics and phonology, the selection of their objects of documentary and analytical interest, and the development of their theoretical machinery (Ladefoged 1968, Goldsmith

Matthew Faytak, Ivo Forghema Njuasi, Nicholas Mori & Angelique Griffith. 2026. A speech corpus of Kom verbal arts and its applications. In Vicki Carstens, Katherine Russell, Olawale Akingbade, Deborah Morton & Michael Diercks (eds.), *Pamoja tena 'Together again': African linguistics after COVID*, 443–467. Berlin: Language Science Press. DOI: ??

1976, Olson & Hajek 1999, Casali 2008, Lionnet & Hyman 2018). In current phonetics and phonology, research increasingly relies on the collection and analysis of large, annotated, machine-readable corpora of speech, with added benefits accruing in the form of a much larger number of observations, a deeper consideration of variability, and improved naturalness over elicitation (Liberman 2019). Most African languages, however, lack the resources needed for researchers to carry out this corpus-based work.

This leads to an impasse: large-scale corpus-based research is in many ways an important part of the future of phonetics and phonology, but a shortfall in the amount, quality, and type of data available threatens to leave African languages out of this development entirely. Text- and transcription-based corpora of African languages *are* already constructed to a high standard (e.g. Prinsloo & de Schryver 2001, De Schryver & Nabirye 2018), but are not suitable for carrying out phonetic analysis without time-alignment of the transcriptions. Few existing data collections – not only for African languages, but for less-documented languages in general – have data voluminous, well-annotated, and well-recorded enough to be useful for large-scale instrumental phonetic analysis. In a recent survey of 20 corpora deposited in various archives, more than half were found to be unsuitable for automated, large-scale phonetic work due to incomplete or inconsistent metadata, missing annotations, and other structural impediments to large-scale processing (Babinski et al. 2022, Yi et al. 2022).

As annotated, machine-readable corpora of African languages are in short supply, particularly for the lowest-resourced African languages and those which lack writing systems, the construction of such resources should constitute a long-term priority for the field of linguistics. In this article, we suggest some ways in which interested researchers can advance towards this goal by leveraging proverbs and other short-form verbal arts. These are held in common by many African speech communities; are readily accessible to both community members and linguists; and allow for the production of culturally appropriate and easily processed materials that spark community interest.

1.2 Motivating the current work: a double function of verbal arts

In this article, our focus is on *read-speech* corpora in which a fixed set of materials, usually a written set of sentences, is read by participants. Corpora of verbal arts need not be produced from writing, and so the label of *read* speech does not entirely suit the materials discussed here. They might better be called *controlled* speech to distinguish them from *spontaneous* speech, in which speakers' utterances make no attempt to follow a conventionalized form (Simpson 2013), without making reference to literacy as a mediating factor. The predictability of controlled speech greatly facilitates transcription and annotation of the recorded speech material.

Spontaneous speech, by contrast with controlled speech, must be manually transcribed and parsed after the fact, and the occurrence of particular words, phones, or sequences of items is not guaranteed (see Xu 2010). This poses a major impediment to primary data collection work sometimes known as the *transcription bottleneck* (Seifart et al. 2018). In part because of their relative ease of processing, corpora of read or controlled speech have a long history in acoustic and articulatory phonetics (Zue et al. 1990, Garofolo et al. 1993, Wrench 1999), and are frequently used as training data for natural language processing applications (e.g. Black 2019, Sikasote & Anastasopoulos 2021, van Niekerk et al. 2017). In addition to speeding up annotation, selection of certain predictable material also allows researchers to approximate phonetic balance, i.e. ensure similarly frequent occurrence of all phones or phone sequences of a given length.

We wish to make the case that proverbs and other short-form verbal arts are an especially suitable material for constructing audio corpora of spoken African languages. Although verbal arts are not typically performed from writing, they overlap in form with typical read-speech corpus materials (a series of controlled sentences) and do double duty as a repository of cultural information that generates community interest. Because they are based in oral tradition, they are accessible to literate and illiterate speakers alike. Proverbs take relatively predictable forms: they are accessible common knowledge, and as such are expected to be produced in a relatively conventionalized form, facilitating transcription and parsing. Furthermore, proverbs are a common target for anthropological, sociological, philological, and linguistic research (Yankah 1989, Nwachukwu-Agbada 2012, Tadi 2016), and published collections of proverbs for particular languages often exist which may be leveraged to rapidly elicit and parse a large amount of speech material.

These resources are also suitable for future development of speech technology. The languages of Africa are dramatically under-served by existing speech technologies and data resources (Blasi et al. 2021), in spite of ongoing efforts to expand access (e.g. Orife et al. 2020). Most of the training data for novel speech technology applications takes the form of read-speech corpora (van Niekerk et al. 2017, Sikasote & Anastasopoulos 2021, Gauthier et al. 2016, Schlippe et al. 2012, Abdillahi 2014, Gutkin et al. 2020), but these corpora typically use collections of neutral, randomly chosen sentences that are disconnected from any documentary function or community interest. Short-form verbal arts provide material which has greater relevance to community documentary priorities; taking this approach also does not require that all participants in the corpus creation process are literate.

In this paper, we put forward a corpus composed of proverbs and riddles in Kom, collected over a single field season in 2022, as a proof of concept for a corpus of verbal arts which can both support scientific and technological work and document a community's cultural heritage. The remainder of this paper is structured as follows: an overview of Kom, Kom verbal arts, and their cultural significance follows in Section 2. The construction of a spoken verbal arts corpus is described in Section 3, and two exploratory case studies of how such a corpus can be used for phonetic work are included in Section 4. Section 5 concludes with some discussion of ethical considerations and future developments planned for the corpus.

2 Kom verbal arts

2.1 Kom: background

Kom (*itaŋikom*) is a Grassfields Bantu language spoken by roughly 300,000 people in the North West region of Cameroon (Eberhard et al. 2023, Hammarström et al. 2023), along with a sizeable Kom-speaking diaspora elsewhere in Cameroon and in Europe and North America. It is spoken by the Kom people, who are mainly agriculturalists living on hilly terrain, with systems of matrilineal and patrilineal descent and a centralized political structure under a paramount chief (*fòyn*).

While some phonological, morphological (Hyman 1980, 2005, Kießling 2010, Akumbu 2019, Njuasi 2022), and morphosyntactic (Chia 1976, Shultz 1997, Kawuldim 2014) work has been carried out on Kom, no corpus of the spoken language has been collected or maintained. However, in addition to a New Testament translation and a sizeable lexicon (Jones 2001) in a standard orthography, a collection of proverbs (Loh 1997) has been published by SIL Cameroon, and a number of small press books make use of written Kom.

2.2 Genres of Kom verbal art

The two major genres of verbal arts in Kom focused on in this paper are *ngàynsɨ* 'proverbs' and *mɨtɨtɨ̂* 'riddles' (singular: *ngàyn*, *fɨtɨtɨ̂*, respectively). These are both typically short utterances, with riddles additionally paired with one or more accepted answers. The format and social function of these genres is reviewed below.

Proverbs or *ngàynsɨ* are short utterances usually mostly phrased as indirect advice, admonishments, or lessons. Proverbs are characteristic of the speech of older adults; even among adults, usage of proverbs increases with age and the attendant social responsibilities. The language used tends toward the highly metaphorical, which enhances the perception that proverbs are linked to wisdom and places them in an elevated register compared to everyday speech.

Riddles or *mɨtɨtî* have two components: first, a short overt question to be answered or a cryptic description of an object or concept; second, one or more expected responses or answers. Riddles may be used with the goal of teaching reasoning to the young, and they occasionally serve a similar didactic function as proverbs; elders may tell younger relatives riddles and stories aimed at educating them on good morals, critical thinking, problem-solving, and resilience. If shared between age-mates or siblings, riddles may simply have entertainment as their function. Traditionally, riddles were most often exchanged in the evening, before longer stories, as an activity before sleeping (Jick & Ngam 2016).

If the audience cannot guess the answer to a riddle, the poser of the riddle has the right to earn a royal title from the audience as "currency" in exchange for the answer. This may come in the form of a request to be addressed by a title such as *mbæ̀*, which is normally reserved as a polite address for chiefs. The riddler may also demand compensation in the form of the names and titles of chiefs or subchiefs of chiefdoms in or around Kom, typically with the request *fu fòyn* 'give a chief'; the riddler may reject offers deemed not prominent enough for the answer to the riddle. Conversely, if a riddle is guessed by a member of the audience, the winner may demand a royal title from the riddler.

The Kom verbal arts encompass far more than the short utterances typical of *ngàynsɨ* and *mɨtɨtî*: Jick & Ngam (2016) make mention of folk tales and explanatory historical texts as two important long-format genres which we do not consider here. Musical verbal art forms such as songs are also not considered here due to the very different nature of the task and the interference of the musical accompaniment in addressing linguistic questions. We do not deny the validity of these genres as topics of study, but rather focus on spoken proverbs and riddles as more suitable for the construction of corpora for research into phonetics and phonology.

Kom *ngàynsɨ* and *mɨtɨtî* demonstrate two key affordances for linguists and community members: they are repositories of cultural information *and* sources of compact, easily parsed linguistic material. Proverbs and riddles compare well in some respects to the materials used in read-speech corpora: they are recalled in a fairly consistent form across speakers, and may already have a written form, although speakers typically do not need to read a proverb or riddle to recall it.

They differ from typical read-speech corpus materials in other respects. They are in an elevated register and formal in tone: because of their didactic function and association with the wisdom of elders, younger speakers may also produce proverbs in a less natural way than would be desirable. By contrast, read-speech corpora are typically designed to be neutral in content. Because true neutrality is actually quite difficult to achieve in corpus materials, however (Tripp & Munson 2021), we view this as a concern which is not particular to verbal arts.

2.3 Cultural significance

Language and culture are inextricably linked, and the connection is emphasized by the "double function" of the corpus materials as sources of both linguistic and cultural information. We explore the latter here as manifested in some of the proverbs which make up the corpus. For instance, the topography and natural physical features that characterize Kom land are reflected in many sayings. The proverbs in (1) use a shared understanding of the natural environment of Kom, which is characterized by hills, cliffs, rivers, and streams, toward their didactic ends. In (1a), the well-known stream Ngeyngeyn, which meanders as it travels downhill, stands in for an aimless person in need of direction; a cliff represents a dangerous situation in (1b); and the streams and dry hills in (1c) are used to illustrate a request made from a position of abundance to one of deprivation.

(1) a. Ngeynɨ̂ ta Ngeyngeyn ngeynɨ Ijɨ̀m.
twist.PROG like Ngeyngeyn twist.PROG Ijɨm
'Twisting like Ngeyngeyn twists at Ijɨ̀m.'

b. Wù bœ̀ a mɨ̂wùm mɨ̀ bo.
3SG.SUBJ escape PREP edge ASSOC cliff
'He has escaped at the edge of a cliff.'

c. Ànkem a kèyn fvɨ̀ a jvâ na loyn mu sɨ̂
crab ASSOC this come.out PREP river PROG beg water PREP
fɨ̀siŋgè fɨ a kfɨyn.
beetle ASSOC PREP hill
'This crab is coming out of the river, begging water from the beetle on the hill.'

Likewise, in the riddles shown in (2), the guesser may be guided to a correct answer by their experience of the land: in (2a), a small cave or spring on the side of a hill is the ear on the head, and in (2b), the eyes are cued by the sensory experience of gazing across a valley.

(2) a. Q: Fɨjela fɨ ibɨ i kfɨyn. A: Atuŋlɨ.
landmark ASSOC hole ASSOC hill ear
Q: 'A landmark of a hole on a hill.' A: 'An ear.'

b. Q: Dyèyn n-dyɨfa. A: Isɨ.
distance N.be.far eyes
Q: 'The distance is far.' A: 'The eyes.'

The built environment is likewise reflected in proverbs and riddles such as those in (3). The references to the use of shelves below the thatch ceiling to store food in (3a), the back-and-forth movement of the door latch in (3b), and the height of the doorpost in (3c) are accessible only with knowledge of how traditional Kom houses are constructed.

(3) a. Fɨ lu a ngwo ɨ chem a lumi.
3SG.SUBJ go PREP ceiling CONSEC fall PREP pot
'A bit has come out of the ceiling and fallen into the pot.'

b. Q: Afo ki a kɨ chwò ko', kɨ chwò kàlì. A: Ifal
thing REL 3SG.SUBJ only pass go.up only pass go.down ?
i ndo.
ASSOC house
Q: 'The thing that goes up, (and) comes down.' A: 'The traditional door.'

c. Q: Afo ki a kɨ cho' afol atu nɨ foyn. A: Fɨmbe'sɨ.
thing REL 3SG.SUBJ only knock.off cap head POSS chief doorpost
Q: 'The only thing that knocks the cap from the chief's head.' A: 'The doorpost'

Proverbs and riddles also reveal some Kom values and beliefs (see also Tosam 2014, Njwe 2015). The saying in (4a) is used to admonish people who, on failing to support a needy person materially, cannot even offer sympathetic words. It is a common practice among the Kom people to offer gifts to people who have suffered misfortune; it is expected that those who lack material things should at least visit with the person. The proverb in (4b) shows the typical view that if you cannot succeed at achieving something, you should not impede those who can.

(4) a. Wà wò afo ɨ̀ wò gɨa.
2SG.SUBJ lack thing CONSEC lack voice
'You lack a thing and also lack a voice.'

b. Wà wo àtem ka wa n-fsɨ̀ jva.
2sg.subj lack calabash proh 2sg.subj n-foul river
'If you lack a calabash, don't dirty the stream.'

The Kom proverbium also reveals the typical belief in magic spells, divine justice, supernatural medicine, and the self-righting nature of society and the world. The saying in (5a) echoes the belief in magic spells invoked to punish wrongdoers with illness; an innocent person who has not profited from wrong actions (the "empty belly") is unaffected by such magic. The saying in (5b) echoes the Kom belief in retribution: if you taunt someone because of a condition they suffer from, you are sure to be struck by something worse. The belief in powers held by diviners or oracles is seen in the saying in (5c), which acknowledges the power of a fortuneteller to direct actions towards health and prosperity.

(5) a. Chɨ̀sɨ nɨ n-ku wi ɨlvâ ɨ yûm.
spells prog n-catch neg belly assoc empty
'Spells don't catch an empty belly.'

b. Wà chye' àdzi wa li àntàŋ.
2sg.subj laugh sore 2sg.subj take smallpox
'If you laugh at scabies, (may) you take smallpox.'

c. À n-bì ndo ndà wu fi se'
impers n-be.fortunate house who 3sg.subj do.again set.out.for
ifɨeŋ i fɨnya' a?
divining assoc god q
'Whose house was fortunate and he went out again to consult a diviner?'

3 Corpus creation

3.1 Data collection

Recruitment of speakers was carried out in consultation with the local chapter of KOMCUDA (Kom Cultural and Development Organization) in Douala. Knowledgeable speakers were selected from KOMCUDA membership and invited to contact the investigators if they wished to participate in recording. No speakers nominated in this way had self-reported speech or hearing disorders, and all reported being frequent users of Kom in their daily lives.[1]

[1]Although located very distantly from the Kom homelands, participants in community meetings such as KOMCUDA are active and regular users of Kom in the home and in numerous

Speakers were recorded in the same quiet room in a suburb of Douala, typically in pairs but sometimes alone, with Shure SM10A head-mounted cardioid condenser microphones for each speaker. Audio was recorded onto a Zoom H4N portable solid-state recorder, using separate channels to record pairs of speakers. The room was not sound-treated, but due to the microphones used, very little reverberation and background noise can be perceived in the recordings, and the signal-to-noise ratio of recordings is consistently high.

Proverbs were generally elicited from one of two published sources. The vast majority, in excess of 100, were elicited using a compendium of proverbs published by SIL Cameroon (Loh 1997). Several more pertaining to cultural practices of the chief and traditional council were elicited from Lo-ah (2018). Although the source materials were written, only two participants were literate in Kom, and as such the proverbs were prompted verbally in Kom or English translation by the first author from the written materials. Speakers then produced them in the specific formulation they were familiar with, which often differed from the source materials and from other speakers' formulations. Speakers often spontaneously volunteered conceptually related proverbs, all of which were included in the corpus; many of these are attested in writing for the first time in our corpus, though some can be found in other published work on Kom proverbs (Njwe 2015, Jick & Ngam 2016, Tosam 2014, Chenwi & Che 2022). Riddles were elicited entirely from speaker memory, since no published work on Kom riddles exists except for the notes in Jick & Ngam (2016).

After being prompted to produce a saying, speakers were asked to repeat the utterance five times to increase the overall amount of material. This is fairly standard practice in fieldwork situations in which phonetic material is collected (Ladefoged 2003). Occasionally, repetition yielded variations in the utterance's form; further variation emerged for proverbs spontaneously volunteered by multiple speakers, who often recalled relatively verbose or shortened versions of the same saying, shown in (6) below for one very commonly volunteered proverb. We included all such variants in the corpus under the shared lemma of the longest version of the proverb; including the shorter variants provides data on conventionalization of fixed phrases in the language which otherwise tends not to be reflected in formalized collections of proverbs. Occasionally, multiple answers to riddles were provided, which were all recorded for similar reasons.

(6) a. Tɨm avɨ, kfɨ kɨ iti.
shoot foot die only stone

interactions outside the home, including the KOMCUDA meetings themselves; speakers thus easily maintain high fluency in the language.

b. Wa tɨm avɨ, a kfɨ kɨ iti.
2sg.subj shoot foot res die only stone

c. Wa tɨm avɨ a kɨa, a kfɨ kɨ iti.
2sg.subj shoot foot assoc 2sg.poss res die only stone

d. Wa tɨm avɨ a kɨa a ngo', a kfɨ kɨ iti.
2sg.subj shoot foot assoc 2sg.poss prep rock res die only stone
'If you stub your foot on a rock, may only the stone break.'

3.2 Corpus characteristics

The corpus contains contributions from 17 speakers (7F, 10M) producing roughly 133 minutes (2.23 hours) of labeled Kom speech. A total of 30,269 word tokens were produced across all speakers, representing 772 unique word types. These were produced in 4,305 utterances (i.e. continuous spans of diarized speech). A total of 146 proverbs and 14 riddles were elicited, with 39 proverbs and all riddles volunteered by speakers from memory and the remainder drawn from published sources (Loh 1997, Lo-ah 2018). Some speakers contributed more utterances than others, and some speakers uttered a wider variety of sayings than others. The resulting corpus contains 81,917 phone instances covering the entire phonemic inventory of the language in various prosodic positions, though certain phonemes are markedly more or less frequent than the average. This appears to be mediated by word frequency; for instance, the front vowels /e ɛ y ø/ and the "fricative vowel" /z̩/ occurred four times less frequently than the most common vowels, /a u/, apparently driven by the frequent appearance of the latter set in high-frequency function words, e.g. *wà* '2sg.subj'; *a* 'in, at'; *wùl* 'person, someone'; *bu* 'neg.perf'.[2]

The Kom corpus falls within the lower end of the range of speakers, word tokens, and duration of speech material attested in existing annotated audio corpora of African languages (e.g. van Niekerk et al. 2017, Sikasote & Anastasopoulos 2021, Gauthier et al. 2016, Schlippe et al. 2012, Abdillahi 2014, Gutkin et al. 2020). The specific corpora cited here were collected as training data by practitioners of natural language processing (NLP) and vary widely in their particulars; speaker counts range from a minimum of five to a maximum of 102, the number of utterances ranges from 1,300 to more than 20,000, and duration of target

[2]Archaic expressions which might be expected in proverbs did not appear to drive any especially high word or phone frequencies. However, a fuller comparison to a spontaneous or conversational speech data set (which does not yet exist) would be necessary to confirm the presence or absence of such an effect.

language speech ranges from 80 minutes to 24 hours. The Kom corpus is also somewhat smaller, in terms of word tokens, than typical linguist-produced text corpora of African languages (e.g. Prinsloo & de Schryver 2001, De Schryver & Nabirye 2018). However, the number of speakers and amount of material clearly exceed an acceptable minimum for research applications in both linguistics and NLP: it contains more material and speakers than several of the NLP-related corpora (van Niekerk et al. 2017), and more material compared to some read-speech corpora traditionally used in phonetics, such as the original TIMIT corpus (Zue et al. 1990, Garofolo et al. 1993).

3.3 Processing and semi-automatic annotation

Transcripts of recorded material were written and diarized in Praat TextGrids (Boersma & Weenink 2023) with the help of the third and fourth authors, working from the published text for each utterance and referring to the first author's field notes when no published version existed. Modifications to written forms in published sources were often necessary to reflect the actual utterances produced, since speakers typically recalled a very similar proverb but varied in the precise way it was recited. For instance, tense-aspect or person reference often differed from the published version, mainly through presence or absence of verbal prenasalization *n-*[3] and use of second-person subject *wà* versus impersonal *ghɨ*. Presence or absence of sub-constituents of complex phrases, including occasional *pro* drop, also exhibited inter-speaker and inter-token variability. An example of a time-aligned transcript for a pair of speakers is shown in Figure 1.

Orthographic forms were standardized to eliminate ambiguities, mainly relating to whether *y* represented the semivowels [ɨ̯] or [i̯]. These standardized transcripts were used to produce word- and phone-level alignments to the audio using the Montreal Forced Aligner v3.0.0 (McAuliffe et al. 2017). Forced alignment aligns an expected string of words and segmental phones to an audio signal, providing the landmarks necessary for large-scale automatic analysis. A sample word- and phone-level alignment on the present data is shown in Figure 2. The process of forced alignment requires a pronunciation dictionary that maps between the orthographic form and the string of phones to be aligned to that word's portion of the sound file. An example of the pronunciation dictionary for this corpus is given in Figure 3 below. Because lexical tone is not fully marked in the official orthography (see Bernard et al. 2002), it is omitted from the pronunciation

[3] This verbal prefix is analyzed in Shultz (1997) as a progressive aspect marker. In our own corpus, it is not always clear that preverbal *n-* should receive this reading, though it is clear that its presence or absence affects tense-aspect semantics.

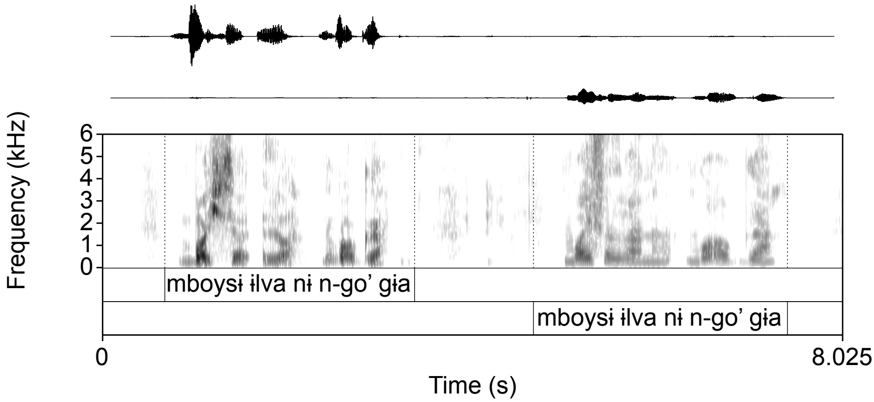

Figure 1: A sample diarized transcript of the proverb *Mboysɨ ɨlva nɨ n-go’ gɨa* ‘Peace in the belly is only a voice’ as uttered by two speakers in a dyadic recording.

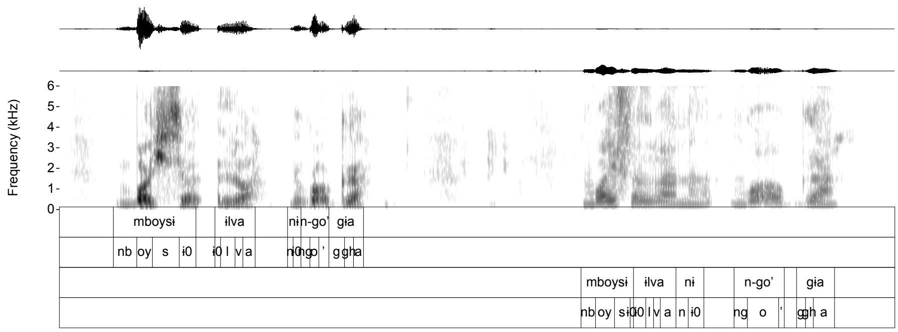

Figure 2: A sample word-level alignment (first and third tiers counting from the top) and phone-level alignment (second and fourth tiers counting from the top) for the recording shown in Figure 1.

dictionary. This omission is not expected to have a substantial negative effect on the quality of segmental alignment (Yuan et al. 2014), but it precludes research on tone using the corpus, which is an acknowledged shortcoming of the present materials. We stress that future versions of the corpus annotations will include a full tonal representation for the segments to be aligned. Other suprasegmentals *are* included in the pronunciation dictionary: stem prominence was marked by

```
n-cha'    nch a '
fɨnchɨa   f ɨ0 nch gh a
iŋwa'i    i0 ŋ w a ' i0
kɨjem     k ɨ0 j e m
mɨchi     m ɨ0 ch i
abɨl      a0 b ɨ l
akɨa      a0 k gh a
kia       k i ae
nyamsɨ    ny a m s ɨ0
ghayn     gh ay n
n-ko'     nk o '
ɨyi       ɨ0 y i
ngaŋ      ng a ŋ
asɨ       a0 s ɨ
iyvɨ      i0 y vɨ
```

Figure 3: A sample of the Kom pronunciation dictionary used for forced alignment. Orthographic forms used for transcription at left; phones to be aligned at right.

adding a ‘0’ to vowel phones occurring outside of stems (see Section 4.1 for an at-length discussion).

4 Case studies

In this section, we provide a phonetics case study (Section 4.1) and a phonology case study (Section 4.2) which may be carried out on this corpus in spite of its relatively small size and lack of selection for phonetic balance. Probabilistic information about frequencies and distributions of words and phones can also be collected through sufficiently large corpora of connected speech, such as the present corpus. We do not present frequency and distributional data from the Kom corpus here for reasons of topicality and space, rather than a lack of feasibility: more typical African language corpora may reach the hundreds of thousands or millions of tokens, particularly if they are purely text-based (e.g Prinsloo & de Schryver 2001, De Schryver & Nabirye 2018), but African language corpora *smaller* than the Kom corpus (e.g. Bostoen & de Schryver 2015) have also been shown to be effective sources of comparable data.

4.1 Stem-based prominence and the vowel space

In Grassfields Bantu, prosodic prominence often does not manifest in the same way as a cross-linguistically typical stress system (Franich 2021). Elsewhere in the world's languages, prominence is often associated with hyperarticulation, or more extreme vowel articulations (Lindblom 1990, De Jong 1995); sonority expansion, in which vowels are produced lower to increase the radiated sound energy (De Jong et al. 1993, Cho 2005, Mo et al. 2009); or some mixture of both. Stem-based prominence in Grassfields Bantu generally, and the Ring family to which Kom belongs specifically, is known to result in an asymmetry in contrast distribution *without* obvious hyperarticulation or sonority expansion: the stem-initial syllable bears the most segmental and tonal contrasts (Hyman et al. 2019, Franich 2021, Faytak & Akumbu 2021).

This same pattern holds in Kom (Shultz 1993, 1997, Faytak 2017, Hyman 2005). Looking specifically at vowels, Kom's vowel inventory consists of eleven monophthongs per an update to earlier analysis in Shultz (1997):[4] the qualities /i y e ø ɛ a o u ɨ/, joined by two "fricative vowels" typically transcribed as /z̩ v̩/ (Faytak 2017), may occur in the first stem syllable. In closed syllables, /e/ is realized as [ə]; otherwise it is realized as [e]. Outside of the initial stem syllable, only /i ɨ a/ occur. Based on the impression of the authors, the *prominent* vowels /i ɨ a/ which occur in stem-initial syllables are produced differently from those *non-prominent* vowels produced elsewhere. To facilitate comparison with other Grassfields Bantu languages such as Medəmbɑ, we use the assembled corpus materials to investigate whether prominence in Kom conditions hyperarticulation, sonority expansion, or neither.

To assess vowel quality using the acoustic data available, the frequencies of the first, second, and third formants (F1, F2, and F3) were extracted from all aligned phones in the corpus using Parselmouth (Jadoul et al. 2018). Prominence, coded by the addition of a '`0`' to non-prominent vowels (located outside of the first stem syllable), was also extracted. Each formant was *z*-score normalized by speaker. The resulting F1-F2-F3 space for Kom is shown in Figures 4 and 5. Not all vowels are displayed here: it was not possible to reliably extract formant frequencies from the fricative vowels /z̩, v̩/ due to fricative noise frequently obscuring the formants. An apparent effect of prominence is clearly visible in Figure 5: small differences in the typical F1 and F2 of each vowel which occurs in prominent and non-prominent environments can be seen. Based on inspection of the data

[4] The analysis in Shultz (1997) and earlier in Shultz (1993) derives the vowels /y ø ɛ/ through coalescence of VV sequences, and incorrectly treats the fricative vowels /z̩ v̩/ as CV sequences.

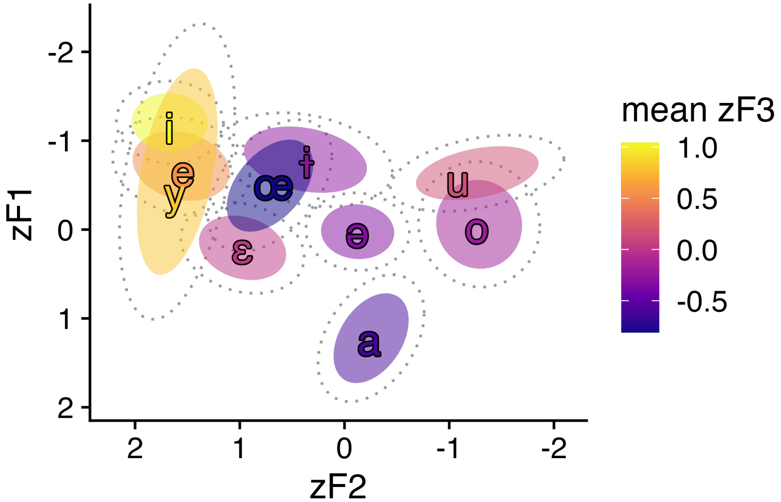

Figure 4: F1, F2, and F3 frequencies of the Kom vowels. Data more than one standard deviation from vowel-speaker means is removed; dotted ellipses include 75% of remaining data, and solid ellipses include 50%.

in Figure 5, only F1 and F2 are considered for the quantitative analysis: F3 does not appear to contribute substantially to the stress difference.

To quantitatively assess this apparent effect of stem prominence, linear mixed-effects regressions were carried out in R using *lme4*; *p*-values were calculated using the *lmerTest* library (Bates et al. 2015, Kuznetsova et al. 2017). Models assessed formant frequency with respect to fixed effects of vowel, prominence, and their interaction, with random intercepts for speaker and word.[5] In both the models run for F1 and F2, non-prominent /a/ is taken as baseline.

The models indicate that prominence mediates the formant frequencies of all vowels examined here. The main effects of vowel type are highly significant in both the F1 and F2 models (all $p < 0.001$). This is unsurprising, given the large differences in formant frequencies expected among /i ɨ a/. The main effect of prominence reaches significance in the F1 model ($\beta = 0.122$, $p < 0.001$), indicating that regardless of vowel type, stress has an effect on F1 frequency, and specifically that prominence increases F1 for the baseline vowel /a/. The interactions of stress and vowel identity reveal the specific effect of stress for the other vowels: for /i/, the interaction is strongly negative relative to baseline ($\beta = -0.288$, $p < 0.001$), as is the interaction for /ɨ/ ($\beta = -0.442$, $p < 0.001$). That is, prominence *decreases* F1 for

[5]Using the *lme4* syntax: `formant ~ vowel * prominence + (1|speaker) + (1|word)`.

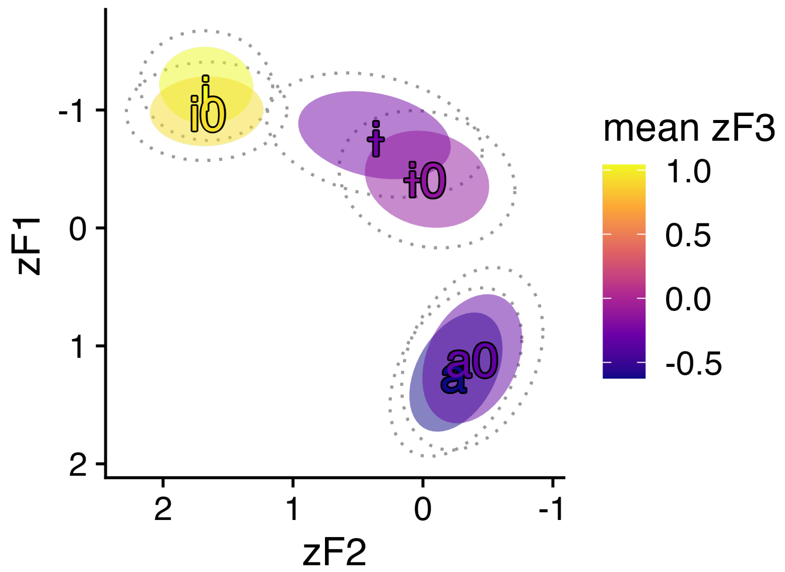

Figure 5: F1, F2, and F3 frequencies for the Kom vowels which occur in both prominent and non-prominent positions. Non-prominent position is indicated with '0'.

the high vowels, the opposite of its effect on /a/. The main effect of prominence also reaches significance in the F2 model (β = 0.138, $p < 0.001$), indicating that prominence slightly increases F2 for all vowels and the baseline in particular. The interactions of vowel type and prominence on F2 all reach significance. The interaction for /i/ is negative and almost the same magnitude as the main effect (β = -0.122, $p < 0.001$), indicating little difference for this category specifically. The interaction for /ɨ/ is strongly positive (β = 0.239, $p < 0.001$), indicating that prominence increases F2 more than baseline.

To further assess the differences conditioned by prominence, post-hoc comparisons of estimated marginal means were carried out using *emmeans* (Lenth et al. 2019), in order to compare the formant frequencies for prominent and non-prominent vowels of each quality. The comparisons, summarized in Table 1, confirm that prominence significantly affects the formant frequencies of all three vowels considered. The sole exception is that prominence has no significant effect on the F2 of /i/.

To sum up, unlike some other Grassfields Bantu languages such as Medəmbɑ (Franich 2021), positional prominence in Kom does have an effect on vowel quality, rather than merely restricting the inventory outside of prominent syllables.

Table 1: Planned comparisons of estimated marginal means for /i a ɨ/ under prominent and non-prominent conditions.

F1 comparison	estimate	*p*
a-a0	0.122	< 0.001
ɨ-ɨ0	-0.320	< 0.001
i-i0	-0.166	< 0.001
F2 comparison	**estimate**	***p***
a—a0	0.138	< 0.001
ɨ—ɨ0	0.377	< 0.001
i—i0	0.016	0.920

This effect is best characterized as hyperarticulation rather than sonority expansion, given the relative centralization of non-prominent vowels. This observation provides a starting point for further research on prosodic modulation of vowel quality in Grassfields Bantu, with other factors known to be implicated in hyperarticulation such as duration, intensity, and fundamental frequency left for future research.

4.2 Monophthongization of /Vn/

We next explore the usefulness of the corpus for assessing variable phonological processes which may otherwise only be approached less naturally through elicitation. In Kom, coda /n/ before pause and consonants (/Vn#/, /Vn#C/) co-occurs with diphthongal vowels; the diphthongs which may occur are realized as [ɨi̯n əi̯n ai̯n oi̯n ui̯n]. (Recall that in closed syllables generally, /e/ is realized as [ə].) Njuasi (2022) observes that these diphthongs are realized instead as monophthongs (i.e. [ɨnV ənV anV onV unV]) when the segment following the /n/ is a vowel (i.e., /Vn#V/). The pattern is summarized in (7). [6]

[6]The underlying sequences at issue have been treated in a variety of ways: as underlying diphthongs /ɨi̯n əi̯n ai̯n oi̯n ui̯n/ (Njuasi 2022) and as vowels followed by palatal nasal codas /ɨɲ eɲ aɲ oɲ uɲ/ (Shultz 1993, 1997). We assume here a more straightforward representation of /ɨn en an on un/, based on evidence from the alternations themselves (ruling out the option with coda /ɲ/) and because the diphthongs are best viewed as conditioned variants of monophthongs, which occur in a more diverse set of environments than diphthongs.

(7) a. fòyn
chief
'chief'

b. fòyn kom
chief Kom
'chief of Kom'

c. fòn atum
chief foreign.place
'foreign chief'

d. fòn ìdyèyn
chief Noni
'chief of Noni'

The reappearance of the underlying monophthongs in the pre-vowel context may relate most straightforwardly to syllabification of the /n/ if the process is treated as operating within-syllable: when a vowel follows, syllabification as /V.nV/ occurs, depriving the vowel of the trigger for diphthongization. As a categorical phonological process sensitive to upcoming content, monophthongization in Kom bears some resemblance to other phonological processes such as French liaison (Kilbourn-Ceron 2017) and North American English flapping (Kilbourn-Ceron & Goldrick 2021). Both of these processes, however, are variable in their application, due to variation inherent to planning speech in real time (Eddington & Elzinga 2008, Smolensky et al. 2020, Kilbourn-Ceron & Goldrick 2021). It remains to be seen whether the behavior in Kom is totally categorical, or also varies in a similar way. As such, to make a first step toward better understanding of the process, we examine whether speakers actually categorically engage in monophthongization in the /Vn#V/ sequence.

A largely automatic analysis of the corpus data was carried out as follows: all /Vn#V/ sequences were extracted from the corpus, a total of 417 tokens. The forced aligner which produced word- and phone-level alignments (see Section 3.3) was given the choice of aligning either a diphthong [V̯in] or a monophthong [Vn] to any /Vn/ sequence. The interval corresponding to the /VnV/ sequences was automatically checked for the presence of an aligned phone [i̯] and the outcome recorded.[7]

[7]Future work on this topic will involve a more labor-intensive manual determination of whether the outcome is a monophthong or a diphthong.

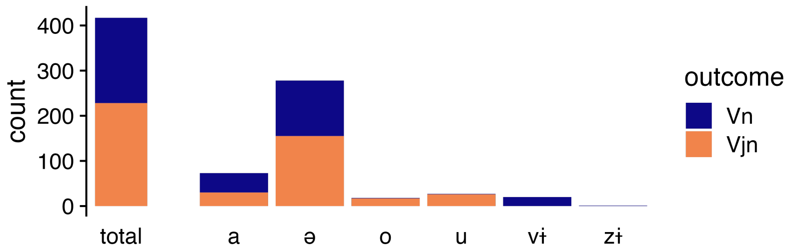

Figure 6: Incidence of [V̯in] versus [Vn] pronunciations of /Vn/ in /Vn#V/ sequences in the corpus.

Count results for each outcome are shown in Figure 6. Counter to initial expectations from the description in Njuasi (2022), the output is not entirely categorical; both expected [Vn] and unexpected [V̯in] pronunciations are observed in the [Vn#V] environment. Variation in this phonological outcome, paralleling the cases from French and English above, suggests the need to investigate the role of factors such as lexical frequency, frequency of particular sequences of words, and prosody (such as the intrusion of pauses or phrase boundaries, which the aligner may not have recorded) in determining a probabilistic outcome.

5 Discussion

This article has aimed to present a proof of concept for the collection and annotation of proverbs and other verbal arts as an efficient means of investigating the sound structure of African languages, with benefits accruing to both the community and the analyst. Proverbs act as a repository of cultural information and hold deeper relevance to community interests than neutral corpus materials might; because they are short and consistent in content, they are also easily transcribed and parsed, facilitating annotation and processing of the data at a large scale. The case studies demonstrate that one field season's worth of materials is sufficient to carry out quantitative and qualitative studies of sound structure. It is hoped that the article can provide a model for interested researchers, which may support the creation of resources suitable for phonetic, phonological, and NLP work for a wider variety of languages, particularly smaller and less-documented African languages.

Future plans for the corpus include the addition of tone annotation, which presents a non-trivial challenge given the complexity of Kom's tonal phonology (Jones 1997, Hyman 2005) and its systematic undermarking in the official orthography (Bernard et al. 2002). Given the need to annotate tone, annotation of materials at a scale sufficient for technological development likely hinges on the contributions of experts in the Kom community who are literate in the official orthography. Collection of spontaneous speech materials, though subject to the "transcription bottleneck" (Seifart et al. 2018), are also planned in order to complement the corpus' controlled speech and allow for analysis of different speech production styles.

Acknowledgements

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Chapter 20

Headless relative clauses and anti-agreement in Kirundi

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Anti-agreement effects have been widely documented in the Bantu language family where they occur in the specific context of Class 1 subject extraction. This paper looks at an instance of apparent Bantu AAEs that occur in an even more restricted context; in Kirundi, AAEs arise only in headless relative clauses but not other Class 1 subject extraction constructions. Drawing on novel fieldwork, I argue that the restricted distribution of AAEs in Kirundi is the result of the restricted distribution of overt relative markers. Both only occur in headless relative clauses. I link this correlation to unique properties of headless relative clauses, specifically proposing that such relative clauses are headed by empty nouns which require a relative operator in D. This proposal, in the spirit of Jenks et al. (2017), provides a unified account of variation in the structure of relative clauses in Kirundi, and the presence of AAEs, with greater implications for the understanding of relativization and anti-agreement in related Bantu languages.

1 Introduction

Anti-agreement effects (AAEs) occur when the expected agreement relation between an argument and a verb is suppressed or altered. Within the Bantu language family, AAEs have been widely documented; see e.g. Schneider-Zioga (2000, 2007) for Kinande, Diercks (2009, 2010) for Lubukusu, Burns (2013) for Abo, Cheng (2006), Henderson (2007, 2009, 2013) for Bemba, and Baier (2018) for Kikuyu. Notably, Bantu AAEs exhibit several shared properties that further distinguish them from other anti-agreement phenomena: they are expressed with a

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special morpheme (instead of default agreement or no agreement) and their distribution is restricted, occurring only in the extraction of local Class 1 (i.e. human singular) nouns.

These properties are demonstrated below using examples from Bemba. In a standard clause with a Class 1 subject, the canonical agreement morpheme *a-* appears (1a). When this subject is relativized, standard agreement is not allowed (1b); instead, a special morpheme *u-* is required (1c). This does not occur with subjects of any other noun class.

(1) Bemba (Cheng 2006: 197)

a. Umulendo **a**-ka-belenga ibuku.
   1boy **1.AGR**-FUT-read 5book
   'The boy will read the book.'

b. *umulendo ú-**a**-ka-belenga ibuku
   1boy 1.REL-**1.AGR**-FUT-read 5book
   'the boy who will read the book'

c. umulendo ú-**u**-ka-belenga ibuku
   1boy 1.REL-**1.AAE**-FUT-read 5book
   'the boy who will read the book'

One school of thought, including Henderson (2007, 2009, 2013) and Diercks (2010), has argued that Bantu AAEs result from a special relation between C and T wherein agreement in C determines agreement in T in local extraction contexts. They argue that agreement in C lacks [person] features so when the local subject moves to C and triggers [person]-less agreement there, a C-T relation then results in agreement in T lacking [person] features as well. This impoverished agreement is the source of AAEs.

One implication of this proposal is that languages in which subject extraction does not trigger AAEs must either have full ϕ-agreement in C, with all features included, or lack C-agreement altogether. The latter analysis has been proposed by Henderson (2013) for Kirundi, a language in which standard relative clauses never exhibit AAEs. Both matrix clauses with Class 1 subjects and subject relative clauses with Class 1 heads share the same canonical Class 1 agreement morphology and no relatives display overt relative markers, as seen in (2).[1]

[1]Unattributed examples come from the author's fieldwork conducted in collaboration with four native speakers of Kirundi, three of whom currently live in North America and one who lives in Burundi. Three speakers are male and one is female. Two grew up in rural areas and two in the main city of Bujumbura. The research adopts standard theoretically-driven fieldwork methodology (see, e.g., Matthewson 2004, Bowern 2008, Bochnak & Matthewson 2020).

(2) a. U-mu-gabo **a**-tēka u-mu-ceri.
AUG-1-man **1.AGR**-cook AUG-3-rice
'The man cooks rice.' (Matrix)

b. u-mu-gabo **a**-tēká u-mu-ceri
AUG-1-man **1.AGR**-cook.DEP AUG-3-rice
'the man who cooks rice' (Headed relative)

However, despite past claims, I provide new data which show that Kirundi does *not* lack AAEs entirely. In this paper, I demonstrate that a subset of extraction constructions – headless relative clauses – *does* exhibit the morphology of AAEs and that these AAEs align with the defining properties of Bantu AAEs. As seen in (3), a headless relative clause in Kirundi requires alternative subject agreement which exactly resembles the AAE morpheme found in other Bantu AAE languages.

(3) u-**u**-têka u-mu-ceri
AUG-**1.AAE**-cook.HRC AUG-3-rice
'the one who cooks rice ' (Headless relative)

These data are puzzling for two reasons. First and most crucially, they suggest that headed relative clauses and headless relative clauses may differ structurally such that AAEs are triggered only in the latter. This is surprising if we assume that both are formed through the same relativization process. Second, the presence of Kirundi AAEs casts doubt on Henderson's (2013) claim that Kirundi lacks ϕ-features in C, if we take that the widely accepted stance that AAEs are linked to agreement in C.

In addressing these issues, this paper has two main goals: (1) to document AAEs in Kirundi, correcting past analysis of the language as lacking AAEs; and (2) to present an account of relative clauses in Kirundi that explains why anti-agreement morphology occurs in a smaller subset of possible contexts compared to other Bantu AAE languages. Ultimately, I argue that Kirundi exhibits AAEs only in headless relative clauses due to variation in the realization of elements of the relativized nominal. In the spirit of Jenks et al. (2017), I propose that D elements in Kirundi are in complementary distribution with the relative marker. I analyze this relative marker as a relative operator, akin to English *which* that only appears overtly when it occurs with empty head nouns (Panagiotidis 2003). It is this makeup that gives rise to the unique form of headless relative clauses in Kirundi. This proposal explains both the AAE facts in Kirundi as well as related facts regarding the presence of overt relative markers in non-local headless relative clauses.

The structure of this paper is as follows. Section 2 presents an overview of relative clauses in Kirundi, demonstrating that both headed and headless relative clauses satisfy diagnostics for relative clauses despite exhibiting different morphological forms. Section 3 assesses Kirundi's AAEs in the context of Bantu AAEs, showing that, despite their restricted distribution, Kirundi AAEs pattern exactly like AAEs in other Bantu languages, including the presence of agreeing relative markers. Section 4 sketches a syntactic analysis of Kirundi relative clauses which accounts for both the relative marker and AAE facts in the language. Finally, Section 5 concludes with a brief summary and discussion of further areas for research.

2 Relative clauses in Kirundi

Kirundi is a Great Lakes Bantu language, spoken by ~12.5 million speakers in the country of Burundi where it is the national language. It is part of a dialect continuum with Kinyarwanda (Rwanda) and smaller contiguous communities in Tanzania, Uganda, and the DRC (Bastin 2003). Linguistically, Kirundi exhibits many common traits of Bantu languages, including a rich noun class system, subject agreement on verbs and concord on nominal modifiers. Phonologically, it possesses a binary H/L tonal system which is important in making both lexical and morphosyntactic distinctions.[2]

This section builds off the author's fieldwork with first-language Kirundi speakers to provide an overview of relative clauses in the language. Section 2.1 presents data on the most common form of relative clauses in Kirundi: *headed* relative clauses. These relative clauses lack overt relative markers and AAEs but nevertheless satisfy morphosyntactic criteria for relative clauses. Section 2.2 examines *headless* relative clauses, which lack overt nominal heads but exhibit both AAEs and agreeing relative markers. These properties – anti-agreement and overt relative markers – have not been included in past analyses of Kirundi relative clause structure. Section 2.3 summarizes the Kirundi relative clause paradigm at the heart of this paper's puzzle.

2.1 Headed relative clauses

Headed relative clauses are those that contain overt nominal heads. In Kirundi, these constructions lack overt relative markers and AAEs. This is demonstrated

[2] Tone and vowel length are represented in all examples in line with the Kirundi Utwâtuzo system (Nkengurutse 2024). For example, a long /a/ vowel is expressed as *ā*; with high tone on the first mora as *â*; and with high tone on the second mora as *ǎ*.

below in comparing a matrix clause (4) with corresponding subject and object relative clauses (5). Unlike headed relatives in other Bantu AAEs, exemplified by the Bemba example in (1), the relativization of a Class 1 local subject in Kirundi results in canonical Class 1 agreement indicated by the morpheme *a*-.

(4) U-mu-gabo **a**-tēka u-mu-ceri.
AUG-1-man 1.AGR-cook AUG-3-rice
'The man cooks rice.' (Matrix)

(5) a. u-mu-gabo **a**-tēká u-mu-ceri
AUG-1-man 1.AGR-cook.DEP AUG-3-rice
'the man that cooks rice' (Subject headed relative)

b. u-mu-ceri u-mu-gabo **a**-tēká
AUG-3-RICE AUG-1-man 1.AGR-cook.DEP
'the rice that the man cooks' (Object headed relative)

One thing that is immediately apparent in Kirundi is that relative clauses lack overt relative markers in headed relative clauses. Relative markers, a term used to refer to morphemes that correspond to relative pronouns in English, have been analyzed as both relative complementizers (see, e.g., Demuth & Harford 1999, Schneider-Zioga 2007, Henderson 2009) and relative operators (Jenks et al. 2017). Notably, in the vast majority of Bantu AAE languages, AAEs always occur with overt relative markers, driving past analyses which link Bantu AAEs to agreement in C, expressed as an agreeing relative complementizer. Nevertheless, there are a few languages (such as Kinyarwanda) which possess AAEs but lack overt relative markers in headed relative clauses. In this sense, Kirundi patterns differently than previously documented Bantu AAE languages, lacking both AAEs and an overt relative marker in headed relative clauses.

Despite the lack of an overt relative marker, evidence that these constructions are indeed relative clauses can be found in both language-internal and cross-linguistic diagnostics. Externally-headed relative clauses, such as those seen above, are typically defined as dependent clauses with external nominal heads that are linked to a missing constituent in the clause (see, e.g., Caponigro 2021: 23–24). Within Kirundi, relative clauses possess a handful of properties that distinguish them as dependent clauses in line with this definition.

First, dependent clauses in Kirundi exhibit a different tonal patterns than matrix clauses. When a verb with an underlyingly low tone, as seen in the matrix clause in (6a), is used in an embedded context, such as with a complement clause, a high tone appears on the second syllable of its root (6b). This is also true in relative clauses, as seen in (6c).

(6) Dependent clause tone

a. U-mu-gabo a-tēka u-mu-ceri.
AUG-1-man 1.AGR-cook AUG-3-rice
'The man cooks rice.' (Matrix)

b. Tū-zi kó [u-mu-gabo a-tēká u-mu-ceri].
we.know COMP AUG-1-man 1.AGR-cook.DEP AUG-3-rice
'We know that the man cooks rice.' (Complement)

c. Ndakūnda u-mu-gabo [___ a-tēká u-mu-ceri].
I.like AUG-1-man 1.AGR-cook.DEP AUG-3-rice
'I like the man who cooks rice.' (Relative)

Secondly, dependent clauses differ from matrix clauses in the position of negation morphology (see Chaperon 2023 for further discussion of negation in Kirundi). Though matrix clauses express negation before subject agreement (7a), dependent clauses express it after subject agreement (7b). This is shown for a relative clause in (7c).

(7) Dependent clause negation

a. A-ba-gabo **nti**-ba-tēka u-mu-ceri.
AUG-2-man NEG-2.AGR-cook AUG-3-rice
'The men don't cook rice.' (Matrix)

b. Tūzi kó [a-ba-gabo ba-**ta**-tēká u-mu-ceri].
we.know COMP AUG-2-man 2.AGR-NEG-cook.DEP AUG-3-rice
'We know that the men don't cook rice.' (Complement)

c. Ndakūnda a-ba-gabo [___ ba-**ta**-tēká u-mu-ceri].
I.like AUG-2-man 2.AGR-NEG-cook.DEP AUG-3-rice
'I like the men who don't cook rice.' (Relative)

Finally, dependent clauses differ from matrix clauses in the availability of the disjoint morpheme *ra*-. The disjoint marker appears in matrix clauses, such as (8a), in which the postverbal element is not focused. It is prohibited in embedded clauses, as shown in (8b). Relative clauses pattern like dependent clauses in prohibiting an overt disjoint marker (8c). For more on the conjoint/disjoint alternation in Kirundi, see Ndayiragije (1999) and Nshemezimana & Bostoen (2017).

(8) Dependent clause disjoint

a. A-ba-gabo ba-**ra**-tēka.
AUG-2-man 2.AGR-DJ-cook
'The men cook.' (Matrix)

b. Tūzi kó [a-ba-gabo ba-(*ra)-tēká].
we.know COMP AUG-2-man 2.AGR-DJ-cook.DEP
'We know that the men cook.' (Complement)

c. Ndakūnda a-ba-gabo [___ a-(*ra)-tēká].
I.like AUG-2-man 2.AGR-DJ-cook.DEP
'I like the men who cook.' (Relative)

In addition to this morphological evidence, Kirundi relative clauses also satisfy common diagnostics of Ā-movement (see, e.g., Safir 2016). Firstly, they can establish long-distance dependencies, by-passing intervening subjects, as in the object relative in (9a). Relative clauses can also be formed across longer distances; both embedded subjects (9b) and embedded objects (9c) can be relativized.[3]

(9) a. Ndakūnda u-mu-ceri [ba-ryá ba-gabo ba-tēká ___].
I.like AUG-3-rice [2-DEM 2-man 2.AGR-cook.DEP]
'I like the rice [those men cook ___].'

b. Ndakūnda ba-ryá ba-gabo [a-vugá kó [___ ba-tēká u-mu-ceri]].
I.like 2-DEM 2-man [1.AGR-say.DEP that [2.AGR-cook.DEP AUG-3-rice]]
'I like those men [she says [___ cook rice]].'

c. Ndakūnda u-mu-ceri [a-vugá kó [ba-ryá ba-gabo ba-tēká ___]].
I.like AUG-3-rice [1.AGR-say.DEP that [2-DEM 2-man 2.AGR-cook.DEP]]
'I like the rice [she says that [those men cook ___]].'

Secondly, Kirundi headed relative clauses are island-sensitive. For example, neither the subject (10) nor the object (11) of a relative clause can move out of said relative clause.

(10) a. Ndakūnda u-mu-ceri [a-ba-gabo ba-tēká ___].
I.like AUG-3-rice [AUG-2-man 2.AGR-cook.DEP]
'I like the rice [the men cook ___].'

[3]As noted by a reviewer, it is not the case that these sentences involve multiple relative clauses because the presence of the complementizer *kó* can occur in complement clauses but is disallowed in relative clauses.

b. *Ūzi a-ba-gabo [nkūndá u-mu-ceri [___ ba-tēká
you.know AUG-2-man [I.like.DEP AUG-3-rice [2.AGR-cook.DEP
___].
]]
Intended: 'You know the men that [I like the rice [___ cook ___]].'

(11) a. Ndakūnda a-ba-gabo [___ ba-tēká u-mu-ceri].
I.like AUG-2-man [2.AGR-cook.DEP AUG-3-rice]
'I like the men that [___ cook the rice].'

b. *Ūzi u-mu-ceri [nkūndá a-ba-gabo [___ ba-tēká
you.know AUG-3-rice [I.like.DEP AUG-2-man [2.AGR-cook.DEP
___]].
]]
Intended: 'You know the rice [I like the men that [___ cook ___]].'

In sum, Kirundi headed relative clauses pass standard tests for relative clauses, patterning as dependent clauses and satisfying diagnostics for Ā-dependencies. However, unlike prototypical Bantu AAE languages, they lack alternative agreement and overt relative markers.

2.2 Headless relative clauses

Headless relative clauses are relative clauses that lack overt nominal heads. An example can be seen for Kirundi in (12) which compares a headed relative clauses with its headless counterpart. The two relative clauses differ in that the headed relative (12a) includes an overt nominal head, the noun *banyêshŭre* 'students' which is made up of a noun class marker and a noun root. This noun is absent in the corresponding headless relative clause in (12b) which consists solely of an augment vowel followed by an inflected verb.

(12) a. a-ba-nyêshŭre ba-shākā gutāha
AUG-2-student 2.AGR-want.DEP go.home
'the students who want to go home' (Headed)

b. a-ba-shâka gutāha
AUG-2.AGR-want.HRC go.home
'the ones that want to go home' (Headless)

While some accounts of Bantu relatives clauses have proposed that this first vowel on the headless relative is a relative marker (see, e.g., Cheng 2006 and

Diercks 2010 for Bemba and Lubukusu, respectively), I argue that this is really just the augment in Kirundi.[4] Crucially, this morpheme patterns like the augment in the rare contexts in which Kirundi nouns can appear unaugmented. When a noun occurs with the negative existential *nta*, the augment cannot appear, as seen in (13a). Similarly, headless relative clauses also lose their initial vowel in this context, as seen in (13b).

(13) a. Nta [(*a-)ba-nyêshǔre ba-shāká gutāha].
there.is.no AUG-2-student 2.AGR-want.DEP go.home
'There are no students that want to go home.' (Headed)

b. Nta [(*a-)ba-shâka gutāha].
there.is.no AUG-2.AGR-want.HRC go.home
'There are none who want to go home.' (Headless)

Assuming then that this morpheme is an augment, we see that non-Class 1 headless relative clauses look exactly like headed ones except for the absence of an overt head noun. The similarity between the two types of relative clauses is further supported by the fact that headless relative clauses satisfy the same criteria for dependent clauses and Ā-dependencies as headed relative clauses. As dependent clauses, headless relatives exhibit non-matrix tone (14a) and dependent clause negation (14b), and disallow the disjoint morpheme (14c).[5]

(14) Headless relative clauses are dependent clauses

a. a-ba-**têka** u-mu-ceri
AUG-2.AAE-cook.HRC AUG-3-rice
'the ones who cook rice' (Non-matrix tone)

b. a-ba-**ta**-têka u-mu-ceri
AUG-2.AAE-NEG-cook.HRC AUG-3-rice
'the ones who don't cook rice' (Dependent negation)

[4] In the Bantu tradition, the augment, or pre-prefix, refers to the initial morpheme of a DP which is typically associated with definiteness or existence (Katamba 2003). In Kirundi, it appears on almost all nouns outside of negative existential constructions, vocatives, R-expressions, incorporated nouns, and some locative expressions (Ndayiragije et al. 2012).

[5] Notably, the tonal pattern in headless relative clauses is different from those in standard dependent clauses. While a high tone appears on the second syllable of the verb root of a headed relative clause, for instance, it appears on the first syllable of the verb root in headless relative clauses. For now, I take both of these tonal patterns to distinguish relative clauses from matrix clauses but do not present an analysis of why headless relative clauses further differ in tone from other dependent clauses.

c. a-ba-(***ra-**)têka u-mu-ceri
AUG-2.AAE-DJ-cook.HRC AUG-3-rice
'the ones who cook rice' (No disjoint morpheme)

Additionally, headless relative clauses can have long-distance dependencies (15a) and are sensitive to islands (15b). Notably, in these contexts, an overt relative marker is required. This marker *-ó* displays agreement morphology which indicates the number and noun class of the resulting headless relative construction.

(15) Headless relative clauses involve Ā-dependencies

a. a-b-ó [a-vugá kó [___ ba-tēká u-mu-ceri]]
AUG-2-REL [1.AGR-say.DEP that [2.AGR-cook.DEP AUG-3-rice]]
'the ones [she says that [___ cook rice]].' (Long-distance)

b. *a-b-ó [nkūndá u-mu-ceri [___ ba-tēká ___]]
AUG-2-REL [I.like.DEP AUG-3-rice [2.AGR-cook.DEP]]
Intended: 'the ones that [I like the rice [___ cook ___]' (Island)

Despite these similarities, Kirundi headless relative clauses differ from their headed counterparts in two striking ways. First, they exhibit AAEs. Subject headless relative clauses that denote Class 1 individuals possess the same alternative Class 1 agreement morpheme *u-* on the verb as seen in other Bantu AAE languages, as demonstrated in (16a). Second, they have overt relative markers. Object headless relative clauses require the presence of a relative marker where the head noun of a headed object relative would occur, as seen in (16b). This relative morpheme *-ó* is identical to that found in several other Bantu languages such as Kinande (Schneider-Zioga 2007).

(16) a. u-**u**-têka u-mu-ceri
AUG-1.**AAE**-cook.HRC AUG-3-rice
'the one who cooks rice ' (Subject headless relative)

b. **u-u-ó** u-mu-gabo **a**-tēká
AUG-3-**REL** AUG-1-man 1.AGR-cook.DEP
'the one that the man cooks' (Object headless relative)

Before continuing, a note on the interpretation of Kirundi headless relative clauses is in order. Headed relative clauses are defined semantically by the individual property introduced by the head noun. For instance, the headed relative in (17a) must refer to a dog because the head noun contains the root for 'dog'. In headless relatives, however, there is no head noun. As a result, the entity which they denote is only defined by the noun class which appears in its agreement. Thus, the headless version of (17a), seen in (17b), can refer to any entity as long as it is a class 9 entity. Because class 9 typically denotes animals, this may lead to an intuition that the speaker is referring to some animal that likes Muco but it is also possible that the construction could refer to any number of other non-animal class 9 entities.

(17) a. i-m-bwá i-kūndá Muco
AUG-9-dog 9.AGR-like.DEP Muco
'the dog that likes Muco' (Headed relative)

b. i-i-kûnda Muco
AUG-9-9.AGR-like.HRC Muco
'the one that likes Muco' (Headless relative)

Because English lacks relatives of this type (akin to free relatives), I have chosen to translate Kirundi headless relative clauses with the English 'one' in order to approximate their usage across all types of entities and their lack of inherent referentiality. I will return to the interpretation of Kirundi headless relatives in Section 4.2.

2.3 Empirical summary

Both headed and headless relative clauses in Kirundi satisfy commonly accepted criteria for relative clauses. They pattern as dependent clauses and are shown to involve Ā-dependencies. Despite these similarities, variation in agreement and relative morphology, as summarized in Table 1, shows that a more complex situation exists.

Table 1: Kirundi relative clauses

HEAD	REL	SUBJ	T	OBJ
SUBJ			AGR	OBJ
OBJ		SUBJ	AGR	
SUBJ$_{\varnothing}$			**AAE**	OBJ
OBJ$_{\varnothing}$	**REL**	SUBJ	AGR	

While headed relative clauses lack relative markers and AAEs, headless relative clauses, represented in Table 1 with null HEAD positions, possess both of these features. This suggests that properties of the relativized head – e.g. overt vs. null – affect the relativization strategy.

3 Anti-agreement effects and relative markers in Kirundi

Having established the empirical landscape of relative clauses in Kirundi, this section looks at the unique features of Kirundi headless relative clauses. Unlike standard relative clauses in the language, headless relative clauses possess both anti-agreement effects and, in the case of object relatives, agreeing relative markers.

Section 3.1 assesses Kirundi AAEs against the commonly accepted properties of Bantu AAEs, demonstrating that AAEs in Kirundi pattern exactly like those in other Bantu AAE languages, albeit with a restriction to headless relative contexts. Section 3.2 proposes an additional property of Bantu AAEs: the presence of agreeing relative markers. This aligns with observations from across the Bantu family as well as the presence of overt, agreeing relative morphology in Kirundi object headless relatives. Building on this, Section 3.3 makes the case for the presence of relative markers in Kirundi *subject* headless relatives, arguing relative markers do occur in these constructions but that their presence is obscured by the effects of morpho-phonological processes such as Kinyalolo's Constraint (Kinyalolo 1991). This aligns Kirundi AAEs with past proposals for the occurrence of agreeing relative markers in Bantu AAE constructions generally (see, e.g., Henderson 2013). Section 3.4 revises the Kirundi relative clause paradigm to account for these claims, paving the way for a formal analysis of the difference between headed and headless relative clauses in the language.

3.1 Kirundi AAEs are Bantu AAEs

AAEs have been documented in a broad variety of languages since Ouhalla's (1993) initial work on the topic, including Bantu languages. Crucially, within Bantu languages, AAEs pattern very similarly, suggesting a common phenomenon. The properties of Bantu AAEs are listed in (18).

(18) Properties of Bantu Anti-Agreement Effects
   a. Special form – anti-agreement is expressed via a unique morpheme
   b. Limited distribution – AAEs are only present with Class 1 nouns
   c. Local – AAEs only occur with subject extraction

These same properties characterize AAEs in Kirundi. Firstly, Kirundi subject headless relatives exhibit alternative agreement morphology when they denote Class 1 arguments. As seen in (19), Class 1 subject agreement is typically realized as *a*-. However, in headless relative clauses, the *u*- morpheme appears instead. This is in line with many Bantu AAE languages which have *u*- AAE morphemes.

(19) a. U-mu-gabo **a**-ra-soma i-bi-tabu.
AUG-1-man **1.AGR**-DJ-read AUG-8-book
'The man reads books.' (Matrix)

b. u-**u**-sóma i-bi-tabu
AUG-**1.AAE**-read.DEP AUG-8-book
'the one who reads books ' (Headless relative)

Like other Bantu AAEs, Kirundi AAEs are limited to Class 1. Headless relative clauses of other classes do not exhibit alternative agreement morphology, as seen in (20) and (21).

(20) a. a-ba-gabo **ba**-somá i-bi-tabu
AUG-2-man 2.AGR-read.DEP AUG-8-book
'the men who read books' (Headed – Class 2)

b. a-**ba**-sóma i-bi-tabu
AUG-2.AGR-read.HRC AUG-8-book
'the ones who read books' (Headless – Class 2)

(21) a. i-n-ká **zi**-rí mu mu-rimá
AUG-10-cow 10.AGR-be.DEP in 3-field
'the cows who are in the field' (Headed – Class 9)

b. i-**zi**-rí mu mu-rimá
AUG-10.AGR-be.HRC in 3-field
'the ones who are in the field' (Headless – Class 9)

Finally, Kirundi AAEs, like other Bantu AAEs, are limited to local subject extraction. They do not appear on the lower verb of embedded subject extraction constructions (22) or compound tense constructions (23).

(22) No AAEs in embedded agreement

a. *u-u-ó [tūzí [kó **u**-kūndá Muco]]
AUG-1-REL we.know.DEP that 1.AAE-like.DEP Muco
Intended: 'the one we know likes Muco'

b. u-u-ó [tūzí [kó **a**-kūndá Muco]]
AUG-1-REL we.know.DEP that 1.AGR-like.DEP Muco
'the one we know likes Muco.'

(23) No AAEs in lower compound tense agreement

a. *u-**u**-ríko **u**-ra-soma
AUG-1.AAE-PROG.DEP 1.AAE-DJ-read
'the one who is reading'

b. u-**u**-ríko **a**-ra-soma
AUG-1.AAE-PROG.DEP 1.AGR-DJ-read
'the one who is reading'

Taken together, these data strongly suggest that Kirundi AAEs pattern like other Bantu AAEs do and should be considered as part of the same phenomenon.

3.2 Bantu AAEs involve agreeing relative markers

Along with the characteristics presented in (18), I argue that Bantu AAEs are further characterized by their co-occurrence with agreeing relative markers. Almost all past descriptions of Bantu AAEs include the presence of such morphemes, typically analyzed as relative complementizers expressing agreement in C. Along with Bemba, shown in (1), AAE constructions in Kinande and Lubukusu have also been described as including agreeing relative markers. This is shown in (24) for Kinande and (25) for Lubukusu.[6]

(24) Kinande Subject Relative (Schneider-Zioga 2007: 417)
omukali **oyo** u-anzire Kambale
1woman 1REL 1.AAE-like Kambale
'the woman that likes Kambale'

(25) Lubukusu Subject Relative (Diercks 2010: 115)
o-mu-seecha **o**-o-eba e-ndika
AUG-1-man 1.REL-1.AAE-stole 9-bicycle
'the man who stole the bicycle'

There are only a few documented exceptions to this co-occurrence and in these cases, this apparent divergence can be explained via morpho-phonological

[6] I will look closer at Lubukusu in Section 3.3.2 because new data suggest the need for a revised analysis of its relative clauses, in support of a similar approach for Kirundi.

processes. Two exceptional Bantu AAE languages are Kinyarwanda and Dzamba which display AAEs but appear to lack overt relative markers, as shown in (26) and (27), respectively.

(26) Kinyarwanda anti-agreement (Zorc & Nibagwire 2007: 235)

u-mū-ntu	u-sába
AUG-1-person	1.AAE-ask

'the person who asks'

(27) Dzamba anti-agreement (Bokamba 1976)

o-mo-to	ó-kpa-aki	imundondo
AUG-1-person	1.AAE-took-IPFV	jug

'the person who took the jug'

Regarding Kinyarwanda, I follow Henderson (2013) in assuming that an agreeing relative morpheme exists underlyingly but is not spelled out phonologically. This is in line with the fact that headed relative clauses never have overt relative markers and agreeing complementizers do not appear in any contexts in the language (Zorc & Nibagwire 2007). This could be driven by the Doubly-Filled COMP Filter (Koopman & Szabolcsi 2000) in that the presence of a relativized element in Spec,C prevents the spell-out of C itself. Ultimately, I assume that an agreeing relative morpheme still accompanies the presence of Kinyarwanda's AAEs underlyingly even if not realized on the surface, allowing for Kinyarwanda AAE facts to remain within the larger Bantu AAE phenomenon.

Regarding Dzamba, I similarly assume an unrealized relative marker in standard relative clauses; however, in this case, I propose that the absence of such a morpheme is driven by the wider Bantu morpho-phonological process called Kinyalolo's Constraint (Kinyalolo 1991, Carstens 2005), also analyzed as φ-feature dissimilation by Oxford (2018). Kinyalolo's Constraint states that "AGR on a lower head is inert iff its features are predictable from AGR on a higher head" (Carstens 2005: 253). In other words, when adjacent heads agree with the same goal, only the higher one is spelled out. This morphological constraint is found in many Bantu languages where it affects both C/T and T/Asp realization.

Kinyalolo's Constraint can be seen in Dzamba in non-AAE contexts such as (28). When nothing intervenes between C and T, there is only one morpheme expressing agreement in between the relativized subject and verb of a subject relative construction (28a). However, when negation intervenes, both a relative marker and a subject agreement morpheme – representing two heads – are spelled out (28b). A similar situation occurs in Sotho where a high adverb intervenes between C and T, as demonstrated in (29).

(28) Kinyalolo's Constraint in Dzamba (Bokamba 1976)

a. mukanda mú-tom-aki omwana
5letter 5-AGR-send-PFV 1child
'the letter that the child sent'

b. ma:kenge **má**-ta-mà-bung-aki o kalasi emba
6slate 6REL-NEG-6.AGR-lose-PFV at school NEG
'the slates which were not lost at school'

(29) Kinyalolo's Constraint in Sotho (Demuth 1995)

a. batho bá-pheháng dijó
2person 2.AGR-cook.DEP 8food
'people that cook food'

b. batho **báo** kajéno bá-pheháng dijó
2person 2REL today 2.AGR-cook.DEP 8food
'people that today cook food'

Crucially, Kinyalolo's Constraint also affects *AAE constructions* in these languages. And when it does, AAEs are unaffected. If negation is added to a relative clause with AAEs in Dzamba, the resulting construction will contain both a relative marker and a subject agreement morpheme expressing anti-agreement. This is demonstrated in (30) where both the relative marker and subject agreement appear.

(30) o-mo-to **ó**-ta-**ò**-kpa-aki imundondo emba
AUG-1-person 1.REL-NEG-1.AAE-took-IPFV jug NEG
'the person who did not take the jug' (Bokamba 1976)

Having shown that Kinyalolo's Constraint is in effect in Dzamba, we can now revisit the apparent lack of a relative marker in (27). Assuming that Kinyalolo's Constraint applies here, just as it does in other subject relatives without intervening elements between C and T, what we are seeing is not the absence of a relative marker, but the obliteration of subject-verb agreement as the *result* of a relative marker. Therefore, the morpheme glossed as AAE here is actually the relative marker itself, bringing Dzamba and languages with similar relative constructions back into the wider cross-Bantu picture in which AAEs and relative markers go hand in hand. In fact, with this revision, Dzamba patterns strikingly like Kirundi.

3.3 Kirundi AAEs also involve agreeing relative markers

In conjunction with the overt presence of agreeing relative markers in many Bantu AAE constructions, I take the morpho-phonological facts from Kinyarwanda and Dzamba to support the larger claim that agreeing relative markers are a necessary component of Bantu AAE constructions. By extension, I propose that AAE constructions in Kirundi must also possess agreeing relative markers even if it doesn't seem like such a morpheme is realized. To motivate this, I propose that Kinyalolo's Constraint applies to Kirundi subject headless relatives such that only one of the two underlying agreement heads – REL and AGR – is realized when the heads are adjacent. This derivation is represented step by step in (31).

(31) u-u-têka
AUG-**1.AAE**-cook
'the one who cooks'

a. AUG-**1.REL** **1.AGR**-cook (Underlying)
b. AUG- 1-REL ~~**1.AGR**~~-cook (Kinyalolo's Constraint)
c. AUG-**1.REL**-cook (Surface)

Note that what has been glossed in past examples as AAE in the surface form is in fact the realization of the relative morpheme, not the subject-verb agreement morpheme. In other words, what appears to be AAEs in Kirundi is the realization of a relative marker in the absence of a subject-verb agreement morpheme from T. I look closer at what this means for AAEs at large in Section 4.4.

Within the Bantu family and within Bantu AAE languages, it is not uncommon for Kinyalolo's Constraint to obfuscate the full array of elements involved in AAE constructions, as seen in the previous section. In languages like Dzamba, alternations between relative clauses with and without high negation or high adverbs provide clear evidence that Kinyalolo's Constraint is in effect. Therefore, it would not be surprising if Kirundi were to pattern similarly. However, unlike these languages, Kirundi does not appear to possess any elements which intervene between the two agreement heads in questions. To my knowledge, the language does not possess any high adverbs and negation in relative clauses always occurs after the subject agreement morpheme. As a result, these sorts of diagnostics cannot be used to prove that Kirundi exhibits Kinyalolo's Constraint.

That being said, support for the extension of Kinyalolo's Constraint to Kirundi subject headless relatives still exists. Drawing on both language-internal and cross-linguistic data, I argue that the presence of the augment on Kirundi subject headless relatives provides evidence that the proposed approach is on the right track. Specifically, I claim that the anti-agreement morpheme in Kirundi subject headless relatives must be a relative marker and not a subject agreement morpheme because the augment can only occur with nominal morphology such as noun class markers, complementizers, and relative markers.

3.3.1 The augment in Kirundi headless relatives

In Kirundi, the augment primarily occurs with nominal constructions that are phonologically realized and include nominal noun class morphology. This includes all the DPs we've seen so far but it does not include proper nouns or pronouns, which lack noun class prefixes (32).

(32) No augment without noun class markers

a. ***u**-Yohǎni
AUG-John
Intended: 'John' (Proper noun DP)

b. ***u**-jēwé
AUG-me
Intended: 'me' (Pronoun DP)

Syntactically, this aligns with a long line of work which argues that the augment is a D element (Ndayiragije et al. 2012) that selects for *n*/Num (spelled out as a noun class prefix) (Fuchs & van der Wal 2021). The presence of this noun class prefix is important because it is from this morpheme that the augment gets its form, via the copying or sharing of phonological features (Ntihirageza 2001, Niyondagara 1993).

These facts predict that the augment cannot occur when no overt nominal morpheme is realized. This prediction is borne out in examining the distribution of the augment in deverbal nominalizations and CP arguments.

Regarding deverbal nominalizations, the augment can appear with verb roots if there is also a noun class prefix, as in the case of the deverbal nominalization in (33a). However, it is impossible for the augment to appear directly on a verb root (33b) or on a verb inflected with subject agreement (33c). Though noun class prefixes and subject agreement morphemes both express the features of a noun, clearly they differ in their ability to co-occur with the augment. Only a noun class prefix is able to support an augment.

(33) Deverbal nominalization

a. **u**-ku-rima
AUG-15-cultivate
'the farming' (Deverbal nominalization)

b. ***u**-rima
AUG-cultivate
Intended: 'the farming' (Verb without noun class prefix)

c. ***u**-a-rima
AUG-1.AGR-cultivate
Intended: 'his farming (the farming by him)' (Verb with subject agreement)

Based on this, it would be surprising for the augment to appear as it does in Kirundi subject headless relatives if the anti-agreement morpheme in such constructions were the realization of a verbal T head. A subject agreement morpheme, despite its similarities with a noun class prefix, is not enough to support the presence of the augment.

Evidence that the augment must appear as the result of a relative marker, instead of T agreement, comes from instances in which complement clauses function as DP arguments. Complement clauses in Kirundi typically occur with the complementizer *kó*, as shown in (34). This complementizer can be broken down into two pieces: the noun class prefix *ku-*, denoting default Class 15, and the complementizer morpheme *-ó* (Morgunova 2023).

(34) A-ra-vuga [**kó** Yohăni a-ryá ibitōke.]
1.AGR-DJ-say that John 1.AGR-eat.DEP 8banana
'She says that John eats bananas.'

When a complement clause functions as an argument, as in the passive sentence in (35), it appears with an augment which takes its form from *kó*'s noun class prefix *ku-*. The result is a DP argument formed from a CP. When a complement clause is used as an argument, it is not possible for the augment or the complementizer to occur without the other.

(35) Augment with CP (Morgunova 2023: 23)
Kagabo a-á-tangāj-w-e n(a) *(û)-[*(kó) Kēza
Kagabo 1.AGR-PST-surprise-PASS-PFV PREP AUG-that Kēza
a-á-koze icǔmba cîwé].
1.AGR-PST-work.PFV 7.room 7.POSS.3S
'Kagabo was surprised by (the fact) that Kēza cleaned her room.'

This example demonstrates that even CPs can occur with the augment as long as they have a morpheme that can host a noun class prefix at their left edge. It is not insignificant that the complementizer morpheme in complement clauses resembles the relative marker morpheme in Kirundi's object headless relative clauses (as well as relative markers in many other Bantu languages) in deriving from an *-ó* form. I take these similarities to support the claim that the augment in subject headless relatives is allowed because the apparent AAE morpheme is a relative marker and not an agreement morpheme.

In sum, though Kirundi lacks the straightforward evidence for Kinyalolo's Constraint which exists Dzamba, there is nevertheless a case to be made for the application of Kinyalolo's Constraint to Kirundi headless relative clauses. Based on the properties of the Kirundi augment, it would not be possible for the augment to appear in subject headless relatives as it does unless the morpheme between the augment and the verb is the relative marker, expressing nominal morphology like the complementizer *kó*, and not a subject agreement morpheme expressing verbal morphology.

3.3.2 The augment in Lubukusu

Additional support for this claim comes from Lubukusu in which the application of diagnostics for Kinyalolo's Constraint reveals that the form of relative clauses in Lubukusu is strongly analogous to the Kirundi data. In Lubukusu, subject relative clauses possess two morphemes between the relativized subject and the verb. In past work such as Diercks (2010), these have been glossed as the relative marker and the subject agreement morpheme, as shown in (25) and repeated here.

(36) Lubukusu Subject Relative (Diercks 2010: 115)

o-mu-seecha	**o**-o-eba	e-ndika
AUG-1-man	1.REL-1.AAE-stole	9-bicycle

'the man who stole the bicycle'

While Diercks analyzes the first *o-* morpheme here as a relative marker, I argue instead that it is better explained as an augment, like we see at the left of Kirundi headless relatives. Notably, this morpheme has the same form as the Lubukusu augment and, as noted by an anonymous reviewer, the construction can function as a headless relative clauses if the head noun *omuseecha* is removed. We would expect any headless relative clauses denoting a definite entity to have an augment and the fact that the augment remains in headed contexts aligns with documented determiner spreading in relative clauses in other Bantu languages (see, e.g., Schneider-Zioga 2007, Asiimwe et al. 2023).

An additional point of evidence for this re-analysis is the application of Kinyalolo's Constraint diagnostics to Lubukusu relatives, presented in (37). When a high adverb appears directly before subject agreement, as in (37b), the standalone relative marker of object relatives is used and not the alleged relative marker *o-*. In other languages in which both the relative marker and a subject agreement morpheme co-occur (such as Kinande), we do not see the relative marker change form because of an intervening adverb but we *do* see this in cases where Kinyalolo's Constraint affects the realization of these agreement heads (see, for instance, (29) from Sotho, where the relative marker is *bao* when standing alone and *ba-* when adjoined to the verb).

(37) Kinyalolo's Constraint in Lubukusu (Michael Diercks, p.c.)

a. Wafula **o**-o-akaata ekhaafu
Wafula 1REL-1.AAE-slaughtered 9cow
'Wafula who slaughtered the cow'

b. Wafula **niye** wakana o-lakaata ekhaafu
Wafula 1REL perhaps 1.AAE-slaughtered 9cow
'Wafula who perhaps slaughtered the cow'

Assuming then that this first morpheme in Lubukusu relatives is the augment, I conclude that Lubukusu subject relative clauses have exactly the same form as their counterparts in Kirundi. Both include adjacent agreement heads – the relative morpheme and the subject agreement morpheme – which are reduced to just a relative morpheme as the result of Kinyalolo's Constraint. The remaining construction occurs with the augment which gets its form from the nominal morphology of the relative marker, as shown in (38). The only difference between the two languages is that this structure occurs in both headed and headless subject relatives in Lubukusu but only headless ones in Kirundi.

(38) Lubukusu Subject Relative Revised
o-mu-seecha **o**-o-eba e-ndika
AUG-1-man AUG-1.REL-stole 9-bicycle
'the man who stole the bicycle'

3.4 Summary

Kirundi AAEs pattern with other Bantu AAEs in that they satisfy all criteria of Bantu AAEs from previously documented analyses, as summarized in (18). Additionally, Kirundi headless relative clauses contain agreeing relative markers, just like other Bantu AAE constructions. One reason this claim has not been made in the past is that the application of Kinyalolo's Constraint to Kirundi, as in other Bantu AAE languages, makes the empirical picture in regards to anti-agreement morphology harder to assess on the surface. Nevertheless, morpho-phonological evidence from Kirundi and related languages supports the claim that Kirundi's headless relatives contain relative markers. This claim is noteworthy because it brings Kirundi AAEs under the umbrella of Bantu AAEs more generally.

With this in mind, a revised overview of Kirundi relative clauses is presented in Table 2.

Table 2: Kirundi relative clauses (revised)

NP_{REL}	REL	SUBJ	T	OBJ
SUBJ			AGR	OBJ
OBJ		SUBJ	AGR	
$SUBJ_{\varnothing}$	**REL**		**AGR**$_{\varnothing}$	OBJ
$OBJ_{\varnothing}$	**REL**	SUBJ	AGR	

In the next section, we turn to the question of why only *headless* relative clauses contain relative markers, and consequently AAEs.

4 An account of relative markers in Kirundi

One question that arises with headless relative clauses is whether a nominal head truly exists in the syntactic representation. In the case of Kirundi, answering this question is linked to determining what the syntactic status of the relative marker is. If it is a complementizer, located in C, then headless relatives look truly headless, composed of a D element selecting a CP with no head noun. If the relative marker is instead nominal, this points towards the presence of a nominal head, albeit one that differs in form from those found in headed relatives. In this section, I draw on evidence of the complementarity between determiners and relative markers to argue that the Kirundi relative marker is indeed a nominal element, in the spirit of Jenks et al. (2017) on Basaá. This claim paves the way for a head-raising analysis of Kirundi's relative clauses from which the restricted distribution of relative markers, and by consequence, AAEs, follows.

To support this proposal, Section 4.1 examines the distribution of D elements in Kirundi relative clauses, ultimately demonstrating that the relative marker is in competition with demonstratives and the augment. I propose that this complementarity gives credence to an approach in which the relative marker is an overt relative operator, like English *which*, which competes with determiners for a place in D. Drawing on this, Section 4.2 addresses why overt relative operators only appear in headless relative clauses. Taking up a head-raising analysis of relative clauses, along the lines of Kayne (1994) and the majority of past work on Bantu relative clauses, I argue that the realization of an overt relative operator in D is a consequence of whether said operator takes an overt or null NP as its complement. Section 4.3 presents a full sketch of the derivation of both headed and headless relative clauses based on the established claims. This sketch demonstrates one viable manner in which to generate the expected distribution of relative markers and AAEs in Kirundi. Finally, Section 4.4 returns to the issue of anti-agreement, discussing the implications of the overall proposal for the status of AAEs in Kirundi in relation to prominent past analyses of the phenomenon.

4.1 Complementarity of determiners and relative marker

In their analysis of Basaá, Jenks et al. (2017) show that the relative marker is in complementary distribution with demonstratives. They argue that this complementarity is a result of the fact that demonstratives can function as relative operators (in line with numerous accounts of grammaticalization from demonstratives to relative marker in Bantu; see, e.g., Diessel 1999, Heine & Kuteva 2002) and that both demonstratives and relative operators occupy the same syntactic position in the noun phrase. In this sense, the relative marker of Basaá functions like English *which* when analyzed as an overt relative operator.

A similar observation occurs in Kirundi where the relative marker is in complementary distribution with demonstratives as well as the augment. Headless relative clauses cannot appear with demonstratives (39b). If a demonstrative is used, it is illicit to have a relative marker (39c); in other words, demonstratives can only appear in headed relative clause constructions.

(39) a. u-u-ó Muco a-kūndá
AUG-1-REL Muco 1.AGR-like.DEP
'the one that Muco likes' (REL)

b. *uwo u-ó Muco a-kūndá
1.DEM 1-REL Muco 1.AGR-like.dep
Intended: 'that one that Muco likes' (*DEM + REL)

c. uwo Muco a-kūndá
1.DEM Muco 1.AGR-cook.DEP
'that one that Muco likes' (DEM)

As evidenced by the examples above, the demonstrative and relative marker morphemes look quite similar. In fact, the only phonological distinction is the presence of a high tone on the relative marker. That being said, despite their similarities, the two are clearly distinct in their distribution. The relative marker can only appear in relative clauses while demonstratives can occur in any context. Based on this observation, I follow Jenks et al. (2017) in positing that the relative marker in Kirundi is a relative operator in complementary distribution with demonstratives. Like a demonstrative, it appears with NPs; however, unlike standard demonstratives, it only occurs with NPs that head relative clauses, giving credence to its status as a relative morpheme.

One aspect in which Kirundi differs from Basaá is that demonstratives and the augment are also in complementary distribution in Kirundi. In Basaá, demonstratives can occur pre- and post-nominally and when they occur post-nominally, an augment appears on the noun. Jenks et al. (2017) argue that this is because demonstratives are generated post-nominally but can move to Spec,DP to provide focus readings. When this occurs, an augment in D is not expressed due to the Doubly-Filled COMP Filter.

In Kirundi, demonstratives can never occur post-nominally and can never appear with an augment, as seen in (40). Note that this same pattern is true for all demonstratives in the language, not just the one form represented here.

(40) a. uwo mū-ntu
1DEM 1-person
'that person' (Pre-nominal DEM)

b. *uwo u-mū-ntu
1DEM AUG-1-person
Intended: 'that person' (*DEM + AUG)

c. *u-mū-ntu uwo
AUG-1-person DEM
Intended: 'that person' (*Post-nominal DEM)

To support this data, I follow Ndayiragije et al. (2012),Carstens & Mletshe (2016), Gambarage (2019) and many others in locating both the augment and demonstratives in the same position, D. Crucially, this places the augment, demonstratives, and the relative operator in competition. This claim has a number of benefits.

First, as seen in the previous section, the presence of the augment in headless relatives suggests that the relative marker is nominal. Second, treating the morpheme as a functional head allows it to function like other functional heads in Bantu languages which trigger movement and partake in agreement.[7] This proposed complementarity is borne out in attempts to create relative clauses which include both the augment – on an overt NP – and the relative marker, as demonstrated in (41). In these examples, we see that if a relativized noun already has an augment, the relative operator cannot occur.

(41) a. *u-mū-ntu u-ó Muco a-kūndá
AUG-1-person 1-REL Muco 1.AGR-like.DEP
Intended: 'the person that Muco likes' (*AUG + REL)

b. u-mū-ntu Muco a-kūndá
AUG-1-person Muco 1.AGR-like.DEP
'the person that Muco likes' (AUG)

In sum, I place the augment, demonstratives, and the relative marker all in the same syntactic position, specifically D, to account for their complementarity. Additionally, I analyze the relative marker as an overt relative operator which originates with the NP head in the relative CP. These claims give us the ingredients needed to account for the distribution of Kirundi's relative markers within a head-raising analysis of relative clauses. However, before presenting a full analysis of the derivation, there is one additional question to address: why overt relative markers are *only* found in headless relatives.

4.2 Overt relative operators in headless relative clauses

As it stands, the current proposal accounts for the lack of an overt relative operator when the nominal head of a relative clause includes another D head. In these cases, a relative operator is in competition with demonstratives and the augment. However, this set-up does not account for the fact that overt nominal heads *cannot* appear with an overt relative operator. To resolve this issue, I make two additional claims. First, I propose that both headed and headless relative clauses

[7] It could also be the case that demonstratives are generated below D and then move to a Spec,DP position, as proposed by Carstens (1991) and others, or that the augment is only spelled out when D is empty and N finishes N-to-D movement. For the purposes of this paper, it's not crucial exactly where demonstratives originate as long as they are in competition with relative markers and the augment. This can be accounted for in a number of ways depending on one's approach to Bantu DP structure.

in Kirundi have nominal heads but that these heads differ in whether they are phonologically realized. In the case of headless relative clauses, they are overt nouns (noun class prefix + root). In the case of headless relative clauses, I suggest that the heads are empty nouns, in line with Panagiotidis (2003), where an empty noun is a functional nominal category which encodes syntacto-semantic features such as noun class. In this sense, an empty noun includes the Num and *n* heads of a standard Bantu noun but lacks a lexical root. This categorization facilitates the open interpretation of headless relative clauses as any entity within a given noun class while allowing for the empty noun to take part in agreement. It also explains how the relative operator can appear in headless relative clauses; the operator takes an empty noun as complement and moves to Spec,CP.

One implication of this claim is that an empty noun is able to appear with other determiners besides the relative operator. In the sense that an empty noun functions like an NP, any D head can take it as a complement. Based on this, we would expect to see relative heads that only consist of a demonstrative or augment. The first is indeed possible, as seen in (39c) where a demonstrative heads a relative clause with no operator morpheme. However, the second option, in which the augment appears with an empty noun, is not found. I propose that this construction is illicit because the augment requires an overt nominal element in order to appear, barring co-occurrence with an empty noun.

The second additional claim aids in restricting the presence of overt relative operators to relative clauses with empty noun heads. Following Jenks et al. (2017), I take the relative operator to trigger movement of the head noun to its specifier.[8] This movement can be justified in a number of ways. In the Bantu fashion, we could say that the operator hosts an [EPP] feature linked to φ-agreement, resulting in both movement and the apparent agreement we see on the operator's *-o* morpheme (Carstens 2005).[9]

From here, I argue that the presence or absence of the relative operator is determined by whether the head noun in its specifier is phonologically realized or not. In line with a general Doubly-Filled COMP Filter, the operator can only be spelled out when its specifier appears empty. When an overt noun appears with

[8]To avoid movement that is too local, I take that some intervening projection also exists. A projection like Agr is common in the Bantu tradition so there could be such an intervening projection here. For the purposes of simplicity, I will leave this level out in the representations below.

[9]Alternately, we could propose that the NP moves higher than Spec,DP, for instance, to a higher CP shell (Zwart 2000), a higher *n* head (de Vries 2002), or to some reprojection above CP (Bhatt 2002). Regardless of the exact mechanics, what's crucial is that the relative operator triggers movement and featural agreement with the head noun.

the relative operator, its realization is blocked but when an empty noun appears with the operator, its realization is allowed (and required). These two structures are represented in (42).

(42) Realization of the relative operator with overt and empty nouns

a. $[_{DP}$ mūntu$_i$ $[_{D'}$ Op $[_{NP}$ t_i]]] (Overt head noun)

b. $[_{DP}$ $\varnothing_i$ $[_{D'}$ -ó $[_{NP}$ t_i]]] (Empty head noun)

With this in place, we can now account for the distribution of relative markers in Kirundi as a result of the selection of different head nouns by different D elements. Because the relative operator is in competition with the augment and demonstratives, it cannot occur with nouns that are selected by these D elements. When it does select a noun, it is only realized if that noun is phonologically null. As a result, only empty head nouns occur with relative markers. These facts are summarized in Table 3 which shows the distribution of D elements with head nouns in Kirundi relative clauses.

Table 3: Distribution of D elements in relative head nouns

	Overt head noun	Empty head noun
Augment	+	-
Demonstrative	+	+
Overt Operator	-	+
Covert Operator	+	-

4.3 A head-raising analysis of Kirundi relative clauses

Having motivated the distribution of relative markers in Kirundi relative clauses, I turn now to a full derivation of their structures, following a head-raising analysis.[10] First, I present the derivation of a relative clause that lacks an overt relative operator – in other words, headed relative clauses. This includes all relative clauses in which the head noun occurs with a demonstrative or augment. The structure shown in Figure 1 includes a relative head that was selected by a demonstrative.

[10] Unlike other approaches, such as the operator-movement and matching analyses, head-raising lends itself uniquely to our claims about the complementarity of demonstratives and relative operators because head nouns must originate within the relative CP.

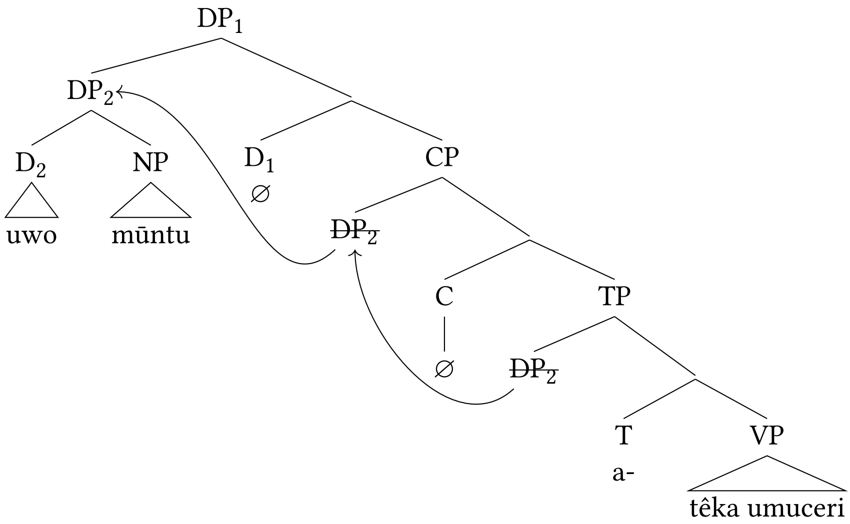

Figure 1: Headed relative clause

Under a head-raising analysis, a D head (D_1) takes the relative CP as its complement. The DP which includes the relativized nominal moves to the specifier of CP as a result of Ā-movement. C is not spelled out due to the Doubly-Filled COMP Filter. At this point, the nominal head (NP) must move again because D_1 requires a noun in its minimal domain (Bianchi 2000) and, at the moment, the presence of the demonstrative in D_2 is preventing this.[11] Consequently, the whole DP moves to the specifier of the higher D (Spec,DP_1). This satisfies the minimal domain requirement and also prevents the spell-out of an augment in the D below it, again as a result of the Doubly-Filled COMP Filter. The resulting structure generates the desired form for relative clauses which lack relative markers.

In the case of headless relative clauses, or those which possess an overt relative operator, the derivation is slightly different as a result of differences in the makeup of the relativized nominal. The overall structure of a headless relative clause is shown in Figure 2. In this representation, an empty noun is selected by a relative operator. If this noun were overt, the same structure as shown here would occur but the operator would not be realized, as proposed in Section 4.2. Unlike in the previous derivation, the occurrence of a relative operator in this structure drives movement of the head NP to a higher position, in this case, Spec,DP_2. As a result, once the relativized nominal reaches Spec,CP, the minimal

[11]"The minimal domain of a head X includes all categories that are immediately dominated by, and do not immediately dominate, a projection of X" (Bianchi 2000: 128).

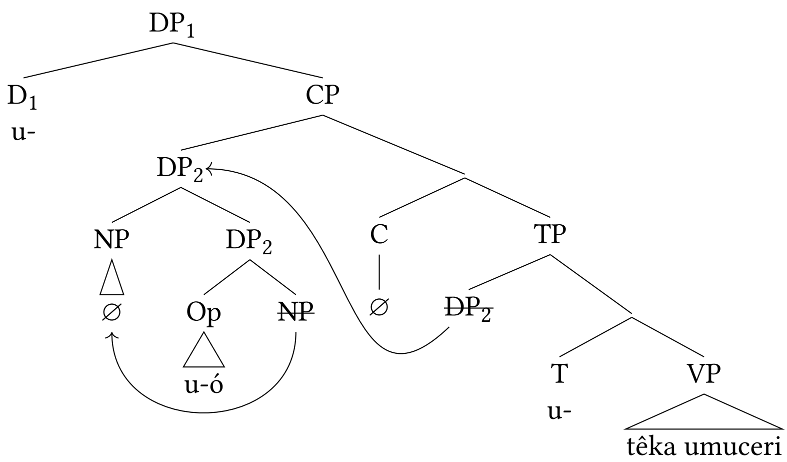

Figure 2: Headless relative clause

domain requirement of D_1 is met without further movement. The augment in D_1 is realized and the structure is complete, resulting in the headless relative clause form expected. While explaining the complementarity of demonstratives and relative markers in Kirundi, the above analysis also accurately accounts for the asymmetrical distribution of relative markers, and by extension, anti-agreement effects.

4.4 Kirundi AAEs and implications

With the structure of Kirundi relative clauses in place, we can return to the issue of anti-agreement effects. As the puzzle at the start of this paper noted, it is surprising that Kirundi appears to exhibit AAEs as this has been argued against in the literature and as these AAEs seem far more restricted than those in other Bantu AAE languages. However, the presence of such morphological facts falls out naturally from the structure adopted here. In addition, this proposal has the potential to fit within past analyses of the phenomenon.

As shown in Section 3.2, the key to this analysis is Kinyalolo's Constraint. In examining other Bantu languages, Kinyalolo's Constraint was applied to duplicative agreement on adjacent heads in C and T. In addition to this, I propose that the D head instantiated by the relative operator in Kirundi headless relative clauses is also a viable agreement head for the purposes of Kinyalolo's Constraint. As a result, when this head appears with no overt intervening elements between it and T, as in the structure presented for Kirundi headless relative clauses above, and these two heads agree with the same element, Kinyalolo's Constraint results in the obliteration of the lower morpheme. This accounts for the surface form of subject headless relative clauses which only possess a relative morpheme and no subject-agreement morpheme. It is this surface form that is interpreted as anti-agreement effects. This also accounts for the absence of AAE morphology in subject headed relative clauses where the relative operator is either blocked by the presence of another D element or is not realized due to the presence of an overt NP in its specifier.

Notably, the result of Kinyalolo's Constraint in this case actually covers up any evidence of true AAEs, as they are defined as the suppression of argument-verb agreement. Because there are no elements that intervene between the two agreement heads in Kirundi, Kinyalolo's Constraint always gets rid of the subject-verb agreement morpheme of T. The only agreement head that is actually realized in these headless relatives is the relative operator which always expresses Class 1 agreement with the morpheme *u-*, in line with other D elements and adnominal modifiers in the language. This is different from languages like Lubukusu where the obliteration of T in subject relatives obfuscates AAEs when nothing intervenes between C and T but the presence of intervention reveals that agreement in T *also* takes the alternative form that agreement in C does. As a result, evidence is ultimately inconclusive on whether true AAEs do exist for Kirundi.[12]

That being said, the analysis proposed above opens the door for the derivation of AAEs in Kirundi and can in fact fit into the major past analyses proposed to account for AAEs in Bantu and beyond. To illustrate this, let's take a quick look at how the structure from this proposal can facilitate the AAE analysis of Henderson (2013) and Baier (2018), two of the most prominent works on AAEs. Henderson (2013) argues that Bantu AAEs are the result of agreement between the ϕ-features in C and T where C's [person]-impoverished features overwrite those in T. C-T agreement must occur when a local subject moves from one position that requires spell-out to another. In the Kirundi picture, this approach can be accommodated if we extend the movement chain in question to include

[12] A reviewer asks why AAEs do not appear on all verbs in compound tense constructions, as seen in (23) if Kirundi does possess true AAEs. In fact, the absence of alternative morphology on lower verbs in subject relative compound tense constructions is also found in other Bantu AAE languages. I take this cross-linguistic pattern to be evidence for a raising approach to compound tense, as proposed by Carstens (2008), wherein the subject raises through the specifier of each verb. In the case of AAE contexts, anti-agreement only occurs after the verb has reached T and lower agreement has already occurred. The absence of AAEs on lower verbs is strong evidence against the competing concord approach, proposed by Henderson (2011) wherein agreement in T triggers concord across all lower verbs. This would predict the spread of AAEs across the entire construction.

the D head of the relative operator as well. In this way, the necessary spell-out of features in the relative operator leads to the same features, lacking [person], being passed down to C and T. In the case of Kirundi, however, neither C nor T are actually realized in these constructions. In general, then, this analysis, and the importance placed on the spell-out of features above T, meshes well with the current proposal.

However, there is one element of Henderson's account that requires revision due to the Kirundi data. The account suggests that Kirundi lacks features in C entirely because it does not possess overt relative markers or AAEs. The data above provide a strong case that this is not true, especially in the context of agreement between adjacent heads. If we accept the potential for D-C-T agreement in Kirundi, C would need to have features in order to participate. If we reject the theory proposed in this paper and take that Kirundi's headless AAEs are a result of C-T agreement, C would still need to have features. Additionally, such a claim runs into issues with feature inheritance theory (Chomsky 2008) whereby C must always have features since it passes its features down to T. As a result, it seems necessary to posit the presence of features in C in Kirundi regardless of the theory adopted.

In addition to accommodating the most prominent Bantu AAE theory, the present proposal also fits well within the Ā-driven feature impoverishment approach of Baier (2018) which aims to account for AAEs across all languages. Baier argues that probes copy back both φ- and Ā-features. When feature bundles contain both such features, Ā-features can trigger feature impoverishment with [person] features being the first to be deleted. In the case of Kirundi, this analysis can be applied if we assume that the DP-internal movement triggered by the relative operator is associated with a unique type of Ā-feature which drives feature impoverishment. If so, the copy-back of this feature to T would result in AAEs. Crucially, the feature involved in Ā-movement to Spec,DP must be different than that involved in Ā-movement to Spec,CP. In Kirundi, the DP Ā-feature triggers anti-agreement but the CP one does not; then, in other Bantu languages, the inverse is true.

In sum, then, while it is hard to determine definitively if Kirundi has true AAEs or just the effects of Kinyalolo's Constraint with a nominal functional head, there is room to account for AAEs under past analyses.

5 Conclusion

This paper presented an overview of relative clauses in Kirundi with a focus on the unique properties of headless relative clauses. Unlike headed relative clauses which lack relative markers and exhibit standard subject agreement, Kirundi's headless relative clauses have both relative markers and apparent anti-agreement effects. I demonstrated that the presence of relative markers in subject headless relatives in Kirundi is obfuscated by the occurrence of Kinyalolo's Constraint, a common morpho-phonological process in related constructions across Bantu languages. To account for the distribution of relative markers in Kirundi, I proposed an analysis based on the complementarity of relative markers and determiners, arguing that Kirundi's relative markers are in fact relative operators in D which are realized only when they agree with empty head nouns. Because only headless relative clauses contain empty head nouns, it is only such relatives that exhibit agreeing relative markers. Having established this syntactic structure, I connected the presence of agreeing relative operators to apparent AAEs, concluding that what appears to be Bantu AAEs in Kirundi could also just be the application of Kinyalolo's Constraint.

The focus of this paper contributes to the typology of relativization in Bantu as well as our understanding of the anti-agreement cross-linguistically. The presence of AAEs in such a subset of subject extraction contexts has not been documented for any other Bantu language. This empirical picture brings past analyses of anti-agreement effects into question and suggests that the phenomenon may occur in a wider range of contexts than originally expected.

Finally, this paper brings up several fruitful avenues for future research. First, though I have briefly addressed the connection between the proposed structure and anti-agreement, a deeper analysis would be beneficial. Second, due to the parametrical context of Bantu syntax, it would be interesting to see if other languages which exhibit complementarity in relative markers and determiners may shed more light on the anti-agreement puzzle. Finally, there are elements of Kirundi headless relative clauses which are still not fully understood, such as the alternation between headed and headless relative tone.

Abbreviations

Abbreviations used in glosses are as follows: AAE – anti-agreement morpheme; AGR – agreement morpheme; AUG – augment; DEP – dependent clause; DJ – disjoint; HRC – headless relative clause. Noun class is marked by number.

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Chapter 21

Deriving OV word order in Kono

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The Mande languages are known for their strict SOVX word order (Gensler 1994, Creissels 2005, Nikitina 2009). Following analyses laid out in Koopman (1992) and Smith (2022, 2024a), we argue that Kono's surface OV word order is derived from underlying VO order. Evidence includes a class of complex predicates whose internal arguments obligatorily follow the verb, CP complements which occur post-verbally, and coordinated direct objects, in which the coordinator and second conjunct can occur post-verbally. The internal arguments of canonical verbs raise for case-licensing, while the internal arguments of complex predicates remain post-verbal as they are case-licensed by their encoding adposition.

1 Introduction

The Mande languages of West Africa have been noted for their strict SOVX word order (Gensler 1994, Creissels 2005, Nikitina 2009). In this paper we suggest that Kono (ISO 639-3 kno), a Mande language spoken in Sierra Leone, has canonical O-V word order, which surfaces only in a specific (albeit frequent) context, namely with DP objects of canonical verbs. We show three contexts in which the verb precedes its internal argument, including particle-verb-like constructions (V- [O – Prt]), verbs with CP complements, and split direct objects (O_1 - V – *and* O_2). We argue that these non-canonical constructions point towards an underlying VO order.

Typical of the Mande family, Kono has canonical OVX word order, as in (1) where the object *Mɛli* precedes the verbs *yian* 'introduce' and *yen* 'see.' Dative

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objects always appear post-verbally, as in (1a), while (1b) shows a post-verbal temporal adverb, and (1c) shows a post-verbal locative.[1]

(1) a. SOVX - Dative
Pìtà à **Jɔ̀n** yìàn Mɛ̀lì yà.
Peter PFV John introduce Mary to
'Peter introduced John to Mary.'

b. SOVX - Temporal Adjunct
Pìtà nàà **Mɛ́lì** yén-fán kùnù.
Peter PFV Mary saw-FOC yesterday
'Peter saw Mary yesterday.'

c. SOVX - Locative Adjunct
Pìtà nàà **Mɛ́lì** yén màkìtì yɔ̀.
Peter PFV Mary saw market at
'Peter saw Mary at the market.'

The data in (2) and (3), however, show that the verbs *chian* 'fear' and *beya* 'follow' obligatorily take their internal argument in a post-verbal position, encoded in a phrase headed by the "particles" *ma* and *fɛ* respectively, which otherwise function as postpositions in the language. The data in (2b) further illustrates that the internal argument cannot precede the verb, whether *ma* surfaces with it or not, while the data in (3) show that the internal argument occurs post-verbally, but can surface above or below various adjuncts.[2]

(2) Complex Predicate

a. Pìtà nàà chìàn-fán **Mɛ́lì** mà.
Peter PFV fear-FOC Mary PRT
'Peter feared Mary.'

b. Pìtà nàà {*Mɛ́lì (mà)} chìàn-fán {**Mɛ́lì** mà}.
Peter PFV Mary PRT fear-FOC Mary PRT
'Peter feared Mary.'

(3) Complex Predicate

[1]The third author is a native Kono speaker, and we use his grammaticality judgments. Given that this is new research on the language, we do not have every word glossed.

[2]Following Kaiser (2011), we mark *fan* as FOC, a focus marker. Further evidence that *fan* is a focus marker is that it cannot co-occur in wh-constructions or with negation, as seen in (10) below. See Smith (2024a) and Smith (2024b) for a similar analysis in Mende.

a. Pìtà à béyà {kùnù} Mɛ̀lì fɛ́ {kùnù}.
Peter PFV follow yesterday Mary PRT yesterday
'Peter followed Mary yesterday.'

b. Pìtà à béyà {mákìtì yɔ} Mɛ̀lì fɛ́ {mákìtì yɔ}.
Peter PFV follow market at Mary PRT market at
'Peter followed Mary at the market.'

c. Pìtà à béyà {núndɔ} Mɛ̀lì fɛ́ {núndɔ}.
Peter PFV follow secretly Mary PRT secretly
'Peter secretly followed Mary.'

Descriptively, we propose that Kono has two classes of verbs: the first is represented by verbs like *yian* 'introduce' and *yen* 'see,' while the second is represented by *chian* 'fear' and *beya* 'follow.' We label these classes as canonical verbs and complex predicates respectively. We argue that these classes are distinguished by how the verb's internal argument is case-licensed. Empirically, our objective is to demonstrate that Kono is canonically, but not strictly, OV. Analytically, our objective is to argue that Kono has an underlying VO word order and that its surface OV word order is derived via case-driven, leftward movement (Kayne 1994, Kandybowicz & Baker 2003).

The analysis for which we argue is set out in Figure 1. In brief, we propose that the verb's internal argument raises from a post-verbal underlying position, surfacing in the specifier position of a Kase Phrase. The subject subsequently raises into a higher position.

The remainder of this paper is structured as follows. Section 2 is background data on Kono, as well as a brief review of analyses of OV word order in Mande languages. Section 3 considers canonical verbs, while Section 4 investigates complex predicates. Section 5 looks at CP complements, and Section 6 introduces coordinated direct objects. Section 7 is a conclusion.

2 Background data

There are roughly 340,000 Kono speakers, mostly in eastern Sierra Leone. Kono is a Western Mande language with two major dialects – Northern Kono and Central Kono (Eberhard et al. 2024) – and is closely related to Vai (Kaiser 2011), though the two languages are geographically separated by Mende. Kono is not taught in the public school system of Sierra Leone, and Kaiser (2011) claims that it is commonly spoken in the home in rural areas, while not spoken as frequently at home in urban areas. Given the pressure to speak English in the country, along

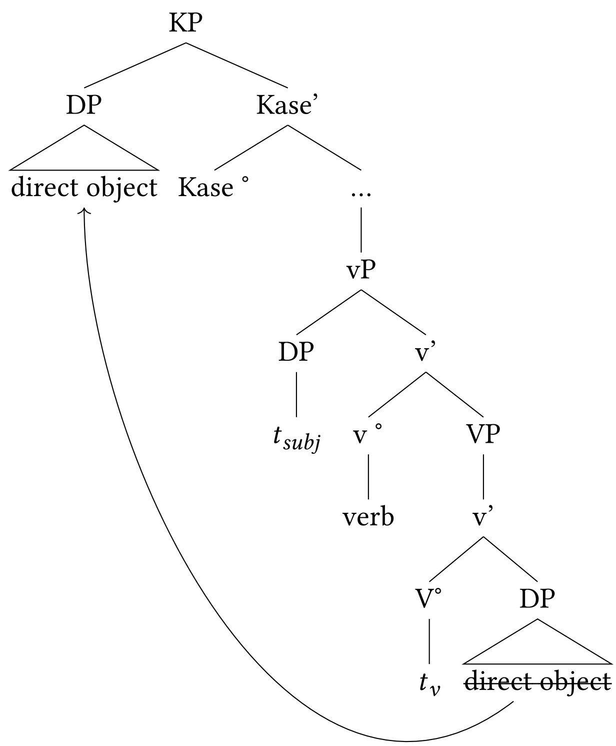

Figure 1: Canonical verb structure

with the prevalence of Krio and English, the language is at risk of losing speakers due to language shift (cf. Kanu n.d., Gellman 2020 on language loss in Sierra Leone). Manyeh (1983) also asserts that Kono is much less studied than other Sierra Leonean languages such as Mende, Themne, Limba, and Krio, noting that it has a short and poor literary history.

To this point there has been no syntactic analysis of Kono. Descriptive work includes a grammar by Kaiser (2011). Most other work has focused on phonology (Manyeh 1983, Foday-Ngongou 1985), and tense (Jimissa & Mansaray 2022).

Within the broader Mande language family, there is also a dearth of syntactic analyses of word order. We highlight two analyses to give perspective on how Mande OV word order has been analyzed. The first analysis is Nikitina (2019), an analysis of Wan, a Mande language spoken in the Ivory Coast. Typical of Mande languages, Wan has OV word order.

(4) Wan (Nikitina 2019: #1)
È só klā [PP sógò tā].
3SG cloth put:PST horse on
'He covered the horse with cloth (= put cloth on the horse).'

Nikitina argues that Wan is head-final, such that the DP object *sɔ́* 'cloth' merges into a pre-verbal position. She further argues that all post-verbal material, such as the PP *sógò tā* 'on the horse' adjoins at the IP level and does not form a constituent with the verb phrase. Her analysis is laid out in Figure 2.

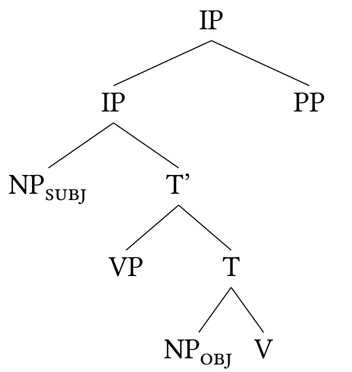

Figure 2: Clausal structure of Wan (adapted from Nikitina 2009: #19)

We argue that this analysis cannot work for Kono. In Kono, a PP modifier of the internal argument can in some cases appear in a pre-verbal position adjacent to the argument (5a) or can surface in an extraposed position to the right of the verb (5b). As seen in (5c), however, the 3rd person plural pronoun *an* can replace the entire DP object, that is, the pre-verbal DP object *mango-nu* 'the mangoes' and post-verbal PP modifier *tebu ma* 'on the table'.[3] Based on this, we argue that the DP is a constituent.

(5) Constituency
a. Jɔ́n á **mángò-nù tébù mà** à àn dàún fán.
John PFV mango-PL table on A AN eat FOC
'John ate the mangoes on the table.'

[3] See Smith (2024a) for similar data and analysis in Mende.

b. Jɔ́n á **mángò-nù** à àn dàún fán **tébù mà**.
John PFV mango-PL A AN eat FOC table on
'John ate the mangoes on the table.'

c. Jɔ́n á **àn** dàún fán.
John PFV 3PL eat FOC
'John ate them.'

The analysis that we set out in this paper is based on Koopman (1992), an investigation of Bambara, and Smith (2024a), an investigation of Mende. Koopman argues that Bambara is underlyingly VO and that its OV word order results from case-driven movement, with the object moving to SpecVP and the subject moving to SpecIP respectively.

She presents data on two types of verbs – those that select for an NP object and those that select for a PP object. Verbs such as *min* 'drink' in (6a) select an NP object which raises from its post-verbal position into SpecVP to be assigned accusative case. Verbs such as *bò* 'visit' select for a PP, which does not need to receive case and remains post-verbal (6b).

(6) Bambara (Koopman 1992: 6a and 7a)[4]

a. Den ye **jin** min.
child PFV water drink
'The child drank water.'

b. N bò-ra **i** ye.
1SG visit-PFV 2SG at
'I visited you.'

Based on Koopman's analysis of Bambara, Smith (2022, 2024a) proposes a similar analysis for Mende. Mende is canonically OV but has two classes of complex predicates. The verb *lɔ* 'see' is a canonical verb with a pre-verbal internal argument (7a), while the verb *ja* 'touch' takes its internal argument in a post-verbal phrase headed by the preposition *a* (7b), and the verb *lema* 'forget' takes its internal argument in a post-verbal phrase headed by the postposition *ma* (7c).

(7) Mende

[4] Koopman suggests that the two allomorphs of the PFV aspect marker *ye* and *ra* are conditioned by the licensing of the verb's internal argument. The marker *-ra* appears with transitive verbs that select a prepositional phrase encoded DO and which do not assign structural accusative case, while the dummy INFL marker *ye* appears with transitive verbs that do assign it.

a. Kpàná **níkè-í-síà** lɔ̀-í lɔ̀.
Kpana cow-DEF-PL see-PFV NF
'Kpana saw the cows.'

b. Kpàná jà-í lɔ̀ à **níkè-í-síà**.
Kpana touch-PFV NF at cow-DEF-PL
'Kpana touched the cows.'

c. Kpàná lemà-í lɔ̀ **níkè-í-síà** mà.
Kpana forget-PFV NF cow-DEF-PL on
'Kpana forgot the cows.'

Smith (2022, 2024a) argues that the objects of canonical verbs raise for Case, while the objects of complex predicates remain in a post-verbal position, where they are assigned Case by the encoding adposition (cf. Stowell 1981, McFadden 2004, Asbury 2008). In this paper, we extend this analysis of Mende to Kono, proposing that the objects of canonical verbs raise for Case, while the objects of complex predicates do not raise. Instead, they are Case-licensed by their adposition.

3 Canonical verbs

In this section, we investigate canonical verbs, that is, verbs whose internal argument precedes them. In doing so, we compare the variation between OV and VO word order in Kono with Gungbe and Dutch, languages in which an OV – VO variation also surfaces. Crucially, in the tense and aspect configurations that we have studied in Kono, the internal arguments of canonical verbs always precede the verb.[5,6]

(8) a. Perfective
Jɔ́n nàà **mángò-nù** dàún fán.
John PFV mango-PL eat FOC
'John ate the mangoes.'

b. Progressive
Jɔ́n ɛ́ **mángò-nú** dàún dà.
John COP mango-PL eat VSUXF
'John is eating the mangoes.'

[5]While we have not yet investigated every tense / aspect construction, in those that we have, when there is a canonical verb, the order remains OV. See Smith (2024a) for a similar argument for Mende.

[6]We follow Kaiser (2011) in marking ɛ́ in (8b) and (8c) as being a copula.

c. Future
Jɔ́n ɛ́ **mángò-nú** dàún dà wán.
John COP mango-PL eat VSUXF FOC
'John will eat the mangoes.'

This differentiates Kono from the Gbe languages Aboh (2004) and Nupe Kandybowicz & Baker (2003), where the aspectual status of the clause determines whether the object precedes or follows the verb.

(9) Gungbe (Aboh 2004: 192)
a. Imperfective (OV)
Kɔ̀jɔ́ tò [$_{DP}$ àmì lɔ́] zân.
Kojo IMPERF [oil SPF[+DEF]] use-NR
'Kojo is using the specific oil.'
b. Perfective (VO)
Kɔ̀jɔ́ tò [DP àmì lɔ́].
Kojo use-PRF [oil SPF[+DEF]]
'Kojo used the specific oil.'

In Kono, neither an indefinite object (10a), negation (10b), nor a wh-construction (10c), affect the word order. Note, further, that in (10b) and (10c), there is no focus marker, and the OV word order remains.

(10) a. Indefinite
Jɔ́n nàà **mángò-nù** dàún fán kùnù.
John PFV mango-PL eat FOC yesterday
'John ate mangoes yesterday.'
b. Negation
Jɔ́n nàà má **mángò-nú** dàún-ní kùnù.
John PFV NEG mango-PL eat-NEG yesterday
'John did not eat the mangoes yesterday.'
c. Wh-question
Jɔ́n nà **fén** dàún kùnù.
John PFV what eat yesterday
'What did John eat yesterday?'

Distinguishing Kono from the Germanic languages, whether the verb surfaces in a matrix clause (e.g. (10a)) or an embedded context (11), the internal argument of a canonical verb precedes the verb.

(11) Embedded
Mɛ̀lì àà mìn mbɛ́ John á nàà mángò-nú dàun fán.
Mary 3SG.PFV hear C John 3SG.PFV PFV mango-PL eat FOC
'Mary heard that John ate the mangoes.'

Contrastingly, in Dutch the internal argument follows the verb in a main clause (12a) but precedes it in an embedded clause (12b).

(12) a. Main Clause VO (Zwart 2012: 22)
Jan kust Marie.
John kisses Mary
'John kisses Mary.'
b. Embedded Clause: OV (Zwart 2012: 24)
dat Jan Marie kust
that John Mary kisses
'… that John kisses Mary.'

In summary, we suggest that word order for canonical verb constructions in Kono is not influenced by tense, aspect, negation, wh-questions, definiteness/indefiniteness, or whether it surfaces in a matrix or embedded clause. Whenever there is a canonical verb with a non-coordinated DP object, the DP object will occur in a pre-verbal position.

4 Complex predicates

While the internal arguments of canonical verbs in Kono always occur in a pre-verbal position, the internal arguments of complex predicates always occur in a post-verbal position, encoded in an adpositional phrase. The data in (13a) shows that the verb *nyina* 'forget' has a post-verbal internal argument encoded in a phrase headed by the particle *ma*. (13b) shows that the internal argument cannot appear pre-verbally, whether or not the particle surfaces with it, while (13c) demonstrates that the internal argument cannot appear pre-verbally with the particle stranded in a post-verbal position.[7]

[7] Additional complex predicates in Kono include *beya* 'follow,' *chinjinda* 'forgive,' *tuwa* 'touch,' *fɛnebiwa* 'surprise,' *tawa* 'visit,' *ayinachiwa* 'remember,' *tumuwa* 'need,' *komu* 'obey,' *ka* 'avoid,' *biya* 'chase,' and *suawa* 'defend.' We suggest that these verbs are lexically unpredictable, though there is a great degree of overlap with the two classes of complex predicates in Mende, which include *lema* 'forget,' *to* 'follow,' *manu* 'forgive,' *gɔla* 'surprise,' *folo* 'visit,' *kpei*, 'need', and *kolo*

(13) Complex Predicates

a. Jɔ́n à nyìnà wàn **Mɛ́lì** *(ma).
John PFV forget FOC Mary PRT
'John forgot Mary.'

b. *Jɔ́n à **Mɛ́lì** (ma) nyìnà wàn.
John PFV Mary PRT forget FOC
'John forgot Mary.'

c. *Jɔ́n à **Mɛ́lì** nyìnà wàn (ma).
John PFV Mary forget FOC PRT
'John forgot Mary.'

As was the case with canonical verbs, tense, negation, and wh-constructions do not alter the position of a complex predicate preceding its DP internal argument.

(14) a. Present Tense
Jɔ́n nyìnà dìmù **Mɛ́lì** mà.
John forget DIMU Mary PRT
'John forgets Mary.'

b. Negation
Jɔ́n ní má nyìnà **Mɛ́lì** mà.
John NI NEG forget Mary PRT
'John did not forget Mary.'

c. Wh-Question
Jɔ́n nà nyìnà **nyɔ̀** mà.
John PFV forget who PRT
'Who did John forget?'

d. Embedded Clause
Mélì à mín mbɛ́ Pìtá bèyà wàn **dèn mùsù nɛ́** fɛ̀.
Mary PFV hear C Peter follow FOC female child NƐ PRT
'Mary heard that Peter followed the girl.'

Thus far, we have argued that there are two distinct classes of verbs in Kono – canonical verbs and complex predicates, distinguished by the context in which

'obey', which all take post-verbal objects headed by a postposition. The verbs *ja* 'touch' and *ngi* 'remember' both take post-verbal objects headed by a preposition (see Smith 2022, 2024a for more on the two classes of complex predicates in Mende). Neither the list of Kono nor Mende verbs is comprehensive, and further description and analysis should be fruitful.

their internal argument surfaces. We turn next to the particle, investigating its lexical contribution, as well as its relationship to the verb and the argument.

We first highlight that different verbal predicates occur with different particles. For instance, the verb *chian* in (15) can only encode its internal argument with the particle *mà*. The verbs *butuneche* 'attack' (16) and *beya* 'follow' (17) likewise use the particles *à* and *fɛ́*, respectively. This points to a selection relationship between the verb and the particle.

(15)

Jɔ́n	chìán	dìmú	**Mɛ́lì**	mà	/ *à	/ *fɛ́.
John	fear	DIMU	Mary	PRT	PRT	PRT

'John fears Mary.'

(16)

Jɔ́n	à	bùtùnɛ̀chɛ́	yáá	à	/ *mà	/ *fɛ́.
John	PFV	attack	lion	PRT	PRT	PRT

'John attacked the lion.'

(17)

Jɔ́n	a	béyà	**Mɛ́lì**	fɛ́	/ *à	/ *mà.
John	PFV	follow	Mary	PRT	PRT	PRT

'John followed Mary.'

Crucially, each of these particles also functions as a postposition in the language. In the following data, we see that when they function as postpositions *mà* means 'on' (18), while *fɛ́* means 'beside' (19) and *ɔ̀* means 'at' (20).

(18)

Jɔ́n	nàà	mángò-nú	yén	fán	[tébù	**mà**].
John	PFV	mango-PL	see	FOC	table	on

'John saw the mangoes on the table.'

(19)

Jɔ́n	nàà	Mɛ́lì	yén	fán	[màchìtì	**fɛ́**].
John	PFV	Mary	see	FOC	market	beside

'John saw Mary beside the market.'

(20)

Jɔ́n	nàà	Mɛ́lì	yén	fán	[màchìtì	**ɔ̀**].
John	PFV	Mary	see	FOC	market	at

'John saw Mary at the market.'

As noted earlier, we assume that adpositions are able to assign Case. As such, we propose that the objects of complex predicates do not raise for Case, as their encoding particle is a postposition and assigns Case within its functional structure instead (Stowell 1981, McFadden 2004, Asbury 2008).

Our analysis of canonical and complex predicates is laid out as follows. The initial portion of the derivation of the sentence in (21) with the canonical verb *yen* 'see' is shown in Figure 3. We propose that the DP internal argument merges as the complement of the verb before raising into SpecKaseP for Case-licensing.[8]

(21) Pìtà nàà Mɛ́lì yén fán.
Peter PFV Mary see FOC
'Peter saw Mary.'

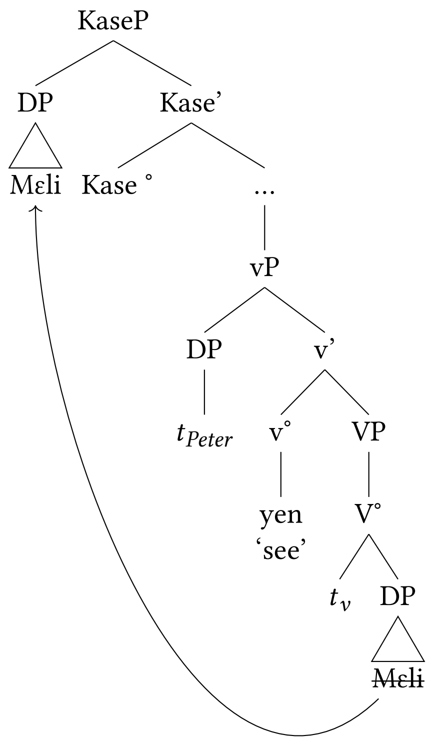

Figure 3: Derivation of canonical verb in Kono

The complex predicate *chian* 'fear' in (22) takes a post-verbal internal argument, encoded in an adpositional phrase. Paralleling the movement of the DP

[8]Two reviewers raise the question of where the Kase Phrase is located and how the DO arrives in that position. We acknowledge that the details of this need to be worked out. Our goal at this point is to simply propose that it is Case-driven movement which generates the OV word order. More research is necessary to understand the specific process which generates that order.

object of the canonical verb, we suggest that the DP object of a complex predicate merges as a complement to the encoding postposition, before raising into a specifier of its extended projection, a KasePhrase for Case licensing (Figure 4). Since it no longer needs to raise above the verb for Case-licensing, it yields a V [O Prt] surface structure.

(22) Pìtà nàà chìàn fán Mɛ́lì mà.
Peter pfv fear foc Mary prt
'Peter feared Mary.'

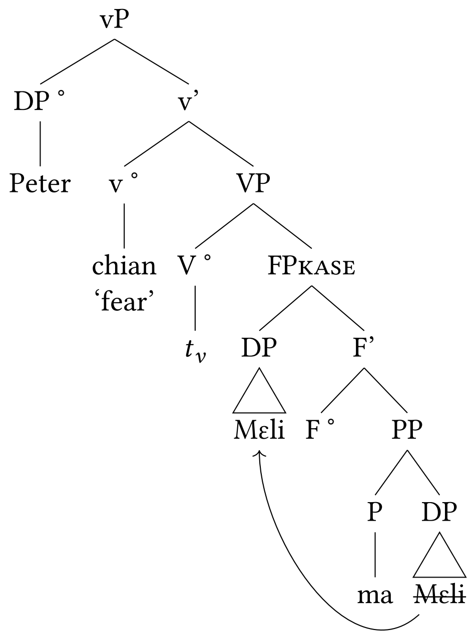

Figure 4: Derivation of complex predicate in Kono

5 CP complements

Thus far we have sought to account for the distinction between canonical (OV) and complex (VO) word orders in Kono. In this section, we turn to CP complements, showing that they obligatorily occur in a post-verbal position. The verb *min* 'hear' is a canonical verb, and (23) shows that its DP internal argument *faniya* 'the lie' must surface in a pre-verbal position.

(23) DP Object

Mɛ́lì	à	fànìyà	mín	fán	{*fànìyà	mà/à/fɛ́}.
Mary	PFV	lie	hear	FOC	lie	PRT

'Mary heard the lie.'

When the verb's complement is a CP, instead of a DP, it must surface in a post-verbal position (24). Adopting Stowell's (1981) Case Resistance Principle, we suggest that since the CP is [+tense], it does not need to receive Case.

(24) CP Object

Mɛ́lì	à	{mín}	[mbé	Jɔ́n	ànná	mángò-nú	dàún]	{*mín}.
Mary	PFV	hear	C	John	3.PFV	mango-PL	eat	hear

'Mary heard that John ate the mangoes.'

The verb *nyina* 'forget,' being a complex predicate, has a post-verbal object (25). When the verb in a complex predicate takes a CP complement, the CP surfaces in a post-verbal position, and, significantly, there is no postposition (26) .

(25) DP Object

Mɛ́lì	à	{*mángò-nú	mà}	nyìnà	{mángò-nú	mà}.
Mary	PFV	mango-PL	PRT	forget	mango-PL	PRT

'Mary forgot the mangoes.'

(26) CP Object

Mɛ́lì	à	nyìnà	[mbé	Jɔ́n	ànná	mángò-nú	dàún]	(*mà).
Mary	PFV	forget	C	John	3SG.PFV	mango-PL	eat	PRT

'Mary forgot that John ate the mangoes.'

These data are significant for two reasons. First, the post-verbal position of the CP aligns with our proposal that all verbal complements begin in a post-verbal position. Second, the absence of the postposition suggests that its purpose is Case-licensing. If the postposition Case-licenses the verb's internal argument in complex predicate constructions, there must be a mechanism to Case-license the internal arguments of canonical verbs, which we suggest is a KasePhrase whose null head licenses the DP argument that moves into its specifier. We propose similar structures for canonical and complex predicates with CP complements in Figure 5. Whether the verb is canonical, e.g. *min* 'hear' or complex, e.g. *nyina* 'forget,' it selects a CP complement, and since the CP does not need to be Case-licensed, it remains in a post-verbal position.

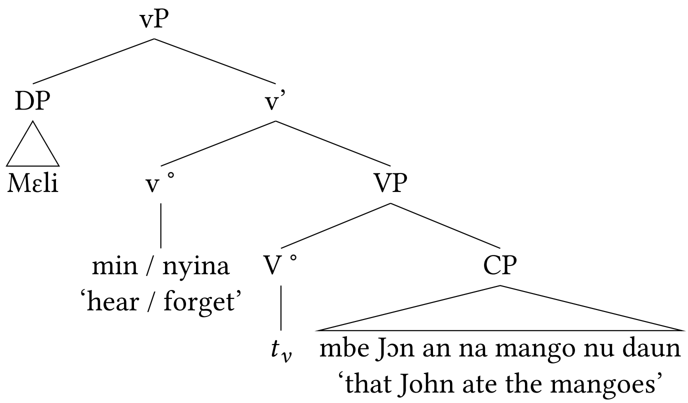

Figure 5: CP complement

6 Coordinated direct objects

We conclude our argument for the underlying head-initial structure of verb phrases in Kono by looking at some intriguing data related to coordinated direct objects. The verb *daun* 'eat' is a canonical verb, and in (27) it has a coordinated direct object *mangu-nu ni dumbi-nu* 'mangoes and oranges.' As expected, the entire coordinated DP can surface in a pre-verbal position in (27a). Intriguingly, the DP object can also be split, with the first conjunct surfacing pre-verbally, while the coordinator and second conjunct remain in a post-verbal position, as seen in (27b). It is ungrammatical for the entire coordinated direct object to remain in a post-verbal position (27c).

(27) a. Pre-verbal Object

Jɔ́n à **mángò-nù ní dùmbí-nù** dàún fán.
John PFV mango-PL and orange-PL eat FOC
'John ate the mangoes and oranges.'

b. Split Object

Jɔ́n à **mángò-nù {*ní}** dàún fán **{ní} dùmbí-nù**.
John PFV mango-PL and eat FOC and orange-PL
'John ate the mangoes and oranges.'

c. *Post-verbal Object

*Jɔ́n à dàún **mángò-nú ní dùmbí-nú**.
John PFV eat mango-PL and orange-PL
'John ate the mangoes and oranges.'

These data present an interesting puzzle. The basic facts are that one or both DPs can surface in a pre-verbal position, which points towards either the first DP or the entire coordinated phrase raising for Case via phrasal movement. This data aligns with our proposal that DP objects of canonical verbs begin in a post-verbal position and raise for Case. We suggest the derivations in Figure 6 and Figure 7, based on the data in (28). In both cases the canonical verb *daun* 'eat' takes a coordinated DP complement to its right. In Figure 6 the specifier of the coordinated phrase, that is the first conjunct *mangu-nu* 'the mangoes' raises for Case-licensing itself, while in Figure 7 it pied-pipes the entire coordinated phrase.[9]

(28) Pre-verbal Object
Jɔ́n à **mángò-nù** {**ní dùmbí-nù**} dàún fán {**ní dùmbí-nù**}.
John PFV mango-PL and orange-PL eat FOC and orange-PL
'John ate the mangoes and oranges.'

7 Conclusion

In this paper, we argued that Kono should be classified as canonically, but not strictly, OV. Though most verbs surface with a pre-verbal internal argument, we introduced a class of verbs whose internal argument obligatorily surfaces in a post-verbal position, encoded in a postpositional phrase. Based on work by Koopman (1992) and Smith (2022, 2024a), we suggested that OV word order is derived via leftward movement of the internal argument into the specifier of a KasePhrase. The internal argument of complex predicates does not need to raise, as it is Case-licensed by its postpositional head. Further evidence that this particle is a postposition is that it does not surface when the verb takes a CP complement, which does not need Case. We showed that complex noun-phrase-internal arguments of canonical verbs behave as predicted with the DP portion raising into a Case position, while the CP remains in-situ in a post-verbal

[9] A reviewer raised the question as to whether this analysis in which only one DP raises out of the coordinate structure points to a violation of the Coordinate Structure Constraint (e.g.(27b))? While we don't have a clear answer for that, we note that this type of construction with a split coordinated DP object occurs in multiple Mande languages, e.g. Mende (Smith 2024a). Mende permits wh-movement, and coordinate structures are strong islands (Smith 2024b). It could be that in Mande languages the CSC holds for Ā-movement but not A-movement, or it could be that these are not true coordinate structures, but rather comitatives. A further question is how the second conjunct is Case-licensed in this construction. Further research is necessary in order to tease out these distinctions.

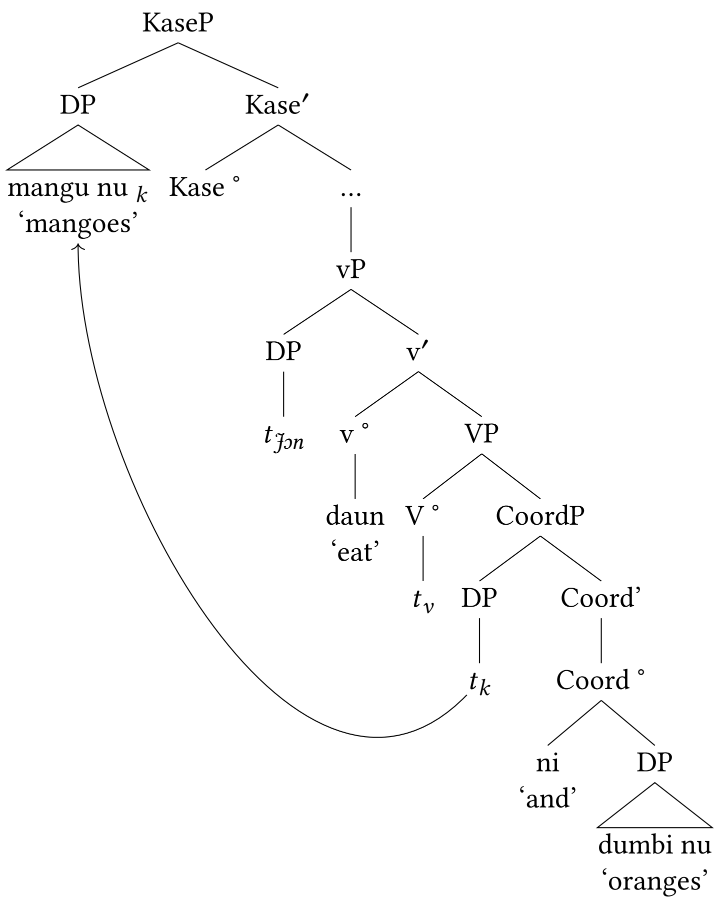

Figure 6: Split coordinated object DP

position. We concluded by looking at coordinated direct objects, where we show that at least one conjunct must raise into a pre-verbal position, ostensibly for Case-licensing. It is possible for both conjuncts to raise via pied-piping or for the coordinator and second conjunct to remain in a post-verbal position. How the post-verbal conjunct is Case-licensed is a subject for further research.

This data in this paper aligns with constructions found in many Mande OV languages, as seen in the following examples. In each case, the verb’s complement surfaces in a post-verbal position with a postposition.

(29) Mandinka (adapted from Creissels 2024: #3)

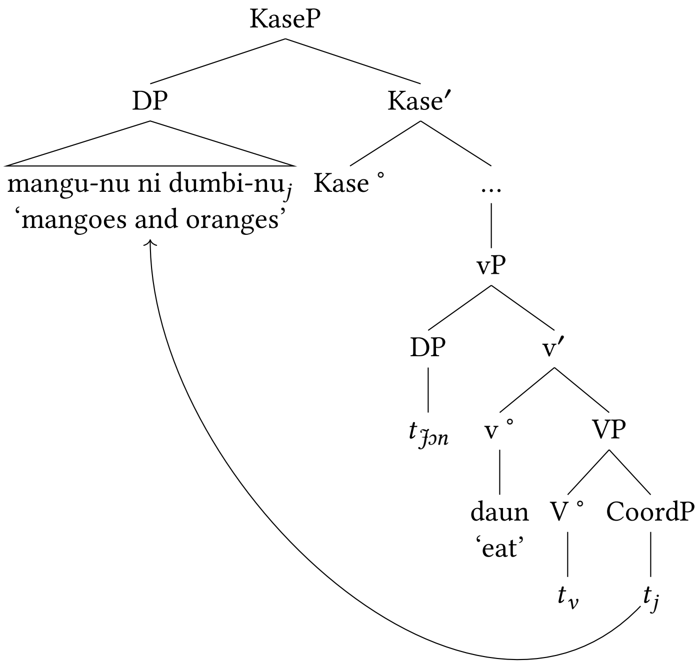

Figure 7: Pre-verbal coordinated object DP

Kèw-óo làfí-tà **kódòo** lá.
man-D want-CPL money-D P
‘The man wants money.’

(30) Lorma (adapted from Dwyer 1981: 81-82)[10]
Gè wɛ́lɛ́ **másságì**-và.
1SG see chief-P
‘I saw the chief.’

(31) Susu (adapted from Duport 1865)
Um nemu **a** ma.
1SG forget 3SG P
‘I forget him.’

(32) Mende

[10]Dwyer refers to the verb’s object as ‘the object of a postposition.’

Kpàná lémà-í lɔ́ **níkè-í-síà** mà.
Kpana forget-PST NF cow-DEF-PL P
'Kpana forgot the cows.'

We argue that this data points to a consistent structure across Mande languages, one which greatly resembles Germanic particle verbs, particularly the Danish data in (33b).

(33) Germanic (Larsen 2014, citing Svenonius 1996)

a. Norwegian
Vi slapp {ut} hunden {ut}.
we let out the.dogs out
'We let the dogs out.'

b. Danish
Vi slapp {*ud} hunden {ud}.
we let out the.dogs out
'We let the dogs out.'

c. Swedish
Vi slapp {ut} hunden {*ut}.
we let out the.dogs out
'We let the dogs out.'

In light of this striking similarity, we will be working on a detailed comparison of particle verbs in Germanic and Hungarian with similar constructions in the Mande languages.

Abbreviations

1	first person	NF	neutral focus
2	second person	NR	nominalizer
3	third person	PFV	perfective
C	complementizer	PL	plural
COP	copula	POST	postposition
DEF	definite	PRT	particle
FOC	focus	SG	singular
IMPERF	imperfect	SPF	specific
NEG	negative	VSUFX	verbal suffix

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Chapter 22

Focus and the association with focus sensitive particle *yerane* in Kasem

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This paper investigates the syntax of focus and the focus sensitive particle *yerane* 'only' in Kasem (Mabia, formerly Gur). I begin by discussing focus realization in Kasem, and sketched an analysis for it. Next, I discuss the syntactic position of the exclusive particle *yerane* by comparing it with the German's *nur*, and argue that unlike German's Extended Verbal Projection (EVP) claim by Büring & Hartmann 2001, Kasem's *yerane* is adnominal in nature.[1] However, I show that Büring & Hartmann's (2001) EVP analysis for *nur* may be extended to Kasem, but with an important additional assumption: that *yerane* is licensed by a high covert exclusive operator (Op_{only}) which occupies an EVP position, and is responsible for the exclusive semantics. Thus, following earlier studies, I assume that this is a bipartite analysis which requires that the exclusive interpretation comes from the high exclusive operator [Op] instead of the adnominal-*yerane* itself (see Bayer 1996, 1999, 2018, Lee 2004, Barbiers 2010, Quek & Hirsch 2017, Sun 2021, Branan & Erlewine 2023: a.o.). In order to link the high exclusive Op with the adnominal-*yerane*, I assume a purely syntactic agreement system.

1 Introduction

Because of their significance to the study and understanding of focus system and realization, focus-sensitive particles have received considerable attention, cross-linguistically. However, most of these studies are geared towards Indo-European

[1] The term *adnominal* is used interchangeably with *adfocal* in this paper. However, the latter is used in a more wholistic way, i.e. when discussing adjunction to both nominals and non-nominals.

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languages such as English, Dutch, German, French, Greek, and so on (cf. Rooth 1985, Rooth 1992, Bayer 1996, 1999, Kayne 1998, Barbiers 2010, 2014, Iatridou & Tatevosov 2016, Quek & Hirsch 2017). Some non-Indo-European languages that have equally received some amount of attention include Mandarin Chinese, Vietnamese, Japanese, and so on (cf. Lee 2004, Hole 2017, Erlewine 2014a,b, Erlewine & Kotek 2018, Sun 2021). African languages have received very little attention with regard to the nature of the focus-sensitive particles like *even*, *also* and *only*. See for instance, Hartmann & Zimmermann 2008, Grubic 2015 and Carstens & Zeller 2020 for studies on Bura, Ngamo and Nguni respectively. When it comes to the syntax of focus sensitive particles, African languages, or at least West African languages present interesting questions. This is partly because of the kinds of focus strategies that these groups of languages employ. For instance, while languages like English and German mark focus prosodically, most West African languages primarily mark focus morphosyntactically. There is usually a dedicated morphological focus marker that is present when an element is focalized. A focus constituent can also be fronted to the left periphery for interpretive purposes (Kiss 1998, Cruschina 2009, 2012, Bianchi et al. 2016). These different strategies and different focus positions leave us with questions such as what is/are the syntactic position(s) of focus sensitive particles like *even*, *also* and *only*?[2] Does the syntactic position of the focus sensitive particles suffice for the syntactic requirement of association with focus? This study aims to answer these questions by focusing on the exclusive focus sensitive particle (FSP) *yerane* 'only' in Kasem.

Let us begin with focus, which is the foundation for AwF. Consider the sentences below. The interrogative sentence in (1a) has a *wh*-phrase whose answer is a focused constituent in (1b).[3] The focus indicates the presence of other contextually determined set of alternatives such as [bread, avocado, banana, fries, …]. In other words, John could have eaten something else apart from a SANDWICH.

(1) a. What did John eat?
    b. John ate A SANDWICH.

If we add an FSP like *only* to the focus answer as in (2b), we get the assertion and the truth condition in (3). The presence of the FSP *only* does not change the assertion of the focus sentence. It however affects the truth condition of the

[2] Beaver & Clark 2008 categorizes these three FSPs as *Conventional Association Particles* because, according to them, they require the presence of a focus constituent in their scope. (cf. Erlewine 2014b: a.o.)

[3] Small caps is used to represent the constituents that are focused in the English translation.

sentence: the sentence is true iff John ate a sandwich and nothing else. FSPs act upon the salient alternatives that the focus evokes. What happens with *only* as an exclusive particle is that it excludes all other salient alternatives except the focus constituent.[4]

(2) a. Did John eat a sandwich and a banana?
    b. John ate *only* a SANDWICH.

(3) a. Asserts: John ate a sandwich
    b. Entails: John ate nothing else (apart from a sandwich)

Kasem is a Mabia (formerly Gur) language that is spoken predominantly in Northern Ghana, Burkina Faso, and Togo. It is an SVO language with noun classes. Interaction between the noun classes and number is reflected on the choice of the determiner (see Awedoba 1996). The sentence in (4b) shows a canonical word order in Kasem. The canonical word order structure is represented in (5); showing the post-verbal position of the internal argument and the pre-verbal position of the external argument respectively. The T head can be filled with past tense morphemes. When the aspect is imperfective, it is morphologically realized as a suffix to the verb as *-id* as in (4b). However, in perfective clauses, there is no overt morpheme. The left periphery of the clause (between ForceP and TP) houses the informational structural phrases like TopP and FocP, hence the three dots [...] in the structure in (5).

(4) a. Tell me something (that I need to know).
    b. Adam pa'a kɔdig-id nua.
       Adam IMM.PST slaughter-IPFV fowl
       'Adam was slaughtering a fowl.'

[4]The other conventional association particles (additive *also* and scalar additive *even*) do not affect the truth-condition of the proposition as in (i). They however presuppose that apart from the focus constituent (sandwich), there is at least one other alternative that is contextually relevant that John ate. The only difference between the two has to do with scalarity. While *also* is non-scalar (i.a), *even* is a scalar particle (i.b). For example, in (i.b), the presence of *even* presupposes that the focus associate is ranked high on the scale of what John is likely not going to eat (in comparison to other salient alternative(s)). In other words, while the focus associate is high on the relevant scale, the alternatives are ranked low. Thus, the lower we go on the relevant scale, the more likely the alternatives become (see a.o., König 1991, Jacobs 1983, Beaver & Clark 2008, Hole 2017).

(i) a. John also ate a SANDWICH .
    b. John even ate *only* a SANDWICH.

(5) [$_{ForceP}$ ... [$_{TP}$ DP$_{EA}$ [$_{T'}$ T [AspP V+v+Asp [$_{vP}$ ⟨DP$_{EA}$⟩ [$_{v'}$ ⟨V+v⟩ [$_{VP}$ ⟨V⟩ DP$_{IA}$]]]]]]]

Lastly, there are a few linguistic studies on Kasem, which makes it a "fertile ground" for linguistic research. See Awedoba 1996 and some of the references therein for a list of studies on Kasem. See also Bodomo 2020 and the other studies in that collection for a survey of Mabia languages in general.

Having given a brief introduction to focus, AwF, and the syntax of the language, the remaining part of this paper is based on the following sections. Section 2 begins with a description of focus realization in Kasem, and then an analysis would be proposed which sets the stage for AwF. In Section 3, I discuss the syntactic property of *yerane* and show that it indeed has an *adnominal* realization. A bipartite analysis with agreement between a high exclusive Op and *yerane* is provided in Section 4. Lastly, a summary and areas for further research are presented in Section 5.

2 Focus realization

2.1 Focus marking strategy

In Kasem, a constituent is marked as focused with the use of a morphological focus marker *mo*. *Mo* forms a constituent with the focus DP, PP, VP and adjunct. All these focus constituents can be realized either in their canonical position (in-situ) or at the left periphery of the clause (ex-situ). For instance, when an object *wh*-question like (6a) is asked, the answer can be in the form of the sentence in (6b) where the focus answer remains in its canonical position followed by *mo*, or it can be fronted to the left periphery as in (6c). The latter case also has *mo* following the focus constituent.[5]

(6) a. Bɛ *(**mo**) Adam go-a?
what FOC Adam slaughter-DISJ
'What did Adam slaughter?'

b. Adam go [chworo]$_F$ *(**mo**).
Adam slaughter fowl FOC
'Adam slaughtered A FOWL.'

[5] I will use the square bracket plus F-marking [XP]$_F$ to distinguish the focus constituent from the background. The focus marker **mo** will be in bold and it is obligatory.

c. [Chworo]$_F$ *(**mo**) Adam go-a.
fowl FOC Adam slaughter-DISJ
'Adam slaughtered A FOWL.'

The same strategy is used with regard to adjunct focus. The adverbial focus answer is followed by *mo*. The resulting constituent can be in an in-situ position (7b) or an ex-situ position (7c), respectively.

(7) a. Maŋ-kɔ̄ **mo** Adam go chworo kom?
time-which FOC Adam slaughter fowl DEF
'When did Adam slaughter the fowl?'

b. Adam go chworo kom [diim]$_F$ **mo**.
Adam slaughter fowl DEF yesterday FOC
'Adam slaughtered the fowl YESTERDAY.'

c. [Diim]$_F$ **mo** Adam go chworo kom.
yesterday FOC Adam slaughter fowl DEF
'Adam slaughtered the fowl YESTERDAY.'

With respect to V(P) focus, the focus marker *mo* immediately follows the verb if it is an intransitive verb (8b), i.e. without any internal argument. However, if the verb is transitive, *mo* follows the direct object instead (8c). The consequence of this is ambiguity. The focus could be on the verb itself, the whole verb phrase, or even the direct object. The only way to distinguish between what seems to be ambiguous on the surface is through the current question that the structure answers.

(8) a. Bɛ **mo** Adam kea dīīm?
what FOC Adam do yesterday
'What did Adam do yesterday?'

b. Adam [toŋe]$_F$ **mo** dīīm.
Adam worked FOC yesterday
'Adam WORKED yesterday.'

c. Adam [go chworo kom]$_F$ **mo** dīīm.
Adam slaughter fowl DEF FOC yesterday
'Adam SLAUGHTERED THE FOWL yesterday.'

The verb and/or verb phrase cannot be fronted in Kasem, whether the verb is doubled (9b) or not (9c). They remain in their external merged position when they are focused.[6]

(9) a. Bɛ mo Adam kea?
what FOC Adam do
'What did Adam do?'

b. *[Toŋe]$_F$ **mo** Adam toŋe.
worked FOC Adam work
'Adam WORKED.'

c. *[Toŋe]$_F$ **mo** Adam *t* .
worked FOC Adam
'Adam WORKED.'

2.2 Focus analysis

For the focus analysis, I assume that *mo* forms a constituent with the focus phrase. Consider the sentences below. They show the different possible placement of *mo* in the clause; it obligatorily follows whatever is the focus constituent. *Mo* follows the direct object in (10a), the indirect object in (10b), and the adverb in (10c). On the first glance, this examples suggest that *mo* forms a constituent with the focus. As a reviewer pointed out, the post-focal position of *mo* can still be derived via various movements in examples (10a) and (10b). For instance, assuming that *mo* is the head of a VP-internal FocP and the focused DP moves to Spec,FocP, we could derive (10b) by assuming that the direct object moves to a position higher than FocP.

(10) a. Adam pɛ [sabu kom]$_F$ **mo** John diim.
Adam give money DEF FOC John yesterday
'Adam gave THE MONEY to John yesterday.'

b. Adam pɛ sabu kom [John]$_F$ **mo** diim.
Adam give money DEF John FOC yesterday
'Adam gave JOHN the money yesterday.'

c. Adam pɛ sabu kom John [diim]$_F$ **mo**.
Adam give money DEF John yesterday FOC
'Adam gave John the money YESTERDAY.'

[6] This is unlike other Mabia languages like Kusaal and Likpakpaanl that have verb focus fronting only if the fronted verb is norminalized and have an overt copy in its base position.

While this is a plausible analysis, the challenge with such an analysis would be to determine the position that the direct object moves to. We would need to assume probably an FP to accommodate the direct object in order to derive the linear order. Such movements would be unmotivated.

Perhaps, example (10c) offers a better evidence for the constituent status of *mo*. Here, we have a case of adverbial focus with *mo* following it. Again, if we wish to derive such a structure by assuming a VP-internal FocP, then we would need two positions to move both the direct object and the indirect object into.

Ex-situ focus offers convincing evidence for our constituency claim here. The fact that each of the focus phrases that are in-situ in (10) can be fronted, as in (11), with an obligatory *mo* following them supports the view that both *mo* and focus phrases form a constituent.

(11) a. [Sabu kom]$_F$ **mo** Adam pɛ John diim.
money DEF FOC Adam give John yesterday
'Adam gave THE MONEY to John yesterday.'

b. [John]$_F$ **mo** Adam pɛ sabu kom diim.
John FOC Adam give money DEF yesterday
'Adam gave JOHN the money yesterday.'

c. [Diim]$_F$ **mo** Adam pɛ sabu kom John.
yesterday FOC Adam give money DEF John
'Adam gave John the money YESTERDAY.'

In order to derive the in-situ focus structure, I use the focus answer in (12b) as a template. The assumption is that *mo* adjoins to the focus DP and then projects a functional phrase called *mo*P (Figure 1).[7] Nothing hinges on the name of this projection. It can as well be called a DP. It is within the nominal domain, and the features on the focus DP are present at *mo*P.

(12) a. Bɛ *(**mo**) Adam go-a?
what FOC Adam slaughter-DISJ
'What did Adam slaughter?'

b. Adam go [chworo]$_F$ *(**mo**).
Adam slaughter fowl FOC
'Adam slaughtered A FOWL.'

[7]The number [3] is only an index notation that represent feature valuation. We can as well use *val* (which equally stand for a valued feature), à la Pesetsky & Torrego 2007.

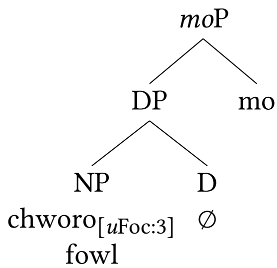

Figure 1: *Mo*-adjunction

Next, the structure is built up to the TP level. Then, a FocP is projected in the left periphery (Figure 2). The head of the FocP is null but has an interpretable unvalued feature. It is interpretable because it licenses the focus phrase and is responsible for the focus semantics. *Mo* is more or less the exponent for the valued feature on the focus DP. The unvalued feature on the focus head probes for valuation. Foc finds the focus DP which has a valued feature, and then Agrees with it. This results in a shared feature value [3].[8] I assume that in in-situ focus, Foc lacks an EPP. So, only agreement took place. The focus does not move to Spec,FocP, at least overtly.

A reviewer wonders how an unvalued feature can be interpretable. This may only seem to be an odd assumption based on Chomsky's (2000, 2001) initial view of Agree (specifically in the Minimalist Inquiries/Derivation by Phase framework) which is summed up with the *biconditional* statement below.[9] Based on (13), the prediction is that an uninterpretable feature cannot be valued, and an interpretable feature cannot be unvalued.

(13) **Biconditional of Valuation & Interpretability** (Chomsky 2001: 5)
A feature F is uninterpretable iff F is unvalued.

However, subsequent work (beginning notably with Pesetsky & Torrego 2007) has argued for the possibility of an intepretable but unvalued feature by eliminating the biconditional view of Chomsky. Pesetsky & Torrego's types of feature combination is giving below. The types in boldface are the novel types they added, and only the two types in the lower line can act as probes. In short, only unvalued features act as probes. Therefore, the assumption in this paper falls ex-

[8] Notice that the valuation index [3] is underlined. This is after valuation has taken place. Before valuation, it looks like this: [iFoc:_].

[9] MI = Minimalist Inquiry; DbP = Derivation by Phase

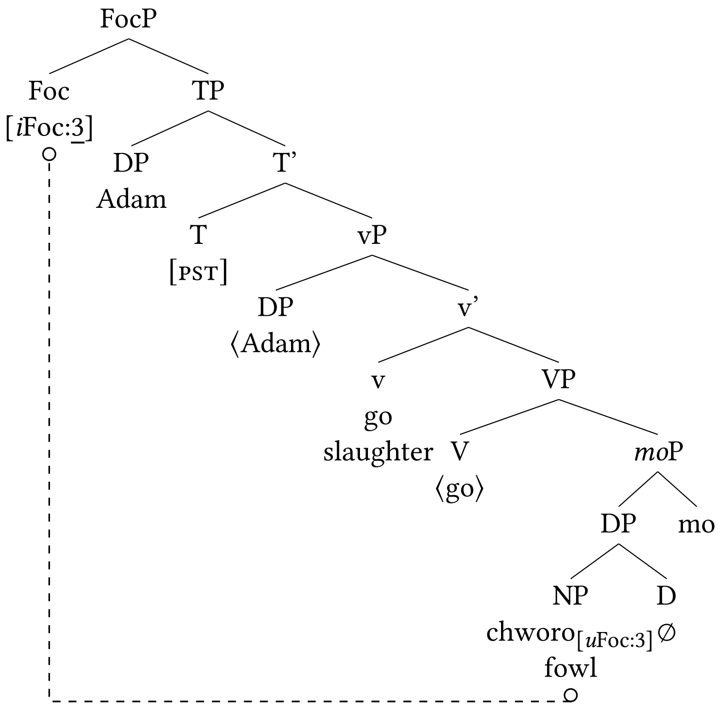

Figure 2: In-situ focus structure

actly within the purview of Pesetsky & Torrego's view of Agree (also Chomsky's based on unvalued feature = probe).[10]

(14) **Pesetsky & Torrego's (2007) types of features**
  ***u*F *val* uninterpretable**, valued   *iF val* interpretable, valued
  uF [ ] uninterpretable, unvalued   ***i*F [ ] interpretable, unvalued**

For the ex-situ focus strategy, the head of the left peripheral FocP is specified with an EPP feature which requires the fronting of the focus constituent. Consider the repeated ex-situ example below.

(15) [Chworo]$_F$ *(**mo**) Adam go-a.
  fowl FOC Adam slaughter-DISJ
  'Adam slaughtered A FOWL.'

[10]See Pesetsky & Torrego 2007, and the references they cited, for empirical evidence supporting such a wider view of Agree possibilities.

The structure is given in Figure 3. The head of FocP is still covert but has both an unvalued focus feature and an EPP. The unvalued focus feature causes the head to probe downward and Agrees with the goal (object focus). Next, the whole of *mo*P moves to Spec,FocP in the left periphery. So, agreement feeds movement.

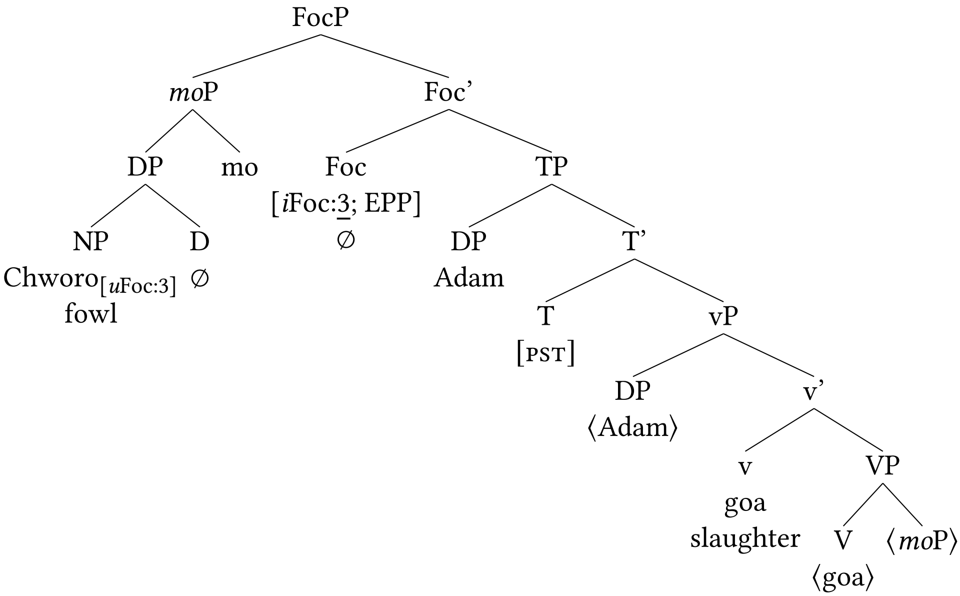

Figure 3: Agreement and focus fronting

So far, I have discussed the focus construction in Kasem and provided an analysis for it. The stage is now set for the next section which concerns the association with focus, with the exclusive focus-sensitive particle *yerane* ('only') as the case study.

3 FSP *yerane* and the association with focus

3.1 *Yerane* associating with focus

The exclusive focus-sensitive particle *yerane* and its focus associate are contiguous. *Yerane* follows the focus associate but precedes the focus marker *mo*. This position is constant regardless of whether the associate is a subject, an object, or an adjunct.[11] With a context like (16a), the answer involves a subject focus

[11]The focus categories that show a different distribution of *yerane* are VP and TP. For instance, when the focus is on a transitive verb, both *yerane* and *mo* follow the direct object, and not the

(16b) with the exclusive *yerane* occurring between the subject focus and *mo*.[12] The presence of the focus marker is obligatory in all cases.

(16) a. Wɔ mo toŋe? Adam *(mo) naa John *(mo)?
who FOC work.PFV Adam FOC DISJ John FOC
'Who worked: Adam or John?'

b. [Adam]$_F$ **yerane mo** toŋe.
Adam only FOC work.PFV
'Only ADAM worked.'

Association with an object focus has the same structure of [associate < yerane < mo]. This structure is retained when the focus associate is in a post-verbal position (17a) or when it is fronted (17b). The same thing applies to association with adjunct focus as in (18). This leads to the proposal that *yerane* further forms a complex constituent with the local FocP that was derived in Section 2. I will come back to this soon.

(17) a. Adam go [chworo]$_F$ **yerane** *(**mo**).
Adam kill.PFV fowl only FOC
'Adam only slaughtered A FOWL.'

b. [Chworo]$_F$ **yerane** *(**mo**) Adam go-a.
fowl only FOC Adam kill.PFV-DISJ
'Adam only slaughtered A FOWL.'

(18) a. Adam toŋe [diim]$_F$ **yerane** *(**mo**).
Adam work.PFV yesterday only FOC
'Adam only worked YESTERDAY.'

[[Diim]$_F$ **yerane** *(**mo**)]$_i$ Adam toŋe ___$_i$.
yesterday only FOC Adam work.PFV
'Adam only worked YESTERDAY.'

verb (i.b).

(i) a. Adam yeigi lɔɔre dem mo naa o kwei lɔɔre dem o yeigi mo.
Adam buy car DEF FOC DISJ 3SG pick car DEF 3SG sell FOC
'Did Adam **buy** the car or **sell** the car?'

b. Aawo, O [yeigi]$_F$ lɔɔre dem **yerane mo**.
no 3SG buy car DEF only FOC
'No, he only BOUGHT the car.'

[12] Just like the focus marker, I will use a boldface for the exclusive particle **yerane**.

3.2 Is *yerane* adnominal, adverbial or both?

Based on the structural position of the exclusive particle *only* in the clause, there are two groups of linguists who posit that *only* can either have an *adverbial* property (Jacobs 1983, 1986, Büring & Hartmann 2001) or a *mixed* property (Bayer 1996, 1999, 2018, Reis 2005, Barbiers 2014, Smeets & Wagner 2018). In other words, while the former involves *only* adjoining to an Extended Verbal Projection (henceforth EVP), the latter involves both an adverbial (EVP) and an adnominal adjunction of *only*. In German, for instance, Büring & Hartmann 2001 have argued that *nur* 'only' adjoins to the EVPs regardless of whether the focus is a DP argument or a non-DP argument (VP, TP, CP) (19) (see also Mursell 2021).

(19) Ich habe [$_{VP}$ **nur** [$_{VP}$ [$_{DP}$ einen Roman]$_F$ gelesen]]
I have only a novel read
'I have only read A NOVEL.' (Büring & Hartmann 2001: 231)

Due to the V2 property of German, the auxiliary verb *habe* occupies the V2 position, and causes the main verb *gelesen* to be in a sentence-final position. This makes it look as though *nur* is adnominal on the surface in the example above.[13]. However, according to Büring & Hartmann 2001: 233 if *nur* adjoins to a DP complement of a preposition (20) or inside a complex DP (21), the constructions are ungrammatical. The reasoning behind this is that these constructions should be fine if *nur* is adnominal.

(20) a. *mit **nur** Hans
with only Hans
'with only Hans'

b. *[$_{PP}$ P FP DP]

(21) a. *der Bruder **nur** des Grafen
the brother only the-GEN count-GEN
'the brother of only the count'

b. *[$_{DP}$ D FP DP]

In contrast, English, for example, uses the mixed-analysis where *only* can either adjoin to the focus associate itself (22a) or an EVP (22b) (König 1991, Erlewine 2014b).[14]

[13]This view of *nur* in German is not without its own challenges. In fact, there are others who have argued that German rather involves a mixed-analysis- adverbial and adnominal (see Reis 2005, Meyer & Sauerland 2009, Bayer 1996, 2018, Hole 2015, Smeets & Wagner 2018: among others). I will not go through their interesting arguments here.

[14]See also König 1991 on Bengali.

(22) a. John ate **only** [rice]$_F$.
b. John **only** ate [rice]$_F$.

Kasem's *yerane* is strictly adnominal with regard to its position. If we adjoin *yerane* to the DP complement of a preposition (as done for German above), the result is a grammatical structure (23b).

(23) a. Adam di wodiu kom de dúrú de seo diim na?
Adam eat.PFV food DEF with spoon CONJ knife yesterday Q
'Did Adam eat the food with a spoon and a knife yesterday?'
b. Aawo, o di wodiu kom [$_{PP}$ de [dúrú]$_F$ **yerane mo**]
no 3SG eat.PFV food DEF with spoon only FOC
diim.
yesterday.
'No, he only ate the food with A SPOON yesterday.'

However, this does not tell us much because of the position of *yerane* and *mo*: they seem to be in the clause-final position. It could be argued that they adjoin to an EVP. This is unlike English and German where *only* and *nur* precedes the focus DP but follows the preposition. Nevertheless, there is other evidence that supports the adnominal nature of *yerane*. First is the fact that the same complex constituent that we find in the postverbal (in-situ) position is also present in the left periphery of the clause (24).

(24) [[Chworo]$_F$ **yerane mo**] Adam go-a.
fowl only FOC Adam kill.PFV-DISJ
'Adam only slaughtered A FOWL.'

Secondly, the focus associate cannot be fronted leaving *yerane* stranded (25).[15]

(25) *[Chworo]$_F$ **mo** Adam go **yerane**.
fowl FOC Adam kill.PFV only
'Adam only slaughtered A FOWL.'

[15] A reviewer highlighted an important point that needs to be addressed here. In English, where *only* can adjoin to the EVP, if a constituent is moved out of *only*'s c-command domain (as in i), this would only mean that association between *only* and the dislocated element is lost (see Erlewine 2014a,b on why there is a lack of backward association with *only* in English). However, *only* can still associate with the verb or even the indirect object. But the same is not true for Kasem because *yerane* is adnominal in (25). Once its focus associate is fronted, stranding *yerane* does not result in association with the verb *go* 'to kill' in (25).

(i) A new bicycle John **only** loaned to Mary.

In contrast, the focus associate can move to the ‘prefield’ position in German, leaving *nur* behind. This should not be possible if both form a constituent. Consider the German sentence from Mursell 2021: 196 (citing Jacobs 1983: 43) below.[16]

(26) Der MENSCH interessiert ihn **nur**, (nicht die Arbeit).
the human interest.3SG him only not the work
‘Only the HUMAN interests him, not the work.’

Finally, there is indirect evidence from the language with regard to the behaviour of another FSP *maŋe* ‘even’.[17] The additive scalar particle *maŋe* behaves like an adverbial-*even* in English because it does not form a constituent with the focus associate (27b). In fact, it allows the focus associate to move to the left periphery of the clause while it remains in its pre-verbal position (27c) (see Erlewine 2014a and the references therein for the arguments on why backward association with *even* is possible in English). It does not move to the left periphery with the focus associate (27d). It behaves more like the German *nur* case, and not like *yerane*.[18]

(27) a. Napog nɛ vara dedɛ diim.
Napog see.PFV animal many yesterday
‘Napog saw many animals yesterday.’

b. O **maŋe** o na [teiri]$_F$.
3SG even 3SG see pig
‘He even saw A PIG.’

[16]Again, the whole picture is not as simple as it seems. Several studies have argued for the presence of ad-focal *nur* in German (see Reis 2005, Meyer & Sauerland 2009, Hole 2015: a.o.). But for the sake of our analysis here, I follow Büring & Hartmann 2001 to assume an adverbial *nur*.

[17]In the presence of the additive scalar particle *maŋe*, the focus marker *mo* is optional when the focus associate is in an in-situ position, but obligatory when the focus associate is fronted. Compare the examples (27b) and (27c).

[18]An interesting phenomenon is observed in association with *maŋe* ’even’, which is the presence of a lower copy subject pronoun. Why this is so is unclear at this point. A reviewer asked whether *maŋe* could be an auxiliary. This is a possibility because modal auxiliaries in the language have similar structures. For example, the sentence in (i) with the modal *wae* ‘may’ involves a pronoun which resumes the subject. This is an interesting phenomenon which I leave for a future research.

(i) Adam wae *(o) go chworo.
Adam may 3SG kill.PFV fowl
‘Adam may slaughter a fowl.’

c. [[Teiri]$_F$ *(**mo**)]$_i$ o **maŋe** o na ___$_i$.
pig FOC 3SG even 3SG see
'He even saw A PIG.'

d. *[[Teiri]$_F$ **maŋe** *(**mo**)]$_i$ o ___ o na ___$_i$.
pig even FOC 3SG 3SG see
'He even saw A PIG.'

So far, I have shown that the exclusive particle *yerane* is adnominal in Kasem. Next is the proposed analysis.

4 Proposed analysis

In this section, I will propose a syntactic analysis for *yerane*. I will show that Büring & Hartmann's (2001) EVP proposal may still be extended to Kasem. The only thing is that we need to assume that *yerane* is an ad-focal (embeds the adnominal property) operator whose semantics depends on a covert exclusive operator (Op_{only}) which can occupy the EVP positions proposed by Büring & Hartmann 2001.[19] I will begin by showing that this idea is technically a bipartite or concord analysis which is similar to the negative concord analysis (cf. Zeijlstra 2004: a.o.), and has also received cross-linguistic considerations (Quek & Hirsch 2017, Sun 2021, Branan & Erlewine 2023: a.o.).

4.1 Introducing the bipartite analysis

Considering languages like English that have a mixed-analysis for *only* viz: *adverbial*-only and *adnominal*-only, it has been argued that there is a higher position for *only* in the form of an exclusive operator (henceforth Op_{only}) that could be (c)overt. This is similar to what has been proposed for negation for example, where a higher Neg Op Agrees with a lower Neg(s) in languages with negative concord and/or negative indefinites (see a.o., Zeijlstra 2004, 2008, 2011, Penka 2011). This is a case of doubling. In the case of *only*-doubling, it is argued that the complex constituent of the *adnominal*-only plus its focus associate raises to the Spec of a higher Op_{only} which is seen as a scope position. This movement could either be *covert* or *overt* as shown in Figure 4 (cf. Bayer 1996, 1999, Kayne 1998,

[19] In fact, this is also the position of Hole 2015 for German. Hole 2015 argues that the German *nur* have instances of both adverbial (v-level, in his words) exclusives and ad-focal exclusives where either of them can be covert depending on a rule he called *First Come, First Spell-Out*. See particularly Hole 2015 section 3.3, for nice arguments.

Erlewine & Kotek 2018, Sun 2021). Thus, the exclusive interpretation actually comes from the Op$_{only}$, and the *adnominal*-only is only a semantically vacuous concord marker.

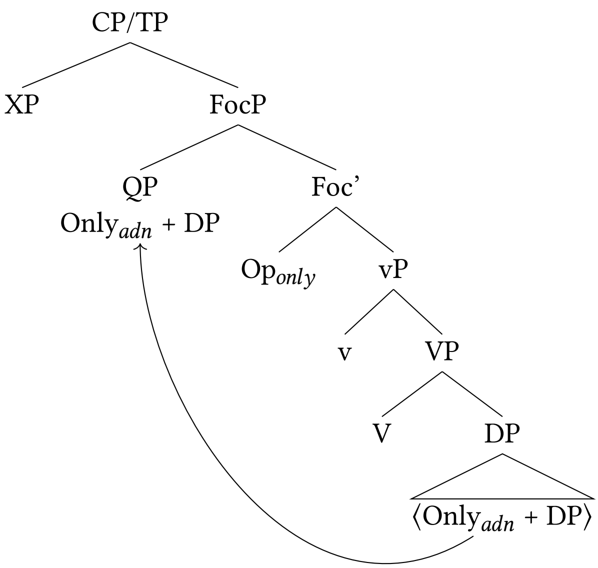

Figure 4: (C)overt raising of *adnominal*-only + DP

Another approach that has been proposed for the bipartite analysis is (28). Under this approach, the *adnominal*-only enters into the syntax with an uninterpretable exclusive feature [*u*Excl], and Agrees with the Op$_{only}$ which has an interpretable exclusive feature [*i*Excl]. The agreement occurs under a Spec-Head relationship (28) (Sun 2021, Hole 2017, Quek & Hirsch 2017, Yip to appear). The advantage of these approaches is the provision of a unified analysis in which 'only', as a propositional operator, is really adverbial and is interpreted in an EVP position in the sense of Büring & Hartmann 2001. The adfocal particles are more or less concord-like elements.

(28) [CP/TP [FocP [QP only$_{[u\text{Excl}]}$ + DP]$_i$ [Op$_{[i\text{Excl}]}$ [vP … t$_i$]]]]

One argument for these approaches is the ability of the *adnominal*-only to realize an ambiguous scope reading in the presence of modals (cf. Taglicht 1984, Bayer 1999, Quek & Hirsch 2017). Although, on the surface, the position of the *adnominal*-only is lower in the clause, it can scope not only below the modal, but also above it. The sentence in (29) can have the two readings in (30), depending on the scope of *only*.

(29) John *may* slaughter *only* A HEN. [only > may; may > only]

(30) a. [may > only]:
John is allowed to not slaughter any animal other than a hen.
b. [only > may]:
John is not allowed to slaughter any animal other than a hen.

Not only modals, but also other quantificational elements like negation and quantificational adverbs can trigger a ambiguous scope reading for *adnominal*-only. With this in mind, it therefore seems that there is a null Op in the clausal spine that accommodates the observed ambiguity. This has been observed in other languages such as Vietnamese, Mandarin, Cantonese, German, and Dutch (see a.o., Bayer 1996, 1999, Lee 2004, Barbiers 2010, 2014, Hole 2017, Sun 2021, Yip to appear).

4.2 Extending the bipartite analysis to Kasem

The bipartite analysis can be extended to the Kasem AwF case. The adnominal-*yerane* can have an ambiguous scope reading. First, I will discuss the ambiguous scope reading in relation to adnominal-*yerane* with a modal and a higher predicate. Secondly, I will adopt a downward Agree system to the bipartite analysis (and not Spec-head Agree), as this will account for the post-verbal AwF context.

4.2.1 Adnominal-*yerane* with modals

In (31), the presence of the modal auxiliary *wae* 'may/can' creates an environment for the ambiguous scope reading. The exclusive particle could either scope above or below the modal verb (32). This supports the claim for the presence of an empty exclusive *Op* in a higher position, which means that it is not *a what you see is what you get* case. The ambiguous scope suggests that the exclusive interpretation comes from a higher position instead of the adnominal position of *yerane*.[20]

[20]It is important to point out here that the ambiguous scope property of *yerane* may also be analyzed with a quantifier-raising approach, instead of a bipartite agreement analysis that is proposed in the next section. However, Aremu to appear argues against such a QR-approach with strong evidence. See Aremu to appear for an extended discussion.

(31) [only > may; may > only]
Adam wae *(o) go [chworo]$_F$ **yerane** *(**mo**).
Adam may 3SG kill.PFV fowl only FOC
'Adam may slaughter only A FOWL.'

(32) a. [may > only]:
Adam is allowed to not slaughter any animal other than a fowl.

b. [only > may]:
Adam is not allowed to slaughter any animal other than a fowl.

4.2.2 Adnominal-*yerane* with higher predicates

The ambiguous scope is also realized when an adnominal-*yerane* occurs within the infinite CP complement of a higher predicate like *require* (à la Taglicht 1984). In the data in (33), *yerane* adjoins to the object of the embedded non-finite verb. The two possible readings that are derived are given in (34).

(33) [only > require; require > only]
Adam jege se o di [mumuna]$_F$ **yerane mo**.
Adam require to 3SG eat rice only FOC
'Adam is required to eat only RICE.'

(34) a. [only > require]:
The only requirement is that Adam eat rice (no other requirement).

b. [require > only]:
The requirement is for Adam to eat only rice.

4.2.3 Agreement

I will now propose an agreement system for my analysis. The agreement mechanism provides the possibility of linking different syntactic positions for interpretive reasons. Usually, these positions carry identical feature(s) which can be *interpretable* or *uninterpretable*. When agreement takes place, the result is that two (or more) syntactic elements or positions share the same feature that Agree acted upon. An important implication of Agree is that it creates non-local syntactic relations (Chomsky 2000, 2001).

In order to apply the agreement mechanism, there is a need to first propose a syntactic structure for AwF. I will use the in-situ object focus in (35) as a template.

(35) Adam go [chworo]$_F$ **yerane mo**.
Adam kill.PFV fowl only FOC
'Adam only slaughtered A FOWL.'

As indicated earlier, I assume that while the adnominal-*yerane* projects a **Particle Phrase (PrtP)** above the focus associate, but below *mo*P, the exclusive Op$_{only}$ heads an **Exclusive Phrase (ExclP)** that is merged between the CP/TP and the vP Figure 5. In Figure 5, the Prt head is specified with a valued uninterpretable exclusive feature [uExcl:2]. The Excl head (Op$_{only}$) on the other hand, has an unvalued interpretable exclusive feature [iExcl:__]. The idea is that the exclusive interpretation comes from the null Op$_{only}$. The agreement is on the basis of feature valuation. Thus, it is the unvalued feature on Excl that serves as the probe. It probes downward for a suitable goal and finds the valued Prt head. The latter then values the exclusive feature on the Excl head.

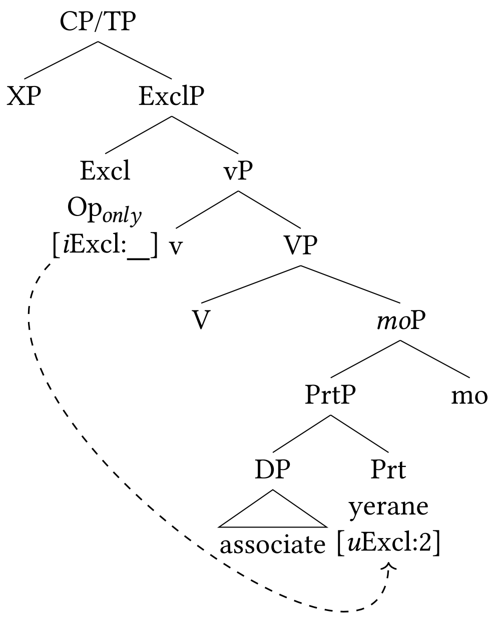

Figure 5: Agreement of Op$_{\text{only}}$ and *yerane*

The tree in Figure 5 shows when the probe searches for the goal, and Figure 6 shows when the value has been copied to the probe in the form of [2]. The agreement mechanism here maintains a downward Agree as in the focus derivation.[21] Similarly, feature valuation serves as the probe.

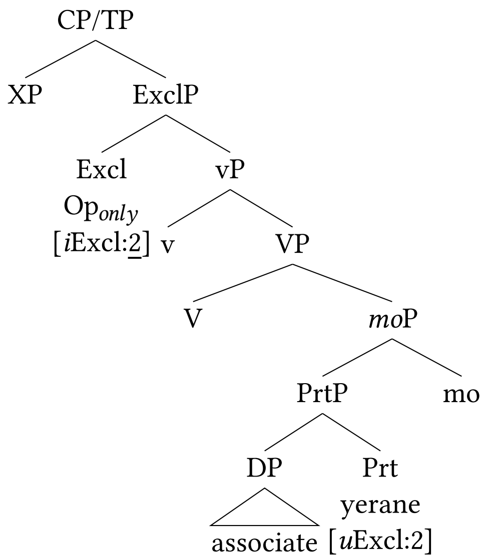

Figure 6: AwF structure after feature valuation

The tree in Figure 7 shows what a full TP structure with an association with direct object focus looks like.

4.3 Extending to other Mabia languages

Although this study situates Kasem in a pool of other studies that have adopted the bipartite analysis, it is the first of its kind among the Mabia languages as far as I know. The prediction is that the analysis that is proposed here for Kasem may be extended to the other Mabia languages. For example, Kusaal, another Mabia language, behaves similarly to Kasem with regard to the behaviour of 'only'. In Kusaal, the adnominal-'only' is *ma'aa*. It always follows the focus associate whether in an in-situ position (36a) or in an ex-situ position (36b).[22] The only difference is that while the in-situ focus marker follows the focus associate

[21]Unlike a few of the previous studies (cf. Sun 2021, Yip to appear: a.o.), here there is no movement of the adnominal-*yerane* to Spec,ExclP for Spec-head agreement, or an upward Agree between *yerane* and Op$_{only}$.

[22]In Kusaal, the in-situ focus marker is *nɛ*, while the ex-situ focus marker is *ka*, as seen in the examples.

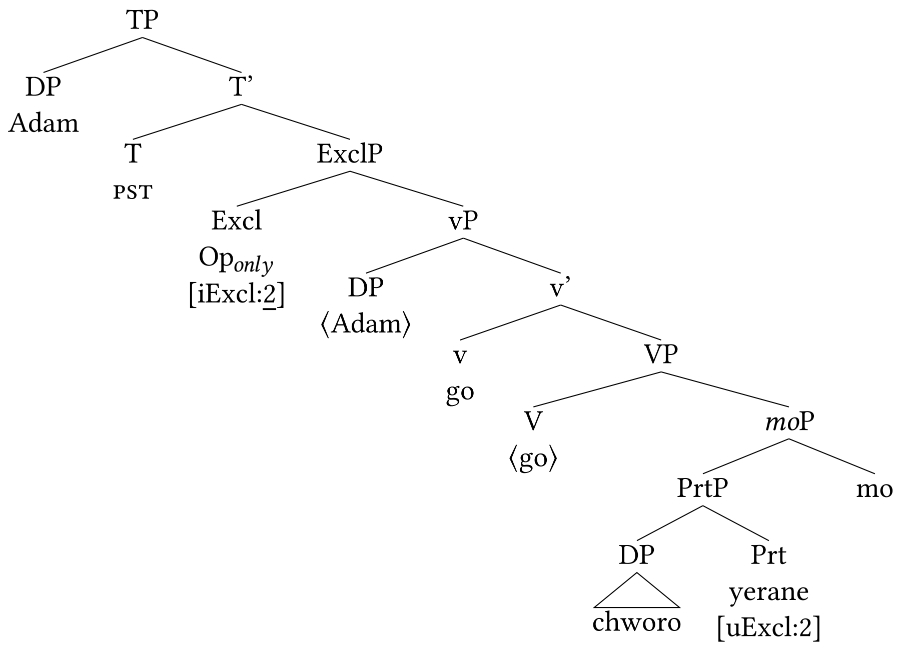

Figure 7: Full AwF structure

and the exclusive particle in Kasem, the in-situ focus marker precedes the focus associate and the exclusive particle in Kusaal (36a). I do not think that this constitutes any problem for the bipartite analysis that I assume in this paper. So, I propose that the same analysis can be extended to Kusaal, and other Mabia languages as well.[23] I leave this for a future study.

(36) a. Adam kɔdig **nɛ** [nua]$_F$ **ma'aa**.
Adam kill.PFV FOC fowl only
'Adam slaughtered only A FOWL.'

b. [Nua]$_F$ **ma'aa ka** Adam kɔdig.
fowl only FOC Adam kill
'Adam slaughtered only A FOWL.'

[23] See Abubakari 2018, 2019 for some descriptions of the Kusaal focus system and association with focus.

5 Conclusion

In this paper, I have described the syntactic property of the exclusive focus sensitive particle *yerane* in relation with narrow focus. As a way of setting the stage for the discussion on association with focus, I first described the focus marking strategies in Kasem, and went further to propose a syntactic analysis for focus in the language. I argued that the contiguous nature of the focus marker *mo* in the language allows for a constituent analysis with the focus DP. In other words, the focus marker forms a complex constituent with the focalized element. Having done this, I introduced the data on the exclusive particle *yerane* and argued that it has an *adnominal*-only syntactic positioning. This is in contrast to the *adverbial*-only analysis that has been proposed for *nur* in German, for example (Jacobs 1983, Büring & Hartmann 2001, Mursell 2021). I argue that we can maintain the EVP proposal only by assuming that *yerane* is an ad-focal particle which is licensed by a high covert exclusive Op_{only}. This leads us to a bipartite analysis which means that there is a high exclusive Op which heads an ExclP and is responsible for the exclusive interpretation. *Yerane* on the other hand is a semantically inert functional head of PrtP.[24] Evidence for such a claim comes from the ability of the *adnominal*-only to trigger an ambiguous reading in the presence of modals and higher predicates. Thus, in order to link the Op_{only} to *yerane*, I assume an Agree system which requires the latter (which is uninterpretable but valued) to value the former (which is interpretable but unvalued). At the end, I predicted the possibility of extending the analysis that was developed for Kasem to the other Mabia languages. Finally, this study provides cross-linguistic support (from a West African language) for the bipartite approach to the syntax of *only*.

Abbreviations

AwF = association with focus
DEF = definite
DET = determiner
DISJ = disjunct
EPP = extended projection principle
EXCL = exclusive
FOC = focus
IMM = immediate
IPFV = imperfective
N = noun
NEG = negation, negative

[24]Whether the adfocal particle is indeed semantically inert is a topic for a future research. But, for now, I leave it as such.

PFV = perfective
PL = plural
PLA = principle of lexical association
PRT = partcle
PST = past
Q = question particle
QP = quantifier phrase
SG = singular
TOP = topic
3 = third person

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Pamoja tena ‘Together again’

The papers in this volume were selected from those presented at the 54th Annual Conference on African Linguistics, which was held at the University of Connecticut in June, 2023. The volume contains 22 papers with diverse topics within phonology, morphology, syntax, language acquisition, and historical linguistics. The African languages that are investigated in this volume include Kom, Igbo, Dagaare, Igala, Kalenjin, Rɨ̀kpá, Ekhwa Adara, Hausa, Nduga, Maa, Lugwere, Emai, Seenku, Kirundi, Chichewa, Tiriki, Kono, and Kasem.